

<div>
<id>p001r_1</id>
<ab>
Master Nicolas Coste[a][b], in the <pl><fr>Rue de la Heaumerie[c]</fr>, at the sign of St Claude</pl><figure></figure>Mirrors<lb>
Master Jehan Cousin[d], who resides in the <pl><fr>Faubourg St Germain</fr></pl>, knows about the master<lb>
Master Jehan Garnier, in the <pl><fr>Rue des Escrivaains</fr></pl> by <pl><fr>St Jacques de la Boucherie</fr></pl>, <pro>currier</pro>. Try <m>woad[e] flowers</m>
</ab>
</div>

<div>
<id>p001r_2</id>
<ab>
The <pro>harvester</pro> that leaves some ears is not blamed[f].<lb>
<la>Sacra Eleusinae deae[g] propalare nefas[h].</la><lb>
Trumpets, see the book of funerals[i]</ab>
</div>

<div>
<id>p001r_3</id>
<la><ab>Vvolfgangus Lazius[j]<lb><[k][l]
Petrus Appianus mathematicus[m][n] Ingolstadiensis, Comment. <pl>urb. rom[o][p].</pl></ab>
<la><ab>Hieronimo Ruscelli[q]<lb>
Hermolaus Barbarus[r][s]<lb>
Angelius Bargaesus, De <al>aucupio</al> et <al>venatione[t][u]</al><lb>
Nicolaus Damascenus[v]</ab>
<la><ab>Caes {ariensis}[w], De rebus Persarum[x]<lb>
Isidorus[y][z][aa]</ab>
<la><ab>Orosius[ab]<lb>
Eupolemus, historicus gentilis qui de rebus Davidis & Salomonis scripsit[ac]</ab>

<ab><figure></figure><fr>Catalogue[ad] des villes[ae]<lb>
Calendrier des <pro>bergiers[af]</pro><lb>
Grammaire <pl>italie</pl>ne[ag]<lb>
Arithmetique de Savonne[ah]<lb>
Instruction pour le fait des finances[ai]<lb>
Questions aenigmatiques[aj]<lb>
Des praeceptes d'agriculture[ak]<lb>
<figure>+</figure>Le[al] secret des finances à <pl>Lyon[am]</pl></fr><lb>
Synesius[an][ao]<lb>
Olaus Magnus[ap]<lb>
<figure>+</figure>Master[aq] Bernard Palissy[ar], <pro>inventor</pro> [as]of rustic figurines to the king and the queen mother.</ab>
</div>

<div>
<id>p001r_4</id>
<la><ab><al>Aquatilium animalium</al> historiae, Hypolito Salviano Typhernate authore, <pl>Romae</pl> 1554[at].</ab>
</div>

<div>
<id>p001r_5</id>
<fr><ab>Les Annales de <pl>Normandie[au]</pl></ab>
<la><ab>Alexander Aphrodisaeus[av]<lb>
Polydorus Vergilius[aw]<lb>
Appianus[ax]<lb>
Athenaeus[ay]<lb>
Pausanias<lb>
Statius Thebaidos[az]<lb>
Servius, In Aeneidem[ba]<lb>
Macrobius[bb]<lb>
Aulus Gellius<lb>
Alexander ab Alexandria<lb>
Festus[bc]<lb>
Nonius[bd][be]<lb>
Magius, Miscellanea[bf][bg]<lb>
Pollux, Onomasticon[bh]<lb>
Hyginus[bi]<lb>
Berosus[bj][bk]<lb>
Suetonius<lb>
Valerius Maximus<lb>
Cornelius Tacitus<lb>
Xenophon<lb>
Seneca<lb>
Dionisius Halicarnassensis[bl]<lb>
Sabellicus[bm]<lb>
<add>cum permultis<lb>aliis </add>Julius Capitollinus[bn][bo]<lb>
Budaeus[bp][bq]<lb>
Spartianus[br]<lb>

Blondus[bs]
Volaterranus[bt]
Herodotus
Paulus Manutius[bu][bv]
Strabo
Julius Firmicus[bw][bx]
Quintus Curtius[by]
Dio[bz][ca]
Cornelius Nepos
Flavius Vopiscus[cb]</ab>

<fr/><ab>Monsieur[cc][cd]</ab>
</div>

[a]Pamela has his ID

[b]Nicolas Couste, the grandson of Jacques Couste a "master painter and bourgeois of Paris," and son of Nicolas Couste, also a painter. Jacques Couste was the brother of Christine Rousseau, Jean Cousin (pere's) wife. Nicolas Couste, fils was an armorer in the workshop of Jean Daussonne (also a family relation of Jean Cousin, fils). From: Maurice Roy, Jehan Cousin père, sculpteur : la statue de l'amiral Chabot et le jubé de la chapelle de Pagny, par Maurice Roy ? 1909; Maurice Roy, Artistes et Monuments de la Renaissance en France: Recherches Nouvelles et Documents Inédits, 1929; Henri Zerner, L'Art de la Renaissance en France (Flammarion, Paris: 1996); Guy Michel-Leproux, La Peinture a Paris sous le rene de Francois I ch. 4 "Les debuts de Jean Cousin a Paris"; Jules Guiffrey, Artistes Parisiens des XVI et XVIIe Siecles, 1915. Research by Etienne Stockland, 2012.

[c]Close to Rue des Écrivains mentioned below.

[d]Jean Cousin the younger, famous painter, c.1522-c.1594 (check dates)

[e]lydiamayhansell: "Isatis Tinctoria" (commonly called 'dyers' woad")

[f]This phrase and the following occur together later (p166r_b1a) as parts of the text "pour la Boutique"

[g]Ceres / Isis

[h]Cf. Cornelius Agrippa, De Occulta Philosophia, Cologne 1533, lib. III, cap. 2, "De silentio et occultatione eorum quae secreta in religione sunt" (?): "Numenius etiam quidam, occultorum curiosior, offensam numinum contraxit, quod Eleusinae deae sacra inter-pretando evulgavisset"[http://bivio.filosofia.sns.it/bvWorkPage.php?pbSuffix=64%2C866217]

[i]Claude Guichard (1545?-1607), Funérailles et diverses manieres d'ensevelir des Romains, Grecs, et autres nations, (Lyon) Jean de Tournes, 1581, p. 48-52: detailed discussion of trumpets in ancient funerals.

Note that this passage of Guichard mentions many of the authors listed below: Statius (Theb.), Virg (Aen.), Pers., Appian., Hygin, Macr., Alex. Aphr., Polyd., Festus, Spart., Volaterr., Florus, Corn. Nep., Val. Max., Tac., Dio, Suet. ...

[j]lydiamayhansell: Latinized version of Wolfgang Laz (1514-1565), Austrian physician and humanist

[k]Re-establishing the line breaks between items to convey the list format.

[l]of course, no idea who removed them

[m]lydiamayhansell: Petrus Appianus, Tutor to Charles V

[n]Some possibility of a confusion between the father, Peter, and the son, Philipp, an Ingolstadt-born mathematician

[o]? Unidentified. Unrelated to Appianus?

[p]Maybe Raphael Volaterranus, Commentariorum Urbanorum libri (a widespread encyclopedia, with sections on pigments etc.)

NB Several recipes on p046v are referenced with "Urb". Try and identify in Volaterranus

[q]Girolamo Ruscelli (c.1518-1566), a prolific polygraph. Probably mentioned here for his Secreti del reverendo donno Alessio Piemontese, Venice, Sigismondo Bordogna, 1555, etc.; French translation, Les secrets, contenant remedes contre plusieurs maladies, playes et autres accidens, avec la maniere de faire distillations, parfuns, confitures, teintures, couleurs et fusions, Anvers, Christophe Plantin, 1557, etc.

[r]lydiamayhansell: (1454-1493) Italian Renaissance Humanist. Authoritative work on Pliny with 5,000 corrections

[s]Ermolao Barbaro (1454-1493), a prolific Venetian humanist. See especially, Naturalis scientiae totius compendium ex Aristotele et aliis philosophis, Francfort, Christian I Egenolff, 1550, etc.

[t]Pietro Angeli da Barga, De aucupio liber primus..., Florence, heirs of Bernardo I Giunta, 1566. Maybe mixed up with (or bound with?) another book on (aucupium &) venatio.

[u]The other book may have been the De Venatione libri IIII, a translated version of Oppianus' text by Jean Bodin (Paris: Michel de Vascosan, 1555). This text, like Pietro Angeli da Barga's text, is quarto size. Alternatively, Girolamo Ruscelli (mentioned two lines above in the MS) was involved the production of a different version of the De Venatione libri IIII translated by Natale Conti (Venice: 1551). The full title page reads: Natalis Comitum Veneti De Venatione, Libri IIII Hieronymi Rvscellii Scholiis revissimis illvstrati. However, Conte's De Venatione is octavo size, eliminating the possibility that it was bound with Pietro Angeli da Barga's text.

[v]Nicolaus Damascenus, historian and philosopher (1st c. BC), Ex Nicolai Damasceni Universali historia, seu De moribus gentium excerpta JOHANNIS STOBAEI collectanea, [Geneva, Jacob Stoer for] Pierre de Saint-André, 1593. FIND earlier edition.

[w]Ms. has: Cares.

[x]Procopius Caesariensis,

De rebus Gothorum, Persarum ac Vandalorum libri VII, Basel, Hervagius, 1531 (first published as De bello Persico, Rome, 1509, 1516). USTC 686905 jumbles up the title.

[y]lydiamayhansell: Isidorus of Seville

[z](Usually Isidore rather than Isidorus in English)

[aa]Should names be Anglicized and/or book titles be translated?

[ab]Ms. has: Osorius.

Paulus Orosius (5th c.), presumably his Historiae adversus paganos, Augsburg, Johann Schüssler, 1471, etc.; many editions, mainly under the title Adversus paganos historiarum libri septem.

Context makes this misspelling more plausible than a reference to the Portuguese theologian and historian Jerónimo Osório (although the translation of his Histoire de Portugal was printed in Paris and Geneva in 1581).

[ac]Eupolemus, Greek Jewish historian (2nd c. BC) Source of quote should be identifiable. Cf. Andreas Masius, Josuae imperatoris historia illustrata atque explicata, Antwerp, Plantin, 1574, p. 635 (index): "Eupolemus scripsit acta Davidis & Salomonis"

http://books.google.fr/books?id=A7a5_Dqe-IsC&pg=PA635

[ad]lydiamayhansell: - indicated before the start of this word

[ae]Gilles Corrozet, Le catalogue des villes et citez assises es troys Gaules, Paris, Denis Janot, 1539; 1540, 1543, 1575.

Rather than his Le catalogue des antiques erections des villes et citez, 1st edn no place no date; 1536, 1539, 1540, 1543, 1551

[af]Paris, Guy Marchand, 1491, and later editions from various printers. throughout the 16th c. Illustrated calendar.

[ag]Possibly Jean-Pierre de Mesmes, La grammaire italienne composée en françoys, Paris, Gilles Corrozet, 1548 ; 1567, 1568, 1581.

[ah]Pierre de Savonne, Arithmetique, Paris, 1563 ; 1565 ; 1571, 1585 (etc.).

[ai]Philibert Boyer, Instruction pour le faict des finances, Paris, Ambroise Drouart & Guillaume Le Noir, 1581

[aj]Antoine Du Verdier, Questions enigmatiques, recreatives et propres pour deviner et y passer le temps aux veilles des longues nuicts, avec les responses subtiles, & autres propos joyeux, Lyon, Benoît Rigaud, 1568.

[ak]Cassianus Bassus, Selectarum praeceptionum de agricultura..., Basel, Hieronymus I Froben & Nikolaus I Episcopus, 1538 (NB the French translation, Paris & Lyon 1550, has "enseignemens", not praeceptes. so he is presumably referring to the Latin edn)

[a]lydiamayhansell: + indicated before this word

[am]N. Froumentau, Le secret des finances de France, decouvert, & departi en trois livres..., no place, 1581.

[an]lydiamayhansell: Synesius of Cyrene

[ao](370?-413?), bishop of Cyrene. Many works.

[ap]1490-1557. Main publication: Historiae de gentibus septentrionalibus libri XXII, Antwerp, Jean Bellère, 1552 ; 1555, 1558, etc. etc. ; Fr. transl., Antwerp 1561, etc

[aq]lydiamayhansell: Cross indicated before this word

[ar]lydiamayhansell: Bernard Palissi (c.1510 - c.1590) - French Huguenot potter

[as]This is how Palissy calls himself (without "maistre" and with a minor variant: "...royne sa mere") on the title page of his Discours admirables..., Paris, Martin Le Jeune, 1580

[at]Ippolito Salviani, Aquatiliū animalium historiae, liber primus, cum eorundem formis, aere excusis, Rome, the author, 1554.

[au]? Possibly Jean Nagerel, L'histoire et cronique de Normandie, Rouen, Martin Le Mégissier, 1581, etc. (and earlier edns not in USTC), among other histories of Normandy, none of which seem to bear the title Annales. — Needs further investigation

[av]Alexander Aphrodisaeus, Problemata, Venice, Antonius de Strata, 1488-89, 1501, 1505 ; and under Alexander Aphrodisiensis, various commentaries on Aristotle. — (This comment not final.)

[aw]Polydorus Vergilius (1477-1555), humanist and historian, presumably mentioned here for his most popular work, De inventoribus rerum, Venezia, Christophorus de Pensis, 1499, various other edns and translations. But also published other books on the Latin language, Proverbs, etc.

[ax]lydiamayhansell: Appianus of Alexandria

[ay]lydiamayhansell: Athenaeus of Naucratis

[az]lydiamayhansell: Publius Papinius Statius

[ba]Maurus Servius Honoratus, 4th-c. commentator of Vergil. Numerous editions of his work in various forms (notably editions of the Aeneid with his comment).

[bb]lydiamayhansell: Macrobius Ambrosius Theodosius

[bc]lydiamayhansell: Sextus Festus (late Roman historian)

[bd]lydiamayhansell: Nonius Marcellus

[be]Grammarian and lexicographer, active c300 AD

[bf]lydiamayhansell: Girolamo Maggi

[bg]Girolamo Maggi, Variarum lectionum, seu Miscellaneorum libri IV. in quibus multa auctorum loca emendantur, atque explicantur, et quæ ad antiquitatem cognoscendam pertinent, non pauca afferuntur..., Venice, Giordano I Ziletti, 1563; 1564.

[bh]Julius Pollux, Onomasticon. Vocabularium, Venice, Aldo I Manuzio, 1502 ; Florence, 1520; Bâle, 1536, 1541.

[bi]lydiamayhansell: Gaius Iulius Higinus

[bj]lydiamayhansell: Pseudo author

[bk](Works forged by Annius of Viterbo.)

[bl]lydiamayhansell: Dionysos of Halicarnassus

[bm]lydiamayhansell: Marcus Antonius Coccius Sabellicus

[bn]lydiamayhansell: A group of six scribes (scriptores) who wrote the Historia Augusta: Aelius Spartianus; Iulius Capitolinus; Vulcacius Gallicanus; Aelius Lampridius; Trebellius Pollio; Flavius Vopiscos

[bo]Marc Smith: authors rather than scribes

[bp]lydiamayhansell: Born of a noble family in Paris, (1467-1540)

[bq]Marc Smith: More importantly, a leading humanist

[br]Another of the authors of the Historia Augusta

[bs]Flavio Biondo from Forlì (1388 or 1392-1463), humanist & historian. Roma instaurata, Rome, [1471]; 1481; Roma triumphans, Mantua, [1473]; 1482, 1531, 1559; ; Italia illustrata, Rome, [1474]; Historiarum ab inclinatione Romanorum imperii decades, Venice, 1483; 1484, 1533.

[bt]Raffaele Maffei from Volterra

[bu]lydiamayhansell: Paulo Manuzio, Venetian print maker (1512-1574)

[bv]you mean printer (and publisher), son of Aldo Manuzio (and Paolo with o)

[bw]lydiamayhansell: Iulius Firmicus Maternus

[bx]author of: Mathesis (treatise on astrology)

[by]lydiamayhansell: Quintus Curtius Rufus, Historiae Alexandri Magni

[bz]lydiamayhansell: Cassius Dio

[ca]Lucius Cassius Dio, Historiae Romanae

[cb]Another of the authors of the Historia Augusta

[cc]lydiamayhansell: Italic hand

[cd]A common sort of "essai de plume" with no specific meaning.

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<page>001v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f8.image</image>

<div>
<id>p001v_1</id>
<head>For easing the belly</head>

<ab><m><pa>Prunes</pa> of <pl>Saint Antonin</pl></m>, and if you like you can put leaves of <m><pa>malva</pa></m> and <m><pa>viola</pa></m>, adding some <m>sugar</m> and, if you like, some <m><pa>cinnamon</pa></m> for the stomach.</ab>

<ab>Or in a <m><al>chicken</al> broth</m>, add some <m><pa>marshmallow</pa> root</m>. When fresh it is more mollifying.</ab>

<ab>Stirring together <m>cirop de <pa>guindoles</pa> doulces</m> with <m>water</m> and taking some of this in the morning eases the belly.</ab>
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<page>002r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f9.image</image>

<div>
<id>p002_1</id>

<head>Books to recover</head>
<ab>The chronicles of Sigebert[a][b]
Rufinus[c][d]

Irenaeus[e], In [Haereses[f][g]]
Paulus Aemilius[h]
Paulus Jovius[i][j]
Polydorus[k]
Bergomensis[l][m]
Philippe de Commynes[n]</ab>
</div>
<figure>-----</figure>
<div>
<id>p002r_2</id>
<head>Against pain</head>
<ab><m>Turpentine oil</m>, <m>oil of Jacob</m> and <m>salt from <pa>cabbage</pa> ashes</m>.</ab>
</div>
[a]Sigebert de Gembloux, c.1028 Gembloux-1112. The chronique de Sigebert was popular not only in the Holy Roman Empire but also in France, with 35 manuscripts and the Paris editions of 1513, 1575, 1583 and 1608.
<http://www.archivesdefrance.culture.gouv.fr/action-culturelle/celebrations-nationales/recueil-2012/litterature-et-sciences-humaines/sigebert-de-gembloux>
[b]Sigebert de Gembloux, Chronicon ab anno 381 ad 1113, cum insertionibus ex historia Galfridi et additionibus centum et tres sequentes annos complectentibus, nunc primum in lucem emissum, Paris, Henri Estienne & Jean Petit, 1513; etc.
[c]Tyrannius Rufinus Aquileiensis c.354-c.410. Maybe the Apologia quam...? http://www.documentacatholicaomnia.eu/20_40_0345-0410-_Rufinus_Aquileiensis.html
[d]Probably mentioned here for his more famous work, the Latin translation and continuation of the Historia ecclesiastica of Eusebius of Caesarea
[e]Irenaeus c.120-c.200, Asia Minor.
[f]the author wrote in exegesi, which is obviously a mistake; the only work from Irenaeus that reached us is de Haereses.
[g]Irenaeus Lugdunensis, Opus in quinque libros digestum, in quibus mire reteggit & confutat veterum haereseon impias ac portentosas opiniones..., Basel, Johann Froben, 1526; 1528, 1534, 1545, 1548, 1560, 1563, 1567, 1569, 1570, 1571, 1575, 1576. Variant titles include such forms as "contra haereses" (starting Paris 1545) or "adversus haereses" (starting Paris 1563), apparently never "in" (meaning the same), so the title here is in the author's own words. [Not "de", merci François de corriger ci-dessus]
[h]Paolo Emili (1460 Verona-1529 Paris). De rebus gestis francorum...many Paris editions from 1518. USTC
[i]Paolo Jovio, Paulus Jovius, (Como 1483- Florence 1552). Historiarum sui temporis? Vitae virorum illustrium? (Catholic Encyclopedia).
[j]NB Paolo Giovio is the correct Italian form, Paulus Jovius in Latin, Paul Jove in French.
[k]Polydorus Vergilius, (1470 Urbino?-1555 Urbino). Polydore's De Inventoribus rerum, an encyclopaedia of inventors and inventions mainly from the classical past, but also Jews, Egyptians, and Asiatics, was first printed at Venice in 1499, as three books dealing with material inventions, natural philosophy, medicine, and other profane matters. http://www.ilab.org/catalog_view/1725/1725_16006%20Polydorus%20Vergilius.pdf
[l]Iacobus 1434-1520. Supplementum chronicarum?
[m]Giacomo Filippo Foresti from Bergamo, priest and chronicler. Presumably mentioned for his historical work: Novissime hystoriarum omnium repercussiones... que Supplementum supplementi cronicarum nuncupantur..., Venice, Albertino da Lessona, 1503, 1513, etc.; Supplementum chronicorum, Paris 1535; 1538. Italian translations, several editions.
[n]Philippe de Commynes (Commynes). 1447-1512. Cronique et histoire. The title Memoires was used for the first time for a 1552 edition.

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<page>002v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f10.image></image>

<div>
<id>p002v_1</id>
<it/><ab>Practica, cioè nova inventione di contegiare, stampata in <pl>Brescia</pl> per Vincenzo Sabio[a].</ab>
</div>

<div>
<id>p002v_2</id>
<add><head><m>Emeralds of Brissac[b]</m></head></add>
<ab>I took <m>minium</m>, 26 <figure>3</figure>, <m>crystal</m> <add>[c][...] and ground on marble</add> 12 <figure>3</figure>, <m>verdigris</m>3 <figure>3</figure>. Incorporated all together. Put in a crucible covered by another well luted crucible with a hole on top. Melting: 7 hours without blowing. The mass came out a fine green.</ab>
<ab>I took <m>minium</m>, 12 <figure>3</figure>, <m>crystal</m>, 6 <figure>3</figure>, <m>verdigris</m>, 2 grains. Melting: moderate, vi hours, baked 24 hours. The mass came out green, yellow and red.</ab>
</div>

<div>
<id>p002v_3</id>
<head><m>Sapphire</m></head>

<ab><m>Pebbles</m> or <m>crystal</m>, 8 <figure>3</figure>; <m>minium</m>, 16 <figure>3</figure>; <m>prosmith</pro>'s salt</m>, 4 <figure>3</figure>; sparkling <m>coryal[d]</m>, 4 grains. Melting: 6 [?] hours.</ab>
</div>
[a]Prattica cioè nova inventione di conteggiare. Ridotta a modo tanto facile che ogn'uno potrà far ogni gran conto, si in vender come in comperare, sia a misura, o a peso, o a qual si voglia altro modo, ad ogni sorte di precio, & moneta, per tutte le parti del mondo..., stampata in Brescia, per Vincenzo Sabbio, 1589; 1600, etc. (USTC has only 1600).

A commercial handbook, very scarcely preserved, possibly no extant copies from a pre-1581 edition. Otherwise, would be evidence of the Ms. being post-1589
[b]Might be a place or a person
[c]see TC
[d]i.e. coral? Reading pretty uncertain — and coral would be a strange ingredient for a sapphire

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<page>003r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f11.image></image>

<div>
<id>p003r_a1</id>

<head>Imitation coral</head>

<figure>+</figure>

<ab>One must first make the branches from <m>wood</m> or take a fantastical <m>thorn branch</m>, then melt a pound of the best possible <m>clear</m> <m>pine resin</m> and add one ounce of finely ground <m>vermilion</m> together with <m>walnut oil,</m> and if you add a little <m>Venice</m> <m>lake</m> the color will be all the more vivid, and stir all together into the <m>resin,</m> molten over a <m>charcoal</m> fire, not over an open flame, lest it catch fire. Then dip in your branches with a swirling motion. And should there remain any filaments, turn the branch over the heat of the <m>charcoal</m>.</ab>

<ab>
<margin>left-top</margin>
<m>Colophony</m> is nothing other than <m>resin</m> that has been cooked again. To do it well, you take a <m>leaded</m> <tl>pot</tl> and melt the <m>resin</m>, boiling it over the brazier for a good hour until it appears not thick but clear and liquid like <m>water</m> and it easily runs as a thread off the end of a stick, which you use to crush and test it. Then pour it through a coarse <m>canvas</m> or <m>tammy cloth</m> so that it falls into the strongest <m>vinegar</m> you can find, because the <m>vinegar</m> makes it strong and makes it less brittle. Repeat this two or three times and it will be fine and well purified. To imitate your <m>coral</m>, you can mix a fourth part of <m>mastic</m> with your purified <m>resin</m> to make it more solid and finer, and if you should use just one drop of <m>mastic</m>, it would be all the better, but it would take too long.
</ab>

<ab>
<margin>left-middle</margin>
<m>Sulfur</m> and <m>vermilion</m> have the same effect.</ab>

<ab>
<margin>top</margin>
<m>Coral</m> made of <m>red enamel</m> withstands filing and polishing.</ab>

<ab>
<margin>right-top</margin>
It is made like <m>cement</m>, which is stronger when mixed with crushed <m>glass</m> rather than with <m>brick</m>. In the same way, together with the <m>vermilion</m>, one mixes in <m>opaque red enamel</m>[a], finely ground. It is the same way with all <m>enamel</m> colors.</ab>

</div>
<div>
<id>p003r_a2</id>
<head><m>Varnish</m> for paintings</head>

<ab>Take a pound of <m>Venice turpentine</m> and heat it in a <tl>pot</tl> until it simmers, put in half a pound of the <m>whitest turpentine oil</m> you can find and stir all together well on a <m>charcoal</m> fire and take it off immediately after, and it is done. But if you find it too thick, add more <m>oil</m>, whereas if it is too clear, you can thicken it by putting a little <m>turpentine</m>. And so you will give it whatever c[b]onsistency you want. It could be made without <m>fire</m>, but it is more desiccative when heated. It is appropriate for panel paintings and other painted things, without distorting the colors or yellowing, and dries both in the shade and under the sun, overnight, and during the summer as well as in the winter. It is usually sold 15 sols a pound.</ab>

<ab>
<margin>left-bottom</margin>
You need a little more <m>turpentine</m> than <m>turpentine oil</m> to thicken the varnish, which you need to apply with your finger in order to spread it thinner and less thick because when it is thick, it turns yellow and gathers [together]. <m>Varnish</m> is not used to make paintings shine, because it just takes the light out of them.
</ab>

<ab>
<margin>bottom</margin>
But it is used to enhance colors which have soaked in and to keep them from dust. <m>Mastic varnish</m> does not resist rain, whereas <m>oil</m> and <m>rosin varnishes</m> do.</ab>

</div>
<div>
<id>p003r_a3</id>
<head>Thick <m>varnish</m> for floorboards[c]</head>

<ab>There is a type of <m>varnish</m> that takes a long time to dry and is still damp up to two months after it has been applied to the floorboards. But the following <m>varnish</m> does not remain damp like previous ones, which were made of <m>linseed oil</m> from <pl>Aux</pl> boiled with <m>garlic</m> to extinguish them[d] and rid them of fat, and also with <m>wheat</m>. And this old one yellowed over time and made the blue color of paintings greenish. This one is made like former ^{<m>varnishes</m>} except that one puts in <m>common thick turpentine</m></ab>

<cont/>
</div>
[a]"en corps" seems to refer to opaqueness: see e.g. <ab id="p038r_b4">, "l'ambre transparent et a l'autre qui a corps"
[b]'corps' in French. Perhaps the translation needs to be reconsidered?
[c]Note: planchers can also mean ceilings, i.e. any large surface covered in "planches" (boards). Only French plancher is horizontal, English boarding is vertical...
[d]literally. NB estaindre, not estendre

<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f14.image</image>

<div>
<id>p004v_a1</id>
<head>Black varnish for sword guards, ^{<m>metal</m>} bands for chests, etc.</head>

<ab>Take <m><pa>linseed</pa> oil</m> or, for a cheaper option, <m><pa>walnut</pa> oil</m>, and rid it of fat with <m><pa>garlic</pa> cloves</m> and <m><pa>hog's fennel</pa></m> (some also add <m>bread crusts</m>), which you will boil in it for a good quarter of an hour. Then add to one pound of the <m>oil</m> boiled in this way a piece of <m>black pitch</m> the size of a walnut and a <ms>double handful</ms> of <m><pa>wheat</pa></m>, without removing the <m><pa>garlic</pa></m> and <m><pa>onions</pa></m>, and allow to boil together for a good quarter of an hour. Once the <m>pitch</m> is well melted and the <m>oil</m> takes on body, you can remove it from the fire. Then, to varnish, place your <m>iron</m> over a warm <m>charcoal</m> fire and apply with a <tl>feather</tl> or a <tl>brush</tl>. When you see that it no longer smokes, it is done and your varnish is dry.</ab>
<ab>
<margin>left-top</margin>
For an excellent black varnish, add two or three <m>paternoster beads</m> to the rest.</ab>

<ab>
<margin>left-top</margin>
Some consider <m><pa>walnut</pa> oil</m> to be better.</ab>

<ab>
<margin>left-top</margin>
If there is a quantity of varnish, it needs to boil for at least half a day, because the more it boils, the better it is. Should the varnish boil over, it is dangerous for it to catch fire and it is difficult to extinguish. So do this in a <env>courtyard</env> or other <env>open space</env>.</ab>

<ab>
<margin>left-middle</margin>
In five or six lbs. of <m>oil</m>, one must put one lb. of <m>galipot</m>[a], which costs 4 sols, and some peeled <m><pa>garlic</pa> cloves</m>. This varnish is not black in and of itself, but turns black when exposed to fire.</ab>

<ab>
<margin>left-middle</margin>
See below, around the 3rd part of this book, after sands, in the chapter on <pro>furbishers</pro>.</ab>

</div>
<div>
<id>p004v_a2</id>
<head>Black varnish without fire, without disassembling harnesses or removing ^{<m>metal</m>} bands from chests.</head>

<ab>Take <m><pa>spike lavender</pa> oil</m> varnish and mix it with <m>charcoal black</m> or <m>lampblack</m> and, without fire, it will mix in by itself. Apply the varnish with a brush and it will soon be dry. <m>Turpentine</m> varnish would work quite well but it does not dry as quickly.</ab>

</div>
<div>
<id>p004v_a3</id>
<head><pro>Iron engravers'</pro> varnish.</head>

<ab>Take <m><pa>linseed</pa> oil</m> or <m><pa>walnut</pa> oil</m> and instead of <m>black pitch</m> you will add some <m>pine resin</m>[b][c] and you must cook it like <m>black varnish</m> on fire. And to apply it, heat your <m>iron</m> and apply just one layer of varnish, and when it no longer smokes, it is dry. Then engrave with a <tl><m>steel</m> point</tl> whatever you want. Then take some <m>salt</m> and <m>verdigris</m> and soak it with as much of one as of the other in very strong <m>vinegar</m>, and leave it for XX4 hours before using it, and the whole will be soaked. Then spread this liqueur or sauce on the engraved object with a <tl>sponge</tl></ab>

<cont/>
</div>

[a]This can be either "pine resin" (which we have used until now for "poix resine"), or a mixture of this with various greasy substances, mainly used to make ships watertight.
[b]can also be found spelled as "rosen"
see Cotgrave
<http://www.pbm.com/~lindahl/cotgrave/search/742l.html>
[c]this should be resin, not rosin (which the Ms. calls "colophonie")

.....

<page>005r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f15.image</image>

<div>
<cont/>
<id>p004v_a3</id>
<ab>or <m><pa>linen</pa> cloth</m> and leave it thus for xx4 hours and it will be engraved. But if you should want to engrave it quickly, varnish all your work and boil it in the aforementioned liquor, and this will engrave it quickly.</ab>

</div>
<div>
<id>p005r_a1</id>
<head><m>Steel</m> mirrors</head>

They are called ^{mirrors} because in the past they were usually made of ^{steel} polished with ^emercy paste. But more easily, various kinds are made of ^{cuiivre franc}, which is ^{rosette} and ^{tin}, because these can be cast in a mold and made round, hollow, convex or in whatever various shapes you want to represent.

So take half ^{copper de rosette} and half soft ^{tin}, i.e. fine and yet unused. Place them in a crucible, and first melt the ^{copper}. Once it has been well melted, add the ^{tin} and mix them together and cast that in a ^{white stone} mold with no hole, and grease the mold, which should be moderately warm, with ^{oil}. Then, once your mirror has been molded, you can polish it in this manner.

Set one ^{mirror} in ^{plaster} to hold it firmly, then put the other one over it with fine sand between them, and rub one against the other, whether they are hollowed or flat, and thus you will polish both at once. If you want to polish them on both sides, you only have to switch them, which means putting the one that was polishing in the ^{plaster} and [using] the one which was in the ^{plaster} to polish. Once they have been polished with ^arene you can soften them with ^{tripoli} of Venice that should not be ^{sand}y, and then with ^{paste}. You use ^{water} with ^{arene} to polish, but polishing with ^{tripoli} and ^{paste} is done dry. Once ^{the mirror} has been polished, you can set it.

^{p005r_a2}

Concave mirrors

A concave mirror shaped in the aforesaid form produces an endless number of illusions which seem like magic. If one wants to see oneself from behind, you have to place the concave mirror on the ground on its foot, at a slant, and then look at it one step further. But if you look at it from a closer position, it shows you the right way up but with the face quite large and the hairs of your beard as thick as

[a]potee desmery

[b]arene, rather than sable, leave it in French

^{005v}

^{http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f16.image}

^{005r_a2}

a string and will reflect a ^{nipple} [a]as large as a plate, and women can see the secret places they do not want to show to ^{surgeons}. It projects [b]the reflection outwards, and if you touch with your ^{finger} the eye of the reflection, another ^{finger} will come against yours.

You will be able to provide light at night for someone quite far away, if you set the concave mirror behind a candle placed on a window sill. You will be able to read and write quite far from the candle if you set it behind the candle and turn the concave face towards you. If you put the mirror at the far end of a table with a candle on it, it will reflect two lights on the table for you. To see what is happening in a street, shut the window, or the more you can shut the better, make a hole in the bottom ^{pane}, the size of your ^{little finger}, perfectly round all the way through. Then apply a sheet of paper onto the window and let the middle of the sheet be over the hole. Pierce the sheet in the same place as the hole and of the same size. Then turn the concave face of the mirror towards the hole, and anyone passing by in the street will be reflected on the sheet of paper. Do the same with the pierced door of a closed room. You can make wavy ^{mirrors} and triangular ones, such as are described by Ptolemy and Archimedes. The concave sort can also set fire to straw or light a candle from afar, using the sun. By reverberating [c] fire too, it can provide warmth.

^{p005v_a1}

For bronzing in yellow and white

Take ^{mirror tin} [d][e] and melt it. On half a pound of this, put half an ounce of ^{quicksilver}, and remove it immediately from the fire and grind it until it is like ashes. Wash it thoroughly in ^{clean water}, then grind it on a ^{marble} slab and soak it with water of ^{glue de retailles}. Then apply that on your wooden candlesticks and any other works you like, and burnish with a ^{tooth} [f]. You can grind ^{pin filings} and apply them with the said ^{glue} ^{de retailles}.

[a]A form for "teton" rather than a "teston" coin, since he is looking at body parts.

[b]Suggesting a 3D illusion

[c]representer seems to mean "reflect" an image, whereas "reverberation" seems to mean to reflect heat?

[d]tin used for mirrors, mirror maker's tin...?

[e]tin used for backing mirrors and for the sand of hour glasses because it is thin, fine and dry, and does not react to changes in weather: from Cotgrave "estaim de glace"

[f]Wolf teeth were sometimes used for polishing

^{006r}

^{http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f17.image}

^{p006r_1}

<head>To lay down and set burnished <m>gold</m> and give [a] it a red or green or blue ^{hue}</head>

<ab>
<m>Ceruse</m> and <m>lead white</m> [b] is not appropriate for polished white nor for burnishing because it is fat, but it is quite good for matte <m>gold</m>, which is made with <m>oil</m> by mixing it with yellow <m>ochre</m> and <m>mine</m> and tempering the whole with <m>oil</m>, and this matte <m>gold</m> so applied lasts in the rain, like gilded <m>leadwork</m> and similar things. Therefore, for burnished <m>gold</m> take good <m>chalk</m>, quite white, well ground and soaked with <m>distemper glue</m> and do four coats of it, one after the other, on the <m>wood</m>. And when the last one is dry, rub it with <m><pa>prele</pa></m>, which is a kind of grass otherwise named <m><pa>horsetail</pa></m>, to render it well polished. Afterwards, take fine <m><pl>Armenian </pl>bole</m> and <m>sanguine</m>, as much of one as of the other, a bit of <m><al>lamb </al>tallow</m> the size of a <m>bean</m> or a <m>pea</m> depending on the quantity of <m>bole</m>, and a little <m>willow charcoal</m>, or as much as the <m>tallow</m>, and half a <m><pa>walnut</pa> shell</m> filled with half-burned <m>saffron</m>; some put in a little <m>candi sugar</m>. Grind everything together with <m>water</m> and apply it without <m>gum</m> or <m>glue</m>, and let it dry. And rub the place that you would like to <m>gild</m> with a piece of white cloth to better unify ^{it}, and when the rubbed place is a little shiny it is a sign that the <m>gold</m> will behave well. After rubbing, wash the place that you want to gild with a clean <tl>brush</tl> soaked with clear <m>water</m> and apply the <m>gold</m> at once, which you will burnish when dry. And if you want to set <m>rouge clair</m> and glaze with it, grind <m><pl>Venice </pl>lake</m> on the <m>marble</m> with <m>walnut^{oil}</m> or <m>linseed^{oil}</m>. After grinding, mix ⁱⁿ <m>turpentine</m> varnish or <m>spike lavender</m> and apply on <m>gold</m> with the <tl>brush</tl>, <m>Brazilwood</m> and <m>laque ronde[c][d][e][f]</m> fade. For green, temper <m>verdigris</m> with <m><pa>walnut</pa> oil</m> or <m><pa>linseed</pa>^{oil}</m> and grind it, then mix in some <m>turpentine</m> varnish; not <m>aspic</m>^{varnish}, which is not suitable for <m>verdigris</m>. If you want to glaze with <m>azure</m> [g] it must be set on burnished <m>silver</m>. Take <m>azure enamel</m>, and without grinding temper it with <m>turpentine</m> varnish and apply it.

</ab>
</div>
[a]cannot make sense of "donner" here (mistake for dorer? meaning "to add"?), so kept the translation literal.
[b]Are these two different things or does he mean to use the & to actually say 'or.' The verbs are singular, indicating that he may have thought of 'them'/'it' as the same thing?
[c]research failure, don't know what this is. "ronde" probably in contrast to the "platte" of "laque platte de Venise," translated above as "Venice lake" as done in 3r
[d]Some info, albeit not particularly useful for translation http://portail.atilf.fr/cgi-bin/getobject_?p.66:46./var/artfla/encyclopedia/textdata/image/
[e] _Marked as resolved_
[f] _Re-opened_
[g]unclear: "with azure" (the pigment) or "in blue"?

.....

<page>006v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f18.image></image>

<div>
<id>p006v_a1</id>
<head>For cages</head>

<ab>You can embellish them with <m>thin</m> <m>enamel</m> <m>canes</m> [a][b][c] of various colors by covering the <m>yellow latten</m> or <m>iron</m> wire with said canes. These you will break neatly into the length that you want if you make a small notch with a cutting <tl>file</tl> at the point where you want to break them, and they will not break anywhere else. You can bend them with a <m>wood</m> en model over a brazier or by the heat of a <tl>lamp</tl>. A cane can be stretched out as long as you want in a small <tl>furnace</tl> made like a <tl>reverberatory oven</tl> but with openings on both sides. And once the large cane is red, they seize the hot end of it with <tl>small pincers</tl> that have long beaks, with one end of the beak inside the end of the cane, so that it may be stretched without becoming blocked[d]. The other end of the cane is held with the hand, as it is not hot. Once the cane is stretched enough, the one who is working seated with his <tl>stove</tl>, the size of a carnation pot, placed in front of him, breaks it off and carries on. This is for making canes for capes, which may be cut, as already mentioned, with a <tl>file</tl>. <pro>Glass-button makers</pro> also use the said <tl>stove</tl>.</ab>

<figure>
<id>fig_p006v_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5eVFfRyJCTmczmM></link>
</figure>

<ab>
<margin>left-middle</margin>
Under the door is a grill that supports the burning <m>charcoal</m>. The <m>ash</m> is emptied by turning the <tl>stove</tl> upside down [e].</ab>

</div>
<div>
<id>p006v_a2</id>
<head>For stamped ornaments[f] used for embellishing and inserting into or covering the edges of mirrors, the tops of chests, or the friezes of bed valances</head>

<ab>Etch with <m>aqua fortis</m> on <m>iron</m> or <m>copper</m> whatever you have pounced and drawn there, next make it neat with a <tl>burin</tl> or <tl>chisel</tl>. Then pour <m>soft</m> <m>tin</m>, yet unused, onto polished <m>marble</m> and flatten it, making it quite thin, with a <m>wood</m> en board. Or else pour it on a table as is done with <m>lead</m>, or put it through a <tl>roll-press</tl>. Next lay your <m>tin</m> plate over the engraving, and over the <m>tin</m> plate put a piece of <m>felt</m> and strike it with a <tl>hammer</tl>. Then gild it in the following manner.[g][h][i]</ab>
</div>

[a]I have decided to keep the original French word as it does not translate into a specific English term (Cotgrave: a little cane, reed, pipe or hollow straw)
[b]this seems to refer to hollow canes of enamel that are being used to cover the wires of the cage
[c]it does. Or pipes?
[d]? At any rate not clogged, which would mean something (a clog) has got inside.
[e]The French phrase is unclear.
[f]Estampes are made by a process of hammering tin into an engraved surface, and could be translated as "stampings," but we chose stamped ornaments because of their use as specified on fol. 7r.
[g]The first entry on the page immediately following is for gilding.

[h] _Marked as resolved_
[i] _Re-opened_

.....

<page>007r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f19.image</image>

<div>
<id>p007r_a1</id>
<head>To gild with <m>gold color</m> and <m>tinsel</m></head>

<ab>Once you have colored ^{or[a]} made your stamped ornament[b] in <m>tin</m> (as previously described[c]), or in <m>iron</m> or <m>copper</m>, you must not put <m>glue</m> as on <m>wood</m>, but just have some fatty <m>oil</m>, which is made in seven or eight days in the sun from <m>walnut oil</m> and <m>white lead</m>, stirring often, or cooked on fire if one should be in hurry. Then, with the <m>oil</m> thus made fatty, grind a little <m>white lead</m>, <m>massicot</m> and <m>black graphite</m>, at discretion, as much of one as of the other. <m>Minium</m> will color the <m>gold</m>. Then apply an even layer of that onto your stamp, taking care you do not fill the hollows. And once it is almost dry, apply the <m>gold</m> leaf on it using <m>cotton</m>. Such <m>gold</m> will withstand rain on houses or elsewhere. And if you have made your gilt with <m>tinsel</m>, color it with <m>smoke [d][e] from a partridge</m>[f][g] or from yellow or red <m>cloth[h][i]</m>, and it will look as fine as pure <m>gold</m>. You can cover trunks, mirrors, bed valances and posts in colored <m>velvet</m> or <m>satin</m> and then apply the gilded stamped ornament on them with strong <m>glue</m>.</ab>

</div>
<div>
<id>p007r_a2</id>
<head>To color stamped ornaments for trunks</head>

<ab>The stamped ornament made from sheets of <m>copper</m> or <m>latten</m> are made with <m>wood from a <pa>service tree</pa></m>[j] engraved and cut, should you want to spend less. Next, the stamped piece is colored with the aforesaid lacquer colours of <m>lacquer</m>, <m>verdet</m>, <m>blue enamel</m>, and soaked in <m>turpentine</m> varnish. But wherever you want to make it azure, apply some fer blanc, which is more appropriate for an azure background.</ab>

</div>
<div>
<id>p007r_a3</id>
<head>Doublets</head>

<ab>Good <m>dragon's blood</m> soaked in <m>spirits</m> produces its own <m>mastic</m> or <m>glue</m>. So do <m>vert de vessie</m> and <m>saffron</m>.</ab>

</div>
<div>
<id>p007r_a4</id>
<head><m><al>Fish</al> glue</m> or <m>usblac</m> and <m>mouth glue</m></head>

<ab>It is made from <m><al>codfish</al> skin</m> that has been boiled rather than <m>salted</m>. <pro>Joiners</pro> use it on their masterpieces and <pro>guitar makers</pro> use it for their more delicate works. It needs to be strongly whipped, then soaked gently in barely boiling <m>water</m>.</ab>

<ab><m>Mouth glue</m> is made of <m>parchment</m> scraps and used to glue <m>paper</m> or similar things without fire, by wetting it with one's <tl>
<bp>mouth</bp></tl>.</ab>

<ab>
<margin>left-bottom</margin>
It is whipped and left to soak in <m>white wine</m> for one night, then melted over a low heat. Others soak it in <m>spirits</m>.</ab>
</div>

[a]Text unclear
[b]these gilded tin ornaments are produced by a process of stamping--see fol. 6v--thus we are translating estampe as stamped ornament.
[c]on p006v
[d]"fumee de perdrix ou de drap rouge ou jaulne". According to Cotgrave "fumee" can mean smoked but also fattened with dung.
[e]fattening: only fattening soil
[f]Probably color? partridge
[g]burning partridge feathers??
[h]"fumee de perdrix ou de drap rouge ou jaulne". According to Cotgrave "fumee" can mean smoked but also fattened with dung.
[i]fattening: only fattening soil
[j]Sorbus domestica

.....

<page>007v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f20.image</image>

<div>
<id>p007v_a1</id>
<head>To give the color of all kind of <m>metals</m> and <m>woods</m> and other things</head>

<ab>Take <m>touchstone</m> and <m>pumice stone</m> as much of one as of the other and grind them together, mix that with a <m>glue de retailles</m> stronger than that used for painting, and apply a layer of it on whatever you like. Once dry, rub this layer on any <m>metal</m> whatsoever, then burnish it.</ab>

</div>

<ab>Take a dozen dried <m>common <pa>walnut</pa>s</m> and throw them one after another onto a good <tl>brazier</tl> where they may catch fire, and take them out with <tl>tongs</tl>, and let them burn and flame well in the air, and extinguish them in a glass of good <m>wine</m>. Then let the <m>wine</m> cool and filter it, then pour half of it, since six extinguished <m>nuts</m> make one dose. An excellent cure against suffocation of the matrix.</ab>

<ab>Take half a pound of finely powdered golden and yellow <m>marcasite</m>, half an ounce of <m>storax</m>, 4 lb. of <m>urine</m>, incorporate well together, little by little, in a <tl>mortar</tl>, then boil all together quite strongly. But the pot needs to be well covered so the smoke does not escape. Then distill the imbibed <m>urine</m>, separated by inclination, in an <tl>alembic</tl>, well luted and covered with a <m>copper</m> head. And soak a cloth with the said <m>water</m> and apply it lukewarm on the pain.</ab>

<ab>Cook i. lb. of <m>old smiths' water</m>, i. 3 of <m><pl>Armenian</pl> bole</m> reduced in the finest powder, and .iii. 3 of <m>common honey</m>, until the <m>honey</m> stops foaming. Once cooled, strain with great pressure and use the results of filtration by injection.</ab>

<div>
<id>p008r_1</id>

<ab>Write with <m><pa>fig tree</pa> milk</m> then let it dry a little, afterwards apply the <m>gold</m> in leaf and, once quite dry, rub with <m>linen</m> and only the lettering will hold.</ab>
</div>

<ab>Put into the <m>powder</m> one sixth as much of <m>quicksilver</m></ab>
</div>

<ab>Heat the <m>lead</m> bullet as much as you can without melting it, and once it is hot dip it into the strongest <m>vinegar</m> that you can. And do this four or five times. Afterwards, mince some <m>lard</m> very finely and some <m>linen</m> also very delicately, mix all this together and use it as wadding, making it go into the arquebus or pistol with a little strength and very tightly. And before shooting, if possible, the bullet should be hot.</ab>

<ab>Having made a hole in a wall as you know, put one mortar inside, charged with <m>powder</m>, almost flat, and another upright, then fill in the hole and fire.</ab>
</div>

<div>

<id>p008v_1</id>

<head>To polish a <m>ruby balay</m></head>

<ab>Unlike others, you don't polish it with <m>tripoli</m>, but with <m>marcasite powder</m> and <m>oil</m>.</ab></div>

<div>
<id>p008v_2</id>

<head>Ground <m>gold[a]</m> and <m>silver</m></head>

<ab>You have to calcine your <m>pure gold</m> with <figure><m>☿</m></figure> and then let your amalgam dry well and the said <m>mercury</m> fly off. Then crush it on <m><tl>marble</tl></m> with <m>gummed water</m>. As for <m>silver</m> you have to use it in its raw form[b] and corrode it with <m><la>aqua fortis</la></m>, then take it out with a <m>copper</m> blade and rinse it well and then grind it with <m>gum water</m>.</ab></div>

<div>
<id>p008v_3</id>

<head>Polishing <m>stones</m></head>

<ab>There are three wheels for this purpose, one of <m>tin</m>, one of <m>lead</m>, and one of pure <m>copper</m>. On the <m>copper</m> wheel, only <m>diamonds</m>, <m>rubies</m>, <m>sapphires</m> and oriental <m>jacinths</m> are usually polished. The <m>tin</m> wheel is for softer stones such as <m>emeralds</m>, <m>amethysts</m> and others.</ab></div>

<div>
<id>p008v_4</id>

<head><pro>Pewterers</pro>' mixture</head>

<ab>Because they often mix in with the pure <m>tin</m> half the quantity of <m>lead</m>, which renders the <m>tin</m> deaf. To correct this, they mix in some <m>latten</m> filings to make it resonant.</ab></div>

<div>
<id>p008v_5</id>

<head>Perfect amalgam</head>

<ab>Take the finest <m>sandstone</m> you can find and make a cavity in the middle. And put in it an eighth part of <figure><m>☿</m></figure> and rub it and wear it down while adding some <m>water</m> with <figure><m>☿</m></figure> or <figure><m>☿</m></figure> until the material becomes paste-like and even harder. Then, if you want, corrode it with good <m><la>aqua fortis</la></m>, if it is <figure><m>☿</m></figure> or <figure><m>♀</m></figure>. Then melt it again, having taken it out as you know. The <figure><m>☿</m></figure> will have some color, and after being melted it will leave some flecks of <figure><m>☿</m></figure>, and the <figure><m>♀</m></figure> will be very fine and will have almost no covering.</ab></div>

[a]i.e. ormolu (but we can't use it here because of the simultaneous mention of silver)
[b]uncertain translation

.....

<page>009r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b9059316c/f22.item</image>
<div>
<id>p009r_1</id>

<head><pro>Plowman</pro></head>

<ab>Since <m><pa><fr>millet</fr></pa></m> is naturally very dry, <pro>harvesters</pro> thresh it most often in the cool of night and in moonlight. Which could not be done with another, wetter, <m><pa>grain</pa></m>, because you must let the sunbeams pass over it, and one rarely threshes it before midday. Immediately after the <m><pa><fr>millet</fr></pa></m> is harvested, the earth needs to be plowed because its roots eat and make lean ^{the soil} as much as if the <m><pa><fr>millet</fr></pa></m> was still standing. The ground where <m><pa><fr>millet</fr></pa></m> will have been sown becomes quite diminished in fertility, which you can tell next time it is sown. But more than any other the <m>grain[a]</m> called <m><pa><fr>pomole</fr></pa></m> in <pl>Gascony</pl> or <m><pa><fr>baillard</fr></pa></m> in <pl>^{northern} France[b]</pl>, makes the earth shake, according to the common saying, seven years later. <m><pa>Broad beans</pa></m> improve and enrich the soil, provided that one lets the roots and stems rot in it.</ab>
</div>

<div>
<id>p009r_2</id>

<head><pro>Painters</pro></head>

<ab> After laying down their colors, some smooth it with the back of a <tl>feather</tl> but the large <tl>brush</tl> is best.</ab>
</div>

<div>
<id>p009r_3</id>

<head><pro>Merchants</pro></head>

<ab>They commonly have four books. The tiro, which is a formulary for apprentices, by which they can know how much every item of merchandise costs and for how much it should be sold. The second is the waste book, where they put down in rough what they sell daily. The third is the sales book, where they put down the clean copy and detailed account of what is contained in the waste book. The fourth is the ledger, which mentions the sale, the balanced account, the bill and the term of payment, and this book is accepted as evidence in court.</ab>

</div>
[a]a kind of barley
[b]As opposed to Gascogne. "France" is often used meaning the central and northern parts of France now limited to what is known as the "Ile-de-France" region)

.....
<page>009v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f24.image</image>

<div>
<id>p009v_1</id>
<head><pro>Painters</pro></head>

<ab>They melt <m>orpiment</m> entirely in a <m>glass</m> bottle on very hot <m>ashes</m>, but it would be more appropriate to sublimate <m>orpiment</m> in a long-neck <tl>matrass</tl> [a]as for making <m>garnets</m>. The <m>orpiment</m> thus turns red like <m>red enamel</m> and it has to be ground first in distemper, because once entirely melted it is very brittle and difficult to grind. Then, once dry, you can thin it with <m>oil</m> and it will make a deep and saturated yellow. But to prevent <m>orpiment</m> from fading ^{in contact} with other colours and make it compatible with these and to make it dry quickly, a most rare secret is to <m>calcine</m> common <m>salt</m> and to grind it together ^{with the <m>orpiment</m>}. <m>Green verditer</m> is not used with <m>oil</m>.</ab>

@
<div>
<id>p009v_2</id>
<head><pro>Merchant</pro></head>

<ab>The order that <pro>merchants</pro> are accustomed to keep in their books, which are the waste book, the sales book, and the ledger, also called the account book.
</ab>
</div>

<div>
<ab>Waste book</ab>

<ab>First, François du Cros owes from 2 September 1581, for 1 canne[b] of <m>batiste</m> at 8 livres per canne, total <figure>--</figure> 8 livres tournois</ab>

<ab>
<margin>left-middle</margin>
Moved to the sales book at r. 25[c]</ab>

<ab>François du Cros owes from 2 September 1581, for 1 canne of <m>batiste</m> at 8 livres per canne, as shown by the waste book at r. 25</ab>

<ab>
<margin>left-middle</margin>
Moved to the account book at r. 55[d]</ab>

<ab>François du Cros owes from 2 September 1581, for the merchandise taken by himself, as specified in detail in the sales book, No. 7, at r. 55, the sum of 8 livres, of which the aforesaid du Cros has made a bill on the aforesaid day, to be payed on Saint Andrew's day next, total <figure>--</figure> 8 [...] r.</ab>
</div>
[a]In French written as "mattelas". It would make more sense if it was 'matras', a long-neck glass receiver used in Alchemy. or Materas, cf Colgrave. (in English: matrass)
[b]Uncertain about translation - unit of measure almost equivalent to 'toise'
[c]Uncertain
[d]Uncertain

.....
<page>010r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f25.image</image>

<div>
<id>p010r_1</id>
<head>Imitation <m>jasper</m></head>

<ab>Take <m>horn</m> as is used to make lanterns, quite thin, and underneath imitate your <m>jasper</m>, <m>cornaline[a]</m>, and other <m>stones</m>. That will

look more appropriate than doing it on glass, which is too shiny. And the <m>horn</m> gives a lustre and a fatty [b]polish similar to <m>jasper</m>.</ab>

<ab>You know how you can imitate <m><pa>rose</pa></m>s with scrapings of the said <m>horn</m>. The colors for this <m>jasper</m> need to have as a base clear <m>turpentine</m> or <m>spike lavender</m> varnish. And matte, opaque colours are not appropriate, no matter how fine they are. You have to <m>oil</m> the unpainted underside with <m>spike lavender oil</m>.</ab>

<ab><margin>top</margin>
Thin <m>glass</m> looks very fine for this effect</ab>

<ab><margin>left-top</margin>
You can inlay beds with it, and on the joints you can throw <m>talc</m> or <m>metal filings[c]</m> on the fresh cement of the said joints. You have to bond them with <m>gum amoniacum</m> soaked in <m>vinegar</m>. To better imitate <m>marbled jasper</m>, apply heavy <m>yarn</m> dyed various colors and intermingled. After applying all the colors, scrape oblique lines into them then apply <m>gold</m> and <m>silver</m> leaves. If you apply colors made of <m>turpentine</m> on the <m>horn</m>, give it a backing of <m>silver</m> or of <m>tin</m> leaf<m>.</m> You can also file <m>horn</m> and mix it with strong <m>glue</m> and apply it to the joints of the <m>horn</m> piece and finish with a <tl>joiner's plane</tl>.</ab>
</div>

<div>
<id>p010r_2</id>

<head><m>Scudegrun[d]</m></head>

<ab>It is made in <pl>Lyon</pl> from the <m>juice of weld</m> mixed with <m>chalk</m> or better yet with <m>ceruse</m>, which is appropriate for <m>tempera[e][f]</m> and <m>oil</m>.</ab>
</div> <pro>

</pro><div>
<id>p010r_3</id>
<head><pa>Rose</pa>s</head>

<ab>These are imitated either with the scrapings of <m>horn</m> used for lanterns, or with very light, fine and dyed <m>parchment scrapings</m>, used as you know.
</ab>
</div>
<div>
<id>p010r_4</id>
<head>Purple color</head>

<ab><pro>Painters</pro> make it beautifully by first applying an undercoat of ordinary <m>azur</m>, or better yet <m>azure enamel</m>, and then they glaze it with <m>lacquer</m>, which will be more appropriate for this if you mix in <m>alum</m>, which will give it a violet tint depending on the quantity that you mix in.</ab>
</div> <pro>

</pro><div>
<id>p010r_5</id>
<pro>
</pro><head>Powder for hourglasses</head>

<ab>It must be made very fine and not subject to rust and with enough weight to flow. Taking i lb[g] of <m>lead</m>, melt it and skim and purify it from its filth, then pour into it four <figure>3</figure> of finely ground <m>common salt</m>, and take care that there are no stones or earth. And immediately after pouring it, stir continuously very well with an <m>iron</m> ^{tool} until the <m>lead</m> and <m>salt</m> are quite incorporated, and take it immediately off the fire, stirring continuously. And if it seems too coarse, grind it on a <tl><m>marble</m> slab</tl> and pass it through a fine <tl>sieve</tl>, then wash it as many as times as necessary until the <m>water</m> runs clear, throwing out the fine powder that will float on it, renewing the <m>water</m> as many times as necessary until it is completely cleared.</ab>
</div>

[a]<http://www.micmap.org/dicfro/search/cotgrave/cornaline>
[b]fat as in what Pamela Smith describes as the binary of fat/lean.
[c]from pins? (to make them as fine as possible)
[d]See TLF for stil de grain. Interesting form here, Scudegrun, closer to the presumably German etymology of the word, Schüttgrün

PS What is wrong with stildegrain, why keep French scudegrun?
[e]We think that here destrempe is the noun for tempura and not the verb, since it is comparable to oil.
[f]tempura? hmm, time for Japanese dinner. But I agree, tempera vs oil painting.
[g]meaning 1 lb - this is not a typo

<page>010v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f26.image></image>

<div>

<id>p010v_1</id>

<head>For painting in <m>oil</m> on <m>taffeta</m> without the <m>oil</m> running</head>

<ab>One needs to make a layer of <m><fr>batture</fr></m>, which is made of <m><fr>colle forte</fr></m> soaked for 24 hours in <m>water</m> then gently boiled so it is not too strong. Then stir in a little <m>honey</m> to soften it, and bring it all to a boil. And on top of this layer, which will soon be dry, you will be able to paint in <m>oil</m>, which will soften the layer even more, and which will also serve to make an underlayer for <m>gold</m>. <m>Alum water</m> also keeps <m>oil</m> from running.</ab>
</div>

<div>
<id>p010v_2</id>

<head>Underlayer for <m>gold</m> leaf on <m>parchment</m> or <m>paper</m></head>

<ab>Make very clear <m>starch glue</m>, showing little body. And apply six or five layers, and when the last is almost dry, apply your <m>gold</m> to it.</ab>
</div>

<div>
<id>p010v_3</id>

<head>For cleaning <m>gold</m> leaf applied to iron</head>

<ab>Rub your weapons or other gilt objects with <m>dry <al>horse</al> dung</m>.</ab>
</div>

<div>
<id>p010v_4</id>

<head>For whitening <m>ceruse</m></head>

<ab>Hard-boil some <m>eggs</m>, cut them in half and take the <m>yolk</m>s out, and between the two halves of the <m>white</m> put a bit of <m>ceruse</m>, and tie the two halves together with thread. Then boil in clear <m>water</m>, and it will become black, and thus the <m>ceruse</m> will be purified. Some reheat it over fire and it becomes very white.</ab>
</div>

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<page>011r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b9059316c/f26.item</image>

<div>
<id>p011r_1</id>
<head>Painting <m>enamel</m> azure in <m>oil</m></head>

<ab>This is a secret that is hardly known to common <pro>painters</pro>. Some take the most delicate they can & crush it with <m>ceruse</m>, which binds it, and next prick with an awl in several places the area they want to paint with <m>azur d'esmail</m>, such that the <m>oil</m> enters & does not cause the <m>azure</m>, which in itself is heavy, to run. Others lay the panel flat & apply the <m>azure</m> on it, which is also done <fr>à destrempe</fr>. The main thing is to crush it well on <m><tl>marble</tl></m>, and before that, to have washed it thoroughly. Some grind it with thoroughly with an <m>egg yolk</m> & then wash it in five or six lots of water and lay it on not with a <tl><fr>painceau</fr></tl>, which would be too soft, but with a <tl><fr>brousse</fr></tl> h[a]ighly softened & bent, & laying it on thickly as if one were applying it with a <tl>trowel</tl>; settling down it evens out and flattens. I have e[b]xperienced that crushing <m><fr>azur d'esmail</fr></m> with <m>egg yolk</m> & then washing it in several lots of <m>water</m> is good, although it loses a little of its vividness in the crushing of it. I have also washed it in several lots of <m>water</m> &, when it had settled a little, I removed the <m>water</m>, still q blue, with a <tl>sponge</tl> and squeezed it into another <tl>vessel</tl> thus <add>where it settled, & from the residue</add> I had the <m>ash</m>, flower and subtlest part of the <m>azure</m> without crushing it, which is the best, for in the crushing of it, it loses some of its color. Those who make it in <pl>Germany</pl> compose it like <m>enamel</m>, in large pieces which they pestle, & pass through several <tl>sieves</tl> & wash.</ab>

<ab>To make <m>azures</m> beautiful, they wash or soak them in a <m>r[c]ock water</m>, as they call it; it is a <m>water</m> distilled from mines where <m>azure</m> or <m><fr>vert d'azur</fr></m> is found, which distills naturally through the mountain's veins or is distilled with an <tl>alembic</tl> <fr>par</fr> from mineral stones of <m>azure</m> or <m>copper</m>.</ab>

<ab><m><add>Azure</add> ashes</m> are only good for landscapes because they die in <m>oil</m>. Only true <m>azure</m> lasts. <m>Azur d'esmail</m> cannot be worked if it is too coarse. Try it, therefore, on the <bp>fingernail</bp> or the <m>oil</m> palette. If it happens to be sandy, do not crush it except with the <m>egg yolk</m> or, better yet, wash it in clear <m>water</m> & with a <tl>sponge</tl> remove the <m>colored water</m> after i[d]t starts to go to the bottom, and in this manner you will extract the quite delicate flower, which will be easy to work with.</ab>
</div>
[a]Possibly referring to a flat brush "softened and bent" by use.
[b]'experimenté'
[c]water from a rocky spring

[d]'il' in original, which could refer to (masculine) 'azur d'esmail' rather than to the (feminine) 'eau colorée' (as in this case the pronoun should be 'elle').

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<page>011v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f28.image</image>

<div>
<id>p011v_1</id>

<head>Applying the color of <m><pl>Damascene</pl> steel</m> on <tl>knives</tl></head>

<ab>Rub the <m>steel</m> with <m>common <la>aquis fortis</la></m> mixed with <m>earth</m>. After, rub everything with <m>ashes</m> or <m>sand</m>. Do with the <m>gold</m> and <m>silver</m> as you would do on a^[a]<tl>touchstone</tl>.</ab></div>

<div>
<id>p011v_2</id>

<head>Getting rid of red <bp>eye</bp>s or black <bp>eye</bp>s</head>

<ab>Make two small licks of <m>raw <al>mutton</al> flesh</m>, and with a blindfold, apply it at night on your <bp>eye</bp>s.</ab></div>

<div>
<id>p011v_3</id>
<head>Applying unpolished <m>gold</m></head>

<ab>There is nothing better than the black varnish of <pro>sword-makers</pro> to use as a base layer because it dries quickly and makes the <m>gold</m> appear very beautiful.</ab></div>

<div>
<id>p011v_4</id>
<head>Making <tl><m>diamond</m> points</tl> for engraving</head>

<ab>These are for sketching what you want to draw on <m>stones</m> and for engraving. When these <tl>points</tl> are too dull and cannot engrave, it is necessary to rub <m>diamond</m> against <m>diamond</m>; one can rub them with <m><tl>pumice stone</tl> powder</m>. It is necessary to strip what you want to engrave on a <tl><m>lead</m> wheel</tl> or with a <m><tl>stone</tl></m> or with <m><tl>glass</tl></m>, because the polishing is like varnish or <m>glass</m> and stops the <m>diamond</m> or <m>sapphire</m> from engraving well. This is why it does not engrave easily on <m>horn</m>, which is polished and fat. But when the polish has been taken off, it is easy to engrave.</ab></div>

<div>
<id>p011v_5</id>

<head>Polishing a <tl><m>copper</m> wheel</tl></head>

<ab>Instead of polishing your <m>stone</m>, it is necessary to remove the grease by rubbing the flesh side of a piece of <m>leather</m> ^{on the <tl>wheel</tl>}, all while spinning the <tl>wheel</tl> if you have not worked on it in a long time.</ab></div>

[a]We think this means something like "apply to the knife as you would to the touchstone," i.e. in fine strokes.

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<page>012r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f29.image</image>

<div>
<id>p012r_1</id>
<head>For molding from <m>sulfur</m></head>

<ab><m>Sulfur</m> is made beautiful by mixing in <m>soot black</m> or powdered <m>sanguine</m>, which makes it harder and stronger. After letting it melt entirely until it becomes liquid and similar to <m>oil</m>, ^{then} mixing it with <m>verdigris</m>, you can use it for casting a <al>lizard</al> or something else in <m>plaster</m>, very cleanly.</ab>

<ab>
<margin>left-top</margin>
You must not cast it until it has cooled down and lost all its bubbles and eyes, and its surface has fallen and become flat as <m>water</m>. <m>Soot black</m> gives it a fine luster and makes it neater. Use yellow <m>sulfur</m> of the best kind, as the greyish natural sulfur is no good. Don't cast it in the wind and cold for it would fill with bubbles.</ab>
</div>

<div>
<id>p012r_2</id>
<head><m>Cimolee</m></head>

<ab><m>Cimolee</m> or <pro><oc>fuller</oc></pro>s' clay (fullers meaning those who finish cloth) is an excellent material for molding hollow or in relief. If you want to reheat it, it should first be warm and you should reheat gently from a distance, and little by little, otherwise it will crack. To reheat the figure put it in a <tl>pot</tl> in an <tl>oven</tl>, or in a <tl>closed oven</tl>. It comes out very smooth, clean and fine. With <m>cimolee</m> you can make a hollow mold of an animal and cast it in <m>lead</m>.</ab>

<ab>
<margin>left-middle</margin>
The work needs to have dried for 4 or 5 days before you bake it. When you are molding and making a hollow in <m>cimolee</m>, don't press all at once, but gently, or else it would split.</ab>
</div>

<div>
<id>p012r_3</id>
<head><m>Paper</m></head>

<ab>The whitest and thinnest is the best. And when the hollow mold is quite clean, as when it is made of <m>sulfur</m> or baked <m>cimolee</m>, <sup>the
paper</sup> comes out very clean. You can apply a layer or two of <m>white</m>, with a border of <m>gold</m>, to imitate <m>alabaster</m>. And after applying the
<m>white</m> you may burnish it with a <tl>tooth</tl>. Yet in order for it to take burnish you should soak the <m>lead white</m> with <m>egg white</m> and
<m>peelings from the fig tree</m>. Or better still, varnish your work with <m>white varnish</m>. Also while you are molding with <m>paper</m>, as it starts drying,
burnish the back of it with the <tl>tooth</tl>.</ab>
</div>

<div>
<id>p012r_4</id>
<head><m>Plaster</m></head>

<ab>Mountain <m>plaster</m> is greyer, and that from the region of <pl>Albi</pl> is whiter. It has to be heated over a closed fire, such as a <tl>reverberatory
furnace</tl> or <tl><fr>fours de barbiers</fr></tl>. The more freshly heated, the better it is to work with. It should be finely crushed on <m><tl>marble</tl></m>. After
preparing your hollow mold of <m>sulfur</m> or another [material] and oiling it and surrounding it with a circle, soak your <m>plaster</m> in <m>water</m>, not too
thick, and rub it well with your <tl><bp>finger</bp></tl>, and if it makes bubbles, throw on more powder of the said <m>plaster</m> and crush it with your <tl>
<bp>finger</bp></tl> until <figure>+</figure></ab>

<ab>
<margin>left-bottom</margin>
<figure>+</figure> it makes no more bubbles. Then once more throw and sprinkle <m>plaster</m> powder on top and leave to set well, then scrape the powder off.</ab>
</div>

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<page>012v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f30.image</image>

<div>
<id>p012v</id>
<head>Moulding <m>stucco</m> promptly</head>

<ab>Crush and pulverize finely <m>brick</m> or <m><pl>Armenian</pl> bole</m> or <m>sanguine</m> and incorporate it with melted <m>wax</m>, and thus melted,
cast like the others on a relief medal, and so you will have a hollow mold where you will be able to cast ground <m>paper</m> or <m>cimolee</m>.</ab>

<ab>
<margin>left-top</margin>
Finely <tl>sieved</tl> <m>brick</m> is better, since <m>bole</m> is too fat</ab>
</div>

<div>
<id>p012v_2</id>
<head>Cleaning yellow <m>pearls</m></head>

<ab>Gently boil them in a <pro>goldsmith</pro>s <tl><fr>bouteure</fr></tl> then, with <m>tripoly</m> and a small piece of <m>leather</m>, sprinkled on the flesh
side with the powder of the said <m>tripoly</m>, rub and polish the <m>pearl</m>.</ab>
</div>

<div>
<id>p012v_3</id>
<head>To whiten <m>enilanroc</m></head>

<ab>Some believe the <m>ceruse</m>-color crust that is on top to be natural. Others say that it is a lost secret of ancient <pro>lapidaries</pro>. Others say that it is
<m>enamel</m> that has been reheated. And, from this crust, one can form multiple ciphers, letters, circles and other bizarre motifs. As to the means, I proceeded thus: I
once wanted to add a layer of <m>arsenic</m> ground on <n><tl>marble</tl></m>. However, I did the experiment without it, and I put the above mentioned thing,
<m>enilanroc</m>, in a small <m>iron</m> <tl>casket</tl> on the fire of my <tl><pro>goldsmith</pro>s forge</tl> with three or four small half-burnt <m>coals</m>
and blew on it only with my mouth. However, the thing caught fire and became all white, not only on its surface but also inside. And then I let it cool on its own near the
fire, since otherwise, exposing it suddenly to cold air, it would have cracked. While it was all white, I passed it through my <tl><m>lead</m> wheel</tl> [a] and I found it
to be as hard as it was before. And after scraping off a little of the white, I found it to be of a clear fleshy tone. Finally, I polished it and observed that it took quite a

beautiful polish luster and that it could compare with a quite beautiful <m>agate</m> for the purpose of engraving it with some beautiful face and applying it to a table of

</ab>

<cont/>

</div>

[a]check meaning

<page>013r</page>

<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f31.image</image>

<div>

<cont/>

<id>p012v_3</id>

<ab>

<m>agate</m> of various colors. But because this overall whiteness did not correspond with the white crust on the surface that I was seeking, leaving the rest of the thing its natural color, I made a hole in a <m>brick</m>, precisely the same size as the object, and put it inside. Then I heated two <tl><pro>glassmaker</pro>'s soldering irons</tl> until they became red hot and, when they were red hot, I applied them one after the other onto the surface of the object until I got the white crust I was looking for, on which I made the drawing that I wanted, revealing the red background, with a <tl><m>diamond</m> point</tl>, and I polished it with <m>hog bristles</m> and <m>tripoli</m>. I don't know if it would be better to reheat it under hot <m>ashes</m>, and if it would be good to encase it in <m>alabaster</m>, which is quite cold, as I did in the <m>brick</m>.</ab>

</div>

<div>

<id>p013r_1</id>

<head>Flesh color with <m>arsenic</m></head>

<ab><m>Sublimated arsenic</m>, that is to say, the white kind sold as stone, when powdered on <m><tl>marble</tl></m>, mixed with <m>vermilion</m> or <m>laque</m> or <m>mine</m>, produces a beautiful flesh color that always shines. <m>Yellow arsenic</m> has a very beautiful color, the white one works well in <m>oil</m> and goes well with <m>lake</m>.</ab>

</div>

<div>

<id>p013r_2</id>

<head>To dye</head>

<ab>Mix <m>sal ammoniac</m> and <m>vitriol</m> and boil them together. Then mix in <m>lake</m> or <m>verdigris</m> and <m>azure</m> or similar ^{colors}, and dye. This will not come off unless the <al>animal</al> sheds. @<la>Non bona</la>.</ab>

</div>

<div>

<id>p013r_3</id>

<head>Polishing of <m>stones</m></head>

<ab>Engraved <m>stones</m> are not polished on a <tl><m>copper</m> wheel</tl>, but with <tl>brushes</tl> and <m>tripoli</m>. Yet faceted and flat stones are polished on the said <tl>wheel</tl>.</ab>

</div>

<div>

<id>p013r_4</id>

<head>Fine <tl>sieves</tl> of raw <m>silk</m></head>

<ab>Tammy of raw <m>silk</m> is made on a weaver, to make very fine and delicate <tl>sieves</tl>. And for that effect, you must not choose raw <m>silk</m> whitened by <m>sulfur</m> smoke: this covers the <m>silk</m> in a sticky vapor that would hold the flour and in the end would prevent it from</ab>

<cont/>

</div>

<page>013v</page>

<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f32.image</image>

<div>

<cont/>

<id>p013r_4</id>

<ab>

passing through so well. But for this effect, choose yellow and natural raw <m>silk</m> because it is stronger and rejects <m>flour</m> as <m><al>horse</al>hair</m> does.</ab></div>

<div>

<id>p013v_1</id>

<head>To recognize good <m>a[a]zur dust</m> for <m>oil[b]</m></head>

<ab>The one that piles up in small clods and is lumpy is the best because it is finer. Also the one that is also very pale in color, because <m>oil</m> darkens it. Certain sophists mix them together, but you will find this out if you pour some onto a piece of <m>paper</m> and lay it flat and spread it with your <tl><bp>finger</bp></tl> since, if it is mixed, it will be found varied and so to say with light and darker stripes, but if it is whole it will be even and of one color.

</ab>

</div>

<div>
<id>p013v_2</id>
<head>Whitening <m>pearls</m></head>

<ab>When they have been applied in embroidery on some garment, they are greasy and may be cleaned in <m>soapy water</m>.
</ab>
</div>

<div>
<id>p013v_3</id>
<head><m>Toadstone</m></head>

<ab>That which comes from the head of the <al>toad</al> and has the shape of the <al>toad</al>, painted naturally as you have seen, is the most excellent. It is said that if powder made from it is put on a <tl>brazier</tl> in someone's room at night, they will be unable to move, speak, or stop <pro>thieves</pro>.
</ab>
</div>

<div>
<id>p013v_4</id>
<head><al>Snakes</al></head>

<ab>It is said that if one calls a <al>snake</al> in Greek, saying <greek>ΟΦΗ ΟΦΗ</greek>, it will flee. Similarly, that if one calls a <al>swine</al> in Greek, <greek>ιον</greek>, it will come.
</ab>
</div>

<div>
<id>p013v_5</id>
<head>Candles</head>

<ab><pro>Candlemakers</pro> never make good candles when the <oc><env>autan</env></oc> wind blows because they always melt, however good is the <m>tallow</m> they use in them.
</ab>
</div>

[a]See Annotation Spring2016 Chang&ClemensShi on Azure Enamels
[b]for oil paints?

.....

<page>014r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b9059316c/f31.item</image>

<div>
<id>p014r_1</id>

<head>For <m>earth</m> walls and rustic construction</head>

<ab><al>Swallows</al> have taught us this craft by building their nests out of <m>mud</m> mingled with <m>wisps</m> and bits of <m>hay</m> or <m>straw</m> to make it bond. Therefore, in places where <m>stones</m> and <m>bricks</m> are lacking, one can use <m>earth</m> to make walls thin and thick. The best for this is light <m>earth</m>, which does not form clumps when plowed, or better, that which in some way mingled with <m>arene</m>, because it can be most easily beaten and tamped down. It is true that it needs to be dampened and cut into clumps with a <tl>ditch-spade</tl>, and put and placed in this form. This one lasts more and does not require that much effort, and dryness doesn't make it split or crack. However, since such kinds of earth are not found everywhere, those who live on good and fertile land, after drawing the width and length of their foundations with a <tl>rope</tl>, drive in long <tl>poles</tl> and <tl>rafters</tl> along the edges on each side to support boards between which they throw the <m>earth</m>, making each layer about one foot thick, interspersed with <figure>S.S.S[a]</figure>-shaped branches of <m>
<pa>heather</pa></m> or similar things, then they tamp it down and beat it with beaters of three different forms. One is called the mall, which has a triangular form like A, and this is used firstly to tamp down the <m>earth</m>. Then one uses another made of pointed <tl>woodblocks</tl> fitted onto a thick <tl>stick</tl>, and this one is used to properly press the <m>earth</m> at the wall's ends and edges where they adhere to the boards, and is called [blank]. The other one is called the bat and is used to flatten and beat the <m>earth</m> for the last time, as shown in C. Then one makes another layer of <m>earth</m> and <m><pa>heather</pa></m> and beats as has been said, and continues thus until completing the wall, which is covered with <m><pa>heather</pa></m> and then with <m>earth</m>. Some intersperse the said wall with rows of <m>bricks</m>. They also make walls with a sloped foot, adding width to the foundations according to how high they want to build the wall. When it is old it whitens, showing that it has <m>saltpeter</m> in it. Which is why, when they fall down, <pro>gunpowder makers</pro> profit from them.</ab>

<figure>
<id>fig_p014r_1</id>
<margin>left-bottom</margin>
<link>https://drive.google.com/open?id=0B9-oNrvWdIO5b2IVN19SU1Z6UkU</link>
<!--A-->
<!--Nicolas Misery:text refers also to a "C" part of the figure, which is absent in the actual figure.-->
</figure>

</div>

[a]Uncertain. Author seems to refer to a S-shaped movement of the hand, described with the letters "S.S.S.". To verify.

.....

<page>014v</page>

<image><http://gallica.bnf.fr/ark:/12148/btv1b9059316c/f33.item></image>

.....

<page>015r</page>

<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f35.image></image>

<div>

<id>p015r_a1</id>

<head><m>Damask Cloth</m></head>[a][b]

<ab>You can make <m>damask cloth</m> [c][d][e] of two different colours and imitate embroidery without adding anything else to it, as follows. Once it is is <m>dye</m> d yellow, pounce onto it such a pattern as will please you. Then you will sew some <m>string</m> or a bigger <m>cord</m> loosely [f] onto the pattern and throw it into a <m>dye</m> of <m>woad</m> or <m>pastel</m> and it will become green, except that which is beneath the <m>string</m>, which will remain yellow because the <m>green dye</m> will not have penetrated there. And you can do the same with other colours and, instead of <m>string</m> or <m>cord</m>, add some pieces of poor quality <m>cloth</m> cut in Moorish shapes on top of the first colour. In that manner, you will have cheap embroidery.</ab>

</div>

<div>

<id>p015r_a2</id>

<head>Casting <m>metals</m></head>

<ab>Candlesticks and small works are cast in a <tl>frame</tl> with sand. Having stamped the work, sprinkle it with <m>flour</m> in order to make the <m>copper</m> or <m>latten</m> run better. When the sand has been in use for one month, it is necessary to take some new ^{sand} because that which has been used, being reheated <figure>+</figure>, dries & loses its binding power. Yet it is used mixed with the new ^{sand}, because it makes the work less porous. Large works such as artillery, bells and similar things are cast in <m>earth</m>, and <m>copper</m> cast in <m>earth</m> makes less <m>crust</m> [g] and is whiter than when cast in sand. The <m>earth</m> is <m>sandy clay</m> [h] mixed with <m>horse dung</m> and <m>cloth</m> waste. The <m>earth</m> that has been used for casting, which is black, baked and as if burnt, is mixed with artificial sand, and is very good. For softening and making the <m>copper</m> run, once it is melted, one throws in some <m>lead</m>, which does not form an alloy but is found on the surface of the cast.</ab>

<ab>

<margin>left-middle</margin>

<figure>+</figure> by the heat of molten <m>metal</m></ab>

</div>

<div>

<id>p015r_a3</id>

<head>Casting <m>gold</m> and <m>silver</m></head>

<ab>The <m>sand</m> must be made from something very dry and arid and reheated well because, if it were humid, like <pro>founder</pro>'s sand, the <m>gold</m> and <m>silver</m> would leap out ^{of the mold} and sustain damage. It is also necessary for the <m>earth</m> to absorb the <m>metal</m> because cast <m>gold</m> or <m>silver</m> become very porous [i]. That is why it must be beaten out again, otherwise it is brittle, as seen in <tl>spoon</tl> handles.</ab>

</div>

[a]French original is tl "drap damasse", tcn "drap damassé", which better translates as a cloth made IN THE FASHION OF DAMASK.

see def. DMF: DAMASSÉ, adj.

"Tissé à la façon du damas"

def. Cotgrave

damas = damaske

[b]Yes, so we suggest "Damasked Cloth" instead of "Damask Cloth." The recipe is not for making authentic damask cloth but its imitation by dyeing (damask is woven material). For details, please see our annotation "AnnotationFall2015_YiKok_15r

[c]The French original uses an adjective (=damasse, tcn damassé)) which better translates into "Damasked Cloth" "to make a cloth that looks like damask or to make a cloth that is patterned/appears as damask" see definitions DMF

[d]Or we could translate the passages as "you can damask cloth" instead of "you can make damask cloth"

[e]Excellent! I agree.

[f]We suggest that this should be translated as provisionally instead of loosely, because for French "faufiler, arrived from original faulfiler" we found the following definition which makes much more sense meaning to sew something provisionally onto a piece of cloth with big stitches.

<http://cnrtl.fr/definition/faufiler>

[g]Need to check this technical meaning

[h]Clay used to stabilise the sand: http://foundry101.com/new_page_7.htm

[i]spongieulx

.....

<page>015v</page>

<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f36.item></image>

<div>

<id>p015v_1</id>

<head>Tablets</head>
<ab>Some are made from <m>boxwood</m> and inlaid with <m>ebony</m>, <m>sandalwood</m>, <m>ivory</m>, or from <m>gold</m> and <m>silver</m> like <m>damascening</m>. After that one writes with <tl>silverpoint</tl> and then erases by rubbing them with a <m><al>cuttlefish</al> bone</m>.</ab>
</div>

<div>
<id>p015v_2</id>
<head>Making letters easier to read</head>
<ab>One cuts the largest pieces of <m>beryl</m> or <m>crystal</m> round on one side and flat on the other, then one fits it with a little handle and puts the flat side on the letter</ab>
</div>

<div>
<id>p015v_3</id>
<head>Stamping</head>
<ab>If you have a <m>bronze</m> medal and you want to make it very light, make a <m>lead</m> cavity, then spread upon it a thin plate of <m>gold</m>, or sheets of <m>lead</m> or reheated <m>silver</m>, place your <m>bronze</m> medal on top and hammer it with a <m>wood</m> <tl>mallet</tl>.</ab>
</div>

<div>
<id>p015v_4</id>
<head>Softening <m>horn</m></head>
<ab>
Those who mount acou <la>Agnus Dei</la>s and make circles from <m>horn</m> for certain little boxes soften the said circles by soaking them in hot <m>water</m> and then shape them on a round or oval-shaped <tl>triblet</tl>.</ab>
</div>

<div>
<id>p015v_5</id>
<head>Ears</head>
<ab>When a defluxion occurs in the <bp>ear</bp>, be very careful not to put anything in it, and following the <a] proverb, <bp>ear</bp>s and <bp>eye</bp>s should only be touched with your <bp>elbow</bp>. However, to an affected <bp>ear</bp>, it is good to apply musky <m>cotton</m>, that is to say, <m>cotton</m> kept in <pa>musk</pa>, which comforts quite well.</ab>
</div>

<div>
<id>p015v_6</id>
<head><bp>Tooth</bp>ache</head>
<ab>Some put a clove of <m>garlic</m> in the <bp>ear</bp> on the side of the aching <bp>tooth</bp>, and after two or three hours they feel better. Others put in their <bp>nostril</bp>, on the same side as the <bp>tooth</bp> that hurts, some green skin scraped from the stem of <fr>b[b]isaube</fr>, namely that which is under the grey one that looks like a vine.</ab>
</div>

<a]English proverb: "You shouldn't put anything in your ear that's smaller than your elbow," but doesn't mention eyes.
<b]Unclear until we understand what bisaube (?) is

.....
<page>016r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f37.image</image>

<div>
<id>p016r_a1</id>
<head>Melting <m>soft iron</m></head>

<ab>It is a common opinion that once <m>iron</m> has been melted, it cannot be remelted because they only heat it in small <tl>forge</tl>s, in which it only catches fire. <pro>Alchemists</pro> are confident they can melt it with <m>realgar</m>, <m>lead</m> or <m>orpiment</m>. But without all that, some have found a way to melt not only brittle <m>iron</m>, like that used for <m>iron</m> pots, but also soft <m>iron</m> such as pig or ingot <m>iron</m>, which are the most difficult ones [to work with]. For this result they make a <tl>furnace</tl> that is one <ms><fr>pan</fr></ms> and a half wide and two <ms><fr>pan</fr></ms>s deep, and the <fr>tuelle</fr>, which is the barrel marked A where the <tl>bellsows' pipes</tl> fit, has to be placed midway through the depth of the furnace so that there is one <ms><fr>pan</fr></ms> of the <tl>furnace</tl> opening on the <fr>tuelle</fr> and one <ms><fr>pan</fr></ms> under it.</ab>

<ab>
<margin>left-top</margin>
Each square <ms><fr>pan</fr></ms> of the <tl>furnace</tl>'s opening can hold one <ms>quintal and a half</ms>, and the round <tl>furnace</tl> <ms><fr>pan</fr></ms> can hold two <ms>quintals</ms>.</ab>

<figure>
<id>fig_p016r_1</id>
<margin>left-middle</margin>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5aUw0eThJNEVTelk</link>
<!--José Beltrán Coello: Drawn before the writing of at least the second half of the first paragraph-->
</figure>

<ab>
<margin>right-middle</margin>

The <pro><m>iron</m> smiths</pro>, to make the <m>iron</m> run, at the opening of the <tl>forge</tl>, place two or three handfuls of <m><pa>wallwort</pa></m> ^{in the direction} they want to make the <m>cast iron</m> run, and that makes the <m>iron</m> wonderfully ductile and flowy.</ab>

<ab>It is necessary that the <fr>tuelle</fr> is inserted ^{through} to the middle of the <tl>furnace</tl> opening, which is a main clue for the secret because the wind will hit the edge and the wall, which makes up the center of the <tl>furnace</tl> and in that manner, it spreads equally everywhere, as well as above, like flames in a reverberatory oven, and thus heats much more; because if the <env>wind</env> hits the melted or ready to be melted matter perpendicularly, it would cool it down and it would stop it from running and melting. It is also necessary for the <tl>bellows</tl> to be activated by jumps and by the <m>water</m> stream as ^{happens} at <tl>forges</tl>, because in this way, <tl>bellows</tl> are activated with great precision[a] and great speed, which the strength of <pro>workers</pro> could not duplicate. As you can see, the wall is thus raised up and the <tl>furnace</tl> ^{is laid} on the ground, at the edge of which you dig a pit, like at other <tl>forges</tl>. In order to put the moulds you want to cast inside, and for this purpose, you unstop the opening that you made at the bottom of the <tl>furnace</tl> to allow the melted matter to flow. You will be able to melt two <ms>quintals</ms> of <m>iron</m> each time and, in order to do this, you will choose the biggest <m>charcoal</m> you will find and put a measure of it at the bottom on the surface area of the furnace and pile up from the bottom to the top of the wall, in the shape of ^{an upside down V}.</ab>

<ab>
<margin>left-bottom</margin>
Some make a mould of the same size as the piece, then immerse it, then beat a soft <m>iron</m> piece cut to size, and when this bit is red hot, they beat it into the mould and round it out with a large <tl>file</tl>. These bullets are thought to be stronger than the melted ones which, being made of hard <m>iron</m>, are more prone to break.</ab>

<cont/>
</div>

[a]Faire toutes choses par règle & par compas, ou par compas & par mesure, pour dire, Avec une grande exactitude, une grande circonspection.
<http://artflsrv02.uchicago.edu/cgi-bin/dicos/pubdicollook.pl?strippedhw=compas>

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<page>016v</page>

<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f38.image></image>

<div>
<cont/>
<id>p016r_a1</id>
<ab>which is usually of 3 <fr><ms>pan</ms></fr>s high. But first of all, you have to reheat your <tl>furnace</tl> with some big <m>charcoal</m>, so the bottom is red hot and then you will put the said <m>charcoal</m> measure in the middle of which you will put your <m>iron</m>, not at once but ten or fifteen <ms>pound</ms>s each time. And since it will be swallowed at the <tl>furnace</tl> bottom, keep putting some more. And add three or four <ms>shovel</ms>s of bigger <m>charcoal</m> and avoid it from getting consumed too quickly by wetting[a] it. And when you will see your <tl>furnace</tl> full of two <ms>quintal</ms>s of matter, or less if you don't have that much, you will leave the <m>charcoal</m> consume itself. And when the <m>charcoal</m> is almost dying at the <tl>furnace</tl>'s level, you will be able to pour in the moulds and <m>iron</m> or <m>metal</m> shells, which is even better for one <m>metal</m> will bite the other. And the inside of the mold has to be covered with soaked <m>ashes</m>, so it doesn't adhere to it.</ab>
</div>

<div>
<id>p016v_a1</id>
<head>Against Gonorrhea</head>

<ab>Soak some <m><pa>quince</pa> seeds</m> in some clear <m>water</m> and inject this viscous <m>water</m></ab>

</div>
<div>
<id>p016v_a2</id>
<head>To take fine forehead hair off</head>

<ab>Have a loose <m>silk</m> <tl>needle[b]</tl> and put it tight through hair bulbs and they will bite the <m>silk</m> like fine <m>cotton</m> wisps.</ab>

</div>
<div>
<id>p016v_a3</id>
<head><m>Silk</m></head>

<ab><m>Crimson silk</m> is more frequent than all the other ones because its colour is not as expensive as blue or green ones which are, also, good bargain for the <pro>worker</pro>. <m>Black silk</m> is less frequent because it costs a lot.</ab>

</div>
<div>

</div>
[a]billart (RC): a small ant thick truncheon

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<page>017v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f40.image</image>

<div>
<cont/>
<id>p017r_1</id>
<ab>The large canon is for great batteries and usually weighs fifty five or 60 <ms>quintal</ms>s. The breech is two cannonball and three quarters each. The mouth is one cannonball and three quarters each. It is thirteen or fourteen <fr><ms>pan</ms>s</fr> in length. But they are very uneasy to drive. The proper battery, to be used rapidly and to have great impact, is a hundred fifty to two hundred <ms>pace</ms>s. It is true that a fair impact can be had from three or four hundred <ms>pace</ms>s, but more <m>powder</m> is needed. Its usual load is 20 <ms>pound</ms>s of <m>powder</m>; its cannonball weighs 40 <ms>pound</ms>s, and 25 <al>horses</al> are required to carry it. When one needs a longer range than the usual one, half a <ms><tl>linstock</tl></ms> of canon <m>powder</m> is added. 80 or a hundred shots can be made with this canon per day, but it needs to be cooled down after each nine or ten shots if the battery is continuously used; if there are some breaks, it is not necessary to refresh it as often. For each <ms>quintal</ms> of <m>copper</m> - or one <ms>quintal</ms> and a half for big cannons - a <ms>quintal</ms> of <m>metal</m> is added. The <m>metal</m> is composed of eight <ms>pound</ms>s of <m>tin</m> for each <ms>quintal</ms> of <m>rosette</m> and even less for big bells for which we use only six <ms>pound</ms>s of fine <m>tin</m> for a <ms>quintal</ms> of <m>rosette</m>, to give it a louder voice. Because the more <m>tin</m> there is, the clearer the sound is. For canon founding, if the material and <m>charcoal</m> is provided as it is usually done because <pro>master</pro>s cannot afford it, it costs 10 to 12 <cn>pound</cn>s per <ms>quintal</ms>. And when the <pro>master</pro> provides everything, you give him 40 <cn>pound</cn>s per <ms>quintal</ms> for big pieces such as cannons, according to the King's ordinance. And 50 <cn>pound</cn>s for smaller pieces. For the more matter there is, the more the <pro>master</pro> makes profit. Another kind of 30 <ms>quintal</ms>'s fowler is founded which is longer than the others, and usually 8 <fr><ms>pan</ms>s</fr> long, which is used to attack fortifications and <fr>cassemattes</fr> by displaying them, at night, in groups on the moat's edge.</ab>

<figure>◯</figure>

<ab>
<margin>left-top</margin>
Each side of the breech's opening is half a cannonball thick, plus a third of a cannonball.</ab>

<ab>We give it two <ms><tl>linstock</tl></ms> of cannon <m>powder</m> for its load, and half a <ms><tl>linstock</tl></ms> of arquebus <m>powder</m> and the same for the others.</ab>

<ab>
<margin>left-middle</margin>
<figure>#</figure>
The alloy of cannons from <pl>France</pl> is made of one <ms>quintal</ms> of metal for two of <m>rosette</m>. But those of <pl>Toulouse</pl> and <pn>Ponce</pn> put 3 <ms>quintal</ms>s of <m>rosette</m> for one of metal.</ab>

<ab> The <m>rosette</m> used to remelt is better than <m>rosette</m> for cauldrons, for the latter get all dirty.</ab>

<ab>
<margin>left-bottom</margin>
The older pieces are composed of almost as much of each, that is to say a part of <m>rosette</m> and one of metal. This alloy can be recognized with a <tl>burin</tl> because the matter is acid and the part scraped off with the <tl>burin</tl> is yellow and white.</ab>
<cont/>
</div>

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<page>018r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f41.image</image>

<div>
<cont/>
<id>p017r_1</id>
<ab>
Culverines are big batteries of forty <ms>quintal</ms>s and eighteen <fr><ms>pan</ms></fr>s long. Their bullet of the <ms>King's standard</ms> and battery weighs 30 <ms>pound</ms>s and is therefore lighter than the canon's one. Therefore, It does not carry so many munitions because fifteen <ms>pound</ms>s are enough for its load. The canon has a bigger mouth due to the size of its cannonball, but the colverine is more precise and goes faster, having greater range due to its length. Its breech is two bullets and a third thick, the front is a bullet thick. Culverines are used for fighting fortifications from far away when it is not possible to easily approach them. And then canons come closer. They are used also to support the battery. Fifteen or sixteen <al>horse</al>s are necessary to carry it. They are made of the same metal alloy than the canon, like all others smaller pieces ^{of artillery}. For these, we add a little bit more metal in order to make the melting run better. And for two <ms>quintal</ms>s of <m>rosette</m>, you add six unit of metal per pound for smaller pieces. They range from 8 or 9 hundred <ms>pace</ms>s to a thousand <ms>pace</ms>s if the <m>powder</m> is strong and in the air half a <ms>league</ms>.</ab>

<ab>
<margin>left-top</margin>
Some invented loading cannons with cartouches.</ab>

<ab>
<margin>left-top</margin>
Some don't fill the canon with <m>powder</m> at once but twice and each time ramming the <m>powder</m> in, saying that each time you ram it, you raise it and give an <ms>inch</ms> more. But this is not sure for big pieces for they are loaded with a lot of <m>powder</m>.</ab>

<ab>
The <fr>bastarde</fr> is a middle-sized piece ^{of artillery} which weighs thirty <ms>quintal</ms>s and its cannonballs weight 15 <ms>pound</ms>s and is loaded with 10 or 12 <ms>pound</ms>s of <m>powder</m>. Its proportions are two cannonballs and a third part of a third one thick at the breech and one cannonballs and three parts of two at the front. They are used for fighting against less important defences such as gabions or sentries, topped with a tower, or similar thing. It is thirteen to fourteen <fr><ms>pan</ms></fr>s long like the great cannon. Ten <al>horse</al>s are necessary to carry it. It goes with the culverine for shooting because it carries small munitions. </ab>

<ab>
<margin>left-middle</margin>
Some make the breech three cannonballs thick and the front two cannonballs.</ab>

<ab>
The <fr>bastarde culverine</fr> weights 35 <ms>quintal</ms>s and is 25 <fr><ms>pan</ms></fr>s long. The breech is three cannonballs thick and the front two cannonballs. Its cannonball is like the bastard one, weighing 15 <ms>pound</ms>s. These are fixed pieces which cannot be carried on a carriage. They are for city defences. Some make cannons like these which are 27 or 28 <fr><ms>pan</ms></fr>s long, like <pl>La Rochelle</pl>'s <fr>vache</fr>, but such pieces are strengthened at the breech with a width of three cannonballs. Their range is one <ms>league</ms> and a half. Its load is like the <fr>bastarde</fr>'s one, and if one wants to hit <pro>horsemen</pro> very far way, more <m>powder</m> is added. After the cannonball there is a trace of smoke which drives your trajectory to where the cannonball is going. This is understood for cannons and the <fr>culverine</fr>, but not for small pieces.</ab>
<cont/>
</div>

.....
<page>018v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f42.image</image>

<div>
<cont/>
<id>p017r_1</id>
<ab>
The average weights 18 <fr><ms>quintal</ms></fr> and is xii <fr><ms>pan</ms></fr> long. Its cannonball weighs eight or 9 pounds and its load is six pounds of <m>powder</m>. The breech is three cannonballs thick and the front is two. It is more appropriate against cities' defence rather than in battery. It is however sometimes used either to break a barricade or to support the battery after cannons have shot, in order to prevent the assailed from rebuilding after the cannons have played and shot. Four <al>horses</al> can carry it. It has a range of eight or nine hundreds <ms>paces</ms> and almost as many as the bastarde.</ab>

<ab>
<margin>left-top</margin>
To cannons and big pieces we give as much <m>powder</m> as a cannonball and a half's diameter in proportion to the <fr>escusson</fr>. For medium pieces, such as campaign pieces, and the ones smaller, we provide them with powder the equivalent of two cannonballs have a quantity of <m>powder</m> equivalent to two cannonballs thick in proportion to the <fr>passevolant</fr> and smaller ones are given three cannonballs thick of <m>powder</m>. Usually, all pieces are loaded up to the escusson. We load all pieces under 4 <fr><ms>quintal</ms></fr> with the <tl>linstock</tl>, two <tl><ms>linstock</ms></tl> of cannon powder or one and a half lantern of arquebus powder. For smaller ones, we only use one <tl><ms>linstock</ms></tl>. The pieces of one <fr><ms>quintal</ms></fr> as loaded with a little charge.</ab>

<ab>
A battlefield piece weighs ten or twelve <fr><ms>quintal</ms></fr> and is ten or 12 <fr><ms>pan</ms></fr> long. Its cannonball weights 6 or 7 pounds and its loaded is 4 pounds of <m>powder</m>. The breech is three cannonballs thick and the front is two, like all the pieces under the mid-sized. The breech is thicker for they are in proportion longer and also because they are more frequently used than big pieces against houses or elsewhere. Their caliber is also small, so we give their breech a thickness of three cannonballs. They are used to follow immediately a camp and for the defence of cities and houses, settling them on walls or in a tower. Three good <al>horses</al> are required to carry it.</ab>

<ab>
The passevolant weighs 6 <fr><ms>quintal</ms></fr> and is eight or nine <fr><ms>pan</ms></fr> long. Its cannonball weighs two pounds, and its load is a pound and a half of <m>powder</m>. The breech is three cannonballs thick and the front two. Two <al>horses</al> can carry it because one single horse can't carry a piece. It is used for houses' defence or to be brought amongst the infantry to break a horsemen rank.</ab>

<ab>
A <fr>faulconneau</fr> weighs 4 <fr><ms>quintal</ms></fr> and is 10 <fr><ms>pan</ms></fr> long. Its cannonball weighs a pound and a quarter, its load is half a pound of <m>powder</m>. The breech is three cannonballs thick and the front two. To carry it, two <al>horses</al>, although these pieces are seldom moved since they are manly used to defend houses. The ones which are carried over for battery or a fight, or a siege, are culverines, medium-sized bastardes and pieces</ab>
<cont/>
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<page>019r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f43.image</image>

<div>
<cont/>
<id>p017r_1</id>
<ab>
of campaign. <fr>Faulconneau</fr> is to commonly loaded with a <tl>linstock</tl>. There are also some other <fr>faulconneaux</fr> weighing three <fr><ms>quintal</ms></fr>, and which are nine <fr><ms>pan</ms></fr> long. Their cannonball weighs half a pound. Their load is a quarter pound of <m>powder</m>. They can be loaded with a <tl>linstock</tl>, but more commonly they are loaded with a charge. The breech is three cannonballs thick and the front two. The small pieces which are under three <fr><ms>quintal</ms></fr> are at least three cannonballs and a twelfth of a cannonball thick at the breech, and the front thickness is sometimes reduced to get the proportion according to the length.</ab>

<ab>
<margin>left-top</margin>
The strength of the piece is at the level of the <fr>tourillon</fr>, where the lighted <m>powder</m> starts.</ab>

<ab>
<margin>left-top</margin>
The strength of the <fr>berche</fr> is at the <fr>maslée</fr>, behind the breech.
</ab>

<ab>
A double musket weighs 2 <fr><ms>quintal</ms>s</fr> and is seven <fr><ms>pan</ms>s</fr> long. Its cannonball is a quarter of a pound or less. The breech is three cannonballs thick, and the front two. It is loaded with as much <m>powder</m> as cannonballs, to the top of the touch line, which is equivalent for such small pieces to 3 or 4 arquebus <m>powder</m> cannonballs, because if canon <m>powder</m> is used, it can be loaded five cannonballs high.</ab>

<ab>
<margin>left-middle</margin>
This has to be understood as <m>iron</m> cannonballs. When a metal cannonball is fired, we overload it with a quarter of <tl><ms>linstock</ms></tl> because if a cannonball weighs 40 pounds, a metal one weighs 60 pounds. This is why the sight is higher when a metal cannonball is fired. Because if an <m>iron</m> cannonball is fired directly, a metal cannonball is shot from six lines higher. A metal cannonball, mixed with <m>copper</m> to make it more brittle, is more effective at close range than the <m>iron</m> one. But an <m>iron</m> one is more effective at long range. </ab>

<ab>
A simple musket weighs one <fr><ms>quintal</ms></fr> and is six <fr><ms>pan</ms>s</fr> long. We disregard the ones that are less than two <fr><ms>quintal</ms>s</fr>, and which usually throw lead cannonballs by its weight, and [concentrate on those] which are loaded by caliber. However, those [muskets] that can be loaded with metal or <m>iron</m> cannonballs will work better, for they will do more damage than the <m>lead ones</m>. It is loaded up to the touch line, which is 4 cannonballs thick. </ab>

<ab>
Arquebus à croc weighs 60 pounds, the bigger one is five <fr><ms>panel</ms>s</fr> long, it is loaded up to the touch line, which is 4 cannonballs thick; it is loaded with <m>lead</m> cannonballs, and is for house's defence. These great and small arquebuses are used for to make <fr>orgues</fr>, appropriate for an assault outside and inside. They are casted separately in order to be used in various ways. See the 4th leaf marked.

<figure>
<id>fig_p019r_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5WWY3VjdIVktqZVlk</link>
<!--Philip Cherian: Continues into p022v-->
</figure>
</ab>
</div>

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<page>019v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f44.image</image>
<div>
<id>p019v_1</id>

<head>
Mathematical figures without a <tl>ruler</tl> or <tl>compass</tl></head>

<ab>
you know how to draw a circle with a <tl>quill</tl> and otherwise with your right <tl><bp>thumb nail</bp></tl> and your folded <tl><bp>big finger</bp></tl> if you don't have any <tl>ruler</tl>, fold a <m>paper</m> sheet in five or six folds to use it. And if you want to draw a column, arch, oval, cartouche correctly without a <tl>compass</tl>, fold your <m>paper</m> so the fold is a straight line from which you draw within the relevant distance a point or a line and fill it with some <m>ink</m>. Then fold again the <m>paper</m> and rub it on its reverse and it will stamp whatever you have made. In that way, A is the line without <m>ink</m> made by the <m>paper</m> fold, B is the line you have drawn, C is the stamped one.

<figure>
<id>fig_p019v_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5SUUzLXQ4UVIGR1U</link>
<!--Philip Cherian: Figure under text of block, placed after In this way-->
</figure>

<figure>
<id>fig_p019v_2</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5cy0tdXF5MGJTLU0</link>
</figure>

<figure>
<id>fig_p019v_3</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5R0VMdFQ3VC1lQmM</link>
</figure>

<figure>

<id>fig_p019v_4</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5TGJOZGVvTmV1LW8</link>
</figure>
</ab>
</div>

<div>
<id>p019v_2</id>
<head>To write as easily from the left as from the right</head>

<ab>
Write in the best manner possible with some well gummed <m>ink</m> on as many cards as many words you want to write and when your letter is full of <m>ink</m>, apply your <m>paper</m> and rub with a <tl>tooth</tl> the back of the cards.</ab>
</div>

<div>
<id>p019v_3</id>
<head>Enemas</head>

<ab>
They used to mount them with a <tl><m>leather</m> handle</tl> or <tl>pocket</tl>, the best one being of <m><al>cat</al> pelt</m> which is more padded than any other else. And then they started to fold the <tl>handle</tl> from one side and then they kept folding it and turning it on itself and in that way enema runs slowly, but this manner takes more time and is less easy than a <tl>syringe</tl> which has been discovered since then with which a man can give to himself an enema. It is true that it always gives wind at the end.</ab>
</div>

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<page>020r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f45.image</image>

<div>
<id>p020r_1</id>
<head>@For making <m>millas</m></head>

<ab>You must have some <m>millet</m> soaked[a] in order to remove the <m>chaff</m> and then clean it well. Next you must <tl>grind</tl> it again quite finely and pass through a cloth <tl>sieve</tl>. One must soak the <m>flour</m> with fresh melted <m>butter</m> and some <m>milk</m> so that it is very light, like the <m>pastry</m> to make <m>beignets</m>, and add <m>egg yolk</m> depending on the amount of <m>flour</m>, so that there are two <m>egg yolks</m> for each <m>millas</m>. Then you will put in some <m><pa>saffron</pa></m> if you want to give them a little colour.</ab>

<ab>Afterwards you must have some <tl>mold</tl>s, which must be made from <m>terra^{cotta}</m>, in the shape of the bottom of the catholic hat[b], but it must be open at both ends. And then, having made a good fire, clean where you want to put your <m>millas</m> and then take your <tl>mold</tl>s and grease them very well so that the <m>pastry</m> does not stick to them when baked. Once you have done that, put your <tl>mold</tl>s in a place in the <tl>oven</tl> where you clean and sprinkle a bit of <m>flour</m> on the bottom and fill them with some of the aforementioned <m>pastry</m>. Then you will cover the said <tl>mold</tl>s, with a <tl>lid</tl> made like the <tl>mold</tl>, but it must be bigger and not open at the top. Then once done, you will put some <m>straw</m> on the said <tl>lid</tl> and a lot of hot <m>charcoal</m>, and set a fire around it, once done, you will open up one of them after a bit of time and will see when it is baked, it will be hard.</ab>
</div>

<div>
<id>p020r_2</id>
<head>@<pro>Glazier</pro></head>

<ab><m>Glass from <pl>Lorraine</pl></m> is smoother and more even than <m>plate glass</m>. It is more durable and enduring. But commonly it is tainted with veins as if with straight lines and wrinkles, which happens because the <pro>glassmakers</pro> have made it put it still hot on <m>straw</m> which therefore stains it. One can, however, rub it off with some varnish, <m>salt</m> and other <m>drugs</m> used by <pro>glassmakers</pro>. <m><pl>Lorraine</pl> glass</m> is sold by <ms>band</ms>s and each <ms>band</ms> is composed of three <ms>panel</ms>s. A <ms>band</ms> costs usually X <fr><cn>soul</cn></fr>s. The <m>plate glass</m> is made in <pl>France</pl>, it is whiter and clearer but it does not last as long as that from <pl>Lorraine</pl>. It is also commonly blown And in that case, it is better to cut little lozenges rather than large square pieces in the shape of a frame because they cannot sit themselves well evenly.</ab>

<ab><margin>left-bottom</margin>
The <m>plate glass</m> is sold in <ms>set</ms>s. Every set contains 4 <ms>plate</ms>s that are usually sold at 10 or twelve <cn>lb</cn>s.</ab>
</div>

[a]my translation, but uncertain (pain fraisé is soaked bread)
[b]See annotation by Chiostrini and Palframan on Making Millas.

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<page>020v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f46.image</image>

<div>
<id>p020v_1</id>
<head>To whiten the <bp>face</bp></head>

<ab>Pound <m><pa>puffball</pa></m> in <m>cistern water</m>, and no other, and wash with this whitened <m>water</m>. This is highly regarded. And I believe that making it from <m><pa>wheat</pa> starch</m> and using that would be even better.</ab>
</div>

<div>
<id>p020v_2</id>
<head>Against windiness, colic, etc.</head>

<ab><m>Sap squeezed from <pa>orange</pa> peel</sup></m> or <m>water distilled from <pa>orange</pa> peel</m> is excellent against windy colics. <m>Candied peel</m> is also excellent for <bp>stomach</bp> winds. And to try it, having poured <m>sparkling wine</m> in a <tl>glass</tl>, press <m><pa>orange</pa> juice</m> on the <m>wine's foam</m>, which comes only from vapor and wind, and you will see that the foam will immediately disappear. Also, if you squeeze <m>juice</m> against a candle it will burn brighter.</ab>
</div>

<div>
<id>p020v_3</id>
<head><m><fr><pa>Eschervis</pa> racine</fr></m></head>

<ab>They want to be planted in a very humid place or such that, with ^{water from} wells or fountains, they can be frequently watered, because this makes them tender. Otherwise a hard core forms inside, which takes away all their goodness.</ab>
</div>

<div>
<id>p020v_4</id>
<head>Against burns</head>

<ab>Pound an <m><pa>onion</pa></m> with <m>verjuice</m> and leave it to soak like this, then apply it. Or better, apply <m>black soap</m> on the burn. Experimented.</ab>
</div>

<div>
<id>p020v_5</id>
<head>A common saying</head>

<ab>Steeple of <pl>Rodez</pl>, church of <pl>Albi</pl>, bells of <pl>Mende</pl>. However, the bell of <pl>Toulouse</pl> named Cardaillac is held to be one of the most beautiful in <pl>France</pl>, all the more since it weighs two hundred and fifty <fr><ms>quintal</ms></fr>s.</ab>
</div>

<div>
<id>p020v_6</id>
<head>A form of regimen</head>

<ab>Every <tmp>morning</tmp>, take two or three <m>eggs</m> laid on the same day or the <tmp>day</tmp> before. Heat them up but do not cook them. Eat the <m>yolk</m>s well dusted with <m>sugar</m>. And afterwards drink one <ms><bp>finger</bp></ms> [a]of <m>wine</m>. And in addition to being nourishing, it is good for the <bp>stomach</bp>.</ab>
</div>

[a]Iara says this is actually current bartender-speak

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<page>021r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f47.image</image>

<div>
<id>p021r_1</id>

<head><pro>Cannoneer</pro></head>

<ab>As for small cannons that are not loaded with a linstock, one loads them with <m>powder</m> all the way up to the touch hole, which is placed at a point on the cannon that indicates the appropriate measure.</ab>

<ab>To point a cannon, that is to say to fix the target, one needs to set the sights, in other words, take aim, from the sides rather than the top, that is to say the top of the cannon. By taking aim from the top, you can find the line of fire that leads to your target, but you will not find out if the cannon leans more to one side or the other. Then

take your sights first from one side and then from the other and adjust your cannon to point at your intended target. Then set your sights from the top of the breech, which will be quickly done. Next, use your discretion to lower the cannon a little if you are within firing range, because the force of the <m>powder</m> blast usually raises the barrel. If you are outside of the range in which the cannon can fire in a direct line, you should consider that the weight of the cannonball will lower the barrel.</ab></div>

<div>
<id>p021r_2</id>

<head>Shooting a cannon at night</head>

<ab>Some keep their cannons loaded andin the daylight hours. Then, so that the <pro>cannoneer</pro> can shoot into the breach where the besieged are perhaps repairing [the breach], the assailants raise a false alarm so that the besieged throw torches or <m>artificial fire</m> into the moats or around the breach, light by which the <pro>cannoneer</pro> can take aim. Sometimes by using the reflection off <tl>mirrors</tl> or <tl>jars</tl> full of <m>water</m>, the assailants can redirect the light onto the breach. The method that you know, using a sundial and <m>lead</m> line, is also very good. Others use strong <m>iron</m> pegs to fix two or three planks onto a platform of <m>wood</m> made to carry a battery of cannon, and leave some notches empty where the cannon wheels can pass perfectly. And with this method you will always put the cannon in such a position so that it doesn't tilt to the right or the left. And so that it isn't too high or low when you fire the following day, you place a marker that is well fixed in the ground, that just touches the bottom edge of the canon after it has been aimed and adjusted for firing.</ab>

<ab><margin>left-bottom</margin>If the platform, the wheels or wedge parts break or come loose, this invention holds no advantage.</ab>
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<page>021v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f48.image</image>

<div>
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<id>p021r_2</id>
<ab>Others place two well-staked poles into the earth in such a way that the one is as far from the edge of the cannon's mouth as the other. And underneath they place the above-mentioned ruler. Then when want to fire the cannon at night, they push their cannon straight towards the above-mentioned ruler and make it so that the edge of the mouth rests on the end of the aforesaid ruler. Next, they measure with a ruler, or a similar tool with measured intervals, from one pole to the side of the cannon and if it is too close or too far, they adjust the canon and make it the same distance as from the other pole.</ab></div>

<div>
<id>p021v_1</id>

<head>Unspiking your cannon</head>

<ab>Some people say that one should remove the cannonball if the cannon is loaded and introduce fire into the mouth of the cannon. But it is probable that the fire will leave again through the mouth rather than seek its exit through the touch hole which is spiked and restricted. The most reliable way is that the <pro>cannoneer</pro>, who is often a <pro>founder</pro>, drive in a <m>nail</m> and turn back the point of the <m>nail</m> that the enemies placed therein, and with a <tl>trephine</tl>, make another touch hole at the side of the spiked one, which can be done in an hour. And if with time the violence of the <m>powder</m> loosens the <m>nail</m> that the cannon was spiked with, you will make a nut in the second touchhole and place a <m>screw</m> in it, so perfectly that it will never come undone.</ab></div>

<div>
<id>p021v_2</id>
<head>Defense of a town</head>

<ab>There is nothing more efficient for defending a breach than mortars placed at the front line, as soon as the initial assault. They carry a lot of ammunition and are filled with cart <m>nails</m>, big <m>steel</m> dice, bits of <m>chains</m> and similar items. And when one senses that the enemy is close, one lights everything on fire. Grenades, well-constructed and thrown at the right moment, also cause much damage. And to make them worse, one can add some roughly crushed <m>glass</m>, either in the mold or in the crucible, when one casts them. This worsens the wounds. The grenades must not be too brittle, but it is good that they be somewhat alloyed with the material of the grenade so that instead of breaking up, they resist shock better and inflict more violence.</ab></div>

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<page>022r</page>
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<div>

<figure>
<id>fig_p022r_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5dWtBam9HWEpiaHc</link>
<caption>The one depicted is a bit too big, see the following one.</caption>

<!--Francois V. Pageau: The whole page seem to be rejected, saying the circle is too big and refers to the following one. Might be useful to reconstruct foliation.-->

<caption>Cannon ball, weighing 40 <ms>lb</ms> and being 12 <fr><ms>lignes pied de roy</ms></fr> long.</caption>
</figure>

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<page>022v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f52.image</image>
<div>
<id>p022v_1</id>

<figure>
<id>fig_p022v_1</id>
<margin>left-top</margin>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5RnQ2Ri1PenA1aDA</link>
<!--Philip Cherian: Continues from p019r-->
</figure>

<head>Orgues</head>

<ab>Then they are affustés on a rather thick board close to each other and notches are made onto the board this way <figure><id>fig_p022v_2</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5RnQ2Ri1PenA1aDA</link></figure>.
And through these notches is fit the harquebus's hook which has a hole through which we pass, from under, <m>iron</m> pegs. When you want to aim, you affuste the middle and side ones. If you want to make double, triple or four ranks or more, you have to put similar board rank one on the other, like the first one. This manner of making an orgue fait faute @[a] and makes it very stable. Because the other ones made of one massive and solid piece have cannons of only one and a half pan long. They therefore are used only for a salvo at an entrance or to protect a gate.</ab>

<ab>
A small hooked harquebus a-crocque weighs 40 <ms>pounds</ms> and is 4 <fr><ms>pans</ms></fr> long, and is to be loaded to the touch hole, its breech is three cannonballs thick, its front two.</ab>

<ab>
The double cannon is nothing different from the big ordinary cannon, apart from the fact that its cannonball has one <ms>line</ms> of thickness - or of diametre - more. A <ms>line</ms> is a <ms>twelfth part</ms> of a cannonball. At the breech, it has a thickness of one <ms>line</ms> more than the cannon, and the mouth is half a <ms>line</ms> thicker. Its load is six or seven <ms>lbs</ms> of <m>powder</m>, more than a cannon, that is to say of two big linstocks. Its cannonball weighs 56 <ms>lb</ms>. Thirty <al>horses</al> are necessary to carry it. It weakens the walls more than a cannon. But it is very uneasy to drive. This is why it is unused today and is more appropriate to be brought to a nearby place for close combat or for a [ceremonial] entry, rather than for an ordinary execution. They are called basilics, and they are unmovable pieces.</ab>
</div>

[a]fausser: could mean pierce

.....
<page>023r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f51.image</image>

<div>
<id>p023r_1</id>

<ab>A mortar is made so that it weighs three <ms>quintals</ms> and carries a cannonball flat on the side from which it must exit and round on the side of the cannon, as if it was a cannonball cut in half. It is two pans long. It has to be loaded with xxv <ms>lb</ms> of powder because it has to be full to the opening and has to fire. Its bottom is not thicker than the opening and is made of one single piece. Its material has to be better than other artillery pieces, and for 4 <ms>quintals</ms> of fine <m>copper</m> there must only be one quintal of <m>metal</m> so that it absorbs the shock, and so that, when fired with more force, it is more efficient. It has to be used against a door with a large <m>iron</m> cross before the bullet, and once loaded, needs to be covered with a firmly sewed rough <m>canvas</m> which should be completely smeared with <m>turpentine</m>. Four handles need to be added to it while it is cast, it is thus easier to place. The <m>iron</m> cross is fixed to the opening with the <m>canvas</m> covering it. The handles have to be held at its opening's edge, as you can see. To place it, you need three or four <m>iron</m> <tl>pegs</tl> of one <fr><ms>pan</ms></fr> long and as thick as a <ms><bp>finger</bp></ms>, which have their point like a <tl>gimlet</tl>, and their body like a screw as they are strongly pulled, and a ring on the other side to turn them with a short <tl>stick</tl> that fits into the <tl>ring</tl>. And the <tl>pegs</tl> are fixed on the door, not straight, because they would not have any strength, but crooked as if you wanted to fix them towards the middle of the <m>mortar</m>, and to do the holes of the handles need to be quite large. In that way, the shooting <m>mortar</m> pushes the <tl>pegs</tl> back and forth into the door, and makes more damage. Once in place, you need to have <fr>saucisse de <m>buckram</m></fr> @[a] made in this manner: take eight or nine <fr><ms>canes</ms></fr> of <fr><m>buckram</m></fr> selvaige or more if the gap is larger, which have to be as large as four or five <ms><bp>fingers</bp></ms>, have it sewn in such a way so that it is like an intestine where a <tl>stick</tl> as thick as a <ms><bp>finger</bp></ms> can fit. Fill it completely with good</ab>

<ab>
<margin>left-top</margin>
This one is to be put below an undermined tower with its opening towards the top. One casts two large <m>iron</m> <tl>rings</tl> and, with a <tl>stick</tl> or two, four men carry it. They are also used for placing within wall breaches but just half a charge is necessary, that is x <ms>lb</ms>, and fill the rest with some <m>pebbles</m> and <m>cart stones</m></ab>.

<figure>
<id>fig_p023r_1</id>
<margin>left-middle</margin>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5RzRNM284T3Q0eTQ</link>
</figure>

<ab>

It needs to be covered entirely with a waxed canvas and rubbed with turpentine and other combustible things. This cover is made to ensure that the bullet does not fall and so that, when the cover is given fire, the powder does not fail. Upon the fuse, you will need to put a good quantity of powder. Some put on the ball a cross of iron which is longer than the mouth of the mortar by two pans. Others only put the ball.

<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f54.image></image>

p024v_1

To bring a canon over land

It requires a hundred roadworkers or more to even gap tracks and others obstacles such as trees, stone piles and similar things to make gabions on the road. Once one is close to the battlefield, the gunner goes at night to locate the most convenient place to display the battery and if he does find one, a false alarm should be raised so the ones inside throw torches. Very often, you shoot as strong as possible so the assailed despising the battery do not fix it. Usually, there are not many gun-turrets or other defenses. And when a breach is made at the strongest and the cannon replaced, it will surprise the assailed. One should approach fortified towns at night, but [illegible]. The workers bring gabions and once these are displayed, some boards or plank are put behind gabions to keep the workers safe while behind, they fill the gabions with some earth by means of their shovels. And while this is being performed, another false alarm is raised from another side. And once you have shot a cannon ball at the defense, the gabions the workers display the boards to make a plain ground or a platform of four fingers or half a foot thick. And they are set into the earth. The platform is five or six feet larger than the cannon's wheels on each side if there is just one cannon. But if there are more, platformes are side by side. The space between the pieces is equivalent to the width of two gabions. As long as the battery lasts, the gunner's sleep and eat close to a cannon. In small spaces, one does not bother to dig trenches to reach the pieces but display many high gabions and run towards them. In large places one digs trenches. There are always in trenches or behind barricades units on guard duty to defend the pieces. A small powder container is kept containing three or four separate cags.

You have to know the French containers for the artillery.

It is good if the platform is slightly oriented upwards because the piece is more quickly put in action [& tient plus coup] and hits harder.

p024v_2

margin-left-top

Lined cannons

are those with a lining in the powder container space. The bullet rolls only up to the opening of this chamber. These ones are not so efficient and are not easy to load but they do not light up so quickly.

p024v_3

margin-left-middle

Steel touch-hole

is to be put as an anchor in the canons so it will not be damaged during the shooting unlike the others very quickly did. But these anchors are prone to jump off and injure the canonner. The only thing is to use an appropriate touch-hole.

p024v_4

margin-left-middle

A touch-hole at the bottom level is the best. That way the cannon does not come backwards.

.....

025r

imagehttp://gallica.bnf.fr/ark:/12148/btv1b10500001g/f55.item

cont

p024v_1

a hundred feet away from the battery and should be surrounded with barricades or trenches. And this is where they go and get powder in bags or baskets. And each piece has its own basket to it three or four times. Then, when the cannon is placed on the platform, one puts an arquebus bulletproof board between two gabions. Then, with a lantern, a gunner reloads it, and with the ramer which is at the other end of the lantern, two men press as hard as they can, nine of ten pushes, until the powder is dense Then another one adds some hay or straw and the others ram for a while. Another one loads the cannon ball. And the one with the hay adds the same amount and presses again. Another one adds a wood hung but this might damage the piece. And if one has to shoot a lot, one would need many hungs which are heavy to bring. When the cannon is loaded, xx or xxv workers push it forward with the sticks by the side, behind the gabions without taking off the board until the cannon reaches it. Once the cannon is forward, the gunner adjusts it with the hooks at the back, pointing at the target on each side and then in the middle. Then another one nearby sets fire. Each gunner needs 3 cannoners: one to load, one to point at the target and one to set fire. Sometimes, against a camp where breaches are made, one shoots a cartridges, as you find elsewhere, or one may use a basket full of stones, cars' nails or similar things. A cannon must not be carried unless it is with two wood or iron carriages. And you need seven quintals and a half of iron to fill the cannon's wheels, flasques and other equipage. The best battery is, according to some people, made as a curtain which sideways, because by doing so you unsettle more stones.

figure-----figure

<ab>
<margin>bottom</margin>
Some add small wheels to a piece and a thick board to the end of the piece so it can be moved quicker and without noise. Others add some leather around the wheels and also add soles or leather shoes to the horses' hooves so they cannot be heard.</ab></div>

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<page>025v</page>

<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f56.item</image>
<div>

<id>p025v</id>
<head>Gunner[a]</head>

<ab>There are three rings on a cannon, one at the breech[b], one in the middle, one at the mouth. When founders want to work on their piece or have a cannon or another lighter piece and easier to carry, they make it from the breech ring to the middle one in the same size and proportions as the bullets described below. But they taper it more or less one line on each side from the middle ring to the mouth ring, always using a compass for their measures from the straight line which they draw in the <fr>
<tl>modelle en <m>papier</m></tl></fr> whatever pattern. And this reduction is consequently equivalent to seven or eight <ms>quintals</ms> less. And this does not make the piece less secure. This is done for big pieces, but for pieces smaller than the mid-one, one has to keep the aforesaid proportion and make the piece in one straight line from one end to the other. Usually the trunnions[c] are situated one foot away from the middle ring in the direction of the breech.

But since some gunners prefer a piece heavier at the front for it is easier to adjust and for that purpose, it is sufficient to place the trunnions closer to the breech and further away from the middle. On the contrary if you want to make it heavier at the bottom, you will place the trunnions closer to the middle of the piece or forward in direction of the mouth. When you drill your piece make a fluted cast <m>iron</m> bar like a bed poster and there must be sixteen flutes or notches in which you will fix sixteen well-even blades[d], so altogether they will cut and scrape similarly. Because should some not cut or scrape, there would be some waves and spaces in the cannon which would be bent and the piece might break in its middle. In order to avoid this the drill has to go evenly from the breech to the mouth and the piece's mouth must be evenly hard so the <m>powder</m> will be released at once with more strength. On the contrary, if it is blocked it goes sideways and makes the piece break.</ab></div>

[a]Previous translation referred to the word "cannonier", we changed it into "gunner." The word "gunner" is used to describe a professional maker of canons in modern sources. See for instance : John ROBERTS, The Compleat Cannoniere, London, 1652 , p. 25.
https://en.wikipedia.org/wiki/Cannon_operation#/media/File:Cannoniere_1652.jpg
[b]"Breech" used for "culasse". John ROBERTS, uses this exact term to describe this bottom part of the cannon, The Compleat Cannoniere, London, 1652 , p. 25,
[c]Uncertain. The author seems to refer to an axis/element which allows one to adjust the angle of the shooting. John ROBERTS, uses the term "tunnion", The Compleat Cannoniere, London, 1652 , p. 25
[d]French says "costeau", which might refer to a specific tool. The word "blade" rather than "knife" seems more appropriate to suggest the use of the tool.

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<page>026r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f57.item</image>

<div>
<id>p026r_1</id>

<ab>
<margin>top</margin>
<rub>This is the right bore measure but the drill always rubs a bit more off.</rub></ab>

<figure>
<id>fig_p026r_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdIO5SHVaaE9oQjRySIE></link>
<caption>Cannon ball</caption>
</figure>

<cont/>
</div>

<div>
<id>p026r_2</id>
<head>Mortar</head>

<ab>They are better when shaped as the crucible with a narrow bottom and a larger mouth. You have to strongly crush the powder and add either some sand or some <m>cork</m> or even better a <m>pure pierced wax plate</m> because it seals well up and is more efficient. A bullet just makes a hole, the <m>powder</m> only is more efficient. One knows that they are stronger when kept in a dry place for fifteen days.</ab>

<figure>
<id>fig_p026r_2</id>
<margin>left-bottom</margin>

<link><https://drive.google.com/open?id=0B9-oNrvWdlO5UGxoanZwVEw2akE></link>
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<page>026v</page>
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<figure>
<id>fig_p026v_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5TmZXTkNLdDFjWms></link>
<caption>Culuerin bullet </caption>
</figure>

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<page>027r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f59.image></image>

<div>
<id>p027r_1</id>

<figure>
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<link><https://drive.google.com/open?id=0B9-oNrvWdlO5S3loampQRU55UjQ></link>
<caption>Cannonball of a bastarde cannon</caption>
</figure>

<figure>
<id>fig_p027r_2</id>
<margin>left-bottom</margin>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5cE1VQIVIV2ZWakE></link>
<caption>Medium-sized cannonball</caption>
</figure>

<figure>
<id>fig_p027r_3</id>
<margin>right-bottom</margin>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5M2Ftb2g0bG1FQ3c></link>
<caption>Cannonball of a campaign artillery piece</caption>
</figure>

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<page>027v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f60.image></image>

<div>
<id>p027v_1</id>

<figure>
<id>fig_p027v_1</id>
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<link><https://drive.google.com/open?id=0B9-oNrvWdlO5bW1tdUg0czEzTjg></link>
<caption>falcon</caption>
</figure>

<figure>
<id>fig_p027v_2</id>
<margin>right-top</margin>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5OVloUjZocVpuVTg></link>

<caption>Four quintals falcon or <fr>pasveolant</fr> reach 8 <fr>pans</fr></caption>
</figure>

<figure>
<id>fig_p027v_3</id>
<margin>left-middle</margin>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5OGJ5MXhrMVfVTVA></link>
<caption>3 quintals ^{falcon} reach as long as 9 <fr>pans</fr></caption>
</figure>

<figure>
<id>fig_p027v_4</id>
<margin>middle</margin>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5OWVRTXVjNTljRnM></link>
<caption>2 quintals reach a <fr>cane</fr> which is 8 <fr>pans</fr> in <pl>Montpellier</pl></caption>
</figure>

<figure>
<id>fig_p027v_5</id>
<margin>right-middle</margin>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5T2k5QjJ0ZGtfc1U></link>
<caption>1 quintal reach six <fr>pans</fr> and a half</caption>
</figure>

<figure>
<id>fig_p027v_6</id>
<margin>left-middle</margin>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5WmlyWDktZUhka1E></link>
<caption>1 quintal and a half reach 7<fr> pans</fr> and a half</caption>
</figure>

<figure>
<id>fig_p027v_7</id>
<margin>right-middle</margin>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5Z3BiOF9jTFIzdA></link>
<caption>A 60 lb musket reaches 5 <fr>pans</fr> and a half or 6</caption>
</figure>

<figure>
<id>fig_p027v_8</id>
<margin>bottom</margin>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5MVh1MEdHdDZLWmc></link>
<caption>5 quintals reach eleven <fr>pans</fr> and a half</caption>
</figure>

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<page>028r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f61.image></image>

<div>
<id>p028r_1</id>
<head>The measures of the calibers of pieces</head>

<figure>
<id>fig_p028r_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5eUZXU293VGImdVU></link>
<margin>left</margin>
<caption><margin>right-top</margin>
The numbers and dots show how many <ms>quintals</ms> a piece marked with that caliber weighs.</caption>
</figure>

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.....

<page>028v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f62.image></image>
<div>
<id>p028v_1</id>

<head><pro>Pewterers</pro></head>

<ab>They think that the best <m>tin</m> is the one that comes in <fr>saulmons</fr>, which has not been remelted since it came from the mine, because the masters remelt it on the grill, weighing two or three pounds, to more easily cut it and sell it by the piece. And in this they often mix in left-overs from bars of <m>soldering</m> and common <m>tin</m>. And to recognise the best one, it is the most lustrous, which is almost burnished, because it is softer. Sometimes in their ingots they find pieces of <m>iron</m>, <m>stones</m>, or similar troublesome additions, made to cheat on the weight. The <m>tin</m> from <pl>England</pl> is so hard that the <pro>miners</pro> add <m>lead</m> to soften it. The one that comes from <pl>Germany</pl> is softer. Usually, the master <pro>pewterers</pro>, sworn in to the good cities, add 6 or 8 pounds of fine <m>lead</m> to a quintal of fine <m>tin</m>. The others, who work in the fields, add fifteen or twenty ^{pounds} or as much as they can, and to disguise the darkness and the softness of the <m>lead</m>, they put in <m>looking-glass tin</m>, 4 lb. per quintal, <figure>+</figure>, which whitens and hardens, and six or eight lb. of <m>rosette</m>, to give the plate a good ring. To cast platters and plates, they make them out of <fr>pierre morte</fr>, that is from <m>grès</m>, and they form and polish them on the ^{potter's} wheel. They melt their <m>tin</m> in an <m>iron</m> posnet on a <m>charcoal</m> fire, and with an <m>iron</m> spoon that holds the quantity sufficient for a platter, they cast in their cold molds, which they keep joined and squeeze between their knees. Soon, they open the molds so that they do not heat up, and having taken out the casting which is on the female side, and which breaks easily. Then, with a <tl>piece of cloth</tl> soaking in <m>water</m>, which they always have at hand, they rub the middle of the back side of the plate and around the edge, so that it comes out better, and they rub the <tl>female</tl> mold with it.</ab>

<ab>
<margin>left-middle</margin>
<figure>+</figure>

When the <m>tin</m> is fine, one adds less <m>glass-looking tin</m> to it, that is 4 lb. per quintal, but if the <m>tin</m> is low^{-quality}, that is allied with a lot of <m>lead</m>, one adds at least five or six lb. of <m>looking-glass tin</m> to it. If there is not a lot of <m>looking-glass tin</m>, one adds about two or three lb. per quintal. One adds eight lb. of <m>rosette</m>. But if there is a lot of <m>looking-glass tin</m>, about 4 or 5 lb., one only adds six of <m>rosette</m>. And commonly, for one quintal of <m>tin</m>, one adds x lb. of both.

</ab>
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<div>
<id>p028v_2</id>
<head>Go to the fifth page.</head>
</div>

<page>029r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f63.image</image>

<div>
<id>p029r_1</id>
<head><m>Stucco</m> for molding</head>

<ab>Take <m>tragacanth gum</m> and put it to soak until, having absorbed its <m>water</m>, it is swollen and becomes like jelly. Then crush it quite forcefully on the <m>marble</m> and then take rye flour, which is better than <m>wheat [flour]</m> because it is more humid and does not make the paste as rough, and dust your <m>tragacanth gum</m> with it and thus continue to pound and mix in the very finely sieved flour, little by little. And knead as if you wanted to make <m>bread</m>, until you perceive that it has enough body and is as firm as bread dough that is ready for the oven. This is perceptible when it can stretch enough without breaking. And if it is not strong enough, it will not come off [the surface] properly. It being thus prepared, rub the hollow [part of the mold] with <m>oil</m> with a brush, so that the <m>oil</m> penetrates everywhere to make it come off better, and press the paste inside quite forcefully. And if it does not come off well, mix in more <m>flour</m> until it has enough body. With this you will mold very neatly such work, either masks or garlands, as you like, and they will be dry within one day. Afterwards, you apply them with <m>strong glue</m> or <m>paste glue</m>, as you like, and you will be able to paint and furnish them with <m>gold</m> and all colors. In <pl>Rome</pl> they make floor ornaments with it. You can make bed ornaments with it. <head>If you want that the work remains white</head>, it is better to mold with <m>plaster</m> instead of <m>flour</m>. It is true that it is also more brittle and firm, but it must be prepared like this: mix it while it is a powder in a good amount of <m>water</m> so that it is light-colored and pound it several times a day for fifteen days. Then pour water by inclination, collect the <m>plaster</m> and crush it fine on the <m>marble</m>, and put it in a clean lead vessel, so that no dust or dirt falls into it, and leave it in the open <fr><env>au serain</env></fr> for fifteen days with its <m>water</m>, and it will become matte, strong, white and light, very suitable to become the groundlayer for burnished gold. And you can mix some of this powder instead of <m>flour</m> with <m>tragacanth gum</m>, and your work will be beautiful. Instead of plastre you can mix in well ground <m>chalk</m> or <m>lead white</m>, and try <m>bole ^{armeni}</m> and similar things. This <m>stucco</m> with the <m>tragacanth gum</m> has the peculiarity that it is pliable before it is dry, it fits on either round or flat things, as you like. It is to make an ornament at little expense.

<ab>
<margin>left-top</margin>
The <m>flour</m> is not good in this, but <m>chalk</m> or <m>lead white</m> [is].
</ab>
</div>

<page>029v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f64.item</image>

<div>
<id>p029v_1</id>
<head><m>Color of gold</m> without <m>gold</m> on <m>silver</m></head>

<ab>Color your applied <m>silver leaf</m> with some <m>terra emerita</m> and once dry, spread a handful of varnish of <m>aspic</m> and <m>sandarac</m> and it will be more beautiful than fine <m>gold</m>.</ab></div>

<div>
<id>p029v_2</id>
<head><m>Brown gold layer</m> on <m>paper</m></head>

<ab>Prepare your underlayer of <m>starch</m> soaked in <m>water</m> and your <m>gold</m> will turn brown very well. Clear <m>starch water</m> laid down on <m>paper</m>, then dried, and so by three times, is a good base for <m>brown gold</m> on <m>paper</m> and is not thick.</ab></div>

<div>
<id>p029v_3</id>
<head>Removing stains</head>

<ab>A good <m>eau de vie</m> takes it off if you rub your <m>cloth</m> with it.</ab></div>

<div>
<id>p029v_4</id>
<head>Layer of <m>gold</m></head>

<ab>Ground <m>terra emerita</m> with some <m>saffron</m>, altogether mixed with some strong very white <m>glue</m> and sieved through a <m>cloth</m>.</ab></div>

<div>
<id>p029v_5</id>
<head><m>Oil</m> extracts</head>

<ab><pro>Apothecaries</pro> say that anything with does not stick to a mortar while ground is oleaginous.</ab></div>

<div>
<id>p029v_6</id>
<head><m>Dragon's blood</m></head>

<ab>Have a well chosen drop of it which shows its transparent red and put it in a <m>glass bottle</m> with enough quantity of the best <m>eau de vie</m> possible and close it diligently so it does not get exposed to air[a] otherwise it will be worth nothing and leave it so for a long time, because the longer it stays, the more beautiful and better it will be and it will dissolve if it is good otherwise it will become like <m>lees</m>. When you want to use some, make a small hole in the bottle stopper and pour some and close it again each time and apply it on gold. Good <m>dragon's blood</m> can be found in grands lopins like <m>cattle cake</m><add>this one has no worth if it is sophisticated</add> and broken it shows on its edges some transparent scale as light red enamel, it is also lumpy in some points like small rubies. The <m>eau de vie</m> must be very strong and plusieurs passes[b].</ab>

<ab>
<margin>left-bottom</margin>
The darker <m>dragon's blood</m> is the best and the larme is the most dying one, which you can find in lapin like <m>pitch</m> and <m>big nuts</m> which look like
<figure>
<id>fig_p029v_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5cXJfWVISSGIKOGs</link>
<margin>right-bottom</margin>
</figure>
</ab>

<ab>
<margin>left-bottom</margin>
I put it in some ordinary <m>eau de vie</m>, adding some <m>aqua fortis</m> to give it strength. At the beginning the <m>water</m> turned slightly colored, but at the end it just looked like some tragacanth <m>gum</m> with which they sophisticate I think the <m>dragon's blood</m>.</ab>

<ab>
<margin>left-bottom</margin>
When it is applied on <m>gold</m>, it is prone to break, this is why some apply on it some <m>turpentine oil</m>. Cold stops the <m>water</m> from pouring and taking the color off. And for that, you can hold it by the fire.</ab></div>
[a]There is an ambiguity between the meaning of "esventer" (which means "expose to air" and which we chose) and "esserver" (which means "to drop").
[b]uncertain how to translate

.....

<page>030r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f65.image</image>

<div>

<id>p030r_a1</id>
<head><m>Lead</m></head>

<ab>The best, fattest and softest, is imported from Germany and is not in sow[a]s but in bands, of around an arm long and three inches wide. It is imported in Lyon and they use it in Nuremberg to make the [lead] leaves for small mirrors.</ab>

</div>
<div>
<id>p030r_a2</id>
<head><pro>Pewterers</pro> </head>

<ab>They also make their molds out of <m>metal</m> which last longer and cast more precisely than <m>earthenware</m>[b] [molds], because the vessel is cast smoothly and evenly. But this is for the rich because a mold costs fifteen or sixteen francs. These <m>metal</m> molds are usually made for basins and ewers and salt cellars and similar things which are figured.</ab>

<ab>For pints[c], molds are made with <m>white clay</m> mixed with <m>horse dung</m> and <m>wool</m>[d] and well beaten. Because the pints are deep half-circles, they will not be easily removed from the <m>earthenware</m> moulds, unlike the <m>clay</m> ones, which are softer.</ab>

<ab> Vents [souspirals[e][f]] are made to <m>metal</m> molds, and we cast all at once, so that there is no smoky black line in the middle of the dish which comes from the <m>metal</m>'s steam and smoke when it is casted. This means that in that particular spot, the work is knotted and very often pierced. However they can fix this by soldering, as I have said. This smoke is found in the middle right half of the cast, and in <m>earthenware</m> molds.</ab>

<ab>The vessel can be hammered so that it will sell better, but it will not last as long.</ab>

<ab>Depending on the size of their dishes and plates, they have natches on the spoke of their <m>iron</m> wheels. And when they want to apply</ab>
<cont/>
</div>

[a]Saulmon: a great sow of lead. RC.
[b]<http://www.micmap.org/dicfro/search/cotgrave/grais>
[c]According to Cotgrave : "the French, or Parisian pinte, somewhat less then a fixt part short of our Quart (for in weight it is about 27 ounces, our Quart 32)" Pinte could also mean small pewter vessel - both work in this case
[d]bourre is wool or hair wadding or stuffing or cloth waste (Cotgrave)
[e]a breathing hole
[f]cotgrave lists 'vent' as a possible translation, which would fit perfectly (<http://www.micmap.org/dicfro/search/cotgrave/souspirails>)

.....
<page>030v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f66.image></image>

<div>
<cont/>
<id>p030r_a2</id>
<ab>any bowl or plate, they enlarge and break with the strike of a hammer the <m>iron</m> wedges that has play in the spokes of the wheel. And, because they are on the notches and points where they should be, they tighten the aforementioned wedges [by] striking them with the hammer.</ab>

<ab>When their plates or bowls have been cast, soldered if needed, and purged anew of the burrs around with the hot soldering iron, and then with a big file. They adapt it on the wheel, and first of all wrap their kind of esgrusouer with a rope, as thick as the little finger, and lean it firmly on the crossing bar to secure it well. And while somebody else quickly turns the wheel, they guide the sharp edge of the esgrusouer by hand towards the edge of the round hollow, and gently guide it all the way to the center of the circle. And this esgrusoue is for removing the first rough and lumpy burrs of the work. And then, they unite it with a cutting iron called a plane, the sharp edge of which they rub with a bit of <m>putty</m> on <m>leather</m> nailed with a piece of <m>wood</m>. And this in such a way that the flesh or flower is on the outside, because if the sharp edge of this plane was not rubbed and polished, it would not burnish and polish the <m>tin</m>, which would remain white instead of black and polished as a mirror. Moreover, they repair with this plane for the second time the work as they did with the esgrusouer. Then, with a knife or other cutting iron, they scrape the edges of the plates or dishes to smooth them so that they do not get cut while handling them.</ab>
<cont/>
</div>

.....
<page>031r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f67.item></image>

<div>
<cont/>
<id>p030r_a2</id>
<ab>Their hammers must be quite even and polished, and if by some accident rust or other usage has damaged them, they polish them first with <m>stripping powder</m>, and then they finish them off by burnishing and softening with rough <m>metal</m>. Otherwise they would not strike neatly. It is also necessary that the plane of the anvils be level.</ab>
</div>

<div>
<id>p031r_1</id>
<head>A new vine</head>

<ab>When [the vine] has taken root well, if you want it to grow many cordons, having dug furrows [next to it], put two handfuls of <m>pigeon guano</m> next to each rootstock.</ab>
</div>

<div>
<id>p031r_2</id>
<head>Colored water</head>
<ab>Distilled <m>urine</m>, <m>distilled vinegar</m> and <m>eau-de-vie</m> take on a blue and green color from powdered and finely ground <m>burnt bronze</m>. And the residue from this tincture will be <m>red copper</m> if you distill the things indicated with a filter.</ab>
</div>

<div>
<id>p031r_3</id>
<head>Water-resistant varnish</head>
<ab><pl>Flanders</pl> varnish made with <m>turpentine</m> and its <m>^{turpentine} oil</m>, or with <m>mastic ^{oil}</m>, can be taken off and doesn't subside in the rain. But that made with <m>white nut oil</m>, as you do it, holds in the rain and is very clear and fine and dries quickly. This is why one uses it for painted standards and signs that one carries into the rain.</ab>
</div>

<div>
<id>p031r_4</id>
<head>Fruit from <m>sugar</m></head>
<ab>Don't paint them with a brush like other color-moistened things, for the <m>sugar</m> would melt. But rub them with color with a finger.</ab>
</div>

.....

<page>031v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f68.image</image>
<div>
<id>p031v_1</id>

<head><pro>Painter</pro></head>

<ab>Scribes make darker shades of <m>lake</m> and other colors for clothing with <m>egg yolk</m>, but this is a fraud and does not last.</ab>

<ab>Others glaze <m>rouges de mine</m> and other [colors] that are not beautiful in <m>oil</m> using <m>egg white</m> so that they seem to be done in <m>oil</m>. But humidity damages all that.</ab>
</div>

<div>
<id>p031v_2</id>
<head>Brushes</head>

<ab>Take some that have been used a little by <pro>white-limers</pro>, because the harshness of the <m>quick-lime</m> and usage make them easier to handle.</ab>
</div>

<div>
<id>p031v_3</id>
<head>Lifelike painting</head>

<ab>One should not attempt such working in poor sunlight because you might make your skin tones browner than they should be.</ab>
</div>

<div>
<id>p031v_4</id>
<head>Painting on glass</head>
<ab>For blue, they take some of the roughest <m>azure d'email</m> and grind it on shale, and mix in two or three rosary beads' worth of <m>lead rock</m> or more, according to the quantity of work. And having done their drawing with <m>noir d'escaille</m> (which is painted towards the light with the piece of <m>glass</m> upright), they lay all their pieces down over white <m>paper</m> so that by the whiteness of the <m>paper</m> they can judge the lighting of their scenes and put down all colors of the same kind at the same time. Then they reheat.</ab>
</div>

.....

<page>032r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f69.image</image>
<div>
<id>p032r_1</id>
<head>The body of the cuirass</head>

<ab>The high lames of the spaulder in a vambrace @[a] are like a master's chef d'œuvre, because these pieces have to be very precisely hollowed and adjusted so that movement may be free. The bodies of cuirasses must be very evenly beaten and worked with a hammer.</ab>

<ab>Morions from <pl>Milan</pl> are all in one piece and thus better than which are of two.</ab>

</div>

<div>
<id>p032r_2</id>
<head><pro>Painters</pro></head>

<ab>As soon as the colors of paintings are well dried, the <pl>Flemish</pl> varnish them so they do not fade any further and remain in that state.</ab>

<ab>Those who know well how to work in distemper will work well in <m>oil</m>.But on the contrary, those who know well how to work in <m>oil</m> will not be able to work in distemper.</ab>
</div>

<div>
<id>p032r_3</id>
<head>To lay down <m>gold</m> in distemper</head>

<ab><pro>Scribes</pro> and local <pro>painters</pro> make <m>batture</m>, which is <m>joiner's glue</m> tempered in <m>water</m> over the fire, somewhat clear, mixed with very little <m>honey</m>, that is a few drops to make it thicken, and then they use it to draw letters or whatever they want to <m>gild</m> with a brush, and immediately after lay down the <m>gold</m>, but the work they do is never quite neat, and if there is a lot of <m>honey</m> it dries only with great difficulty. This layer will be damaged in the rain. </ab>

<ab>Others do better, they temper <m>candy sugar</m> in <m>water</m> and mix it with <m>sanguine</m> that they call cocon, very crushed, and add a little <m>soap</m>. This makes it neat, and renders <m>gold</m> beautiful if one uses it as the underlayer.</ab>
</div>

[a]Apparently fr. 640 is somewhat incorrect here, since the brassard (vambrace) should protect the forearm. Here it seems to include the various pieces covering the whole arm.

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<page>032v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f70.image></image>

<div>
<id>p032v_1</id>
<head><pro>Mat Makers</pro></head>
<ab>Two kinds are made in <pl>Toulouse</pl>, one to cover rooms' walls which are finely woven, almost like <m>straw</m> hats worn by villagers, and are made in long rolls, some 10 <m>straws</m> wide, others 13. And they work on them mainly in summer and winter. Then when they prepare it they sew it, but beforehand they dye it in usually three colours: green, red and sometimes purple. The green one is made with only a <m>pastel tincture</m> since green is made from yellow and blue, so the <m>pastel</m> dyes the dark yellow <m>straw</m>. It becomes bright green. For red they use some <m>alum</m> and <m>brazil wood</m>, for purple they use <m>pastel</m> and some <m>copperous</m> which darkens blue with its black tincture.</ab>
</div>

<div>
<id>p032v_2</id>
<head><pro>Glassmakers</pro></head>
<ab>They have no way to produce a perfect red that needs to be annealed. Try, however, the <m><pl>German</pl> red</m> which is scaled red. They produce their ordinary red with some <m>sanguine</m>, <m>iced pewter</m>, <m>lead rocks</m>, and some <m>iron file dust</m>. This red is to be applied on both sides of the <m>glass</m> because it is more colourful. If it were applied on one side only it would look pastel orange.</ab>
</div>

<div>
<id>p032v_3</id>
<head><pro>Founders</pro></head>
<ab>For 20 pounds they buy a quintal of <m>rosette</m> which is harder to melt than <m>brass</m> because it is milder. The milder great <m>metals</m> are, the harder they are to melt. Because the <m>pewter</m> used to make bells is fine and sour, it is easier to melt than <m>lead</m>, which is mild. <m>Brass</m> soured with <m>calamine</m> melts more quickly than <m>red copper</m>. The <m>metal</m> that is used to make bells is mixed with some <m>tin</m>, is very sour, and can be quickly melted. The more <m>silver</m> that is mixed in, the quicker it melts, and this is why <m>solders</m> are made with it. In <pl>Germany</pl> candelabras are made very light because they rotate by means of <m>water</m>, but they are breakable. A quintal of fine <m><pro>coppersmiths</pro>' rosette</m> is sold for 30 or 40 pounds. Another type used by founders is sold for 12 or 15 pounds for a quintal of six pounds of <m>metal</m>.</ab>

<ab>
<margin>left_bottom</margin>
+ <m>Gold,</m> <m>silver</m> <m>copper</m><lb/>
<m>leton</m> <m>iron</m>.</ab>

</div>

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<page>033r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f71.image></image>

<div>
<id>p033r_1</id>
<head>Knife to cut a nose or a finger</head>

<figure>
<id>fig_p033r_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdIO5RWIDQlc4cU5HN3M></link>
</figure>

<ab>This is a well-furbished knife, assembled with a nail in the middle so that it can easily swivel to one side or the other and be hidden, alternately on side A or on side B, inside its handle marked C. The two stops you see on the tips of the knife are used to block it on one side or the other against a nail which is at the butt of the aforesaid handle, so that when one wants to press hard on something in order to cut it, it does not move from the handle. You might as well allow the side marked B to cut, only not during your performance, but the side that is notched and marked A must not be sharp, it should only be furbished. And the notch must also be as wide as the edge of a knife so that it cannot hurt. And you will only show the part where the knife is not notched, for the side with the notch must be hidden inside the handle. And when you want to cut a nose or a finger, pretend to sharpen your knife on your thigh and at the same time turn it deftly, and the part with the notch, which you will cover with one finger, will come into your hand, and you will place the notch, colored with <m>brazilwood rosette</m> or <m>black cherry juice</m>, upon the nose.</ab>
</div>

<ab>
<margin>left-middle</margin>
The conjuror must not amuse himself by watching what he is doing, but must look at the spectators while entertaining them with hocus-pocus words to make them look at his face and not his hands.</ab>

<div>
<id>p033r_2</id>
<head>To relight an extinguished candle in your hands without blowing</head>

<figure>
<id>fig_p033r_2</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdIO5SFNtME0xWURubzA></link>
<margin>left-bottom</margin>
</figure>

<ab>Take a twig of well dried <m>wicker</m> and cut a small piece like a toothpick, and put one of its ends between your index and middle fingers and clasp both your hands together, fingers well joined, and move them deftly around the lit candle as if you wanted to cover it, and your <m>wicker</m> will light. And instantly remove your hands thus clasped, and instantly put out the candle. And then put your hands around it again as if you wanted to cover it, together with your little light from the glowing <m>wicker</m>. By means of the smoke that you are holding in your two palms, the candle will light. And then instantly extinguish your <m>wicker</m> and secretly discard it.</ab>
</div>

.....
<page>033v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f72.image></image>

<div>
<id>p033v_1</id>
<head>To make it seem to someone that he has a piece of silver on his forehead</head>

<ab>Take a token or a piece of <m>silver</m> and wet it and hold it against your forehead. Then, say to your assistant that if you put it thus on his forehead, he would not be able to make it fall without using his hands, no matter how he shakes his head and, making it seem as if you are placing it ^{the coin} on his forehead, keep it in your hand and wet your thumb with your <m>saliva</m> and press it quite strongly on his forehead and, thinking he has the piece on his forehead as a result of the coolness of the <m>saliva</m>, he will shake his head and will be able to do anything.</ab>
</div>

<div>
<id>p033v_2</id>
<head>For making blood or wine appear on someone's forehead or on a wall</head>

<figure>
<id>fig_p033v_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdIO5MXk2RzNQdF9aeDg></link>
<margin>left-middle</margin>
</figure>

<ab>Take a pipe or funnel of <m>white iron</m> which is double-walled in the body but not at the tip. At the top edge, make a small hole and another slightly bigger on the inner wall that will be a bit higher than the tip, just as you can see in the opposing example. Then when you want to use it, ^{put} in <m>wine</m> or liquid <m>rosette of Brazilwood</m> or <m>black cherry juice</m>, and blocking the hole at the bottom tip with your little finger, make sure that the pipe is well-filled so that the <m>wine</m> can enter inside the double wall through the hole at the edge, and if it does not flow inside well, act as if you are tasting the <m>wine</m>, sip and draw in a little air through the little hole. After, make a spectating neighbor drink the rest of the <m>wine</m> or throw it out or leave it to flow through the end, but before that, you must have pressed your thumb well against the hole on the top edge, because in this way, by the compression of air, the wine will be held inside the double wall. Then with a dagger dulled at the tip, pretend that you pierce the forehead of a person who holds a glass in his hand, and while you pierce, you conceal your act with the pipe that you hold close to</ab>

<ab>
<margin>left-bottom</margin>
<figure>#</figure>
his forehead, then removing your thumb from on top of the hole on the edge near the top, the air, being free, will make the <m>wine</m> flow out from the double wall through the hole marked B, and fall through the tip marked C into the glass. But you must bend the head of the spectator well in order to do it properly.</ab>
</div>

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<page>034r</page>

<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f73.image</image>

<div>
<id>p034r_1</id>
<head> To tell someone that if he keeps holding his finger on his forehead, you will be able to stop him from going out of a room</head>

<ab>Make him hug a bedpost or something similar, and with the same arm, make him hold a finger to his forehead.</ab>
</div>

<div>
<id>p034r_2</id>
<head>Bet someone that when he walks to a certain place and returns, he cannot say boot without spurs four times in a row</head>

<ab>If he tries to say it himself, be sure that he speaks loudly, so when goes and returns you tell him that he lost because he needed to say boots without saying spurs four times, for this was your bet.</ab>
</div>

<div>
<id>p034r_3</id>
<head>To hang a candlestick on a wall without making a hole</head>

<ab>Make a groom hold it against the wall.</ab>
</div>

<div>
<id>p034r_4</id>
<head>How not to break a glass with a log or a large stick</head>

<ab>You will cut a piece as small as a toothpick from a log or a large stick, and thus you will give it to him and he will not be able to break the glass.</ab>
</div>

<div>
<id>p034r_5</id>
<head>Cunning writing</head>

<ab>Cut some long rolls in fine <m>parchment</m> and mark them in order with A B C et cetera, then sew them inside the selvage of a rough <m>cloth</m> shirt, like for a messenger, who will not notice anything if you want to steal his shirt from him.</ab>
</div>

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<page>034v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f74.image</image>

<div>
<id>p034v_1</id>
<head>For melting or transforming a jewel placed in a box</head>

<figure>
<id>fig_p034v_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5VTA5cWtrWktEajQ</link>
</figure>

<ab>
Take a cylindrical ringed vessel carved in wood, as you see, into which you will insert another small vessel that easily fits, as you see in the one marked B, and make it so that it does not touch the bottom of the larger one marked A, but so that there is some distance. Whatever is entering the larger vessel must be planed, but that the rim is made in a circle so that, being joined, they will appear as if they are the same vessel. You must also have a round cover of <m>leather</m> of the same length as the vessel, as you see marked C, which will fit easily and surround the aforesaid vessels thusly, placed one within the other. But before you do your trick, you must put <m>gold</m> or <m>silver</m> powder or <m>mercury</m> amalgamed with <m>silver</m> at the bottom of the larger vessel, and then place the little vessel marked B on top. After, holding the vessel with your index finger on top, present it, and only showing the bottom of the smaller vessel marked B, put a ring or something similar in it. This being done, while placing the vessel on the table, cover it with the leather cap, and making the requisite facial expressions, you will remove the leather cap without showing the interior, and put it on the table and ask them to recall the ring that you placed in the bottom of the smaller vessel marked B. Then, place the leather cover gently back on top and after saying inhonorificabilitudinitatibus, take the outside of the leather cover with two fingers, and squeezing it more firmly than before, then you will lift only the smaller vessel that is inside and contains the ring and you will place both carefully on the table. Then you will pour out the powder or amalgam that is in the bottom of the large vessel, then replace the leather cover with that ^{vessel} inside it, as it was at the start. Then, removing the leather cover without removing the smaller vessel, you will show the ring that returned to the bottom ^{of the vessel} in its previous state, holding your index finger on the rim of the vessel so that it does not fall out.</ab>
</div>

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<page>035r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f75.image</image>

<div>
<id>p035r_1</id>
<head>To tell someone that you will teach him something he does not know, and neither do you</head>

<ab>Take a string or a small stick and measure from the tip of his ear to the tip of his nose and show it to him. Thus you will teach him something you did not know, and neither did he.</ab>
</div>

<div>
<id>p035r_2</id>
<head>To support a bucket of <m>water</m> on the points of three knives laid flat without touching the ground</head>

<figure>
<id>fig_p035r_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5SGstUTZ6TFRnUjg></link>
</figure>

<ab>According to the strength of your knives, either kitchen or table knives, you will make them support a large weight on their tips laid flat arranged in a triangle, edge to edge, as you can see, and not touching the ground. And after arranging them this way, if you turn them the other way without disassembling them, the handles can also support a large weight without touching the ground. You can as well adapt this to other uses with pikes, with timbers, or to quickly make a tripod in a camp with three halberd points. If the edge of the knife marked A is turned to the left, the butts of the handles will rise, but if the edge of the same knife is turned to the right, the tips will rise.</ab>
</div>

<div>
<id>p035r_3</id>
<head>To boil an <m>egg</m> in cold <m>water</m> without a fire</head>

<ab>Empty an <m>egg</m> from both ends so that nothing remains inside, then fill it with <m>quicklime</m> and <m>natural sulphur</m>, then plug the holes with <m>wax</m> and put it in <m>water</m> so that it floats two or three fingers [above the surface].</ab>
</div>

<div>
<id>p035r_4</id>
<head>To make <al>rabbits</al> come out of a burrow</head>

<ab>Have some <m>embers</m> in a pot, and after putting <m>sulphur</m> on top, place it inside the burrows and block them with something light.</ab>
</div>

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<page>035v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f76.image></image>

<div>
<id>p035v_1</id>
<head>For preventing someone from eating ^{a piece of} <m>meat</m></head>

<ab>Dry some <m>calf's foot root</m>, otherwise <m>iarus</m>, and dust the meat with it. There is no danger in this. See @[\[a\]](#) Mattioli. </ab>
</div>

<div>
<id>p035v_2</id>
<head>For making <m>grain</m> pass from one vessel to another</head>

<figure>
<id>fig_p035v_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5WEY2UUhMRTg4Y00></link>
<margin>left-middle</margin>
</figure>

<ab>Take two small <m>wooden</m> cylindrical vessels @[\[b\]](#) of the same size, hollowed on the bottom exterior by about the width of one knife or more. One of the two will remain empty, and on the bottom of the other you will glue <m>grain</m> with <m>starch</m> so that it will be covered entirely with <m>grain</m>, and will appear to be filled with it. Also, take a <m>wooden</m> bell cover into which you will place as much grain as one of the vessels can hold, and over the top place a piece of plain <m>leather</m> that fits well inside the bell cover. Put all into a bag or a napkin or a folded handkerchief, unless you want to use a conjuror's pouch. First show the empty vessel, then in front of the audience fill it with grain, then replace it in the napkin. After, leave that one there and take the other where the grain has been glued with starch and it will appear to be the first one, filled with grain. Place it under a hat and place the bell cover on the table, gently this first time. And if you like, quickly and without stopping show the inside that is covered with white leather of the color of the wood. Then, pretending that you are showing the vessel that is under the hat, turn it deftly to the side that is empty and leave it covered, and then you will command that by invisibilim the grain pass into the bell cover, which you will have previously tapped on the table a little strongly, and the grain will fall to the bottom and cover the piece of leather. Then raise the hat, the vessel will be found to be empty and the bell cover full of grain, which you will scrape deftly with the piece of leather
</ab>

<cont/>
</div>
[a]MARC: Commentaires de M. Pierre André Matthiole medecin senois, sur les six livres de Pedacius Dioscoride Anazarbeen..., Lyon 1579 (and earlier edns), pp. 366-368. Also provides the Arabic name "iarus". Contains illustrations.

NB The root is edible, but bitter, and more so in certain regions.
[b]MARC: A boisseau is cylindrical, which is necessary so you can't tell which way up it is. (Literally "bushel", but that suggests a specific — and large — measure).

The trick must be done with very small vessels, more like cylindrical cups. (Except the shape of a "cup" in the 16th cent. is quite different). This is more like a mug (or an espresso cup).

.....

<page>036r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f77.image</image>

<div>
<cont/>
<id>p035v_2</id>
<ab>
that cannot be seen, and you will skillfully and gently put all of it back into the bag or pouch.</ab>
</div>

<div>
<id>p036r_1</id>
<head>To make a ball change into a fine thimble</head>

<figure>
<id>fig_p036r_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5UmZaWjV2VE82QjQ</link>
<margin>left-top</margin>
<caption>de cuir</caption>
</figure>

<ab>Make a small case of plain <m>cow leather</m> or <m><pl>Spanish</pl> goat's leather</m> quite thick, as high as three fingers, sewn like the finger of a glove and large enough that two fingertips can fit inside and which, at the tip, is a little creased, and has a round hole as large as a double liard. Take two <m>boxwood</m> balls that look the same, and one should be hollowed and opened by a round hole only on one side so that the thimble can enter. Thus, when you want to perform, you will place your <m>leather</m> case on the table with the ball that is not hollow. Put a stick into the <m>leather</m> case to show that there is nothing inside it, also show the round ball, and hold the hollow one which has the thimble inside it between your little and ring finger, and do this with your right hand with which you hold the stick. After, take the <m>leather</m> case and place it over the hollow ball which is between your fingers and pretend you put some <m>goldibus powder</m> upon it, and replace the <m>leather</m> case that contains the ball on the table, then you take the large round ball remaining on the table, and order it to enter from upon the table into the <m>leather</m> case. Then, lifting the <m>leather</m>, it will seem to be the same, although it is the hollow one. Then, cover it and command that it should become invisible, and lifting the case while pressing it, you will remove it at the same time as the ball, and meanwhile discreetly put it back in the case. In its place will be found a thimble for the ladies whose bottom hurts, that is the bottom of the needle.</ab>

</div>

.....

<page>036v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f78.image</image>

<div>
<id>p036v_1</id>
<head><pro>Founder</pro></head>

<ab>Kitchen pots are high quality made, with the same metal used for bells, in order not to give meats a bad taste. It is true that <pro>founders</pro> add <m>latten</m> to pots to make them more yellow, in order to make them more saleable; but <m>latten</m> itself, just by touching it, is stinking and bad smelling.</ab>

<ab><m>Latten</m> does not lose (or just very slightly) its <m>calamine</m> in an air-furnace when it is melted in a crucible -- nor does it lose it in a wood furnace -- but it does lose it in a bellows furnace because bellows produce intense flames.</ab>
</div>

<div>
<id>p036v_2</id>
<head>Air furnace</head>

<ab>Its mouth has to be narrower than its bottom; it is sufficient if the crucible can get in, and if there is space enough to remove it with pincers.</ab>
</div>

<div>
<id>p036v_3</id>
<head><pro>Glassmakers</pro>' glass</head>

<ab>It is said that in <pl>Lorraine</pl> and <pl>Flanders</pl> <m>linking glass</m> is made of <m>fern ashes</m> and <m>pebbles</m>. First they blow up a long still that another worker breaks off and cuts vertically with big shears. Then this long still expands by being placed on a stone or large platine in a furnace slightly colder than one for melting. Furthermore, they flatten it by rolling ^{over} it a big and long iron stick. Then they take it out of the annealing furnace. Similarly, they make some in <pl>England</pl> that are quite beautiful. Close to <pl>Rouen</pl> in <pl>France</pl>, flat glass is made with some <m>saltworth</m> and <m>pebbles</m> and is whiter and softer than the <pl>Lorraine</pl> one, because it can be melted with a candle, unlike the <pl>Lorraine</pl> one. This flat <m>glass</m> is blown up in a long still -- the end of which someone else cuts and blows up whilst turning it, then flattens it using a plane which is on the ground, and then reheats it. Thus the middle of the still, where it began, always stays the same.</ab>

<ab><margin>left-middle</margin>The <m>glass</m>, when wet, can be broken again with the flame of a candle, but not as precisely as with hot iron.</ab></div>

.....

<page>037r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f79.image</image>

<div>
<id>037r_1</id>
<head><pro>Founder</pro></head>
<ab>The more it is melted, the whiter the <m>metal</m> becomes because the <m>tin</m> does not come off, and so it is more mixed. And, once it has been held in the fire for a long time, the <m>dust</m> is eaten away, which is what makes it sour. If you want to separate the tin from the <m>copper</m>, after it is well heated, throw <m>salpeter</m> on it often. Only this [saltpeter] and not the fire separates it, and purifies it, and eats away the <m>dust</m>. <m>Metal</m> can be cast very neatly. <m>Copper</m> is prone to swell, but because it is soft, it can be fixed with a hammer.</ab></div>

<div>
<id>037r_2</id>
<head>Medicine for the stomach which warms it [stomach] and unstops the liver</head>
<ab>Have some <m>wormwood powder</m>, dissolve it in some <m>capilli veneris syrup</m> or <m>preserves</m>, and make an opiate that you will take once a week on a piece of <m>bread</m> dipped in <m>wine</m>. Then you will be able to drink a bit of fully dipped <m>wine</m>. This clears the phlegm and gas which come from there. You can take six <m>pepper</m> grains, and grate their <m>rinds</m> so they are smooth, and swallow them without chewing them. This heals the stomach without warming the liver.</ab></div>

<div>
<id>037r_3</id>
<head>Sheets for precious stones</head>
<ab>Once they have been made, do not keep them in any <m>iron</m> or <m>metal</m> box because this stains them. But in some box of</ab></div>

<div>
<id>037r_4</id>
<head>For preventing candles from dripping and making them [candles] white</head>
<ab>Once they have been made, put in a mould some fresh <m>water</m> whipped with some <m>bran</m>, which should by no means be purged again of its <m>flour</m>, so that the <m>water</m> is white. And dip your candles in this. Then leave them to dry. And do so thusly two or three times.</ab></div>

.....

<page>037v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f80.image</image>

<div>
<id>p037v_1</id>
<head><pro>Gardener</pro></head>
<ab>
To sow melons properly, it must be the end of a moon [cycle], because otherwise they would sprout too many leaves and not bear much fruit. Every two fingers, make small holes with a small pickaxe and, in each, put two or three seeds. Then, take very fine <m>earth</m> and lightly fill the holes with it. Afterwards, spread the thickness of one thumb of 'hanets,' that is wheat chaff, over everything, so that when the rain comes, it does not beat the earth and prevent their birth. 'Hanets' attract <al>field mice</al> that eat the seeds, and to prevent this, they must be moistened with a decoction of <m>wormwood</m>.
</ab></div>

<div>
<id>p037v_2</id>
<head><pro>Glass-maker</pro></head>
<ab>
Some do not lay <m>gris d'escaille</m> on the <m>glass</m> to work on it, but trace straight on the <m>glass</m> with <m>noir à huile</m>. However, it is important that the <m>wood</m> be <m>degreased</m>, because if it has even a little <m>grease</m> [on it], the color will not take at all. And even, if the working <pro>glass-maker</pro> has a stink from his nose or his mouth, and he breathes on the <m>glass</m>, the color will not take on it. Those who came up with the invention of working in small works of softened <m>enamels</m> use only <m>azure enamel</m>, which is blue, and <m>esmail colombin</m>, which is the color of purple, which they soften with <m>rocks</m> or <m>lead glass</m>. As for yellow, they make it from <m>silver</m>, red from <m>sanguine</m>, as is said elsewhere, black and gray and shadows from <m>scale black</m>, either strong or weak, carnation from light <m>sanguine</m>. Green is made first from yellow, then they overlay <m>azure enamel</m>, either strong or weak, depending on whether they want to make it bright or dark.
</ab></div>

<div>
<id>p037v_3</id>
<head><m>Yellow Amber</m></head>

<ab>
It is carved with a file, then one passes a certain pulverized <m>salt</m> on it, which an <pl>Englishman</pl> called desramonct. But I believe that this was pulverized <m>pumice stone</m>, because it had as much grit as <m>gravel</m>. And with a stretched cord, he polished his <m>amber</m>, then passed <m>tripoli from <pl>Bretagne</pl></m> on it with his finger, others [did it] with <m>leather</m> or a cane. <m>Amber</m> loses its color if a person in ill health wears it, and becomes whitish. But to get it back, it must be soaked for a night in <m>urine</m>, then boiled a little in it.
</ab>
</div>

.....
<page>038r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f81.image</image>

<div>
<id>p038r_1</id>
<head><m>Sapphire</m></head>

<ab>There are <m>sapphires</m> that one calls trellis ^{sapphires}, for they are pierced and it is said that a certain king had a certain ornament in the shape of a trellis made out of these, as perhaps the S[**a**][**b**][**c**]reen of Charlemagne, as it is named, would be, in the treasure of <pl>Saint Denis</pl>, in which the gemstones are set without foil, so as to enjoy their luminosity from both sides, & to show their vividness. I have a white one that seems to be rough & pierced, and is all over spotted with blue blemishes; I am of the opinion that these are artificial & made of <m><fr>taffre</fr></m> or very clear azure <m>enamel</m> fused on the <m>sapphire</m>. <add>A <tl>file</tl> bites into it as it does with <m>beryl</m>.</add></ab>
</div>

<div>
<id>p038r_2</id>
<head><m><pa>Saffron</pa></m></head>

<ab>It is imitated & cut with half dried <pa>marigold</pa> leaves, & twisted like a thread, & placed in the hottest sun to dry, & is mixed, & the aforesaid <pa>marigold</pa> even gives some color.</ab>
</div>

<div>
<id>p038r_3</id>
<head><m>Sapphire</m></head>

<ab><pro>Stone cutters</pro> sometimes choose old pieces of antique <m>glass</m> in church windows, which are much thicker than those of today & are of more vivid colors. If it is for <fr>souf</fr> <m>sapphire</m>, they choose beautiful blue & from such a place that there are no pieces grains, if it is possible. And having cut it into squares with <m>emery</m>, they cut it in bevel & polish it. In this manner, they imitate very beautiful <m>sapphires</m>. The old azure <m>enamel</m> for <m>silver</m>, verging on aquamarine, was very suitable for imitating <m>sapphires</m>, but it is scarcely found. <m>Aquamarines</m> are imitated with white <m>glass</m>, but they take it from the bottom of the glass.</ab>
</div>

<div>
<id>p038r_4</id>
<head><m>Amber</m></head>

<ab>The orange color of transparent <m>amber</m>, and of the other one which has body, is not internal, for i[**d**]t is whitish on the inside. But it acquires this reddish crust with age or through the wearing of it. This is why those who cut it with a file or on the wheel do not remove this crust if it is possible. But rather they only polish it, rubbing it with a <m>willow</m> stick or other soft wood dipped into <m>water</m> & dusted with <pl>Brittany</pl> <m>tripoli</m>, and it takes a beautiful polish. If it is too straw yellow, & you want to intensify its color, hang it inside a chimney where there is much <m>soot</m> & smoke, and it will take on a reddish-orange color. Try to take the most whitish transparent one, and put it under <m>manure</m> to turn it green, as with bones. Or smoke it in a closed space, in <m><fr>safre</fr></m> or <m>manganese</m> smoke & other drugs that you know of. Or in <m>urine</m> and <m>distilled vinegar</m> mixed with colors, or in the vapor of <m><la>aqua fortis</la></m> boiling in <m>copper</m>, or <m><la>aqua fortis</la></m> boiling with <m>silver</m> & <m><lat>sal ammoniac</lat></m>; & some cut facets in the <m>amber</m> on a wheel of <m>soft wood</m> with <m>putty</m> instead of <m>emery</m>, & <m>jet</m> also. The salt called <fr>de Langlois de armonie</fr> is a mineral salt that resembles <m>marble</m> stone and is a very hard mineral with large pieces, like that of <pl>Cardona</pl> and <pl>Monserrat</pl>.</ab>

<ab>
<margin-left-bottom</margin>
I have e[**e**]xperienced that it turns reddish on the surface when boiled in <m>lye</m> or <m>corrosive water</m>. And if it is scrubbed with <m>fir</m> and <m>soft wood</m> before cooling down, it is easily cut.</ab>
</div>

[a]<http://expositions.bnf.fr/carolingiens/grand/063.htm>
[b]<http://expositions.bnf.fr/carolingiens/grand/064.htm>
[c]this is the image of the top bit that still survives, where you can see the sapphires with their spots
[d]i.e. the amber 'with body'
[e]'experimenté'
.....

<page>038v</page>

<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f82.image</image>

<div>
<id>p038v_1</id>
<head>Black color for dyeing</head>

<ab>Take <m>lye made from quicklime</m> and <m>white lead</m>, mix and leave to soak and you will have a dark brown dye, and reiterating the same you will make black. Try other colours with the <m>lye made of lime</m>.</ab>
</div>

<div>
<id>p038v_2</id>
<head>Against nose bleeding and for dyeing</head>

<ab>Pound some of the kind of <m>vinete</m> or <m>lapathum acutum</m> that is red-veined, which is called <m>dragon's blood</m>, and apply it on the bleeding person's forehead. This herb is a strong dye & makes beautiful violet.</ab>
</div>

<div>
<id>p038v_3</id>
<head>Beautiful artichokes</head>

<ab>You need to trim them every year so that only one stem is left, and water them a lot. Also plant some more every year, because the second year they will bear very beautiful fruits.</ab>
</div>

<div>
<id>p038v_4</id>
<head>Planting trees</head>

<ab>It is said that rarely a tree planted on Saint Paul's day will fail to take. To cover the young shoots, oxen manure is very appropriate, for it doesn't come apart in the rain like earth.</ab>
</div>

<div>
<id>p038v_5</id>
<head>Merchants</head>

<ab>Those who retail velvet and other materials do not keep double-entry books because they sell in small quantities and ^{recording} these details would be too much effort. They only have their sales book and account books. But those who sell in bulk and those who traffic in woad have a double-entry book.</ab>
</div>

<div>
<id>p038v_6</id>
<head>Crimson</head>

<ab>Because one aulne costs seven or eight lb. to dye, they use ^{cloth} worth seven or eight francs. But if one wants something beautiful, one should buy some white cloth worth fifteen francs an aulne and dye it with some pure crimson woad & a little cochineal. Black fabrics are very fine because the dye is inexpensive.</ab>
</div>

.....
<page>039r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f83.image</image>

<div>
<id>p039r_1</id>
<head><pro>Goldsmith</pro></head>

<ab>They bind <m>silver filings</m> with <m>salt peter</m> which refines it & does not make it brittle. But <m>gold filings</m> are bound with <m>borax</m> or, to save the <m>borax</m>, with <m>lead</m>, which refines the <m>gold</m> & softens it, for the <m>salt peter</m> would make it brittle, which it does not do to <m>silver</m>. This is why, in order to save, <pro>goldsmiths</pro> use it to bind it in order to save <m>borax</m>, which costs viii <cn><fr>sols</fr></cn> per ounce & <m>salt peter</m> costs x <cn><fr>sols</fr></cn> per lb.</ab>
<ab>When <pro>goldsmiths</pro> have thus bound their <m>silver filings</m> with <m>salt peter</m>, a <m>red enamel</m> vitrifies at the bottom of the crucible. I do not know if the <m>copper</m> mixed in with the <m>silver</m> is the cause for this. <rub>Try for <m>enamel</m></rub>.</ab>
</div>

<div>
<id>p039r_2</id>
<head><m><pa>Dyers' woad</pa></m></head>

<ab>It is grown in <pl>Lauragais</pl> where the deep soil is so fertile that every year wheat was grown there, it would lie flat for being too vigorous. This is why <m>dyers' woad</m> and wheat are grown there alternately. For cultivating <m>dyers' woad</m>, the soil is ploughed with <tl><m>iron</m> shovels</tl>, as <pro>gardeners</pro> do. Next, it is harrowed with <tl>rakes</tl>, & broken up finely as for sowing some <pa>kitchen herbs</pa>. It is commonly sown on <tmp>Saint Anthony's day in January</tmp>. Eight harvests are made. The first ones are better. The best <m>dyers' woad</m> of <pl>Lauragais</pl> is the one from <pl>Carmail</pl> & the one from <pl>Auragne</pl>. And sometimes the <m>dyers' woad</m> is good in one field & in the one close by it will hardly have worth. The goodness of the <m>dyers' woad</m> can be recognized when put in the mouth it gives the taste of <m>vinegar</m>, or when crumbling & breaking it, it has some mold-like veins which are as it were golden or silver. It is assayed in the <tl><pro>dyers</pro>' vat</tl>, and to fill a <tl>vat</tl> with it, six <ms>bales</ms> of it are needed. There several flocks of <m>wool</m> are dyed, and if it dyes fifteen times, it is said to be worth 15 <cn>florins</cn>, if it gives xx dyings, xx <cn>florins</cn>. The good kind dyes up to 30 times & commonly up to xxv or 26.</ab>

<page>039v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f84.image</image>

<div>
<id>p039v_a1</id>
<head><m>Enamel</m></head>

<ab><m>Enamel</m> takes more readily on <m>copper</m> than on <m>silver</m>. It is true that the cut needs to be well hollowed out & rough. <m>Azure</m> in body & the <m>red</m> called gules, white <m>enamel</m> & thick <m>green</m> take very well. Having <m>enamelled</m>, one gilds the <fr>[a]euillages</fr> of the cutting[illegible]. <m>Copper</m> has a similar hardness when cut as pure <m>silver</m> or <cn><fr>pistolet</fr></cn> <m>gold</m>.</ab>

<div>
<div>
<id>p039v_a2</id>
<head>Colors for illumination on <m>glass</m></head>

<ab>In order that your <m>turpentine</m> colors shall not run, & become unified, mix in a little <m><fr>larne de mastic</fr></m> together with the <m>turpentine</m>.</ab>

<div>
<div>
<id>p039v_a3</id>
<head>Tracing a history on <m>glass</m></head>

<ab>If you want to trace a history in intaglio on <m>glass</m>, you can do so in different ways. Place your <m>glass</m> pand**[b]** on, as thin as possible, over the printed history, & having cleaned the <m>glass</m> well with <m>lye</m> & <m>ashes</m> so that it is not at all greasy, trace over the lines visible to you with <m>oily black</m> or <m>scales black</m> with a <fr>pinceau</fr>, if you want to paint with colors in the manner of <pro>glaziers</pro>, who spread a wash of <m>scales black</m> all over their <m>glass</m> pane & then scratch & uncover what they want to coat with color, leaving that which is necessary for shading. But if you want to make gilt histories on <m>glass</m> with a background of colors, which imitates the b**[c]**asse-taille of <pro>goldsmiths</pro>, gild the whole <m>glass</m> pane with <m>gum water</m> or <m>garlic juice</m> or <m>fig-tree milk</m>. Then moisten your printed history with two wet <m>linen cloths</m> & lay it down on the gilt <m>glass</m>. Then with a <tl>pin</tl> mounted on a small <tl>stick</tl>, follow the lines of your history as if you wished to pounce it, & thus you shall vividly trace it on the gilding of the <m>glass</m>. & next you shall uncover the background & what needs to be empty with a quite pointy <tl><m>steel</m> awl</tl>, & and neatly follow once more the lines & accomplish your work & fashion your faces & flesh tones in pounded <m>silver</m>; then fill the background with <m><fr>azur d'esmail</fr></m> or <m>verdigris</m> or fine <m><fr>l[d]aque platte</fr></m>, tempered with <m>clear turpentine</m>, mixed with a little <m><fr>larne de mastic</fr></m> if you want the colors to be more unified & not to run. Next, apply on the back of the <m>glass</m> & over the colors a <m>white tin</m> sheet. And once this is dry, you can cover the <m>tin</m> sheet of with color to hide your secret. The <m>tin</m> sheet gives light to the colors. Thus you will be able to paint without being</ab>

<cont/>
</div>

[a]i.e. the uncut surfaces of the metal cut in champlevé technique.
[b]taille douce? steel engraving? steel as <m>
[c]an enamelling technique
[d]perhaps one form in which lake was sold

<page>040r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f85.image</image>

<div>
<cont/>
<id>p039v_a3</id>
<ab>being at all expert in painting. If your <m>glass</m> pane is bulging as if taken from the belly of a jar, it will show better. When you apply your <m>turpentine</m> colors to your <m>glass</m> panes, first place them on a hot tile &, once they are hot, spread your colors & leave it a while on the tile, then lay down your <m>tin</m> sheet.</ab>
</div>

<div>
<id>p040r_1</id>
<head>Dye</head>
<ab>Some take the root of <pa><la>Lapathium acutum maius</la></pa>, which looks like <pa><pro>monk</pro>'s-rhubarb</pa> or <pa>sorrel</pa>, & using the root, which is yellow in summer, they dye threads & similar things.</ab>
</div>

idp040r_2

Aqua fortis

Some put ~~on~~ for four pounds of ~~aqua fortis~~ matter ~~four~~ that is in the ~~retort~~, four ounces of common water in the container, which is better than putting it into the ~~retort~~. The alum is de-phlegmed & calcined such that the water does not have as many dregs. Many make it without de-phlegming the alum.

idp040r_3

Vinegar

One takes for granted that ~~fr~~ heat red-hot the mineral salt that looks like marble & that is called in Catalonia & at the border of Spain Cardona salt, & throwing it ~~in the~~ red hot or quite hot in wine, it turns it into very good vinegar. Some make it with water poured on pomace soured after being pressed by grape pickers, but it will not keep & spoils in heat & thunder storms.

idp040r_4

Silver gilt buttons

Because silver gilt does not fear fire, it is cut into lozenges flat~~[i]~~ on one side, then they are joined together in a star shape with a paste of crushed enamel, which is then melted & the enamel is gilt with gold leaf and then reheated~~[b]~~.

idp040r_5

Grottos

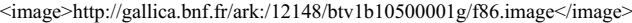
To fill some empty place that cannot be laden with hanging rocks, one puts a piece of thick parchment close to the fire, which shrinks & crumples. Then one paints it with distemper, then in oil. Next one affixes it.

[a]deletion in the middle of the word "plat"

[b]This was likely done on a copper base which would form the button.

.....

page040v



idp040v_1

Cross of the commanders of Malta

The fine rouge clair which is the background of the white enamel cross is of fine dragon's blood drops soaked with eau de vie or Indian laque plate, which I believe is made in Flanders, distempered with clear turpentine and mastic drops and applied on a silver leaf, not the one used by painters but a thicker one, which is burnished by the makers of foil backings for gemstones or by goldsmiths, and that gives it its fine brilliance.

idp040v_02

Latten and calamine

Founders do not melt latten in a bellow furnace but in crucibles, for in a bellow furnace and amid the charcoals, the calamine would disappear.

idp040v_03

Metal

Tin needs to be pure because if there is any lead, it will go up in smoke whilst melting. Alloyed tin withstands fire for a long time but in ashes it will come apart.

idp040v_04

Aqua fortis

If your retort is well luted you should not surround it with ashes, which would only prevent the vapors from escaping properly. But when, towards the end, you fire it intensely, surround it with charcoals lit elsewhere beforehand, such as those in the furnace should be, so they do not crackle or smoke. Place said lit charcoals around the body of the retort, not close to the neck, which does not need to be heated as much. The top grate, on which you place your retort, should be sufficiently far, one ms or half a foot, from the bottom grate where the charcoal is placed, since thus you will not waste as much aqua fortis.

<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f87.image</image>

<div>
<id>p041r_1</id>
<head><m>Casting earth</m> for <pro>founders</pro></head>

<ab>It becomes fat by being beaten and also mixed with <m><al>horse</al> manure</m>. <m><pro>Potter</pro>'s earth</m> would be too soft and would break and wouldn't resist the fire. But [your soil] has to be mixed with one half of sand and a quarter or a fifth of <m><al>horse</al> manure</m>. And leave it to dry, then reduce it to powder, then put it through a sieve to make it fine and free from gravel, which would prevent it from casting cleanly. The <m>manure</m> makes the <m>earth</m> more amenable [to casting] & easier to work with, but it should be free of <m>straw</m> and other things. And when the <m>earth</m> is very fat, you have to add more sand and more <m>manure</m>. But one does find fat varieties of <m>earth</m> [that are] naturally mixed with sand. If they are not, make them so artificially. One should always cook the <m>soils</m> again before casting.</ab></div>

<div>
<id>p041r_02</id>
<head>Garden lily</head>

<ab>If it is broken during its first blossom, it will not grow or bloom until the following year, and I believe it is the same for bulbous flowers [in general].</ab></div>

<div>
<id>p041r_03</id>
<head>Sand</head>

<ab>The sand to be used for casting should be chosen such that it is not too dry for it won't hold together, nor should it be too fat. And although you find some in nature, however, it is not everywhere. And if you are in a place where it is not found, you can put it together yourself but not from fat <m>earth</m>, because the sand should contain none at all, since it makes it porous. But you can make it bind by mixing it with <m>brick</m> well ground on <m>marble</m>, or <m>plaster</m> or <m>calcinated alabaster</m> or something similar, or the <m>burned marrow of <al>ox</al> horn</m> or burned <m>asphalt</m>. If you grind it quite finely on <m>porphyry</m>, it binds better & then you can burn it with <m>asphalt</m> or mix it with a quarter of <m>tripoli</m>. Make sure no <m>bread</m> falls into your sand because this makes it porous.</ab>

<ab>
<margin>left-middle</margin>
Try mixing in soot black.</ab></div>

<div>
<id>p041r_04</id>
<head><al>Ducks</al></head>

<ab>Young domestic ones do not grow for a month after hatching but remain in this state. But after, they soon grow up, even if they go into the water. They are fed boiled <m>millet grains</m>, to which are added crumbled <m>bread</m> and finely chopped <m>lettuce</m>.</ab></div>

.....

<page>041v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f88.image</image>

<div>
<id>p041v_1</id>
<head><pro>Glassmaker</pro>'s <m>black</m></head>

<ab><m>Filings of iron</m> taken from bars that have been in the fire for a long time, which are thick, are much better than the common soft ones that fall under the anvil in the forge, because they imitate <m>niello</m>. To said <m>black</m>, described elsewhere, some people add a little <m>minium</m>.</ab>
</div>

<div>
<id>p041v_2</id>
<head>Sheared <al>ewes</al></head>

<ab>If the <pro>shearer</pro> wounds them at some point, he applies the very excrement of the ewe on it. When they are fat, they are easier to shear and do not get hurt as easily.</ab>

<ab>
<margin>left-middle</margin>
If the <pro>shearer</pro> wants to 'viscorter' them, that is to castrate them as [they] are one or two years old, he must not have been with his wife, because this would cause the <al>sheep</al> to die.</ab>
</div>

<div>
<id>p041v_3</id>
<head>Colors and <m>gilt</m> on <m>glass</m></head>

<ab>You should not put as strong <m>gummed water</m> on <m>glass</m> for the <m>silver</m> layer as for the <m>gold</m> layer because a <m>silver</m> leaf is twice as strong as <m>gold</m> leaf. Also, <m>silver</m> is harder and therefore does not scratch as easily. And if <m>gum</m> was not a little [too] strong for <m>silver</m>, it would not crack so clearly. When you work, the exhalation of your <m>breath</m> humidifies the leaf lying on the <m>glass</m>, and therefore it would be good, when working, to heat it up a few times. You have to very thoroughly wash what you want to be empty and used as a surface painted with colors, because if it is not really clean of the <m>gum</m>'s <m>grease</m> and viscosity, and of other things, the colors will not be so clear on it. To advance your work you can smooth with a <m>pumice stone</m>, or [you can do this] to better lay down <m>gum</m> and leaf on a cut <m>paper</m>. This way, you will have to do very little repairs. If you want a <m>gold</m> color there without <m>gold</m>, mix soaked dried <m>saffron</m> with a little bit of <m>massicot</m>.

</ab>
</div>

.....
<page>042r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f89.image</image>

<div>
<id>p042r_1</id>
<head><pro>Founder</pro></head>

<ab>The <m>clay</m> mould should be baked until sometimes it becomes red hot, since otherwise the <m>metal</m> would be unstable and be no good at all. Similarly, the inner core must be baked very hard, and mixed with <m>charcoal powder</m>.</ab>

</div>
<div>
<id>p042r_2</id>
<head><m>Wax</m> for seals and impressions</head>

<ab>For large <m>wax</m> seals, you should always have tepid water handy, and keep your <m>wax</m> in it. Yet previously you should have kneaded it with your hands, so as to make it very smooth and even, for otherwise the water would penetrate the <m>wax</m> and prevent it from holding together. Afterwards, you can imprint whatever you want. Cover the <m>wax</m> in three or four sheets of <m>paper</m>, and, with a round and smooth stick similar to a pestle, roll it out as if to polish it. It will then stick to the <m>paper</m>, helping you to take it off from the seal. In this way you will imprint better than if you were dripping molten <m>wax</m>. You can carve figures and color them in gold, in silver, or paint them with <m>couleurs à vernis</m>, and transfer them onto a <m>glass</m> pane painted with <m>couleurs à tourmentine & mastic</m>. And if you want to apply these designs by inlaying, use <m>gum ammoniacum</m> mixed with <m>vinegar</m>, and it will stick well.</ab>

</div>
<div>
<id>p042r_a3</id>
<head>Casting in <m>plaster</m></head>

<ab>Melt some <m>wax</m>, and with a big brush coat it over the relief of which you want to have the cave, as if you wanted to paint it. And a light crust will be formed upon which you will cast <m>plaster</m>, to give it shape and consistency. Afterwards, in the cave of the <m>wax</m>, cast your <m>plaster</m> as well to get the relief, and it will come out very easily, because of the <m>wax</m>. This is done more often for large pieces in haut relief than for others.</ab>

</div>

.....
<page>042v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f90.image</image>

<div>
<id>p042v_a1</id>
<head>Pounce for scratching <m>satin</m></head>

<ab>After pouncing with crushed <m>flour</m> or <m>chalk</m>, in order that the dots of the pounce not be erased, go over the trace with <m>egg white</m> beaten with the <m>milk</m> or <m>bark of a fig tree</m>, which will immediately make the <m>egg white</m> clear up like water, without becoming opaque, it will maintain the trace that you will go over again and scratch with a piece of glass or penknife.</ab>

</div>
<div>
<id>p042v_a2</id>
<head>Canvas for <m>oil</m> painting without breaking</head>

<ab>In order that your <m>oil</m> painting be not broken or ruined as you bend it, make your (preparatory) layer with <m>honey</m>, a bit of <m>oil</m>, <m>water</m>, and <m>flour</m>.</ab>

</div>
<div>
<id>p042v_a3</id>
<head><m>Oil</m> painting on <m>taffeta</m></head>

<ab>In order that the <m>oil</m> not run, make your first layer with <m>honey</m>, <m>water of alum</m>, and <m>essence of amidin</m>.</ab>

p042v_a4

Casting sand

The key is to grind it well upon marble, and that it be extremely fine; mix in a bit of calcined alabaster.

p042v_a5

For decorating a painting

Go over the lines with lacquer ground together with olive oil, which will not dry out.

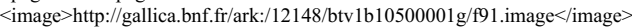
p042v_a6

White varnish on plaster

Put down two or three layers of white glue for painting. Afterwards, varnish with varnish of sandarac, aspic oil, and a bit of mastic. Put it into a vessel at night, everything ground well together without fire, which would make it turn yellow. Then, with a paintbrush, it becomes dry immediately. Filter the oil that will have absorbed the substance.

.....

043r



p043r_a1

Mericoton and pavis

One should graft them before Christmas so that they will not freeze too much.

p043r_a2

White bronzing on plaster

One must apply four layers of painters' distemper glue onto the plaster portrait, leaving one layer to dry at a time. Then apply gently some burnt English point ground on a marble slab, or some lead burnt, ground and distempered with a bit of gum or glue. When dry, rub gently with a little cotton and the plaster portrait will look like lead. Others grind minium with quicksilver.

p043r_a3

Purpurine

Take half an ounce of soft tin, melt it in a spoon. Once it is melted, throw in an 3 of sulfur, mix together. Once they are cold, grind on a porphyry slab. Then take an 3 of sal ammoniac and an 3 of the yellowest sulfur that may be found, grind both. And then mix very well all the aforementioned materials. Then put all together into a glass sublimatorium. Hold this over a small fire for an hour, and for an hour over a stronger fire, and for an hour over a very strong fire, and it will be ready. Then, to use it, apply resin black with the glue used by painters to paint, two or three times, until it is quite black. Then apply a bit of varnish. Once it is dry, apply the purpurine dry with a finger where you want. The more you apply, the finer it will look. Then, if you like, you can apply varnish on top.

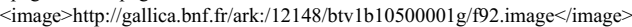
p043r_a4

White bronzing

Apply some black with glue on the medal, as described above, and then varnish it. Once dry, rub it with English point using a brush or your finger. The said rubbed point embellishes lead medals.

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043v



<id>p043v_1</id>
<head>Varied and transmuted <m>wine</m></head>

<ab>Grate <m><pl>Brazilwood</pl></m> very fine, let it soak for one or two hours in <m>clear water</m>. Then take this <m>tinted water</m> and add some clear water and you will make <m>wine</m> as clear as you like. If you please, put a drop of juice of <m>lemon </m> or of <m>orange juice</m> in it and it will immediately become white. You can drink it without danger.
</ab>
</div>

<div>
<id>p043v_2</id>
<head><m>Pearls</m></head>

<ab>It is said that pulverised <m>talc</m> blown into a lamp makes them so.
</ab>
</div>

<div>
<id>p043v_3</id>
<head><pro>Arquebusier</pro></head>

<ab>To shoot your arquebus precisely the end of the breech must come precisely on the edge of the light, because like this the <m>powder</m> catches [fire] and burns all at once without blowing and has more strength and does not recoil. On the contrary, if the breech is made hollow, as they commonly are, the <m>powder</m> catches more quickly in this place, makes it recoil, and blows. This is clearly shown [by] a arquebus with a chamber which recoils more than another. And since the cannon is bigger at the breech than at the muzzle the sights are also uneven, because that at the breech is higher than that at the muzzle. Additionally, the thickness of the cannon is greater than the caliber by about one line, and the sight by another. It would therefore be necessary to either make the cannon all of one size or to raise the barrel from the end and towards the muzzle, and push it in and lower towards the breech. The weight of the <m>powder</m> must be one third of the [weight of the] bullet.
</ab>
</div>

<div>
<id>p043v_4</id>
<head>Hail shot for the arquebus</head>

<ab>If you want that it stays together put a piece of <m>felt</m> or <m>leather</m> or <m>paper</m> on it, depending on the distance, and let the piece be made with a form precisely cut depending on the caliber of the arquebus.
</ab>
</div>

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<page>044r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f93.image</image>

<div>
<id>p044r_a1</id>
<head>The reach of an arquebuse</head>

<ab>The proportion of a fowling piece is: 4 king's feet long and the <m>bullet</m> weights xviii pennyweights, the expulsion vi pennyweight of <m>powder</m>, its range iiiii-by-xx pans and 3 and a half feet and two thumbs and around a line, which is the Paris aulne. The medium arquebuse, which is the usual one and the easiest, admits xv pennyweight of <m>bullet</m>, v pennyweight of <m>powder</m> & reaches lx paces.</ab>

</div>
<div>
<id>p044r_a2</id>
<head><pro>Pewtermaker</pro></head>

<ab>They mix viii or ten pounds of <m>lead</m> for each hundredweight of <m>tin</m> in the cities that have regulated guilds. But elsewhere, they add as much as they can. One also uses ii. lb. of <m>estaim de glace</m> to bind it and three lb. of broken <m>brass</m>, that is, <m>copper</m> scrap that <pro>kettlemakers</pro> make. This makes the vessel more reverberant and it is not as breakable.</ab>

<ab>
<margin>left-middle</margin>
This <m>pewter</m> is called ordinary <m>pewter</m>.</ab>

</div>
<div>
<id>p044r_a3</id>
<head>Arquebuse</head>

<ab>To keep it from backfiring, clean your arquebuse every eight days, and rub it with <m>oil</m>, and when you draw out, wet some <m>linen</m> in <m>oil</m> & put it in instead of <m>paper</m>.</ab>

</div>
<div>
<id>p044r_a4</id>
<head><m>Lacquer</m></head>

<ab>To test it, soak it and apply it to <m>paper</m>, and if one or two hours later it has not died at all, it's fine and well-done.</ab>

</div>
<div>
<id>p044r_a5</id>
<head>Dyes from <m>flowers</m></head>

<ab><m>Red poppies</m> that grow amongst wheat make a very beautiful columbine on white leather. The <m>boufain</m> makes a very beautiful blue. An <m>herb which grows in hedges, which has a stem similar to flax, long and broad leaves like little bugloss, which has a violet flower verging on blue and looks like the fleur de lys, </m> makes a quite beautiful turquin, better than azure. Another <m>columbine flower of the shape and size of the bugloss flower, which has a leaf like that of the pansy</m>, also makes a very beautiful turquin. It grows in wheat in light earth.</ab>
</div>

.....
<page>044v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f94.image></image>

<div>
<id>p044v_1</id>
<head><m><al>Horse</al>hair sieves</m></head>

<ab>They are made often in <pl>Normandy</pl> with <m><al>horse</al>tail hairy</m> that they clean with some washing <m>water</m> because they pick them out with their mouth and separate short and broken ones. They do not make the <m>thread</m> longer than the sieve and it is round. They attach the black or white <m>hair</m> at the two edges of the thread, according to the piece they want to make. And the thread is woven above and below like any other fabric. And passing a small flat stick of ii or three fingers wide between two, they pass between after two bits of hair at each step strike and weave with three steps. The entire sieve is made of xvii bits of <m>hair</m>. They sell them by dozens and each xxx sols. They bring them to <pl>Toulouse</pl> to send them then to <pl>Spain</pl> and there, they exchange it with <m>silk</m>.</ab></div>

<div>
<id>p044v_2</id>
<head>Excellent <m>water</m> against the pest by the Master of Montorsin</head>

<ab>Take some <m>theriac rue</m>, dittany and good <m>vinegar</m>, put the whole in a <m>glass</m> bottle and leave it half-open and well attached. And put it in a bath for three or 4 days. Then take some of your brew from its container when you need it and pour some, when necessary, on a red hot tile and inhale the steam and also perfume your clothes. This is a very precious and proved secret.</ab></div>

<div>
<id>p044v_3</id>
<head><m>White soporific oil</m></head>

<ab>Have x or 12 <m>Mandragore apples</m>, put them in quarters together with some <m>olive oil</m> in a glass bottle which you will cook in a bain marie for two days and your <m>oil</m> will turn white with which you rub the sole of your feet and you will quickly feel sleepy.</ab></div>

<div>
<id>p044v_4</id>
<head>Stucco</head>

<ab>To stick stones, some do not use gem and pitch <m>black rosin</m> because it is too greasy, but they use as much <m>rosin</m> and <m>sulphur</m> and the same quantity of <m>wax</m> and then add finely ground <m>brick</m>. Others some white <m>chalk</m> and crushed and powdered and sieved white <m>stone</m>.</ab></div>

.....
<page>45r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f95.image></image>

<div>
<id>p045r_1</id>
<head>Loading a <m>pellet</m> arquebus</head>

<figure>
<id>fig_p045r_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdIO5bkhmQklfNHJoRnc></link>
</figure>

<ab>It is necessary to make a <m>wooden</m> pipe 4 or 5 fingers wide that is perfectly hollow, according to the open end of your barrel, of the form represented in A. Then, having cut some <m>paper</m>, which has to be as strong as that of large printed books, in the form indicated in D and as wide as you want your cartridge to be long, wrap the <m>paper</m> around the baton B and at the top, where the pointed tip F will be around the aforementioned <m>paper</m>, and the cartridge will be formed. But, in order to make the bottom more uniform, put it into pipe A, then introduce baton B and, in the other end of the pipe, which is marked G, insert baton C and pack so as to compact and press the folded tip of your cartridge, then take it out, and the charge in this, put three or four drageons or larmes. Then, with a punch, marked E, of the same caliber of your arquebus or pistol, punch round pieces on <m>cardboard</m> or <m>felt</m> or <m>waxed cloth</m>, and put this round piece into your cartridge, and push it until the drageons with the baton B, the more drageons, 3 or 4, then a round piece. And continue like that until the cartridge is full, the put it into the arquebus. If you want your pellet to scatter earlier, make round pieces out of <m>paper</m>. If you want it to reach further without scattering, make the piece out of <m>cardboard</m>. If you want it to reach even further, make the round piece out of <m>waxed cloth</m>, or either of <m>leather</m> or <m>felt</m>, and a piece of <m>waxed cloth</m> on the top. And if you make the cartridge in part with <m>waxed cloth</m> or <m>waxed paper</m>, it will reach one hundred paces and will make a big hole, and the <m>pellet</m>, barely scattering, will produce a big opening. But if you make your cartridge with <m>waxed cloth</m> or a material stronger than <m>paper</m>, it cannot be so long, with the <m>cardboard</m> tip being like in D, because it is enough if it wrapped around twice. In such a way, the <m>pellet</m> barely scatters and makes a big hole in porte-corps armor or others.</ab>

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<page>045v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f96.image</image>

<div>
<id>p045v_1</id>
<head>Loading an arquebus</head>

<ab>You must have different charges depending to the range that you want your arquebus to have, all of which have a screw on the end that fits in the nut of the the said arquebus' ramrod. So when you put the powder in the charge, which is in the nut, hold the ramrod upright. And raising your arquebus up, insert the said ramrod until the charge is in the bottom of the arquebus, then straighten the arquebus, the mouth of the barrel up, and remove the ramrod. That way, your powder will now be entirely in the breech without any grain or dust that lines the sides of the arquebus, which always has some grime in the chamber. That way it will not recoil, and you will shoot more true.</ab>

<figure>
<id>fig_p045v_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5ZIVWTkcyU2FfanM</link>
</figure>
</div>

<div>
<id>p045v_2</id>
<head>How to adjust a bent gun barrel</head>

<ab>Take off the breech and put it [the barrel] on a well-leveled table set perpendicular to the floor, and insert a very smooth string without knots, on each end of which there is a plumb hanging from each side. Then look in the barrel along the string, and if it does not touch equally all over, note the place where it does not touch and hit it with a hammer on the exterior of this this side in the same place, and this will make it straight. Do this all around it.</ab>

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<page>046r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f97.item</image>

<div>
<id>p046r_1</id>
<head><m>Oil of sulfur</m> for the teeth</head>
<ab>Some people whiten them with compositions of <m><fr>eau forte</fr></m>, however one says that this corrupts them afterwards and causes them to blacken. One says that <m>oil of sulfur</m> is excellent, but it needs to be applied in this way: take as much <m>clove oil</m> as can be held in a <m>nutshell</m>, as much <m>rose honey</m>, and seven or eight drops of <m>oil of sulfur</m>, and put it all together, and after having cleaned the teeth with a small burin, touch them lightly with a good bit of <m>cotton</m> dipped in the aforementioned oils and leave it for a little while, then spit it out or rinse the mouth with tepid water, and repeat two or three times. <m>Oil of sulfur</m> penetrates and is corrosive, and l but the <m>clove oil</m> and the <m>rose honey</m> correct it. Therefore use it with discretion.</ab>

<div>
<id>p046r_2</id>
<head><m>Wheat oil</m></head>
<ab>Applying it to hair makes it fall out and keeps it from growing back.</ab>
</div>

<div>
<id>p046r_3</id>
<head>Against the falling sickness</head>
<ab>Have fresh <m>crow eggs</m>, either four or five, and the over the space of four or five mornings cook one, and having mixed in it a bit of powdered <m>gentian</m> and a drop of <m>oil of sulfur</m>, give it to the patient. It is believed that the pain will not return.</ab>

<div>
<id>p046r_4</id>
<head>Against cold gouts</head>
<ab>Put <m>Guaiaic oil</m> together with <m>oil of sulfur</m> and apply.</ab>
</div>

<div>
<id>p046r_4</id>
<head>Tree flowers</head>
<ab>Those that are seen with every new <fig>🌳 </fig> are hardly profitable. Only two buds need to be left on the graft.</ab>
</div>

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<page>046v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f98.image></image>
<div>
<id>p046v_1</id>
<head><m>Oil of sulfur</m> for the writer</head>

<ab>If the writer wants to quickly clean his quill of the thickness of the <m>ink</m> which dried on it, it only needs to be dipped in <m>oil of sulfur</m>, and immediately it will be white and clean. Urb.</ab>
</div>

<div>
<id>p046v_2</id>
<head><pro>Shoemaker</pro></head>
<ab>If he wants to do some tanned work on black <m>leather</m> shoes, he can dip in <m>oil of sulfur</m> a writing quill and paint with it what he would like on shoes, boots, and colets[a][b]. And he will obtain a reddish color which no effort will erase. Urb.
</ab>
</div>

<div>
<id>p046v_3</id>
<head>Erasing a letter</head>
<ab>
Dip a writing quill in fresh and good <m>oil of sulfur</m> and, thus, retrace all the letters with the dipped quill and they will disappear from the parchment. Urb.
</ab>
</div>

<div>
<id>p046v_4</id>
<head>Writing without <m>ink</m></head>
<ab>
Write with <m>oil of sulfur</m> and heat [it], and the letter will become as black as this one. Urb.
</ab>
</div>

<div>
<id>p046v_5</id>
<head>Making à jour letters in <m>paper</m> and other work</head>
<ab>
If you write d or trace something on the <m>paper</m> and heat it soon after, and immediately it will become black; then by rubbing with a sponge, the letter will be cut out because it will be burned. Urb. Or retrace the letter with a wiped quill of.
</ab>
</div>

<div>
<id>p046v_6</id>
<head>Making someone's boots burn</head>

<ab>
Rub the abovementioned <m>oil</m>, and as he warms himself they will burn without a flame.Urb.
</ab>
</div>

<div>
<id>p046v_7</id>
<head>Black letters on <m>stone</m></head>

<ab>
The same <m>oil</m> on letters engraved in <m>stone</m>, once heated, becomes black and penetrates.</ab>
</div>

[a]A type of shoe/footwear?
[b]could also be a part of the shoe (on the *cou-de-pied*), a type of leather (see collet in <http://tictac-cordonnier.blogspot.com/2010/10/glossaire-lexique.html>) or laces ("prendre au colet")

.....

<page>047r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f99.image></image>

<div>
<id>p047r_1</id>
<head>For teeth</head>

<ab><m>Sal ammoniac</m> i <figure>3</figure>, <m>rock salt</m> 1 <figure>3</figure>, <m>alum</m> half <figure>3</figure>. Make <m>water</m> with the
cornue[a], and no matter how little of it you touch the tooth with, the tartar and blackness will go away. It is true that it has a bad odor, but you can mix it with <m>rose
honey</m> and a little <m>clove oil</m> or <m>cinnamon^{oil}</m>.
</ab>
</div>

<div>
<id>p047r_2</id>
<head><m>Antimony oil</m></head>

<ab>It yellows quite strongly, even <m>silver</m>, if you put it on it once heated.
</ab>
</div>

<div>
<id>p047r_3</id>
<head>Against falling sickness or dizziness</head>

<ab>Wear, hung to your collar, some root of <m>paeonia</m> which has been plucked when the moon is waning.
</ab>
</div>

<div>
<id>p047r_4</id>
<head>Crucibles</head>

<ab>Those from <m>Bayonne</m>, which seem to be made of <m>grès</m>, are better because they last, without comparison, longer than others. And since they are not
very thick, one wastes less <m>charcoal</m> to melt <m>gold</m> and <m>silver</m> because they are quickly heated. They do not absorb <m>silver</m> as much as
others, and for this reason, there is not much laveure [b]to be found in them. They are also not as likely to fall over as others that have a flat bottom.
</ab>
</div>

<div>
<id>p047r_5</id>
<head>Against diarrhea and dysentery</head>

<ab>Against diarrhea, it is good to use preserve of <m>symphytum</m>, also [known as] <m>consolida maior</m>. And if it is against dysentery, you can rub the
temples, the hollow of the hands, and the sole of the feet with <m>wax oil</m> and take a drop of two of it with a spoonful of <m>broth</m>. The root of <m>consolida
maior</m>, crushed and put between pieces cuts in a piece of <m><al>beef</al></m>, then boiled, rejoins them, as it is said.
</ab>
</div>

[a]Very likely a retort, a small glass vessel with a corne (or horn), made in one piece.
[b]See, perhaps, "laveure des orfèvres" in Cotgrave, i.e. leftover metal found in the Goldsmith's sop, which is then washed and remelted?

.....

<page>047v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/fl00.image</image>

<div>
<id>p047v_a1</id>
<head>A beverage which tastes like wine</head>

<ab>Take <m>tartar</m> from large vessels, dilute it in <m>water</m>, and it will taste like wine.</ab>

</div>
<div>
<id>p047v_a2</id>
<head><m>Melting</m></head>

<ab>Some to make it flow neatly, put half <m>metal</m> and half <m>copper</m>. Others take a part of <m>potin</m> and a part of copper<m>latten</m>. Others take a part of <m>latten</m> from candlesticks and a part of <m>latten</m> from a basin. And among basins, the <m>latten</m> from larger ones is harder than that from skillets because the material needs to be softer in order to give them such a small circumference. From the large ones, from which they take the more ample circumference, the harder material can be taken.[a] The harder the <m>latten</m>, the better it flows and the neater it casts, but it must be cast very neat hot. One As one wishes to cast, one put[b]s <m>lead</m> on the material[c], which makes it flow and run and goes up in smoke and be careful that the mold does not get damaged. And if you want to cast pieces on <m>fire</m>, mix more <m>rosette</m> than <m>latten</m>, so that it is less brittle. The crucibles used to melt

<m>metal</m> should be better than those used by <pro>glassmakers</pro> because glass is not as heavy as <m>metal</m>. And in this instance, the crucibles need to be luted with <m>earth</m> and crushed <m>glass</m>.</ab>

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</div>
<div>
<id>p047v_a3</id>
<head>Crucibles[d]</head>
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<ab>Those made during the winter and humid weather are not good. Those used by founders should be thick and should be coated.</ab>

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</div>
<div>
<id>p047v_a4</id>
<head>Cast</head>
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<ab>Finely-calcinated <m>pumice stone</m> casts 22 carat <m>gold</m> without ^{needing} repair, but it needs to be ablaze and red as the metal. And it will survive many casts.</ab>

[a]Check translation
[b]Gecter's meaning seems to be ambivalent and varies between cast and throw/add/put. My choice here reflects what I think is most coherent with the rest of the sentence
[c]I believe the 'material' refers to the latten mix abovementioned
[d]Cotgrave also offers the term "cruzet" for "creuset," but I don't think that's an English word anymore. "Crucible" is more familiar.

.....

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<page>048r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f101.image</image>
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<div>
<id>p048r_a1</id>
<head><pro>Founder</pro></head>
<ab>The hot [part][a]<lb/>
<figure>
<id>fig_p048r_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5YXhXTUJpMk44eTQ</link>
</figure>
A furnace[b] to cast a large quantity of <m>metal</m></ab>
</div>
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```
<div>
<id>p048r_a2</id>
<head>Curing dogs from mange</head>
```

<ab>Give them a pill of precipitate[c], that is to say 3, 4, or up to 5 <m>grains</m>, mixed with their <m>food</m>>[d]. Having given them one for one[e] week, give them another seven or eight days later.</ab>

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</div>
<div>
<id>p048r_a3</id>
<head>Excellent <m>mustard</m></head>
```

<ab>Dry some <m>bread</m> in an oven, then stick <m>cloves</m> & <m>cinnamon</m> into it and put it to soak in good <m>wine</m>. Then, being well crushed, pass everything through a <m>cloth</m> strainer and incorporate it with your <m>mustard seeds</m>.</ab>

```
</div>
<div>
<id>p048r_a4</id>
<head>Keeping birds and animals</head>
```

<ab>Take a measurement from [the part of] their body that is fleshy and more susceptible to decay, with some <m>canvas</m> that you shall cut according to their size and width. Having filled this with <m>cotton</m> and stitched it, skin them. But leave the head, neck, wings and feet of birds, and head, legs, feet and tail of animals, because those dry easily. Then use that <m>skin</m> to dress the mold made of <m>canvas</m>. Small ones can be dried in an oven or covered in <m>sel ammoniac</m> and <m>(?)</m>.</ab>

[a]? unsure how to translate. Is it just "heat" for diagrammatic purposes
[b]No consensus in Vocab list about whether "four" is to be translated as "furnace", "stove" or "oven". The same equipment is also mentioned in the mustard recipe below so not sure if term must be the same for melting metal and making mustard...
[c]Cotgrave defines this as either an unspecified "precipitate" or "the red, poisonous, corroding powder of burned quicksilver"
[d]Is "meat" perhaps a more appropriate translation here?

[e]'Une' is repeated twice in the transcription but translated only once here.

.....

<page>048v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f102.image</image>

<div>
<id>p048v_a1</id>
<head>Regulus extracts</head>

<ab>Take one pound of <m>antimony</m>, half a pound of <m>black soap,</m> and half a pound of <m>Montpellier tartar</m>. Once it is all pulverized, you have to boil it in a <m>varnished earthenware pot</m> on a big <m>fire</m> as you stir it with a <m>piece of wood or iron</m>. The <m>soap</m> will burn off and be consumed by the <m>fire</m> and the rest will stay red. The quantity has to be reduced to a third. Then throw that on a <m>tile</m> and it will turn blackish. You will have to melt and pour that on a <m>tile</m> four or five times and then you will have <m>regulin from antimony</m>.</ab>

<ab>Others pulverise the <m>antimony</m> and mix it with equal quantities of <m>salt peter</m> and powdered <m>tartar</m> and heat a <m>pot</m> or <m>crucible</m> until it is red hot, then add a little bit at a time, turning their back to it to avoid the smoke, and keep doing so until everything is added. They maintain the <m>fire</m> until everything is completely melted and melt that several times.</ab>

</div>
<div>
<id>p048v_a2</id>
<head><m>Lead tin</m></head>

<ab>They crack if they are often melted again and thicken and burn, so that even if a salmon is made, the <m>tin</m> becomes thick at the bottom if it is not stirred often during casting. It is better for casting to melt some <m>lead</m> and some <m>tin</m> and to renew it each time. The <m>antimony</m> makes them breakable, the <m>iced tin</m> makes them whiter. <pro>Pewterers</pro> mix a pound of <m>iced tin</m> with a quintal of <m>fine tin</m> and two and a half or three pounds of <m>red, mild cauldron copper</m>, which is better than <m>latten.</m> There is <m>fine and mild tin</m> and <m>sour tin</m>. <pro>Pewterers</pro> cast the sour as grills to sell it better. They go better and more cleanly in <m>stone</m> moulds than in <m>copper</m> molds because <m>copper</m> is fat and draws^[a] the <m>mild tin</m> when heated. Or when casting a large work, three quarts of <m>lead</m> with one pound of <m>mild tin</m> produces a very liquid mixture which is appropriate for casting and is as shiny as a mirror. Laxunge^[b] <m>glass</m> lightens them. All things that lighten <m>metals</m> a lot and remove their thickness and natural density and make them as liquid as water make them appropriate for casting because their thickness stops them from flowing. The <m>mild tin</m> is more even than the <m>sour tin</m>, which is whiter and seems to have been polished like a mirror.</ab>

<ab>
<margin>left-bottom</margin>
To properly alloy the <m>tin</m> and the <m>lead</m>, you have to melt them together, Then, mix them together a little bit at a time and throw it often on a <m>marble tile</m>. And when it is going well, you will see it becoming unified and shining as if it were burnished. Because sometimes it becomes spotted due to too much <m>lead</m> and sometimes due to too much <m>tin</m>. There is some <m>tin</m> allied with more <m>lead</m>. Ordinary <m>tin</m> is allied with some <m>lead</m>.</ab>
</div>

^[a]Perhaps this means it binds with it?
^[b]I couldn't find this word or place.

.....

<page>049r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f103.image</image>

<div>
<id>p049r_a1</id>
<head><m>Lead</m> casting</head>

<ab>Those who cast these small items, sold in front of the churches, melt half a pound of <m>lead</m> on a pound of <m>tin</m>, and cast in a <m>stone</m> frame. And if they want their work [to be] whiter, they use half of <m>tin</m> and half of <m>lead</m>, but the first mixture is the best, and does not make bubbles. You have to cast it hot enough, otherwise it won't flow. Some [people] smoke their molds with a <m>rosin</m> <m>candle</m> for neat casting. The <m>antimony</m> makes the matter harsh and breakable. The <m>looking-glass</m> <m>tin</m> makes the work whiter, but does not make it flow more than with <m>tin</m> that is correctly made with pure <m>lead</m>. Those who make very precise <m>lead</m> castings use puncheons^[a]. <pro>Tintiers</pro>^[b] cast with <m>copper</m> molds. It is said that one makes <m>lead</m> adhere to <m>glass</m> by using <m>resin</m>. Some people put a quarter of <m>tin</m> on <m>lead</m>. <m>Glass</m> <m>tin</m> makes the work stronger.</ab>

<ab>
<margin>left-top</margin>
Poncet^[c]^[d].They cast by using the souldure the <pro>glass-makers</pro> use. Lump [of metal] of [...] Calcinated <m>oyster shell</m>.</ab>

</div>
<div>
<id>p049r_a2</id>
<head><m>Sand</m> for <m>lead</m> casting</head>

<ab><m>Rye straw</m> ashes well boiled, dried and then well sieved. alloy it with <m>egg white</m>. One can add <m>copper</m>, <m>latten</m> and others.</ab>

</div>
<div>

<id>p049r_a3</id>
<head>Other</head>

<ab>Burnt and calcinated <m>pumice stone</m>, burnt and calcined <m>cuttlefish</m> <m>bones</m>, in the same quantity; and ashes of <m>walnut</m> <m>tree</m> or <m>vine</m>, boiled, dried and finely sieved alloy with <m>egg white</m>.</ab>

</div>
<div>
<id>p049r_a4</id>
<head><pro>Pewterers</pro></head>

<ab>They add one lb. of <m>glass</m> <m>tin</m> to one quintal of <m>fine</m> <m>tin</m>, which makes the <m>tin</m> ticker. There is two kinds of <m>tin</m>, one which flows better, like lead<m>,</m> while the other one is sour[e] and can be made thicker. They are casted in thick and solid <m>tin</m> molds or, to cast more neatly, in chisel-engraved <m>copper</m> molds, or <m>stone</m> or <m>earth</m> [molds]. <m>Latten</m> scrapings mixed in make the <m>tin</m> more breakable and harder, and more difficult to work with. They cast in the very hot molds some very hot <m>tin</m>. They perfume their <m>tin</m> molds with <m>resin</m> <m>candles</m>.</ab>

<ab><m>Sour</m> <m>tin</m> is often mingled with <m>salmons</m>[f][g], easy to cut but difficult to work with and melts if it is not mingled with the other <m>soft</m>[h] one. And thus it would become leftover[i][j].</ab>
</div>

[a]I replaced maker's mark. Cotgrave gives punction for "poincon".
[b]Should this be translated as "Pewterers"?
[c]Could not find a translation. Might be a small pumice stone, "ponce" with the diminutive suffix "et".
[d]From Pascal Julien, Univ. of Toulouse: ""ceux" signifie les fondeurs d'artillerie et "poncet" est l'un de ces fondeurs qui travaillait à Toulouse, dans l'arsenal de la ville. Les Poncet sont des fondeurs de cloche et d'artillerie, actifs des années 1560 à 1640. Tous deux, père et fils, se prénomment Pierre. ils sont originaires de Mirepoix, en Ariège.
[e]Among other choices, Cotgrave gives "brittle" for sour. Here the author talks about two kinds of tin, one "aigre" and the other one "doux". This might be a reference to the acidic and alkaalin binary. In Cotgrave, the tin "doux" is also called "estain de Cornouailles", tin from Cornwall.
[f]Saulmon: Masse de métal telle qu'elle sort de la fonte. DMF
http://atilf.atilf.fr/scripts/dmFX.exe?
LEM=SAUMON;MENU=menu_dmf;AFFICHAGE=2;ISIS=isis_dmf2012.txt;OUVRIR_MENU=2;s=s133b1c38;LANGUAGE=FR;XMODE=STELLa;FERMER
[g]The link above has since timed out.
[h]Again, contrast between tin that is "aigre" and tin that is "doux".
[i]"Deschets": Ce qui tombe d'une matiere qu'on travaille; reste, chute. DMF
[j]waste?

.....

<page>049v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f104.image</image>

<div>
<id>p049v_1</id>
<head><al>Birds</al></head>

<ab><al>Calandra larks[a]</al> and all <al>birds</al> taken from the nest and which have not known freedom are better, because those taken with a net never sing as well. However, <al>wild birds</al> give much pleasure with their chanting, but they must be taken before Michaelmas, otherwise they do not live long. All get a small impostume on the fork of their tail, sometimes, which some call "gaillardise", which makes them sick and sometimes kills them, especially <al>calandra larks</al>. One can tell when they are sad and do not sing. It must be pierced, not with a pin, but very delicately by turning with the tip of a quite sharp <m>feather</m>. The <al>calandra lark</al> should not be without <m>river sand</m>, and grey sand, a little rough, is the best. It rubs itself against it and purge[b]s itself [with it].</ab>
</div>

<div>
<id>p049v_2</id>
<head>Grain of <m>lead</m></head>

<ab>Take a playing card, pierced with a pin or needle, folded over on all four sides. Hold it with the end of a small split stick and, holding said stick by the other end, put your card three or 4 fingers away from the <m>water</m> that you will have put in a platter or a similar vessel. Then pour your <m>lead</m>, which should not be too hot, into the card and continually tap on the stick that holds it. And thus your <m>lead</m> will turn into round grains. And pass it through a large sieve to separate the larger grain from the smaller. The larger one will carry 25 to 30 paces.</ab>
</div>

<div>
<id>p049v_3</id>
<head><al>Birds</al></head>

<ab>Those that are caught at Michaelmas are good to keep, but those that are caught in March die, because they are starting to fall in love. <al>Ortolans</al> are fed <m>oats</m> all summer so that they eat without getting fat, to be more suitable for hunting and helping to catch others. Then, when one wants to fatten them to sell, one gives them <m>millet</m>. They sing at night. They must be caught after mid-July and before Michaelmas, because after the period when they have their young in this country, they leave as <al>turtledoves</al> do. <al>Siskins</al> should also be caught after All Saints' Day and before Christmas, because afterwards they leave for the mountains to brood.</ab>

<ab>
<margin>left-bottom</margin>
<al>Ortolans</al> are ventriloquists, so that, singing without opening their beaks, they seem to be...
</ab>
</div>

[a]'calendres' -- also the mythical bird, in English referred to by the Latin 'caladrius'
[b]'purge' -- double meaning of cleaning and purging in English

.....

<page>050r</page>

<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f105.image</image>

<div>
<id>p050r_1</id>

<head>Preserving fruit for the entire year</head>

<ab>Take a rounded <m>glass</m> phial, with a large enough opening to receive cherries and plums and whatever fruit you want. Put in some <m>hot water</m>
^{and leave it} for two hours, and once your <m>water</m> is cold, throw it out and turn the phial upside down onto its opening on a perfectly even table and in a room where there is not a bit of draft or wind. Then prepare a stopper of fresh <m>wax</m>, wrapped in <m>oilcloth</m>, and adapt it precisely to the <m>glass</m> phial so that it will be ready to stop it once you have put in the fruit. And once you have hand-picked the fruit, only the non-rotten fruit, and on a warm and dry day, withdraw to the room, well closed so that no wind or draft can get in, and put it gently and deftly in the bottle. Then stop the bottle well with the stopper and lute it, making your lute, if you want, with some <m>quicklime</m> and <m>oil</m>, so that no <m>water</m> gets in. Then put your bottles into a tub full of <m>water</m>, in a cellar, during the summer, and in the winter put your bottles into a basket filled with some weights and lower it to the bottom of a deep well. Because in the winter the <m>water</m> in the tub would be too cold if your cellar is not warm enough, because the <m>water</m> needs to be like <m>river water[a][b]</m>.</ab></div>

<div>
<id>po50r_2</id>

<head>Molding</head>

<ab>If you want to quickly mold in demi-relief anything you come upon, fold some <m>paper</m> five or six times over, and place it on the medal and make sure the <m>paper</m> is folded around the edges of the medal so it is very secure. Next take a stick, broad at one end and with a well-dulled point at the other, and rub firmly on the <m>paper</m>, and retrace the lines with the point of the stick until you reckon that your impression is well done. Then, at your convenience, rub <m>oil</m> onto the <m>paper</m> lightly with a brush, and cast some <m>tallow</m> or <m>wax</m> or <m>sulphur</m> into it. And the <m>paper</m>, without burning, will give you a neat design that you can then mold in <m>plaster</m> or <m>tripoli</m> and then in <m>lead</m> and other <m>metals</m>.</ab>

<ab><margin>left-bottom</margin>Polished <m>cardboard</m> of little thickness and slightly humid is appropriate. Then if you want, strengthen it with some <m>paper</m> glued on the reverse side.</ab></div>

[a]We think "like river water" means a similar temperature and liquid state.
[b]yes

.....

<page>050v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f106.image</image>

<div>
<id>p050v_1</id>

<head><al>Little Birds</al></head>

<ab><al>Buntings[a]</al> sing during the night if they are left in the cool open air, but they are better suited to fattening for fine tables than for singing.
<al>Chaffinches</al> are prone to blindness, and their eyes swell at the beginning of August. <al>Goldfinches</al> love <m>lettuce seeds</m>, and you can catch them in gardens on seedy lettuce using two <m>lime-twigs[b]</m>. But because such a daily fayre would be too expensive for them, they are usually fed with <m>hemp-seed</m>. The <al>owsell[c]</al> is purged with a <al>spider</al>, which you must give it once a week, so that it doesn't become too strong. The <al>lark[d]</al> is purged with <m>dry mortar</m>, composed of <m>lime</m> and sand, to keep it from becoming too strong and fat. Small young birds are purged with the same <m>dry mortar</m>, by placing small lumps of it in their cages. The <al>cuckoo</al> lays its eggs in the nest of the <al>bunting</al>, otherwise know as <al>verdaule[e][f]</al>. This bird is very simple-minded, I have previously let two go, which after a few days returned to their cage. To feed young <al>goldfinches</al>, it is necessary to crush <m>almonds</m> with very fine <m>breadcrumbs</m>, and <m>lettuce seeds</m> or <m>hemp seeds</m>. For <al>linnets</al>, some <m>rapeseed</m>.</ab></div>

<div>
<id>p050v_2</id>

<head>Hot-worked <m>steel</m> and files</head>

<ab>The <m>steel</m> that <pro>blacksmiths</pro> and other <pro><m>iron</m> workers</pro> commonly use is not refined like that from <pl>Germany</pl> or <pl>Biscay[g]</pl>, which is hardened in a <m>molten iron</m> bath, but among the <m>iron</m> in bars which is transported by flatboat [h]from the forges of <pl>Foix</pl> and elsewhere, some is harder, whiter and more refined than the rest, as it comes from the mine. And the <pro>workers</pro> choose it and use it like the

other <m>steel</m>. And because it consists of <m>common iron</m>, they call it strong <m>iron</m>. But it is not of such good quality as purified <m>steel</m> like that from <pl>Germany</pl> and <pl>Biscaye</pl>, which is sold in small beads. Some hot-work their <m>steel</m>, giving it a heating [i] and then dipping it into a large quantity of <m>water</m>, then forging it. And fine <m>steel</m>, which is brittle, does break and crumble, whereas <m>iron</m> can be pulled out. Thus they separate the finest <m>steel</m>, which with another heating they make into a mass. The <pl>Germans</pl> make their files from strong <m>iron</m>.</ab>

<ab><margin>left-bottom</margin><m>Steel</m> is applied to tools not on both sides, but underneath the part where one sharpens and whets them, and this part must be made of very soft <m>iron</m>.</ab>

<ab><margin>left-bottom</margin><pl>Levantines</pl> refine our own <m>steel</m> because their country provides them with none, and they reheat it in a pot with <m>bitumen</m> etc.</ab></div>
[a]In this text we found that venarris was referred to as ortolan/ Venarris might be a gascon name for the bird : <http://dzt-isto.chez-alice.fr/evesche.htm> (reference found in DE LA LANDE, CHAP VII)
[b]definition from RC.
[c]According to Cotgrave "A little black-browne Owsell (or bird like an Owsell) thats euer alone, or if in any companie, with Sparrowes, among stone walls, or on the tops of houses, where sometime she singeth prettily."
[d]in Estienne, defined as "a type of lark"
[e]this is a bit surprising - cotgrave has 'verdaule' and calls it hedge sparrow - a different family of birds than the ortolan bunting.
[f]Verdaule here is presumably not the hedge sparrow, as suggested by Cotgrave, but a local name for a variety of bunting showing at least some greenish color. Maybe the yellowhammer (Emberiza citrinella)?
[g]A nortnen province of Spain, rich in iron ore of excellent quality.
[h]Another tempting interpretation would be: in the form of flat bars, but I don't think the expression "en plates" allows this.
[i]"donner une chaude" is a technical term used by goldsmiths as well as glass makers and iron workers.

.....

<page>051r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f107.image></image>

<div>
<id>p051r_a1</id>
<head>Copying off patterns</head>

<ab>Make some <m>soap water</m> and with it, rub and moisten the <m>engraved stone</m> [a] that you want to transfer the pattern from, then lay a white sheet of <m>paper</m> on it and rub strongly against it with a sharp piece [b][c] or the bottom of a <m>glass</m> bottle and you will transfer your stamped piece. It is true that it will be on the left side, but if you apply <m>aspic</m> or <m>turpentine oil</m> on your <m>paper</m>, it will be stamped on the right on the other side. Then trace these lines with a paint brush or a quill, then heat the <m>paper</m> and the <m>oil</m> will evaporate [d] and it will leave your <m>paper</m> white. And if you don't want that to be known if, by chance, you borrowed the <m>cut stone</m>, soak the <m>paper</m> and the polishing that the <pro>polisher</pro> made on the back side, which distinguishes what has been made, will not be recognizable. The <m>soap water</m> will turn the piece yellow, but well <m>gummed wate</m> [e][f]r, which has the same effect, does not do this. If you want to, for the same effect, make <m>gummed wate</m>r then, dissolve some <m>soap</m> in it and do as is said.</ab>

</div>
<div>
<id>p051r_a2</id>
<head>Cutters of printing [g][h]plates</head>

<ab>To clean the <m>copper</m> plates or to have used ones print better, boil them for four or five hours in a good <m>detergent</m> with quite used <m>laundry water</m>. Then make your <m>ink</m> with some <m>linseed oil</m> instead of <m>walnut oil</m> and press with the rollers. The <m>copper</m> plates are ready sooner than the <m>wood</m> ones, but they are not so clean to print promptly. The <m>wood</m> ones are tedious but would have sooner printed twenty sheets against two [sheets] of the other. To print with <m>wood</m>, the secret is firstly to poach, that is to say to lay the transferred pattern or drawn piece down on the <m>wood</m> plank and to make sure that the drawn side is stuck to the <m>wood</m>. Once dry, you will gently rub with a moist <m>handkerchief</m> the reverse side of the <m>paper</m> which, by being rubbed, will become so fine that what will remain will almost only be the drawn part, which, after, one should trace while cutting the pattern. You could do that to ornamented <m>glass</m> and <m>glass</m> layered with <m>black</m> cut pieces, to then scrape and layer your <m>colors</m> on the uncovered [side]. To make <m>ink</m> for <m>copper</m> plates</ab>

<ab>
<margin>left-bottom</margin>
These rollers are good for printing promptly with thick sheets of <m>paper</m> cut in different kinds of pastes.[i][j]</ab>

<ab>
<margin>left-bottom</margin>
One can place the plates among the linens when the <m>detergent</m> is rather fine, or also in a pot.</ab>

<cont/>
</div>
[a]Cotgrave uses "free-stone", recherche under "tailler"
<http://www.pbm.com/~lindahl/cotgrave/search/895r.html>
[b]"Dent" of an object
Élément pointu de certains instruments (p. ex. une fourche) ou de certains mécanismes (p. ex. une roue dentée)"
<http://www.cnrtl.fr/definition/dmf/dent>
[c]Dictionnaire du Moyen Français, via CNRTL
[d]Used this for "laissera"
[e]Où l'on a dissous de la gomme. Eau gommée.
<http://artflsrv02.uchicago.edu/cgi-bin/dicos/pubdico1look.pl?strippedhw=gomme>

More specific:
http://fr.wikipedia.org/wiki/Eau_gomm%C3%A9e
[f]Littré: Dictionnaire de la langue française (1872-77), via ARTFL. See also comment on "planche" from GC on p051v

[g]<http://www.pbm.com/~lindahl/cotgrave/search/732r.html>
[h]for "planche" - Cotgrave, via Lindahl, has "also, the Till of a Printers Presse"
[i]Needs revision
Paste could also be "pastel"

[http://atilf.atilf.fr/scripts/dmfX.exe?](http://atilf.atilf.fr/scripts/dmfX.exe?LEM=paste2;XMODE=STELLA;FERMER;;AFFICHAGE=0;MENU=menu_dmf;;ISIS=isis_dmf2012.txt;OUVRIR_MENU=2;s=11183738;LANGUE=FR;)
LEM=paste2;XMODE=STELLA;FERMER;;AFFICHAGE=0;MENU=menu_dmf;;ISIS=isis_dmf2012.txt;OUVRIR_MENU=2;s=11183738;LANGUE=FR;
[j]from Dictionnaire du Moyen Français, via ATILF, not publicly accessible.

<page>051v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f108.image></image>

<div>
<cont/>
<id>p051r_a2</id>

<ab>[a]which is different from the printers' one, you have to boil for a long time some <m>walnut oil</m> or, even better, some <m>linseed oil</m>, and after boiling for a long time, you will add <m>garlic</m> cloves and <m>bread</m> crusts to skim them and leave again to boil on big fire, then eventually you will add a measure of <m>lampblack</m> that you would have ground with some <m>oil</m> previously on the <m>marble</m>. And lastly you will mix everything and until you see that the plate[b][c] prints well. And when you will have spread the <m>ink</m>, dry the plate with a white sheet of <m>paper</m> and do that until the <m>paper</m> looks clean and then print. If you dry the plates with a <m>cloth</m>, it would remove the <m>ink</m> off. <m>Linseed oil</m> gets thicker while boiling and becomes similar to a varnish and risks catching fire as soon as you boil it, so make sure that when you boil it, it is in an open space or where the flame cannot damage anything. Some burn some <m>scale</m> until it turns black and grind it with some non-boiled <m>linseed oil</m> or <m>walnut oil</m> that some consider to be better. This black is beautiful but the <m>lamp smoke</m> one is blacker. When you want to print, take with your finger's end and reasonably spread some of the said <m>black</m> all over the plate previously cleaned with some <m>detergent</m> as said, then, rub it with a white sheet of <m>paper</m> until the paper is completely white and do not touch it with your bare hand but with a <m>paper</m> applied on it. And after cleaning the plate perfectly, rub again the sides and edges with a folded sheet of <m>paper</m>. Then, have [available] a small even table and on it a piece of <m>felt</m>, then a few sheets of <m>paper</m> and eventually the one you want to print on, moistened between two humid <m>napkins</m>. And on this sheet of <m>paper</m>, put your sheet then some more <m>paper</m>, and at least another piece of <m>felt</m>. Then put it through the rollers and you can print a dozen pieces consecutively while always charging the plate with <m>ink</m> and cleaning it as you did. But if you stop printing, the rest of the <m>ink</m> will dry out in the plate's lines and so you will have to let it boil in <m>detergent</m> or <m>urine</m>, as said, to clean it. The rollers must not be too loose.</ab>

</div>

[a]continued from previous page
[b]This translation has to be checked. I could not find a better one for "planche" understood as planche d'impression:
Feuille de métal ou morceau de bois plat sur lesquels on grave. Se dit aussi quelquefois des caractères d'imprimerie formant les feuilles d'impression.
See definition #5 here:
<http://www.littre.org/definition/planche>
[c]for "planche", via Littré

<page>052r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f109.item></image>

<div>
<id>p052r_1</id>
<head>The Work done in Algiers</head>

<ab>Have a <al>colt</al> of three of four years and feed it on <m>barley</m> and <m>straw</m> cut in the same way as that which they use to feed horses in <pl>Spain</pl>. Let it drink <m>water</m> from a good fountain or river ^{<m>water</m>}. I do not know if it would be good sometimes to let him drink the <m>water</m> of <m>sulfur</m> springs, and to sometimes give him <m>fenugreek</m> or other hot foods, for the intention of the <pro>worker</pro> is to use the heat of his <m>manure</m>, and the climate here is cooler than that of Algiers. Keep it in a warm place and use it and make sure it loses none of its <m>manure</m> or <m>urine</m>, of which you will make a mass or two so that while one cools the other will keep its heat and be suitable to continue. Also have a large flask as thick as possible, one finger thick if it can be so made, and with a capacity of one pitcher or <m>clay</m> jug, and around the feast of St John place a dozen and a half <m>chicken eggs</m>, that is to say, the <m>egg</m> without the <m>sup>egg</sup> white</m> and the <m>germ</m>. Others say sixty <m>yolks</m>. And with this dozen and a half of <m>chicken egg yolks</m> put one half ounce (others say sixty <m>eggs</m> and a half pound) of <m>female silk worm eggs</m>. And after carefully luting the flask (I do not know if air will be needed for the generation) and bury it in the heat of the <m>manure</m> up to the neck, and leave it there until several <al>worms</al> are engendered and then remove the flask and do not bury it in the <m>manure</m> anymore. But put it on the hot layer of the <m>manure</m> until all the <al>worms</al> are eaten and consume one another by shriveling and stirring, and only one remains. Once this has happened, you must lure it at regular intervals, day and night, with the assistance of two men, who will care for it in shifts, and you will lure it with an <m>egg yolk</m> covered in <m>gold</m> leaf or with a liquid <m>yolk</m> into which the <m>gold</m> leaf has been incorporated. And be careful that it does not miss such fodder (some say one <m>egg yolk</m> per hour, others say three, but the thing itself will demonstrate the practice). So nourished in this way it will grow in two month or seven weeks and will become like a <al>serpent</al>, one span and four fingers long, and one pound in weight, and as the wings will begin to develop, you must kill it, doing so with a <m>charcoal</m> fire in a ring around the bottle one span away from it, and at that time lute the bottle well so that it does not exhale.</ab>

<cont/>
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<page>052v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b9059316c/f110.item></image>

<div>
<cont/>
<id>p052r_1</id>

<ab>or to be safer, go away until the fire is completely blown out and that everything is cold again. Because the exaltation may be dangerous. And for that moment, when you take[a] it with some tongs, do cover your mouth with some good <m>vinegar</m> and have a protection and cover yourself with it. Once it has died out, put it into a cloth or a canvas made of <m>silk</m> and fold it and hold it to a board exposed to air and the sun will dry it. Once it is well dry, powder it in a mortar and keep this powder cautiously Because one ounce of this one thrown on 3 pounds of melted <m>δ</m> transforms it into a finer <m>⊙</m> than the other one. But there is not so much pitch[b]. You also have to choose the oldest <m>δ</m> possible which has often melted before and finely hammered into blades or other works, and at first purify it by melting and throwing it into some <m>honey</m> and <m>vinegar</m>. The completion time for such work is nine months from Saint John's day until the 25th April.

</ab></div>

<div>
<id>p052v_1</id>
<head><m>Linseed</m></head>

<ab>If one wants to have it strong and very fine, one has to gather it before the seed is completely ripe. Because if you wait for the seed to be completely ripe, the base is so dried by the sun that it is brittle and hard.It is true that like this, the seed reduces by a third.</ab>

</div>

<div>
<id>p052v_2</id>
<head>Colors</head>

<ab>If you want to keep them moist and prevent them from losing their initial color, pour some <m>white wine</m>, the whitest and the clearest you can in a bottle with some <m>tragacanth gum</m>, and once you have corked the bottle, mix and shake it very strongly, and leave it still until the <m>tragacanth</m> is dissolved. Add also some fine <m>white earth[c]</m> to lighten that. Some mingle the <m>white wine</m> with some <m>spring water</m>, and with this composition they mix their colors without grinding them, and they show very beautifully on miniature works.</ab></div>

[a]"apastes" in manuscript. Uncertain about meaning and appropriate translation.
[b]Uncertain about meaning. Rather: "But it does not weight as much"?
[c]In the manuscript, he clearly indicates "sol" which might rather be "sel", as "white salt" would fit in this context

.....

<page>053r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f111.image</image>

<div>
<id>p053r_a1</id>
<head>Soldering a vise</head>

<ab>It is good that the jaws of the vise should be high for use in making a long piece, and that they join well to clamp a thin object. In order to solder the nut, after you have made the bolt, you will make a long <m>iron</m> strip of the right thickness to fit into the notch of the bolt when red hot, and you will bend it all around [the bolt] by striking it with a hammer. Once it is well wrapped around, you will insert the bolt around which the nut is wrapped.</ab>

</div>
<div>
<id>p053r_a2</id>
<head><m>Copper</m></head>

<ab>If, while in the fire and while being cast, it touches the <m>iron</m>, that <m>iron</m> will be so sour that it will be impossible to forge.</ab>

</div>
<div>
<id>p053r_a3</id>
<head><m>Lead</m> casting</head>

<ab>According to some it is mixed, half <m>tin</m> and half <m>lead</m> and, in order to heat it, a little <m>arsenic[a]</m> is mixed in. It is cast well in small sizes in a <m>cuttlefish</m> <m>bone</m>, provided it is good.</ab>

</div>
<div>
<id>p053r_a4</id>
<head>Molding from <m>paper</m></head>

<ab>Over hot <m>ashes</m>, boil some <m>cotton</m> in <m>aqua fortis</m> mixed with <m>sal ammoniac</m> like <m>aqua regia</m>. And the cotton will become very fine, like <m>powder</m>. Then soak it with gummed water and you will cast quite delicately.</ab>

</div>
<div>
<id>p053r_a5</id>
<head>Almond trees, apricot trees</head>

<ab>They grow quite straight if they are grafted. And every kind of tree [whose fruit has] pits, such as nectarines, peaches, clingstone-peaches, apricots etc., comes best by shield budding on an almond tree by.</ab>

</div>
<div>
<id>p053r_a6</id>
<head><m>Lead</m> and <m>copper</m> casting</head>

<ab><m>Lead</m> and <m>tin</m> come out well in <m>white chalk</m> but the softer it is the better. That from Champagne wins first prize, and it is transported to Lyon. Burnt and calcinated <m>horse bones</m> cast very neatly.</ab>

<ab>
<margin>left-bottom</margin>
I believe that the <m>marrow</m> from the <m>horns</m> of oxen or sheep, that is to say, the <m>spongy bone</m> from inside, casts quite neatly and is better than <m>bone</m>.</ab>

<ab>
<margin>bottom</margin>
Note that any sour <m>metal</m> comes out better than a fat one. Also, lean <m>sand</m> receives it and absorbs it better than dense <m>sand</m>.</ab>

</div>
[a]Cotgrave indicates that "sublimé" is "arsenic," but the DMF offers the alternative definition: "Empl. subst. ALCHIM. "Résidu (toxique et caustique) de la sublimation ; corps toxique et caustique (en partic. bichlorure de mercure)"

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<page>053v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f112.image</image>

<div>
<id>p053v_1</id>

<head><al>Silkworms</al></head>

<ab>They are produced from grain, that is eggs, which are sold by the ounce, which is commonly sold in Languedoc 3 lb. and 5 s. The one from Spain brought by merchants is considered to be the best, because the worms coming from it are not so subject to illnesses and produce more silk. In Spain, one ounce of grain gives worms that commonly make 15 lb. of silk. But from one once produced in France, they do not make but 10 or 12. Three ounces of grain are to produce such a quantity of worms, with which you will be able to furnish a room with three or 4 shelves of wide boards. They begin to shed their skin on their own around Easter. And to do this, one has to put them in a pine box, like the ones in which we put pellet, warmly among feather cushions. And in the beginning, they shed their skin as little black ants, and as soon as there are two or three without skin, they have to be given white mulberry leaves. And then arrange them on the boards. And three times per day, it is necessary to change the leaves for fresh ones. And if during the day there is any storm or rainy weather, cloudy and cool, one needs to keep in the room three or 4 embers and with glowing coal, and to light incense until the room is filled with its smoke. And when the weather is warm and serene, they produce more and better silk. Some worms make it whiter, others more yellowish. And even if it is white, it can be yellowed when it is extracted with hot water. From their birth until the moment they make their cocoons and prisons, worms sleep and rest 4 times, and each time they remain 4 or five days resting without eating, as if they were dying so as to be born again, because each one sheds their skin and begins by uncovering the head, then consequently, on different days, the rest of the body, and they go from white to grey, and from grey to white. And if one of them has some sickness and does not have the strength to shed, one needs to help it and to be careful not to squash it, because if it gives off a yellow liquor, it is no longer worth anything. And they do not even serve much if one handles them. Around Pentecost, they begin to want to climb on the dry heather branches that we prepare and attach</ab>

<ab>
<margin>left-top</margin>
See Marco Girolamo Vida, Bishop of Alba and Cremonesi, wrote a poem on the nature of silkworms.[a]
</ab>

<ab>
<margin>left-middle</margin>
How they are [illegible]
</ab>

<cont/>
</div>
[a]In Latin in the original.
.....

<page>054r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f113.image</image>

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<cont/>
<id>p053v_1</id>

<ab>to some of the upper boards, and one can tell when they want to climb up when, on the leaf, they stretch out and raise their heads and a part of their bodies when one

takes them to heather branches where they stop and begin to spin their prison, which we call cocoon, generally the size of a pigeon egg, although there are some which are much bigger because it sometimes happens that two or three and up to 11 <al>worms</al> put themselves in a cocoon, which is hairy and cottony, around which ball is filoselle or floret[a], and of the cocoon, which is a white, solid, continuous and firm skin, <m>silk</m> is made. The cocoon is so hard that it is cut with difficulty with a fingernail. And yet to leave its prison, the <al>worm</al> eats away at it on one end, and after having stayed inside, living on its own juices for three weeks, it comes out, reduced in size by half. Because when it begins to spin, it is as long as a ring finger and has eight legs, and when it comes out it is less than half as long and only has four legs. On the other hand, it has become a butterfly and has wings; however, it does not fly. There are males and females. As soon as they come out of the cocoon, the male mates with the female, and they are put on a piece of white linen where they lay their eggs, which will not be good and viable if the male was not given to her. When the male has detached himself from a female, one must get rid of it because it would not be good to give it to another female. They finish spinning and laying eggs in three weeks and around Saint John's Day. And then one keeps their eggs and grain until Holy Week, as mentioned. Some [worms] spin among the leaves and make their cocoons there without climbing high.</ab>

<ab><margin>top-left</margin>La <m>soye</m> des</lb>
coquons ou il y a</ab>

</div>

[a]verify

.....

<page>054v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f114.image</image>

<div>
<id>p054v_1</id>

<figure>
<id>fig_p054v_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5N1FIRklFdZIxUzQ</link>
</figure>

<ab>To dry jams that are [not?] liquid[a], have an open tub pierced on each side with a small, square window, and pass sticks through these from one side to the other, and place on these a sieve made of strings in the form of a net. And on this, you will neatly put your jam containers. And having put at the bottom of the tub a hot coal or two, cover the tub with a cloth folded in two.</ab>
</div>

<div>
<id>p054v_2</id>
<head>To lead a horse</head>

<ab><figure>+</figure>One has to give it sweet bread and it will recognize the one who will give him this treat.</ab>

<ab>And in order to keep it good for leading, make it use a good gait from the beginning of the path and continue it.</ab>
</div>

<div>
<id>p054v_3</id>
<head>Dogs</head>

<ab>To keep them from biting, it is good to have <m>pitch</m> balls, formed like those of cobblers, to throw to them when they yap at you. They will bite on the balls so much that their teeth will get stuck together and they will not be able to harm you.</ab>
</div>

<div>
<id>p054v_4</id>
<head>To disguise a horse</head>

<ab>Cut his ears and tail and, with burning <m>hay</m>, singe the hair on his head and elsewhere.</ab>
</div>

<div>
<id>p054v_5</id>
<head>Boots of <m>hay</m></head>

<ab>Lacking boots, you can make a rope out of <m>hay</m> and wrap it around your legs, and rain will never pass through.</ab>
</div>

[a]"Pour faire seicher les confictures qui ne sont liquides" in the original.

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<page>055r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b9059316c/f115.item</image>

<div>
<id>p055r_1</id>

<head>To shoot a gun without noise</head>
<ab>Warm up the barrel and close the lumiere d'embas[a], and take a piece of the freshest <m><al>pork</al>suint[b]</m> that you can find and as long as the phalanx of your little finger and the size of the diameter of the canon, and push it back and forth until it is completely melted. Then, load a cartridge which fits on the top of the barrel, as you know, and it will only make a little whistling.</ab></div>

<div>
<id>p055r_2</id>

<head>Dipping for a bullet that will reach far away and will compromise the healing</head>

<ab>Mix some <m>suint</m> and strong <m>vinegar</m> and dip the bullet in this mixture while it is still hot.</ab></div>

<div>
<id>p055r_3</id>
<head>Against wounds</head>

<ab>Hurt a <al>chicken</al> or a <al>dog</al> to test and put some <m>sap</m> and ground herb which is called <m>sempervina</m> on the wound. It is the one with the leaves like small grains that some call <m>vernicularis</m>. It well-known that it will not die.</ab></div>

<div>
<id>p055r_4</id>
<head>Oneneu elbirro trauf no eheram<lb/>
rus enn elbat no [illegible] ueirtse<[c]>head>

<ab>Get in the month of June and July, many of the biggest <al>snails</al> you can find with their shell, put them in a <m>glas</m> bottle with a sufficient quantity of very hard and very boiled <m>egg yolks</m> with some good <m>vinegar</m>. Cork it and put it under some <m><al>horse</al>muck</m> for fifteen days. The <al>snails</al> will first survive thanks to the <m>egg yolks</m>, then with the heat they will come out of their shells and eat live on it. Finally, it will come a <m>white salve</m>. For your own safety you will not cut the bottle, keep it well closed and put it in strong sun and dawn for[d]</ab>

<cont/>
</div>
[a]Most certainly a part of the gun. Translation uncertain.
[b]Manuscript reads "suin". Uncertain.
[c]"To kill someone with a horrible poison spread on a board over which the person walks or on stirrups"??
[d]Translation uncertain.

.....

<page>p055v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f116.image</image>
<div>
<cont/>
<id>p055r_4</id>

<ab>for fifteen days. Next you will break it [sc. the bottle] from a good distance & with a long stick without looking at it & your eyes covered, you will put it where you want, or rub a little in a running river or on a tree or on grass that is on the bank.</ab></div>

<div>
<id>p055v_1</id>
<head>Another for <m>waters</m></head>

<ab><al>Toad</al> in a pot[a], with quicklime, who will consume all this powder, next and.</ab></div>

<div>
<id>p055v_2</id>

<head><m>Damascus steel</m></head>

<ab>Distil some worms[b][c] on one side & some horseradish leaf[d] on the other & mix these two <m>waters</m>, as much one as the other, and do the tempering in that.</ab></div>

<div>
<id>p055v_3</id>

<head>To know one's course in open sea</head>

<ab>Soak a piece of <m>cloth[e][f]</m> in <m>oil</m>, then attach it to the poop of the boat so that it is dipped in the water, and it will make a trail that will be visible for ten leagues since the <m>water</m> parts wherever the <m>oil</m> has passed.</ab></div>

<div>
<id>p055v_4</id>

<head>For casting</head>

<ab><m>Chalk</m> most recently drawn from the quarry is good for <m>lead</m>. But do not wet nor[g] crush it, make it fine only by grinding or grating it and passing it through a sieve. The same applies to <m>pumice</m>, which one must not crush with <m>water</m> nor corrupt its nature. <m>Lead</m> must be mixed with little <m>tin</m>, just to give it some strength, and two parts of <m>lead</m> for one of <m>tin</m> or more.</ab></div>
[a]Manuscript reads "Tuaparc nad nu top".
[b]Cotgrave for "lumbrique"; text says "lumbri[s]" for which we take lumbrique
[c]Uncertain!
[d]Uncertain.
[e]AK: from Cotgrave: "A certaine weight; (among Goldsmiths, seven graines, among Lapidaries, two carrats) also, the stuffe, or thin woollen cloth, called Wadnall."
[f]Both Cotgrave and La Curne spell the word felin or fellin. Defined by La Curne (after Oudin), as "espèce de toile". None of the other main dictionaries has this meaning.
[g]This apparently means: do not crush it wet, as with the pumice.

.....

<page>056r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b1050001g/f117.image</image>

<div>
<id>p056r_a1</id>
<head>Excellent temper for breastplates</head>

<ab> Have the clearest and the most running <m>river water</m> you can, and heat it up slightly more than warm. Then have a bunch of <m>reseda[a][b],</m> take its seeds and dip them into the said <m>water</m> and bring it to a rolling boil[c] it while stirring it with a <m>stick</m>. Then pour it into a muid[d][e] or any other vessel, then throw into it two pecks[f] of <m>salt</m>. </ab>

<ab>Then take a big cauldron full of the said <m>river water</m> and heat it up slightly more than warm. Take three or four times the quantity of <m>red greasy earth</m> you can hold in two cupped hands, dip it and throw it into the cauldron filled with the said <m>water</m>. Have as much <m>pigeon manure</m> as of the said <m>earth</m>, as much <m>horse manure</m> and as much <m>iron scrapings</m>. And mix each of these separately, throw into the caldron and leave for two or three days. Then throw it into the said muid and stir all strongly altogether. And the older this temper is, the better it will work.</ab>

</div>
<div>
<id>p056r_a2</id>
<head><m>Varnish</m> for tempera[g]</head>

<ab>You can make <m>tempera[h] marble[i]</m> with <m>lacquer</m> or <m>rose of Ghent</m>[j] and <m>chalk</m>. Once dry, glaze it with <m>lacquer</m> mixed with <m>wine</m>, for <m>glue</m> will dull and blacken it. And the whole will firstly appear red, but the <m>varnish</m> you will apply, which will penetrate it, will make dark and light parts appear as they should. The <m>varnish</m> is made like this: mix with clear <m>Venice</m> <m>turpentine</m> some <m>aspic oil</m> until it becomes clear and liquid, and this is done without fire. This is for things done in tempera, and the <m>turpentine varnish</m> that you know is for panel [k]paintings. Pure <m>lavender spike varnish</m> is not good for panel paintings because <m>lavender spike oil</m> [l] is too penetrating and makes colours shine unless it has been made long beforehand.</ab>
</div>

[a]reseda luteola; also called "dyer's rocket" or "dyer's weed"; translation from the French "gaude".
[b]elsewhere: weld
[c]OK in context?
[d]a kind of vessel
[e]Muid: "Mesure de capacité pour les liquides et, plus souvent, les matières sèches (de valeur variable selon les marchandises et les régions)"; DMF. Désigne par métonymie le contenant d'un muid.
[f]Picotin:
1. "a (French) Pecke, or, the fourth part of a Boisseau", Cotgrave;
2. "petit panier servant à mettre la ration d'avoine d'un cheval", et, par analogie, "bonne quantité de quelque chose" (avoir un picotin de qqc); DMF.
[g]?
[h]?
[i]"mabre" in French.
[j]?
[k]Check
[l]"huile daspic" translated both as "aspic oil" and as "lavender spike oil"

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<page>056v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b9059316c/f118.item</image>

<div>
<id>p056v_1</id>

<head>To paint</head>

<ab>The <m>lacquer</m> and <m>rosette from <pl>Ghent</pl></m> and elsewhere lose their color and die in contact with the air.</ab>

<ab>Any <m>marble</m> on which a knife can [illegible] is not worth to grind fine colors.</ab>

<ab>The <m>chalk</m> is not thick with <m>oil</m>, the <m>ceruse</m> is more appropriate but even better is the <m>white lead</m>. The <m>ceruse</m> is the whitest and is to grind firstly with some <m>water</m>, the white lead is grayish. But turns perfectly white with some <m>oil</m>.</ab>

<ab>When you grind your colors, start by cleaning your workshop because if you raise some dust while walking that will damage your colors which will never be beautiful if they are not clean.</ab>

<ab>The <m>Flor[a]ey[b]</m> must not be mingled with some <m>azur enamel</m> or anything else for it turns it into green.</ab>

<ab><m>Ivory</m> is excellent to make a painter's' palette Knots of the fir tree The pear-tree and if it is walnut tree make sure to cut with the grain of the wood.</ab> [c]

<ab>You always have to stamp on <m>wood</m> to <m>oil paint</m> in order to fill holes and unevenness. And stamp[d] with some [illegible[e]] and <m>ceruse</m> mixed with oil, then soften with a feather which evens more than a brush. Or when the stamp is dry, rub it strongly off with a knife.</ab>

<ab>To use some <m>azur enamel</m> with some <m>oil</m>, you have to choose the finest one and to make it thinner, do not grind for it turns white but wash it and because the bigger goes at the bottom, choose the one at the top of the <m>water</m> or pour the cloudy <m>water</m> out and pile the <m>azur</m>.</ab>

<ab>
<margin>left-top</margin>
Someone who operates with <m>oil</m> will not work properly with <m>distemper</m> if he has not been trained well during his youth. The manner of working is indeed very different because the brush point always has to be moisten for working with <m>distemper</m>. While when one cleans the brush in some <m>oil</m> to soften it, one should dry it well. Otherwise the work would drip and soon be damaged.</ab>

<ab>
<margin>left-middle</margin>
The Italians hatch with a flatten rough brush which makes some serration.</ab>

<ab>
<margin>left-middle</margin>
And do not apply their shades at once like the Flemish but apply them by hatching from the lighter towards the light then a darker and then an even darker to make more contrast.</ab>

<ab>
<margin>left-bottom</margin>
With <m>distemper</m> do not mingle your various colors together for they will die but use each separately and in order to stop them from drying and to have time to soften, moisten the back of the canvas.</ab>

<cont/>
</div>

[a]"The blew scumme of woad boyling on the dyers lead, which fleeted off and dried unto powder, serves painters." RC.
[b]See Kirby, Jo, Susie Nash and Joanna Cannon (eds.), Trade in Artists' Materials: Markets and Commerce in Europe to 1700 (London, 2010, p. 449.
[c]I) Punctuation seems a bit tricky. See also tcn!

II) Manuscript reads "noue", which might refer to a knot in the wood. Uncertain of how to translate.
[d]In French "imprimer". Is "stamp" in this context the appropriate translation? "Stamp" here is understood as "to lay" for the early modern French world like "empreindre"
[e]In French, we inserted "fond" but it is not very legible. Verify.

.....

<page>057r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f119.image</image>

<div>
<cont/>
<id>p056v_1</id>

<ab>
<m>Lake</m> takes a long time to dry in <m>oil</m> and for that reason you have to crush <m>glass</m> with it, but you have to choose <m>crystallin</m> because it is neater. And since it would be too difficult to be crushed by itself, you have to heat it red hot, then, when it is entirely red, throw it into cold <m>water</m> and it will crumble and pulverize itself to then be crushed easily. After being crushed with a lot of <m>water</m>, it looks like crushed <m>lead white</m>, but for this reason it has no body. I think it could be good for casting.</ab>

<ab>
<m>Lead white</m> is made with <m>lead</m> sheets beaten gently and put under some manure.</ab>

<ab>
White <m>turpentine</m> varnish or <m>aspic oil</m> and <m>turpentine</m> is colored with powdered <m>turmeric</m> boiled together with it. It gives a <m>gold</m> color on <m>silver</m> which is even more beautiful when burnished. It dries within a quarter of an hour. <m>Aloes</m> would made an even brighter color but it takes time to dry whereas the other one dries within a quarter of an hour in winter as well as in summer.</ab>

<ab>
Good <m>lake</m> moistened with <m>saliva</m> quickly darkens. That from <pl>Florence</pl> is mixed with too much <m>gum</m>.</ab>

<ab>
If you apply one layer of <m>printers' ink</m> on <m>velvet</m> and apply <m>gold</m> leaves and then stretch it, it looks grainy as if it was powdered with <m>gold</m> sprinkled on it.</ab>

<ab>
<m>Vermillion</m> crushed by itself is wan and pale but is more beautiful when it is crushed with [a] <m>lake</m>.</ab>

<ab>In order to remove the <m>grease</m> off <m>marbles</m>, one crushes <m>ordinary ashes</m> on it, which afterward are good to make the first primer coat [b] of a painting to be done in <m>oil</m> in order to seal the cracks and veins of the <m>wood</m>. They are thicker than <m>chalk</m> and are quite greasy. They are mingled with the aforesaid <m>chalk</m> or with colors collected from the vessel that is used to clean brushes, and they are desiccative and spare the colors. Once this primer coat has been made on the <m>wood</m>, it is scrubbed with a knife until even. After, one makes a second coat with <m>ceruse</m> or with the poorest colors mixed together. On an <m>oil</m> canvas painting one only applies one coat and the mixed <m>ashes</m> can be used. Also, after crushing a color, one crushes the <m>interior of a large loaf of bread</m> in order to remove the <m>grease</m> off the <m>marble</m>.</ab>

</div>

[a]the french is "after" but it does not make sense
[b]uncertain about how to translate "impression" here

.....

<page>057v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f120.item</image>

<div>
<id>p057v_1</id>

<head><pro>Tanner[a]</pro></head>

<ab><pro>Tanner[b]s</pro>use small <m>skins</m> of <al>lambs</al> and young <al>goats</al> nearly as delicate as <m>parchments</m> and they keep them in some humid places and keep wetting them. When they want to print some stories on it they lay them in their molding hollows of their figure[c] and then on top they spread some greasy fine <m>earth</m> used for trimming <m>wool</m> fabrics, well pressed and beaten and slightly humid and soft and then put on the <m>earth</m> a small one-piece board and put the whole under a press and leave that dry there. After the <m>earth</m> the <m>leather</m> only remains very neatly printed [missing word] of the size of a pea. And then apply on the <m>leather</m> two or three <m>black copperas</m> and <m>iron scale</m> layers one after the other like the <pro>tanners[d]</pro> do. And since this tincture is astringent and shrinks the leather will be stronger more more easily printable. Once dried stick on the reverse some <m>fabric</m> with strong <m>glue</m>, and by doing so one can quickly imitate big statues and very fine medals and paint them and [illegible] and are light and can be carried and last for a long time. What can be printed on a relief can be made differently and can be reheated with a hot <m>iron</m> tool.</ab></div>

<div>
<id>p057v_2</id>
<head><pro>Painter</pro></head>

<ab>Good <m>crayons</m> are not made with <m>glue</m> but with <m>women's milk</m>.</ab>

<ab>Images made of <m>card</m> have to be dipped in some very fine and clarified melted <m>rosin</m> once they have dried. This strengthens them or they will turn flaccid in wet weather.</ab>

<ab><pl>Venice</pl> masks are made with a molding hollow and the face covered with <m>copper</m>.</ab>

<ab>The <pl>Flemish</pl> do not use other white colors for <m>oil</m> painted skin colors after some <m>white lead</m> for the <m>ceruse</m> turns them yellow.

</ab>

<ab>The 4 or 5 years old and clear <m>nut oil</m> is the best anti-dust color. The one slowly squeezed from the press similar to <m>almond oil</m> is white even if the <m>nuts' skin</m> is peeled off.</ab>

<cont/>

</div>

[a]Appropriate translation of "gainier"? Modern French would alternatively refer to "mégissier" and "tanneur" to describe this whole process. What should we do?

[b]Appropriate translation of "gainier"? Modern French would alternatively refer to "mégissier" and "tanneur" to describe this whole process. What should we do?

[c]Would "molding hollow" be enough or is "figure" necessary to understand the meaning of the sentence? See tcn.

[d]See above.

<page>058r</page>

<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f121.image</image>

<div>

<cont/>

<id>p057v_2</id>

<ab>You must make at least three layers of flesh color in order to paint faces in <m>oil</m>. And to begin, one places the black and shadow where it is appropriate. After, the highlighting with <m>lead white</m> must not be placed on the black. Flesh colors and [missing word] containing <m>ceruse</m> will yellow in five or six months, but <m>lead white</m> does not change.</ab>

<ab><m><pl>Florence</pl> lake</m> is better than ^{<m>lake</m>} from <pl>Flanders</pl> because in <pl>Florence</pl> they make the best dyes. To make a beautiful flesh color, the reddest and brightest <m>lake</m> is the best, because those that are purple and violet as a result of the addition of too much <m>alum</m> look like the flesh of someone that is very cold. That is why women who want to color their cheeks crush <m><pl>Florence</pl> lake</m> very finely, then fill a little <m>cotton</m> with it, which they then wrap in a little fine <m>cambric</m>. And thus they rub the <m>lake</m> on their <bp>cheeks</bp> and then, with another clean <m>cotton</m>, they soften it.</ab>

<ab><m>Crystallin</m> having been crushed in <m>water</m> appears to have body, but in <m>oil</m> it does not have any. It is crushed with <m>lake</m> and with <m>bitumen</m>, which would not dry for a very long time otherwise.</ab>

<ab><m>Aspic oil</m> is commonly put with <m>lead white</m>, not entirely pure but mixed with a little <m>walnut oil</m>. The said <m>aspic oil</m> will not be good for <m>lake</m> and colors that do not have body because it will make them crack, but with those that have body and are somewhat greasy, it is quite appropriate.</ab>

<ab><m>Verdigris</m> and <m>orpiment</m> must be first crushed with <m>urine</m> rather than thinned with <m>oil</m>. Thus they are beautiful and do not fade.</ab>

</div>

<page>058v</page>

<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f122.item.</image>

<div>

<id>p058v_1</id>

<head>Blacks</head>

<ab>Coal black, ordinary black lead black, burnt ivory black, peaches stones black, black smoke, black light, burnt oxes' feet bones black</ab></div>

<div>

<id>p058v_2</id>

<head>Shadows</head>

<ab>When you are making a painting with several persons shadows as well as skins have to be different</ab></div>

<div>

<id>p058v_3</id>

<head>Frescoes </head>

<ab><m>Azure</m> is not good for that but [illegible] and you have to work with long brushes. Frescoes cannot be made on <m>wood</m></ab></div>

<div>

<id>p058v_4</id>

<head><m>Azure</m></head>

<ab><m>Turpentine oil</m> makes it very beautiful, try some <m><la>palma christi</la> seed oil</m>. <m>Nut oil</m> in <pl>Flanders</pl> costs at least a hundred sols a pint. The <m>azur</m> requires some greasy <m>oil</m> for it is not thick</ab>

<ab>

<margin>left-middle</margin>

<m>Enamel azur</m> hates more than any other to be ground, even with some <m>water</m> for it dies and loses all its color. Because however it cannot be used if it is

big, grind it not with some water but with some oil and grind it thickly and so it will not die</ab>

<ab>
<margin>left-middle</margin>
Always choose the thinnest one</ab>
</div>

<div>
<id>p058v_5</id>
<head>Brushes </head>

<ab>When the color has dried inside them and you want to clean them soak them in some <m>spike lavender oil</m> and they will turn immediately soft again as before, then you will finish to clean them in some <m>nut oil</m>. <m>Nut oil</m> is not as appropriate to soften them as the <m>spike lavender</m> one which is clear like water and penetrates and is not as thick as the <m>nut oil</m>. Brush handles are made by the quirky ones of <al>porcupine</al> hairs, by some others with fine branches of <pl>Turkish</pl> <m>wood</m> with which they also make small sticks to rest their hand when they are painting</ab>

<ab>
<margin>left-bottom</margin>
To work properly on small scale you need some very fine brushes with a strong point. And because the <m>hair</m> taken from a <al>squirrel</al>'s tale is <fr>mollesin</fr>, the most quirky ones use the of oldest <m><al>rats</al>' hair</m> and even of <al>dormouses</al> if they can find some and put two or three in the middle of a brush. These brushes draw a straight line like a quill and all the other <m>hairs</m> stick to them. The <m>hair of a <al>stone marten</al></m> or of a <al>weasel</al> and small animals used to make some <m>musk</m> are even better for just one <m>hair</m> is necessary in a brush</ab>
<cont/>
</div>

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<page>059r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f123.item</image>

<div>
<cont/>
<id>p058v_5</id>
<ab><m>Lake</m> and <m>white lead</m> and <m>ceruse</m> are easy to work with some <m>oil</m>, but every kind of <m>azure</m> is difficult. And to make a beautiful <m>azure</m>, one has to lay it not with big strokes of the brush but with fine strokes of the end. Not only for <m>azurite ash</m> and <m>azur</m> from <pl>Acre</pl> but mainly for <m>enamel azur</m> which has to be very fine, otherwise you will not be able to work with it unless with hard work and you have to mingle it with some <m>turpentine</m> to thicken it and and to mingle it with some <m>white lead</m>. Any <m>azure</m> must not neither be ground nor washed for it loses its color and become pale. But put some of it powdered on your palette and you moisten it bit by bit with some <m>nut oil</m> or <m>turpentine</m>, dip a knife point in some <m>oil</m> and then mix it bit by bit on the said palette</ab></div>

<div>
<id>p059r_1</id>
<head>Shadows</head>

<ab>First shadows which are closer to the daylight have to be light and very faint then the last ones very dark to enhance strongly the relief. The <pl>Italians</pl> usually make three shadows, the first one very faint, the second one darker and the third one very dark, then put these three shadows together by hatching them from the darkest to the lightest</ab>

<ab>
<margin>left-middle</margin>
The varnish is more beautiful on a painting when the color has been completely soaked up</ab>
</div>

<div>
<id>p059r_2</id>
<head><m>Azure</m></head>

<ab><m>Azure</m> is more beautiful when on the painting it has soaked up some <m>nut oil</m> with which it has been firstly thinned without any <m>oil of spike lavender</m>. If you want to know whether it is dry, breathe on it and it will not shine and will seem to be very soaked up, otherwise it will shine</ab>

<ab>Soaked with <m>oil</m>, <m>azure of email</m> leaves it and comes back to its first natural state if you dip it in some <m>water</m></ab>

<ab>Colors for small scale works have to be very strongly ground and to be worked with a brush point if you want your work to be very fine</ab>

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<page>059v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f124.item</image>

<div>

<id>p059v_1</id>

<head>Painting weapons[a]</head>

<ab><m>Black charcoal</m> mixed with <m>white lead</m> is very appropriate for painting weapons, also mixing in a little <m>azure</m> if you wish. <m>Charcoal</m> by itself is somewhat bluish.</ab>
</div>

<div>
<id>p059v_2</id>
<head>Painting crepe</head>

<ab>First you need to make a greyish background using <m>charcoal black</m> mixed with <m>white lead</m> and a little <m>azure</m>. Then, after drying, highlight with strokes and lines in <m>white lead</m>.</ab>
</div>

<div>
<id>p059v_3</id>
<head>Folds in clothing</head>

<ab>You must take care not to represent any ^{folds} falsely, but rather imitate natural effects. Thick cloth hardly makes any folds, <m>taffetas</m> and <m>silk</m> cloth make more, and <m>crepe</m> more still. Observe which folds go lengthwise and which across.</ab>
</div>

<div>
<id>p059v_4</id>
<head>Siccatives</head>

<ab><m>White lead</m> and <m>massicot</m> are most siccative, nevertheless they need a good two days. If you want to verify whether an <m>oil</m> is siccative, use it to distemper <m>white lead</m>, and if it soon produces a crust it means that it will dry.</ab>
</div>

<div>
<id>p059v_5</id>
<head>Double layers</head>

<ab><m>Azures</m>, flesh tones and reds are applied twice. Other ^{colors} are not.</ab>
</div>

<div>
<id>p059v_6</id>
<head>Mending cracks in a panel</head>

<ab>If a painted panel breaks you can glue it well on the reverse, but to fill in the cracks properly in order to paint and repair it, you must not use any <m>glue</m>, which weakens in damp weather and would swell when touched with the <m>oil</m> of the painting. But have some <m>white wax</m>, which is harder than the other sort. The <m>oil</m> mixed into the <m>wax</m> prevents it from melting in the sun. The <m>wax</m> must not be hard, but soft as if recently cooled. Apply it on cracks and crevices with the tip of a knife, then scrape.</ab>
<cont/>
</div>
[a]arms? armory?

.....
<page>060r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f125.item</image>

<div>
<cont/>
<id>p059v_6</id>
<ab>and to prevent it from breaking, mix in a little <m>oil</m> and melt them together and fill the cracks and make them even and paint on them. And the <m>oil</m> mixed with <m>wax</m> will help colors to hold since they are also distempered with <m>oil</m>.</ab></div>

<div>
<id>p060r_1</id>

<head>First whitening of the painting</head>

<ab>Apply two or three layers of <m>chalk</m> distempered with <m>glue</m> on the painting, not with a paintbrush but with a ^{large} brush as if you intended to rub it down, and leave to dry. And repeat this two or three times, then make the last layer quite even using a knife, <add>then apply a layer of <m>glue</m>

</add> upon which you will later be able to put your underlayer and then paint. But make sure that your first white layer is not too thick for it would break easily. Flemish painters have such paintings [a]made by the dozen.</ab></div>

<div>
<id>p060r_2</id>
<head>Working neatly</head>

<ab>When you are at work make sure you do raise no dust, and when you leave your work cover it so that the said dust will not fall upon it. Also apply your colors as thinly as possible, for if they are thick you will be unable to soften them.</ab></div>

<div>
<id>p060r_3</id>
<head>Oil</head>

<ab><m>Nut oil</m> extracted as if from [b]<m>peeled almonds</m> is very white, ^{like} <m>Palma christi ^{oil}</m>.[c] And when the <m>oil</m> is a little thick the colors are softer, for if the <m>oil</m> is too clear the colors flow and do not bind, even those that are not that thick. Greasy <m>oil</m> that is not easily soaked up is appropriate for varnish. The <m>oil</m> is sufficiently desiccative when it dries as quickly as common varnish. <m>Oils</m> do not dry as quickly in cold countries as in hot countries. <m>Oil</m> left in the sun becomes very clarified but becomes greasy if you add some <m>ceruse</m> or filed <m>lead</m> or <m>white lead</m> or <m>glass</m> powdered and calcined. To avoid this, put the flask in clear <m>water</m>.</ab>

<ab>Once <m>oil colors</m> are dry, they sometimes do not easily receive second colors. To avoid this, breathe on them and the color will hold.</ab>

<ab>
<margin>left-middle</margin>
Do not apply color on color if they are not similar, such as white on black, but rather flesh over flesh, and so on. And leave empty spaces for shadows or different colors. By doing so, laying each sort of color directly on the underlayer, they will not fade and your work will be neat.</ab></div>
[a]Meaning, I think, they buy their canvases/panels (?) already prepared like that.
[b]French unclear here
[c]It seems that a word is missing here. Author seemingly making a comparison between the appearance and quality of nut oil (from almonds) and palma christi oil. Should we mark ^{like is}?</ab>

.....
<page>060v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b9059316c/f126.item</image>

<div>
<id>p060v_1</id>

<head>Varnish that dries within an hour</head>

<ab>Take white <m>turpentine oil</m> and <m>turpentine</m> and powdered and finely sieved <m>mastic</m> and boil them together whilst continuously stirring with a stick until dry. And add two liards of <m>eau de vie</m> and if you take the <m>mastic</m> larme off[a], it will be whiter and clearer. There is no need to put <m>turpentine</m> but only white <m>turpentine oil</m> and <m>mastic</m>, pulverized as much as one wishes[b], until it is sufficiently thick.</ab></div>

<div>
<id>p060v_2</id>
<head>How to clean paintings[c][d][e]</head>

<ab>Some clean them with soapy <m>water</m>, others with <m>urine</m>, others with <m>white wine</m>, for dust damages the colors.</ab></div>

<div>
<id>p060v_3</id>
<head><m>Wood</m> color</head>

<ab>
Apply one <m>bister[f]</m> layer and then a layer [g]of varnish.</ab></div>

<div>
<id>p060v_5</id[h]>
<margin>left-middle</margin>

<head><m>Aspic oil</m></head>

<ab>Do not put any into colors because it is so strong and penetrating that it makes the colors flake and afterwards they come off. And that is why painters use it to clean their brushes when they are dry, because it softens and cleans them immediately by penetrating the dry color which makes them solid. Sometimes painters who are jealous of the work started by another, secretly, in the evening, pour a few drops of <m>aspic oil</m> on the top edge of the painting so that, running down, it produces a stain that penetrates as far as the <m>wood[i]</m> and makes the colors come off. St that, in order to make their work even and uniform, they need to redo it completely and so they lose their efforts.</ab></div>

<div>
<id>p060v_4</id>
<head>Flemish work</head>

<ab>They do all their <m>oil</m> paintings with the tip of the brush, like good illuminators, and grind their colors very finely, avoid any dust and often clean off the fragments of hair released by their brushes, for if these should remain on the painting it would be difficult to work neatly, which they do with great care. Thus their work appears very soft, especially small paintings, which require more diligence since one sees them closer. They usually complete the forehead, then the eyes, then the nose, finally the mouth and the rest. But they do not proceed like some others who apply two or three different flesh tones, one yellowish and one darker, because the colors always mix and end by fading. They just make their underlayer carefully.</ab>

<ab>
<margin>left-top</margin>
<figure>+</figure>
which you will know when it is placed on a knife in the wind and does not flow. This one is excellent for panels and is dry within an hour and does not stick like
<m>turpentine</m>.</ab>
<cont/>
</div>
[a]"Tqking off" uncertain, more like "drawing", pulling out
[b]or: pulverized mastic, as much as one wishes (?)
[c]Here and elsewhere, I believe tableaux means (prepared) panels or canvases, not the actual paintings
[d]I'm not sure because it says that dust damages the colors, which would only be true for already painted paintings
[e]We need to check other instances. One thing I seem to remember is that tableau should refer to panel paintings rather than paintings on canvas. True in this case if he were talking of cleaning the panel before you use it for a painting, he would probably say so more explicitly
[f]Middle French "bistre" refers to a pigment, modern English "bister" seems to refer to a colour only. How to translate?
[g]Not clear whether couche and main have different meanings. (Main might be a thinner layer?)
[h]The order of recipes is not kept on purpose because a <id>p060v_5</id> is in the margins and <id>p060-4</id> continues on the next page and its <div> can't be closed.
[i]there: tableau is a PANEL

.....
<page>061r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f127.image</image>

<div>
<cont/>
<id>p060v_4</id>
<ab>and unite it well and being quite dry, they draw their portrait and apply their natural skin color, leaving the areas of shadow like the forehead and the side [a] of the cheek and the empty space around the eyes and drying until they have filled the rest. Afterwards they apply separately the shadows which they do not make as dark as they could naturally be. In sum, they don't apply shadow on skin color nor skin color on shadow. So white is on white and black is on black and each ^{color} is separate, and in this way their work is neat and colors don't fade away.</ab>
</div>

<div>
<id>p061r_1</id>
<head>Skin colors</head>

<ab>It is necessary that you make it of two kinds, one more red to make the main layer, the other more pale for the highlights, as around the eyes. And then on this last skin color, you will touch lightly the main lights with a little <m>white lead</m>. But avoid applying too much of it because this will look like a face of death. The beautiful <m>pl>Florence</pl> lacquer</m> makes a beautiful vivid skin color that resembles the complexion of <m>alexandrine pink</m> and <m>incarnadine</m>.</ab>

<ab>
<margin>left-top</margin>
Certain colors need to be ground like <m>minium</m> or <m>massicot</m>. The ashes don't need to be ground at all.</ab>
</div>

<div>
<id>p061r_2</id>
<head>Making ^{things} round</head>

<ab>If you want to model well, soften round things by rounding them off with the brush point and the rest with a flat part if it is flat, and thus for the others according to their nature, and lightly with the dry and flattened brush point and with patience.</ab>
</div>

<div>
<id>p061r_3</id>
<head><m>White lead</m></head>

<ab>Without this, you can't work on a small scale for which it is necessary to soften with great care [b]. But you can't do it with <m>ceruse</m> because it does not have enough body.</ab>
</div>

<div>

<id>p061r_4</id>
<head>Lights</head>

<ab>Your brush shows it to you by making some shadow which must always follow the back of your hand, not in a straight line like this, <figure/> because the light would be too crude and too harsh, but obliquely and as an incline like this. <figure/><figure/> It is necessary that the painting is not facing the light, but be half turned back against it, and above all look for a soft light, for it makes a ^{soft} shadow and soft work, like a harsh light ^{makes} a harsh work.</ab>

[a]"couste du front & de la joue"
[b]curiosité (dictionnaire moyen français): Attention (intriguée), intérêt, Soin mis à qqc., empressement, assiduité

.....

<page>0061v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f128.image</image>
<div>
<id>p0061v_1</id>

<head><m>Water</m> to make light for <pro>painters</pro></head>

<ab>Distill some <m>vine water</m> and put it into a big bottle. Put your <m>candle</m> behind that, and it won't disturb your vision.</ab>
</div>

<div>
<id>p0061v_2</id>

<head>Chassis used by Germans</head>

<ab>Germans who work in small scale make 'chassis" not of <m>glass</m> but of <m>canvas</m> <m>oiled</m> with <m>clear turpentine varnish</m>, that is half of <m>turpentine oil</m> and almost half of <m>turpentine</m>, because such light, being less bright than <m>glass</m>, makes the lines appear thicker. And when they want to make something more delicate, such as veins and similar things, they use brushes made of two or three <al>rat</al> hairs.</ab>
</div>

<div>
<id>p0061v_3</id>

<head><m>Azure enamel</m> with <m>oil</m></head>

<ab>One must choose the finest possible, for if it is coarse one cannot work in <m>oil</m>. And if you cannot find any that is fine enough, you may as well grind it, not with <m>water</m> but with <m>oil</m>, and grind it thick. Then lay it on your palette and mix in some <m>turpentine</m> but not much, to make it bind, and make it as thick as <m>butter</m> or <m>mortar</m>, and then, with a large enough brush, work it while always moving the brush freely. Then, to soften it, make jagged hatching ^{movements} with the tip of the brush. The highlights will be made using the same diluted with <m>ceruse</m>, which makes it bind, making it easier to work. I have seen it used thus. It must be very thick, so that you almost need an effort to spread it with your brush. And it will be all the better if you lay down your painting. All these difficulties do not occur when it is very fine and fluid without being crushed, and it doesn't run.</ab>

<ab>
<margin>left-bottom</margin>
<m>Azure</m> needs to be applied neatly, which is why it always fades when it is applied over old, already tarnished, <m>azure</m> to mend an old painting. In such cases, it is better to scrape off the old layer and prime again and then apply the <m>azure</m>. It is the same for almost all other colors. Also azure mixed with <m>oil</m> always remains shiny, which is not good for <m>azure</m> because that makes it fade.</ab>
</div>

<div>
<id>p061v_4</id>

<head>Grinding colors</head>

<ab>A slab of <m>glass</m> one inch thick is more appropriate than anything else for grinding colours neatly, especially for <m>lacquer</m> and for <m>whites</m>.</ab>
</div>

[a]Mentioned in C. Estienne, Agriculture et maison rustique
[b]not frames for paintings: rather an apparatus through which you look at your subject?
[c]"luy mesme" is a mistake, refers to "les rehaussements" (highlights) -- would be "eux-mêmes"
[d]no, it refers to the color
[e]"pouce" = inch, but perhaps more research needed to determine exact measurement

.....

<page>062r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f129.image</image>

<div>

<id>p062r_1</id>
<head>Perspectives</head>

<ab>Painting perspectives in <m>oil</m> takes a long time because of the many lines that need to be drawn with a ruler, and for that reason one generally paints them in distemper. One chooses the ^{vanishing} points at will. The main thing is to know ^{where} the point is. One commonly sets two or three, sometimes ^{up to} five.</ab>
</div>

<div>
<id>p062r_2</id>
<head><m>Ocher</m></head>

<ab>This is used for faces, hair, skulls and rocks.</ab>
</div>

<div>
<id>p062r_3</id>
<head>Distant people and animals</head>

<ab>First one roughs them out in gray or in purple, which is made of <m>azure</m> ash and <m>lake</m>. Once dry, one highlights and finishes with flesh tones and other colors and white. And it will look better and is quicker to do than in black and white. Armies are painted the same way.</ab>
</div>

<div>
<id>p062r_4</id>
<head>Drawing</head>

<ab>After you have primed your panel and scraped it with a knife to make it quite even, you will start drawing with the longest possible piece of <m>charcoal</m>, because with a short one you would not see your line so well and you would make it rougher. Let the tip of the <m>charcoal</m> be sharp, and to prevent it from quickly becoming dulled and blunt, drag the tip flat, thus you will constantly sharpen it. Hold your <m>charcoal</m> as far as possible, and accustom yourself to drawing with a light touch. Because if you become accustomed to drawing delicately with <m>charcoal</m>, you shall do the same with colors. And whoever is rough[a] with <m>charcoal</m> is never exquisite with colors. And by a line of <m>charcoal</m>, masters assess their apprentices. First make the contour of your drawing, that is the outline, lightly and without too diligent work, rather with boldness. By so doing, you will learn to be an artist, and if you need to to erase anything, you will not waste as much time as if you had elaborated it. Afterwards work on all the details, not standing too close to your panel, but sometimes stepping back in order to better assess the proportions. Once you are satisfied with the first drawing, retrace all the lines with your paintbrush in pink or another <m>gummed</m> or <m>distempered color</m>. Thus you will work more surely [b]with colors, and with less trouble.</ab>
</div>

[a]meaning: unrefined
[b]or confidently?
.....

<page>062v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b9059316c/f130.item</image>

<div>
<id>p062v_1</id>
<head>Portraits</head>

<ab>To become an artist, it is necessary to draw by eye, without compass or ruler. Masters do not allow apprentices to use them.</ab>

<ab>When the priming has been done a long while ago, it becomes greasy. It must be scrubbed with <m>ashes</m> and <m>water</m>.</ab>
</div>

<div>
<id>p062v_2</id>

<head>Perspective</head>

<ab>In order to tell where the ^{vanishing} point is one must lay a ruler over the lines, and the point will be where the lines meet[a]. Some make a hole at the end of their rulers in order to fix the point by that hole while moving the ruler about. Others lay a ruler across the panel,, then on this ruler they place the tip of another ruler which, being attached to the former by a screw, will move about and reach as far as necessary without losing the point. The ruler and compass without discernment of the eye will cause errors. Perspective is very difficult.</ab>

<figure>
<id>fig_p062v_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5ZkRNOG9hZmpMT2M</link>
</figure>

<ab>
<margin>left-middle</margin>
To paint a perspective in oil, you should not use a ruler, for you would smudge everything. Instead, in applying your colors you must keep and follow the lines of your first drawing.</ab>
</div>

<div>
<id>p062v_3</id>

<ab>Also, to make plumb lines, that is lines from the top to the bottom of the panel, you need to have a thread with a piece of <m>lead</m> attached to one end and a little hook to the other end for hanging the said thread from the top of the painting.</ab>
</div>

<figure>
<id>fig_p062v_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5X0hKa0FueWdjMFk></link>
<margin>left-bottom</margin>
</figure>

<div>
<id>p062v_4</id>

<head>Painting</head>

<ab>In order to draw and to apply the colors, especially in <m>oil</m>, you must keep your panel as straight as possible on the easel, since if it is tilted there is a danger that dust or impurities might stick on it. Keep the place where you paint clean and free of dust.</ab>
</div>
[a]based on interpretation: but *demenée* suggests "unquiet motion" etc. (Cotgrave)

.....
<page>063r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f131.image></image>

<div>
<id>p063r_1</id>
<head><m>Amaranth</m> color</head>

<ab>Although it is said to be immortal, <m>white wine</m> removes its coloration and makes it light red and the dried <m>flower</m> ends up white. Note that <m>wine[a]</m> does so, and I have tried it. <m>Spirits</m> have the same effect.</ab>
</div>

<div>
<id>p063r_2</id>
<head>Drying of <m>oil</m></head>

<ab>If the <m>oil</m> does not dry, put in <m>massicot</m> and <m>minium</m>. There is no better <m>oil</m> than <m>walnut oil</m>, because it does not dry out too quickly nor too slowly.</ab>
</div>

<div>
<id>p063r_3</id>
<head>Casting</head>

<ab>The Germans use <m>lead</m> from <pl>Flanders</pl> because it is very "doux". And to cast better they take <m>lead ore</m> and melt it, then separate the molten <m>metal</m> from impurities and <m>"mine"</m>, and pour it into a separate vessel, then they make their cast with it. This is because the <m>lead</m> must not have been used before, but must be completely new. Some people cast into forms of <m>iron</m> or <m>copper</m>.</ab>

<ab>
<margin>left-middle</margin>
or new <m>lead</m> coming straight from the mine. They melt this twice in a crucible. The first melting cleans it of impurities. With the second, they make the cast.</ab>
</div>

<div>
<id>p063r_4</id>

<head>Shadows</head>

<ab>German <pro>painters</pro> shade their flesh tones with crushed <m>jet</m>, <m>stil de grain yellow</m> and <m>ochre</m>.</ab>
</div>

<div>
<id>p063r_5</id>

<head><m>Bister</m></head>

<ab>It is only employed in distemper, and is used for the color of <m>wood</m>.</ab>
</div>

<div>
<id>p063r_6</id>

<head><m>Verdigris</m> and another very beautiful, gay green</head>

<ab>One must not crush it with <m>water</m> alone, because that makes it fade. To make it beautiful in distemper, some crush it with <m>vinegar</m>, but that makes it turn pale and become whitish. To make it beautiful, crush it with <m>urine</m> and leave it to dry. Then, whenever you like, crush it with <m>oil</m>. And after you have collected it with the palette, before you finish cleaning the <m>marble</m>, crush it ^{there} with <m>stil de grain yellow</m>. And you will have a very beautiful green.</ab>
</div>

[a]something missing here? red wine maybe?

.....
<page>063v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b9059316c/f132.item</image>
<div>
<id>p063v_1</id>

<head><m>Velvets</m> and <m>blacks</m></head>

<ab>One must make the main layer very thick, and the folds and highlights of the meste[a] lighten a lot with white and on the ends of its light, you apply a white line. For blue and green velvets, you shade [b][c] with coal made from peach pits which is very black. Concerning the <m>lacquer</m>, the <m>carbon black</m> that produces a reddish black on lacquer for velvets. The common charcoal produces a whitish black.</ab>
</div>

<div>
<id>p063v_2</id>

<head>Armour</head>

<ab>Soft wood charcoal appears bluish. It is good[d] for depicting armour.</ab>
</div>

<div>
<id>p063v_3</id>

<head><m>Scudegrun</m></head>

<ab>It is made with the broom flower well boiled in water, putting in it enough alum, then some ceruse.</ab>
</div>

<div>
<id>p063v_4</id>

<head>Red copper</head>

<ab>Some people ^{add} <m>chalk</m>, but the best ^{kind} is composed of <m>white lead</m> and <m>Brazil dye</m>.</ab>
</div>

<div>
<id>p063v_5</id>

<head>Flanders blue-green[c] </head>

<ab>In the month of May, one puts some putrified [f]cow dung under horse dung. Then mix them with iron[g].</ab>
</div>

p063v_6

Snow-covered landscapes

One only uses three colors: white, black and bister.

p063v_7

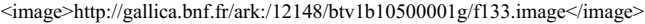
Softening

If you have worked and prepared your palette with some drying color, such as minium [\[h\]](#) and similar colors, complete softening when the work is fresh, because if you wait until tomorrow it will be dry and you will not be able to complete it properly.

[\[a\]](#)can't find what it is exactly in ant dictionnary : a pigment ? we call "arbre de mestre" the mast of a ship
[\[b\]](#)meaning incomprehensible. Check ib this 6/12
[\[c\]](#)At the end of the day draw a white line where there is blue and green velvet. You paint the fold recesses with ground peach stone, which is very black, and which increases the blackness of the lacquer when applied over the folds. Carbon black, which has a reddish tinge, is used for velvets
[\[d\]](#)or proprer, or appropriate
[\[e\]](#)translation of "Pers"de Flandres
[\[f\]](#)problem of orthography : "pourrir" whereas it should be written "pourri"
[\[g\]](#)ferrée
[\[h\]](#)refer to the definition of the Dictionnaire du Moyen Français

.....

page064r



p064r_1

Working neatly

If you can avoid it, never apply two colors on top of one another. But after making your drawing with great care, keep the shadowed areas separate for themselves and the lights and highlights as well, particularly without smoothly laying a color, then add to this highlighting or shading. And in this way, you further your hard work, spare your colours and work cleanly, which as a result, the colors are not muddled nor mixed together, they do not decay and you soften then better, thus finding it not so thick.

p064r_6

marginleft-top

head<m>Yellow ochre</m>

You need a bit in every flesh tone.

p064r_7

marginleft-center

headSoftening

One softens in the same way on <m>oil paper</m> as on <m>wood</m>. But it is easier to soften on <m>canvas</m>, because the softening must be coarse there.

p064r_2

headGreasy <m>oil</m>

Greasy <m>oil</m> does not work well with colors because it makes them thick, and as difficult to work as <m>azur enamel</m>. Thus use the clearest and freshest <m>walnut oil</m> that you can.

p064r_3

head<m>Vermilion</m>

<ab>It must not be kept in <m>water</m> when crushed with <m>oil</m> because it loses its color. It is better to choose <m>whitish vermilion</m> than <m>dark vermilion</m> or or a <m>blackish vermilion</m>. Because <m>vermilion</m> is commonly mixed with a little <m>lake</m>, otherwise it would hardly be different from <m>minium</m>. But the pale one shows more vividly than the dark one. It does not dry by itself, so for that effect one mixes in <m>calcined crystal</m>.</ab>
</div>

<div>
<id>p064_8</id>
<margin>left-bottom</margin>
<ab>
<head>Eyes</head>

An eye must always follow the circle of the compass and not be flat or square.</ab>
</div>

<div>
<id>p064_4</id>
<head><m>Oil colors</m> in <m>water</m></head>

<ab>Commonly, after crushing them one puts a piece of <m>tin leaf</m> on top, and one puts them into <m>water</m> to keep them from drying out. But this is more appropriate for <m>white lead</m>, <m>minium</m> and <m>massicot</m> than for the others, for <m>lake</m> fades and loses its color, as does <m>azure vermillion</m>.</ab>
</div>

<div>
<id>p064_5</id>
<head>Double layer</head>

<ab><m>Verdigris</m> does not fade, and therefore does not need to be applied twice. But <m>lake</m> and others, especially flesh tone, need two layers. Colors hardly change once they are dry.</ab>
</div>

.....

<page>064v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f134.image</image>

<div>
<id>p064v_1</id>
<head>Apprenticeship of the <pro>painter</pro></head>

<ab>He is first made to draw an <m>egg</m>, in which he must draw half a face, then the entire ^{face} with the ears, then the neck, then the parts of the body separately, then joined, then one figure, then two and three, and finally a narrative scene, teaching him how to hold the <m>charcoal</m> by the point, and also the paintbrush. Once he knows how to draw, he is taught how to apply colors.</ab>
</div>

<div>
<id>p064v_2</id>
<head>He is also shown these strokes and lines</head>

<ab>The figure of the <m>egg</m> is the main pattern for faces and for curled-up bodies, as the cross is the model for a full, straight figure[\[a\]](#). Without these strokes you will never do anything well.</ab>
</div>

<figure>
<id>fig_p064v_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5NmZDTmZwc1Jqb2M</link>
</figure>

<div>
<id>p064v_3</id>
<head>Softening</head>

<ab>To soften well, you must not lay the colors thick, but rather lay them twice, except for <m>azure enamel</m>.</ab>

<ab>
<margin>left-middle</margin>
All colors that lack body in <m>oil</m> also lack body in distemper. In distemper they even sink to the bottom. <m>Distempered colors</m> need to be fat, which you can tell when they stick to the palette after being distempered in <m>oil</m>.</ab>

</div>

<div>
<id>p064v_4</id>
<head>Straight lines</head>

<ab>You can use the ruler, but do not lay it flat on the painting, but somewhat lifted and resting on the edges of the panel. Otherwise you would smudge everything, and also you would not see the strokes well.</ab>

</div>

<div>
<id>p064v_5</id>
<head>Distemper</head>

<ab>It is necessary that <m>distempered colors</m> are kept darker while you distemper them, because they become lighter when they dry. But <m>oil colors</m> remain the same colors.</ab>

</div>
[a]plant is used elsewhere meaning position, here it seems to refer specifically to a full human figure, as opposed to curled-up bodies (and single body parts)

.....

<page>065r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f135.image></image>

<div>
<id>p065r_1</id>
<head>Shadows</head>

<ab> Because blacks appear in different hues, some reddish black, others bluish, and others greenish, choose those verging on yellow in order to obtain beautiful shadows in <m>oil</m>, for shadows are yellowish, especially those of men. And for this effect use very strongly crushed <m>jet[a]</m>, which you mix with a bit of <m>yellow ocher</m> and <m>white lead</m>. Or else, after you have crushed your <m>white lead</m> and gathered it with the <figure/[b]>, crush the <m>jet[c]</m> into it. Thus it ^{the black} will be more desiccative, and a yellowish black on its own. When mixed with a bit of white, it will be perfect for men's shadows. Blacks which appear greenish black are appropriate for women's shadows. Take then some black of <figure/>[d], a little <m>sap green</m> and some <m>bistre</m>, and you will have a perfect shading for [flesh tone] in distemper for a woman.</ab>

<ab>
<margin>left-top</margin>
The powder of orberé[e][f][g][h] <m>grain</m> is duller[i] than <m>umber<[j]/m> and when you lack <m>umber</m>, the aforesaid powder may be used, but it lacks body.</ab>

</div>

<div>
<id>p065r_2</id>
<head>Flesh color</head>

<ab>You must know how to mix the colors and the appropriate shadows well, but above all make sure to soften [k] them well. And note and observe that an object that appears flat needs to be softened in straight lines, such as from right to left like

<figure>
<id>fig_p065r_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5eUhrRmdpaWNCZlk></link>
</figure>
, and whatever needs to be represented round must be softened in circles.</ab>

<ab>
<margin>left-middle</margin>
One cannot lay down <m>oil</m> on cut <m>paper</m> or make a stencil [l] as in distemper because the colour will run. Thus, in order to <m>gild</m> with matte <m>gold</m>, one has to pounce and then lay in the colors with the brush.</ab>
</div>

<div>
<id>p065r_3</id>
<head>Mirror</head>

<ab>When you paint flesh, it is useful to have a <m>crystal</m> mirror, which represents a lifelike image, and in which you will see if your shadows are soft enough or too crude. But do not look at it using a candle, because the light of the flame will make the shadows look reddish. The painter also learns for himself with a mirror, because he sees in it what a lifelike image can produce.</ab>

</div>

<div>
<id>p065r_4</id>
<head>Greasy colors</head>

<ab>After the laid-down colors are absorbed, if some part remains shiny and does not seem dry, it means that this part is greasy, and that the second colors you lay down would not adhere either, unless you scrub this part with <m>soap</m> or breathe on it, because humidity will make the colors adhere.</ab>

<ab>
<margin>left-bottom</margin>
All greasy colors, such as <m>ceruse</m>, <m>minium</m>, <m>massicot</m>, <m>ochre</m>, and <m>white lead</m> are good for making <m>gold</m> color.
</ab>

<ab>
<margin>left-bottom</margin>
Any color or thing that thickens when mixed with <m>water</m>, while being crushed [m]has body. But those that lack it, such as crushed <m>glass</m> and <m>lake[n]</m> etc., become clear.</ab>

</div>
[a]uncertain - check
[b]blank
[c]uncertain - check
[d]gap left within the line as if he wanted to add something else. Unsure about the way we encoded it.
[e]Apparently 'orbière' is the fallow side of the field that is not worked, see p. 205 Glossaire étymologique anglo-normand, 1884 - added to the Vocab.
[f]We are not entirely satisfied with the translation though
[g]"Orvière" may also be a city, but I was unable to check if it could have been located near Toulouse
[h]orbere (or rather orberé) clearly refers to a variety of grain — unable to find which — or to a way of processing it
[i]or darker?
[j]Better earthly umber or keep "terre d'ombre"? In tc it says "terre dombre" to not confuse it with the use of "ombre" we should not ommit "terre" here.
[k]blend?
[l]or "pattern" (Cotgrave), but pattern in the current sense seems too generic
[m]or: has body when crushed?
[n]Colgrave says 'lacque' is always a red pigment but looking at the pigment glossary it could also mean 'a brown resin-like material' (Lac)

<page>065v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f136.image</image>

<div>
<id>p065v_1</id>

<head>Mid-bright colors</head>
<ab>When colors are absorbed into the <m>canvas</m>, it means that it has been primed only once, and for this reason the colors become matte rather than shiny. But that is best, as long as you retouch it twice, for in this manner, the colors, having more body, will not fade and are much more beautiful, especially <m>azure</m> and <m>lake</m> and those that lack body. But those who want to dispatch their painting carelessly prime twice. Thus the colors will not be absorbed, therefore they will not last as long. The second layer of these colors will not be absorbed and will remain shiny.</ab>
</div>

<div>
<id>p065v_6</id>
<margin>left-top</margin>
<head>Lifelike</head>
<ab>There are some who paint lifelike images on <m>oiled paper</m>, and if they do not complete their work in one go, they put their <m>paper</m> and unfinished[a]
painting in <m>water</m> so that it will not dry. </ab>
</div>

<div>
<id>p065v_2</id>
<head>Painting large figures</head>

<ab>You must attach your <m>charcoal</m> to the end of a stick in order to make your first drawing from a distance because up close, you would not be able to judge proportions as well as from a certain and sufficient distance. Also, when you paint up close, hold your <m>charcoal</m> as far as you can, practically with your fingertips. For in that way you will acquire an artist's touch and make your hand light. </ab>

<ab>
<margin>left-middle</margin>
When painting on <m>paper</m>, <pro>illuminators</pro> distemper their colors with <m>gum</m> and mix in a little <m>soap</m> to make them run better.</ab>
</div>

<div>
<id>p065v_3</id>
<head><m>Glazing</m></head>

<ab>One usually glazes with colors lacking body, such as <m>lake</m> and <m>verdigris</m>. However, in order to use other sorts, one mixes in plenty of calcined and ground <m>crystal</m>, which also has no body and reduces their thickness.</ab>

<ab>
<margin>left-middle</margin>
<pro>Painters</pro> should all learn how to paint from life, because it is their provision and annuity and means to earn their passage when travelling the land with only their <m>"cocon"[b][c]</m>.

</ab>
</div>

<div>
<id>p_065v_4</id>
<head>White and black</head>

<ab>It is useful that an <pro>apprentice</pro> should work with black and white for two or three years in order to become an <pro>artist</pro>.</ab>
</div>

<div>
<id>p065v_7</id>
<margin>left-bottom</margin>
<head>Priming</head>

<ab>You must be quite careful about this, and do not do it, as some will, using <m>gold</m> color made by cleaning the <m>oil</m> brushes, because <m>verdigris</m> and other corrosive colors included will end by fading the colors that are laid in after. It is useful to do it with <m>ceruse</m>, <m>yellow ocher</m>, and a bit of <m>massicot</m>, and make it scarcely thick so that it will not crack.</ab>
</div>

<div>
<id>p065v_5</id>
<head><m>Lead white</m></head>

<ab>It does not fade and has a lot of body.</ab>
</div>

[a]the French is "commancé" but "started portrait" was very awkward in English; uncertain
[b]We think this is an idiomatic expression meaning that they carry their security with them everywhere they go.
[c]"cocon" is also used on 32r and refers to a mixture of fine sugar, water, and sanguine as a base for laying gold leaf. Maybe it implies a special secret?

.....

<page>066r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/fl37.image</image>

<div>
<id>p066r_1</id>
<head>Double layers</head>

<ab>The <pl>Flemish</pl> & those who paint paintings by the dozen only apply one layer & finish in one go. But the colors soon die. And when they are laid twice, they will not die. One needs to lay them lightly & with minimal thickness, in order to soften them.</ab>
</div>

<div>
<id>p066r_2</id>
<head>Shadows</head>

<ab>For women: <m>asphaltum</m>, <m>umber</m> and a little <m>lake</m>.</ab>
</div>

<div>
<id>p066r_3</id>
<head>Painting in distemper on <m>wood</m></head>

<ab>Because it is quite tedious to paint in distemper on <m>wood</m>, and difficult to paint a face well, some mix their colors with <m>egg white</m> seeped through a sponge, or mixed with the <m>^{egg} yolk</m>, <m>water</m>, and well beaten with the <m>peeling of the fig tree</m>. With this they paint & soften on the <m>wood</m>, like with <m>oil</m>, and this supports the varnish, but this does not last.</ab>
</div>

<div>
<id>p066r_4</id>
<head>Colors in <m>oil</m> that are imbibed</head>

<ab>It is best that <m>oil</m> colors are imbibed, that is to say they do not remain shiny after they have dried for they will not die. But if in some places they are shiny, it means that the fattiness of the <m>oil</m> has remained in that place which will kill the colors. The varnish mends all this & unites them & renders each place as alike to one another.</ab>

<ab>
The <m>gold</m> color needs to be laid thickly, for if it is clear it imbibes and runs.</ab>

<ab>On <m>canvas</m> & in distemper, one <m>gilds</m> with <m><fr>bol</fr></m>, & <m>honey</m>, & a bit of <m>garlic juice</m>.</ab>
</div>

<div>
<id>p066r_5</id>
<head>Gilding painting mounts with <fr><m>or</m> mat</fr></head>

<ab>See the other side of this folio this mark: <figure>*[a]</figure></ab>

<ab>The <pl>Flemish</pl> give one layer of distempered <m>glue</m> on the mount, then they mix <m>lamp black</m> or <m>soot black</m> with the same <m>glue</m> & leave to dry. Next, they pounce some <fr>moresque</fr> designs in the corners & paint them with <m>minium</m>, <m>massicot</m>, <m>ochre de ru</m>, and a bit of <m>yellow ocher</m>, mixed with the <m>fatty oil</m> in which they clean their <m>oil</m> brushes, because the three aforesaid colors are gripping and dry immediatly. And next, they apply the <m>gold</m>, then varnish on top of everything. And it seems by this way to some, that all of the black is painted in <m>oil</m>, but that would not be good because the <m>gold</m> attaches everywhere, and not only on the <fr>moresque</fr>. Go back to the top to <figure>*[b]</figure></ab>

<ab>
<margin>left-top</margin>
<figure>*</figure>
<rub><fr><m>Or</m> mat</fr></rub>
<ab>It is made with <m>massicot</m>, <m>minium</m>, <m>ochre de ru</m>, & <m>yellow ocher</m>, so that the composition resembles <m>gold</m>. Gild the day after <fr>un jo</fr> it is applied, if you in the made the groundlayer the ^{same} evening as the color. Next, having gilded, leave it to dry & rest for one day. Next, rub the gilding with a <tl>feather</tl> & <tl>cotton</tl>, so that no small flakes remains, then varnish with <m><pl>Flanders</pl> varnish</m>, which you make, mixed with a little <m>spirit</m>, to make it desiccative.</ab>

<ab>The varnish will dry in an hour. It will enhance the <m>gold</m> color. And this <fr><m>or</m> mat</fr> withstands rain, even if one rubs it, but it must be well dry, for eight or ten days.</ab>

<m>Gold</m> color that is made of various colors cleaned from <m>oil</m> brushes is not so good & with time tarnishes the <m>gold</m> because of the <m>verdigris</m>. One must not touch with the <bp>finger</bp> the layer made for the <m>gold</m> because that will prevent it from attaching itself there. The <m><fr>or moulu</fr></m> spoils if <m>water</m> touches it, but <fr><m>or</m> mat</fr> withstands well the rain & in <m>water</m>. One must not burnish it, because the <tl>tooth</tl> would remove it. Collect the small flakes & small bits that the <tl>cotton</tl> removed when the <m>gold</m> is dry and that you clean it, for as little as an <ms>ounce</ms> is still <m>gold</m>. In <pl>Flanders</pl>, women make <m>gild</m>. It is more beautiful when it is a little thick but in <pl>France</pl> they beat it too delicately.

<ab>Breathe on the <m>gold</m> color, and if it loses its shine, it is dry enough, but if does not welcome the vapor of the breath, it is not yet dry enough.</ab>

<ab>One should not do <fr><m>or</m> mat</fr> gilding[c] after with having applied <m>gold</m> color[d], but wait one day and one night, but take heed to not and, when it appears to be dry, it will grip the gold. Next, one varnishes.</ab>

</ab>
<cont/>
</div>

[a]refers to text continuing on folio 66v - left-middle margin also indicated with an asterisk. This previous sentence an interlinear addition underneath the heading.This is presumably the very last sentence written on this page.

[b]this asterisk refers to the not on the left top, also indicated with an asterisk, and which follows below in the tc.

[c]dorer

[d]dorer

.....
<page>066v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f138.image</image>

<div>
<cont/>
<id>p066r_5</id>
<ab>
<margin>left-top</margin>
When <m>gold</m> color starts to grip, it signals that in ten to twelve hours it will be dry and ready for gilding.</ab>

<ab>
<margin>left-top</margin>
<figure>*</figure>
To make <fr><m>or</m> mat</fr> beautiful, put in a little varnish or <m>fatty oil</m>, not the kind that is made from cleaning brushes, but the pure kind, that is made fatty in the sun, or mixed with <m>ceruse</m>.</ab>

<ab><fr><m>Or</m> mat</fr> of this kind is scarcely less beautiful than burnished <fr>es</fr> <m>gold</m>, and lasts longer in the rain & the like, & ^{is} made more quickly. Burnished <m>gold</m> cracks with time and comes apart in the rain.</ab>

</div>

<div>
<id>p066v_1</id>
<head>Preventing the swelling of breasts[a] d <add>or to make those that became too large smaller</add></head>

<ab>Take large <m>loafs of bread</m> freshly drawn from the oven and cut them in half &, as hot as possible, lay them on & ilz & do this 3 times a day & continue 4 or 5 days. Next, make a plaster with <m><pl>Venice</pl> turpentine</m> or, even better, <m>common turpentine</m>. Mix in <m>sumac</m>, <m>sloe</m>

from bushes, <m>quince seeds</m>, <pa>pomegranate flowers</pa>, <m>leaves of olive trees</m>, & the like, cooked & mixed with the <m>turpentine</m>. But, I forgot, one needs, after having laid on the hot <m>bread</m>, which is soggy & makes one sweat, to lay on linens soaked with <m>water from a beehive</m>, which is to say <m>honey</m> and <m>wax</m> extracted all together from the beehive.</ab>
</div>

<div>
<id>p066v_2</id>
<head>Attracting <al>pigeons</al></head>

<ab>Fry in a pan with <m>oil</m> some <m>hempseed</m> & give some to the <al>pigeons</al>.</ab>
</div>

<div>
<id>p066v_3</id>
<head>Against the bruises of eyes</head>
<ab>At night, apply very thin sheets <m>lead</m>. Singular remedy.</ab>
</div>

<div>
<id>p066v_4</id>
<head><fr>Pinceaulx</fr></head>

<ab>In order to make them well, cut the <m>hair from the tail of a <al>squirrel</al></m>, as much as can be held in one go in a card rolled like a hollow straw. And placing them thus in the aforesaid folded card, tap & shuffle it so that the <m>hair</m> gathers together & becomes of the same length. Put one or two <m>hairs of a <al>rat</al>'s whiskers</m> in the middle, then grab & squeeze it tightly with the thumb and index finger. Dip the tip of all this <m>hair</m> well in <m>water</m>, then, moving yourself close to</ab>
<cont/>
</div>

[a]Would be worth checking if it can also refer more generally to some outgrowth or swelling of the skin.

.....

<page>067r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f139.image</image>

<div>
<cont/>
<id>p066v_4</id>
<ab>
daylight, pull out the <m>hairs</m> which make the tip too long with the ends of your fingernails, until the tip appears good to you, & so that you notice in it the <m><al>rat</al> hairs</m> that are entirely black and the <m>hairs of the same <al>squirrel</al></m> that are at the whitish from the root to the middle. When it pleases you, bind and tighten it very well with a thread in two places, then cut what is superfluous & fit it in a quill of appropriate size, of a <al>duck</al> or a <al>crow</al> for the small ones. Good brushes are those that, once dipped into <m>water</m>, do not bulge when you trace on your hand.</ab>
</div>

<div>
<id>p067r_1</id>
<head><m>Oil</m></head>

<ab>Every <m>oil</m> that imbibes into <m>paper</m> &, when heated by fire, evaporates & leaves the <m>paper</m> clean, such as <m>turpentine</m> & <m>aspic oil</m>, is good for making varnish.</ab>
</div>

<div>
<id>p067r_2</id>
<head>Sand</head>

<ab>All things crushed and tempered <fr>s[a]oufle</fr>, according to some.</ab>
</div>

<div>
<id>p067r_3</id>
<head>F[b]urnace with bellows</head>

<ab>Before you put in the <m>metal</m>, the back ^{of the furnace} needs to be red like a <m>charcoal</m> and well lit. Next, you [illegible] fill it up & cover the <m>charcoal</m> & adjust the bellows, otherwise the <m>metal</m> on top will melt as a result of the bellows & its bottom will be curdled & cool & will not run. But if you proceed in this manner you can melt everything you like.</ab>
</div>

<div>
<id>p067r_4</id>
<head>Varnish on <m>paper</m></head>

<ab>The <pl>Germans</pl> make boxes p covered with painted <m>paper</m> & varnish it with <m>egg white</m>, mixed with <m>gum</m> and some <m>oil</m>, not <m>aspic</m>, but another odorant which looks like <m>olive oil</m>. Every work done with <m>egg white</m> sustains <m>oil</m>. It is with this that <pro>painters</pro> trick the poor peasants, painting their bands of <m>taffeta</m> with this <m>egg white</m> to be done faster. But the first rain washes away everything.</ab>
</div>
[a]needs translation
[b]four

.....

<page>067v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f140.image</image>

<div>
<id>p067v_1</id>
<head><tl>Files</tl></head>

<ab>If they are not <fr>trempées à paquet</fr>, that is, in the manner described, with <m>soot</m>, <m>salt</m>, & <m>vinegar</m>, they will not be good. The large <tl>square files</tl> used by <pro>locksmiths</pro> to draft their work are only made of <m><fr>fer doulx</fr></m>, But the <tl>soft files</tl> should be made of <m>steel</m>. <pro>Founders</pro> whose work is to repair <m>latten</m> & <m>copper</m> should have their files coarse otherwise they fill up with <m>copper</m> which makes them smooth & soon renders them useless. <m>Latten</m>, which is more brittle, welcomes softer files than <m>copper</m> does.</ab>
</div>

<div>
<id>p067v_2</id>
<head><m>Ox hooves</m> for sand</head>

<ab>Once they have been burned well twice & pulverized, they mold very neatly as sand & will not want to be [a]ecooked, but simply heated with the flame of <m>straw</m>. But if you mold them <fr>en noyau</fr>, give it its first layer, simple & very thin, with a <fr>pinceau</fr>, & leave to dry at ease. Next fortify the following layers with <m>wadding</m> mixed with the aforementioned tempered <m>bone</m> sand.</ab>

<ab>
<margin>left-middle</margin>
It is the neatest sand that can be found for <m>copper</m>.</ab>
</div>

<div>
<id>p067v_3</id>
<head><m>Essential oils</m></head>

<ab><m><add>Walnut</add> oil</m> mixed with an equal quantity of <m>turpentine</m> and distilled with an alembic produces an <m>essence</m> whiter than common <m>water</m>. But this has no body & does not unite colors, which are immediately imbibed, and then color is left unable to grip and fades away. Imbibed thus, they would not let you work with or soften them, anymore than in distemper. But you can rectify this flaw by giving it a little body with <m>turpentine</m>, ^{but} not as thickly as you would use for varnishing. And thus you can work with <m><fr>azur d'esmail</fr></m>, & make perfect <m>lead white</m>. <m>Turpentine varnish</m> made from this <m>oil</m> dries within one hour.</ab>
</div>
[a]reheat

.....

<page>068r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f141.image</image>

<div>
<id>p068r_1</id>
<head>Fanciful tables</head>

<ab>You can make various grooved compartments & on these, paint <al>fishes</al> naturally & with color on simple <fr><m>carton</m></fr> & if you wish, on <m>silvered</m> and burnished <m>paper</m> to represent the scales. And next, cover them with very clear <m>lantern horn</m>. You can apply the very same to other works.</ab>
</div>

<div>
<id>p068r_2</id>
<head>Planting <pa>trees</pa></head>

<ab> <pa>Trees</pa> need to be planted in dry weather, & a lot of earth gathered all around their feet, like a mound, so that the rain won't fill the holes & drown the <pa>trees</pa>.</ab>
</div>

<div>
<id>p068r_3</id>
<head>Casting</head>

<ab>I have tried four kinds of sand for <m>lead</m> and <m>tin</m>: <m>chalk</m>, crushed <m>glass</m>, <m>tripoli</m> and burnt <m>linen</m>, all four are excellent. As to the <m>chalk</m>, it needs to be of the softest kind available, like the <m>Champagne chalk</m> used by <pro>painters</pro>. It releases very cleanly, needs not be dampened with <m>magistry</m> or with anything else, but is used in its natural dry state and finely pulverized.[a][b] The first cast is always cleaner, however it will sustain two or three castings. Should you only cast once, keep in mind, while remaking your box mold, to take powdered <m>chalk</m> that has not yet been used, since the one previously used in the box has dried out and will not bind and hold together as well as fresh <m>chalk</m>. Crushed <m>glass</m> can be made from ordinary <m>glass sand</m>, however <m>cristallo glass</m>[c] is more excellent, because <m>common glass</m> contains <m>saltwort</m> only, while <m>cristallo glass</m>[d] contains both <m>salt of tartar</m>[e] and <m>saltwort</m>. Both of them help the fusion, during which the glass is calcinated and reduced to its prime matter. In order to calcinate it perfectly, throw your pieces of <m>glass</m>, of whichever sort, among the largest possible glowing <m>coals</m>, unless some other violent heat source is available. And once the glass is red hot, throw it into <m>water</m>.</ab>

<ab>
<margin>left-middle</margin>
<m>Putty</m> is considered excellent for these two <m>metals</m>.</ab>
<cont/>
</div>

[a] I can't help but notice the similarities in vocabulary between this passage and what is said in this video about sandcasting at 5'35". <http://bit.ly/1w5UKKa> Here too, the sand needs to be smooth and dry. Relate to similar passages throughout the ms. about the smoothness and dryness of the material.
[b] link to video: Chris Follows, "Sand Casting - INTRODUCTION Philip White & jenny dunseath", YouTube
[c] like the one made in Murano, PHS
[d] like the one made in Murano, PHS
[e] There is an entry "Tartar" in CAMEO <http://cameo.mfa.org/wiki/Tartar>
Salt of tartar is a potassium carbonate. <http://chemistry.about.com/od/chemistryglossary/g/Salt-Of-Tartar-Definition.htm>

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<page>068v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f142.image></image>

<div>
<cont/>
<id>p068r_3</id>
<ab>and immediately you will be able to crumble it between your fingers and to easily grind it in a <m>metal</m> mortar, or better yet, in an <m>iron</m> one. After, it will be easy to crush it, not on a <m>common marble</m> slab that would be damaged, but on <m>porphyry</m>, and it must be finely crushed with <m>water</m> so that when you place it on your fingernail, you will find it soft without any roughness like the colors that the painters crush for <m>oil</m> ^{painting}. You can use this one instead of sand, when it is dry, for the box mold, without humidifying it with anything and without reheating it. If you do not like it, fumigate the hollow of the mold with <m>sulfur</m> smoke or with the tip of the flame of a <m>wax</m> candle that barely smokes, and with pure <m>lead</m> and <m>tin</m> ^{the} mold</sup> will make a very clean, shiny, and smooth work. I would say the same for finely pulverized <m>rotten stone</m>, which does not want to be reheated or humidified. Using burned <m>cloth</m> is easier than anything because it does not need to be reheated, nor to be humidified, nor to be cast and removed subtly or very cleanly, like the others I mentioned, and it stands up to several castings. First you must burn it with a flame, then leave it to rot and be consumed with its fire until it is completely burned, then refine it on <m>marble</m> or <m>paper</m>. And if you reheat it in a red hot crucible all of it will be rendered fine and insubstantial[a], I believe that this make it even better. But because a lot of <m>cloth</m> is reduced to very little when it is burned, if you want to save some, sprinkle and cover only the medal that you want to cast with it, and fill the box mold with crushed <m>slate</m>, that also molds very neatly, but note that the first cast is always the most beautiful and the cleanest. By the way, if you cast the <m>soft tin</m> that <pro>pewterers</pro> sell, the kind that appears to be burnished on these little turning wheels, very well cleaned and approaching the color of <m>silver</m>, without being mixed with anything else, it is true that it must be cast quite hot and that the molded object must be four fingers from the cast. In order to know the temperature you must first melt it quite hot so that if you put a <m>paper</m> or <m>straw</m> in it when it is simmering it quickly burns it. Then remove it from the fire and leave it to rest a little and cast the <m>tin</m> mixed
</ab>
<cont/>
</div>

[a] I suggest to translate as "impalpable" as in French original, because this is a specific technical term for very fine powders (see Biringuccio's Pirotechnia, Dover edition 1990, p. 324)

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<page>069r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f143.image></image>

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<id>p068r_3</id>
<ab>with <m>half lead</m> works almost better than <m>tin</m> or <m>pure lead</m>, even though in any case they work well. Nevertheless heed that you cast <m>lead</m> not nearly as hot as <m>tin</m>. In big works it is necessary to wet the <m>sands</m> with <m>magistra</m> or <m>egg white</m>.</ab>

</div>
<div>
<id>p069r_1</id>
<head><pro>Joiner</pro></head>

<ab>Three things take the longest, leaves, hair and garlands.</ab>

</div>
<div>
<id>p069r_2</id>
<head><m>Sand</m></head>

<ab>For a cast, do not choose that [<m>sand</m>] that is in any way lean & arid, such that it does not bind at all, like that from the sea or from sandy roads dried out by the

sun. But choose preferably that from quarries or rock caves, since that one is best, as long as it is very fine. You will know the vein if, in the greyish rocks or quarries that seem to be made of sand, after rainy weather, you see certain little heaps like dust attached to the said rocks, or if the humidity causes them to shed stones that crumble easily between your hands. You will also find sand in lean soils, which comes off in large lumps like tuf stones, or in boulbène[a][b][c] soils, which sands are much better than those from fat and strong soils. And do not take it from the surface, lest it be mixed with common soil, but three or 4 feet below and closest to the tuf or stone. When it comes off in large lumps it appears to have started to form itself into stone, but once wet it comes apart easily. When breaking it up, make sure that it is quite granular and that in crushing it between the fingers, it becomes really very fine, keeping nevertheless its sandy asperity, without muddying the fingers like clay. One dries it slowly on the <m>fire</m>, then pounds it and passes it finely through a double sieve or a <m>linen</m> sleeve, then wets it with <m>wine</m> or <m>magistra</m> etc. /<la>Verte</la></ab>

<ab>
<margin>left-middle</margin>
The Toulouse <pro>molder</pro> reheats it strongly then grinds it finely on a <m>marble</m> slab and passes it through <m>linen</m> and wets it with <m>wine</m>. He makes the cast very large and flat and hardly deep. He casts <m>pure brass</m> for scales and similarly thin objects. He casts very hot.</ab>

<ab>
<margin>left-middle</margin>
Try calcinated <m>vitriol</m>.</ab>
<cont/>
</div>

[a]AK: ms has "bolvenes", probably a variant of "boulbène": "PÉDOL: Terre composée principalement d'argile et de sable, composant le sol de la région du Sud-Ouest de la France, plus particulièrement de la vallée de la Garonne." TLFi, CNRTL <http://www.cnrtl.fr/definition/boulb%C3%A8ne>
[b]The Garonne flows from Bordeaux through Toulouse, Wikipedia has a good map <http://fr.wikipedia.org/wiki/Garonne#mediaviewer/Fichier:MapGaronne.jpg>
[c]actually the other way round: from the Pyrenees to Bordeaux via Toulouse. Not sure the explanation or the Wpedia reference are necessary.

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<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f144.image></image>

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<cont/>
<id>p069r_2</id>
<ab>The <m>sand</m> near my area is very excellent. But in order to make an excellent work, take it fresh, hardly having been used. Because it dries out after having done a casting several times & is arid & has no body at all. If you cast some fine work which has hardly any thickness, your <m>copper</m> matter must be very hot so that it penetrates and runs. Some people mix <m>lead</m> within this molten <m>metal</m>, but this is for large works & not for small ones.</ab>

<ab><m>Sand from bulls' feet</m> twice-burned and finely ground melts more neatly with <m>copper</m>, & without a coating[a], than <m>sand</m> that I've seen. I cast a medal with quite high relief in it, and slender, with the thickness of a knife blade or a card. And there being a hollow on one side, the obverse, with the relief being on the other side.</ab>

<ab>Lean <m>sand</m> needs to be more moistened than others, that's to say with <m>magistra</m> or good pure <m>wine</m> or <m>wine boiled with elm tree roots</m> or something similar. But very fine <m>sands</m>, like <m>burned linen</m> which is fat and soft of its own accord, want to be applied dry.</ab>

<ab>All moistened <m>sand</m> needs to be very beaten and tumbled in order to make it fine & flattened of the little globules that it makes in itself when it is drenched.
</ab>

<ab>The <m>olive oil</m> that some people mix in with beaten <m>egg white</m> makes it puff out.</ab>

<ab><m>Sand of calcined glass</m> lasts for many fusions. But there are only the first ones.[b] It also puffs out.</ab>

<ab><m>Latten</m> works well on its own, but it charges[c]. It is good to mix it with some <m>copper</m>, about a quarter part, with <m>frying pan material</m>.&br/></ab>

<ab><pro>Founders</pro> cast box frames well up to about 30 or 40 pounds. But no more.</ab>

<ab>
<margin>left-middle</margin>
It is good for big work. But for small it is troublesome to take away. This is because it crumbles. It would be good for it to be a little glued together with something fatty that binds, like molded <m>tripoli</m> or burned <m>felt</m> or <m>salt ammoniac</m> or <m>tripoli</m> & similar things.</ab>
<cont/>
</div>
[a]"chape": see "chappe," 086v
[b]We don't understand the sense of this sentence. It translates: "Mays il n'y ha que les premieres."
[c]"il charge": meaning unclear.

.....

<page>070r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b20500001g/f145.image></image>

<div>
<cont/>
<id>p069r_2</id>
<ab>Once you have molded, it is good to reheat your mold with the <m>smoke</m> of the material you are melting, because the cast would absorb the quality of the <m>metal</m>, then this <m>metal</m> will flow more easily in something that takes after itself.</ab>

<ab>The <m>human bones</m> are the best for casting when they are calcined.</ab>

<ab>Your material needs to be quite hot in order to cast clean. To achieve this, when your material is melted, add some <m>iron shavings</m>, or <m>iron filings</m>, or [~~<m>iron</m> in other form~~][a], because it heats <m>copper</m> very much and it cleans it of its fat. At the end, when you want to cast, add some <m>saltpeter</m>, and place your preparation between a folded sheet of <m>paper</m> in order to hide the composition of your mix, to keep it secret.</ab>

<ab><m>Sheep foot bones</m> are even better than the <m>ox foot bones</m>.</ab>

<ab><m>Oil</m> and <m>tallow</m> make fill with bubbles[b], and <m>crushed glass</m> and <m>copper</m> alone.</ab>

<ab><m>Ashy soil</m> molds very well.</ab>

<ab><m>Ashes</m> don't have enough body to hold <m>copper</m>.</ab>

<ab>It is better not to mix [different types of] <m>sand</m> at all, but to fill the chassis[c] with one type [<m>sand</m>] only.</ab>

<ab><m>Brass</m> is always fat, and does not mold very clean. It is believed that it turns out better being alloyed with a quarter <m>copper</m>, but it must be cast very hot.</ab>

<ab><m>Copper</m> is good mixed with a bit of <m>metal</m>. If you mix <m>metal</m> with <m>brass</m>, it will be harder and more difficult [to handle].</ab>

<ab>A slightly coarse <m>sand</m> has more body.</ab>

<ab><m>Rock's sand</m> is always better, because it looks like tuff in small pieces which has a beautiful and very fine grain, a bit fat. Grind it, then dry it in a frying pan on the fire, until it stops smoking. Then filter it through a fine double sieve and use it to mold.</ab>

<ab>
<margin>left-middle</margin>
<m>Copper</m> or <m>brass</m>[d] cannot come out well if the medal does not sufficient thickness and if it does not have it, give it some with some <m>wax</m>.</ab>

<ab>
<margin>left-middle</margin>
Some mold through a hole done in the middle of the reverse side of the medal.</ab>

<ab>
<margin>left-middle</margin>
Some want to mold large works in <m>brass</m>[e] mixed in the <m>sand</m> some ground <m>glass</m> to give the <m>soil</m> unity. But it makes it filled with bubbles.[f] And you have to repair the work.</ab>

<ab>
<margin>left-bottom</margin>
One puts a good amount of <m>lead</m> within a large work to make it flow, but not in a small one because it leaves grime all around the work.</ab>
<cont/>
</div>

[a]needs revision
[b]needs revision
[c]frame
[d]ms has "leton" here (elsewhere on this paige laicton) - translate as "Latten"? See comment on p069r for difference
[e]ms has "leton" here - translate as "Latten"? See comment on p069r for difference
[f]?

.....

<page>070v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f146.image</image>

<div>
<cont/>
<id>p069r_2</id>
<ab><m>Copper</m> and <m>latten</m> are ready to be cast and are at the proper temperature when, as you stir them in the <m>crucible</m>, they emit a very white flame and the bath is very liquid.</ab>

<ab>Some cast in well-crushed <m>ceruse</m>, and others in <m>flour</m> moistened several times with <m>oil</m> and dried in the sun.</ab>

<ab>Others, for casting in <m>lead</m>, use old printers' letters or types for the setting of letters.</ab>

<ab>Others add to the <m>tin</m> or <m>lead</m> a material of fixed <m>quicksilver</m> which makes it run.</ab>

<id>p071v_2</id>
<head><m><al>Sheep</al> fat</m></head>

<ab>In a certain part of <pl>England</pl>, the <al>sheep</al> that graze there have very yellow <m>fat</m> and are nevertheless just as good as others.</ab>
</div>

<div>
<id>p071v_3</id>
<head><al>Spider</al></head>

<ab>There are no ^{spiders} in <pl>Ireland</pl>, & if one touches them with some <m>wood</m> that they have there, they die. That is why some rich people from <pl>England</pl> make their floors with this <m>wood</m>, and in this way, they do not have any spiderwebs.</ab>
</div>

<div>
<id>p071v_4</id>
<head>Sand</head>

<ab>The mold is very good for casting in <m>copper</m>. But do not use that from cutlery, because it's just <m>sludge</m>, but rather that of those who mold giant shears.</ab>

<ab><m>Filières</m>[a][b] stones</m>, when smoothed by those who whet them, mold really neatly when scraped for <m>lead</m>. This is usually slate-colored. One frequently carries it from <pl>Carcassonne</pl> to <pl>Toulouse</pl>.</ab>
</div>

[a]ED: ?
Pierres de filière: pierres extraites de ces filons, proches des veines métalliques?
[b]ms has "filières"; Cotgrave gives "ranke, file, row", Cotgrave, Dicfro, <http://www.micmap.org/dicfro/search/cotgrave/letton>

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<page>072r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f149.item></image>

<div>
<id>p072r_1</id>

<head>Printer's letters</head>

<ab>The people who make the letters mix with the <m>lead</m> nails <m>heads of nails</m>, <m>old horseshoes</m>, and <m>antimony</m> to help melting. This mixture is good for printing, and holds steady. It flows and is well suited to casting in <m>lead</m>.</ab>
</div>

<div>
<id>p072r_2</id>

<head><m>Copper</m> casting</head>

<ab>The box mold should be even-shaped and well fitted onto the board where it is placed, so that it will hold steady and even when one moulds. The sand should be clean and carefully selected, well baked, and regarding the sand you will pour directly over the medal and press the harder while molding, it should be very finely pulverised on the <m>marble</m>. Be careful not to oil your medal or the <m>metal</m> will be brittle. Bake slowly and thoroughly, and let cool. Moisten your sand with <m>wine</m> boiled with <m>elm root</m>, and in it, cast <m>well-flowing latten</m> (that is, <m>latten</m> from scales and similar thin things). Cast as widely as possible, and although the stream [of <m>metal</m>] will not be very thick, it will not be any worse. Still, make sure it enters the mold well. Cast from the foot of the medal up, so that the face, further away from the heat, comes out better and the violent effect of the heat is softened. And if your box mold does not have furrows, draw some in the sand, from the borders of the mold[a] towards the hollow where you will be casting[b]. Cast very hot; you will know [if you are at the right temperature] by throwing a bit of your reserved material into what is already very molten, and if it melts right away, it is a sign that [your metal] is hot enough. Then maintain a vigorous heat in your furnace with <m>bellows</m> of an appropriate size, while holding the mouth of your furnace well covered with a large <m>tile</m> or something similar that needs to be red before the casting. When you want to cast, have your mollettes[c] red hot, and the iron hook as well, to clean out the charcoals [fallen] in your material. Having cast, scrub your work with a <m>latten</m> wirebrush. The residual powder[d][e][f] from large scissors and large knives is very good for <m>copper</m>.</ab>

<ab>
<margin>left-middle</margin>
It is said to be a great secret to add <m>persicaire</m> to [the cast][g], in order to make it as fluid as lead. Try to extract <m>salts</m> from it.</ab></div>

[a]"la pièce" = the box mold as a whole
[b]"le gect"
[c]Despite the context, I can't determine the appropriate translation for "mollettes." Cotgrave via Dicfro, "the rowell of a spurte."
<http://www.micmap.org/dicfro/search/cotgrave/mollete>
[d]DMF: "Poudre de pierre et de fer qui tombe de la meule à aiguiser et qui sert de teinture." Powder made from stone and iron that falls from whetstones and is used for dyeing.

[e]It is interesting to note how similar this process is to that of obtaining cimolee (not the naturally-occurring kind), Littre "dépôt qui est produit par l'usure des meules à aiguiser". Also the term "molée" and the term "chimolée" could very well has the same origin. "Fire", by one of the Brueghels, shows how knife grinders would use buckets to gather the mixture of iron, stone and water dripping from under the whetstone.
[f]Dictionnaire du Moyen Français, via ATILF

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<page>072v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f150.image</image>

<div>
<id>p072v_1</id>
<head>Casting</head>
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<ab><m>Lead</m>, which is soft and heavy, wants to be cast much hotter than <m>tin</m>. When it is not hot enough, it makes lines in the medal. <m>Straw</m> burns in it, if it is hot ^{enough}. It makes a solder so soft and runny that it can be melted in a <m>tin</m> dish. It is composed of one part out of <m>looking glass tin</m>, of one part out of <m>soft tin</m>, and another part of <m>lead</m>. It ^{the solder} runs very cleanly and casts in sheets, but the work is very rough and breaks. The best solder, for casting well, is the common one, but it leaves certain coarse crumbs. Combine <m>lead</m> with <m>tin</m> so that the ingot that you are going to cast becomes smooth and lustrous and polished, and doesn't make any eyes or bubbles except for a small point in the middle. And this sign will tell you that there is enough <m>tin</m>, otherwise the <m>lead</m> dominates too much. The sand is good for <m>lead</m> and <m>tin</m>. Good ^{sand} is well dried and fine and thin, which however can be picked up between the fingers. The sand wants to be recooked rather than put to work ^{directly}. And to mold with it, it wants to be well dampened and then recooked, not all at once nor under a lot of pressure, because this spoils it, and makes it shrink and also makes it crumble. But if you reheat it, and take your time, it will make it very firm, to be good for molding. It is necessary to filter it through a shirt sleeve in order to make it the more fine, with the aim of putting it first on the piece to be cast.</ab>

<ab>One makes a solder with <m>quicksilver</m>that is white, but it is thick. Make sure your box is joined well and that one does not see any daylight between the joints.</ab>

<ab>The Germans cast <m>lead</m> very thinly, because they think it is better than very thick, but as it is too pliable, it is so thin they mix a little <m>tin</m> with the aforementioned lead which, otherwise, would fold like <m>wax</m>.</ab>

<ab>Some say the blackest <m>lead</m> is the best and the softest and flows the best. You recognize the goodness of it by rubbing it with your finger, which will blacken immediately.</ab>

<ab>The good alloy for the flow of <m>lead</m> and <m>tin</m> is [made] from one lb of <m>lead</m>, and one and a half lb of <m>tin</m>. It is solder which flows well that is good for casting, but it is rough.</ab>

<ab>
<margin>right-top</margin>
To make the <m>lead</m> flow well, mix in a little <m>quicksilver</m>.</ab>

<ab>
<margin>left-top</margin>
Rub a <m>cuttlefish bone</m> on a table in order to make it flat and one against the other and against the two <m>bones</m> with pegs to keep them in place.</ab>

<ab>
<margin>left-top</margin>
<m>Cuttlefish bone</m> molds <m>lead</m> better than [any]thing there is, but try to see if it needs to be recooked. One must mold the back of the <m>bones</m> and along the side of them because the <m>marrow</m> there is more delicate[a].</ab>

<ab>
<margin>left-middle</margin>
The <m>marrow</m> near the mouth is more delicate and does have not as many fibers and molds more cleanly. The <m>shell</m> which has the <m>marrow</m> ^{which} is calcinated is good for making sand.</ab>

<ab>
<margin>left-middle</margin>
When you cast <m>lead</m>, you must cast it somewhat hot, and not at once and shake your frame and cast two or three times. However, if it is too hot, cast it, it will cast swollen and with bubbles. When it is very hot, it becomes blue; leave it then until the color passes and let it rest a little before casting.</ab>

<ab>
<margin>left-middle</margin>
Some, such as <pro>printers</pro>, mix <m>iron filings</m> or <m>pins ^{filings}</m> with <m>lead</m> in order to render it hard. But this makes it rough and it breaks not under a hammer.</ab>

<ab>
<margin>left-bottom</margin>
Have a scrub brush for cleaning your molds.</ab>
</div>

[a]This has been crossed out with one diagonal line through it.

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<page>073r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f151.image</image>

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<id>p073r_1</id>
<head>For making <m>blue florey</m> varnish or <pl>Flanders</pl> ^{varnish}</head>

<ab>Take the <m>blue florey</m> and <m>quicklime</m>, and put on around four fingers' worth of <m>water</m> and leave it to soak for one day. Take the <m>water</m> where the aforesaid <m>quicklime</m> has soaked, and put your <m>blue ^{florey}</m> with it, and put it on the <m>wood</m>.</ab>
</div>

<div>
<id>p073r_2</id>
<head>For orange color</head>

<ab>Take the <m>water</m> where the <m>quicklime</m> will have soaked and some <m>turmeric</m>, and put them one with the other, and then you will put them on your <m>wood</m>.</ab>
</div>

<div>
<id>p073r_3</id>
<head>For degreasing <m>bone</m> or <m>horn</m></head>

<ab>They need to be boiled in <m>quicklime</m>, and be sure that there is always <m>water</m>, and take a well-cleaned pan, and when they are clean, boil them in the <m>ashes of vine shoots</m>, and put it inside the hottest pot you can, and put some <m>verdigris</m> into the <pro>goldsmith's</pro> pot of <m>water</m>, and leave for some time and you will see a most beautiful green color.</ab>
</div>

<div>
<id>p073r_4</id>
<head>For making gray <m>wood</m></head>

<ab>Take <m>alum</m> with <m>river water</m> and boil them together. Then take <m>river water</m> with some <m>bran</m>, pour in the <m>bran</m> and crush them together. Take <m>iron filings</m> and <m>sulfur</m> and <m>verdigris</m> and <m>fuller's earth</m> <gap>, and boil it all together, and crush <m>maplewood</m> which will have been soaked in <m>water</m> of <m>alum</m> and <m>rainwater</m> <a>, and put it to boil together. All of this can be done better in a large crucible.</ab>
</div>

<div>
<id>p073r_5</id>
<head>To make bronze with a <m>gold</m> color</head>

<ab>Take one ounce of <m>sal ammoniac</m> and crush it in an <m>iron</m> mortar. When it is well crushed, take one ounce of <m>sulfur</m>, and
</ab>
<cont/>
</div>

<a>Literally "l'eau depuis" in the transcription, but we think he was thinking phonetically as "l'eau de pluye"
Or 'des puits', i.e. from a well?

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<page>073v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f152.image</image>

<div>
<cont/>
<id>p073r_5</id>

<ab>crush it separately. When it is well-crushed, take <m>sal ammoniac</m> and again crush them together. Then take one ounce of soft <m>tin</m> and one ounce of <m>mercury</m>, then melt your <m>tin</m> and add your <m>mercury</m> drop by drop into your <m>tin</m> while melting, and stir with force until it turns to powder.</ab>
</div>

<div>
<id>p073v_1</id>
<head>For coating</head>

<ab>Take some <m>parchment glue</m> and apply it with the <m>white of one egg</m>, then take a little <m>saffron</m> and <m>vinegar</m> and the <m>sap of a fig tree</m>, and do another coat. Then take the <m>bronze</m> and apply it on top.</ab>
</div>

<div>
<id>p073v_2</id>
<head>Varnish</head>

<ab>Take two ounces of <m>aspic oil</m> and one ounce of <m>sandarac</m>. Take a clean pot and warm it and then take it out of the fire, and after, put the drugs inside, then put them into a vial and apply it ^{the mixture} on the <m>wood</m>.</ab>
</div>

<div>
<id>p073v_3</id>
<head>Another recipe for making varnish</head>

<ab>Take one ounce of <m><pl>Venice</pl> turpentine</m>, and one quarter ounce of <m>petroleum oil</m>, and one ounce of <m>sandarac</m>, and one of <m>aspic oil.</m></ab>
</div>

<div>
<id>p073v_4</id>
<head>The color of burnished <m>copper</m></head>

<ab>Take one ounce of <m>sal ammoniac</m> with one ounce of <m><pl>Armenian</pl> bole^a</m>, and half an ounce of <m>copper filings</m> with one ounce of <m>sulfur</m>, and half a septier^b of <m>vinegar</m>. Push all of it into a small tiled oven^c and around it make a small <m>charcoal</m> fire and put your pot into the aforesaid furnace for the whole day.</ab>
</div>

[a]Armenian bole was already in the vocabulary under "bol armeni" but a quick search demonstrated that Armenian bole was also called "broullaminy." I have added it to the vocab.

[b]Cotgrave explains that this is a measure but says that it can be various sizes according to what it is measuring. He does specify that the "septier" used by physicians is barely three ounces (added to vocab.)

[c]In working on the translation we decided that he was referring to a square tiled oven--we found pictures on google under the search "tiled oven." It is a large oven about the shape of a fireplace covered with decorative tiles, used for heating the home and for cooking. This is our best guess!

.....

<page>074r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f153.image</image>

<div>
<id>p074r_1</id>
<head>To bronze in white^a</head>

<ab>Take <m>looking-glass tin</m> and put it to soak in <m>gum water^b</m>, and pound it in a pestle. And afterwards, you will put it on the <m>marble</m>.</ab>
</div>

<div>
<id>p074r_2</id>
<head>To make <m>vermillion</m></head>

<ab>Take two pounds of <m>sulphur</m>, and melt it, and add two pounds of <add><m>mercury</m></add> to it. And if this material ignites, cover it quite neatly, so that the air does not pass into the pot, then put everything for some time after into a <m>leaded</m> pot, into ^cthe fire. In the space of twenty-four hours, you will have good <m>vermillion</m>.
</ab>

</div>

<div>
<id>p074r_3</id>
<head>To make varnish</head>

<ab>Take one pound of <m>linseed oil</m>, and then put it in a pot of <m>earth</m> mixed with a <m>bread</m> crust and three <m>onions</m>, and put it on a <m>charcoal</m> fire, and you will cook it on a low fire, and let it boil for the space of five hours. You will take half an ounce of <m>flour glue</m>, and will make it boil just as before, and stir continuously^d with a spoon. And then afterwards, you will add two ounces of well-pounded <m>sandarac</m> to it, and will do so as above. And then afterwards, you will take <m>mastic</m> and <m>^{gum} ^{arabic}</m>, two ounces of each, both well pounded, and will put everything together, and will have it boil, always stirring, for the space of five hours. And then you will add <m>rock alu^em</m>, two ounces, and then you will make it boil. And if you add two ounces of <m>^{gum} ^{arabic}</m> to it, if you see that they are not cooked^f enough, have it cook more on a low</ab>

<cont/>
</div>

[a]Unclear. To glaze in white? to bronze it with a whitish cast?

[b]See 78v ("gum water" for "eau de gomme") and VOCABULARY list ("eau gommée" is "gummed water;" is there a difference between gum and gummed waters?

[c]or on the fire? French "au feu," "into" chosen in contrast with "dessus du feu" ("on the fire") below.

[d]This interprets "re-mouvoir" as an intensive of "mouvoir," but note that "remouvoir" literally means "to remove."

[e]seems different from "allum de glace," in English "Roche Alum"--see vocabulary.

[f]There's a slight semantic difference between "cuit" then (which means almost "digested," modified into a single food, baked rather than just "heated through") and cooked, a difference that we were not sure how to render. We did not want to use "boil," used throughout for "bouillir"

.....

<page>074v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f154.image</image>

<div>
<cont/>
<id>p074r_3</id>
<ab>fire until it is cooked enough. And strain it through a cloth that is compact [a] enough. And when you have strained it, leave it to cool a little. And if you see that it is so thick that you cannot strain it, you will add a little of that oil and will make it better.
</ab>
</div>

<div>
<id>p074v_1</id>
<head>To make red varnish</head>

<ab>Take <m>vermillion</m>, and let it soak in quite light [b] <m>gum water</m>, like the other, and make two or three ground layers as is said above.
</ab>
</div>

<div>
<id>p074v_2</id>
<head>To make yellow varnish</head>

<ab>Take some <m>gum arabic</m> and let it soak in <m>water</m>, then take some well beaten saffran and soak said <m>gum</m> [c]. And make it quite light, then make your ground layer on that which you want to varnish, and leave it to dry, and when it will be dry give another ground layer of the same [varnish], and leave it to dry as before, until it will be dry enough. Then take varnish from an <pr>apothecary</pr>, crushed by strokes far enough from one another. Then wash your hands quite well and spread your varnish with their palms.
</ab>
</div>

<div>
<id>p074v_3</id>
<head>Recipe for white <m>gum</m></head>

<ab>Take white <m>wax</m> and <m>ceruse</m> and a little fine <m>pitch [d] </m>, very well pounded together, and put it in a small new pot and make it melt. And when it will be melted you will make a palette of <m>wood</m> and burnish what you want.
</ab>
</div>

<div>
<id>p074v_4</id>
<head>Green <m>gum</m></head>

<ab>Take green <m>wax</m> with a little <m>verdigris</m> and make it melt as said above.
</ab>
</div>

[a]'serrée' -- is he referring to a tight weave or to another property?
[b]grammatically unclear whether the 'bien clair' goes with the vermillion (to make it clear with this process) or the gum water (we thought the latter)
[c]is he repeating himself here or is he doing something with the saffran? Seems likely that he adds the saffran to the soaking gum
[d]see note about 'poix' in the vocabulary -- seems to not necessarily be black

.....
<page>075r</page>

<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f155.item></image>

<div>
<id>p075r_1</id>
<head>Red <m>gum</m></head>

<ab>
Take <m>red wax</m> of vermillion and a little resin pitch, ^{and make} as described above [a].</ab>
</div>

<div>
<id>p075r_2</id>
<head><m>Bronze</m> of <m>copper</m></head>

<ab>
Take some <m>verdigris</m> and <m>clear glue</m> and grind them together. Then apply a layer over your <m>wood</m>, and take some <m>pin filings</m> and grind it on a <m>marble</m> with <m>water</m>, then apply it over the <m>wood</m> and polish it with <m>copper</m>.</ab>
</div>

<div>
<id>p075r_3</id>
<head><m>Bronze</m> like <m>tin</m></head>

<ab>
Take some <m>bismuth</m> and two ounces of <m>quicksilver</m>. Melt the <m>bismuth</m> into a new pot, as soon as it becomes completely melted add the
<m>quicksilver</m> and then remove it from the fire. Leave the aforementioned <m>glue</m> to dry, then take a piece of <m>tin</m>, and burnish it on a
<m>marble</m> ^{slab} with <m>clear water</m>, then coat it over the <m>wood</m> and polish it with a piece of <m>tin</m>.</ab>
</div>

<div>
<id>p075r_4</id>
<head>Making <m>aqua fortis</m></head>

<ab>
Take a demi-sestier of <m>vinegar</m> with one ounce of <m>verdigris</m>, one ounce of <m>copperas</m> and one ounce of <m>solld**[b]**</m>, then pour all the
ingredients into a glass vial.</ab>
</div>

<div>
<id>p075r_5</id>
<head>Ground <m>gold</m></head>

<ab>
Take some fine <m>gold</m> and put it on a very clean <m>marble</m> slab. Add <m>aqua fortis</m> and start grinding. Then take some <m>saltpetre</m> and
<m>ammonia salt</m>, pour it into a large, clean shell.</ab>
</div>

[a]I propose author refers here to the recipe on 074v for making white gum.
[b]a type of plant??

.....

<page>075v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f156.item</image>

<div>
<id>p075v_1</id>
<head>Making grey <m>wood</m></head>
<ab>It is necessary to take three demi-sestiers of <m>iron</m> filings per 18 deniers of <m>rock alum</m>, and for as much ^{18 deniers} <m>green
copperas</m> for six deniers of <m>verdigris</m>, pitch and a quart of <m>currier water</m>; and if you can't find any, use <m>rainwater</m>
[blank] of a slight degree. </ab>
</div>

<div>
<id>p075v_2</id>
<head>Gilding <m>wood</m></head>
<ab>First of all it is necessary to apply a coat of very clear <m>glue</m>; and when it is dry, you will apply a coat of <m>yellow gold</m>; and when this is done you
will take <m>fig tree wood</m> and soak it for a while, and wash the place you want to gild, and cut your <m>gold</m> with a knife near where you will put it; and said
<m>gold</m> must be applied with <m>cotton</m>, and if you can find some <m>fig tree</m>, take an <m>egg white</m> and stir it frequently.</ab>

<ab>
If your <m>gold</m> does not have a nice enough color, you will take a <m>partridge feather</m> burn it in a chafing-dish, and turn the smoke toward what you have
gilded.</ab>
</div>

<div>
<id>p075v_3</id>
<head>Gilding with ground <m>gold</m></head>
<ab>Take a <ms>coquille</ms> **[a]****[b]**of <m>gold</m>, soak it with gum water before you intend to work, and then apply it with a brush, and polish it with a
<m>wolf's tooth**[c]**</m> once it is dry.</ab>
</div>

<div>
<id>p075v_4</id>
<head>Recipe for coloring all <m>wood</m></head>
<ab>To make it black, the wood needs to be soaked in <m>olive oil</m> and tough <m>meat**[d]**</m> in a similar case **[e]**for four or five days, then boil it in where the
wood has soaked for one hour, then take it ^{the wood} out of the oil. Take some <m>natural sulphur</m> if possible otherwise take another</ab>

<cont/>
</div>

[a]"a measure containing five spoonfuls" (Cotgrave)
[b]see entry "or on coquille" = gold ground into microscopic particles from pulverised gold leafs, ground in a mortar with honey or gum water in: Bernard Guineau, Glossaire des matériaux de la couleur et des termes techniques employés dans les recettes de couleurs anciennes, Brepols, 2005. p. 519
[c]Nothing in Cotgrave; see Cennini 82 (ch. cxxxv) on the use of animal teeth (incl. wolf) as burnishing tools.

May also be additional teeth of a horse: "wolf's teeth," in OED, includes the following: "A horse..having two extraordinary teeth, called the Woolfes teeth, which be two little teeth growing in the vpper Jawes, nexte vnto the great grinding teeth." (1566)
[d]carne
[e]"au cas semblablement," seems out of place here.
Full context: "Pour le faire noir il le fault tremper en huile d'olive et la carne au cas semblablement quatre ou cinq jours"

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.....

<page>076r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/fl157.item</image>

<div>
<cont/>
<id>p075v_4</id>
<ab><sup><m>sulphur</m></sup>. Cover the aforementioned <m>wood</m> with this powder, until it cools, and bring it back to a boil. Upon boiling one will see whether or not it is black enough. If it is not, take the aforementioned powder and apply some over the <m>wood</m> and boil it.</ab>
</div>

<div>
<id>p076r_1</id>
<head>Making <m>Wood</m> Green</head>

<ab>
Take one quart of <m>white vinegar</m>, one ounce of green, and one ounce of <m>rock alum</m>. Mix them together and pour all your ingredients into a <m>leaded</m> pot over fire without taking any air. Then put the pot in <m>manure</m> for 15 days, after which time you shall remove and boil your materials for three hours.</ab>
</div>

<div>
<id>p076r_2</id>
<head>Making <m>Wood</m> Red</head>

<ab>
Take one quart of <m>old urine</m>, three ounces of <m>madder</m>, and one ounce of <m>rock alum</m>, put all of the ingredients into your pot. Then follow the previously described green technique.</ab>
</div>

<div>
<id>p076r_3</id>
<head>Making <m>Wood</m> Blue</head>

<ab>
Take one quart of <m>urine</m>, as above, one ounce of <m>alum</m>, and two drams of <m>sal ammoniac</m>, and half an ounce of <m>woad[a]</m>, grind it all to powder. Then follow the previously described green technique.</ab>
</div>

<div>
<id>p076r_4</id>
<head>For <sup>Making</sup> Purple <m>Wood</m></head>

<ab>
Take one quart of <m>urine</m> and one ounce of <m>alum</m> and ten drams of <m>sal ammoniac</m> and one <sup>dram</sup> of lacquer, then do as previously described.</ab>
</div>

<div>
<id>p076r_5</id>
<head>For <sup>Making</sup> Yellow <m>Wood</m></head>

<ab>
Take urine, as above, and two ounces of <m>woad[b]</m>, one ounce of <m>fustet</m> and two <del>ounces</del> drams of <m>sal ammoniac</m>, then do as previously described.</ab>
</div>
```

[a]here he specifically states to use the "pastel" form, which is a finer (ground) form of woad than that indicated by the term "guesde"
[b]here he specifically means the coarser stuff than that defined as "pastel" or "fleur de pastel"

<page>076v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f158.item</image>

<div>
<id>p076v_1</id>
<head>To make a very beautiful and inexpensive golden color</head>
<ab>First of all take a very yellow <m>orange peel</m> and carefully remove the white parts, and pulverize it very well it in a very clean mortar. Take the same amount of <m>sulphur</m>, grind all the ingredients together, pour the mixture into a glass vial, and store it in a cellar or other damp place for eight or ten days. When you want to apply it, the mixture needs to be warmed and applied wherever you like, and you will see a very beautiful color. </ab>
</div>

<div>
<id>p076v_2</id>
<head>Making black gommiche[a]</head>
<ab>
Take some <m>glue</m> and dilute it with <m>vinegar</m> and melt it until it is very clear. Take <m>lampblack</m> or another black, and dilute them together. Then apply the mixture over <m>wood</m>. </ab>
</div>

<div>
<id>p076v_3</id>
<head>For white ^{gommiche[b]}</head>
<ab>Take white wax and ceruse, melt them together and apply the mixture to wood or to anything else. </ab>
</div>

<div>
<id>p076v_4</id>
<head>Gilding iron or tin</head>
<ab>First it is necessary to wash whatever one wants to gild in wine lees[c], and then wash it once again with white wine. Take three ounces of terra merita[d] and one choppine of white wine. Boil the matter for half an hour, and apply your colors over your iron or tin, or whatever else. Then leave to dry on a sheet of paper and do not put your hand in your materials. </ab>
</div>

[a]Couldn't find in Littré, Cotgrave, DMF, Godefroy, blind google search.

Shot in the dark: perhaps gomme + iche, a suffix for lower quality ("à valeur dimin. ou péj.")?

[b]see above note

[c]"lye de vin," dregs or sediment

[d]"A vegetable color, produced by the decoction of an Indian root (curcuma longa). Known in culinary as turmeric" (Vocabulary).

.....
<page>077r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f159.image</image>

<div>
<id>p077r_a1</id>
<head>Against rednesses of the face</head>

<ab>Make a small <m>lead</m> cap and wear it overnight. Excellent secret. Try a <m>lead</m> mask. </ab>

</div>
<div>
<id>p077r_a2</id>
<head>Oriental <m>medicine</m> against all disease</head>

<figure>
<id>fig_p077r_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5ckxxXzJIeWhnMDg</link>
</figure>

<ab>Dry some <m>rosemary</m> in the month of May, then fill a jug with <m>powder</m> made of it [the dried <m>rosemary</m>], and place a burning piece of <m>charcoal</m> on top. And, having closed your mouth quite tightly, breathe in[a][b] the <m>smoke</m> through the mouth, and part of it will come out through your nose. But if you want to clear the head of a catarrhs, colds, and other illnesses, close your nose as well. </ab>

</div>
<div>
<id>p077r_a3</id>
<head><m>Fatty</m> <m>earth</m></head>

<ab><pro>Founders</pro> use [this <m>earth</m>] to make the handles of their bells[c][d][e]. Before it [the <m>earth</m>] is used, they beat it quite hard, [and] then they leave it to dry gradually in the shade for a long time. And they mold it quite neatly. But the <m>earth</m> must not be <m>sand</m>y; rather, the <m>earth</m> must be carefully selected and soft. </ab>

</div>

<div>
<id>p077r_a4</id>
<head>Recipe for making <m>bronze</m> and many kind of <m>varnishes</m></head>

<ab>To make an excellent <m>varnish</m> with <m>clear</m> <m>water</m> which you will use with a brush and which [the <m>varnish</m>] dries immediately for applying to <m>paper</m>, a little table, or any other thing</ab>

<ab>First, you will take five ounces of <m>spirits</m> and one ounce of <m>benzoin</m>[f][g]. Crush them between two ^{pieces of} <m>paper</m> or <m>cards</m> or in a mortar.</ab>

<cont/>
</div>

[a]Given that "recevoir" can mean "recueillir" when making a physiological reference (cf. TLFi "recevoir": "PHYSIOL., PHYS. Recevoir une excitation"), it seems like "breathe in" makes the most sense here.
[b]ms has "recois" (inf. recevoir): TLF via ATILF, <http://atilf.atilf.fr/dendien/scripts/tlfiv5/visusel.exe?12;s=100323375;r=1;nat=;sol=1;>
[c]According to Cotgrave, "cloche" can also mean a "bell-resembling vessel." Does either make more sense here?
[d]bell makes sense
[e]Ms has "cloche," Cotgrave has: A bell, or clocke; also, a little bell-resembling vessel, wherein peares are ordinarily stewed, or sodden. Cotgrave, Dicfro <http://www.micmap.org/dicfro/search/cotgrave/cloche>
[f]I am taking "betjoin" to be "benzoin." From the OED, we find spellings like "belzoin" (1562) and "benione" (1653). Under "benzoin" from the OED: "A dry and brittle resinous substance, with a fragrant odour and slightly aromatic taste, obtained from the Styrax benzoin, a tree of Sumatra, Java, etc. It is used in the preparation of benzoic acid, in medicine, and extensively in perfumery. For scientific distinction it is now termed gum benzoin. Also called by popular corruption benjamin."
[g]OED, <http://www.oed.com.ezproxy.cul.columbia.edu/view/Entry/17809?redirectedFrom=benzoin#eid>

.....

<page>077v</page>

<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f160.image></image>

<div>
<cont/>
<id>p077r_4</id>

<ab>but it should not be too small, then take a glass flask not too big , not too small and into it put a half of each, that means two and a half ounces. Then take the mentioned <m>vetjoin</m>, also roughly crushed, and add it immediately to the aforesaid water and let it rest for one and half day, and then take the aforementioned water and put it into another glass flask and let it soak very well. And on the rest of the aforementioned <m>betjoin</m>, which remains on the bottom of the first flask, add the rest of the water on this should it react well repeat the process you just did and mix the first water with the second, because you use this to make something more noble.
</ab>
</div>

<div>
<id>p077v_1</id>
<head>Other approved varnishes</head>

<ab>Take two ounces of <m>linnen oil</m>, and two ounces of <m>petrol oil</m>, and two ounces of <m>mastic</m>, the whitest you can find, and of <m>roche alum</m> and crush it and take a little bit of <m>white copperoze</m> and put all these drugs together into an <m>earth</m>en pot, which should be new. Put it for a while on hot ashes and you will have a beautiful varnish.</ab>
</div>

<div>
<id>p077v_2</id>
<head>Another varnish</head>

<ab>Take<m>petrol oil</m> and <m>spike lavender oil</m>, in the same even quantity, and mix them together with <m>copperoze</m> and let them melt over hot ashes, depending on the quantity you need. To apply this varnish, put three coats on the work you want to varnish.</ab>
</div>

<div>
<id>p077v_3</id>
<head>Another varnish</head>
<ab>Take one ounce of <m>spike lavender oil</m> and of <m>san<add>darac</add></m> and make</ab>
<cont/>
</div>

.....

<page>078r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f161.item></image>

<div>
<cont/>
<id>p077v_3</id>

<ab>melt the drugs in a new earthenware pot, and when they are melted, you shall pass them through a clean cloth and you will see a beautiful varnish.</ab>
</div>

<div>
<id>p078r_1</id>
<head>Another varnish</head>
<ab>Take one ounce of <m>spike lavender oil</m> and heat it over coals; take a half ounce of <m>sandarac[a]</m> and grind it coarsely, and then put it in the <m>spike lavender oil</m> and mix it with a stick until it has melted, and then put it in a glass vial, and when you want to use it needs to be heated little by little over the coals, and before applying, lay a coat of very clear <m>glue</m>, and then apply it with a paintbrush.</ab>
</div>

<div>
<id>p078r_2</id>
<head>To make red varnish</head>
<ab>Take <m>vermillion</m> and dilute it with <m>gum water</m> as well as with other <m>water</m> and make it like said varnish, and mix this other <m>water</m> the same way as other varnish and then apply it as above to make green wood.</ab>
</div>

<div>
<id>p078r_3</id>
<head>To make green <m>wood</m></head>
<ab>Take very strong <m>vinegar</m>, <m>salt</m>, and <m>rainwater</m>, and mix it together with <m>verdigris</m>, and put it with your <m>wood</m> in a new earthenware vessel, and lay it in some very warm <m>horse dung</m> for eight or nine days.</ab>
</div>
[a]see tcn, "sang de rat/rac" in original, clearly sandarac

.....

<page>078v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f162.item</image>

<div>
<id>p078v_1</id>
<head>To make bronze the color of azur</head>
<ab>Take <m>rock alum</m> and <m>antimony</m>, with twice as much <m>rock alum</m> as <m>antimony</m>, and pound it together in an <m>iron</m> mortar and soak with <m>gum water</m>, then grind it on a <m>marble</m>, and then it needs to be used with a paintbrush, and after letting it dry, polished with <m>
<al>wolf</al> tooth</m> or <m><al>dog</al> tooth</m>.</ab>
</div>

<div>
<id>p078v_2</id>
<head>To make bronze the color of gold</head>
<ab>Take <m>copper</m> filings, <m>ochre</m>, <m>gum</m>, and <m>alum</m>, grind it all together in a <m>stone</m> mortar, and when it is well ground, lay it as needed with a paintbrush, and then polish it with <m><al>wolf</al> tooth</m> or <m><al>dog</al> tooth</m>.</ab>
</div>

<div>
<id>p078v_3</id>
<head>To make <m>bone</m> or <m>horn</m> green</head>
<ab>Take <m>verdigris</m> and soak it in strong <m>vinegar</m>, and soak your <m>bone</m> or <m>horn</m> for an hour, and then apply as needed.</ab>
</div>

<div>
<id>p078v_4</id>
<head>To make ^{it} <m>bronze</m></head>
<ab>Take <m>pin</m> and other <m>brass</m> filings, put them together, and grind it on <m>marble</m>, grind it well with <m>water</m> little by little, until it flows[a], and then put it in a new vessel and</ab>

<cont/>
</div>

[a]"jusques à ce qu'ilz soient distilez" in orig.; not real distilling, but Cotgrave suggests "trill," in the sense of flow, or OED, "To roll, to flow in a slender stream."

.....

<page>079r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f163.item</image>

<div>
<cont/>
<id>p078v_4</id>
<ab>when you want to use it, dilute it with very clear <m>glue</m> and coat what you like two or three times, then leave to dry. But before coating it ^{with the <m>bronze</m> mixture (2 or 3 times)}, apply one coat of very clear <m>glue</m> and leave to dry. And after polish the <m>bronze</m> ^{coat} with a <m>wolf's tooth</m> or <m>dog^{'s tooth}</m>.</ab>
</div>

<div>
<id>p079r_1</id>
<head>Making <m>gray wood</m></head>

<ab>
Take some <m>sandstone[a]</m> powder, ____ ____, <m>iron</m> filings, scrap <m>iron</m>, <m>alum</m>, du <m>noir de courroieur</m>, <m>copperas</m>,
and <m>rock alum</m>.</ab>
</div>

<div>
<id>p079r_2</id>
<head>Making <m>water</m> against illness of the eyes</head>

<ab>
Take some <m>white copperas</m>, heat lightly over fire, then grind it to powder. Put the powder on a white cloth stretched over a glass, then take some <m>fountain
water</m> and filter it through the cloth, while stirring the powder with the finger until all the water has filtered through. Then wash your eyes as often as you wish and,
with the help of god, you will feel well.</ab>
</div>

<div>
<id>p079r_3</id>
<head>Making <m>gold</m> colored <m>bronze</m></head>

<ab>
Take six ounces of tin [...], six ounces of <m>natural sulphur</m>, and <m>sal ammoniac</m>. Melt the <m>tin</m> in a spoon. When it is molten add
<m>quicksilver</m> and cast in a line[b]. Then crush all the drugs together in a mortar, and</ab>

<cont/>
</div>

[a]grais = grès = sandstone?
[b]or cast in ingots (literally, poured in a line to be used later)

.....

<page>079v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/fl64.image</image>

<div>
<cont/>
<id>p079r_3</id>

<ab>and put in an <m>earthenware</m>pot, and ^{make it} that very hot <m>ashes</m> are underneath. Little by little, make the fire underneath grow bigger
and bigger, and the pot must not be moved. And when it is cooked, and if it takes up to six hours, mind that the smoke does not make you ill because it is bad.</ab></div>

<div>
<id>p079v_1</id>

<head>Making <m>varnish</m></head>

<ab>Take some <m>mastic</m>, @[a][b]<m>sandarac</m>, <m>gum arabic</m>, and <m>aspic oil</m>, as much of one as the other,[c] and melt them all
together. Before coating it, lay a coat of quite clear <m>glue</m> and let it dry.</ab></div>

<div>
<id>p079v_2</id>

<head>Antidote against the smoke of <m>metals</m></head>

<ab>In the morning, take a piece of thin <m>toast</m> with <m>butter</m>, neither <m>antimony</m> nor any other vapors will harm you. Or put half a <m>
<al>pig</al>'s bladder</m> in front of your face.</ab></div>

<div>
<id>p079v_3</id>

<head><m>Tin</m> for casting</head>

<ab>They use <m>common tin</m>, which is the one that <pro>pewterers</pro> use for plates, which is composed of 9 or ten pounds of pure <m>lead</m> on one
quintall of <m>tin</m>.</ab>

<ab><margin>left-bottom</margin>Some find that there is nothing better than pure <m>tin</m>.</ab></div>

<div>
<id>p079v_4</id>

<head>Mixture for <pro>printers</pro></head>

<ab>Apply a coat of <m>antimony</m>, and pulverize on a coat of <m>latten</m>, old <m>metal scrapings</m>, or thin plates of <m>iron</m> or the
<m>iron</m> nail heads, and continue in this way until the crucible is full. Melt it inside a four à vent[d]. And then mix in a little <m>tin</m> and fill the rest of the
container with <m>lead</m>. Build up the fire and stir continuously to make sure that the substances are well-allied. The mixture for large letters is harder.</ab>
</div>

[a]'sang darac' = sandarac, surely?
[b]no doubt
[c]I think this is basically saying take one part mastic, one part blood of a copperhead, one part gum arabic and one part oil of aspic.
[d]needs translation

.....
<page>080r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f165.item</image>

<div>
<id>p080r_1</id>
<head>Mortar</head>
<ab>They make them of fine material and fine copper in the shape of reinforced barrel, and which can hold 50 pounds of powder, and having made a clean hole in the
ground at the foot of a wall, they embed the said loaded mortar in this hole, with the mouth on top, which will make a large breach.</ab>
</div>

<div>
<id>p080r_2</id>
<head>Very hard white stucco</head>
<ab><m>White wax</m>, <m>turpentine of Venice</m>, <m>eggshell</m>, and <m>ceruse</m>.</ab>
</div>

<div>
<id>p080r_3</id>
<head>Fountains</head>
<figure>
<id>fig_p080r_2</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdIO5c3dLNVNsem9nTjQ</link>
</figure>
<ab>If you wish to raise <m>water</m> higher than its source, make a pipe descend as in A. to B. to give it push , then from B. to C. make the pipe ascend, but not
quite to the height of the source, here represented by the dotted line, and then make it descend again in a pipe from C. to D., and then let it rise higher than the line
demarcating the height of the source, and do this successively until you reach the desired height. Remark nonetheless that the length of the descending pipe is always twice
as long as the ascending pipe. This cone also fills itself and demonstrates a perpetual fountain which you can adapt into a rock, attracting the water with which it is filled by
the tip here marked, by sucking and breathing in. You can also make a watering can pipe in this way, so long as the bent pipe is just as long as the straight pipe, but does not
extend as far down.</ab>

<figure>
<id>fig_p080r_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdIO5N2U2RXB5UFc3TUE</link>
<margin>left-middle</margin>
</figure>

<figure>
<id>fig_p080r_3</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdIO5R2ZBNWRseEpQdnc</link>
<margin>left-middle</margin>
</figure>
</div>

.....
<page>080v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f166.image</image>

<div>
<id>p080v_a1</id>
<head><pro>Founders</pro> of small <m>tin</m> work</head>

<ab>They usually cast from <m>soft</m> <m>solder</m> the things that should not become hollow, because hollow works require fine and <m>soft</m> <m>tin</m>. However, these things would not otherwise lose their coat or become hollow if they are mixed or include <m>glass tin</m> in them, just like they mix a little
[<m>glass tin</m>] in the <m>soft tin</m>. They carve their work on <m>stones</m> of which the sharpening stones [a]or barber's files are made. These are found in
great flakes near the mountains, and resemble slate. They are found in three colors: one <m>reddish</m>, that is not as perfect as the others because it does not sustain heat

 [another] one is <m>the color of dark slate</m>, and the other is <m>whitish</m>. When they work a relief, first they print it on <m>carton[b]</m>, which is one finger thick, to serve as a pattern. After planing their <m>stones</m> and rubbing them together , they use a compass or little <m>iron [c][d]tools</m> to carve their figures. They make their molds with three or four <m>stones</m>, to make a circle or a square with the <m>stones</m> which join perfectly because the <m>stones</m> are of equal size. Before casting, they rub the mould over with <m>tallow</m>, and it [the mould] absorbs it quickly because it is hot. Then, having [put] fine powder of <m>quicklime</m> into a cloth, they rub the mould using <m>pounce</m>[e][f][g][h] in beating the <m>linen</m> on top, then blow it a little on top; this prevents bubbles [from forming]. The main thing is to make some vents. If the work is big, they do as it is shown [on the picture]. They make a hole in the medal, somewhere where it is least visible, and with a bit[i] they pierce the mold on the side of the medal.

<figure>
<id>fig_p080v_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5VktqeFlScTAyczQ</link>
</figure>

And if they want their work to last, they pierce [it] somewhere and fit in a piece of <m>cork.</m> <m>Lead</m> or <m>tin</m> will not damage it. </ab>

<ab>
<margin>left-top</margin>
Make sure the pegs of your frame fit in easily so it will open with ease without moving anything else. Your <m>box molds</m> should fit well, and the table should be very flat. </ab>

<ab>
<margin>left-top</margin>
Try to carve with <m>distilled vinegar</m>. </ab>

<ab>
<margin>left-middle</margin>
Try powdered <m>oyster shells</m>; they are said to be excellent for molding. </ab>

</div>

[a]coux
[b]in the sense of papier mache
[c]Ferremets: iron fitments?
[d]ms has "petits ferre{mens}," in Cotgrave: Ferrement (m.), An instrument, or toole of iron. Cotgrave, Dicfro <http://www.micmap.org/dicfro/search/cotgrave/ferremets>
[e]ponce: pounce (art)
<http://bit.ly/1pnBMg2>
[f]TC: Poncer in Littré makes connection to pumice in Latin. Poncer in Wordref is to sand.
[g]FP: ms has "poncent," The Grove Encyclopedia of Materials and Techniques in Art has "pounce": (powder or dust of black chalk, charcoal or pumice if support was light; of white chalk, gesso or light pumice powder if the support was dark, The Grove Encyclopedia of Materials of Techniques in Art, <http://books.google.com/books?id=mkJfbdTS--UC&pg=PA527&lpg=PA527&dq=pouncing+technique&source=bl&ots=WI-jivFR78&sig=sqqpDtpSSL9AQz2E02PGQqamytQ&hl=fr&sa=X&ei=RSijU57jDrLOsASg4YHoAw&ved=0CGUQ6AEwCg#v=onepage&q=pouncing%20technique&f=false>
[h]ms has "poncent," Littré gives "ponce": Pierre volcanique légère et spongieuse. Littré, <http://www.littre.org/definition/ponce>
[i]foret: to pierce holes.

.....
<page>081r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f167.image></image>

<div>
<id>p081r_a1</id>
<head><m>Sand</m></head>

<ab>Some excellent [sand] is found for use with <m>lead</m>, <m>tin</m> and <m>copper</m> close to the hill of <pl>Pech-David[a]</pl>, close to <pl>Toulouse</pl>. It should not be reheated much because drying out suddenly, it will lose its binding [properties] and retract [?] at the first inflow of <m>metal</m>, which makes the work become rough and unclean. It is much better to reheat it from a distance, instead of with a <m>straw flame</m>, which generates <m>grime</m>. Instead of filling the frame, you can use finer <m>sand</m> which you will have in a very <m>thin cloth</m>, and polish the medal and then the <m>wet</m> <m>sand</m> that you will put over it<m>,</m> which has more body<m>,</m> will grip it. </ab>

<ab>
<margin>left-top</margin>
<m>Tallow</m> makes things fill with bubbles</ab>

<ab>
<margin>left-top</margin>
<m>Mercury</m> stops the burning when one wants to cast. Allow the metal <m>lead</m> to flow, but if it is pure, it needs to be cast quite hot so it flows and vents. </ab>

</div>
<div>
<id>p081r_a2</id>
<head><m>Earth</m> for moulding</head>

<ab><m><pro>Sheath maker</pro>'s earth</m> or the one <pro><m>potters</m></pro> use to blanch the pots to make the <m>lead</m> flow better on it and to prevent the <m>lead</m> from soaking into them is very good to cast the hollow parts of what you want in relief. It releases better than <m>plaster</m> or <m>sulphur</m> which become hard once they have grasped on [to the mould] because <m>earth</m> sticks. It needs to be beaten vigorously so that it does not break. If the piece you want to mould is very large you can mould different parts separately. If you reheat it, it must be over a <m>closed fire</m>. One finds this <m>earth</m> at <pl>Le Fousseret[b]</pl>, and in another place called <pl>Ox[c]</pl>. </ab>

p081r_a3

Plaster

It needs to be well cooked, which you will recognize when cooking it in an *iron* or *metal* casket. Mix it with an *iron* rod and if it sticks to the *iron*, it is not cooked enough, and if it doesn't then it is ready. The *water* with which you soak it has to be a little warm and the mould of the above mentioned *earth* or other, also should be a little hot. The *plaster* work does not last and the faces and delicate things break if one does not use *glue*.

p081r_a4

Carton[d]

It is necessary to take *paper* from Florence, which is the finest, and crush it and and soak it several times and change the *water* every day, that way it does not smell bad. And after having moulded it into the hollow, put a *cloth* over it, and from the back side, rub the back of your *paper* with a sharp piece, as if you wanted to polish it and it will mould very neat. Then *glue* some cloth from the back side with strong *glue*.

[a]The hill is now part of Toulouse. NB "puy" is the French form of Occitan "pech" and Latin "podium" (hill).
[b]Haute-Garonne, arrondissement de Muret, chef-lieu de canton.
[c]Haute-Garonne, arrondissement et commune de Muret (18 miles SW of Toulouse)
[d]in the sense of papier mache

081v

p081v_a1

Sand

Brick alone, quite finely sifted, and rubbed together, is good for casting *lead* and *copper*, when diluted with *urine*. This *whitish brick* should be used.

p081v_a2

Make *Copper* and *latten* flow

At the end of the casting, as it is very hot, add some *salt ammicke* and *saltpetre*, it is a way to remove *grime*. Mix some *clay* with the *latten*. The *metal* must not touch the *iron* or the *latten* of the frame, but covered with *sand* or *luth*, it makes a very homogeneous cast.

p081v_a3

Sand

Calcined *black pebbles from the river* are good [to use] for *lead*. But they are weakened if used with *metal*.

There is only *ardille clay* with which one can make *tiles*, recooked and reddened and passed through a *long narrow bag* carefully soaked with *raw egg white*.

Willow charcoal is excellent with *sand* [and] *lead*.

Latten flows better.

p081v_a4

Cast of the frame

It is good to make it with *bow wood*. Turn it in order to make it homogeneous, as that is important for the casting. Cover the mouth of the frame with *clay* or *sand*, so that the *molten metal*, *copper* or *lead*, do not touch at all the *iron* or *latten* of the frame, because it would turn it sour, and make it filled with bubbles. Always cast higher than the mold. Pay attention that the cast is not too wide or too deep because the narrowest is the best one. Because when it is large, the weight of the matter which runs breaks and spreads the mold and fills the matter with bubbles. When you have casted, gently hit your frame, so that the matter spreads better. *Black lead [de saulmon]* of the first melting, works very neatly.

</div>

.....

<page>082r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f169.image</image>

<div>
<id>p082r_a1</id>
<head><m>Copper</m> heat</head>

<ab>It is hot enough when you see it ^{is}clean and smooth on top of its impurity.</ab>

</div>
<div>
<id>p082r_a2</id>
<head>Box mold</head>

<ab><m>Iron</m> and <m>copper</m> swell if the <m>metal</m> touches it [the box mold], and the <m>wood</m> warps. Thus, make a mold from <m>wood</m>, and [then] make the box molds from <m>brick</m>s in this mold ^{the}<m>wood</m>en mold</sup> of the thickness that you will want.</ab>

</div>
<div>
<id>p082r_a3</id>
<head><m>Egg white</m></head>

<ab><m>Egg white</m> gives strength to <m>sand</m> so that many casts ^{can be} made from it</ab>

</div>
<div>
<id>p082r_a4</id>
<head><pro>Watchmakers</pro>**[a][b]**</head>

<ab>To distribute the teeth proportionately on their wheels, they do not use a compass because there is no compass so fine that it can draw distances as small as those of the small wheels. But they make a platform**[c]** (as they call it) out of <m>latten</m> ^{that has been} divided into various circles, and each ^{of those} circle^s is divided into equal sections**[d]** marked by a point. And each circle has a certain number ^{of points}, as ^{for example} one ^{has} 30, another 36, and others 40. Then, on the pierced center of the platform, they place the little wheel. And on the center of the wheel ^{there is} a flat line or an alidade which points to the number on the circle that you want to mark on your wheel. Then with a point, they draw a line on the wheel. Then they return the alidade to the following point, and they thus go on marking it ^{the dial}. And to mark the points on the circles of their platform, they divide the circle in three and then divide this third part into as many sections as is necessary with regard to the total division of the circle. And, in this manner, they make the teeth's division and distribution as fine as they please.</ab>

<figure>
<id>fig_p082r_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5MG1jZS1LOUE2QWs</link>
</figure>

</div>

[a]Alternatively, Cotgrave says a "horlogeur" can be a "dyall-maker."
[b]ms has "horlogeurs," Cotgrave has "horlogeur": (m.) a clock-maker, a dyall-maker. Cotgrave, Dicfro, <http://www.micmap.org/dicfro/search/cotgrave/horlogeur>
[c]Should we call this a dial to make it clearer to our modern readers?
[d]Again, the word "distance" doesn't quite seem to point towards what he's implying. (See the image on the page.) As such, "sections" here seems better.

.....

<page>082v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b9059316c/f170.item</image>

<div>
<id>p082v_1</id>
<head><pro>Watchmakers</pro></head>

<ab>Formerly, they dipping their springs by immersing them into <m>molten lead</m>. Today they dip their straight springs, and afterwards bend them, which is a beautiful secret.</ab>

</div>

<div>
<id>p082v_2</id>
<head>Casting <m>lead</m> in <m>lead</m></head>

<ab>Some make their caves with <m>lead</m>, they make them thick, and then cast <m>lead</m> in it.
<m>Lead</m> that is not cast hot enough is white, and darker when cast very hot.</ab>

<ab>
<margin>left-top</margin>
One can make <m>lead</m> molds and, to cast, smoke it with soot black. But that is never very neat.</ab>

<ab>
<margin>left-top</margin>
<m>Lead</m> never comes as neat as <m>tin</m>.</ab>
</div>

<div>
<id>p082v_3</id>
<head><m>Sand</m></head>

<ab>If <m>sand</m> is too fat it should be reheated and sifted again.</ab>
</div>

<div>
<id>p082v_4</id>
<head>Ways of <m>bronze</m> casting</head>

<ab><m>Laiton de potin</m>, which is the most acid, moulds neater than <m>latten of cassole</m> according to many people. All <m>fine latten</m> molds cleaner than <m>red copper</m>. If you want to cast something fragile and tenuous, the secret is casting as hot as possible so that the matter boil[a]s. You will know that it is hot enough when it <m>smoke</m>s a lot and stirring it makes sparks. For heating well you should place your crucible as low as possible, close to the forging mill, so that the <m>bellows</m> blow on the middle of the crucible, thus it heats better than on the grate on which we <add>usually</add>put the crucible, under which the bellows blow. It is true that the crucible can easily break but you can lute it as you know <pro>founders</pro> do. You need to keep feeding <add>the fire</add>with <m>coals</m>, which are between the bellows and the crucible. Make sure to cast all at once and not in increments or drop by drop, otherwise the conduit will be plugged by the matter. It is better to make an <pro>abrevoir</pro> to <pro>abreuver</pro> the medal. It is necessary that you stir the melted matter with a stick, so that the matter at the bottom heats well. Because <m>Iron</m> turns the matter acid. Turn</ab>

<ab>
<margin>left-middle</margin>
Some people add wool stuffing to heat up and make it runny.</ab>

<ab>
<margin>left-bottom</margin>
Place one large <m>iron slab</m> under your crucible, the plate will redden and this will maintain the heat under your crucible.</ab>

<ab>
<margin>left-bottom</margin>
<m>Copper</m> matter is good to cast because when broken, its grain is long and not short, that shows that the matter is soft. The matter is composed half of <m>fine latten</m> and half of <m>red copper</m>. This long grain is called *longue tige* <add>long rod</add>.</ab>
</div>

[a]This syntax resembles that of the tc, but should it be modified for the tl?

.....

<page>083r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f171.item</image>

<div>
<id>p083r_a1</id>
<head><m>Sand</m> for <m>lead</m></head>

<ab>Recipe: ^{Take} grey <m>soot</m> from the <pro>silversmiths</pro>' furnace, <m>quick lime</m>, and <m>flour</m>. Moisten according to [the rules of] the art, in the finest possible way you can.</ab>

<ab>
<margin>left-top</margin>
The <pro>locksmiths</pro>' grey <m>soot</m> which is held in their forge is quite fine when ground. It molds and is removed very neatly.</ab>

</div>
<div>
<id>p083r_a2</id>
<head><m>Sand</m> for <m>copper</m> medals</head>

<ab>Recipe: ^{Take} hat <m>felt</m> burnt on a covered fire, <m>iron dross</m>, and burnt <m>bone</m>, all of which you should grind very finely, crush and spray with <m>water</m> and <m>salt</m>. Make a paste of it, mold it, and wipe it with <m>hay</m> fire. These three pulverized <m>sands</m>, having been very

well refined on the <m>porphyry</m> slab, mold very well and cast well. Know that separately each of them is quite good. Burnt <m>felt</m> molds well and is removed quite neatly.</ab>

<ab>
<margin>left-top</margin>
<m>Plaster</m> molds quite neatly but it swells. <m>Bone</m> does not have much body while <m>iron dross</m> does. <m>Felt</m> allows for an easy removal of the cast[a].</ab>

</div>
<div>
<id>p083r_a3</id>
<head>Other <m>sand</m></head>

<ab>^{Take} <m>charcoal</m> made from <m>vine shoots</m> and <m>clay earth</m>, both well searced[b], and mix them with well-beaten <m>egg white</m>. Next calcinate it ^{the resulting paste} in the oven and to use it, dillute it in <m>vinegar</m>.</ab>

</div>
<div>
<id>p083r_a4</id>
<head>Other ^{<m>sand</m>} for ^{use with} <m>lead</m></head>

<ab>Recipe: ^{Take} finely crushed <m>slate</m> and <m>pumice stone</m> mixed together. Calcinate them three times in a covered and sealed pot in strong fire, and each time dilute them with <m>urine</m>.</ab>

</div>
<div>
<id>p083r_a5</id>
<head>Other ^{<m>sand</m>}</head>

<ab>Recipe: ^{Take} a <m>tile</m> that has not been cooked too much, grind it and diluted it in <m>white wine</m> with burnt black <fr>trasse</fr> <m>paper</m>. If you add burnt horse <m>manure</m> to it, it will be all the better. Moisten with <m>egg white</m>.</ab>

<ab>
<margin>left-bottom</margin>
Trasse paper burnt under a light fire becomes akin to <m>soft black charcoal</m>. It molds nealty and is removed easily when mixed with the others ^{ingredients or sands?[c]}.</ab>

</div>

<div>
<id>p083r_a6</id>
<head>Excellent <m>sand</m></head>

<ab>Burn <m>alabaster</m> in a crusol under charcoal fire until it turns into powder. Once ^{it is} cold, finely pulverise it through a sieve to make it impalpable. With[d] a pound of <m>alabaster</m>, one should use an ounce of <m>sal ammoniac</m>. Mix well and incorporate everything together. Next, put it in a cave or a wet location. And with this paste, mold whatever you need and then dry the mold under a fire. While the <m>sand</m> is hot, you cast whatever metal you may like and your cast will be as neat as the principal. And the <m>sand</m> ^{will} always work if it is kept in a wet location and dried by the fire.</ab>
</div>

[a]To be checked. Could it means: felt generates deposit instead?
[b]as per Cotgrave. One could use strained to make it sound more modern
[c]To be resolved by expert?
[d]Perhaps "for" might be more idiomatic here?

.....

<page>083v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/fl72.item</image>

<div>
<id>p083v_a1</id>
<head>
Excellent type of molding with water</head>

<ab>
Dissolve some <m>laxunge de verre</m> [a] in a moist place and then pour this water over what you wish to mold, having turned it in a circle. Then let it coagulate over hot <m>ashes</m>. Use the same process with <m>vitriol</m> and <m>ferrous sulphate</m> which are very calcinated and reduce in <m>water</m>.</ab>

</div>

<div>
<id>p083v_a2</id>
<head>
Molding one part in relief and one hollow side</head>

<ab>
It is necessary to cast two parts of very clean <m>copper</m>, <m>latten</m> or a similar <m>metal</m> and then when you mold the <m>hollow</m> in <m>sand</m>, leave the <m>metal</m> figure, that you have molded in the box mold, without budging it from its place. Then cast and if there is not enough thickness press a bit and force the figure down into the box mold.</ab>

<ab>
It is wise to always mold the faces not close to the cast but the head in bottom because the strength of the material concentrates at the entry of the medal and where the force of the impurity is. Thus if you molded the face close to the cast in this manner, you could find impurities there. It is better that impurities remain on the clothes which you can repair more easily </ab>

<ab>
<m>Vinegar</m> is better for moistening than <m>wine</m>. It is necessary that the box mold be quite equal, otherwise as you press one side, it lifts up the opposite ^{one}.</ab>

<ab>
This is why the press is better because it tightens more extremely than by hand, and makes the <m>sand</m> become fat which loses its moisture. With soft and fine <m>sand</m> which is powdered, and dry, burnish the medal.</ab>

<ab>
When you mold, always leaves your medal in one of the box molds without stirring it because if your pipe is not clean, it will not mold well. You can repair it thanks to the bolts which prevent it from moving. But first, burnish your medal with <m>charcoal made from willow</m>. Then clean your medal because that degreases [b] it and makes it easier to remove it from the mold afterwards.</ab>

<ab>Do not knock as you are molding because that damage [c] the box mold and the sand.</ab>

</div>
<div>
<id>p083v_a3</id>
<head><margin>left-middle</margin>Bellows</head>

<ab>To cast with a bellows furnace, cast the hottest you can. Do not place your bellows at the bottom around the base of the furnace, lest some coal should enter them. Place the bellows under the grate. It is wise to place your furnace in a place where there is wind so that the heat will grow better. It is necessary to stop carefully the doors [d] of the furnace, that way only the bellows have an opening. At the end of the cast, you will blow slowly, as you did at the beginning.</ab>

</div>
[a]Not sure what this is or if it is a misspelling?
[b]?
[c]Not sure of the original French verb
[d]more lit. 'mouth'

.....

<page>084r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f173.image</image>

<div>
<id>p084r_a1</id>
<head>Impromptu mask</head>

<ab>Mold some <m>paper</m> and put it on the face of somebody who is making an ugly grimace, let it dry, then take your pattern to paint</ab>

[a]
<ab> # and liftings. It is necessary that it not be too wet for the aforementioned reasons. Reheat little by little, and if you left it out to dry slowly in some dry place without any high heat after molding it, it would work better, since if you expose it all of a sudden to high heat, it warps. I think it would be very convenient to reheat it in the mouth of the oven after you take bread out of it. One should not complain about the difficulty of preparing it; it can take as many castings as you wish, because it becomes as hard as marble, you can polish it, and the mold cannot be ruined. Having used it, crush and sift it again, without making a special effort to crush it very finely, since, when sifted through a common sieve, it is stronger and strips better than when it is very fine.</ab>

</div>
<div>
<id>p084r_a3</id>
<head><m>Magistry</m></head>

<ab>Dissolve <m>rock salt</m> or <m>sandever</m> that you have pulverized finely and placed on a <m>marble</m> slab in a pot. It will dry out while you reheat the mold, and will provide it with a binding to enable several castings. Try moistening it with <m>tartar oil</m>. </ab>

</div>
<div>
<id>p084r_a4</id>

<head><m>Flour</m></head>

<ab>Ground during <tmp>Advent</tmp> lasts the whole year.</ab>

</div>
<div>
<id>p084r_a5</id>

<head><m>Sand</m> from a <pl>Toulousain</pl> mine</head>

<ab><m><pl>Toulousain</pl> sand</m>, which is harvested from the Côteau du Puy-David, deep down in the earth, is excellent on its own, but to make it sustain multiple castings, I mix it with pulverized and moistened <m>sandever</m>, which hardens it, holds well together with it, and enables as many castings as you like. Try to moisten the crushed <m>glass</m> and other sands with the <m>essences of the aforementioned salts</m>. Like all <m>natural sand</m>, it needs to be reheated well. Before putting it to use, one pulverises it, because it is in a lump, and then one reheats it in a <m>copper</m> boiler (or similar) until it ceases to smoke.</ab>

<ab>
<margin>left-bottom</margin>
It is necessary to reheat it before using it. It becomes better for casting because it is reheated, but when it has been used too much, it must be refreshed with new sand.</ab>

</div>

[a]There is a line across the page here.

.....

<page>084v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f174.image</image>

<div>
<id>p084v_a1</id>
<head><m>Sand</m></head>

<ab>I tried the <m>hoof of an ox</m>, quite burned, pulverized and very well crushed on <m>porphyry</m>, until it is fine enough to slip through your fingers without being felt. On its own, it makes a very clean mold. But because on its own it is very dry and lean, it demands to be well wet and humidified, with a thick broth with <m>elm root</m>.</ab>

<ab><m>Dross of Iron</m>, very burned <m>ox hoof</m>, and <m>felt</m>, all burn well in a closed fire. All three ground very finely on <m>porphyry</m> mold very neatly in <m>lead</m>, without any need to reheat it much, and by casting in a box mold either hot or cold.</ab>

<ab><m>Felt</m> on its own provides a very clean mold if wet, and comes off easily. And also makes the other two come off easily</ab>

<ab>
<margin>left-top</margin>
These <m>sands</m> can only undergo one cast.</ab>

<ab>
<margin>left-top</margin>
A fat <m>sand</m> which sticks together neatly, makes bubbles.</ab>

<ab>
<margin>left-middle</margin>
<ab>A fat <m>metal</m> needs lean <m>sand</m>.</ab>
</ab>

</div>
<div>
<id>p084v_a2</id>
<head><m>Eau Magistra</m></head>

<ab>Some people think that <m>salt water</m> is not good, because the <m>salt</m> releases gas when heated and as a result causes bubbles. [In this case], there is only <m>wine</m> boiled with <m>elm root</m>.</ab>

<ab><m>Sanding charcoal</m> makes [things] come off well. But one finds that <m>willow charcoal</m> creates bubbles, but <m>oak</m> or <m>beech charcoal</m> does the job without making bubbles.</ab>

<ab>
<margin>left-middle</margin>
Try burnt <m>oysters</m>.</ab>

</div>
<div>
<id>p084v_a3</id>
<head><m>Lead</m></head>

<ab>For casting, there is only <m>soft lead</m> that can be cast at a high temperature. And <m>soft tin</m>.</ab>

</div>
<div>
<id>p084v_a4</id>
<head> <m>Sand</m>, for the most excellent <m>lead</m> of all, for large and small reliefs</head>

<ab> I took <m>white lead</m> and crushed it dry on <m>porphyry</m>, to make it very fine. Then I moistened it a lot with very well beaten <m>egg white</m>, so that it became like a paste, by making it stick together perfectly with the flat of a knife. I let it rest for a while. </ab>

<ab>
<figure><margin>left-bottom</margin>
X
</figure>
</ab>

<ab>
<margin>left-bottom</margin>
Mix it well with the knife.</ab>
<cont/>
</div>

.....

<page>085r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f175.image></image>

<div>
<cont/>
<id>p084v_a4</id>

<ab>and place it on a flat well-clean table, and since it has a desiccative quality, I knew it would dry out. I left it to dry for a while to reduce it to a powder and mold it with <m>sand</m>, it having been ground into small pieces with my fingers and the sharp side of a knife. I <m>oil</m>ed my medal because <m>oil</m> cleans it without damaging it. And after cleaning and drying the medal with a <m>cloth</m> and small bristle brush of <m>pork skin</m>, I lightly <m>oil</m>ed it once more with clear <m>walnut oil</m>, and gently swiped it with a <m>cloth</m> so that [the medal] would not be too <m>oil</m>ed; having already considered that in this way [the medal] should come out easily, because the <m>white lead</m> wet with <m>glair</m>, which would not damage the <m>oil</m>. This worked very well. I molded a medal very cleanly in high relief, without it having any lumps, which a lot of good <m>sands</m> such as <m>felt</m>, burnt <m>bone</m>, and <m>iron dross</m> failed to do on the first attempt. I re-cooked it and my mold became hard like marble. And then I knew that <m>sands</m> used to mold big relief must be very moist with some kind of <m>water</m>, which gives body and firmness, like <m>egg white</m>, <m>gummed water</m>, [or] one [<m>water</m>] boiled with <m>elm root</m>. And lightly <m>oil</m> your medal. It can take as many firings as you want because it is as hard as glass. But soft <m>lead</m> and sour <m>tin</m> must be cast at a very hot temperature.</ab>

<ab>Since that moment, I have realized that even though this <m>sand</m> is excellent, can be cast often and molds very neatly, it is fat and it makes bubbles. In this way, the <m>soft lead</m> does not suit the mold well. But I have tried to mix it with lean <m>sand</m>, such as <m>pumice</m>, <m>metal scrapings</m>, and similar things, in order to give it body, and to help it come off easily, because lean <m>sand</m> [alone] does not come off easily; and yet it takes <m>metal</m> well.</ab>

<ab>
<margin>left-top</margin>
Try to mix <m>white lead</m> or <m>minium</m> with other sands.</ab>

<figure><margin>left-top</margin>X</figure>

<ab>
<margin>left-top</margin>
Oil and smear with <m>aspic oil</m>, which will disappear when reheated, because the <m>oil</m> causes bubbling.</ab>

<ab>
<margin>left-middle</margin>
It is necessary to mold with <m>egg white</m>, and then mix it well. And once it is in the box mold, beat it well on the top with a pestle, or something else clean. This helps to make molding better and come out more easily.</ab>

<ab>
<margin>left-middle</margin>
It is best to fill the box mold in one go, because the mixture made of several <m>sands</m>, and the one from mines, that you use to fill the box mold, damages it.</ab>

</div>
<div>
<id>p085r_a1</id>
<head>Green <m>varnish</m> for <m>copper</m> medals</head>

<ab>After casting them neatly, cover them with <m>glass salts</m>, which we use for sand, and moisten it. Within three or four days they will be green. Oil them after, and place them under some <m>manure</m>.</ab>

</div>

<div>
<id>p085r_a2</id>
<head><m>Glue</m></head>

<ab>The Flemish mend their <m>earthenware</m> pots with <m>or colés</m>[a], that is to say <m>massicot</m>, <m>minium</m> and <m>varnish</m>. </ab>
</div>

[a]We looked for this material and found nothing. We think it is defined in the text.

.....

<page>085v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f176.image</image>

<div>
<id>p085v_a1</id>
<head><pro>Casters</pro></head>

<ab>They mix beaten <m>egg white</m> with <m>earth</m> of which they make the first layer of the crown ^{of a bell} in pieces. ^{They also make} bells and all other said pieces which the said <m>^{egg} white</m>[a]. This makes ^{it} come clean and sets and stabilizes the material. For the core of a small piece, <m>egg white</m> is also good.</ab>

<ab><m>Rosette</m> in order to make it come clean, wants the mold to be a little hot and <m>lead</m> which is mixed with the <m>red copper</m> ^{is} therefore chiefly for small pieces.</ab>

</div>
<div>
<id>p085v_a2</id>
<head>Mortars</head>

<ab>They should be of the finest material possible. ^{In the same way}, it is better to use <m>copper</m> for a cauldron than <m>red copper</m>, because it better resists a blow.</ab>

<ab>The mortars, therefore, used for grinding are stronger and there is less danger of breaking them if they are of fine <m>copper</m>. And for a private home, they do not ring so much and do not make as much noise as those of <m>metal</m>. It is true that those that are of <m>metal</m> ring louder for <pro>apothecaries</pro>.</ab>

<ab>Grenades must be made from fine <m>metal</m></ab>

</div>
<div>
<id>p085v_a3</id>
<head><m>Sand</m> from a mine</head>

<ab>It wants to be well recooked for big <m>metal</m> ^{objects}. Some people burn it inside a furnace until it becomes quite black and they grind it on the <m>porphyry</m>. Other people burn it with <m>asphalt</m>, but when it is too burned, it does not mold so cleanly, because it lacks body, and is too thin. You can give body to the <m>sand</m> with <m>tripoli</m> or burned <m>felt</m>.</ab>

</div>
<div>
<id>p085v_a4</id>
<head><m>Sand</m> Experiments</head>

<ab>X I have tested <m>sand</m> of <pl>Th{oulous}e</pl>, and after seeing it well recooked two times in a skillet, I passed it through a fine <m>tammy cloth</m>, like the double ^{sieve} which <pro>apothecaries</pro> use, without grinding it further on the <m>porphyry</m>, as I have done other times. I moistened it ^{the sand} in infusion[b][c] with <m>elm root</m>, and in it, I molded a big piece of a portrait of Jesus. I found it unmolded very well without having to tire myself out by hitting it from behind and it molded cleanly on one side of the relief and on the other concave. And [it was] of the thickness of a coin of forty sols. I cast ^{this} very hot</ab>

<figure>
<margin>left-bottom</margin>X</figure>

<ab><margin>left-bottom</margin>
<m>Sand</m> from a mine, well chosen and well reheated, is the most excellent of all without trying to find all other mixtures because it receives all <m>metal</m>. It does not like to be used hot because it bubbles the more finely. For large works, it is not the best because it does not have enough body to sustain ^{it}.</ab>

<cont/>
</div>

[a]check translation
[b]boiled with elm root
[c]check translation? ms has "vin bouilly avecq racine d'orme"

.....

<page>086r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f177.image</image>

<div>

<cont/>
<id>p085v_a4</id>
<ab>with the <m>material of a frying pan</m> mixed with a knob of <m>potin</m>. And before that, I had so reheated my molded box frame, now on the flame of my stove, now putting glowing <m>coals</m> beneath it, that it became almost red. I let it cool and cast it. It came out with a quite neat relief on one side & hollow on the other, as much the figure as the letters. It is true that the material was whitish, as though it were metallic, but this was because of the <m>potin</m>. I made another cast with <m>frying pan material</m>, alone[a][b] in the same <m>sand</m>, but not as reheated. It did not come out well.</ab>

<ab>Then I molded burned <m>bull's foot bone</m>, pulverized & strained through a double sieve, and wet with <m>egg white</m> or <m>wine</m> boiled with <m>the root of an elm</m>. I lightly knocked on top of it while molding. While opening the box frame, I found that the figures had come out not at all clean & it left the molds as though they were floury & crumbling. I wet the <m>bone sand</m> more so that it held together well between the figures, and in this way I molded neatly with good come-away. And even though it seemed like the ground <m>bone</m> was lumpy, when casting with <m>frying pan material</m>, it turned out that my figures were quite clean. It is true that I had really really heated up my box frame, it did not admit but one cast. I find that, when a <m>sand</m> is so fine that it gets as heavy as <m>ceruse</m> & even, without finding it arid, thin, sandy & rather spongy, that ^{sic} it molds quite neatly but it does not receive <m>metal</m> very well, as if it were porous and sucking up the material. But, being fatty & even, it puffs up & does not receive subtle impressions at all. I believe that the secret of casting well lies in finding a <m>sand</m> that conforms well to the <m>metal</m>, this one for <m>lead</m>, the other for another, because each has its own particular one, so that it be molded easily and keenly. Then let it firm up on its own for a few days if you have the leisure for it. And afterwards let it really heat up again, not instantly nor with a large fire, but little by little, otherwise it will crumble & always have some defects. Finally you must cast <m>copper</m> or <m>latten</m> or other <m>great metals[c]</m> very hot and if possible with a large quantity of material that has a lot more heat than a small quantity. It's necessary that the box frame be cold & that you cast all in one go. Always lute the opening of your box frame because <m>metal</m> touching fire or <m>metal</m>. </ab>

<ab>
<margin>left-top</margin>
This <m>bone</m> has to be really crushed in a mortar, it must not be reheated very much, because then it would crumble.</ab>
<cont/>
</div>

[a]Or "just" or "simply"? : "seule"
[b]presumably meaning "without the potin"
[c]ms has "grands metaulx" - is this term significant? it occurs several times in the manuscript and has not been consistently translated.

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<page>086v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/fl78.image></image>

<div>
<cont/>
<id>p085v_a4</id>
<ab>[or <m>metal</m>] fills with bubbles. Most of all, make sure that the cast is always higher than the molded thing, since the <m>sand</m> swells very often when reheating if the, even in the middle, and therefore with the molded thing remaining higher than the cast, the <m>metal</m> cannot run easily or at all nor enter at all. Also make sure that the mold & the cast are indeed reheated. Also cast in one go & outside of the wind. And if your medal is really thin, when you want to mold it, put a <m>card</m>, or two or three thicknesses of <m>paper</m>, so that the mold will be lower than the cast. Also cast in the place where your medal will be least thick & where the relief is lowest.</ab>

</div>
<div>
<id>p086v_a1</id>
<head>Excellent <m>sand</m> for <m>lead</m>, <m>tin</m> and <m>copper</m></head>

<ab>D</ab>

<ab>Since[a], I molded with <m>burned bone</m>, <m>dross of iron</m> & burned <m>felt</m>, really crushed and ground on <m>marble</m> and mixed together. I got them very wet with beaten <m>egg white</m>. And, as in the other [casts], having covered the medal and filled the box frame with <m>sable de mine</m>[b][c], I gave it a little tap. I found it really stripped & molded quite neatly. I let it sit for a night. The morning after next I reheated it little by little for over the course of seven or eight hours (because if possible there must not remain any humidity at all in the box frame). Two times, I cast <m>copper</m> alloyed with <m>ꝛ</m> there, as old <m>K</m>[d]. The material came out really lustrous & resonant & without a chappe. And my <m>sand</m> was not corrupted at all. Since[e], I have cast in there many casts of <m>sweet lead</m> and <m>tin</m>, which came out the best and neatest of any others I have yet found.</ab>

<ab>When you mold make some grooves around your mold in the box frame, so that you draw in this manner the matter from all sides.</ab>

<figure>
<id>fig_p086v_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5dWFuOW9Db1ZjZfK></link>
</figure>

<ab><m>Potin</m> from <m>sernique</m>[f] & other works runs yet better than fine <m>latten</m>. But I believe that half <m>copper</m> and half <m>latten</m> is better, which has served & been in very thin works, like frying pans & similar ones. I've seen come out quite well this mixture of half</ab>
<cont/>
</div>

[a]Since what? He doesn't say. Maybe since his last mold.
[b]Sand from the mine?
[c]cf. 87v, "sable de mine de tholouse"
[d]Unidentified material (?)
[e]Unclear referent but may refer to the copper-silver alloy.
[f]We don't know what this is.

.....

<div>
<cont/>
<id>p086v_a1</id>
<ab>In order to mold well, it is necessary to cast several medals together in a large box mold, because when there is a lot of material and the crucible is almost full, it becomes hotter. Then, if you do not succeed with one of the medals, another will be good.</ab>

<ab>It is better to melt with a bellows furnace than with a wind furnace, because it heats more vigorously. It is true that <m>latten</m> melts better in the wind furnace, because it is more brittle than <m>copper</m>.</ab>

<ab>Some <pro>founder</pro>s superstitiously believe that there are only three days in a week that are good for melting, that is, Tuesdays, Wednesdays Thursdays, and Saturdays. For them, the others are unlucky.</ab>

<ab>When you mold, do not excessively pound on the medal that is in <m>sand</m>[a], because that prevents it from being cleanly stripped, and cracks the mold.</ab>

<ab>See to it that <m>sand</m> does not go over the edges of the box mold, because if the molded medal is higher than the cast, <m>metal</m> will not easily enter the form. Therefore, always take care that the surface of your cast surpasses the mold in a straight line. To accomplish this, if you wish, put a piece of <m>cardboard</m> of whatever thickness you please on top of the mold[b].</ab>

<ab>To prevent their large casting works from becoming too porous, <pro>founder</pro>s are careful to heat their molds very well. And to know if the molds are heated enough, they tap them with their finger, and if they start ringing like a pot, they are heated enough.</ab>

<ab>In order to cast their canons cleanly, they mix with their <m><pro>founder</pro>'s earth</m> some fine casting <m>sand</m>, if they can get any.</ab>

</div>
<div>
<id>p087r_a1</id>
<head><m>Sand</m> of <pl>Toulouse</pl></head>

<ab>The ^{<m>sand</m>} generally considered good is the one found in a vineyard near Pech-David. But the best is the one is from the <pl>Touch[c]</pl>, a river close to <pl>Saint-Michel</pl> and <pl>Blagnac[d]</pl>, in a vineyard at a high altitude. This ^{<m>sand</m>} is thinner, and a bit greasier than the other, and better for small works. It should not be overheated.</ab>
</div>
[a]Is this the medal (medaille) before it is taken out of the box mold, or is it the imprint of the medal after it is taken out of the box mold?
[b]Adham's comment was that this whole process was unclear
[c]Selon Wikipédia: Le Touch est une rivière du sud de la France qui coule dans le département de la Haute-Garonne et la région Midi-Pyrénées. : [http://fr.wikipedia.org/wiki/Touch_\(rivière\)](http://fr.wikipedia.org/wiki/Touch_(rivière))
[d]Not exactly clear which Saint-Michel is meant, but Blagnac is a suburb of Toulouse. The Touch flows very close to modern-day Blagnac (Midi-Pyrénées, Haute-Garonne, Toulouse, Blagnac)

.....

<div>
<id>p087v_a1</id>
<head><m>Sand</m>, <m>slate</m>, and burned <m>earth</m></head>

<ab><m>Sand</m>, when overheated, does not hold as well. <m>Slate</m> is baked and will mold clean, yet it is true that often it becomes porous, just like burned <m>earth</m> also does, as well as any kind of greasy <m>sand</m>.</ab>

<ab>
<margin>left-top</margin>
I find that one must not hammer at all on <m>sand</m> ^{already} very finely pulverised, because it would damage it, and then make^{the sand} would not be able to release the cast cleanly. Still, it needs to be pressed hard, and moisten it sufficiently.</ab>

</div>
<div>
<id>p087v_a2</id>
<head><m>Magistry</m></head>

<ab><pro>Founders</pro> harvest the <m>roots of a young elm</m> when it is sappy, and boil it in <m>wine</m>, or better yet <m>vinegar</m>. They prepare a year's worth of it and store it in a cask.</ab>

</div>
<div>
<id>p087v_a3</id>
<head><m>Sand</m> from sand pit from <pl>Toulouse</pl></head>

<ab>^{In order to use} <m>sand</m> from a sandpit for casting, filter it through a double sieve, then having mixed it with molten <m>resin</m> it burns and heats, and becomes all red and hot as iron.[a] Once it has cooled down it is completely black and can be pulverised on <m>porphyry</m> very well. Having prepared it like this and checked the smoothness of the surface with my nail, I moistened it with <m>egg white</m> and beat it vigorously, until it was not a paste anymore but a powder. I found ^{this sand} to release the cast very well, and molded very clean in <m>lead</m> and <m>tin</m> ^{with it}, however it needs to be reheated

and at leisure.</ab>

<ab>Others beat it strongly in a mortar, in small amounts at a time, thus it detaches and comes out very fine; then they bake it moderately, just enough to dry it, and then they crush it dry on <m>porphyry</m>. Thus it is made almost impalpable, and yet not too dry , and it retains the consistency and the binding quality of the <m>earth</m> to which it is related, and is better than when it is burnt. When you moisten it with <m>egg white</m> and pass it through a <m>sponge</m> it releases very clean for a low relief, but not a figure in high relief. Considering this, since then I experimented with moistening it only with strong <m>vinegar</m>. ^{That way} it releases a figure I previously had not been able to release ^{from the mold}. I think that even when you wet the finest sort with <m>egg white</m>, or in a similar fashion, or the thick sort to fill chasses<c>[d]</c><e>[e]</e> with <m>water</m>, <m>salt</m>, or <m>vinegar</m>, they do not mix so well. However, given that they are of one and the same nature, and are similarly moistened, they espouse each other, and hold together.</ab>

<ab>
To mold well, when your <m>sands</m> are prepared, mold in one day. Bake slowly the next day, then cast the other [f].</ab>

<ab>
<margin>left-middle</margin>
It is necessary to choose ^{sand} that like comes in clods and lumps, and deep down in the earth, because commonly the one near the surface is too much akin to earth, whereas the deep one is similar to rock.</ab>

<ab>
<margin>left-middle</margin>
I have sometimes[g] molded with <m>pure lead</m> cast very clean hot, and the example[h] as well as the principal[i] were very clean, but since the <m>vinegar</m> gives it little binding, it held only for one cast.</ab>

</div>
<a>check syntax?
"dont il participe": the earth it comes from, but also the earh whose nature and properties it shares.
<c>I can't make sense of this.
<d>Chasse could be a kind of box, similar to "caisse", DMF.
<e>I've put it in the vocab but with no definition yet.
<f>the other what? il might mean "the other day, the next day".
<g>"j'en ay moulé" the "en" seems to indicate he's done it on more than one occasion
<h>"exemple" and "principal": probably this means the molded reproduction, and the original figure. Not sure why he would say both are very clear, though.
<i>"exemple" and "principal": probably this means the molded reproduction, and the original figure. Not sure why he would say both are very clear, though.

.....

<page>088r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f181.image</image>

<div>
<id>p088r_a1</id>
<head>Vine</head>

<ab>As with a bramble, it can be planted on both sides. And those who plant the larger side facing a slope say that it makes a larger stock. But it only has a higher point, because it grows higher.</ab>

</div>
<div>
<id>p088r_a2</id>
<head>Mortars</head>

<ab>The fine <m>copper</m> mortars do not throw any flames, as long as the <m>iron</m> pestle is not tempered. If the mortar is indeed made from <m>metal</m>, when this is sour, in this case one makes the bottom of large mortars from fine <m>copper</m>.</ab>

</div>
<div>
<id>p088r_a3</id>
<head><m>Varnish</m></head>

<ab><m>Turpentine oil</m>: <m>turpentine</m> & good <m>eau de vie</m>, to dry it out; heat it without mixing on the stove, so that it mixes of its own accord in melting. After, test it on a very clean blade, & you will know from this whether it has enough body and whether it does not flow too much.</ab>

</div>
<div>
<id>p088r_a4</id>
<head>Grafting</head>

<ab>One knows from experience that the young shoot that is enters into the incision on the sunny side takes hardly often at all when it is also bowed.</ab>

</div>

<div>
<id>p088r_a5</id>
<head><pro>Baker</pro></head>

<ab><m>Dust</m> guards wheat against being worm-eaten. And to clean it well when it is stained & like rust, pass some <m>ashes</m> through a sieve and mix them with the <m>wheat</m>, then sift the whole thing with a bolting cloth made from <m>rough cloth</m> or <m>canvas</m>. The <m>ashes</m> will pass through & your <m>wheat</m> will stay nicely yellow & very fine.</ab>

</div>
<div>
<id>p088r_a6</id>
<head><m>Looking-glass tin</m></head>

<ab>It makes <m>material</m> whiter.</ab>
</div>

.....

<page>088v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f182.image</image>

<div>
<id>p088v_a1</id>
<head><m>Sand</m> from pulverised <m>rock salt</m> and <m>sand</m> from a <m>minium</m> finely ground on <m>marble</m> </head>

<ab>After they have been dryly ground and beaten in the mortar, they are ground finely on the <m>marble</m>. I mixed the same quantity of each material, and in order to mix them better, I ground them on the <m>porphyry</m> again, and then filtered this through a double sieve or the sleeve of a <m>shirt</m>. Then, I put them on sheets of <m>paper</m> and stored it on <m>marble</m> in a cellar. In one night, they were both moist enough [that there there was no need] to dampen them further because <m>rock salt</m>, like all other salts, dissolves in dampness. I molded with this very neatly because both should be quite fine. They need to be moist enough so it can be removed easily.</ab>

</div>
<div>
<id>p088v_a2</id>
<head><m>Mineral sand</m></head>

<ab>It does not matter if the ^{sand} color is white or yellow. Above all it ^{the sand} must be lumped [a] ^{together} much like a precious stone or a stone, and the deeper ^{in the earth} one gets it from, the better. The signs of its quality is that it is thus gathered together, that, if the <m>stone</m> is removed as it should be ^{i.e. in the correct way}, it forms pieces and lumps ^{together}, which demonstrates its ^{capacity for} binding, and that it ^{the sand} is not too dry.</ab>

<ab>However, it should crumble in your hands. And the grain must be quite small, fine, and homogeneous. If this ^{sand} is not fine enough, you can filter it and grind it up finely either with <m>water</m>, or through a sieve, when it or on the <m>porphyry</m>. That way it loses its dryness and becomes greasy and well bound. You can mold it in a box mold or in a noyau [b] without wadding, and try it with <m>lead</m> because, if it does not swell and casts neatly, it will bear out well too. Some say that fatty <m>sands</m> do not work for casting <m>metal</m> that is too hot. <pro>Artisans</pro> who undertake a big job and who want to save money do not need to crush ^{sand} and seek the curious ^{properties} of <m>artificial sand</m>; they have more to profit from finding natural sand that has the finest grain possible. And for a small job, they take advantage of finding the finest natural sand they can. For a smaller job, they only filter the sand through the sieve. But those who work on a very small scale crush the <m>sand</m> and grind it finely ^{until it becomes} impalpable, because they do not need as much ^{sand}.</ab>

<ab>
<margin>left-middle</margin>
Ground <m>wheat</m> makes a tawny-colored powder ^{that is} quite fine and quite soft[c]. Once this powder has been mixed, it can mix mold very neatly. Try <m>wheat flour</m> burned in an enclosed fire.</ab>

</div>

[a]I am taking "amatonné" to be a variant spelling of "matonné."
[b]Needs to be clarified in vocab
[c]The word the author uses is "doulce," so the appropriate translation may depend on how we ultimately decided to translate this idea.

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<page>089r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f183.image</image>

<div>
<id>p089r_a1</id>
<head>Powder of <m>ox bone</m> and <m>rock salt</m></head>

<ab>I pulverised them separately and ground them finely on the <m>porphyry</m> as much as I could. Then I mixed all of one with the other and re-ground it[a] on the <m>porphyry</m>. Afterwards I moistened it[b] in <m>paper</m> folded in a moist <m>napkin</m> which is made wet more quickly from the <m>moisture of the night</m>, or the <m>^{moisture of} the cellar</m>. I have never found ^{one} [c] which can be removed more cleanly from the mold than this, though it needs to be quite moist. And if you want to cast small works, make it very hot. It came out very cleanly from the doulx tin, like the main one, and has sustained several castings. For <m>tin</m>, I believe that it is not necessary to look further to find a material that takes to powder better, and even for use with fine <m>lead</m> which has

almost better results than ~~Tou~~. The bone of an ox hoof is always dry, that is why you must mix it with fatty sand, so it will bind like tripoli, salts, felt, ashes and similar materials. ^{If you do not mix ox-hoof bone}, it will not turn out from the mold and will not mold cleanly because it crumbles.

Reheating or wiping a box mold

Always arrange the most delicate parts, like faces or handles, so that nothing will fall which would then disturb your fusion.

Natural sand

This should not be reheated for lead and tin, it is ~~better all new and fresh.~~

Presumably the mixture referred to above?
see comment above. Perhaps better translated here as 'powder'?
? a powder
?
? "mixture" better here

.....

Ammonia salt and alabaster

Dry ammonia salt, ground ~~on a marble slab~~, molds very cleanly and is easily removed from the mold. When mixed with pulverized alabaster, use two ounces per pound of alabaster; it also is easily removed from the mold. It should be humidified in a cellar or in the cool humid evening air, or to have it quickly, in a sheet of paper in a moist towel. Take care not to leave it too long because it will soften, which would not be good if you wanted to use it as a box mold. However you can use it well as a core provided that you dry it gently and away from the fire. Otherwise the heat would make it swell, and push the salt onto the surface, causing lumps. If you want to mold a core, ~~rub your medal with aspic oil~~. It is better to use 4 ounces of ammonia salt per pound. Humidify it in a damp place for two or three days, so that the material can be handled without sticking. You will mold very cleanly. Let the mold dry out and reheat it, that way it will get body, and become as hard as stone and when it is like that it is far easier to remove from the mold. If you do not reheat it, it could crumble when removed from the mold because of its thinness. It is the same with a large relief medal. Once you have molded, pulverize the material like before, and moisten it in a damp atmosphere.

Left-middle
All sands which can be removed easily from the mold have body, well-made fat ammonia, and ~~is sand~~ ^{is sand} that removes easily from a mold. There is nothing better for binding than the salt of the metal itself. When ground into powder, they pick up moisture, dry and reheat together.

White Glassmaker's sand mixed with ammonia salt

You can find this sand in **Cominge**, near the town **Aurignac**. This sand is dry and as white ~~as salt~~. Glassmakers and potters use this sand ground on the porphyry until it becomes as fine as possible. Easy to grind, this sand looks like calcined alabaster, and provides a very clean mold. I have not found another as suitable for delicate bas-reliefs. It is also excellent to mold as a core, without any box mold. You must grind it very finely on a porphyry with gummed water or pure water, then apply a thick coat (as thick as mustard or a little bit thicker) over the medal, which has already been lightly rubbed with oil, walnut oil or, even better, aspic oil. For best results, ~~let it dry by itself, away from fire~~, for one or two days. If you are in a hurry, you can heat it; it should not break if the coat is quite thick. If it is suddenly exposed to fire, some holes will appear; that does not happen if it has been gently dried. Once dry, reheat it, and it should allow for several castings.

.....

Sal ammoniac and iron dross

I pulverized sal ammoniac, ^{previously} well dried in a hot bronze mortar over heat. Then I filtered the salt through a double sieve; it became very fine. But to make it finer, I dry ground it on the porphyry, then I mixed it with iron dross which was pulverized and ground the same way. Both ingredients take a very clean cast of bas-reliefs, without being moistened with serain, or into the hollow mould, or anywhere else with the serviette, and the cast can be cleanly removed.

<div>
<id>p090r_a2</id>
<head><m>Sand</m></head>

<ab><pro>Molders</pro> from <pl>Foix</pl> who cast their medals sideways use <m>crocum ferri</m> and <m>calcinated</m> <m>slate</m>. It is used for very flat things.</ab>

</div>
<div>
<id>p090r_a3</id>
<head>Every kind of sand can be made useful</head>

<ab><pro>Artisans</pro> who do big works, and who need to search for things made by nature to pursue their work, because nature does not want to sell its wares to its children, and in order to save the time they would spend grinding, pulverizing and artificially preparing their sand, look for sand from sandpits, which is not too fat from the <m>soil</m>, nor too arid and, consequently, without any binder effect. From deep in the sandpit, one can find some lumps and gravel that show the sand's natural compaction; it is quite hard to break, with very small and fine grains, quite smooth between the fingers. And since it is found solely close to mountainous regions or arid lands[\[a\]](#), you cannot find any near every good town, where <pro>artisans</pro> live. Thus, if they can't find any close to their home, they prefer to order some of this sand from afar, like Lyon, Venice or Paris, near the Sainte Chapelle, and similar areas, instead of preparing it themselves. Nonetheless, you can be assured that in all locations, you can make sand from the sandpit</ab>

<cont/>
</div>
[\[a\]"participans de l'arene" is missing](#)

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<page>090v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f186.image</image>

<div>
<cont/>
<id>p090r_a3</id>
<ab>and will be good and will mold cleanly. Because the <m>sand</m> is coarse and dry, filter it through a sieve. If it is not fine enough, wash it. And when the <m>water</m> has settled a little, pour the <m>soiled</m> <m>water</m> into another pot. The <m>coarse [grains</m>] will quickly fall to the bottom of the first pot. But the <m>sand</m> from the <m>troubled water</m> pot, having settled, will be much finer. If this <m>sand</m> does not bind enough, dry crush it on the <m>porphyry</m>, thus making it impalpable, and it will bind as chalk does. If you wish you can reheat it, and straight after crush it and blend it with <m>salts,</m> or <m>burned cloth,</m> or <m>felt</m>, or <m>paper ashes</m>, or similar material.</ab>

</div>
<div>
<id>p090v_a1</id>
<head><m><pro>Potter</pro>s' clay</m>[a]<b href="#">[b]

<ab>Grind it in a mortar mustard mill with <m>water</m>, ^{and} it will become as impalpable as possible. Dry it, then moisten it with <m>salt water</m>, which give force to all sands in order to allow several castings.</ab>

</div>
<div>
<id>p090v_a2</id>
<head>Orange trees</head>

<ab>In Italy, ^{particularly} in the coldest areas like Lombardia, people plant the orange trees inside squared <m>wood</m> planters. The planters are larger at the bottom than at the top. People make handles on each side, where they attach straps in order to carry the planter, like one carries the gout sufferers, because if one used wheels to roll them one would damage the garden's paths. Every two year they remember to open the sides of the planters and meticulously cut and trim the ends of the orange tree's roots, which are dry. Otherwise, the roots find the <m>wood</m> [of the planter] and tighten on it, they double over themselves, the ends dry and the tree dies. Once trimmed, the roots gain new space to expand, without encountering resistance from the <m>wood</m> ^{of the planter}. That is why it is better to join the sides of the planter with screws rather than nails, to avoid shaking the <m>soil</m> when one opens them.</ab>

[\[a\]](#)Is the correct translation for "terre fondue" ? or might it also mean here more generally 'cast earth from the Potters' or "Potters earth for casting"?
[\[b\]](#)This is not terre fondu, which has a discussion on 127v; rather it's either founder's earth for casting pots, or potters' earth for casting? In either case, "clay" is not the best translation.

.....

<page>091r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f187.image</image>

<div>
<id>p091r_1</id>
<head>Grafting</head>

<ab>Any tree that has a large pith will be hard to be graft from, because the cutting is difficult to remove. If the pith is damaged, the graft will take in an unstraight manner, like with vines of all peach and apricot trees.</ab>

<ab>Apricot trees have a very thin <m>bark</m>, and so it is necessary to graft them onto young trees whose own <m>bark</m> is not thick yet, such as the young shoots of prune trees or apricot trees.</ab>

<ab>The graft will not take if the cutting gets wet. It is necessary therefore to graft in serein and mild weather, not too cold and not too windy.</ab>

<ab>One must use <m>old wood</m> to graft, if one wants to graft on a tree that is already mature and quite big; because if this is the case, it will compress the cutting,

and if the cutting is of <m>young wood</m>, it will not grow properly. But <m>old wood</m> resists the compression of <m>old wood</m> better than the young, because it is harder than <m>young wood</m>. But if the graft is done with only <m>old wood</m>, it will never grow beautifully, but will be frail and short and will take a long time to grow.</ab></div>

<div>
<id>p091r_2</id>
<head>Molding with <m>cuttlefish bone</m></head>

<ab>Pay attention to keep the <m>bones</m> in a dry place, because they are very prone to dampness. If you have a small medal, divide the <m>bone</m> into two pieces, then smooth the two parts with a knife. And on a dry, smooth <m>rooftile</m> [that has been] pulverized with <m>willow charcoal</m>, rub and polish the two half <m>bones</m>. In this way, the <m>bones</m> soak up the <m>willow charcoal</m> so that the <m>bones</m> will come off cleanly from the cast, and will not stick to it. Then on a <m>fake bone</m>, that is to say a bit of smooth <m>brick</m> of the same width as your <m>bone</m>, place your medal. And then on the medal, put your <m>bone</m> and press it firmly with another stone of the same size. And for a second time, mold it, and instead smooth over the top with <m>willow charcoal</m> and blow slowly, then press as before, and it will cast neatly. If it is for a spoon handle, two whole <m>bones</m> are needed. Any work that is cast is brittle and prone to breaking, because the <m>metal</m> swells when cast, and contracts and condenses when hammered. That is why it is necessary to retrace with a chisel the cast object and in this way the <m>metal</m> will contract again and let it escrouir[a][b]. If the piece you have to cast is in high relief, first trace the mold and the hollow with a penknife, to make a clear path for the medal and then cast it. And if the medal has two sides</ab>

<ab><margin>left-bottom</margin>Rather than casting, heat the <m>bones</m> only to make them lose the coldness and wetness.</ab>

<ab><margin>left-bottom</margin>When the <m>lead</m> gets too hot, it calcinates.</ab>
<cont/>
</div>

[a]We were almost unable to translate this, the only option we felt might make sense was the definition by Furetière for escroui, adj: "est un terme de Monnoye, qui se dit des pieces durcies à la sortie du moulin, et qu'il faut faire recuire. On le dit aussi de l'or, de l'argent et du cuivre long-temps battu à froid, ensorte qu'il fasse ressort." This is still hard to translate

[b]This still needs to be translated.

.....

<page>091v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f188.image</image>

<div>
<cont/>
<id>p091r_2</id>
<ab>Because the two <m>bones</m> are joined, cut around them with precision, and make notches in different places which cross through where the two <m>bones</m> are joined, so as to recognize the place where they were first joined. Or with little shards of <m>wood</m>, you can also pierce the <m>bones</m>, or coat the joints with <m>clay</m>, and leave it to dry. Compress between them it with your two hands in between your knees; and do not use a ^{normal} press, otherwise the <m>bones</m> will break. Mold at your leisure two or three times, until you can see that it has the design come up[a][b][c] and is well imprinted. And on each occasion rather than putting the piece straight back in the mold, smooth with <m>willow charcoal</m> because it helps it to come out easily. Once you have cast, rub your medal with <m>oil</m> and <m>willow charcoal</m> with a brush, to make it dull. Take care that your medal is not greasy or oily when you cast it in <m>cuttlefish bone</m>. Once you have cast, leave it to cool rather than taking the medal out straight away, because when one takes it out hot, some <m>bone</m> will come away with the medal. Cast at medium heat, which you will recognize because the bone should stay almost white. But when it is too hot, the <m>bone</m> becomes brown. The proportion of <m>lead</m> and <m>tin</m> is equal, one to the other. If the piece is difficult to cast, and composed of various pieces, cast it with a forked pattern, of three or four grooves, and make and make these grooves where the <m>metal</m> is the thickest.[d] Also make these grooves around the medal and directly on the head because this attracts the <m>metal</m> and will fill the figure best. Lift your boxmold a little, once you cast, and in this way you will be able to make several clean casts. <m>Tin</m> that is too hot burns the <m>bone</m> which becomes damaged. Moreover, having come out of the mold hot , it risks breaking.

<figure>
<id>fig_p091v_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5Q2hHbzNsX1JIYUU</link>
</figure>
</ab></div>

[a]In French this is arracher.

[b]ms has, "acache(r)," Littré gives, "arracher": (v.) ôter ou enlever quelque chose qui adhère. Littré, <http://www.littre.org/definition/arracher>

[c]Assuming in our translation that the author made a spelling error

[d]Unclear whether this is referring to the gate or the vent.

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<page>092r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f189.image</image>

<div>
<id>p092r_1</id>
<head>SandMixture easy to melt</head>

<ab>One 3 of part <m>soft tin</m>, one part fine <m>lead</m>, one part <m>looking-glass tin</m> and one part <m><fr>argent vif</fr> arrested[a]</fr></m> makes an alloy and a solder, so easy to melt that it could easily be melted in a dish of fine <m>tin</m>.</ab>

</div>
<div>
<id>p092r_2</id>
<head>Molding hollow on one side and in relief on the other</head>

<ab>And for this effect, one casts a medal of fine <m>tin</m>, which is harder to melt. And since it is neat, one molds with it. And one leaves it in one half of the box mold, & presses it a little so that it holds better. And next, you shall cast in your box mold the solder described above, or another more meltable than fine <m>tin</m>. And

thus, the second medal will melt and mold itself on the first one without ruining it. But to be s**[b]**afer, temper <m>lamp black</m> with <m>water</m>, & with a <tl><fr>pinceau</fr></tl>, give a light coat of this to the medal, which remains in the box mold, & leave to dry. Thus it will not melt.</ab>

<ab>But if you have a medal of <m>copper</m> or <m>silver</m>, you can leave it in the casting, if you want to cast with <m>lead</m> or <m>tin</m>; but it is necessary that it be a little hot, for the cold would make the <m>tin</m> contract.</ab>

<ab><m>Tin</m> needs to be cast quite hot to come out neat.</ab>

<ab><m>Tin <fr>doux</fr></m>, which is the best for the cast, is the one that once cast <fr>en grille</fr>, is burnished & shiny & polished like a mirror, & appears to have been burnished, & does not have holes like the one that is brittle, & that is not shiny, as if burnished.</ab>

</div>
<div>
<id>p092r_3</id>
<head><m>Potin</m></head>

<ab><m>Potin</m>, once cast very hot, runs and drips like <m>tin</m>. But the medal becomes whitish and immediately casts off the green, which good substances do not do.</ab>

</div>
<div>
<id>p092r_4</id>
<head>Molding statues</head>

<ab>Half <m>founder’s earth</m>, which <pro>founders</pro> use, and half <m>clay earth</m> renders very neatly.</ab>

</div>
[a]Cf. Cardano's De Subtilitate: in the French trans. by Richard le Blanc (n.d.) fol. 146v, use of "argent vif arrêté." In Latin original, "firmari" or "extinguatur." In the English trans. by John Forrester, 2013, "solidify." The French translation unifies several different Latin names as "arrester", but Cardano says that it is not the same "to solidify" as "to stop it from flowing", something that the French trans. confuses on occasion. For Fr. 640, the point is to find a alloy of low melting point so that it can be used in the casting of the hollow-relief medals below and will not fuse with the medal being used as a model.
[b]pour être certain de ne pas gâter la médaille d'origine : "faire sans risque"

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<page>092v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f190.image</image>

<div>
<id>p092v_a1</id>
<head><m>Sand</m> of <m>river tellins</m> **[a][b]**and <m>mussels</m></head>

<ab> The long <m>shells</m> that can be found in rivers of <m>fresh water</m>, being calcinated, make a <m>white and very fine [impalpable] sand</m> which moulds very clean..</ab>

</div>
<div>
<id>p092v_a2</id>
<head>Freezing <m>mercury</m></head>

<ab>It can be frozen, as I tested, letting it boil one or two hours in a rather large <m>iron</m> pan putting, for each ʒ of ʒ, ii ʒ of <m>vert-de-gris</m> and <m>vitriol</m> **[c][d]**of <m>saltpetre</m>, each in the same quantity**[e][f]**, with enough of the <m><pro>blacksmith</pro>'s old water.</m></ab>

<ab>Or melt 4 ʒ of <m>tin</m>, and when it has cooled, purify it by removing the <m>filth</m> on the top, then make a hole and inside there add i ʒ of ʒ and if you remelt it, it will be like tin but it will be fragile. If you want to know if it is <m>tin</m> or ʒ, redder a shovel and put a small piece of your ingot on the top. If it is <m>tin</m>, the piece will melt but remain, but if it is ʒ, it will melt immediately then, crackling quite hard, it will evaporate. It must be frozen in a ladle or in a hollow crucible, and make the hole medium sized.</ab>

<ab>It can also be frozen, as I tried, that is to say with <m>aqua fortis</m> which before corroded a bit of <m>silver</m>. This, mixed, will make <m>tin</m> and <m>lead</m> fluid.</ab>

</div>
<div>
<id>p092v_a3</id>
<head>Mixture</head>

<ab><m>Pure tin</m>, <m>frozen mercury,</m> smells like tin. <m>Iced tin</m>, <m>pure lead</m>, as much of one as of the other, makes a substance that melts very quickly but that is fragile and white. I think that it would be good for soldering. [Takes a] long time to cool.</ab>
</div>

[a]Tellin: any marine bivalve mollusk of the genus Tellina, having a thin, rounded shell of white, yellow, pink, or purple.

<http://dictionary.reference.com/browse/tellin?s=ts>

[b]ms has "telines," Littré defines "Tellinès": (f.) Genre de coquilles bivalves. Littré via ARTFL, <http://artflsrv02.uchicago.edu/cgi-bin/dicos/pubdictlook.pl?strippedhw=telline>

[c]Vitriol:

Chemistry . any of certain metallic sulfates of glassy appearance, as copper sulfate or blue vitriol, iron sulfate or green vitriol, zinc sulfate or white vitriol, etc.

<http://dictionary.reference.com/browse/vitriols>

[d]ms has "vitriol," Dictionnaire de l'Académie Française gives "vitriol": (f.) Espece de mineral qui est acide. Vitriol commun. vitriol Romain. vitriol blanc. poudre de vitriol. esprit de vitriol. Dictionnaire de l'Académie Française (1st ed.), via ARTFL, <http://artflsrv02.uchicago.edu/cgi-bin/dicos/publicdo1look.pl?strippedhw=vitriol>

[e]Ana: Mot employé par les médecins dans leurs ordonnances, et signifiant autant de l'un que de l'autre.

ÉTYM. Du grec, indiquant répétition

<http://artflsrv02.uchicago.edu/cgi-bin/dicos/pubdicollook.pl?strippedhw=ana>

[f]ms has "Ana," Dictionnaire de l'Académie Française gives, "ana": Ana, est aussi un mot employé dans les ordonnances des Médecins, pour signifier, Quantité égale de drogues qu'on mêle ensemble. Dictionnaire de l'Académie Française (5th ed.), via ARTFL, <http://artflsrv02.uchicago.edu/cgi-bin/dicos/pubdoc1look.pl?strippedhw=ana>

<page>093r</page>

<div>

<id>p093r a1</id>

<head>Olives</head>

<ab>The olives go bad if you do not refresh their <m>salt water</m> once a month, do not touch it with your hands, otherwise they will go bad.</ab>

<div>

<id>p093r_a2</id>

<head>Country folk's observation</head>

<ab>When it freezes on Palm Sunday, it will freeze every month of the year</ab>

<ab>When the first flowers fall because of the frost, the other flowers also start perish.</ab>

<div>

<id>p093r_a3</id>

<head>Grafts</head>

<ab>Do not use a graft from <m>old wood</m>, because it never makes a beautiful tree.</ab>

<ab>Leave only two shoots on the graft and no more, otherwise it will become weaker. The wind torments it and leaves it strengthless to propagate by cuttings.</ab>

<ab>Trees which have big pith, such as peach trees, and all other trees of their kind, and the plum trees, can't be grafted when freezing because the cold damages their pith, which also has to be protected avoiding to cut the bottom of the graft.</ab>

<ab>All grafts must not be done earlier than the sap flows up, otherwise they dry.</ab>

<div>

<id>p093r a4</id>

<head><m>Sand</m></head>

<ab>The <m>white ash</m> of all kinds of <m>wood</m>, which still sticks to the <m>wood</m> while burning, and which has not fallen into the hearth, molds very clean</ab>

<div>

<id>p093r a5</id>

<head><m>Sand</m></head>

<ab>Very fine pulverized <m>soot</m> molds and removes easily from the mold. The best <m>soot</m>, which is greasy and shiny, is the one from the big kitchens</ab>

<ab>Very fine pulverized <m>quicklime</m> works, but mixed <m>flint[al,</m> crumbled into the hollow, does better.</ab>

[a]ms has "pierre vive," Cotgrave has "pierre vive": flint. Cotgrave, Dicfro, <http://www.micmap.org/dicfro/search/cotgrave/pierre>

.....

<page>093v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f192.image</image>

<div>
<id>p093v_1</id>
<head>Faces painted with <fr><m>distemper</m></fr></head>

<ab>Because colors painted with <m>distemper</m> dry quickly and one would not have enough time to complete the shadows and touches on the face, one wets the reverse of the face, on <m>canvas</m>, with a wet <tl>sponge</tl>. Then, with a small <tl><fr>pinceau</fr></tl>, one makes the shadows, which won't come off. Next, one lays the flesh tones, which do not prevent the already painted shadows from appearing. And then, one retouches the more prominent areas with lively flesh tones. And with another <tl><fr>pinceau</fr></tl> one retouches the shadows in hatching. The shadow strokes do not come off because they are made of <m>bister</m>, which stains the <m>canvas</m> like rust. The said <m>bister</m> is good for making shadows with <m>distemper</m>, for in <m>oil</m> it has no body and would dry with great difficulty. One mixes <m>ru ochre</m> in the said <m>bister</m> for shadows with a little <m>sap green</m>. The best <m>bister</m> is the rich, shiny one from the fireplaces of large kitchens. It is difficult to grind and screeches on the <m>marble</m> slab.</ab>

</div>
<div>
<id>p093v_2</id>
<head><m>Azure</m></head>

<ab><fr><m>Azur desmail</m></fr> always wants to be cleaned because the filth that can be seen in the used water ^{would} makes it die. It always needs to be coated twice, & first very thickly, driving the <tl><fr>pinceau</fr></tl> by laying it lengthwise then across. It is better used on <m>canvas</m>, which imbibes it immediately, than on <m>wood</m>. <m>Varnish</m> restores its vigor, because while imbibing, it becomes dark. To try it, <pro>painters</pro> bring their palettes to the <pro>grocers</pro> and distemper and mix @<a> it with a bit of <m>white</m> crushed in <m>oil</m>, for in this way the beautiful shows its <fr>turquine</fr> vivacity, while the mediocre is lavender grey. The thinner one is best to work with. It is made thinner by being washed.</ab>

</div>
<div>
<id>p093v_3</id>
<head><m>Purple</m> and <m>lake</m></head>

<ab>It is made of <m>azure</m> and <m>lake</m> which is also tested on the palette with <m>white</m>. The one which gives a clear red rose color, tending slightly towards purple, on a <m>polished knife</m> or on <m>glass</m>, is esteemed beautiful. The dark red one is not as pleasant.</ab>

</div>
<a>French: allié, in the sense of 'alloy'

.....

<page>094r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f193.image</image>

<div>
<id>p094r_a1</id>
<head><pro>Burnisher</pro></head>

<ab>The different parts of a sword are the handle, [and] the sheath of the sword. What comes next is the heel. The rest is the blade. The sides are the edge and the point. Certain blades, acute in point, have only one side filed in the middle & along the whole length, and are easy to break. Other blades are called of three molds or three grips, which have one good rise in the middle, but this elevation is flat as if it were acute in point but flattened. These ones are most certain. Others are called fluted, which are notched from the middle, & when it is along the whole length, they are just as easy to break as those acute in point and are more troublesome to burnish because the burnishing stick<a> can't get in. But they make one specially that is narrow.</ab>

<ab>Making up the hilt of the sword are the pommel, the branches of the hilt and the cross-guard, which is this <m>iron</m> strip that closes off the guard [and] that is at the end of the heel to stop thrusts from sliding down the hilt. The rings are these two branches in half round that start from the eye of the guard and go all the way to the branch of the cross-guard. The branch that crosses the hilt is called the body and this escutcheon, by which the sword enters the sheath and to which all the branches return, and which holds them, is called the eye of the guard.</ab>

<ab>Following is the <m>wood</m> of the grip which one <m>glue</m>s or more fittingly [uses] some <m>mingled wax</m> which is made of <m>wax</m> and <m>pitch</m>, for <m>rosin</m> would be too hard. He heats it lightly, then rubs the <m>wood</m> of the grip so that the tang takes hold, or the <m>threads</m>, otherwise, when a <m>thread</m> frays off, the whole thing will break all at once. With <m>iron</m> thread or <m>dog skin</m>, one also uses <m>glue</m>. The trimming set on the <m>wood</m>, which is made of <m>silk</m> or <m>thread</m>, is called the cord, which is made from two or three <m>threads</m> twined on the spinning wheel, or 4 if the <m>silk</m> is thin. A cord a little bigger holds better. The rivements, which are also made of <m>silk</m> at both ends of the grip, are called the buttons.</ab>

<ab>Some grips are made of <m>silk</m>, <m>seal skin</m><e>, <m>iron</m> thread reheated with <m>gold</m> and <m>fine</m> <m>and false silver</m> thread & <m>velvet</m> [thread]. <m><d><e>Iron</m> thread is of less price and is most durable. Next is that of <m>silk</m>, if one does not have the convenience of being close to the sea in order to recover some <m>dog skin</m>, which is quite convenient. The good <m>skin</m> costs fifty or lx s{ous} and makes 4 or five dozen grips. This one gives a good grip and a sure hand. To put it to work, if it is too hard, soak it for one or two hours in <m>slightly lukewarm aqua fortis</m>. Because if it were too hot, it would boil and spoil the <m>skin</m>. It is sewn with <m>black thread</m>.</ab>

<figure>
<id>fig_p094r_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5YTIFNHRBSVRrOEUE</link>
<margin>left-middle</margin>
</figure>
<cont>
</div>

<a>fustée: ça veut dire "fusil d'aiguillage"
Can't find an appropriate definition in one of our online resources.

[c]This is "peau de chien de mer"

[d]previous translator used in purpose the singular form, to match with the manuscript, suggests maybe plural form is better?

[e]We took the plural form

.....

<page>94v</page>

<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f194.image</image>

[a]

<div>

<cont/>

<id>p094r_1</id>

<ab>The <m>gold</m> handles are made with some <m>yellow thin cloth</m> set below or other kind of <m>chalk-yellowed fabric</m>. And this <m>cloth</m> is set on with <m>glue</m> or aforesaid <m>gum</m>. This is in order to make the <m>thread</m> hold better on it. These latter are made of cordons like the <m>silk</m> ones and between two cordons are put one or two <m>threads</m> to enhance it. The <m>silver</m> one is identically made. And there is as much difficulty as for the <m>silk</m> one, not because <m>silk</m> is difficult to twisted, but it needs to be twisted twice. A fine <m>gold</m> handle is worth 28 or 20 cents.</ab>

<ab>The best sword point is of <m>sage foliage</m>.</ab>

<ab>This long stick on which the sword is laid down and attached in order to furbish is called the chameau. It is commonly made from <m>rowan</m> which is hard and even.</ab>

<ab>The stick underneath which is folded bow-like is called the arch[b].</ab>

<ab>The stick above is called stick of the fustée[c][d].</ab>

<ab>And this square piece, used on the sword to burnish it, is called the fustée[e].</ab>

<ab>There are two horns, one is called the <m>oil</m> hornet and the other the <m>emery</m> hornet. </ab>

<ab>There is an <m>iron</m> tool with the shape of a halberd point, square and made of very even and well limed <m>steel</m> called the grateau[f] which helps to soften lime strokes on swords hilts and to sharpen blades once they are softened which is better than with <m>stone</m> or other tools[g] which leave some scratches.</ab>

<ab>To make the swords cut better, the edge from the point is also sharpened.</ab>

<ab><pro>Those who create sword hilts</pro> are different craftsmen than <pro>furbishers</pro>.</ab>

<ab>The fustée is a three-finger-thick, square <m>wood</m> tool, made of <m>boxwood</m> which fit in the middle of the fustée stick to furbish.</ab>

<ab>The fresil stick is made of <m>willow wood</m> which is to [...] [h] weapon with some <m>clinker</m> which is the <m>iron scale</m> falling from the <pro>blacksmiths</pro>' forge.</ab>

<figure>

<id>fig_p094v_1</id>

<link>https://drive.google.com/open?id=0B9-oNrvWdlO5Q29MTEVLUGE5RGs</link>

<margin>left-top</margin>

<caption>le banc <lb/>

des fourreaux</caption>

</figure>

<figure>

<id>fig_p094v_2</id>

<link>https://drive.google.com/open?id=0B9-oNrvWdlO5OEQ3OEFSNnEzeE0</link>

<margin>left-middle</margin>

</figure>

<ab>

<margin>left-middle</margin>

<figure/> a<figure/> <figure/>b<lb/>

D<lb/>

<figure/><lb/>

le baston <lb/>

a felinder<lb/>

<figure/><lb/>

c<lb/>

le <m>flin</m> ou <m>pierre</m><lb/>

de fouldre</m></ab>

<cont/>

</div>

[a]Is there a reason behind the order of anonymous blocks, notes, and images on this page? If not, it needs to be changed.

[b]a small bow

[c]?

[d]Lacking translation.

[e]?

[f]a scraper

[g]?

[h]"destrouiller"

.....

<page>095r</page>

<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f195.image</image>

<div>

<cont/>

<id>p094r_a1</id>

<ab>[a]The <pro>furbisher</pro> buys his blades by the dozen which are not finished, [but ready] to be pulled from [their] sheaths since he does this himself in order to add them well at the pommels and guards.</ab>

<ab>Spanish swords are esteemed to be better for they are of better <m>steel</m> and <m>iron</m> but they are not as well-forged as Vienna swords, in Dauphiné. The Spanish ones are not as well sharpened from the forge since they sharpen them with their feet[b] and this is what makes them wavy[c].</ab>

<ab>Vienna blades cost commonly Xviii or 19 lb.</ab>

<ab>The first thing the <pro>furbisher</pro> does to his new blades is to draw them out from the sheath, which he does himself or has it done by <pro>locksmiths</pro> or <pro>blacksmiths</pro> to whom he gives some liart[d] for their <m>charcoal</m>. Afterwards they pass it over a grindstone to cut it down. Then they lay it down on the chameau<m>,</m> and with some powdered <m>emery</m>[e], fine and soft as flour, and soaked with <m>oil</m> to make it take, they polish the sword with the stick used to take off the tracks left by the grindstone and then they clean the blade well with the <m>emery</m>. And they add a drop or two of <m>oil</m> spread with the finger to give it luster. Once the <m>oil</m> is applied, they polish it again on the chameau[f][g][h] with the "felin" which is a thunderstone mounted on the middle of a stick like the "fustée." And this stick is called the "felindel" stick. Once they used the "felin," they polish it again with some <m>chalk</m> and the <m>oil</m> [already] on the sword. After they follow this and going over it again, they polish it with dry <m>chalk</m>. Finally when they are finished with mounting [of the pommel and guard], they give it a sharp edge with a file. They then dry it and sharpen it and furbish it again with some dry <m>chalk</m>. At the end when they are completed and mounted, they give it [the blade] a sharp edge with a grater[i].</ab>

<ab>They are polished on a false mounting with the guards which are present in order to find out if they[j] work well together. And employing a blade of the sort of which all the sizes should be made[k] in the hand, one conjectures whether it will be as strong when mounted.</ab>

<ab>One puts the blade in a vice between two pieces of <m>wood</m>, then with a file, they enlarge [it] if needed for attaching the guard. Then to clench it, one positions the place to be riveted and the pommel on a piece of <m>wood</m>. And with a hammer, one beats on the top to make it very level and to secure the pommel. Then with the hammer, one finishes the rivet when the hammer is well secured and does not shake. The <m>wood</m> is placed so as not to put gashes in the pommel. Afterwards one makes a place for the rivet with a file or chisel. And some make the rivet [or attachment] in a diamond-shape but this is not as good as a round [topped one] because one cuts oneself on the corner of the rivet.</ab>

<cont/>

</div>

[a]This page is a continuation from <title id="p094r_a1">Burnisher</title>.

We have altered encoding protocol here, as the number of anonymous blocks surpasses the number of letters in the alphabet.

[b]I'm assuming they are using some sort of grindstone worked by a foot pedal

[c]or could be translated as "found to be twisted"

[d]Cotgrave: "a brazen coyne worth three deniers, or the fourth part of a sol."

[e]Emery stone, OED. — 1610 P. HOLLAND tr. W. Camden Brit. II. 225 That most hard and sharpe stone Smyris (which we tearme Emerill). — 1646 SIR T. BROWNE Pseudodoxia Epidemica II. iii. 69 The Magnet attracteth..the Smyris or Emery in powder. — 1678 J. MOXON Mech. Exercises I. I. 15 Make it..smoother with Emerick. See also <http://www.mindat.org/min-9093.html>

[f]This could be chamois cloth or the soft skin of a camel <http://www.micmap.org/dicfro/search/dmf/chameau> or <http://www.littre.org/definition/chameau>. I wonder if there was some sort of stand called a chameau or camel from the way the word is used, but I have found no evidence for this. I checked Cotgrave, atifl, and the ARTFL Encycl. (which also listed chameau/camel parts as materia medica)

[g]This is the "chameau" that he defines on 94v = "This long stick on which the sword is laid down and attached in order to furbish is called the chameau."

[h]Thank you! I liked getting the comments on our work.

Merci,

T
[i]One does not want the edge to be too sharp because then it is more liable to break. See <http://bit.ly/1mkILqg>

[j]the blade and guard

[k]comprised

.....

<page>095v</page>

<id>094r_1</id>

<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f196.image</image>

<div>

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<id>p094r_a1</id>

<ab>[a]The end can be bought already made, a dozen of small ones costs six sols[b] but the big ones 2 carolii[c].The end is fixed either with nails which are fixed on the sides of the cutting edge, but this only spoils the sword and rots the scabbard for the water goes into it via the juncture of the nail. The best way is to apply some <m>resin</m> or <m>glue</m> but the <m>resin</m> is better. And the best is when the end is tightly set and so hot when set that powdered <m>resin</m> on the end melts, in that way it doesn't fall and can't be undone unless it is put in fire.</ab>

<ab>Hilts are of different types:

Ornate

Guttered which is with round mouldings

<m>Pearled</m>

Scarfed when the bands are crosswise
Onioned which is with a flat head
In the King's manner, fully covered </ab>

<ab>The <pro>furbishers</pro> buy them by dozens, the dozen of full ones costs 10 lb. the worked one, 30 sols or [more] depending on its nature.</ab>

<ab>The first thing that the <pro>apprentice</pro> does is to furbish as said.</ab>

<ab>And then to adorn the sword and make a scabbard which is the summum of the art.</ab>

<ab>They buy scabbards' <m>wood</m> pieces which are wholly made of <m>beech wood</m>[d], a hundred for 15 or 20 or 30 sols, depending on how far they are made.</ab>

<ab>Theses <m>wood</m> pieces are thus called estelles[e], and have to be very clean and without any knots and are one finger thick.</ab>

<ab>Then the <pro>furbisher</pro> puts them on a small bench, called a scabbard bench and with a small <m>iron</m> tool similar to the <pro>joiners</pro>' bench, they maintain it firmly. Then with a plane, which is like a two-handled knife, they work the <m>wood</m> piece from the top, then use the <pro>joyner</pro>'s plane[f] to flatten it more. Afterwards, the inside is scoured with a</ab>

<cont/>
</div>

[a]This page is a continuation from <title id="p094r_a1">Burnisher</title>.
We have altered encoding protocol here, as the number of anonymous blocks surpasses the number of letters in the alphabet.
[b]10 French sols (sol tournois) = 1 English shilling. RC.
[c]Coin which is worth 10 deniers tournois, used as an account money until the 18th century. RC.
[d]Fayan: Cotgrave gives Beech wood.
[e]A kind of rod
[f]"plane" and "rabot" are translated to joyner's plane.

.....

<page>096r</page>
<id>094r_1</id>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f197.image</image>
<div>
<cont/>
<id>094r_1</id>

<ab>
round <tl>plane</tl>, also two-handed, and then with a small narrow <tl>plane</tl>, it is gutted and scoured for half a foot towards the end of the heel. And a shorter scouring is better so that the sword does not always cut the scabbard.</ab>

<ab>
The opening of the scabbard is called the mouth and on top of it is the shape of the knife.</ab>

<ab>
The stitch which is a <m>leather</m> rim set on the edge of the knife shape is made to stop the sword frogs.</ab>

<ab>
When the small planks are well flattened, the sword is laid down on them and with a <tl>black stone</tl> one marks the width and length of the blade. Then with a <tl>chisel</tl> the excess is cut. And a <tl>plane</tl> is used to even properly following the marks.</ab>

<ab>
Then the two small planks[a] are stuck edge to edge <add>and all around</add> only following the knife shape and not elsewhere. And this is to make the scabbard <m>velvet leather</m> hold.</ab>

<ab>
Then the sword is put in the small planks and it is set in the scabbard which is sewed. And they rub what is glued directly on the knife shape with some <m>tallow</m> to run into the scabbard.</ab>

<ab>
The scabbard is made of <al>calf</al> or <al>sheep</al> <m>leather</m>, the <al>calf</al> ones are in all one piece, but not the <al>sheep</al> ones. When swords are long, they are made of several patches sewn between <m>leather</m> and <m>flesh</m> and then they rub firmly the stitching with a <m>boxwood</m> piece to cover and even it. From a good <al>calf</al> <m>skin</m>, 8 or 9 or 10 scabbards can be made. A <pro>workman</pro> can easily make 6 a day. If they are too thick, they are lightly thinned out with a plane. If the <m>skin</m> is too hard, it is lightly dipped in some lukewarm <m>water</m> but not for long. Next, the same worker who worked <m>wood</m> piece has to adjust the scabbard in order for it to be well done. It is true that for common people, there are some good already all made, but they never fit perfectly. When the scabbard is made and sewed with black <m>thread</m>, it is rubbed with a <m>boxwood</m> piece or a slightly <m>oiled</m> cloth . The good <al>cow</al> <m>leather</m> scabbard is the best of all and costs also more than all except the <m>velvet</m> ones. Under the <m>velvet</m> one usually put plain <m>canvas</m> or a <al>sheep</al> <m>skin</m> scabbard or a <m>parchment</m> layer. But the best scabbard is a waxed scabbard covered with either <m>velvet</m> or <al>calf</al> <m>skin</m>. One covers the wood with thin canvas, which one then waxes, and covers in <m>velour</m> or <al>calf</al> <m>skin</m>. A <m>waxed</m> scabbard is worth at least 30 cents, a <al>cow</al> leather one, 20 cents. </ab>

<ab>
<margin>left-bottom</margin>
From a <al>sheep</al> <m>skin</m>, v or vi.</ab>
</div>

[a]the author has written estoilles, but meant estelles.

.....

<page>096v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f198.image</image>

<div>
<id>p096v_a1</id>
<head>Grey hilts</head>

<ab>When they are white, it is necessary only to clean them well, and then to heat them over a fire, and make them turn blue. And take good care that they do not turn too blue, since neither <m>gold</m> nor <m>silver</m> nor any color would set. To accomplish this, which is called <fr>les faire passer</fr>, have a caldron full of <m>ashes</m> that you cover with some <m>blazing hot charcoal</m>, and heat the <m>ashes</m> thus very well, for an hour or two. And when the <m>ashes</m> are very hot, put the <m>charcoal</m> aside and put the hilt where the <m>charcoal</m> was, and cover it completely with <m>ashes</m>. And when the body, which is the main part of the hilt, is completely colored, flip it onto the other side. In that way, the grey cannot color it too dark, unless some <m>hot coal</m> touches it. Greying can be achieved in an ordinary fireplace, too, but there is a danger that it may sit there too long and make a crust which which will prevent the color from setting. Blue coloring is obtained as soon as the <m>ashes</m> have heated the hilts a little bit, but it will always be grey on some sides. After blue comes grey.</ab>

</div>
<div>
<id>p096v_a2</id>
<head>Color of water</head>

<ab>For the color of water, pass a <tl>soft file</tl> on the hilts to whiten them, then the <tl>burnisher</tl>. Afterwards, you color them blue until they become grey. Next, when they are still slightly hot, pass <m>bloodstone</m> over them, and, as they cool, one turns them to reheat them, since the color of water is achieved only after doing this a few times. And if hilts weren't warm, the <m>bloodstone</m> would scrape them and undo the color. Pay attention that hilts don't become too grey, since they would give off soot that would prevent the color from setting. Almost one day is necessary to prepare, soften, and set the color of water, but as soon as they turn a little blue, it is enough for grey or other colors. Grey is the lesser color, after varnish made of two colors, either yellow or black.[a]</ab>

</div>
<div>
<id>p096v_a3</id>
<head><m>Yellow varnish</m></head>

<ab>One should vigorously whiten and clean the hilts well. After that, with a feather, one coats them entirely, when they are cold, with <m>nut olive oil</m>. Then, turn them over a good <m>charcoal</m> fire like that of the <pro>furbishers</pro>' furnace without letting them touch the <m>charcoal</m> or <m>ashes</m>. And when they stop smoking, it is done.</ab>
</div>

[a]It seems that there are several possible ways of punctuating this part in the manuscript, and that the meaning is significantly changed depending on the punctuation. This could also be: "it is enough. (As) for grey or other colors, grey is the least of them. Afterwards, [use] varnish that is of two colors, either yellow or black."

.....

<page>097r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f199.image</image>

<div>
<id>p097r_1</id>

<head><m>Black varnish</m></head>

<ab>This is explained at the beginning of the book where you can see that it is being made with some <m>galipot [a]</m>
</ab>
</div>

<div>
<id>p097r_2</id>

<head><m>Gilding</m></head>

<ab>One prepares the hilts very well with a soft, ^{indeed} very soft <m>file</m> until there is no blackness left on the hilts, then heat them, and pass them as has already been said. But to <m>gild</m> and <m>gild</m> with <m>silver</m>, they must not be passed ^{change their color} too much, except maybe leaning towards a blue. Then while the hilts are hot, one ties them to a wood block[b] and with some tweezers you apply the gold and fasten it with the <m>sanguine</m> and the <m>sanguine</m> is very neat and has been rubbed with <m>putty</m>. And when the hilts become cold, reheat them, then apply <m>gold</m> as said above. And you must start with giving the hilt two layers, until it is well-covered. And then reheat it as you did at beginning, and give it its third layer of <m>gold</m>. Then burnish it with a <m>sanguine</m>, very strongly and with good strength. But note that if you want to burnish well, your hilt must remain warm. And to know this, you bring it close to your nose and smell it. And because you cannot touch it with your fingers, take it out of the warm ashes with a small <m>iron</m> stick, which has been purposely made to fit into the eye of the hilt. And when you have finished burnishing it well, rub it well with a piece of white <m>linen</m>, and make sure your <m>gold</m> is thicker and stronger than the ones <pro>painters</pro> use. A quarter of a pound is worth fifty or lx s{ols}.</ab>
</div>

<div>

<id>p097r_3</id>

<head><m>Gilding</m> with <m>silver</m></head>

<ab>You must prepare the hilts and whiten them with a <m>file</m>, coarser than the one used for <m>gold</m>. And you want to pass them ^{change their color} to grey, as you have done for the <m>gold</m>. And since they are warm, you must double the leaves of <m>silver</m> and then divide them with a knife, and then apply them to the hilt with tweezers and a burnisher. And then heat them again, and apply ^{the silver}, until they are all <m>gilded</m> in <m>silver</m>. Then you must apply, for the third time, your doubled leaf as has been said, then burnish them very strongly and with good strength, not with the <m>sanguine</m>, but with a strong burnisher, just as you would not use a burnisher on <m>gold</m>.</ab>

</div>

[a]Organic compound, referring specifically to an impure resin of turpentine. It is obtained from pine trees by evaporation of the essential oil and once purified is called yellow pitch, white pitch or Burgundy pitch.
[b]We went for the broadest translation of "estoc."

.....
<page>097v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f200.item</image>

<div>
<id>p097v_1</id>
<head><m>Mastic</m> varnish dried a half hour</head>

<ab>
Some use 2 <figure>3</figure> of <m>mastic</m>, half <figure>3</figure> of <m>turpentine</m> and of <m>turpentine oil</m>, and <m>spirits</m> at your discretion since it evaporates when heated and nonetheless makes a more desiccative varnish. But I made it like this: I use <m>turpentine oil</m>, at my discretion, and add very little <m>turpentine</m>, since it remains humid and gets tacky[a] if we add too much, together with some <m>spirits</m> and heat the <m>oil</m> in a varnished bowl and, as it become very hot I add some finely ground <m>mastic</m> and pass one-third of the <m>oil</m> through a sieve and let this heat up until it is melted, which will occur quickly on hot <m>ashes</m>. Once it is all melted, try it on your knife, and if you see that it has too much body add a bit of <m>turpentine oil</m>, and if it does not, add some <m>mastic</m>. And so it is finished, and keep it well covered so that no rubbish gets in. When you want to make this, take care to separate and choose the <m>mastic</m> that is white and pure of any dirt and dust and black dross. And when you wash it and dry it, make it very white and clean and make it even better, because if you don't purge it of these dregs and marks they will be pulverised into it, remain within the varnish, and when you wipe it onto some white or flesh-tinted ^{surface}, it will seem as if it were spotted and dirty. Having properly selected, pulverise it in a mortar and pass it through a very fine sieve, and afterward mix it in <m>oil</m>, as stated above. But if you want to make it more carefully, take a drop of <m>mastic</m>, as you know, pulverise, pass it, and mix, and you shall have something very singular for small projects. Be well advised when varnishing to not breathe on it for it will make the varnish white and thick.</ab>

<ab>
<margin>left-top</margin>
It almost dries as you are working on it.</ab>

<ab>
<margin>left</margin>
We know that this varnish does not have enough body when it doesn't take well to an <m>oil</m> painting, since it is like <m>water</m>. Add, therefore, powdered <m>mastic</m> and heat it until it is good. This varnish is very white and beautiful not at all like the <m>aspic</m> one.</ab>

<ab>
<margin>left-middle</margin>
For some, instead of <m>turpentine oil</m>, use some <m>spike lavender oil</m>, which is not as good.</ab>

<ab>
<margin>left-middle</margin>
This varnish is to be applied cold on the canvas with a very clean fingertip, and spread quickly.</ab>

<ab>
<margin>left-middle</margin>
The Italians hardly varnish their paintings since they layer their paints very thick and these take a long time to be dry on the inside, even though they have a dry skin and crust on the surface.</ab>

<ab>
<margin>left-middle</margin>
The varnish is applied with one's finger in order to thinly layer it, because it will yellow afterward.</ab>

</div>

<div>
<id>p097v_2</id>
<head>To work nearby</head>

<ab>
It is a strange thing that the workers in this craft can only work if they have good breath, for if they have bad breath their work will crack when refired.</ab>

</div>

[a]see vocabulary document comment for entry on "attaquer"

.....

<page>098r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f201.image</image>

<div>
<id>p098r_a1</id>
<head><m>Varnish</m> for lutes</head>

<ab>They take a little <m>turpentine</m> and some <m>turpentine oil</m>, or <m>spike lavender [oil]</m>, and some <m>amber</m> pulverized and very finely sieved & proceed as with that of <m>mastic</m>, & they add some <m>dragon's blood</m> to color it and turn it red. And others some <m>terra merita</m> [a] to turn it yellow.
</ab>

</div>
<div>
<id>p098r_a2</id>
<head>Shores for a Garden</head>

<ab>You have to work the ground, then then it with a rake, and afterwards if it is dry you must <m>wet</m> it. Then stretch a cord clean on the ground, and along its length hit the cord with the back of a pruning knife, so that the cord marks a right line. Along its length you will make holes with an <m>iron</m> stake, where you will plant rosemary, its head lopped off & cut to measure and thinned out at the foot.</ab>

</div>
<div>
<id>p098r_a3</id>
<head><pro>Locksmith</pro></head>

<ab>To <m>tin</m> nails and other work, they file their work neatly, then in order to clean & degrease it well, it is boiled in <m>vinegar</m>. Then they dry it with <m>linen</m>, afterwards they heat it & pass some <m>resin</m> over it, which induces a shining skin like varnish. This done, they melt some pure & fine <m>tin</m> in their <m>tin</m>ning tool, which is a <m>square little iron box.</m> And when the <m>tin</m> is well melted and moderately warm, they throw some <m>resin</m> in to consume the scum, & then they temper the head of a nail or something similar for a bit, and having been left in for seven or eight pauses, they take it out and shake it, hitting it with some <m>iron</m> to make the the superfluous <m>tin</m> fall off, & clean it with a small stick wrapped in <m>tow</m> [b]. And if it is not well <m>tin</m>ned, you will hit it until it it hot with a piece of <m>resin</m> and do as before before and leave it to get cold. When the work cools down it becomes yellow, which is a sign that the <m>tin</m> is the right temperature. But it becomes blue if the <m>tin</m> is too hot. Things so <m>tin</m>ned are of a duration, & do not come undone at all, like the tinning done by leaves. But at the end of a year they can be clarified & renewed by heating & striking them.</ab>

<ab><pro>Leadsmiths</pro> say that making a lizard die in the <m>melted tin</m> makes the <m>tin</m>ning become very golden. Or else by adding <m>sal ammoniac</m>.</ab>
</div>
[a]turmeric
[b]Or "stuff." Translates "destoupes," a low quality linen-like fabric. Cotgrave notes the expression "Il a tissu lin avec estoupe," meaning "He hath mingled, or iumbled good, and bad together." The "lin" is good, the "estoupe" bad.

.....

<page>098v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f202.image</image>

<div>
<id>p098v_a1</id>
<head>Small dogs</head>

<ab>Small barbet dogs are recognizable thanks to their muzzle which is larger than that of other dogs.</ab>

</div>
<div>
<id>p098v_a2</id>
<head>Salted <m>artichokes</m></head>

<ab>One should put them fill and not too ripe in a small barrel full of <m>brine</m> [a]. Then can be served all year long, with salads, <m>oil</m> and <m>vinegar</m>, raw, as prepared in this way they take long to cook. Make sure to try your <m>brine</m> with an <m>egg</m>: if it is good, the <m>egg</m> will float in it. If it does not float, it means [the <m>brine</m>] is not strong enough and would not conserve the <m>fruits</m>.</ab>

</div>
<div>
<id>p098v_a3</id>
<head>Trees</head>

<ab>Those which carry <m>fruits</m> age early and do not last long.</ab>

</div>
<div>

<id>p098v_a4</id>
<head>Birds</head>

<ab>Those who have been fed at the spit do not sing among or the company of other birds, they do only when they have their own cage. To teach them to sing well, they have to be taken from their nest as young as possible, and be covered with a <m>duvet</m> or <m>rabbit skin</m>. That way, having not seen their father and known their voice, they learn better to sing the way they are taught to.</ab>

</div>
<div>
<id>p098v_a5</id>
<head>Melting</head>

<ab>The <m>potée</m> [b][c] is made of <m>burnt earth</m> which has served for the melting. It is the <m>earth</m> bourré bolvene [d][e][f] in which it [g]has been melted. It should be ground against a large <m>stone</m> slab, and of this one should make the first coat of the mantle [h][i] of the bell, where the letters and ornamentations are to be found. It does not require to be cooked again against <m>hay fire</m> and will mold very neatly.</ab>

<ab>The <m>earth</m> used for the molding of the core is likely to swell if you do not strongly rub the molded piece with a brush or the finger and if it is not well <m>oil</m>ed. And you need to do it one coat at a time rather than all at once.</ab>
<cont/>
</div>

[a]Saulmure (Cotgrave): Pickle; the brine of salt; the liquor of flesh, or fish pickled, or salted in barrells.
[b]Potée (Cotgrave): Brasse, Copper, Tinner, Pewter, &c., burnt or calcinated; also, a pot-full of any thing.
[c]Mélange plus ou moins liquide d'argile et de terres, déposé en couches successives pour fabriquer la carapace des cloches (moule extérieur).
[d]Find a translation?
[e]ms has "bourre," Cotgrave gives, "bourré": stuffed, or stopped with haire, flockes, etc; also, beaten, thumped, cudgelled; also, stopped. Cotgrave, Dicfro, <http://www.micmap.org/dicfro/search/cotgrave/bourr%C3%A9>
[f]Still need translation for "bolvene"
[g]I believe "it" refers to a bell. This section (this paragraph and the next) is concerned with bell casting
[h]Chape: C'est la partie supérieure du moule, celle qui va recouvrir la fausse cloche. Elle est également en terre et formée de couches successives. Les premières couches sont obtenues au moyen d'une terre très fine, presque liquide, que l'on nomme " Potée ". Ensuite, on continuera la fabrication de la chape avec de la terre glaise, plus épaisse, armée de chanvre, qui assurera à l'ensemble une plus grande solidité.
[i]<http://www.paccard.com/fonderie/fr/clochesdeglise.php>

.....

<page>099r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f203.image></image>
<div>
<cont/>
<id>p098v_a5</id>

<ab><m>Earth</m> or crushed <m>wet sand</m> with <m>water</m> needs to be reheated more than <m>sand</m>.</ab>

<ab><m>Latten from a candlestick</m> is not pure since some <m>filings[a]</m> and <m>pot metal</m> are mixed into it.</ab>

<ab><m>Latten</m> needs to be cast warmer[b] than <m>copper</m>, which I do not believe[c].</ab>

<ab>To cast scrupulously, you must not take all of the <m>alloyed metals</m> together, but take them in [their] pure [form] and mix them.</ab>

<ab><m>Iron</m> should not [...] [d]to cast in <m>copper</m> or <m>brass</m>.</ab>

<ab>It is better to mold <m>sand</m> at once, either as a molding box or <m>noyau</m>[e], rather than several casts.</ab>

<ab>
<margin>left-top</margin>

You must reheat a crucible on a charcoal fire which easily ignites for one or two hours with the material inside. Then breathe on it and cover the crucible with large pieces of charcoal and the opening of the stove with bricks, and let it consume two or three lumps of charcoal which shall fill the stove beneath the crucible. Eventually put the molding box on the fire and when the flame is very white and the material is as light as <m>water</m> and spitting some pale blue flames, [make the] cast. The secret to casting well is to do so at a very high temperature and speedily.</ab>

</div>
<div>
<id>p099r_a2</id>
<head>Thin white <m>sand</m></head><ab> without bond is perfect for casting and it becomes thicker[f] if you crush it very finely on <m>porphyry</m> or you calcinate it by making it red-hot several times, and if it becomes <m>vinegar</m> or thinner with <m>water</m>, as you know, it casts as well very precisely in a core, as I did the experiment. But don't soak it with salted water if you want to anneal it quickly for salt boils on fire and creates more dust while it is being pressed by the fire. I think that reheated cores gently would not react like this. I cast very neatly as a core some matter and [...]. white <m>sand</m> smells <m>sulphur</m> when it is reheated, and I think that it would melt. And since the material has been cast into it, it takes in the mold a luster as if it were leaded or vitrified. I think that <pro>glassmakers</pro> could use it.</ab>

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<ab>
<margin>left-midle</margin>
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A very little <m>sal ammoniac</m> must be dissolved in water.</ab>

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</div>
<div>
<id>p099r_a3</id>
<head>
Bell handles</head>
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<ab>
They[g] can be repaired well without remaking the bell, which you must put back into the pit[h] and the mold inside. But it is thought impossible to give back to it its
original tone when soldering a cracked bell.</ab>
</div>
[a]"lame": A thin plate, esp. of metal; a thin piece of any substance, a lamina; spec. applied to the small overlapping steel plates used in old armour.
[b]less literal: "at a higher temperature"
[c]Latin
[d]Need to fix transcription
[e]unsure how to translate accurately in order to distinguish it from "chassis"
[f]"having more body"
[g]? replace with "bell handles" for clarity
[h]is this a technical term
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.....

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<page>099v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f204.image.r=Recueil%20de%20recettes%20et%20secrets%20concernant%20l'art%20du%20mouleur,%20de%20l'
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<div>
<id>p099v_a1</id>
<head>Quince trees</head>
```

<ab>They have to be pruned often; otherwise they die and do not produce good fruit.</ab>

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</div>
<div>
<id>p099v_a2</id>
<head>Cannons</head>
```

<ab>They are very difficult to drill when they are made of <m>metal</m> since the material is rough and brittle. But you must not push the drill too strongly or push all at once, but remove it often. In that case, the cannon mouth is bigger than the bullet in order to avoid drilling, since the material can be easily cast and slides. The drill should not be like a dovetail, or a point, but should be cut in round like a crankshaft. If it breaks, don't leave it to rust X</ab>

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<ab>
<margin>left-top</margin>
X[a] but instantly take it off from the light. What you will do is pour a little <m>oil</m> on it and then turn it down opposite to the light, strike the opposite side, and it
will fall out. You have to drill gently when a piece is made of metal.</ab>
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</div>
<div>
<id>p099v_a3</id>
<head>Bells</head>
```

<ab>They are not made in a pit if they do not fit</ab>

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</div>
<div>
<id>p099v_a4</id>
<head>To gilt frames paintings without <m>gold</m></head>
```

<ab>Some use some leaved <m>orpiment</m>[b] and then powder it and make the leaves look Moorish, burnish them, and the <m>orpiment</m> shines like gold. However, ground metals or ground crystalline or touchstone are the best.</ab>

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</div>
<div>
<id>p099v_a5</id>
<head>Orange Orpiment</head>
```

<ab>Mix it with <m>arsenic</m> and grind it. Some prefer to melt it and then add a little <m>arsenic</m>.</ab>

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</div>
<div>
<id>p099v_a6</id>
<head>Varnish</head>
```

<ab>Some make the <m>mastic varnish</m> with two ounces of <m>mastic</m> and one of clear, white <m>turpentine oil</m> and some <m>spirits</m> as aforesaid; heat that on <m>ashes</m> until it melts and then leave it still and put it in another vessel to remove dust.</ab>
</div>

[a]This isn't really a note, but a direct continuation of the ab.
[b]Orpiment: a deep orange-yellow colored arsenic sulfide mineral (As2S3)

.....
<page>100r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f205.image</image>

<div>
<id>p100r_1</id>
<head><m>Gold</m> color</head>

<ab>To very quickly make a layer of <fr><m>or surmat</m></fr> on <m>canvas</m> or on other <m>oil</m> panel, one [exposes] <m><pa>walnut</pa> oil</m> under the sun or on the fire to make it fatty, as previously mentioned, for it thickens by itself in this way, despite having neither <m>lead</m>, nor <m>ceruse</m>. And with it, crush <fr><m>massicot</m></fr>, <m>mine</m>, & other desiccants, but not <fr><m>vert-de-gris</m></fr> nor <m>orpiment</m>, for they make the <m>gold</m> die. It will quickly be dry. However, if the weather is not <fr>serain</fr> and dry, do not shade it, for if the weather is dewy & humid, it will turn black.</ab>
</div>

<div>
<id>p100r_2</id>
<head>Portrait-painting from nature</head>

<ab>One must carefully <sn>observe </sn>the <bp>eyes</bp>, the <bp>nose</bp>, and the <bp>beard</bp>, since these three things contribute much to the resemblance.</ab>
</div>

<div>
<id>p100r_3</id>
<head><m>Gemstones</m></head>

<ab>Your materials need to be pounded in a thick <m>glass</m> <tl>mortar</tl> coated with <m>mastic</m> & encased into another <m>wooden</m> <tl>mortar</tl>, so that it does not break; this can be useful to <pro>perfumers</pro>. & use a <m>glass</m> <tl>pestle</tl> too.</ab>
</div>

<div>
<id>p100r_6</id>
<margin>left-middle</margin>
<head><m>Glass</m> <tl>mortar</tl> and <tl>table</tl> for crushing.</head>
</div>

<div>
<id>p100r_4</id>
<head>Vitrified <m>salt peter</m></head>

<ab>
Melted with <m>pebbles</m>, it makes the <m>glass</m> very white. But rather, it needs to be calcined on a hot tile or <tl>paddle</tl>. And mix the liquid remains of the v[a]por with as much white, calcined <m>pebbles</m>. It is true that, in the little <tl>oven</tl>, it does not make it very clear. But for big windows, it makes the <m>glass</m> very clear, & white & clean.</ab>
</div>

<div>
<id>p100r_5</id>
<head>Windows through which the sun will not pass</head>

<ab>Take the aforementioned calcined <m>salt peter</m> & expose it to a humid air or place, and it will convert itself into a liquid which, if rubbed on both sides of the windows, the sun will not pass through; rather, it will cast shade. One needs to put the <m>salt peter</m> on any clean place, & put glowing <m>coals</m> on it, and continue all over until it no longer burns.</ab>

<ab>
<margin>left-bottom</margin>
It also serves for melting &for adding in with <m>alkali salt</m> for cleaning <bp>hands</bp>.</ab>
</div>

[a]He means the remains of the gas or vapor, now "condensed" into a liquid state, but “condensation” already implies a kind of atomistic view, so we have avoided using it here.

.....
<page>100v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f206.image</image>

p100v_1

Gemstones

Take white pebbles that are found along rivers, along gravel banks, & paths and that are somewhat transparent. If they are perfectly transparent, it will be better. If not, use the best & whitest ones that you can find. Calcine them three or 4 times in your four à vent & quench them in water or vinegar. Afterward, take a figure of them and pound them in a completely pure copper mortar, with a pestle of pure copper, and grind them until they become very fine and soft powder; this is a sign that they have taken enough of the substance of copper to give them greenness. Then in your figure of pebbles, add in three figure of good minium, unadulterated by brick or something else, and strongly pound everything together again in a bronze mortar. And in all of this, add in a of grosgros of sel de verre, & some and the common people put in sandever that they find it among the pro-glassmakers or pro-apothecaries. But alkali salt, as you know, is better.

left-middle

Emerald

In this way, the vulgaire makes emeralds & casts them in sand. If you do not have a bronze mortar, crush in a cauldron of pure copper.

left-top

The crucibles need to be reheated before putting materials in, and need to be put in the forge or four à vent instead of making & increasing the fire.

fig_p100v_1

<https://drive.google.com/open?id=0B9-oNrvWdlO5QUx4eF9qWU1jS28>

left-middle

They are done in an hour & a half.

left-middle

A of grosgros of salt in four figure of pebbles & minium. When one says: for figure, what is to be understood is: an figure of body & not of salts & minium.

p100v_2

For ruby take gold in leaves

Raise your stove by two tiles all around because more heat is needed to make rubies than emeralds. And take an figure of white, calcined pebbles, and put it in a glass mortar &, having roughly pounded it with the same kind of pestle, mix in the weight of a grain of gold leaves of the kind used by painters for gilding, & crush.

left-bottom

Some say that mixing the gold with pebbles and saltpeter gives the color of a peach tree.

left-bottom

Others say that the gold has to be cemented several times and then beat into leaves.

[a]unsure of translation: fornais

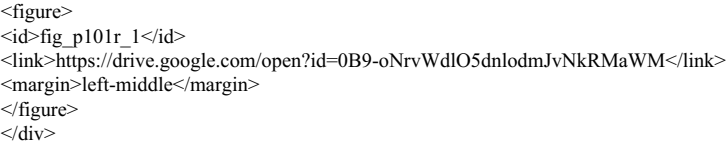
cont

p100v_2

again quite hard, and then add to it three figure of minium, & pound again together & put it in the crucible, which you will cover with an appropriate tile, not too thick, so that heat reverberates better. Then put it on the grill of your furnace, having put a few thick, rounded tiles under it. Next, fill your charcoal furnace to the mouth with charcoal, heaped to the top, and let it light, and always maintain the heat even, without letting the charcoal go down. And to this effect, always ensure that the furnace is heaped to the top & full of charcoal and maintain it like this for one day. The first experiment I made, only a yellow mass emerged, like minium alone when vitrified,

& with gold grains, in a mass at the bottom. Try cemented gold together with antimony.

If you need more heat, put tiles on the edge of the mouth of the glowing furnace, lean them one against the other to make the heat reverberate.

The image shows a topaz gemstone, which is a clear, elongated crystal with a slightly faceted top. It is set against a dark background.

Topaz

The same dose can be observed for all gemstones, which is one part the weight of calcined pebbles with three parts the weight of minium, pounding all separately in a copper mortar for emeralds, & in an iron mortar to make topazes or amber color, with pestles similar to the mortars. Emerald & topaz are of the same heat, an hour & a half, otherwise they could burn. Ruby wants more time & more fire, & colored with gold leaf. I believe that pumice stone or fire-stone for the ruby would be better. See ename[al]s. Also try to mix pieces of colored glass or enamels instead of pebbles. If the mass is not colored enough, pound it further in the iron mortar.

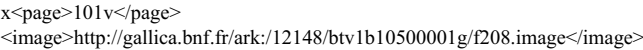
Slightly burnt tartar mixed in makes beautiful yellow, but not much is needed. Arene also makes it more yellow.

Salt for melting

Mix saltpeter & common salt and melt them together & throw them onto copper, molten or in a bain, & it will purify it & makes it run marvellously. First, ordinary salt needs to be decrepitated, which is to say holding it over good heat until it crackles no more; or to melt it better, in an four à vent. Throw it neatly onto a marble slab, then pestle it & crush it very finely, then put it in a crucible on as much saltpeter. Let it boil & mix them together until

It cleans & purifies metal well.

[a]Cross-reference?

x101v
The image shows a page from a manuscript, likely a page from a book on alchemy or metallurgy. It contains text in a historical script, possibly Latin or French, and is numbered 101v.

the crucible is red. Next, throw it on a marble, and you will have a white, hard & uniform matter, like alabaster, from which you will be able to cast medals which will look like marble; but guard them from humidity.

Jacinth

It is made, like rubies, with gold; but you do not need such a great fire. Rubies want fire for a whole day, and if there is not enough fire, it will only have red veins.

Always heat up your crucibles.

One holds that rubified antimony makes jacinth.

<div>
<id>p101v_2</id>
<head><m>Topaz</m></head>

<ab>I melted one part of calcinated & pulverized <m>pumice stone</m> with three parts of <m>minium</m>, the stone having been pulverized in a steel mortar. A very beautiful yellow came to me, without any grain more yellow than the others. It is true that it was strongly charged with colour. I believe it would be better to pulverize the <m>pumice</m> in a glass mortar, because it & the <m>minium</m> make enough yellow by themselves. A mass came to me, the top of which was the beautiful yellow, as mentioned, and the bottom was like a <m>fire-stone</m> without transparency. With which, in mixing other[a]</ab>
</div>

<div>
<id>p101v_3</id>
<head>Varnish</head>

<ab>The Germans thoroughly boil the <m>minium</m> in <m>linseed oil</m>, & to give it the body of varnish, they mix in thoroughly pulverized <m>yellow amber</m>.</ab>
</div>

<div>
<id>p101v_4</id>
<head><m>Ammoniacum</m></head>

<ab>One puts it by small lumps into a bit of good <m>vinegar</m>, & heats it, then passes it through a cloth strainer. All medicinal <m>gums</m> dissolve in <m>vinegar</m>.</ab>
</div>

[a]Here the text is unfinished.

.....

<page>102r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f209.image></image>

<div>
<id>p102r_1</id>
<head>Provisions for the work of Colchos</head>

<ab>7 limbecks of <m>glasse[a][b]</m>
2 lbs. of <m><figure>☿</figure></m>
One measure of coarse <m>salt</m>
6 <m>unleaded</m> <m>earthenware</m> vessel
2 large <m>unleaded</m> pots for calcinating
2 alembics to distill <m>vinegar</m>
4 pots of good <m>vinegar</m>
3 or 4 lamps
2 lbs. of <m>cotton</m>
One pair of scissors</ab>
</div>

<div>
<id>p102r_2</id>
<head>For the furnace</head>

<ab>A chest with a ten inch[c] wide square <m>stone</m>, and of a one-foot thickness, pierced in the middle
2 boxes or two <m>earthenware</m> pots with a lid
2 small pipes of <m>white iron</m> to evacuate the smoke of the furnace
<m>Turnipseed [d]oil</m> for the lamps to heat the furnace
A fire-steel[e][f]</ab>
</div>

<div>
<id>p102r_3</id>
<head><m>Royal cement</m></head>

<ab>One pound and a half of <m>vitriol</m>
As much <m>salt peter</m>
As much <m>rock alum</m></ab>
</div>

<div>
<id>p102r_4</id>
<head><m>Glass</m> vessel</head>

<ab>The <m>glass</m> vessel must be either pear-shaped or round pyramid-shaped, of the same thickness as the small back of a knife, round on the bottom without

folding over on itself, like flasks, its opening of such a size that a <m><pl>Dutch</pl> feather</m> can just about get in, with a lip at the end, in this way of the height of seven fingers' width.

<figure>
<id>fig_p102r_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5MVhNYTRzOXJzR2M</link>
<margin>bottom</margin>
</figure>
</ab>
</div>

[a]See Cotgrave: <http://www.pbm.com/~lindahl/cotgrave/search/242l.html>
[b]"A kind of bending Limbecke (like a Bagpipe) of glasse, varnished, and leaded within; used for th'extracting of oyles out of hard and drie things, as wood, mettalls, &c; also as Malarmat, a kind of Garnet."
[c]Pousse: an inch, sometimes a 1/16 of a foot.
[d]found in the definition for "Navet" in Richelieu: "Espece de graine ronde et noire dont on fait une huile qu'on apelle huile de navette."
[e]See Cotgrave: <http://www.micmap.org/dicfro/search/cotgrave/fusil>
[f]"A fire-stele for a tinder-box; also, the steele wherewith a Butcher whets his knives"

<page>102v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f210.image</image>

<div>
<id>p102v_1</id>

<head>Painting on <m>crystal</m> or <m>glass</m></head>

<ab>They paint without any traces of <m>oil[a]</m>, except for faces where they outline the nose and the mouth in very fine black work. Then they make strokes and highlight them in white, and then they apply a coat of <m>flesh-tinted color</m>. As for the background, they do it in natural <m>azur</m> from <pl>Acre</pl> to make it more beautiful, or <m>lacquer</m> for a quickly-done red, or for a finer red, <m>dragon's blood[b]</m>. But it is necessary to apply it little by little so that it is smooth and of one color, and in the same way for other colors. Then they apply underneath it, a leaf of <m>topaz</m>, <m>gold</m> or <m>silver</m>.</ab>

</div>

<div>
<id>p102v_2</id>

<head>Infusion of <m>anthos[c][d]</m> or <m>rosemary</m></head>

<ab>It is better to cook the flower in August and more suited to <m>oil</m>-making. Take as much as you want. Then put it into a bottle very well corked, and leave them to wilt in the shade for a day. Then add the first substance[e] of <m>wine</m> and leave to settle three or four days; and express it all in another vessel and into this very same substance[f] of <m>wine</m>, add seven or eight infusions made out of new flowers. Then leave the latest infusion in the sun for a month. Then distill it through a still. Then mix two ounces of this <m>water</m> with two or three spoons of <m>white wine</m>; but this is for the elderly. Paul the Third used it.</ab>

</div>

<div>
<id>p102v_2</id>

<head>Casting</head>

<ab><m>Tin</m> casts better when it is thin and delicate, than when it is thick, because when it is thick it contracts in high heat. So if you want to mold a thick piece in <m>tin</m>, mold it only one one side, and if it is possible, make it hollow on the other side. In this way your piece will be neater. And then you can solder two half pieces together. But if you have no choice but to mold it thick, make the shape of it and mold force abreuvouers[g] around the figure in this way.

<figure>
<id>fig_p102v_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5SjZwV0ZHdW93ZEK</link>
</figure>
</ab>
</div>

[a]We think this means pain that contains no oil base.
[b]As Cotgrave says, this is not as ignorant people would have it, i.e. the blood of a dragon crushed by an elephant, but instead red patiens or bloodwort.
[c]the flower of rosemary, Corneille, 1692, "Ce mot est Grec, et veut dire Fleur en general. Cependant tous les Apothicaires entendent par là la Fleur de Rosmarin, comme estant la plus considerable de toutes les fleurs, et meritant d'estre appellée Fleur par excellence."
[d]The OED has anthos as a word with this definition: Formerly applied to Rosemary as ‘the flower’ par excellence.
[e]We don't know if this means the first sediment of wine or the first wine produced of the season.... since he is talking about August.
[f]See note above.
[g]This is the French and we couldn't figure out what it means exactly. We think the image is supposed to clear things up...???

<page>103r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f211.item</image>

<div>
<id>p103r_1</id>
<head>Excellent burn relief</head>
<ab>Heat <m>linseed oil</m> on a light fire, without letting it boil and simmer. But when it is hot put a quarter part^(of the quantity of the linseed oil) of the newest <m>wax</m> you can find. Once it has melted, let it cool and when it begins to curdle, stir it continuously with a new wooden spatula for the length of time it takes to say 9 pater nosters, and while you say them, wash this composition with <m>holy water</m>, always stirring. Having said the first of 9 pater nosters, pour the first <m>water</m> out and add more, then wash and stir the composition for the time it takes to say 8 pater nosters. In the interval it takes to say 3 of the 7 pater nosters, and continuing thusly[a], add more <m>water</m>, as above, until there is only one last new pater noster left[b]. Then you will have a soft white ointment, with which you will apply to the burn for 9 days. But don't apply any more, you must let the flesh grow [c]over. Dress it twice a day and each time you wash your face with water, and mix it together a bit tepidly, without rubbing. But press it with wet <m>linen</m>, and wipe it similarly with fine <m>linen</m>, and apply the ointment. On which you can put <m>binder's string</m>. This will regrow your hair and prevent a scar from forming. A <pro>powder maker</pro> who was completely burnt but had no sign of burns taught it to me.</ab>
</div>

```
</div>
<id>p103r_2</id>
<head>Against dog's gall.</head>
<ab>The English, who fawn [d]over their dogs owing to the care they take of their livestock, have healers for them <sup>the dogs</sup> who hold this particular office,
and who melt <m>pitch</m> in water and, leaving it wet, rub the dogs with this water or with water sublimate.</ab>
</div>
```

[a] meaning, after every third paternoster from here on in. More broadly, he seems to mean you add water on paternosters no. 2, 3, and 6 (of 9).
 [b] or alternatively, "one pater noster of nine" (depending on whether "noeuf" means "new" or "nine").
 [c] could *surcroitre* be rendered in this case as "scab over"?
 [d] I think he means *caresse* in the abstract sense, not literally petting their dogs hard (although the two may not be mutually exclusive).

<page>103v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f212.image></image>

<div>
<id>p103v_1</id>
<head>Enema</head>

<ab>
 In order for the pipe[a] to not hurt children at all, or those with swollen or protruding hemorrhoidal veins, one covers the pipe with a parcel of <m>chicken entrail</m> and fold it over the end, and in this way, one gives the enema.</ab>
 </div>

<div>
<id>p103v_2</id>
<head>Casting</head>

<ab>
<m>Tin</m> recoils if thick, therefore it is better to cast it thin and from one piece, to make two and then solder them together, if necessary. One makes use of a strong piece of <m>taffeta</m> to sieve the fine sand with which one first covers the medal, which is of the thickness of a teston**[b][c]**.</ab>
</div>

<div>
<id>p103v_3</id>
<head>To make colors fluid</head>

<ab>
 Mix some <m>vinegar</m> with a bit of <m>bile</m>, put it into a <m>glass</m> bottle, and if you want to preserve it for a long time, add some <m>salt</m> and mix
 some of this with your colors which will make them flow.</ab>
 </div>

<div>
<id>p103v_4</id>
<head><m>Earwax</m></head>

<ab>
If you take enough with the tip of an ear picker and you sieve it with foam of <m>urine</m>, which often forms in ^{the <m>urine</m>} of those who have a headache, all the foam will dissipate.</ab>
</div>

<div>
<id>p103v_5</id>
<head><m>Enamels</m></head>

To well recognize their difference and their true color by candlelight, it is necessary to put the candle behind a <m>crystal</m> mirror, or a <m>glass</m> globe or jar full of <m>water</m>, because this light is like sunlight.</ab>

[a]pipe? nozzle?
[b]<http://artflsrv02.uchicago.edu/cgi-bin/dicos/pubdico1look.pl?strippedhw=teston>
[c]DAF1: "Teston. s. m. Il faut prononcer l'S. On appelloit ainsi il y a quelque temps une piece de monnoye d'argent, qui n'a plus maintenant de cours, & qui valoit six deniers moins que le quart d'escu, qui avoit la teste du Roy empreinte d'un costé, & les armes de France de l'autre. Les testons n'ont plus de cours. cela ne vaut qu'un teston." via ARTFL

<page>104r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f213.image></image>

<div>
<id>p104r_a1</id>
<head>Corroding and dissolving absolutely pure <m>gold</m></head>

<ab>Put it regular <m>salt</m> with <m>aqua fortis</m> into a violl. Let it rest for two hours, without heating it. Put [in this mixture] <m>tenuous gold</m>, and draw it out as if it had been separated from <m>silver</m>.</ab>

</div>
<div>
<id>p104r_a2</id>
<head>Engraving on <m>iron</m></head>

<ab>One <figure>3</figure> of <m>verdigris</m> and 2 <figure>3</figure> of <m>coarse salt</m> and a bit of <m>sublimate</m>, or crush some <m>ocher</m> with <m>linseed oil</m>, and use it to cover the blade or the very neat and polished <m>iron</m>. And let it dry on the fire or in the sun. Then, to engrave and make it beautiful, take a small coin's worth of <m>verdigris</m> twice that quantity of <m>coarse salt</m> and about four grains of <m>ammoniac</m>, and six grains of <m>sublimate</m>, and the <m>water</m> must not be hot.</ab>

</div>
<div>
<id>p104r_a3</id>
<head>Good mixture to color <m>gold</m></head>

<ab><m>Sulphur</m> and small <m>gravel</m> as much of one as of the other, and the third part of this quantity of <m>salt</m>, and as much <m>turmeric</m> as <m>sulphur</m>.</ab>

</div>
<div>
<id>p104r_a4</id>
<head>How to <m>silver</m> plate <m>copper</m> and <m>latten</m></head>

<ab>Mold the <m>silver</m> as the <m>gold</m>, and put it in the oven in order to get a good <m>gilt</m> pot[a]. And do not let it heat, otherwise the <m>silver</m> would tarnish. Once it will be well spread, make it boil in <m>walnut oil</m>, and then heat it a bit, and put it [into the <m>water</m> the <pro>silversmiths</pro> use to bleach their works][b].</ab>

</div>
<div>
<id>p104r_a5</id>
<head><m>Enamel</m>ing a <m>carnelian</m></head>

<ab>Soak some <m>wheat flour</m> in w<m>hite wine</m>, and slop and cover the whole <m>carnelian</m>. Then cook the <m>violet</m> again, then soak it into <m>white wine</m> for one or two hours, then rub it and leave the part you want to keep on it, then lay on some more on top again.</ab>
</div>

[a]I honestly don't know where does that came from as it is quite difficult (impossible?) to understand what the author meant by "bouliers dorés"...
[b]Was this translation derived from an online source? Does it need to be further substantiated, as it doesn't seem a self-evident translation of "boutteure" in the ms?

<page>104v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f214.image></image>

<div>
<id>p104v_a1</id>
<head>Against dogs' mange</head>

<ab>For normal-sized dogs, take half an <figure>3</figure> of <m>stavesacre</m>, and one <figure>3</figure> for big dogs, as fine powder beaten with two <m>egg whites</m> and one quarteron[a] of <m>oil</m>. They must drink the beverage on an empty stomach, after keeping them locked without any food. Approved[b].</ab>

</div>
<div>
<id>p104v_a2</id>
<head>For casting</head>

<ab>If you've got a big piece which does not allow an easy stripping of casting, mold it with some <m>wax</m>, either on the hollow side or in relief, because it is soft and hollow and so can easily be separated from the piece without leaving anything. But be sure that the melted <m>wax</m> is neither too hot nor too cold. Some people use for molding some <m>wax</m> warmed in <m>boiled water</m>, but while pressing the mold it bends and becomes unusable. Once you have made your <m>wax</m> cav[c]ity, you can cast in some very soft and clear <m>clay</m>, and let it dry. Then, upon it you will make a <m>lead</m> or <m>tin</m> cavity, in which what you will make will easily be stripped.</ab>

</div>
<div>
<id>p104v_a3</id>
<head><pro>Goldsmith</pro>[d]</head>

<ab>Know what it is to work on spangled garments[e], it is small pieces of pure <m>silver</m> and of <m>silver-gilt</m> with which one makes the <pro>archers'</pro> coats.</ab>

</div>
<div>
<id>p104v_a4</id>
<head>Spinet playing by itself</head>

<figure>
<id>fig_p104v_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5cWFSaUU5a2R6N2s</link>
<margin>left-bottom</margin>
</figure>

<ab>Make an axis surrounded with wheels, pierce all around in its thickness [f]and attach some feathers [as you would do] for a timbrel or a spinet and arrange them according to the song you want it to play, leaving a suitable distance. And, turning the axis by yourself or using a clock spring, your invention will work.</ab>
</div>

[a]a quarter of a pound. Cotgrave, via Dicfro gives "a quarter of a pound; also, a quarterne (or the fourth part) of a hundreth; also, a Pinte, or little Quart; also, the halfe of an Emine; also, a Quarter of the Moone. (En Languedoc le Quarteron, common & pour le bled, & pour la terre qui le produit, contient 4 Civadiers, dits aussi boisseuxm divisez par demiz.)" <http://www.micmap.org/dicfro/search/cotgrave/quarteron>
[b]Grégory Champeaud
12:06 PM Jun 23, 2014
I don't know if it means "approuvé" ("approved" or "have been proved") or "à prouver" ("to be proved")??

Marc Smith
11:13 AM Aug 14, 2014
The former, grammatically.

Marc Smith
11:14 AM Aug 14, 2014
Addition (different pen, ink and baseline).
[c]a kind of mold?
[d]ms has "orfevre". Cotgrave defines as a Goldsmith, via Dicfro: <http://www.micmap.org/dicfro/search/cotgrave/orfevre>
[e]"pailleterie" is the practice of ornating clothes. Spangling
<http://dictionary.reference.com/browse/spangle?s=t>
[f]I think it's along the perimeter of the axis of the wheels.

.....
<page>105r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f215.image></image>

<div>
<id>p105r_1</id>

<head><al>Turtles</al></head>

<ab>Males have shells that turn down at the end near their tails, and their underbelly shells have lined marks along the bottom from their tails to their head. And the females have neither the turned down shell near the tail nor the marks.</ab>
</div>

<div>
<id>p105r_2</id>

<head>Drying colors</head>

<ab><m>Soot black</m> and others will not dry in oil unless you add <m>verdigris</m>.</ab>
</div>

<div>
<id>p105r_3</id>

<head>Painting</head>

<ab>Colors that have been applied twice are more gritty by themselves, unless they have a binding agent. Otherwise, paint on <m>wood</m>; otherwise, on
<m>canvas</m>; otherwise on walls.</ab>
</div>

<div>
<id>p105r_4</id>

<head>Gardening</head>

<ab>To graft, it is necessary to take the new growth that was produced within the year by the tree you wish to graft. And cut off a branch in which you can see that beside
the place where the leaves join the stem, there are little buttons we call buds, the initial stages of a sprout. With dexterity you must, with a very sharp pen-knife, cut in the
shape of a graft a small bit of the skin ^{of the tree}, which contains a bud or a sprout with the leaf, as you can see in B. Then on the tender wood, which is full of
sap, and on which you wish to graft, cut quickly two lines as you can see <figure/>, then add in the middle a slit thusly

<figure>
<id>fig_p105r_2</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5T1FZYTdOT1U4Qzg></link>
</figure>

. Then with the point of your pen-knife separate the bark ^{from the tree}, and open it from the slit in the middle, and having neatly lodged your graft so that
nothing is sticking out, only the sprout, binding everything well and wrap the whole thing with a bit of <m>linen</m> or soft string. In this way, there will be nothing
uncovered but the leaf, which in three or 4 days will indicate whether the graft has taken hold, by being green. Keep your graft in this way for seven or eight days. Then
unwrap it, and join it to the bark, keeping the sprout itself straight, and then rewrap the whole thing gently, but not as strongly as before. And if there is any growth, leave it
some room to grow and then three or four times</ab>

<figure>
<id>fig_p105r_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5UWpzdGVweGVvQTg></link>
<margin>left-bottom</margin>
</figure>

<cont/>
</div>

.....

<page>105v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f216.image></image>

<div>
<cont/>
<id>p105r_4</id>
<ab>Then, when the graft has taken hold, cut the top of the tree, which is above your tree, three or four fingers above the graft for the first time. Then, closer so that it
serves. But be advised that when you do your initial grafting there should be no small hole at the back of your sprout, because this means that root is broken and your graft
will never take. Therefore make sure the back is in tact. This type of grafting can be done from mid-May onwards, because the trees have budded and made new growth,
until the beginning of August when the trees are full of sap. Almond trees which are naturally drier lose their sap earlier. Those who want to graft onto them <sup>the
almond trees</sup> mericotons, apricots and nut-tree grafts, which are the best, should do so usually in January. Nut trees, such as perches mericotons should be grafted en
fente.<a> It is said that trees graft by the shield-graft method take longer to bear fruit.</ab>
</div>

<div>
<id>p105v_1</id>

<head>Gardening</head>

<ab>It's not very strange to cultivate in your garden some arnols,<c> because when your soup or broth goes cold, you get <a>worms.</ab>
</div>

<div>
<id>p105v_2</id>

<head>fountain</head>

<ab>Recooked <m>plaster</m> runs when mixed with <m>water</m>, which you will see in the next recipe about wet molds. When put on the joints of the fountain
conduits, ^{it} resists as much as any <m>stucco</m>.</ab>

</div>

<div>
<id>p105v_3</id>
<head>Catching <al>nightingales</al>
<add>S[d]ee the 15th folio <lb/>
after this.</add></head>

<ab>In order to catch them, you must observe them in real life, the food that they favor and their mating season. The <al>nightingale</al>, as long as he is singing, marks his own territory. Therefore approach him by pretending to look for something on the ground. Take some <al>worms</al> from old flour or from beneath what you are kneading or from mills, which <al>nightingales</al> love, and put some on your hat, fastened with a pin or in any other way, so that it can wiggle. And five or six steps away from the hedge where it is singing, dig a hole, and place some <al>worms</al> in it, as well as your device made out of little criss-crossed sticks. As soon as you leave it will be anxious to go and see what you have done. And seeing <al>worms</al> it will enter.</ab>

<ab><margin>left-bottom</margin>Some people sell <al>nightingales</al> on trees.</ab>

<ab><margin>left-bottom</margin>It is easier to catch <al>nightingales</al> in the cool of the morning or evening, near fountains or shaded areas.</ab>
</div>

<div>
<id>p105v_4</id>

<head><al>Nightingale</al></head>

<ab>The <m>iron</m> wire used for its cage must be the thickest possible, so that should it try to leave, it will not hurt itself. At the beginning of its captivity cover its cage in fabric. The first day you will have to forcibly feed it, moving it from its cage in your hand and opening its beak and with a small wooden needle, place ^{its food} in its throat and make it swallow. And continue to do so until it is no longer willful. You must do this to keep it alive, because should it become thin or lose weight it will die.</ab>
</div>

<div>
<id>p105v_5</id>

<head><al>Nightingale</al></head>

<ab>It will need a cage made in the shape of a barn[e], like the ones for <al>larks</al>, lined with green fabric, because it feels the cold. And to get it to eat when it is first put in the cage, you must give it <al>ants</al> mixed with <m>earth</m> in the bottom of the cage to get it used to pecking, and you can give it minced <al>sheep</al>'s heart, and sometimes eggs, and sometimes <al>mealworm</al>.</ab></div>

[a]We think this is another type of grafting but we're not familiar with the details of the diverse ways in which gardeners graft plants.
[b]This could be referring to the split-graft method of grafting?
[c]We're working on this...
[d]Nightingales does in fact continue on 120r-v, fifteen folios on from this entry.
[e]We're not sure of this translation.

.....

<page>106r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f217.image</image>

<div>
<id>p106r_a1</id>
<head>To cast</head>

<ab><m>Asphalt</m> from Germany is the most excellent <m>sand</m> you can find for molding in a box frame, because it is fitting for <m>silver</m> & for <m>gold</m>. And the more you use it and the older it is, the better it is. With it, one molds very slender & subtle things.</ab>

</div>
<div>
<id>p106r_a2</id>
<head>Making <m>gold</m> run for casting</head>

<ab>Because <m>gold</m> cools down very quickly, you have to give it a mixture when it is well melted that will keep the heat. <m>Sublimate</m> softens it nicely. But, since it vanishes into smoke, it won't aid you for very long. Therefore mix this mixture into it when you want to cast it: of <m>sal ammoniac</m>, the best <m>verdigris</m> that you can find, a bit of <m>borax</m> and some <m>salt peter</m>. And this, held over the heat, will become neat. The <m>salt peter</m> makes it clear and heats it. But the key thing in this mixture is the <m>verdigris</m>, which has to be good. This mixture softens your <m>gold</m> so much that it becomes as tractable as lead, even if it is good <m>gold</m>.</ab>

<ab>For <m>sand</m> for molding flowers and leaves & other delicate things, mix in raw <m>plaster</m>, crushed <m>brick</m> and <m>stone alum</m>.</ab>

<ab>To moisten <m>sand</m>, <m>spirits</m> are excellent because it makes <m>sand</m> fine and leaves in a vapor when you reheat your box frame.</ab>

<ab><m>Dilute sal ammoniac</m> is very good, is excellent for moistening <m>sand</m>, but for lots of <m>water</m> you only need a little bit of <m>sal ammoniac</m>.</ab>

<ab>A raw lump[a][b] of <m>metal</m>[c], if you don't have <m>crocum ferri</m>[d], is good for <m>gold</m>.</ab>

<ab>The snakes and the lizards you wish to mold should not be kept for a long time, because if they are alive, they will waste away, and if they are dead, they will decay.</ab>

<ab><m>Plaster of Paris</m> is as hard as stone and very good. When you want to choose some for your <m>sand</m>, take the rawest and the hardest that is possible for you & that does not make any powder. Transparent [<m>plaster</m>] and that which makes powder in giving way is that which is not good for this work. A sign of that which is near perfect is that it is hard & makes shining grains that look almost like sugar. Finally, to mix it into the <m>sand</m> to cast <m>gold</m>, it has to be put in the <m>fire</m> twice, & reheated twice.</ab>

<ab>
<margin>left-top</margin>
<figure>+</figure> When <m>gold</m> is really pure, it is so dry that it can just barely endure a hammer. But this mixture softens it like <m>lead</m>.</ab>
</div>
[a]French "loppe" is the word for unwrought iron as it comes out of the mine.
[b]ms gives "loppe"; Cotgrave via Dicfro gives: "A lumpe of mettall rugged and unwrought, as it comes out of the mine."
<http://www.micmap.org/dicfro/search/cotgrave/loppe>
[c]iron?
[d]"crocum ferri" is a red pigment made from dissolving iron filings in nitric acid - from William Newman, Promethean Ambitions, p. 120

.....

<page>106v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f218.image></image>

<div>
<id>p106v_a1</id>
<head>Casting in <m>gold</m></head>

<ab>To cast in <m>gold</m>, <m>crocum ferri</m> is necessary, and more <m>stone alum[a]</m> than in the following mixture. The <m>stone alum</m> mixture cannot spoil anything. The mold must be red & close it immediately with <m>iron</m> presses.</ab>

<ab>Shells of crabs & [shells of] crayfish are troublesome to burn & have their own proper way of being molded.</ab>

<ab>When molding rats and birds, the feet, being burned, are difficult to recover, because they are small and slender. The body is recovered more easily from the mold. The feet are cast in many pieces.</ab>

</div>
<div>
<id>p106v_a2</id>
<head><m>Tiles</m></head>

<ab>
<margin>right-middle</margin>
those of houses that are outdoors, ones [that are] a bit <m>sand</m>y seem to hold better in fire</ab>

<ab><m>Tiles</m> with which one covers houses, the hardest and surest and recleaned of <m>stones</m> & large <m>gravel</m>, are taken to mix in with the <m>sand</m> for molding. But, rather, it is necessary to reheat them just until they have been quite red for one or two hours, and crush them and pass them finely through a sieve.</ab>

<ab>Little turtles just out of the egg are really delightful to mold.</ab>

<!--[Title: <m>Plaster</m>][b]-->
<ab>
<margin>middle</margin>
If the <m>plaster</m> is mixed with some <m>dust</m> or is not the hard kind, [...] in the fire and make [...]</ab>

<ab>The hardest <m>plaster</m>, as was said, is the best, being taken thusly totally raw from stones of this kind. Crush it as finely as possible and pass it through the most

fine sieve or sleeve that you can. Having been made fine in this way, put a good amount in a cauldron or frying pan over an open flame. & as it starts to heat up, stir it continuously. And heavy and lumpy though it is, it will become so light and so tractable that it will appear to you not to have any resistance to the said stirring, as though it were of water. Continue to keep stirring it continuously until it returns once again to being heavy and thick. And now this is the sign that it is cooked, but do not reheat it at all such that your <m>stone alum</m> brick be reheated & that <m>clay</m> circles form. Because this must be the last reheating, and when everything else is ready, because the less that the <m>plaster</m> remains reheated & the more suddenly it is put into the work, the better, because it takes rather better. It is not permitted to reheat it except when you want to mold. And to reheat it, put it, in fine, totally raw powder, in a cauldron on a clear flame & do not give it too much [heat] in one go nor too fierce. Stir continuously with a long stick, turning around the cauldron to avoid the vapor. And in the middle, in the form of a figure 8 at the beginning, you will find it heavy [and] clayey because it is full of moisture, [and] when it heats up, it will start to throw off [the moisture] in some bubbles on the edges of the cauldron. And finally it will become liquid & mealy and like bran and start boiling in the middle. Continue to stir continuously until you recognize that it has once again become heavy OO[c]</ab>

<ab><m>Stone alum</m> is that which binds well to <m>sand</m>.</ab>

<ab><m>Crocum ferri</m> is fit for <m>gold</m>. It is this which, mixed into <m>sand</m>, breaks it up again & keeps it hot.</ab>

<ab>
<margin>left-top</margin>
<m>Plaster
</m>When you have it in stone form, choose the hardest pieces and those which do not easily break into small pieces with your nail, and clean it well of powder & filth rather than crushing them with it.</ab>

<ab>
<margin>left-middle</margin>
<m>Plaster</m> alone does not sustain [...] but breaks up into lumps.
</ab>

<ab>
Keep reheated <m>plaster</m>, as is said here, one or two months well compacted in a dry place, if it's not the rainy season but rather fresh out it is more excellent for exquisite work.
</ab>

<ab>
<margin>left-middle</margin>
It must maintain its fire and turn red like lit charcoal.</ab>
</div>

[a]ms has "alum de plume," inconsistently translated as "stone alum" and "feather alum"
[b]this entry does not have a title, but it is obviously concerned with plaster.
[c]see matching symbol on 107r for continuatino

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<page>107r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f219.image></image>

<div>
<cont/>
<ab>
<figure>
<id>fig_p107r_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5RFFFT3hSbnlGaVU></link>
</figure>

as heavy as before, and it should be like a paste not very thick, it must not be as easy to stir as a liquid. This state shows you that it is cooked enough. It must boil in the middle, and all around you must have a thick mixture. If you reheat it too much your mixture would be short of body, because if it becomes red the mixture will lose its strength and it will spoil the <m>sand</m>. Let it cool before mixing it to any other <m>sand</m>. [...[a]]</ab>

</div>
<div>
<id>p107r_a1</id>
<head>Catching <m><al>lizards</al></m> and <m><al>snakes</al></m></head>

<ab>Take a stick, pin a net with a slipknot to the top. Whistle, and move the net nearer to the head of the lizard, and pull when it put it's head inside the net. It is more difficult to take a lizard with your hands than a snake, because lizards bite without letting go, and bite as strong as pincers.</ab>

<ab>You can take snakes with your hand, but cover your handle with a woollen cloth, because the teeth of the snake could go through a common cloth. You can recognize dangerous snakes by their blue eyes. They do not bite into water, as it is known by <pro>crayfish catchers</pro>.</ab>

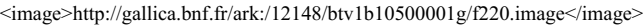
The sand mixture is made of two thirds of ground plaster, reheated as it is said before, and of one part of reddened tiles which are finely ground afterwards. Use also stone alum, half of the quantity of the tiles [b]. Use two full pots of plaster, one pot of tile and half a pot of stone alum. Do not use too much stone alum because that binds the sand, and allow sand to withstand fire without cracking or breaking. [c] Without it [stone alum] sand would not withstand fire, you can use it for all kind of metal, but for gold add a bigger quantity of stone alum, add also crocum ferri, because they [these filings] attract gold.

All the sands you will cast, must withstand fire very well, it must withstand high temperature without spoiling anything.

It is difficult to grind stone alum, and it cannot be sieved, you must grind it on a marble slab. The white [stone alum] powder the apothecaries sell is good, it is easy to crush it into a mortar, use the pestle slowly, that way you will make it [the stone alum] very fine.

Crocum ferri must be heated into a glassworker's furnace during four days.

[a]translation missing here
[b]ms has "brique" here instead of "tuilles"
[c]check translation

107v


Molds for fine things, like wormwood or something similar, can only be used once, dip the molds into water before breaking them. Things twice reheated are dissolve easily into water. [a] If you don't do that your work will be hard to be stripped of the mold without being spoiled.

Good stone alum [b] is white and as shiny as white silk, this stone alum is made of long pieces, as long as a finger, and is very fragile and woolly and fluffy. Stone alum made of stone is harder and not so good. The best quality can be found in France near Ronan. For our sand, stone alum must be crushed into a mortar, and must be ground on the marble again. Don't even think about sieving it because this matter is fat and woolly, it wouldn't pass through any sieve because of the very fine and soft filaments. These filaments give the sand a much greater binding effect than natural wool or even the filler the foundry owner uses for big works, because natural wool and filler burn and stone feather alum resists fire.

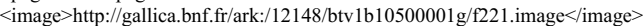
When you cast surround the sand of your mold with very beaten greasy earth. Make a circle around your mold with this earth.

[c][d]Archanum omnibus fere reconditum est in re fusoria,
v {idelicet} res exprimenda formis, sive herba sit sive animal
ut lucerta, inmergatur primum in vini spiritum
aprimere rectificatum, deinde pulvere composito aspergatur
sive illinatur (si pulvis in formam pulvis redactus sit, ut
assolet)

When you want to mold something hollow, the core must be made of the same matter, if the snake or the animal is curved or folded you must do your core in several pieces.

Try [to see] if distilled vinegar is good for eating away, it will dissolve the animal inside the hollow mold.

[a]ED: dissolve the mold in water
[b]"alung de plume," according to Cotgrave is "stone alum," also known as itching powder, out of which candle wicks and matches can also be made that are long burning. Is this asbestos?
[c]This seems to be in Latin because it is great secret of the author's.
[d]translation needed
[e]should be left-middle and left-bottom

108r


To hold the cores, some [people] use wire made of the same metal as that to be cast because it boils down and is made similarly, but when it is melted or pliant the core varies. Other [people] find it serves them better to use sharpened iron wire about the borders because it holds more firmly, and having fine points it does not rise up any more than the point of a needle and one can apply gold or silver on this rather than putting it in works [where] one makes a hole in the work with a small chisel.

Spirits prevent the sand from bubbling and do not make little holes on the borders of the mold if the thing to be molded is very damp, the holes and farts and bubbling does not make it in the place of the mold which is thick but at the borders which are more delicate.

One casts ordinary silver of the sort goldsmiths coarsely work, which is made with mediocre alloys. And when one casts with solder, it runs even better.

<ab><m>Spalt</m> is a whitish stone which can be found in Germany, mainly in Augsburg from which can be made the most excellent <m>sand</m> that can be found for [casting] <m>lead</m>, <m>tin</m>, <m>copper</m>, <m>silver</m> and <m>gold</m>. And the more it is used, the better it is. It is clean for casting flat things in a frame. For rounded things, it is not as attractive [because it] does not hold in the fire as the aforementioned [things] composed of <m>plaster</m>.</ab>

<ab>The snippings[a] of [a] large [piece of] greasy <m>leather</m> are good to cast in melted <m>copper</m> and <m>brass</m> because they clean it and take away from it all its <m>filth</m>.</ab>

</div>
<div>
<id>p108r_a1</id>
<head><m>Spalt</m></head>

<ab><m>Spalt</m> is white like cooked <m>plaster</m> and one can find it in mounds and stones made in long stairs and long veins. It is very soft such that with a fingernail and it makes a powder like that of our <m>chalk</m> from Champagne. And because everything which is provided from the <m>earth</m> is mixed with some other substance, to purify it [the spalt], one boils it with somewhat large <m>gemstones</m> then tempers it with <m>essence of sal ammoniac</m>. One puts [a lump of it] the size of a walnut[b][c] in a large bottle of <m>water</m> and</ab>

</div>
[a]"shreds, clippings, snippings, parings" in Cotgrave
[b]Cotgrave
[c]ms has "noix"

.....
<page>108v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f222.image</image>

<div>
<cont/>
<ab>that way one tempers it & reducing it into small balls, [one] purges it of <m>impurities</m> [by] then putting it on to cook cleanly in a vessel of earth[enware] in a furnace like those in which pots [are made] and one leaves it there to the point at which pots would be fired. After one tempers it more with <m>essence of sal ammoniac</m> & boiling it very strongly and emptying the murky <m>water</m> & putting it in a separate vessel straight after put back the same <m>water</m> over it & one boils & one puts back the murky <m>water</m> again with the other. And one does this therefore until that which is there has passed away[a]. In this way one purges and purifies and makes it a very fine and malleable [matter]. After one has taken away the <m>water</m> by tilting[b] [it] or with a <m>sponge</m>, one takes the residue and dries it. Then moistening it with the aforementioned <m>water</m> the way one did with the other <m>sand</m>, one uses it in a frame[c] or if it shrinks [this] is a sign that it is not fired enough & needs to be fired more. This one is proper for molding all <m>metal</m> and especially <m>gold</m> and <m>silver</m> and the more it is used the better it is. One should put <m>alum</m> which you have cast of <m>lead</m> or <m>tin</m> separately because it makes <m>gold</m> sour[d] if you cast it there[e].</ab>

</div>
<div>
<id>p108v_a1</id>
<head><m>Stone alum</m></head>]

<ab><m>Stone alum[f]</m> must be reheated in a pot covered with a <m>tile</m> over a good <m>charcoal</m>[g] fire so that the impurities [h]which might be in it [are] burned with that which does not burn. This is done either in the <pro>goldsmith</pro> forge surrounded by bricks or in a little oven[i] vented to the fusion fire such that the pot[j] stays red for a quarter of an hour. This is more to purify it than for anything else. It becomes reddish on top and the inside remains white and dries best this way.</figure>#</figure></ab>

<ab>
<margin>left-middle</margin>
<figure>#</figure> Once your <m>stone alum</m> is cooled down, crush it in a mortar, then grind it more finely on <m>marble</m>. And this makes it into very fine woolly dust that gives binding to <m>sand</m> without burning like all other things. [This is] a beautiful invention. Crush it on the mortar to being careful to keep the grindings together because the dust flying can get down your collar or on your face which can have very grave consequences. Put a little each time in the mortar to avoid the dust flying. It is best to grind therefore with the pestle trained on the mortar rather than on the <m>marble</m> where it spreads and on the mortar it gathers [the dust] from all sides and renders it very fine and sweet to the touch.</ab>

</div>
<div>
<id>p108v_a2</id>
<head><m>Clay</m></head>]

<ab><m>Clay[k]</m> used to surround molds, should be very fat and malleable, and should not be used for anything else except to make the contour of the molds. Therefore one should only chose that which is very fat and beat it very well and soak it moderately with <m>water</m> and have it in a pit or <m>earthenware</m> pot and make good holes with a stick with which one replenishes it with <m>water</m> to keep it always fresh and soft and ready to use when needed.</ab>

</div>
[a]or "is gone"
[b]and pouring it off.
[c]or as a frame (this is unclear).
[d]Cotgrave: Aigir = to sharpen, sower (sour); exasperate; aggravate, to grieve, gall, stirre up, or provoke to anger.
[e]in the same place (i.e. gold should not be cast in the same place (same pot, etc.) as where you have cast alum with lead or tin).
[f]Cotgrave: Alun de plume = "A hard, and white Allum, full of streakes, or flakes; we call it, Stone Allum, or itching powder; Looke Plume."
[g]Flame before fire is crossed out.
[h]Cotgrave: "Immondice = f. Filth, uncleannesse, nastinesse, pollution, slutishnesse, ordure." See also DMF, immondice = impurities. From Latin mundus = pure.
[i]Cotgrave: Fourneau = little oven
[j]creuset
[k]in line with text

.....
<page>109r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f223.image</image>

<div>
<id>p109r_1</id>

<head>Working in rough[a] with <m>wax</m></head>

<ab>When the <m>wax</m> is too hard, one mixes in some <m>turpentine</m> or a bit of <m>butter</m>, which renders the <m>wax</m> malleable[b], and cleaner than <m>tallow</m>, which the <pl>Italians</pl> mix in, because oftentimes, it is necessary to put the tools into the mouth, ^{tools} which are better when made from <m>box wood</m> or <m>antler</m>.</ab>

<div>
<id>p109r_2</id>

<head><m>Wax</m> for molding</head>

<ab>Since the <m>wax</m> is molten, they [c] have some <m>sulphur</m> in a spoon or crucible and pour it in the molten <m>wax</m>. And with the <m>sulphur</m> going to the bottom or staying on top, it stays where it is and mixes only its substance in the <m>wax</m>, and renders it more meltable when warming it, in such a way that having molded it, ^{the <m>wax</m>} gently melts in the mold like <m>butter</m>, without leaving any blister or boil. Also be warned to not give it too much heat.</ab>

<ab><margin>left-top</margin>If</ab>
</div>

<div>
<id>p109r_3</id>

<head>Molding <m>wax</m></head>

<ab>So that your mold takes hold without repelling any surface on a form or figure in <m>wax</m><m>,</m> it is necessary to rub the <m>wax</m> with <m>oil</m> and soak your sand in lukewarm <m>water</m>, because cold <m>water</m> repels <m>oil</m> even more and will not make a good layer[d] on[e] the <m>wax</m>.</ab>

<ab><margin>left-middle</margin>Rub the <m>oil</m> on very very lightly, ^{use} so little that it hardly appears at all. Then rub with some <m>spirits</m>. After soak your sand in lukewarm <m>water</m> so that it does not repel the <m>oil</m>.</ab></div>

<div>
<id>p109r_4</id>
<head>Forge bellows</head>

<ab>It is better that at the end with the air chamber the bellows be set in some kind of bit of cloven board[f] because they are more secure this way [g][h]. Afterwards you can lightly secure them from the top. It is necessary that the nozzles be four fingers above the bottom of the forge.</ab>[i]</div>

<div>
<id>p109r_5</id>
<head>Molding <al>snakes</al></head>

<ab>When you get them for molding, if it is possible, do not remove their teeth if you [think you might] want to keep them, because having had their teeth removed, it hurts their gums and mouths, and they can no longer eat. You can keep [the snakes] in a barrel full of <m>bran</m> or even better of <m>earth</m>, in a cool place, or in a <m>glass</m> bottle. And give them some live frog or other live animal because they do not eat anything ^{that is} dead. Also I have noticed that</ab>

<cont/>
</div>
[a]suggestion: 'modeling' (see notes Rowen/Landsman for translation 'esbaucher')
[b]With the idea that the wax becomes more amiable to work with.... I think.
[c]I'm assuming these are the Italians?
[d]Or maybe surface here?
[e]Or maybe "against?"
[f]See Cotgrave: membrure
[g]See Wikipedia entry for bellows...
[h]<http://en.wikipedia.org/wiki/Bellows>
[i]Unsure of this translation vis a vis vocabulary and meaning. However, the image on Wikipedia seems to fit what is generally being said here I think.

.....

<page>109v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f224.image></image>

<div>
<cont/>
<id>p109r_5</id>

<ab>when they want to bite or to eat something, they [snakes] do not go at it head on, but sideways and by sinuous coiling, as do Satan and his henchmen. The snake has a small head, but a very large body; the entryway to sin seems small and inconsequential, but the consequences are great indeed. A snake can abstain from eating for seven or eight days, once he has swallowed a frog, and can swallow three or four frogs, one after the other, and what it has devoured does not rot and is not digested all at once in its

stomach, but each part little by little, i.e. the bones and everything. The rest is just as fresh as when the snake ate it, such that when one bothers and torments it, it spits out what it swallowed up, which, in part, is totally digested, with the other part as fresh as if the frog were alive. It can hold a swallowed frog down for two or three hours and spit it out alive.

</div>

<div>

<id>p109v_1</id>

<head>Toad</head>

If by chance you want to mold this ugly animal and if it should piss in your hand, it will sting your hand and cause it to itch, as if you had been holding nettles. But the very best remedy for this is to put your hand into fresh dirt, and to cover it as if you wanted to bury it. Frogs are not so beautiful when molded because they have shiny skin, and toads have lumpy skin.

<ab>

<margin>left-middle</margin>

You can mold the toad hollow, and leaving a notch open on the underbelly, you will make it jump with the twisted cord of the saw. If it is a small, put it on a magnet cut small to the size of the toad, then put it in good vinegar.

<ab>

<margin>right-middle</margin>

If it is big, mold hollow, and if you want to mold it with its mouth open, put some cotton therein, and then put some melted wax on the cotton.

</div>

<div>

<id>p109v_2</id>

<head>Killing snakes for molding</head>

Some put a drop of aqua fortis into their mouths, which stupefies them, so that the head and the body look dead. But the head is still alive, and when you stick it with a needle in order to fix it to your mold, it moves and spoils and undoes everything. Therefore, to kill your snake, put it into a bottle filled with good vinegar and some spirits. Do the same with lizards and other similar animals.

<ab>

<margin>left-bottom</margin>

If you want to keep them, you can put them in a pipe full of dirt in a garden outdoors, but it is necessary that the pipe be covered with latten wire, because otherwise, the snakes will crawl and climb up quite high.

</div>

[a]Pierre sidérite

[b]?? this needs revision

[c]"Siderite: The yron-like stone Siderites, which, as some imagine, hath power to set men at oddes; also, the Load-stone" Cotgrave, Dicfro, <http://www.micmap.org/dicfro/search/cotgrave/sid%C3%A9rite>

[d]In modern chemical nomenclature, siderite is iron carbonate, also called chalybite. More info at CAMEO: <http://cameo.mfa.org/wiki/Siderite>

[e]ms has "fil d'archal"

[f]"Archal & fil d'archal. Yellow laten wire; or copper wire." Cotgrave, Dicfro, <http://www.micmap.org/dicfro/search/cotgrave/archal>

<page>110r</page>

<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f225.image></image>

<div>

<id>p110r_1</id>

<head>Casting</head>

You could cast from already casted things and animals, but it never works as well as from the natural [original] ones, with which you can make four or five molds.

</div>

<div>

<id>p110r_2</id>

<head>Rock and cavern</head>

You need to grind white and yellow marcasites & various kinds of mines. Wash them thoroughly, so the soiled and chalky end will be clean, in order that only the lustrous grain of the mine will be left, from which you will powder your rock previously covered with colle forte, if your cavern is not used as a fountain and touches water. You will thus obtain a nice work.

</div>

<div>

<id>p110r_3</id>

<head>Snakes</head>

When they feel trapped, they seldom bite, and cannot really hurt unless they are really big snakes. The most important thing is to avoid being bitten before capturing them. Having put your foot as close as possible from the neck, you holds it as close to the head as possible. That way, it will not bite if it cannot stretch its neck and do a sinuous turn, it will not be able to hurt from a frontal attack, nor will it be able to twist into an S. I have found out that the person who taught me how to mold them was

little afraid of snakes. He would catch one and pull it quickly, grabbed it by the neck and, putting it under his shoe, flattened it so it would eject from the behind[c], which is five or six fingers from the tail, everything that is bad and venomous in the belly. After that, the snake has no strength to bite and, if by chance it still bites and ejects from its behind some garbage, he [the man] would dig in the soil with a knife and bury his hand in the fresh and humid soil which, applied to the infected part of the hand, would suck all the venom, without doing anything else. It was the same with toad venom, as I have seen experimented.</ab>

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<ab>
<margin>left-middle</margin>
In three [days] it is as thin as if he had not eaten. Cast it right after catching it is the best, before it loses weight. You should kill it only a quarter of an hour before you want
to cast it.</ab>
<cont/>
</div>
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[a]could not find a translation. Obviously, marcasite is a kind of "mine". Furthermore, "mines" do not like water; maybe they are corrosive, like marcasite.
[b]keep French original?
[c]fondement

.....

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<page>110v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f226.image</image>
<div>
<id>p110r_3</id>
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<ab>When <sup><al>a snake</al></sup> has eaten something; when twisting[a] it, it will vomit <sup>the food</sup>. And if, after having eaten something, <sup>the
snake</sup> is pressed underfoot, this troubles <sup>the snake</sup> a lot and makes it sick. If <sup>the snake</sup> is wounded, it will not eat voluntarily.</ab></div>

<div>
<id>p110v_1</id>
<head><al>Mussels</al></head>
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<ab>There is nothing better for opening them up than to oil them with <m>olive oil</m>, and nothing else. And afterwards, when you want to separate them, soak them in cold <m>water</m>; that is the secret. And you will see that the <m>oil</m><m></m> although it seems to be imbued[b], will separate itself ^{from the water} as ^{does} grease. <al>Mussels</al> become stronger in cold <m>water</m>, and warm <m>water</m> would dissolve them with difficulty, although having been reheated, ^{the mussels} are ^{then} more handleable[c][d] and easy to dissolve in the <m>water</m>.</ab></div>

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<div>
<id>p110v_2</id>
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<head><m>Wheat oil</m></head>
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<ab>Is made on a blade of <m>iron</m> reddened in the fire, and the <m>oil</m> turns into drops, which is appropriate to oil the hair of a butterfly or similar thing, because this <m>oil</m> is instantly dry and makes the rest dry out. It is necessary that the coat or down of any animal that you want to mold be flat, because ^{if left} standing up, it will make the sand raise up[e] and form bubbles.</ab>

<ab><margin>left-middle</margin><figure>#</figure> If you want to mold something delicate, like a pansy, some, to give it a little thickness, more than what is natural, some [f]rub it with <m>butter</m>. But it is best to oil it with <m>wheat oil</m> because it has no opacity[g] and does not block the smaller features as much, and makes the flower firmer.</ab></div>

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<div>
<id>p110v_3</id>
<head>A cast of <m>lead</m> or <m>tin</m></head>
```

<ab>Because you want to cast <m>tin</m> very thinly, if your medal, plant or other thing ^{you want} to mold is thin and fine, make ^{sure} to include more <m>tin</m>, much more than <m>lead</m>, namely less than the fourth part of <m>lead</m> for three of <m>tin</m>. And moreover, one only puts <m>lead</m> in as an alloy. Contrarily, if you want to mold something strong and thick, put [h]a lot more <m>lead</m> in than <m>tin</m>. And for the one and the other you can put in a little <m>looking-glass tin</m>, but only a very little, with a little <m>resin</m>, when you want to cast. Since then, when molding with pure new <m>lead</m>, I put in two ounces of pure <m>tin</m> for every pound ^{of <m>lead</m>}. And when molding with pure <m>tin</m>, I put in two ounces of pure <m>lead</m> for every pound ^{of <m>tin</m>}. I made some plants and snakes as if they were real.[i]</ab>

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<ab><margin>left-bottom</margin>
I cast <m>tin</m> <sup>when it was</sup> nearly red, and the same thing with <m>lead</m>, which however had not remained in the fire for too long, because it becomes brittle and burns up.</ab>
</div>
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[a]Tourmenter can also mean torture in general or applying undue pressure. The glossaire Saintongeais references the Latin root: tormentum.
[b]Or mixed in fully with the cold water....
[c]Cotgrave's translation and I think it's perfect:)
[d]ms has "maniables"
[e]This is perhaps not the best translation but I think it means that the sand would not form itself smoothly around the body and leave spaces around hairs where bubbles could form. Thus hair is too thin to cast, but butterfly wings and insect legs seem to be OK:)
[f]Is the author perhaps forgetting that he has already mentioned the "some."???
[g]This adds to the discussion on "corps."
[h]Could the author have wanted to use mects, but used mays instead? Otherwise I can't make sense of it.
[i]This is in a different ink and may be a later addition to the entry.

.....

<page>111r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f227.image</image>

<div>
<id>p111r_a1</id>
<head>Reheating molds</head>

<ab>Box molds dry out readily, & do not get reheated, because to reheat is properly to turn molds red and inflamed in the fire, and drying out is to let both molds dry, or make them evaporate their humidity, placing them before a fire of flames or one of charcoal. Molds de noyau[a] reheat easily & box molds dry out.</ab>

</div>
<div>
<id>p111r_a2</id>
<head><m>Latten</m> <m>smoke</m></head>

<ab>
It is good to reheat molds for casting <m>latten</m> in it, because this yellow <m>chalk</m>, which appears nevertheless to refill the molds, casts very neatly, being of the same nature.</ab>

</div>
<div>
<id>p111r_a3</id>
<head>For <m>red copper</m></head>

<ab>When it's melted, it is good to throw in a little fine <m>tin</m>, because this makes it run.</ab>

</div>
<div>
<id>p111r_a4</id>
<head><m>Clay earth</m></head>

<ab>This, wet & well beat & kneaded as you know, is necessary for you to make the contour of your molds. But attend that it not be too soft. But likewise [attend that it be] sort of half dry such that it doesn't attack the hands at all, because otherwise it would attack your work. Always keep it in a damp place, and make some holes in it, and fill them with water, to keep it fresh.</ab>

<ab>
<margin>left-middle</margin>
The gray [<m>earth</m>] is best which does not cleave at all.</ab>

<ab>
<margin>left-middle</margin>
It is neater than [illegible] of <m>lead</m>.</ab>

<figure>
<id>fig_p111r_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdIO5X085N09VejNheTQ</link>
<margin>left-middle</margin>
</figure>

</div>
<div>
<id>p111r_a5</id>
<head><m>Iron</m> wire</head>

<ab>You have to be provisioned with different thicknesses according to the molds that you want to make, some large like packthread from [illegible] & some thin, like the fat cords of a spinet, for small works. With it reheated, make clamps to close the molds & make points like needles of various lengths according to your need, to fasten the animal to the mold & hold it subject. Needles would not be good for this since your points must be hollow & of the same thickness throughout, except at the point. And then you have to make them as long as you have need. <m>Iron</m> wire is harder than that of <m>latten</m>.</ab>

</div>
[a]translation?

.....
<page>111v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f228.image</image>

<div>
<id>p111v_1</id>
<head><m>sal ammoniac water</m></head>

<ab>It is enough to use two <m>chestnuts</m>' worth of <m>sal ammoniac</m> pulverized in a pot of <m>water</m>, and that the water tastes lightly salted to your

tongue, because too much is not good.</ab>

<id>p111v_2</id>
<head><m>Sanguine</m></head>

<ab>To clean the various colors of <m>lead</m> and <m>tin</m>, <m>sanguine</m> is pulverized, and it is rubbed on the molded work with pieces of fabric.</ab>

<id>p111v_3</id>
<head><al>Crab</al> and <al>crayfish</al></head>

<ab>
They are among the most difficult to mold, and whoever molds them well will mold many other very subtle things.</ab>

<id>p111v_4</id>
<head><al>Lizards</al></head>

<ab>When one grabs them with two fingers from the sides of the head, at the level of the ears, one makes them open their mouth with the tip of the fingernail, then one grabs the edge of their lower jaw, and then [the edge] of the upper one. He squeezes the fingernail like a pair of pincers, but he cannot go through it, but he squeezes it quite hard; and if he bit the flesh, he would never let go of his hold, and there would be no better remedy than to cut off his head promptly. In the spring, as they have shed their skin, they are better-looking.</ab>

<ab>
<margin>left-middle</margin>
The small <al>lizards</al> that one catches in the summer, the size of small <al>female lizards</al>, grey on the back and green on the belly, are very appropriate to cast in silver, gold, or other metal, because their scales are rougher than those of <m><al>female lizards</al></m> and show better.
</ab>
</div>

<div>
<id>p111v_5</id>
<head>Composition of <m>sand</m></head>

<ab>Everything being baked thoroughly[a][b] as said above, take four 4 molds full of <m>plaster</m>, two 2 of <m>brick</m>, and one of <m>alum de plume</m>. Do not pass through the sieve to mix, because the <m>alum</m> would not go through. Mix it with your hands until neither one nor the other, neither white nor red, may be distinguished from the other. If your <m>plaster</m> is not strong, do not mix in as much <m>alum de plume</m> because it softens molds. Make sure too that your <m>sand</m> is not hot (since it has been baked thoroughly recently) when you will want to mix it with <m>water</m>, because that makes molds too soft and breakable.
</ab>

<ab>
<margin>left-bottom</margin>
You may sometimes decrease the quantity of <m>plaster</m> if you do not find it strong enough to resist fire and if it splits, so that the proportion of crushed <m>tiles</m> sustains [c]it, with the <m>alum de plume</m>, which you can also increase so that it creates a better bond.</ab>
</div>

[a]Re-cuit literally means "baked again" or rebaked, but could also mean "backed through" or "baked thoroughly" (Cotgrave).
[b]Cotgrave: "to harden, as with overmuch baking". It may be interesting to see whether his processes for sand also include various baking sessions or can also exist of one long/thorough one?
[c]strengthens?

.....

<page>112r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f229.image</image>

<div>
<id>p112r_1</id>
<head>To temper the <m>sand</m></head>

<ab>You should have: a large basin full of <m>water</m> to dip the molds into in order to open them, a tankard full of <m>common water</m>, a dish to temper the <m>sand</m>, a small <m>wooden</m> spoon to pick up the mixed <m>sand</m> in the dish.</ab>

<ab>
<margin>left-top</margin>
To make handles for your large oil brushes, if you do not have large enough <m>feather pipes</m>, take two of them, melt them and join them together.</ab>

<figure>
<id>fig_p112r_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdIO5MHphUzNsdW9DQ1U</link>
<margin>left-top</margin>
</figure>

</div>

<div>
<id>p112r_2</id>
<head>Decoction of <m>sands</m></head>

<ab>Already said[a]</ab>

</div>
<div>
<id>p112r_3</id>
<head>Molding en noyau[b]. Mixing of <m>sand</m></head>

<ab>Once you have reheated your <m>sand</m>, that is the <m>plaster</m>, the <m>stone alum</m> and the <m>brick</m>, and once you have finely filtered them through a <m>cloth</m> sieve, <figure>#</figure> mix them the following way: take four measures of <m>plaster</m>, two measures of <m>bricks</m>, and one measure of <m>stone alum</m>. Mix them all together with the hands, so that one may not discern one [ingredient] from the other. Once your <m>sand</m> is ready, you need to kill the animal that you wish to mold in this way.</ab>

<ab>
<margin>left-center</margin>
<figure>#</figure> except for the <m>stone alum</m> which should be ground in a mortar.</ab>

</div>
<div>
<id>p112r_4</id>
<head>Killing the animal to mold</head>

<ab>You must have been careful not to hurt it while capturing it. If you have a stock [of animals] and you wish to keep them for a long time, place it in a <m>glass</m> bottle with <m>bran</m> or in a large barrel half full of damp fresh <m>earth</m>. Give them frogs to eat, live rather than dead, because should you keep them without feeding them, they would wither and become emaciated, and their skin would become saggy and wrinkled. Therefore, the sooner after its capture you mold it, the better it will come out. And kill it in the following manner: place it in a clean bottle or vessel so that [the animal] does not carry any <m>dust</m>; fill [the bottle] with enough good <m>vinegar</m> and <m>urine</m> so as to cover [the animal]. Shake [the bottle] and torment [the animal] inside there until it is dead which should take an hour or half an hour. But if you are in a hurry</ab>

<ab>
<margin>left-bottom</margin>
To know if [the animal] is quite dead, take it out of the <m>vinegar</m> and take it by the tail. If it moves its tail, do not mold it because it still has feelings [c], and when pricking [its members], it would remove itself and harm the mold.</ab>

<ab>
<margin>left-bottom</margin>
Water snakes do not bite much.</ab>
<cont/>
</div>

[a]what is this referring to?
[b]We still have not figured out exactly what this means at that stage and therefore I am keeping the French text here
[c]Should we go for a less literal translation (less poetic than this) such as "it is still alive", considering the rather gruesome context?

.....

<page>112v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f230.image</image>

<div>
<cont/>
<id>p112r_4</id>
<ab>to cast, force feed it ^{your animal} some good <m>spirits</m> or mix a bit of <m>spirits</m> with the <m>vinegar</m>. And soon it will be dead without any wounds that would deform it. As for large animals, for which it is necessary to have a lot of <m>spirits</m>, one uses <m>vinegar</m> and <m>urine</m>, but for small ones, <m>spirits</m> only are needed, and it will be done quicker.</ab>

<ab>When your animal is dead, take it out and clean it very carefully, ^{relieving it} of its old skin if it sheds, of its <al>parasites</al> like <al>lizards</al> have, and then apply it.</ab>

<ab><margin>left-top</margin>in a glass, and submerge, continuously holding the head of the animal. Or if it is not very big, put the whole thing in.</ab>

<ab><margin>left-top</margin>Be advised that before it dies, you must open it eyes, by blocking its eyelids with a bit of <m>wax</m> or something else.</ab></div>

<div>
<id>p112v_1</id>

<head>Affixing and arranging animals</head>

<ab>Having gotten in some fat <m>earth</m> called <m>clay</m>, that should be grey, because that is the best one, or another that will bind well and will be well prepared and beaten, so that it is malleable without being damaging your hands or the work, make a lazagne,[a] or a pancake shape, of this <m>clay</m> equally flattened

with a rolling pin, which <pro>pastry-makers</pro> use so that it is slightly thicker than a poule. [b] And on this pancake, affix your animal, making it as life-like as possible and the way in which it is naturally shaped. And firstly, with a good needle, pierce it from underneath, in the middle of the throat, up to the top bone of the head. Then take the needle out and in the hole, put the point of an <m>iron</m> wire, of such a length that it suffices to maintain the head of the animal as high as it must be, and if the skin gets in when you put in the wire, take it out with the edge of your small pincer[c], and arrange the skin as it was before. Pierce the other end of the wire in the sheet of <m>clay</m>. In this way the head will be held high, which will make it more graceful and proud. Arrange afterwards, the rest of the body, and the legs, and the curling as you think will look best. And so that it is secure and does not move when you put in the sand, secure the legs with little points of <m>iron</m> wire, having already made the first hole with a very sharp needle. Do the same on the body parts that you deem necessary, making sure that the points going through the animal are fixed in the <m>clay</m> sheet, and not so far in front that they jut out beyond the body of the animal by a</ab>

<ab><margin>left-top</margin>Take a needle of fine <m>steel</m>, and which will be the sharpest you can find, so that with it, when you make your first hole in order to place the wire threads, it will pass through without effort, and take care not to rudely insert the wire or take skin with it. Then put a thicker wire in the thick bits [of the animal] and the thinner wire in the thinner bits. But take this secret advice and do not plant your pointed wire in one go through the animal and onto the <m>plaster</m>, but having planted it, take it out of the <m>clay</m> earth but not out of the animal, which is afterwards fastened there. One does this because by fastening the point the first time, it pushes in the skin of the serpent and by taking the point out, it replaces the skin in its original state.</ab>

<ab><margin>left-bottom</margin>When you reuse the points you have already used, make sure to clean them well of rust, so that they do not take away some skin.</ab>

<ab><margin>bottom</margin>The main arrangement of your casting is to keep its head very high and have it looking sideways. For this effect, pierce the head on a solid plank or table, making the wire pass through the bottom of the throat up to the top of the head

<figure>
<id>fig_p112v_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5c3dzVINzcUd6MnM</link>
</figure><lb/>
</ab>
<cont/>
</div>

[a]We kept the french because the word lasagna feels a bit anachronistic. See definition by Godefroy : "Autour de cette place y a des boutiques la ou se vendent des lazagnes, avec lesquelles on fait certaines balotes de chair de beuf. (LEON, Descr. de l'Afr., I, 139, éd. 1556.)"
[b]The measurement of an inch.
[c]a tweezer, see Cotgrave: <http://www.micmap.org/dicfro/search/cotgrave/mollette>

<page>113r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f231.image</image>

<div>
<cont/>
<id>p112v_1</id>
<ab>a bit so that you can take it out if you need to. And be advised that these needle points must be of the thinnest kind of <m>iron</m> wire, with a length that is suitable to the animal; because the smaller your needle is, the smaller the leftover hole will be. However, you must have different sized ones because there are some needles that go on the body and in places that are thicker, like under the throat, and some ^{needles} must be longer and stronger than the ones used on legs and thin places, where the needles must be subtle and thin, almost like the biggest harpsichord strings. And it would be better if they were of <m>iron</m> wire, which is firmer than the one made of <m>latten</m>. And if piece of a tail, a piece of nail or a piece of leg to the body needs to be joined to the body, do not use the needle; it would damage these parts. But apply a bit of <m>wax</m>, as big as a <m>grain of millet</m>, between the body and the end of the tail. Then with a bit of <m>iron</m> wire, flattened and warm, touch the <m>wax</m> and join the two pieces by applying a bit of pressure. In this way, after your animal has been well-placed on the lute-shaped pancake ^{of clay earth}, <figure>
<id>fig_p113r_2</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5YkE1Vm1ZMmRoVm8</link>
</figure>
build around it a wall of the same <m>clay</m>
<figure>
<id>fig_p113r_3</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5aW1sekxQWGJJY0E</link>
</figure> in this way[a][b]. And join everything together. And then with a big knife, with which you have cut the <m>clay</m>, take away the extraneous ^{clay} surrounding the pancake. And place on a piece at the end of the mold to close it off, as you see in B.[c][d] Then, with a small curved instrument, mark on the inside of your mold the thickness you wish, which for a common <al>serpent</al> must be two inches thick; so that when you put in your wet sand, you will see when it is thick enough. When all of this is neatly set out, moisten well the shaped body of the animal everywhere, with good <m>spirits</m>, with a brush. For this is the best-kept secret: because everything that is touched by <m>spirits</m> will come out very neatly and without any small eyes or holes that we call bubbles[e], which usually come not at the back or in the thick sides of the mold, but on the edges. And as soon as you have thusly moistened everything with <m>spirits</m>, put in your wet sand without letting the <m>spirits</m> evaporate, which they do quickly. And soak it thusly.</ab>

<ab>
<figure>
<id>fig_p113r_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5MDEzX0RhXzJGcXc</link>
</figure> and make sure the needle is as long as you need according to how much you want to raise the head, and that both ends are sharp. Having been arranged thusly, affix your <al>serpent</al> on the <m>clay earth</m>, and place the head first and then the rest.
</ab>

<ab>
<margin>left-middle</margin>
If, once you have placed you animal on sheet of <m>clay</m>, you are distracted with other occupations, rub your animal with <m>spirits</m> and under the belly so that it does not stick or attach itself to the ^{al>clay} <m>earth</m>.
<figure>
<id>fig_p113r_4</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5Qm8wSlJPOWxfZWZM</link>
</figure>

ab<div>If, when you pierce the belly of your animal, it releases water on the plaster, clean it well with cotton because the sand that was wet will become lumpy. If it is on top of the animal, clean it very carefully and very neatly and touch it and strongly wet it with spirits.

<ab>Have a porringer[f] of <m>leaded earth</m> of such a size that it can hold the wet sand which will be needed to fill your mold in one go, which is better than in two gos, because if you do not throw the second load of sand in very quickly, there is a risk that while you are preparing it, the first one will take and become solid,</ab>
<cont/>
</div>

[a]There is an image in the text that indicates which way the author means.
[b]close to the left margin
[c]This is another image that is in the text to explain the instructions.
[d]middle of the page
[e]"soufflements"
[f]small bowl

<div>
<cont/>
<id>113r_1</id>
<ab>and this way, while heating it again, the two casts will dissolve. Therefore, have several dishes of various sizes with a pouring beak in order to cast better. In these, pour some regular <m>fountain <m>water</m> and mix into it as much as an egg <figure>#<a></figure> can hold of <m>sal ammoniac water</m>, mixed as described. If you want to cast <m>lead</m> or <m>tin</m> - because one would need more for casting <m>gold</m> or <m>silver</m> - having the <m>water</m> in your dish, add the <m>sand</m> and not the other way around[b] [by adding the water to the sand], and adjust it with such attentiveness that, while mixing it vigorously and wetting it quickly with a <m>wood</m>en<m> spoon, it does not become thicker than mustard. And in such a way cast [i.e., pour] first the clearest part, as is always at the surface, by shaking it a little higher in your mold. And doing so, since your animal is nearly covered, Once your animal is covered, blow on it very strongly to eliminate the small bubbles from your soaked <m>sand</m>. Then pour immediately the rest which is at the bottom and which is thicker than the upper part, fill your mold to the thickness mark you have made.<figure>
<id>fig_p113v_4</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdIO5bEE5QmFaQWxUYm8</link>
</figure> And while casting do not forget to shake your mold, that way your soaked sand runs and get in through every parts. Also tip your mold up on the side of the head, which is higher than the body. That way the contents of your mold will have the same thickness, and you'll spare <m>sand</m>. This being done with care, your <m>sand</m> will harden within a quarter of an hour. Then peel the <m>clay</m> contour which can be used later. Let it dry a little bit, than separate your mold from the slab of earth, and you can mold properly straight after. Cut away the extra parts with a knife, and make it mandore[c] shaped. You can reuse the <m>clay</m> frame. Scrape away what's on the back of the mold, because this part is the weakest; do not forget to make notches around it, that way the second cast would bind much better
<figure>
<id>fig_p113v_5</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdIO5V29QdUxZVmxCUDQ</link>
</figure> and would not move.</ab>

<ab>
<margin>left-top</margin>
<figure>#</figure>
or two <m>eggs</m>, because there is no danger if there is more. If you cast twice, just add <m>sal ammoniac water</m> into the first pouring.</ab>

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<ab>
<margin>left-bottom</margin>
<figure>
<id>fig_p113v_3</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5dEdzS040X3RjbkU</link>
</figure>
If you don't fill your mold with soaked <m>sand</m> at one go, do mix your second cast with the top part of the first cast, stir with a spoon, or the stick you had took to
soak your sand.</ab>
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[a]see p113v c1b

[b]I used this expression instead of translating "& non par l'eau dans le sable" literally.
[c]mandore is a kind of lute

.....

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<page>114r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f233.image</image>
<div>
<id>p114r_1</id>
<head>Second Casting</head>
```

<ab>When you have well molded the first part of the animal, that is to say the back, the spine, and the entire top part, and when the half-mold where the part is encased is hard enough <figure>
<id>fig_114r_2</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5SWVndzB0aEk5MWs</link>
</figure> turn over the bottom and the sides of the belly, sometimes using a knife for the thicker parts, and then with a pen-knife for the places where the more fragile parts are encased, like legs, tail, and similar things. Because the head is higher, you will have to dig deeper to expose it and do a deeper notch that has to be well exposed, by enlarging it from the outside, in order for the second mold that you cast on top of it releases well. A process for which the animal himself can be useful because his nature so permits. However, make sure that you reveal the <al>lizard</al>'s throat more than you would do a <al>snake</al>, because the <al>lizard</al> has big jaws and big bones that do not cast as easily, and has deep ears in which the wet sand is not as easy to remove. But you will avoid all of this if you expose half of the head and those things that you know do not release easily. When everything is well exposed, blow over it and wipe each part again with a cloth made of <al>swine</al> hair, slightly dampened into salt <m>water</m> or clear <m>water</m>, do that gently, that way the lower part of your uncovered animal will be very neat. <figure>#</figure>
<figure>□</figure> Once it is done, dip the covered side of your mold into <m>water</m> for a pater noster. It will suck water inside, that way when you'll rub with <m>oil</m> the uncovered side, the <m>oil</m> will remain on the top, and to prevent the second cast which you will cast on top from sticking. Dip your mold into <m>water</m>, take a particularly fine brush and rub with <m>oil</m> the entire mold but the animal; it should not be touched with <m>oil</m>. Make sure to use you brush between the parts and on all the fragile parts. And if by mistake <m>oil</m> is soaked, dip the mold into <m>water</m>, as it is said above, and drop by drop the <m>oil</m> will come back to the top.</ab>

<ab>
<margin>left-top</margin>
Mold small animals as big ones, however, for small animals, use very fine tacks made of cords of cittern, short and long ones.</ab>

<ab>
<margin>left-top</margin>
<figure>
<id>fig_p114r_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5NVR5Y2FtZHM0ck0</link>
</figure>
remove all the little tacks with the help of very small flat pincers.</ab>

<ab>
<margin>left-top</margin>
<figure>#</figure>
then
<figure>+</figure>illegible</ab>

<ab>
<margin>left-bottom</margin>
<figure>#</figure>However, leave the tack which is pinned into the throat so that it will help you to turn the head over, because it is sunk deeper than any other part, because when you drew the plan, you put higher up. It is at the level of the head that you must make the biggest notch and you should start at that notch to expose and then keep following by looking at the traces left by the tacks. </ab>

<ab><figure>#</figure> and if a small piece of nail, or piece of tail, or other fine things separate from the mold, put a little piece of <m>wax</m>between them, and join them with the tip of a hot <m>iron</m> wire, then smooth it in your mold with a knife, and make notches on the sides in order for the other half of the mold to adhere to it. <figure>□</figure>
<figure>
<id>fig_p114r_6</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5TTRuWU8yVktnc3M</link>
</figure>
</ab>

<cont/>
</div>

.....

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<page>114v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f234.image</image>

<figure>
<id>fig_p114v_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5dmlRQm1lTjc4aXc</link>
<margin>left-top</margin>
</figure>

<div>
<cont/>
<id>p114r_1</id>
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<ab>This thus prepared, put your first half-mold on the disc of <m>clay</m>, surround it with ^{a clay} circle and do a thickness mark similar to the first one. Then, with a rather thick brush, wet with <m>spirits</m> all your animal's exposed half, without, if possible, missing any part, because therein lies the secret to casting neatly. Then, as fast as possible before the <m>spirits</m> evaporate, soak your sand, and cast it, shaking, blowing, and moving the mold, as you did for the first one, making sure that what you pour first on the animal is clearer than the rest, so that it is soaked everywhere and highlights the finer parts. So, leave it for fifteen minutes so it sets, then remove the <m>clay</m> circle and disc. And, with a knife, join you two molds together in a longish form, like a mandore, as we say, because of how you cast. Then soak a little in <m>water</m> the whole mold up to the level of the joints. Because this is another secret to separate them ^{more easily}. Otherwise, you would break everything. Separate them like this skillfully. After this, gently and patiently remove half of the animal still in the mold, pulling it sometimes on one side and sometimes on the other, in order to avoid breaking it, or ruining parts of the mold. The mold, being empty, rejoin both half so that no dust comes in, and put them aside until you want to heat them up, make the runners[a], the vents and put the clamps. As for the animal, to avoid it drying, and in order to be able to use it to mold 4 or 5 of times, put it on <m>water</m>. I believe that it would be better preserved in <m>spirits</m>. However, it would be best to make the 4 or 5 molds the same day because these animals rot and will smell within the day. Having left your molds to dry out, for they will stay humid for at least a month, make the vents for one half. <figure>
<id>fig_p114v_3</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5dkk3MW10YzNIbDg</link>
</figure></ab>

<ab>
<margin>left-top</margin>
<figure>+</figure> Note that once you have made your second casting, if you left your mold without opening it for a day or a night, you would then need to put it into hot <m>water</m> or reheat it, rather than open it, like for the <al>crab</al>.</ab>

<ab>
<margin>left-top</margin>
If after both halves of the noyau have set and hardened, it would not open if you did not dip it into <m>water</m>. But as you dip the back of your mold into <m>water</m>, <m>oil</m> comes back to the top and touches the joints which separates them.</ab>

<ab>
<margin>left-middle</margin>
However, be sure that your second mold is thick enough, and hard enough otherwise it could break while dipping it into <m>water</m> if you want to open it. To avoid this mistake dip the entire mold into <m>water</m> and rub it everywhere, if the mold has set enough it will harden. Then take both molds with a rough cloth, in order to grip it better. Then strongly separate the molds as if you wanted to tear <m>bread</m> apart. But in case it breaks, rejoin them, and fortifies the joints with clamps.
<figure>
<id>fig_p114v_4</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5NmZxVUxKTmhYSXM</link>
</figure>
</ab>

<ab>
<margin>bottom</margin>
Afterwards you can repair the mistakes with a small pair of scissors.
<cont/>
</ab>
</div>

[a]There seems to be another use for 'gect'. In this particular context, it seems logical to assume it refers to 'runners'.

.....

<page>115r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f235.image</image>

<ab>
<cont/>
<margin>left-top</margin>
First cast the tail or the back of your animal, make a cut from the head to the middle to the tail where you cast, and which is the end of the mandore[a][b]. After casting do not forget to make holes, and to notch your casted <m>metal</m>. That way <m>metal</m> will run better than if do not notch it. Indeed <m>metal</m> get damaged into notches, do not run too much and does not swell as much as if you do not make any notch. Put small clamps made of <m>wire[c]</m> to the joints separating both molds it will reinforce the joints of your mold. Clamps prevent molds from distorting and from getting damaged while heating.</ab>

</figure>

<div>
<id>p115r_1</id>
<head>Note about everything above said[d]</head>

<ab>If your snake hasn't eaten for a long time, when you want to kill it into <m>urine</m> and <m>vinegar</m>, it would fill with <m>water</m> and would swell, and then it would dry when you fix it on <m>clay</m>.
Once casted, and removed out of the mold, you could see a small empty line between the animal and the mold where <m>sand</m> will come in when you make the second cast. Try to avoid that if you can, but do not give up your second casting because not much <m>sand</m> comes in this empty space and it will be easy to clear it with the help of a small needle, also this <m>sand</m> does not stick to the first cast.</ab>

<ab>Casts which must be reheated, like casts for <m>gold</m> or <m>silver</m>, or other matters which must be burnt inside the mold, need more <m>stone alum</m> than casts which are stripped of the mold easily, like <m>lead</m> and <m>tin</m>. However do not add too much <m>stone alum</m>.</ab>

</div>
<div>
<id>p115r_a2</id>
<head>Stripping animal</head>

<ab>If you'd made such a twist to your snake that you can not stripped its belly easily of the mold, and if the head and the mouth are thinner, and easier to be removed, do cut away the extra parts of the mold, and do reheat the mold again. The matter inside will burn, then add some <m>quicksilver</m> inside which will attract <m>ashes</m> and burnt <m>bones</m>, the pipe will get wider, and your animal will get easily out of the mold without being damaged. You can strongly pull your molded snake, because it is soft and malleable, and your mold firm enough to keep its print undamaged.</ab>

</div>
<div>
<id>p115r_a3</id>
<head>Reheating molds</head>

<ab>When your molds are reheated, and cleaned, cast as soon as possible, because if you keep your molds it will corrode and become damp. Reheat the thickest part, where the body of the animal is, on high position because if there is something to be burnt inside it will fall down. Mark the high part in order to recognize it.</ab>

<ab>Once well reheated, do not damp the outside of your mold too much because the inside must stay red when you cast.</ab>[e]

<ab>
<margin>left-bottom</margin>
It is the shape of the clamp</ab>

<figure/>

<figure/>
</div>

[a]?
[b]ms gives "mandorre"; Cotgrave via Dicfro gives: "a kitt, small gittern, or instrument resembling a small gittern."
<http://www.micmap.org/dicfro/search/cotgrave/mandorre>
[c]Should be translated as "iron wire"? ms gives: "fil de fer"
[d]Perhaps better translated as "Note on the above"?
[e]tc and ten have 4 anonymous blocks for this entry, but tl only has two entries.

.....
<page>115v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f236.image></image>

<div>
<id>p115v_1</id>
<head><m>Tin</m> casting</head>

<ab>If the <m>herbage</m> or <m>flower</m> you want to cast is delicate and fine, <m>tin</m> must exceed <m>lead</m> in your mixture; on the contrary, if it the flower or herbage is thick, you must add into your mixture more <m>lead</m> than <m>tin</m>. For a fine thing<m>,</m> add one part of <m>tin</m> for four parts of <m>lead</m>. Your mold must not be too hot, so you can hold it with your hand when you cast. Your alloyed <m>tin</m> must be very hot and almost red for casting, and that way it will enter all the small parts of the mold. Otherwise, your tin will cool down before reaching the thin parts of your <m>herbage</m>. Do not forget to mix a little <m>bismuth</m> in your mixture; that way, your tin will run better and be firmer.</ab>

<ab>
<margin>left-top</margin>
With your <m>lead</m> allied with <m>tin</m>, or <m>tin</m> allied with <m>lead</m>, you can add a very little quantity of <m>bismuth</m> and it will look silvery. But do not put too much, because it hardens the matter and makes it sour, and if your work on fine things -- like legs and similar things -- it will not cast properly.</ab>

</div>
<div>
<id>p115v_2</id>
<head>Giving ventilation to your mold</head>

<ab>If your herbage is very fine, in addition to vents, you can drive a thread through the <m>clay</m> contour of your herbage instead of casting with a needle, and thread it also in the herbage. The thread will burn, like the herbage, and the hole will remain, thus giving extra ventilation.</ab>

<ab>
<margin>left-middle</margin>
Do not remove your annealed molds out of the <m>fire</m> before they have cooled down by themselves. Also make sure that your very hot molds do not come into contact with wind or coldness otherwise they would burst. Do not reheat your molds before they are drained and firm again, otherwise they would cast before the vent holes are made and the clamps are fixed.</ab>

</div>
<div>
<id>p115v_3</id>
<head>Annealing core molds</head>

<ab>After your molds have their cast vents and clamps, fix the clamps as above mentioned. Then make a round, made of <m>bricks</m>, at the edge of your forge, or in a clean place. Fill this round with big, half-lighted pieces of <m>charcoal</m>. Put your molds on the round, and leave it during half an hour, that way it will gently heat up and not all at once. Finally, when <m>coals</m> start being white, add big half-lighted big pieces of <m>charcoal</m> and blow with your small bellows until hot. Do as

such, slowly, until the molds redden inside and out. If you want to cast a herbage or an animal which are difficult to strip, and you need to burn what is in the mold, look into the cast and see if it is molds are quite red and inflamed inside; this will be a sign that they are properly annealed. If it is not so, increase the heat and keep at it until it becomes thus. If there is nothing inside the mold to be burnt, this heat is good for <m>tin</m> casting. The molds for <m>tin</m> casting are much better annealed in a reverberatory furnace.</ab>

[image]

<ab>
<margin>left-bottom</margin>
<figure>
<id>fig_p115v_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5dHVja3NnV0dmMlk></link>
</figure>
Do not heat too much at first otherwise your molds could break</ab>

<ab>
<margin>left-bottom</margin>
You know that your molds are annealed enough if the hole in the cast is red, on the contrary if the hole is black the molds are not annealed enough.</ab>

<ab>
<margin>left-bottom</margin>
If you want to cast <m>silver</m>, do not anneal twice, and if something stays in the mold and that must be burnt because no [a] good [...]</ab>
<cont/>
</div>
[a]should be translated as: "it is no good"?
.....

<page>116r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f237.image></image>

<div>
<id>p166r_1</id>
<head><tl>Casting</tl> <pa>plants</pa> and <pa>flower</pa> bouquets</head>

<ab>
One must cast them as soon as they are picked because they wilt and dry out. Dunk them, therefore, right away in good <m>spirits</m> and then put them in your <tl>sand mold</tl>, soaked like <al>snakes</al>. There is nothing to be released ^{from the <tl>mold</tl>s} because while the <tl>mold</tl>s are heating up, the <pa>plant</pa> material burns away, which does not happen with animals, which have <bp>bone</bp>s. For bouquets, the sand must be finer than for <al>snakes</al> because, if it were thick, it would crush the <pa>flower</pa>.
</ab>

<ab>
<margin>left-top</margin>
<figure>
<id>fig_p116r_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5SEdPVDdsS2dEUmM></link>
</figure>
burn the <pa>plant</pa>s, <pa>flower</pa>s, and <al>animal</al> parts that are remaining. Afterward, they must be cleaned and the <m>cinder</m> removed. Then[a] reheat them ^{until} they are red for <tl>casting</tl>. Reheat in the beginning with some <m>charcoal</m> lit over a gentle fire, and put your core <tl>mold</tl>s down flat on the <m>charcoal</m> encircled by <m>bricks</m>. <tl>Molding frames</tl> are reheated on a <tl>grate</tl>.
</ab>
</div>

<div>
<id>p116r_2</id>
<head>Core <tl>molding</tl> the same sand ^{for} <m>wax</m> images, <m>lead</m> medals, and similar things[b]</head>

<ab>Make sure that these are well emptied. If they are not, fill ^{them} with <m>wax</m>, which will not be it, [c] and then smear your <m>wax</m> or <m>lead</m> images with a very thin and light layer of <m><pa>olive</pa> oil</m> in such a way that the <m>oil</m> does not give any thickness or body to the medal. Then heat it with a little <m>spirits</m> and, when it is lukewarm, wet the <m>oiled</m> medal, because if the <m>water</m> is cold, it will resist the <m>oil</m>. It will not hold on to it. After, cast your <m>plaster</m> sand,<m>matton</m>, and <m>alum</m>. Wet it on top having placed your image on <m>clay</m> mold in the shape of a mandore and having given it some contour to give the necessary thickness. These medals <m>oil</m> and rub against themselves in this way with the <m>spirits</m> because they are firm and hard, and they would not ready to be taken out of the <tl>mold</tl>, which is resistant like real <al>animals[d]</al>, which are soft and flexible.</ab>

<ab>
<margin>left-middle</margin>
When you have soaked your sand, do not cast on the middle of the <tl>mold</tl>, but on the edge so that the sand gently settles, so that the first <m>cement</m> be clear and thick afterwards. If your soaked sand sizzles[e] like <m>water</m> falling in a <tl>mold</tl>, it is a sign that it is too light. Make it, then, of an average thickness.
</ab>
</div>

<div>
<id>p116r_3</id>

<head><tl>Casting</tl> <m>gold</m> and <m>silver</m></head>

<ab>
The <tl>molds</tl> must be fire-red and ablaze when you cast <m>gold</m> or <m>silver</m> in them, or when you burn <pa>flower</pa>s and <pa>bouquet</pa>s. The gate must not be very big where the thing being cast enters because the matter will flow better without disturbing the <tl>mold</tl>, and does not become so filled with bubbles.
</ab>
</div>

<div>
<id>p116r_4</id>
<head>A way to <m>enamel</m> very fine <m>gold</m> <m><pa>rose</pa> leave</m>s and others</head>

<ab>
After you have cast or beaten in <m>gold<f></m> the thin <m><pa>rose</pa> leaves</m> or other things, if you want to <m>enamel</m> them, you must solder or braze your fine <m>gold</m> leaves onto <m>silver</m> strips, which will strengthen them to support the <m>enamel</m>. After the things has been <m>enameled</m>, put the work in <m><la>aqua fortis</la></m>, which will eat away the <m>silver</m> and leave all the <m>gold</m> with its <m>enamel</m>. For this, the <m>gold</m> must have been passed...<g>
</ab>
</div>
<a>I have translated this from "secondement" which I flagged in the diplomatic transcription.
Verify translation
Could he be talking about lost-wax casting here?
<c>verify translation
<d>In the original - "animaux naturels"
<e>"crie"
<f>"battu en or"
<g>More text under the tape

.....

<page>116v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f238.image</image>

<div>
<id>p116v_a1</id>
<head><tl>Casting</tl> <al>snake</al>s in all <tmp>season</tmp>s</head>

<ab><al>Snake</al>s hide underground during <tmp>winter</tmp>. Some people feed them a lot in <tl>barrel</tl>s filled with <m>earth</m> and covered with <m>manure</m>. Other people make several <tl>mold</tl>s in <tmp>summer</tmp> because you can make four or five <tl>mold</tl>s with one single <al>snake</al>. Other people <tl>cast</tl> long, natural <al>snakes</al> without coils, using <m>common plaster</m>, annealed as said, with a core and two half-molds. Then they cast it in <m>wax</m>, that way they have a <al>snake</al> as twisted as they want. Then they cast <m>metal</m> as it is said.</ab>

<ab>
<margin>left-top</margin>
If, while twisting your <m>wax</m> snake, some parts become undone, you can rework it if you remove the drips from the <tl>molded</tl> <m>wax</m>.</ab>

</div>
<div>
<id>p116v_a2</id>
<head><m>Tin</m> and <m>lead</m> mixture</head>

<ab>For fine <pa>herb</pa>s, <pa>flower</pa>s and <pa>greeneries</pa>. You need more than 3 parts of <m>tin</m> for one part of <m>lead</m>. If this material is thick and fat, you need 3 parts <m>lead</m>. Heat the mostly <m>tin</m> mixture, which must become red and very hot. When you want to <tl>cast</tl>, remove your <tl>crucible</tl> from the fire, and add two or three grains of <m>resin</m> for one and a half or two <ms>pound</ms>s of <m>lead</m> or <m>tin</m>. With the <m>resin</m>, also some fat <m>looking-glass tin</m>, the size of an <fr>auclane<a></fr> with its shell, mix and cast. Make sure you have more <m>metal</m> than you need; some <m>metal</m> should be set aside. If you haven't enough <m>metal</m>, keep <tl>cast</tl>ing and finish your <tl>cast</tl>, it will set, however it will not be so neat. Dip your <tl>mold</tl> into <m>water</m>, and dismante your <tl>mold</tl> carefully with a <tl>point</tl>. Make an elongated <tl>cast</tl> in order not to damage anything. If you <tl>mold</tl> something very thin, you must make your <tl>cast</tl> with mostly <m>tin</m>.</ab>

<ab>
<margin>left-middle</margin>
If you want to <tl>cast</tl> <m>lead</m> or <m>tin</m> as a core with the sand above mentioned, reheat your <tl>mold</tl> once only if there is nothing to be burnt inside. But if there are <pa>flower</pa>s or <al>animal</al>s to be burnt inside the <tl>mold</tl>, reheat it twice. However do not reheat the second <tl>cast</tl> for as long as the first <tl>cast</tl> for <m>lead</m> and <m>tin</m>. As for <m>gold</m> and <m>silver</m>, those must be red when you <tl>cast</tl>. For <m>lead</m> and <m>tin</m>, let them cool down until you can dip your <tl>bp>finger</bp></tl> into the <tl>cast</tl> without burning yourself. The <tl>cast</tl> must be warm.
</ab>

</div>
<div>
<id>p116v_a3</id>
<head><m>Copper</m> and <m>tin</m> <tl>cast</tl>s</head>

<ab>For <m>red copper</m> you must add <m>sal ammoniac</m>. It will clean it and remove the crust when melted. Some people add big pieces of <m>new leather</m>. Other add peels of <m><al>mice</al> feet</m>. Other add melted <m>common salt</m> only, or <m>salt</m> melted with <m>saltpeter</m>. The main ingredients are <m>sal ammoniac</m> and also a little bit of <m>tin</m>, depending on the <tl>cast</tl> you want to do. <m>Copper</m> is harder to <tl>cast</tl> than <m>tin</m>. But when it is melted it runs better, even if alloyed with <m>tin</m>. Add a quarter of <m>copper</m> to <m>tin</m> and mix it like <m>copper</m>. <m>Calamine</m> especially makes it run well.</ab>
</div>

<a>MM: "I couldn't find an animal with a shell with this name. I looked in Cotgrave, Occitan dictionaries, Medieval French dictionary, dic fro, etc."

[b]Perhaps related to "lucane" (n.m.: Nom générique du cerf-volant, coléoptère remarquable par son dimorphisme sexuel. (Le mâle a des mandibules démesurées, atteignant parfois 8 cm de long ; la larve creuse ses galeries dans le bois des chênes)." <http://www.larousse.fr/dictionnaires/francais/lucane/47975?q=lucane#47895>
Show less

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<page>117r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f239.image></image>

<div>
<id>p117r_a1</id>
<head>A way to mold <pa>flower</pa>s and <pa>herb</pa>s</head>

<ab>It is necessary to pick them in full bloom and color, and only pick them, if it is possible, when you are ready to mold them so that they will not wilt. Or if you need to carry them from afar, put their stems in a bottle of <m>water</m>, or even better in <m>wine</m>. Start with making a <m>wax</m> stick pointed like a big peg, of a size adapted to what you want to cast, as you see in the margin, marked A. At the bottom of this peg, put and prick a bit of <m>iron</m> wire, rather thick and strong. And at the pointed end of the big peg, you will fit your <pa>flower</pa> stem or your <pa>herb</pa>. This done, set it aside. Arrange your <m>clay</m> circle and outline as you did for the <al>snake</al>s, in the shape of a <mu>mandore</mu> according to the size of your <pa>herb</pa>. Then take the <pa>herb</pa> you have prepared to be cast, as we have taught you, and wet and moisten it daintily with good <m>spirits</m> with a <tl>brush</tl>. Then place the wire end through through the end of the <m>earthen</m> mold, shaped like a <mu>mandore</mu>, so that the <m>wax</m> peg, where the <pa>herb</pa> is, touches the edge of the <m>earthen</m> <mu>mandore</mu>. And arrange your <pa>herb</pa> that lies thusly so that it remains in the middle of said <mu>mandore</mu>-shaped mold, without touching any of the side. So that the sand that you put in the mold will be of equal thickness everywhere, and will be two <ms>finger</ms>s higher than your <pa>flower</pa>. And having made sure that the wire is secure, quickly throw your <m>wet sand</m> that will be much clearer than the one used for serpents, in the <mu>mandore</mu>-shaped mold, and the quicker the better, so that your <m>spirits</m> do not evaporate. And adjust your mixture with a bit of <m>essence of sal ammoniac</m>, unlike the one for serpents. And when you make sand especially for casting your bouquets, mix in a bit of <m>feather alum</m>, this will only make it better. In this way, your mold will be in one piece and it will not be necessary to separate it, but you will have to separate it from the <m>clay</m> once it will have set. Let it dry. And then fire it again until the herbs are burnt. Note that if the herb you want to cast has a strong stalk and stem, you can cast it lying down, as said. But if it is fine and delicate, plant up standing up with the <m>wax</m> peg so that the sand will always raise it up, it being very clear. Do not forget to arrange two vent holes with two small <tl>stick</tl>s, which will start at the bottom of the cast, which is the pointed end of the <m>wax</m> peg.</ab>

<ab>
<margin>left-top</margin>
If the <pa>flower</pa>s are so delicate that they come off the top when you are putting in the <m>wet sand</m>, you can stop this and curtail it with a <m>net</m> sewed onto the mold with a <tl>needle</tl>.</ab>

<figure>
<id>fig_p117r_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5WnpBbm8wN2gyTWs></link>
<margin>left-top</margin>
</figure>

<figure>
<id>fig_p117r_2</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5VURRYzIWOFByNzQ></link>
<margin>left-middle</margin>
</figure>

<ab>
<margin>left-middle</margin>
The more delicate the <pa>flower</pa> is, the clearer the sand must be.</ab>

<figure>
<id>fig_p117r_3</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5dVB5bzVUWWJuS0U></link>
<margin>left-bottom</margin>
</figure>

<ab>
<margin>left-middle</margin>
This is a core mold.</ab>
</div>

.....

<page>117v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f240.image></image>

<div>
<id>p117v_a1</id>
<head>To make the ashes of <pa>flowers</pa> and <pa>herbs</pa> leave <tl>molds</tl></head>

<ab>Some people put <m>quicksilver</m> inside. But, if it is a little work, or fine & delicate foliage, that only has a slender issue, they make two errors: the first, that <m>quicksilver</m> by its heaviness can destroy some delicate tract inside in coming out, the other, that some bit will always linger inside that will make [other] <m>metals</m> sour & inhibit the perfection of the casting. It is true that if it is to empty the <tl>mold</tl> of some large animal that has big conduits & passages through which the <m>quicksilver</m> can easily exit, like of a bird or a serpent, one can indeed put some amount of <m>quicksilver</m> to break the calcinated bones of the animal in escaping, because of which the said <m>ȝ</m> will empty out & not remain at all.</ab>

<ab>
<margin>left-top</margin>
The <pa>asparagus</pa> stalk is so hard that most often it perseveres as charcoal. Because of this, do it rather dried out or wet it with <m>sulfur oil</m> &
<m>turpentine</m>, or indeed cast the little branches separately & solder them onto a fat stalk drawn through with a wire.</ab>

<ab>
<margin>left-middle</margin>
If the burnt thing has left behind some filth or ash, let it cool down a little, & with an <m>iron</m> <m>wire</m> wrapped in <m>cotton</m> that can bend with the
cavities that you have, investigate. Clean & blow to a fault, either with a soft <tl>brush</tl> or a clipped <tl>brush</tl>.</ab>

</div>
<div>
<id>p117v_a2</id>
<head><pa>Daisies</pa></head>

<ab>They can be cast well in <m>gold</m>. But if you want to <m>enamel</m> them, you have to make them by hand & <m>enamel</m> them & then attach them.
Otherwise, the <pa>leaves</pa> would be so pressured that it would encrust the <m>enamel</m>.</ab>

</div>
<div>
<id>p117v_a3</id>
<head><m>Sand</m> that has already been used</head>

<ab>Do not cast it. But because it is mixed with <m>stone allum</m>, you can use it in the mixture of other things & it can serve in place of <m>brick</m>.</ab>

</div>
<div>
<id>p117v_a4</id>
<head>Core <tl>molding</tl> <m>wax</m> figures or <m>lead</m> medals</head>

<ab>Rub them with <m>oil</m> with a <tl>brush</tl>. But be sure that it be so softly that your medal be barely wiped with it & that it hardly appear to have been greased.
After rub it with <m>spirits</m> and heat the <m>water</m> with which you will wet your <m>sand</m>, that of <m>plaster</m>, <m>brick</m> & <m>alum</m>, so
that being sort of lukewarm when you cast it, the <m>oil</m> will not refuse it, as it does with cold water, & do not forget to moisten your <m>sand</m> with it, mixing it
always with a little <m>sal ammoniac</m>.</ab>

<ab>
<margin>left-bottom</margin>
Medals <tl>molded</tl> in this way are very fine.</ab>

</div>
<div>
<id>p117v_a5</id>
<head><m>Snakes' blood</m></head>

<ab>If you have to cut some snake inside a <tl>mold</tl> in order to burn it, cut it far from the intake of the <tl>mold</tl> so that no <m>blood</m> at all remains,
because it makes a crust that afterwards would not be taken away by the same <m>quicksilver</m> & removes impressions from your <tl>mold</tl>.</ab>
</div>

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<page>118r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f241.image</image>

<div>
<id>p118r_a1</id>
<head><al>Snails</al></head>

<ab>Those which are found in the <env>ocean</env> and similarly those in which certain small crabs dwell are very beautiful for grottoes, if they have been rid of the first
layer of their shell with <m><la>aqua fortis</la></m>, because they really look to be made of <m>mother-of-pearl</m>.</ab>

</div>
<div>
<id>p118r_a2</id>
<head>For removing flaws and flashing from things <tl>molded</tl> in a <tl>core mold</tl></head>

<ab>If your <tl>molds</tl> are made of good <m>plaster</m> which withstands fire, they will not crack, especially if they are squeezed well between two tables with
<tl>presses</tl> or in a <tl>pot</tl> filled with <m>ashes</m> or sand, and, therefore, they will not develop flaws. If they they develop some ^{flaws}, you can
remove them with a <tl>burin</tl> called <fr><tl>chaple</tl></fr> which has a tip like a small <tl>chisel</tl>.</ab>

<ab>
<margin>left-middle</margin>

The old <pa>vine</pa> stocks</ab>

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</div>
<div>
<id>p118r_a3</id>
<head>For grottoes[a]</head>
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<ab><m>Stone</m>s made from water called <m>stuf</m>; <m>charcoal</m>; the confection of <m>tin</m> and pure[b] <m>brass</m>; <m>paper</m> ground up and mixed with ground up <m>glass</m>; <m>cork</m>; <m>heated parchment</m>, ^{and} <m>white coral</m>, all of them are appropriate for making grottoes. But, once dry, bizarre pieces of <m>wood</m>, which [the bizarre pieces of wood] are found in the forests and among ^{regular} mushrooms and <fr><pa>potirons</pa></fr> ^{growing on} trees, are better than all of them because they are light. Mix in with small pieces of <m>looking-glass tin</m> which have a shiny luster. Grind in a little of a different kind of <m>marcasite</m> in order to clean the <m>earth</m> away, then powder the work, which is very beautiful. If there is no spring in the grotto, glue all of this with strong <m>glue</m>, and then the work is done. You must show all kinds of <m>mines</m>. The <m>sulfurous marcasites</m> which do not have a grain but are even like <m>looking-glass tin</m> are very beautiful.</ab>

<ab>
<margin>left-middle</margin>
<m>Red copper</m> is found sometimes mixed with certain brittle lumps which are pulverized under the <tl>hammer</tl>, which are very beautiful pulverized on grottoes.</ab>

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</div>
<div>
<id>p118r_a4</id>
<head><m>Looking-glass tin</m></head>
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<ab>If one ounce of ^{<m>looking-glass tin</m>} is put in with it ^{pure <m>tin</m>}, pure[c] <m>tin</m> whitens and becomes firm, that is, i
<figure>3</figure> of <m>looking-glass tin</m> in one <ms>lb</ms> of pure <m>tin</m>.</ab>

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</div>
<div>
<id>p118r_a5</id>
<head><m>Salt ammoniac</m></head>
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<ab><m>Salt ammoniac water</m> is naturally quite good for <tl>casting</tl> in <m>gold</m> and in <m>silver</m>.</ab>
</div>

[a]Probably interesting to compare with the work of Buontalenti in his grottoes at Florence: technique of making artificial stalactites and stalagmites.
[b]Under "fin" in Cotgrave: "Wittie, craftie... fine, smalle, prettie... perfect, exact, pure, exceeding good." It seems like either "small" or "pure" could work here, but it seems like "pure" makes more sense.
[c]See comment above regarding "fin."

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<page>118v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f242.item</image>
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<div>
<id>p118v_1</id>
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<head>Casting in <tl>box molds</tl></head>

<ab>The same sand that was used in the reheated <fr>noyaulx</fr>, composed of <m>plaster</m>, <m>brick</m> and <m><fr>alum de plume</fr></m> as is said, is excellent for casting in <tl>box molds</tl>, and I have experienced it thus: I crushed the pieces which had come out of <tl>core molds</tl> in a <tl>mortar</tl>, dragging the <tl>pestle</tl>, because this sand is very <fr>doux</fr>. I did not pass it through the <tl>sieve</tl>, because the <m><fr>alum de plume</fr></m> mixed throughout, which binds it, would not pass; but I ground finely upon <tl>marble</tl> what seemed to me too coarse. And having prepared it thus, I moistened it with <m>sal ammoniac water</m>, made of <m>sal ammoniac</m>, as much as the size of two <ms><pa>walnuts</pa></ms>, in a bottle of <m>common water</m> the same size as a <tl>bottle</tl> in which one boils <m>tisane</m>, or in a good <tl>pot</tl> of <m>water</m> so that you find the <m>water</m> fairly salty. I mixed throughout half a <ms>glass</ms> of <m>sal ammoniac</m>, two <tl>silver spoonfuls</tl> of <m>spirits</m>. Having thus moistened the sand in such a way that it took hold well, nevertheless coming apart easily, I sprinkled my medal with <m>pulverized charcoal</m> with a <tl>file</tl> to rid it of <m>oil</m>, and all other <m>grease</m>. It is necessary to avoid them, for they would hinder good stripping. I blew on my medal & molded it, and the female part of the <tl>box mold</tl> once filled, I marked & made a line on the reverse & edge of the medal, & on the nearby sand as well. In order that the second <tl>box mold</tl> take the imprint thereupon to indicate the place for making the cast, once filledthe female part of the box mold once filled I uncovered the outline of the medal and pounced the whole side with <m>pulverized charcoal</m>, and then filled the male part with sand. Once mad[a] I separated the <tl>box mold</tl> and did not hit the corners of the medal to make it strip, because that knocks the sand and makes it crumble. Rather I struck the back of the <tl>box mold</tl>, retaining the obverse of the medal on the bottom, and it molded very neatly. If it had not stripped thus, I would have waited to remove it until the <tl>box mold</tl> had been dried out over fire. I lit a row of <m>charcoals</m> between two little <tl>trivets</tl> of <m>iron</m> in the form that you see, and put the back & reverse of <tl>box molds</tl> thereupon, & the imprint on top, <figure>
<id>fig_p118v_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5Rm5qaUJCnWVFWlk</link>
</figure>
because in this way, they dry out gently. And if, by chance, they should crack from being too moistened, it is on the back that they take the harshest fire, & the imprint remains safe & whole.</ab>

<ab><margin>left-top</margin>
For the best, it is necessary to reheat the sand that is used in the <fr>noyau</fr> rather than used it in the <tl>box mold</tl>, until it no longer contracts.</ab>

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<head><margin>left-top</margin>
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Excellent sand</head>

<ab>Take a little of the same sand, the finest that you can, for covering the medal[b].</ab>

<ab>
<margin>left-middle</margin>
For medals & flat things, the true heat of <m>lead</m> & <m>tin</m> is when it is melted gently.</ab>

<ab>
<margin>left-middle</margin>
Note that I filled the <tl>box mold</tl> before pressing it, and did not hit it, but rather pressed it with the strength of my <bp>hands</bp> alone, because hitting it makes it go awry. Secure your <tl>box mold</tl> that it does not shift at all, & if you put some moistened sand under it, it will only hold in place more firmly.</ab>

<ab>
<margin>left-bottom</margin>
Make the gate that it is not too thick, so as not to overcharge the medal, but wide enough near the medal that it embraces a third part. Do not forget the vents.</ab>

<ab>
<margin>left-bottom</margin>
Drying <tl>box molds</tl> is to keep them from humidity, that they no longer smoke, once having nevertheless been very hot.</ab>

<ab>
<margin>left-bottom</margin>
<fr>Require</fr> is to redden the box mold, which is done for <m>gold</m> and for <m>silver</m>.</ab>
</div>

[a]:) with irony.... not bronzy...
[b]There is a line underneath this marignal note.

.....
<page>119r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f243.image</image>

<div>
<cont/>
<id>p118v_2</id>
<ab> When I saw that they did not smoke anymore, & scratching the back & the front of the casting & having found that they are rough & firm & hard on both sides, which is a good sign of their being quite dry, I leave them to cool down. I took some fine <m>tin</m>, one lb., & one ounce of fine & new <m>lead</m>. I melted it in a crucible just until it was a bit red. Being in this way quite hot & ready to cast & no sooner, I smoked every side of my box frame with the smoke of a <m>tallow candle</m> & pressed & cast & everything. I set my box frame, well joined, in the press. I drew my crucible from the fire. I let it sit a bit to quench the redness of the crucible's bottom. And wanting to cast, I threw in around two or three grains of rosin, and about the measure of a bean of <m>looking-glass tin</m> & blended it & stirred the crucible a little and cast. And the medal came out as neat as the original. I smoked it with the candle & cleaned it with my <m>coat</m>.</ab>

<ab>
<margin>left-middle</margin>
Always cast through the foot of the medal because the head, which is lower, will come out better, and make the casting somewhat long. And when you cast several medals in a large box frame, they will come out better.</ab>

</div>
<div>
<id>p119r_1</id>
<head>Notice about everything above</head>

<ab> Good <m>tin</m> is that which is hard as <m>silver</m> & soft nevertheless. If your work is fine, it must be almost all <m>tin</m> & alloyed as is said.</ab>

<ab><m>Looking-glass tin</m> must not be mixed until the instant that you want to cast.</ab>

<ab> Nor must the forms be smoked until then.</ab>

<ab> If the <m>sand</m> shrinks in the box frame, this means that it must be reheated & turned red on the fire.</ab>

<ab>Good <m>sand</m> when moistened does not stick at all to the hand when pressed.</ab>

<ab>The perfect <m>sand</m> for the box frame is the <m>asphalt</m> found in Germany, which is soft as wet <m>flour</m>, & almost all the rest are lumpy.</ab>

<ab>
<rub>Nota</rub>that the casting must be fine & hardly thick so that it does not work the material at all, & must not exceed the thickness of a grain of <m>wheat</m> taken crossw-ise, likewise for <m>tin</m> that is going to be cast finely. For <m>lead</m>, a little thicker. There is no need to make air vents very large & deep either.</ab>

<ab>
The <m>sand</m> that you use à noyau for the said mixture is excellent for a box frame. But in washing, crushing & reheating it several times, its nature will be corrupted

& it will no longer be fit to be taken & molded <pro>en noyau</pro>.</ab>
</div>

.....

<page>119v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f244.image</image>
<div>
<id>p119v_1</id>

<head>Way to rework spat</head>

<ab>You find an earth in <pl>Germany</pl>, the color of cooked plaster, but have long and soft filaments which are easy to handle. Because it is mixed with earth and rubbish, one pounds it roughly and forms little balls having soaked it previously in <m>sal ammoniac</m> <m>water</m>, following the above indication. Reheat these balls in a <pro>potter</pro>'s furnace. And then soak them in the same <m>water</m> again. Earth and filth will remain at the bottom, and pure <m>aspalt</m>, which is light, soft and malleable as soaked <m>flour</m>, sticks to <m>water</m> and rises to the surface, making the <m>water</m> cloudy. Pour this cloudy <m>water</m> in another pot. Letting it rests, one empties the <m>water</m> tilting the pot or taking the <m>water</m> out using a sponge. And the pure<m>aspalt</m> remains at the bottom. Dry it and use it in frame, having moistened it with <m>sal ammoniac</m> <m>water</m> Then soak it into <m>sal ammoniac</m> <m>water</m>, and use it for frames in which it shrinks when reheated or dried, that is to say that one needs to reheat it again on a good fire and reddens it, as for example for medals and flat things. Use this one, because it is the best of all for <m>gold</m>, <m>silver</m>, <m>copper</m>, <m>lead</m>, <m>latten</m>, and <m>tin</m>. Because it withstands fire and reddens whenever you need, without deteriorating. The more you use it the better it is, and does not change. In the beginning it is white, and while using it turns grey. However make sure to put aside the one that served for casting <m>lead</m> and <m>tin</m> and <m>latten</m>, because <m>gold</m> would get brittle, and would not come out well. And even better, you will put some of this aside to use it for each metal. </ab>

<ab>
<margin>left-top</margin>
See gesnerus de lapidibus
</ab>

<ab>
<margin>left-top</margin>
It withstands 10 or 12 casts without deteriorating. It withstands fire and reddens. It is well suited for all metals. It is so resistant that if the frame is grooved [a], it holds.
</ab>

<ab>
<margin>left-middle</margin>
<m>spat</m> almost does not set, even if it is reheated and [even if] it is a kind of <m>plaster</m>, which you can easily break with your fingers.</ab>
</div>

<div>
<id>p119v_2</id>

<head>Excellent secret to mould in hollow and very delicately in fine <m>gold</m></head>

<ab>Cast your animal or <al>lizard</al> or other thing with the above mentioned sand, with alloyed <m>silver</m> [b] and it will come out very neat. But make sure to mould it hollow, or at least leave a small hole in it or in another place. Then gild your animal with fine <m>gold</m>, as much homogeneously as possible, three, four or five times, until your <m>gold</m> becomes as thick as a piece of <m>paper</m> or anything similar. And the scales will always come out evenly. Then dip your cast into good <m>aqua fortis</m>, which will corrode the alloyed <m>silver</m> through the hole, and <m>gold</m> will remain hollow, light and wonderful.</ab>
</div>
[a]For "canelé", Cotgrave says "channeled, fluted, furrowed, straked".
[b]argent de billon: a metal where the proportion of other metals is more important than silver

.....

<page>120r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f245.image</image>

<div>
<id>p120r_1</id>
<head>Stamped medals made from <m>wax</m></head>

<ab>You can mold [a] your relief with <m>wax</m> mixed with a bit of <m>resin</m> to make it harder and firmer, whatever relief that you wish, either an animal or a medal. And then, fashion a hollow [b] out of this relief in <m>brass</m> or <m>copper</m>. Or mold your [<m>wax</m>] in relief [c] and strike it in a sheet of <m>tin</m>. And fill [d] [your final product] with <m>lead</m>, and heat it. Try [to use] blades of <m>stone leaf</m> [e] to make the hollow for lizards & [f]</ab>

</div>
<div>
<id>p120r_a2</id>
<head><m>Sand</m> for casting <m>gold</m></head>

<ab>Take <m>common sand</m> made of <m>alum, plaster</m> and <m>brick</m>, according to the composition above-mentioned. Add to it some more <m>feather alum</m>. And add as well something [g] a third part of <m>crocum ferri</m>. However whatever quantity is never harmful, because [<m>crocum ferri</m>] is the one that makes <m>gold</m> shiny and makes it come out of its cast easily. But it is best if your <m>crocum</m> has previously been left for three days and three nights in the furnace of <pro>glassworkers</pro>, in a flat box, where it is not very thick so that it will reheat better.</ab>

<ab>
<margin>left-top</margin>
You can easily cast <m>gold</m> with the <m>common sand</m> of <pro>goldsmiths</pro>, but [make sure] that you add some substance that makes it runny. Before the invention of <m>crocum</m>, one cast flowers in <m>silver</m>, but not in gold. This has only been known in Germany for forty years.</ab>

<ab>
<margin>left-middle</margin>
<m>Sublimate</m> is often used by <pro>goldsmiths</pro> for <m>gold</m>. Some people add <m>sulphur</m>, but they are all wrong, because <m>sulphur</m> makes things sour, even if it heats [well.]. And <m>sublimate</m> gets agitated,boils and bubbles. It is very good to clean <m>gold</m> because its exaltation draws [the bad stuff] out that disappears in the smoke. But to heat up <m>gold</m>, and to keep the heat, there is only the color of <m>verdigris</m>, <m>salt of ammoniacum</m>, <m>salt peter</m>, and <m>borax</m>. This makes it runny and you will be able to throw in a branch from the Santoinge[h].[i]</ab>

</div>
<div>
<id>p120r_a3</id>
<head>Nightingale</head>

<ab>It is necessary for the cage, made like those for larks found in barns[j] and lined with green fabric, to be made with something that slides out from underneath, so that you can refresh the dirt everyday, because the nightingale loves it, and mix in some ants. You can carry an ant-hill with its dirt in a barred full of earth, and keep them there and they will lay their eggs and in this way, you can always have access to them when you need them, should you enjoy feeding nightingales. When you caught it, it was fat and full. And to keep it in such good shape, on the first day, you must take it in you hand and open its beak and place in it, using a sharp twig, some mutton heart or other delicate flesh, chopped up not too finely, so that the belly does something[k], and make sure that it does not get smaller or thinner, until it has gotten used to it. And feed it in this way three or four times a day, and also give it something to drink. The next morning, you will give him in his cage some well-minced flesh mixed with the yolk of a hard-boiled egg, and change this mixture two or three times a day, because it will not eat it if it is hard and not fresh. And If it goes half a day without eating, you must feed it as before[l] and do so until[m]</ab>

<cont/>
</div>

[a]This is probably not mold as in making a mold, but mold as in model or fashion.
[b]This is very unclear - could mean stamping the was relief in a sheet of brass or copper - therefore ending with a stamped sheet with hollows in it.
[c]This is the same process as the beginning of the text - the first step i.e making the wax relief is identical; it is only the second step that differs.
[d]This seems to happen for either step you may have taken to make the metal mold out of the wax relief - although this is not a 100% sure.
[e]Once again, we do not know what this is
[f]The ampersand is in the text - very odd that it should finish this way. The whole recipe is odd and not at the usual standard of precision.
[g]This needs to be retranslated
[h]Santoinge could be "santoline," sometimes confused for santonine. Santoline is "santolina chamaecyparissus," common name "cotton lavender." Santonine is derived from "santonica," or "Artemisia cina," commonly name "wormseed." Both plants have medicinal properties.
[i]We can find no other specific meaning for this other than a branch of from the place of Santoinge/Santonge.
[j]"alouette de grange"
[k]This word is hidden under the strip of paper. It starts with "en" but then we can't read it any further and didn't want to guess.
[l]meaning force feed...
[m]This sentence continues on the next page.

<page>120v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f246.image</image>

<div>
<id>p120r_3</id>
<ab>eat by itself. And so that he likes it better, you can add to the meat and the egg some mealworm, because nightingales love them.</ab>

</div>
<div>
<id>p120v_1</id>
<head><m>Crocum ferri</m></head>

<ab>It is excellent applied [a]to wounds to stop the bleeding. The best one is made of <m>needle filings</m>.</ab>

<ab>
<margin>left-top</margin>
<m>Crocum ferri</m> hardens moulds that have been reheated, the more <m>feather alum</m> is added, the softer and pliant they become.</ab>

</div>
<div>
<id>p120v_2</id>
<head>Anatomy</head>

<ab>You must skin the animal and boil it. Or skin it and put it into <m>quicklime</m>.</ab>

</div>
<div>
<id>p120v_3</id>
<head>Making <m>silver</m> runny</head>

<ab>If it is for a large work, <m>arsenic</m> and pulverized <m>tartar</m> thrown into melted <m>silver</m>, makes the <m>silver</m> runny. This is enough [for

silver.] But for subtle work, you need ~~bronze~~, filings, copper, ~~even antimony~~, ~~sublimate~~, all neatly ground. All of this makes a mass that does not get smaller. If the ~~plaster~~ is good, you do not need to add ~~crocum~~ for ~~silver~~, but one puts in more ~~feather alum~~. It is also necessary to add melted ~~common salt~~, and some ~~saltpeter~~ and the aforementioned drugs: ~~arsenic~~, tartar, bronze, copper filings, antimony &~~d~~

~~Siscitatio dubia~~

A bit of ~~tallow~~, and some ~~arsenic~~.

See the second folio following ~~this~~

Flowers can also be kept just as beautifully in ~~vinegar~~, distilled in a vase, well-sealed so it there is no draft, well-sealed with ~~wax~~ and ~~mastic~~. [For] carnations and roses, the residue of ~~vinegar~~ usually makes them rot. If the ~~sand~~ is dusty, and sticks to the flowers so that it hardly comes off with a brush, it is no good. The leanest ~~sand~~ is the best.

Keeping dried flowers in good condition year round

~~River sand~~, that is washed by the current of water, is good when strained in a cloth to make the powder compact.

This is a rare secret, and which is pleasing for decorating tables, rooms, studies out of season when winter denies you flowers. Be advised to pick them when they are in full vigor and still growing, because if you take them when no longer in bloom or when they are starting to wilt, they will not keep. Having therefore chosen them, take some ~~sand~~, the leanest and driest you can find, that must be very well ground, like the one ~~goldsmiths~~ use to sand ~~enamel~~, or like the one [used] for engraving. But this ~~sand~~ must not be dusty at all, nor must it stay on your hand or leave a trace when you have ground or poured it, because it is~~f~~

Make sure your box is well sealed so that the ~~sand~~ does not get out. Keep it uncovered in sunlight and keep it away from the evening dew, and the moisture of the night, and cover it and keep it in a dry place.

You can not put these aforementioned flowers in big vases, because if you want to take one out, you will take the whole bunch with it.

Be advised to not pick your flowers when it is rainy or humid, but when the sun has been shining on them.

[a]sinapisé: In Cotgrave: a plaster (sometime of mustard seeds) that is used to revive a mortified place i.e something to heal
[b]Is this bronze? See VOCABULARY under aes ustum.
[c]I think he meant copper filings here. There is something crossed out in the original text and instead of brass filings, I think he meant copper filings... instead of just saying "filings, copper...." And I think this due to his note where he restates the ingredients at the end of this same paragraph.
[d]This entire paragraph seems to be crossed out with diagonal bars.
[e]This is Latin and I do believe it means a "dubious citation." If this is the case, it would explain why the entire paragraph has been crossed out.
[f]This sentence continues on the other page.

<div>
<cont/>
<id>p0120v_4</id>
<ab>a sign that there is some humidity and if the flower was also watery it would rot. Moreover the sand must not be rough because its weight would make the flower lose its original shape. Once you have chosen it accordingly, take a <tl>box</tl> in which you first make a layer of sand on which you will display the stalk of the flower laid in a way that the flower does not touch nor the bottom nor the edges of the <tl>box</tl> and stands in the air. Then add more sand on the stalk so that it remains strong and solid. Finally take some of the same sand and sprinkle and subtly display it with two <tl><bp>fingers</bp></tl> on the flower like the flow of an <ms><tl>hourglass</tl></ms>. And once the flower is covered, strike with your <bp><tl>fist</tl></bp> the table on which the <tl>box</tl> is displayed, so that the sand lowers and surrounds everything. Finally cover the entire flower, and lay other flowers, one after the other, as many as the <tl>box</tl> can hold. Then put the <tl>box</tl> in <tmp>warm sun</tmp> for <ms><tmp>several days</tmp></ms>. And as the flower dries, the sand that goes with it continues to press and prevents it from rotting and tightening and consequently, it dries and keeps its original shape. And to do so, make sure to chose boufarcis, <pa>marigold</pa>s, the <pa>yellow meadow's flowers</pa> called <la><pa>rammenlus</pa></la> or <la><pa>palta lupina</pa></la>, <pa>amaranth</pa>, and similar flowers like <pa>broom</pa> and others which you will know from experience.</ab>
<ab>
<margin>left-top</margin>
The sand the <pro>goldsmith</pro>s use to sand <m>enamel</m>s and the white one which the <pro>glassworker</pro>s use and any type of fine sand which does not have much body, put it through a <tl>sieve</tl> made of <m>horsehair</m> because this matter must not be very thin. Then dry it well in the <env>sun</env> for <ms><tmp>several days</tmp></ms> in order to remove the dampness and ventilate it as you would do with <m>wheat</m> so that the dust goes away. Once cleaned of dust and dried, use it as you know.</ab>

<ab>
<margin>left-middle</margin>
<pa>Pansies</pa> can be preserved in the same way.</ab>
</div>

<div>
<id>p121r_1</id>
<head>Keeping fruits throughout the <ms><tmp>year</tmp></ms></head>

<ab>It is necessary to pick them in calm and quite dry weather & by the stem, without <sn>touching</sn> the fruit with the <bp>hand</bp>, and put them into <tl><m>glass</m> bottles</tl> that have wide necks like tall <m>glass</m> <tl>jars</tl> for preserves. Protect them with <m>straw</m> or something else so that they cannot break, and seal them well with <m>wax</m> to keep them from breathing. And if they had <m>glass</m> lids well sealed with <m>wax</m>, it would be all the better. Put them in a <tl>case</tl> and put it in the <env>well</env>, or in the <env>cellar</env>, or in a vat, or a [illegible] of <m>earth</m> full of <m>water</m>.</ab>

<ab>
<margin>left-bottom</margin>
Because where the <bp>hand</bp> <sn>touches</sn>, it will rot, and it is even good not to breathe on them.</ab>

<ab>
<margin>left-bottom</margin>
You can also put <m>vine leaves</m> there.</ab>
</div>

.....

<div>
<id>p121v_a1</id>
<head>Casting in <m>silver</m></head>

<ab>You need to place the animals as said, and make up the same sand out of <ms>four parts</ms> <m>plaster</m>, <ms>two parts</ms> refired<m>brick</m>, and <ms>one part</ms> annealed <m>stone alum</m>. Mix it well, and once it is ready and you wish to mold, take <ms>three parts</ms> of the aforementioned sand in the earthenware <tl>dish</tl> from which the peasants eat their soup. Add pure annealed <m>stone alum</m> that has been pulverized in the <m>mortar</m>, as much as you can grab with your <ms>four <bp>finger</bp>s and <bp>thumb</bp></ms>, or a <ms>small double handful</ms>. Then, mix well and mix in a little of <m>sal ammoniac</m> and the remainder with <m>common water</m><figure>+</figure>
<ab><add><margin>left-middle</margin>Which needs to be placed in the <tl>dish</tl> rather than in the sand, which is put in the <m>water</m>, and not the <m>water</m> in the sand</add></ab>
And stir it with your <tl>spatula</tl> so that it all becomes like a thick sauce or clear <m>mustard</m>. And having rubbed spirits on the animal with the <tl>brush</tl>, cast and blow, and beat the table to shake the <tl>mold</tl> and do as with the others. Do not forget to put in <m>crocum</m>, because it prevents <tl>molds</tl> from cracking, and it is appropriate for all <m>metals</m>.</ab>

<ab>
<margin>left-top</margin>
Alloyed <m>silver</m> is better for casting than fine ^{silver}, provided it is soft.</ab>

<ab>
<margin>left-top</margin>
The <m>stone alum</m> needs be well pulverized and well mixed.</ab>

<ab>
<margin>left-top</margin>
<m>Sal ammoniac</m> gets along well with <m>gold</m> and <m>silver</m>.</ab>

<ab>
<margin>left-middle</margin>
<m>Brass</m> is the enemy of <m>gold</m> and works well with<m>silver</m>.</ab>

</div>
<div>
<id>p121v_a2</id>
<head><m>Silver</m> [used] to cast</head>

<ab>It is not pure but alloyed <m>silver</m>, and it does not become perfectly white on the fire because they [<pro>goldsmiths</pro>] whiten it after the melting to clean up the welding marks that occur when an animal needs to be attached to another or when it needs repairing. [The process] is similar with teston <m>silver</m> and all alloyed <m>silver</m> as long as it is soft and good.</ab>

<ab>Before starting to cast in <m>silver</m>, in order to coat all round and strengthen your <tl>mold</tl>s, you should use <m>earth</m> that can withstand fire, such as this sandy <m>earth</m> mixed with <m>cloth waste</m> which <pro>founders</pro> use to cast their canons, or any good <m>lute</m> that can withstand fire. They need to all be red and to be tied with <tl><m>iron</m> wire</tl>.</ab>

<ab>Rather than anneal the <tl>mold</tl>s, have the mixture necessary so that the <m>silver</m> runs.</ab>

<ab>If <pro>goldsmiths</pro> knew the material to make <m>silver</m> run in their work, they would buy it in quantity. Some buy it five sols for a denier.</ab>

<ab>
<margin>left-middle</margin>
I have used <m>silver</m> from the capital.</ab>

<ab>
<margin>left-bottom</margin>
Before luting and strengthening the <tl>mold</tl>s for <m>gold</m> and <m>silver</m> with the said <m>earth</m>, the cast and the vents must be made and the <tl>mold</tl>s must have been annealed and made quite red to burn the animals, flowers and herbs that are inside. The ashes in the <tl>mold</tl>s should be cleaned thoroughly. Once the <tl>mold</tl>s are quite clean and ready, lute them and bind them with <tl><m>iron</m> wire</tl>. Anneal them straight away until they are quite red.</ab>
</div>

.....

<page>122r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f249.item.r=</image>

<div>
<id>p122r_1</id>
<head>Fixing different animals</head>

<ab>When you discover some twisted <al>lizard</al>s which bite each other are the most difficult thing to mold because of the number of their feet and because of the tip of their tail which are very fine, steady the parts which are uncovered and which look like suspended in mid air, that way the second cast will not vary anything. If you need to steady the parts hanging in mid air, take a bit of <m>wax</m>, rub it and elongate it on a <tl>table</tl>, make it big needle shaped, then cut the part you need with a hot <tl><m>iron</m> wire</tl> then join where it is necessary. Steady also the part which does not stick to the <tl>mould</tl> (when you uncover the half of the womb) with some <m>wax</m> with the point of a <tl>knife</tl> @<a>[a], joined and melted with the <tl>wire</tl>. When you cast, pour only to the spot and at the edge of the <m>wax</m> which being attached to the animal will melt but still lead the cast towards it without damaging anything. This is a secret.</ab>

<figure>
<id>fig_p122r_2</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5TVo3Y3IKWjA0dXM</link>
</figure>

<ab>One can arrange embracing <al>snake</al>s, but only small ones.</ab></div>

<div>
<id>p122r_2</id>
<head>Flower in the mouth of a <al>snake</al></head>

<ab>If you want to put a flower in the mouth of a <al>snake</al> or a branch of a herb which guards against its bite, take the best arranged little branch you can find, and display its stalk into its mouth. Then put <ms>two little pieces of <m>wax</m></ms> around the stalk of the herb. Melt a bit of <m>wax</m> with the point of a warm <tl><m>iron</m> wire</tl>, while you are closing the jaws of the animal with your <bp>hand</bp>. And then you can cut the head to make it burn with the flower. If the herb is strong enough, it will stay in the same position by itself but if the flower is light, like <pa>wormwood</pa>, which droops, the dampness of the sand will lift it up, make it swim and stand by itself, without any support, as well as if you had put some <m>thread</m> under it passing through with a <tl>needle</tl>.</ab>

<figure>
<id>fig_p122r_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5ZDJlZ0g5OXNKcDQ</link>
<margin>left-top</margin>
</figure>

<figure>
<id>fig_p122r_3</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5UDgyYnNSZnlndjg></link>
<margin>left-middle</margin>
</figure>

<figure>
<id>fig_p122r_4</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5elo4VzJZRUpOWc></link>
<margin>left-middle</margin>
</figure>

[a]Manuscript reads "trancheplume". We translated as "knife" but it seems necessary to maintain the specificity of the tool, in connection with writing practices.

<page>122v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f250.item.r></image>

<div>
<id>p122v_1</id>

<head>Cleaning the <tl>bowl</tl> and the <tl>spoon</tl> that one uses to temper sand</head>

<ab>Be very careful to clean, vigorously & quickly after casting, your <tl>bowl</tl> & your <tl>spatula</tl> or <tl>spoon</tl> with which you temper your sand, because if it dries, it crumbles & falls into the fresh sand & makes a hole or fault get made. Also, when the <tl>mold</tl> gets reheated, these little pieces will chip & flake off & prevent neat casting. </ab></div>

<div>
<id>p122v_2</id>
<head>Molding hollow</head>

<ab><m>Wax</m> on its own cools down too rapidly & does not spread well everywhere, & <m>tallow</m> holds its heat & spreads everywhere; but is not good on its own. But when <m>wax</m> and <m>tallow</m> are mixed the work behaves well. The bodies of <m>crawfish</m> & other small animals can easily be molded hollow, but as for the legs, it will be troublesome.</ab>

<ab>
<margin>left-middle</margin>
The first part of the <tl>mold</tl> that is first cast on the plate of <m>clay</m>, cracks more easily in the fire than the second.</ab>
</div>

<div>
<id>p122v_3</id>
<head>Molding medals and flat things</head>

<ab>You are not limited to making the gate with legs thus
<figure>
<id>fig_p122v_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5aTIRdkx2eVdtZXm></link>
</figure>
, but rather in this way
<figure>
<id>fig_p122v_2</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5WFBpaEV5Ti1hRUU></link>
</figure>

, which embraces the medal well. For the other way with legs is only done to accommodate more delicate works and the other, which is wide and in one piece works better for flat medals. But take heed that all gates need to be very thin close to the medal & so as to say not as thick as the medal, if it is not very thin like paper. And then, from the medal out towards the opening of the gate, make it thicker and thicker for it works better thus. If the gate is thick at the entry to the medal, the work never comes out very well. Make that despuys from the middle to the top of the gate be moderately thick, & from the same middle to the bottom very thin. Do not forget to makes grooves in au hau the top of your gate to prevent the <m>metal</m> from spreading furiously.</ab>
</div>

<page>123r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f251.item.r></image>

<div>
<id>p123r_1</id>
<head>How to make <it><m>a[r]gento</m></it> run </head>

Take white arsenic, two ounces, 2 
Orpiment or yellow arsenic, two ounces, 2 
Aes ustum, two ounces, 2 
Fine copper filings moderately well ground, 2 
Antimony, 2 
Rock salt, 2 
Raw Tartar, 2 
Sandever, 2 
Saltpeter, the half amount, namely 1 
Course common salt, melted, the half amount, 1 
Sublimate, the half amount, 1 
Borax, the half amount, 1 
Sal ammoniac, same quantity as borax, 1 

Take these ingredients, the best ones you can find. Weigh them as described above, and crush them separately, keeping your face covered, over the nose & the mouth, from the eyes down, in order to avoid the exhalations of arsenic, sublimate, and orpiment. Mix them well all together, then put them in a good crucible that must be so large that the ingredients can have at least the fourth or fifth part empty. Cover the crucible with a good tile, precisely adapted in its roundness so as to fit the opening. And having connected it tightly with wire from top to bottom and around the sides, lute it with manure or founder's earth, that artillery founders use. And in this, be careful & diligent, & do not forget to mix in crushed glass throughout the lute to fortify your crucible, in a way that it takes no air, for the mixture would be worth scarcely anything, because the ~~4~~ ~~5~~ last ingredients would be exhaled.

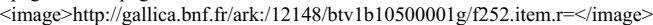
Note that you need to lute your crucible well, so that the ingredients do not take air, and would not be exhaled.

Lute well your crucible, so that it does not break, for if it breaks, the materials evaporate & the fumes from it are dangerous. If within 12 hours it has not melted and intermingled, it will not be well made.

Goldsmiths who know about these things sell a ~~denier~~ of this for 5 ~~sols~~ to other goldsmiths, & especially to those who make dishware and great works, for at the end of their works, they can solder over the first solder by the means of this mixture, in general all those who want to mold and cast something delicate. This mixture should not be divulged, so that it is not abused.

The grain is like broken steel

Written backwards in the original.
1/785Kilo

123v


Your lute once dry, put your crucible in an ~~four a vent~~ in the early morning, & at the beginning, give low fire, as much as you need to heat your crucible as to let the fury of the saltpeter pass gently. Then, little by little & with judgement intensify the fire. And ~~y~~ leave in the fullness of this your ingredients for 12 hours, or a natural day. Make at each go a good quantity of this mixture to not have to repeat it too often, because the fumes, which are dangerous, could hurt you. And before working with it, take a good piece of buttered toast in the morning, and hold in your mouth the said butter, or zedoary, or gold coins, and ~~ada~~ cover your face with a towel from the eyes down. From this mass, the crust will serve to ~~make run~~ clarify ~~the great works of~~ the silver from the metallic mass ~~like~~ snakes & similar things when it starts to melt. ~~But~~ And then, the grain that will be at the bottom of the crucible should be reserved for ~~principally delicate flowers and herbs~~ putting a little in the melted silver when you want to cast it, ~~However it is necessary to always put a little of this grain in the silver when you want to cast~~ it is melted and when you are ready to cast as you do thus for fine tin on copper, & for looking-glass tin on lead and on tin. And just as looking-glass tin makes lead and tin too brittle if you put in too much, likewise the grain composed of the above-mentioned ingredients would make your silver too brittle if ~~l~~ you were to put in too much, and obscure it. This above-mentioned composition will suffice you for a long time, when it has been on a ~~low~~ low fire for a whole day at the beginning & intensified degree by degree until the end. Having thrown one load of charcoal, let it consume itself, & let your crucible cool down. Then break it. You will find two sheets & hard cakes in the crucible. The upper one looks stoned ~~Once~~, once composed of salts, sublimated & mixed together. The lower one is metallic, composed of fillings, aes ustum & antimony, having very small grain. Pulverize the upper cake, made from salts, & put some in to lighten & clean the silver and the metallic grain can be used to put into the melted metal.

<ab>
<margin>left-middle</margin>
Or, after you have pestled your <m>drugs</m> by a rustic, and having put them in your crucible, & the latter luted & dry as said & place into the furnace, have the fire managed by a <fr>boutique</fr> boy, familiar with <m>charcoal</m>. </ab>

<ab>
<margin>left-middle</margin>
One sells this metallic mass to <pro>silversmiths</pro> to soften their solder, because when melting <m>brass</m> exhales. And with a little of this mixture, they solder over the other solder.</ab>
</div>

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<page>124r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f253.item.r</image>

<div>
<id>p124r_1</id>
<head>Hairy animals and very fragile and thin <pa>flowers</pa></head>

<ab>Hairy animals are difficult to mold because hairs raise up & come out looking intermingled & in clumps. At the beginning, it is necessary to keep it flat with a drying agent & that makes it firmer. And in this, the best thing to use is <m>wheat oil</m>, with which you will dawb it. Once cast, you will be able to repair it. The bodies of <m>butterflies</m> and <m>herbs</m> that have a <m>stem</m> & <m>leaves</m> that are flocked with capricious & downy<m>hairs</m> need to be dawbed with the same <m>oil</m> to keep these flocks flat; likewise <m>flowers</m> that have very delicate & thin <m>leaves</m> because dry <m>wheat oil</m> makes them and rigid and firm. And if someone brags about molding anything presented to them, give them the downy head of the herb called <m>dandelion</m> to cast or a <pa>pappus</pa>, which comes from the seeds of <pa>burdock</pa> & takes flight at the slightest sigh of wind.</ab>

</div>
<div>
<id>p124r_1</id>
<head><fr>Noyaulx</fr> for molding hollow</head>

<ab>Animals of <m>gold</m> & <m>silver</m> are generally cast hollow to avoid weightiness, and costs as well, if they are not quite small. But in order to make the core & the <fr>noyau</fr>, the mold needs to be freshly molded and not dry.</ab>

</div>
<div>
<id>p124r_3</id>
<head><al>Spider</al> web</head>

<ab>It comes undone in <m>water</m> & because of this, cannot be molded in a <fr>noyau</fr>, but one casts the <m>spider</m> and then one draws the web <m>filaments</m> that they make in the fields stretched between blades of <m>grass</m> on some <m>carton</m> with a stylus. One makes around it an edge of <m>paper</m>, glued to the <m>carton</m> & one casts with <m>tin</m> m[a][b]ixed with very little <m>lead</m>.</ab>

</div>
<div>
<id>p124r_4</id>
<head>Fine <m>gold</m> </head>

<ab>There is a kind of <m>gold</m> which, even though it is very fine & unalloyed & has been passed through <m>aqua fortis</m> & <m>antimony</m>, is however so brittle that it hardly withstands the hammer. And the composition of <m>verdigris</m> mentioned above to make <m>gold</m> flow, makes it as <fr>doux</fr> as <m>lead</m>.</ab></div>
[a]Original was "alier."
[b]Allier...

.....

<page>124v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f254.image</image>
<div>
<id>p124v_1</id>

<head><m><fr>Rouge clair</fr></m> @ [a] <add><m>enamel</m></add></head>

<ab>Fine <m>gold</m> is unfavorable for it, for [illegible] on the former it is yellowish. But alloyed <m>gold</m> is more favorable, such as that of <cn>escu</cn> & <cn>pistolet</cn>. If <m>Gold</m> in its pale color decays the <m>enamel</m>'s hue. Ma And for this occasion, once cut, one puts it back on the fire to give it a reddish color to make <m>enamels</m> beautiful. Otherwise, they are subdued.</ab>

<ab>There is <m>rouge clair</m> that once worn down with <fr><m>arene</m></fr> loses its beauty.</ab>

<ab>Some <m>rouge claire</m> is found with grains of <m>gold</m> inside. And it is also the opinion of good <pro>goldsmiths</pro> that the good one is made with <m>gold</m>.</ab>
</div>

p124v_2

Casting with gold

Molds are recooked better & more surely than otherwise in a closed fire, such as a ~~pot~~ a reverberatory furnace. So that when goldsmiths want to cast some piece of important work ~~d’or~~ that cost a lot to rough out in wax, they put the molds in a pot & recover them, & fill the pot with earth, sustaining the fire that holds them unified & tight. Then, they recook mold, earth & pot together and when everything is quite red, they cast the gold. Fine gold does not run well, but rather alloyed gold.

Gold and silver do not become brittle: once entirely red and hot, dip them into water.

Gold is a quarter heavier than lead.

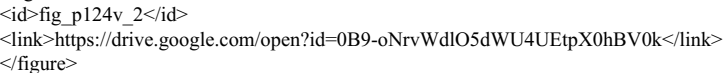
When gold reaches its perfect heat, it is green like an emerald.

Take care that where you want to cast with gold, there must not fall any lead, tin, or lime thereof in the forge.

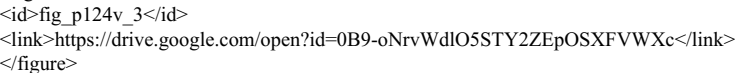
p124v_3

A way to make the gate for small female lizards

Because you always have to make the gate by the tail, and that it is so delicate and thin that the ~~casting scarcely~~ metal would flow with difficulty, especially when the tail is folded, roll wax in little filaments of this size



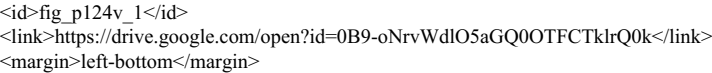
and apply some with the hot wire, as said, one at the end of the tail & others that correspond in the same way, from one ~~end~~ side of the tail to the other, as you see depicted. But take heed to do it with the end of the hot iron so that the end of the wax scarcely touches the animal, for the sand of the second gate will not touch this part. But make the end of the wax arranges itself only at the end of the empty part of the first mold. Make as well channels of wax around the legs &, around the contours of the body, which are a little long & they will serve as a trough for the molded thing.



When the tail, which is delicate and closer to the gate comes out well, the rest will also come out well.

The most important is that the ears of the snake come out well-molded.

Your gate must be very thin at the entrance of the animal & of the thickness of a knife.



[a]hand shift/added later?

.....

<page>125r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f255.item.r</image>

<div>
<cont/>
<id>p124v_3</id>

<ab>and with your pipes and abrevouers @[a], you can also place small threads of <m>wax</m> which are applied to the body so that the <m>metal</m> goes more easily from one part to another and runs quickly throughout. And with these pipes, you can guide your cast and your vents without having ruined anything.</ab>

<ab>
<margin>left-top</margin>
<figure>
<id>fig_p125r_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5ZHJRXJlU2RFZlk</link>
</figure>

small rolls, you make your casts and vents without danger of chipping or [...] any of the mold because the <m>wax</m>, once it has been removed, leaves a thoroughly empty place. Make your vents proceed from the head [of the cast], which is at the bottom, towards the cast. Make your cast so that it is not thick, and make two or three notches inside the passage because this impairs the <m>metal</m> fury. And it makes it flow easily without filling with bubbles, [and] it does not make too much smoke which dams up the passage. You can also divide it into two or three branches thusly

<figure>
<id>fig_p125r_2</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5bDlqQzJZemktSFk</link>
</figure>

when it [...] with the molded thing, and always make holes in the cast.</ab>

<figure>+</figure>

</div>
<div>
<id>p125r_a1</id>
<head><m>Plaster</m></head>

<ab>When you mold something in order to cast with <m>wax</m>, first mold only <m>plaster</m>, you mold with <m>plaster</m> [that is] only reheated after [it] has been pulverized. Because, once [it] has been reheated on the <m>stone</m> slab, the outside is burnt, and the inside remains nearly unburnt. Transparent grey <m>plaster</m> is not strong, but the good one [<m>plaster</m>] becomes white once it has been soaked, and [it] sets well quickly. Nevertheless I have found the grey [<m>plaster</m>] to be quite firm and hard after it has set, but it takes longer [to work with it]. Know the nature of each. You will never mold very neatly if you do not soak until it is quite clear and liquid, your <m>plaster</m> or the core's <m>sand</m>. Soak it straight away after it has been reheated.</ab>

<ab>
<margin>left-middle</margin>
However, if you mold with <m>plaster</m> that has been reheated once as said, coat the first mold well with <m>oil</m>. And when the second cast has set and you have prepared it, soak it in <m>water</m> for a long time. And if it does not release, soak it in hot <m>water</m>, because cold <m>water</m> hardens it.</ab>

<ab>[Something] to know</ab>

<ab>And hot <m>water</m> softens it [<m>plaster</m>] more than cold <m>water</m> [which] does not penetrate it [<m>plaster</m>] as [it does] <m>mixed</m> <m>plaster</m> because it [<m>plaster</m>] is stronger and <m>mixed plaster</m> is spongier. <m>Medal</m>s are cast from this <m>powder</m>ed, reheated <m>plaster</m>, [and the <m>medal</m>s] will be waterproof as though they were <m>varnish</m>ed. In Germany, people hang these <m>medal</m>s on houses. See to it that the <m>water</m> is very hot, and if the <m>water</m> is boiling, it will not endanger [it]. All molds [made] of plaster only or molds [made of] <m>mixed</m> <m>plaster</m> are stripped from it.</ab>

</div>
<div>
<id>p125r_a2</id>
<head>Scimitars</head>

<ab><pro>Workers</pro> from Damascus or from Hungary, neighbors to the Turks, separate <m>iron</m> from the <m>mine</m>s with <m>steel</m>. And they cast the blades of the scimitar in <m>sand</m> with this first <m>steel</m> [that was] first casted from its <m>mine</m>. Afterwards, they [the blades] cut other <m>iron</m> without without great difficulty because all melted <m>iron</m> is harder than soft <m>iron</m> beaten into [...] and bars. Thus is the <m>steel</m> of scimitars, but it is quite brittle. When someone takes the haft of a scimitar off, that person recognizes [it] well by the tip in the haft which is cast in <m>sand</m>.</ab>

</div>
<div>
<id>p125r_a3</id>
<head>Hearing from afar</head>

<ab>Make a small hole into the ground, put your ear against it during the night or during an equally quiet time, and you will easily hear the muffled sound.</ab>

</div>
<div>
<id>p125r_a4</id>
<head>Secret</head>
</div>

[a]The exact translation would be "through" but it might refer to some kind of "Fluid-carrying vessel".

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<page>125v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f256.item.r=.zoom</image>

<div>
<id>p125v_a1</id>
<head>Vipers and snakes</head>

<ab>
I molded a viper, which is much cleaner to cast than any other snake, because it has beautiful & very visible scales on the entire body & principally on the head and under the throat. It has a flat heat, a ruddy snout tending to deep carnation, & thick and short like horned aspic or like the top of a pig's snout, big jowls, eyes very close to the snout, and the mouth wide open where it has double canine teeth on each side coming out of a reed of flesh that covers them and turns back. It also has a reed of flesh in the throat like a dog's penis from which comes its stinger. Other snakes have a double row of teeth.</ab>

<ab>
<margin>left-top</margin>

If you want to mold snakes with open mouth, you must cut off the head & leave it inside, because it will not be able to be stripped.</ab>

</div>
<div>
<id>p125v_a2</id>
<head>Molding turtles</head>

<ab>
Casting snakes & herbs and flowers is a strange matter, since the cavities that are between the two shells require several pieces. To cast in <m>sugar</m> without making a casting, they are molded from plaster, as works for anything you like.</ab>

</div>
<div>
<id>p125v_a3</id>
<head><m>Plaster</m> to cast with <m>wax</m></head>

<ab>
When you want to cast in <m>wax</m> in a <m>plaster</m> mold, you must know this secret, that there is need that your mold be in <m>hot water</m>. The animal will never come out as clean as in metal, because <m>wax</m> grips. But to rough out an animal as close to natural as possible in order to fix it afterwards, you have to remove all the scales, because the <m>wax</m> will permeate them & will not be able to be stripped off. In molding the animal, take off the scales for <m>wax</m>, but on the contrary [when not molding with wax] rub them against their grain so that they raise up, because then these animals [sc. those not molded in wax and prepared in this way] will show better. Do not wait to strip off your wax until it is cooled down at all. But while it is a bit hot, after you have molded the first casting of the animal, uncover it halfway so that it be stripped in so doing. And having as much in one mold as in the other, make very many large castings holding on to the animal in order to fortify them [...] stripping, & afterwards you'll cut it.</ab>

<margin>left-bottom</margin>
<ab>
<figure>+</figure> <m>Alabaster</m> [...] which is <m>plaster</m> in any case, is very hard, but it shrinks quite a bit. It is good for making medals. But it must be very finely strained.</ab>
</div>

.....

<page>126r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f257.item.r=.zoom</image>

<div>
<id>p126r_a1</id>
<head>Molding fruits and animals in sugar</head>

<ab><m>Sugar</m> is fatty, and, with it, round things and large muscles are cast well but fine and delicate things [are cast with] difficulty. However, try well-purified <m>sugar</m>. The <m>plaster</m> mold must be soaked in <m>water</m> for a full night or a full day before casting <m>sugar</m> so that it [the mold] is saturated with <m>water</m> and does not soak up the <m>syrup</m> [the <m>sugar</m> mixture]. The [plaster] mold must also be stripped very well from it [the sugar], because <m>sugar</m> is aigre and brittle. Thus, do not cast anything with <m>sugar</m> which is not stripped easily from it, and which can not be neatly molded in two parts to open as will be needed. If you want to mold a grape, you must get it when it is very fresh; because if it is withered, it [the cast] will look the same. See to it, thus, that you make your molds in the natural season for each thing [fruit]. Grapes that one wants to cast in <m>sugar</m> are man-made, either with <m>wax</m> or <m>earth</m> or with grapes molded with melted <m>wax</m>, on some dish [plaste & chose pleine] in a way so that they are pressed closely together and easily stripped from it. And only a half [of the grapes] should be molded. Or, if you have some of those grapes called chauches or sauvignons which have well-pressed grapes, set half of the grapes in the dish of <m>clay</m>, and cast on the other half, and if any grape is not stripped from it, pluck it out. Note that a grape whose grapes are set apart and separated cannot mold well in either <m>sugar</m> or <m>metal</m> because the ends of the cluster are so fine. Similarly, if the grape is kept, that it cannot hold the bunched grapes. Therefore, a hollow should be cast, which you will not be capable of if the grape is not close together and without having them spread apart.</ab>

<ab>
<margin>top</margin>
to cast and brittle, and [it] breaks when dry</ab>

<ab>
<margin>left-top</margin>
<figure>+</figure> In order to mold pears and apples in <m>sugar</m>, do not cast. Rather, fill half of the mold, and then join the two [halves], and keep turning [it] until the <m>sugar</m> is solidified and cold. Do not mix anything in the mold except the reheated <m>plaster</m>, as you know.</ab>

<ab>
<margin>left-middle</margin>
The mold needs to have been soaked in cold <m>water</m> for one full day and night and [the mold] must be damp when you cast in <m>sugar</m>.</ab>

<ab>
<margin>left-bottom</margin>
The sign that the <m>syrup</m> or the melted <m>sugar</m> has boiled enough in the <m>water</m> for casting fruits is when it makes threads when shaking it. And if it passes this point, it will not be good because it will make [it] damp. If the <m>sugar</m> corrodes itself, throw a bit of <m>amidin</m> in the mold or rub it with an <m>almond</m>.</ab>
</div>

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<page>126v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f258.image</image>

<div>
<id>p126v_1</id>

<head><m>Crocum ferri</m></head>

<ab>It is made more quickly from <m>iron rust</m> than <m>filings</m>, because if you have good <m>vinegar</m>, it will bubble on contact with <m>rust</m> as if it were boiling, as if it were on fire, if you use filings it will not bubble up unless it is placed on the fire or warm ashes. However when you have put your <m>filings</m> through <m>vinegar</m>, and it has evaporated, and heated up and made it very warm, should you do this all for a second time, the <m>vinegar</m> will be a lot stronger and your <m>crocum</m> will be much more subtle and red. Those who use it to cast in <m>gold</m> buy it for forty or fifty sols per ounce.</ab>

<ab><margin>left-top</margin>It does not harden molds, and when you scrape ^{the mold} you cannot tell that it is harder. But it does make the mold more thinner, and it absorbs and attracts <m>gold</m> better.</ab></div>

<div>
<id>p126v_2</id>

<head><m>Plaster</m></head>

<ab>You can mold with it as large a piece as you wish. But if you make a mold upon large works and pieces of <m>wood</m>, it will not release cleanly unless you lather your <m>wood</m> with very hot <m>wax</m>. Because the <m>wood</m> absorbs and drinks, and because of this, it drinks up the <m>plaster</m>, which means it does not come out cleanly. You can assume the same thing will happen for any other large piece of <m>wax</m>. But the cure for this is to lather the work that you wish to cast in <m>plaster</m> with very hot <m>wax</m>. In this way, it will not absorb, and will release very neatly.</ab>

<ab><margin>left-middle</margin>If the <m>plaster</m> shrinks, it will always make flaws, look for harder ^{<m>plaster</m>} and rather than casting it, press your molds well.</ab>

<ab><margin>left-middle</margin>If it shrinks, it is not good <m>plaster.</m></ab></div>

<div>
<id>p126v_3</id>

<head>Plants that are difficult to burn in the noyau</head>

<ab>Any herb that has a hard stem, like <m>wood</m>, is very difficult to burn in the noyau, such as asparagus, thyme, and other similar things, because they remain as solid <m>coal</m> in the small conduits, and if they are not reduced to ashes, it is not possible to remove them from the mold. In this case, some reheat them two or three times. Others in mixing some herbs together in the mold, pass through the herbs and the <m>earth</m> circle, which is there to receive the tempered sand.</ab>

<ab>This is not due to the fact that those herbs have stems with lines in them because rosemary burns well. This is due to the nature of certain herbs.</ab>

<cont/>
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<page>127r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f259.image</image>

<div>
<cont/>
<id>p126v_3</id>
<ab>some thread, as much to secure the herbs so that they do not rise when you throw in your <m>clear sand</m>, as to give them some circulation when they are reheated. Because by doing this the thread burns away and leaves some empty space around the herbs which acts as vents, and places for the air to escape, and places for the air to enter so that they burn more easily; because what keeps them carbonized, is when they burn in a closed fire with no empty spaces because of the fine sprigs. Therefore try to enlarge the main stem with a bit <m>melted wax</m> and leave it to cool. Then mold it. And when the <m>wax</m> melts, it will leave the main stem sitting comfortably and almost loosely so that it burns better. Or you could apply some <m>petrol oil</m>, <m>sulphur</m>, <m>turpentine</m>, <m>brick oil</m>, or similar things, or with <m>aqua fortis</m> or <m>spirits</m>, or make you <m>sand</m> with . Or make your <m>sand</m> with <m>crocum filings</m> of <m>iron lumps</m> or <m>iron esmeri</m> which allow for several days of high fire.</ab>

<ab>Animal bones are not as difficult to burn and reduce to cinders, because their flesh being burned, the bones remain loose and the heaviness of <m>quicksilver</m> makes them break and be crushed, being calcinated and burnt.</ab>

<ab>
<margin>left-top</margin>
Asparagus always remains carbonized like little pins. Try to the take asparagus and thyme when they start to grow, so that the stem is still tender.</ab>

</div>
<div>
<id>p127r_a1</id>
<head><m>Plaster</m> mold for <m>wax</m></head>

<ab>When your <m>plaster</m> mold is done and dried, be sure that your mold comes off cleanly because it sometimes happens that the animal, having been wounded or having lost weight or having withered, has wrinkles where <m>plaster</m> can get in the scales. Having not been rendered well, the animal will attack itself and break and will never be perfect. Be also advised to make your casts for <m>wax</m> very big. In this way, your casts will not be too thick. They are done when the mold of both sides is done and when the animal

<figure>
<id>fig_p127r_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5ZFFkT3I4Vmx2em8></link>
</figure> has been removed from the mold.</ab>

</div>
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<page>127v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f260.image></image>

<div>
<id>p127v_a1</id>
<head>Founder's earth</head>

<ab>This is boulvene earth mixed in with dung or hairs which withstands fire, of which you must always have a large stock hanging around for dawbing the molds that you use to cast in silver and that need reheating. Those where you burn flowers or animals do not need dawbing, unless they have been left to stand once they have been reheated, and once they have been cleaned of any bones and ash that might have remained inside. And having removed the clamps, you open the two halves to see if they have any cracks. Because there is some plaster that does not harden as much in the fire as another one would. Others create crusts on the things that remain to be burned inside. And all of these are imperfections that you must avoid, either by giving the animal a bath of spirits before pouring your soaked sand, or by putting in more or less brick [sand] or alum, or by reheating the brick to higher temperature, or by instead of mixing in brick, mixing in broken bits of crucibles and similar things that resist fire well. Having therefore reheated your mold to burn what is inside, leave it to cool. And if it is neat and unbroken put the clamps on again, and dawb with the aforementioned earth, that is called of the <pro>founders</pro>, dust it with a bit of sifted ashes and leave it to strain. But beware not to dawb your if they are not cold. Those in which you cast silver, where nothing needs burning inside, only need, when the cast and the vents are made, to be dawbed and reheated once. Do not dawb the part of the mold in which you want to cast, but leave it uncovered like the openings of the vents.</ab>

<ab>The molds should crack at the first reheating, because [then] they will not crack at the second reheating anymore.</ab>

<ab>The smallest molds reheat more quickly, therefore take the ones that have finished reheating first when you cast, because you can put several [in your furnace at the same time] to reheat.</ab>

<ab>First light the coals in the forge, or in any other place that is close, and having placed your molds on a layer of good embers, not burning too much, in order to have a slow fire in the beginning, then add in little by little coals that are half lit to the forge.</ab>

<ab>
<margin>left-top</margin>
Earth, of the kind used to braise or solder, which is sandy clay earth. Or else clay mixed with sand after having been dried and sieved, and then composed, is good to dawb your molds. Because this type of earth melts rather than cracks. And any earth that melts cannot be lacking in this.</ab>

<ab>
<margin>left-top</margin>
Use a wire to bind.</ab>

<figure>
<id>fig_p127v_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5S2VzRmhYUWpfbmc></link>
<margin>left-top</margin>

</figure>

<ab>
<margin>left-middle</margin>

Dawb very thickly your molds, and if they are small, you can reheat them immediately in a good fire, even those made out of crocum.</ab>

<ab>
<margin>left-middle</margin>

If the earth has hair in it, and does not crack when heated, and does not separate from the mold, the molds will not crack either and will not run when cast. Reheat them at your leisure in a closed fire and do not expose them when warm to the air.</ab>

<ab>
<margin>left-middle</margin>

Dawb rather thickly in two or three layers, [doing] similarly at the points of the joints, so that the silver does not run out of the mold.</ab>

<ab>
<margin>left-bottom</margin>
Reheat in a reverberatory furnace.</ab>

<ab>
<margin>left-bottom</margin>
To test if your earth is good, rather than putting some on your mold, dawb in that place of the wall of the forge that is around the opening and mouth of the bellows, then turn on the fire and if it tolerates this without cracking, then it is good.</ab>
</div>

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<page>128r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f261.image</image>

<div>
<id>p128r_1</id>
<head>Casting with <m>gold</m> and <m>silver</m></head>

<ab>
To cast <m>silver</m> and <m>gold</m> do always put a little ground stone of <m>borax</m> on the bottom of the crucible, and put your matter on it. That way the acid <m>smoke</m> won't corrupt these two matters, but it is especially efficient for <m>gold</m>.</ab>

<ab>
If you have several molds to cast, do not fill them all in one cast, because <m>metal</m> would get cold. Cast one after another.</ab>

<ab>
When your mold start being red inside, look at the cast, if you do not see any black spot keep the same heat, and if needed, add a few half lighted <m>charcoals</m> with the help of pliers. However, put ground <m>borax</m> at the bottom of the crucible, and put <m>silver</m> on the top of this matter, then put your crucible into the forge, reheat your crucible which is between lighted <m>charcoals</m>, until it becomes red; do not blow before that. When you'll use the bellows, make continuous longitudinal moves, tug a little bit the bellows, push and pull, that way the heat will become greater. From time to time remove your crucible out of the top of the forge with the help of your warm pliers, because if your crucible is close to the tuyere <m>silver</m> would cool down rather than heat up. That why it is better to put the lighted <m>charcoals</m> above the tuelle, three fingers far from the wall of the forge. That way the crucible is heating much better. When your <m>silver</m> starts melting, if you find that it is too acid by noticing that the lumps are cracking and bursting, take the size of a walnut of <m>arsenic</m>, and twice as much of <m>crude tartar</m> which is coarsely ground, that way it will heat better. Pour some of this mixture on <m>silver</m> in the crucible, it will get clearer. But if you have some of that sublimate crust on the <m>metallic matter</m>, and which looks like grains of <m>steel</m>, <figure>#</figure> put some of this matter in your melted <m>silver</m>.</ab>

<ab>
<margin>right-top</margin>
Some people leave the silver to rest a little out of the heat before casting.</ab>

<ab>
<margin>left-top</margin>

<m>silver</m> and <m>gold</m> which are melted with the above matters won't become porous.</ab>

<ab>
<margin>left-top</margin>
For <m>gold</m>, you need not as much <m>crocum</m> as <m>stone alum</m>.</ab>

<ab>
<margin>left-middle</margin>
Small molds can be reheated soon. But you should oven-dry large and small molds because dampness go out of the mold with the heat. The intense heat of <m>charcoals</m> drives the heat from within outside.</ab>

<ab>
<margin>left-bottom</margin>
All <m>silver alloy</m> make chape and all other metals too. </ab>

<ab>
<margin>left-bottom</margin>
<m>Silver</m> should not be peeled off when [still] molten.</ab>

<ab>
<margin>left-bottom</margin>
<figure>#</figure> Coarsely powdered.</ab>

<ab>
<margin>bottom</margin>
Lumps of deteriorated <m>silver</m> vitrify red because of <m>arsenic</m> and <m>orpiment</m>.</ab>
<cont/>
</div>

.....
<page>128v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f262.image</image>

<div>
<id>p128r_1</id>

<ab>
You will see that it will stir it and heat it intensely and soon will make it quite liquid and ready to cast, and lighten it as it should. When it is in that state, very white and polished, shiny like quicksilver, be ready to cast. To that end, have thin, diluted sand in a pot or other vessel, which will be helpful to your mixture. Make a hole in your sand, and with you pincers put your red hot mold in this sand hole. Cover immediately the opening of your mold so that no ash or dust enters in it, and then surround it with sand up to the level of the gate and the vents. This done, peel off you mold and cast you well molten <m>silver</m>, the size of a pea of this metallic grain, which will immediately spread your <m>silver</m> all over, making it boil and cover everything. Cast as soon as you have spread this matter; that is the secret to make <m>silver</m> run, because its crust heats up and lightens it up. You can cast <m>silver</m> thinner than the Capital league [norm], as the royal one. </ab>

<ab>
<margin>left-top</margin>
Don't forget to add first some <m>borras</m> in your molten <m>silver</m>, even if <pro>silversmiths</pro> do not. However, it is good to do it and I have seen it done. Then we add matter from the crust twice, and then some of the metallic thing. Keep it from [illegible] that be laid at the very end of your forge.</ab>

<ab>
<margin>left-top</margin>
Si tu want to blow away the ashes that are around your mold when you hold it with your pincers, hold it upside down and blow.</ab>

<ab>
<margin>left-middle</margin>
When <m>silver</m> is well melted you can expose and blow it with a small bellow, but not continuously like you would do for <m>gold</m>, to get rid of the <m>charcoal</m> and fill it only with matter that make it runny.</ab>
</div>

<div>
<id>p128v_1</id>
<head>Whitening of cast <m>silver</m></head>

<ab>
People, even <pl>German</pl> people, commonly cast <m>silver</m> of poor quality. Because this kind of alloy produces defects and crusts, some <m>silversmiths</m> are eager to whiten their works, including the coarser ones. To do this they mix an equal part of <m>tartar</m> and <m>common salt</m> in their bullitoyre. I saw an excellent German working that way. In my presence, he casted a lizard with a teston league alloy, which had produced a dirty crust. To clean it, this German boiled it in a bullitoyre with <m>tartar</m> and powdered <m>common salt</m> mixed with common <m>water</m> on the fire of his forge. Then he scrubbed it because he thought it was not neat enough. From this crust he burned <m>tartar</m> on a piece of <m>paper</m> until it was black and did not smoke anymore. Then he diluted this <m>tartar</m> into the <m>water</m> form the bullitoyre, and covered all his <al>lizard</al> with this. Then he put the <al>lizard</al> between red hot charcoals in the forge, and he blew a bit. When the <al>lizard</al> became red, he removed it, let it cool, then reheated it with the bullitoyre mixture, and scrubbed it into clear water.</ab>

<ab>
<margin>left-bottom</margin>
<figure>+</figure>
Note that the bullitoyre for <m>silver</m> is not appropriate to use in a crucible because the <m>tartar</m> evaporates. But the vessel, being made of <m>copper</m>, is excellent to whiten <m>silver</m> and for the mixture which colors <m>gold</m>.</ab>

<ab>
<margin>left-bottom</margin>
He made sure that his <m>tartar water</m> would not boil over, because its strength would disappear, so when this first boiling happens, remove it from the fire and put it on again. He held this burned <m>tartar</m> mixture put on low quality <m>silver</m> to be a secret.</ab>
</div>

.....

<page>129r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f263.item.r=</image>

<div>
<id>p129r_1</id>
<head>Painting metal herbs</head>

<ab>If you are in a hurry it would be better to dilute your color with a <m>gum</m> not too thick because its oil takes a long time to dry, and it runs if thinly applied, as you have to do with herbs. If you are still in a hurry, dilute your color with <m>egg white</m> beaten with <m>fig peels</m>; and your work will soon be dry. But do apply it thinly.</ab>
</div>

<div>
<id>p129r_2</id>
<head>Wormwood</head>

<ab> The pallid white of this herb is made from <m>green verditer</m>, <m>white lead</m> or <m>ceruse</m>, a bit of <m>massicot</m>, <m>scudegrun</m>, and <m>cendré d'azur</m>. Mix and arrange your color with a branch you have. </ab>
</div>
<div>
<id>p129r_3</id>

<head><al>Viper color</al></head>

<ab>
It is the most beautiful <al>snake</al> to mold, because it has very beautiful scales, round and transparent. Its natural color is made from good <m>verdigris</m> well ground, with good <m>vinegar</m>, if it is made of <m>lead</m> or <m>tin</m>. And in the darkest parts, season this first color with a little <m>sulfur</m>, as you know. And if you need to lighten and whiten, like under the throat, rub these parts with a rough cloth. Because of its nature, the male ejects a kind of small lump from its backside when pressed, like a half arquebus ball, made in its genitals, and full of highly poisonous prickles.</ab>

<ab>
<margin>left-middle</margin>
<figure>+</figure> Add a bit of <m>looking-glass tin</m> in your <m>tin</m>; your <al>lizard</al> will look as if made of <m>silver</m>.</ab>
</div>

<div>
<id>p129r_4</id>

<head>Molded roses</head>

<ab>
Roses are molded with difficulty because of their petals, which are very delicate, weak, and doubled. To avoid this, rub them with <m>wheat oil</m> which is very dessicative, and once dried it stiffens and firms up the leaves to separate them and for them to withstand soaked sand. Do the same thing with <al>flies</al>, pansies, and other delicate things like capers.</ab>
</div>

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<page>129v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f264.item.r=</image>
<div>
<id>p129v_1</id>

<head>Animals oven dried for a long time</head>

<ab>
Stick your animals on a slab of <m>earth</m>, fix them with sharp pointed iron wires, as if you wanted to mold them, then dry them into an oven as hot as when <m>bread</m> has been taken out. It is the same for <al>snakes</al>, <al>birds</al> and other animals. But they have to be promptly dried.</ab>

<ab>
<margin>left-top</margin>
Note that you have to arrange them and put them into the sand as for flowers, and promptly dry them in an oven as warm as when <m>bread</m> has been taken out. Leave it there one day and one night in order for them to be very dry, otherwise it would stink. However, be sure that the oven is not too hot, it should be as warm as the sun in June, otherwise the animals would cook. Once they are dry, do not soak them; clean them with a brush or a paintbrush.</ab>
</div>

<div>
<id>p129v_2</id>

<head>Animals casted in <m>copper</m></head>

<ab>Once casted in <m>copper</m>, try to burnish them as we do with leaves made from gem, to see if they will get the same color.</ab>
</div>

<div>
<id>p129v_2</id>

<head>Advice about casting</head>

<ab>Be careful that your cast is not too thick, and do not forget to make 2 or 3 holes and notches in the conduit, when your cast is coming near the molded thing, divide it between 3 or 4 parts like fingers, and not deep. Thus the metal will run more gently, without preventing vapors and smoke.</ab>

<ab>
<margin>right-middle</margin>
Always make the entry of you cast close to the medal and the notch, when lumpy, it will lessens the metal's fury.</ab>

<ab>
To cast a <al>spider</al> on a leaf, and to prevent the sand from covering the under belly, fix a bit of melted <m>wax</m> at the end of its tail, and bottom part of the body, with the help of a warm wire as you know. Once the leaf has burnt and the <m>wax</m> has melted, two small holes remain on the leaf, that will be the cast of the <al>spider</al>.</ab>
</div>
<div>
<id>p129v_4</id>
<head>Casting a spider on a leaf</head>

<ab>
Usually big spiders have hairy legs, which are difficult to mold if you do not lay hairs down flat, or if you don't burn them with the flame of a candle, or if you stiffen them and lay them flat by <m>oiling</m> them with <m>wheat oil</m>. Kill them into <m>vinegar</m> and <m>urine</m>, like <al>snakes</al>, or in <m>spirits</m>. Then arrange your them on a beautiful vine leaf, or another leaf, then you can make the undisciplined hair look like tamed hair by rubbing them with mouth glue, or something similar. Their true color is <m>vinegar</m> or greenish. And season with sulphur in random places. Arrange your slab of <m>clay</m> and place your vine leaf on it, your dead spider in the middle. Pierce the middle of the <al>spider</al>'s body with the tip of an <m>iron</m> or <m>latten</m> wire. Then fix diagonally some small <m>latten</m> tacks all around the vine leafs in order to secure it. Then block the end of the legs with a bit of melted wax using a warm <m>iron</m> wire, and smooth it out with the tip of a pair of small pincers. do the same with the other edges of the <al>spider</al>. Return<figure>□</figure></ab>

<ab>
<figure>□</figure>
<margin>left-middle</margin>
<figure>□</figure>

When the wax is cool, scrape away the superfluous parts with the point of a penknife, that way the end of the legs will stay neat. Then arrange the disc of earth around the spider, and cast your soaked sand, as you did for others. In this fashion you will have to burn the vine leaf in the mold. Otherwise, you will not have to. To cast your <al>spider</al> more easily, make sure the animal is dead, avoiding the entanglement of its legs while agonizing. When you have done your first cast, peel this side of the leaf, then do the second cast. </ab>
</div>

.....

<page>130r</page>

<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f265.item.r=</image>

<div>
<id>p130r_1</id>
<head>Drying animals in an oven</head>

<ab>One can skin small <al>cats</al> and remove their eyes and all their guts. Then place a small stick vertically between the teeth in order to open their mouths. Then with <m>iron nails</m> one should attach their feet to a small board, arrange the required position. Once they are tied by their feet with small <m>iron rings</m>, one has to hang them upside down into the sun. By doing so, they will get the right shape and dry, the belly will tighten and the the tail will remain high up thanks to the twist that you made. Once they are already a bit dry, one may twist the head as he wishes and secure the position with any tool. One may finish to dry the animal by placing it in an oven once the bread has been taken off. Then place some realistically painted <m>lead</m> balls or <m>wax</m> balls in the eye-sockets. One should paint them with <m>gummed ink</m> so that they look like jet. One can add a painted tongue, or horns, or wings or anything you may imagine, same goes for <al>rats</al> or any animal.</ab></div>

<div>
<id>p130r_2</id>
<head>Reddening lively <al>crayfish</al> which will look as if they were boiled</head>

<ab>Rub your <al>crayfish</al> with good <m>vinegar</m> diluted with a little <m>eau-de-vie</m> and not much, then you can present them as if they were cooked and yet they will walk.</ab></div>

<div>
<id>p130r_3</id>

<head>Molding a single <al>spider</al></head>

<ab>One has to display it on the jacket of <m>clay</m> as it is said for the <m>vine leaf</m>. Then do the first cast. And once it dry, peel the mold off to the middle of the legs and do the second cast.</ab></div>

<div>
<id>p130r_4</id>
<head>Molding a single <m>vine leaf</m></head>

<ab>Attach it with <m>brass</m> wire points on the jacket of <m>clay</m> and then cast the first mold. Once it is dry peel off the back side and make the second cast. Then peel away the back side, and make your second cast. Once it is dry, you may remove the leaf. You must heat your mold only once, because there is nothing to be burnt inside.</ab>

<ab>
<margin>left-top</margin>
<figure>
<id>fig_p130r</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5cHVSNnNVQ115RnM></link>
</figure>
which surround the fingers of the paws and which stick in the jacket of <m>clay</m>.</ab>

<ab>
<margin>left-bottom</margin>
Also. One cannot mold the hairy legs of big <al>spiders</al> or any other hairy animal if they had not been rubbed with <m>wheat oil</m> before they are laid flat. This will make the <m>hair</m> firmer and will dry very quickly. Hairy things entangle in the sand, and do not burn very well.</ab>
</div>

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<page>130v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f266.item.r=.zoom></image>

<div>
<id>p130v_1</id>
<head>Molding a crab</head>

<ab>It is a secret and a masterpiece to mold [a crab] well because it is necessary to proceed differently [than you would] with other animals, because its shell is quite difficult to [...] and that is why it is necessary to reheat the mold three or four times. And on top of all that, it leaves a hard crust [that is] gray like <m>ash</m>. It is not molded from one piece and one cast as [is done with] plants. Rather, [it is moulded] in two casts as [is done] with serpents which are easily stripped from it [the mold]. After it is opened, nimble remove this crust with the very fine point of a pen-knife {and not with <m>quicksilver</m> which removes nothing from it} and likewise those small crusts of the second cast which get into the cavities that the <m>sand</m> or the molded animal made because it has curved legs. Here is the subtlety and the secret of stripping it. As you molded it [the crab] from the back side in the way [that you did for] the others [animals], remove the stomach and all the curved legs from it. Make your second cast in the way [that you did for] the other [animals], but as it will have set, take care that you not open the mold unless you have first reheated it well. Otherwise, because of its curved legs, you would break the whole thing. Herein lies the subtlety. If, after it has been well reheated and opened, you see that the crust is not burned enough, reheat [it] until it is burned [enough]. </ab>

<ab>
<margin>left-top</margin>
It [a crab] can be painted like a crayfish.</ab>

<ab>
<margin>left-top</margin>
It does nothing.</ab>

<ab>
<margin>left-bottom</margin>
For opening the mold, it is not necessary to soak it because, once reheated, it will open itself by itself.</ab>

</div>
<div>
<id>p130v_2</id>
<head>Stag beetle</head>

<ab>Like a crab, it [a stag beetle] is also difficult to burn, therefore do [stag beetles] as you did crabs.</ab>

</div>
<div>
<id>p130v_3</id>
<head>For molding thinly</head>

<ab>After you have molded the first mold as the core, leave it to dry well before removing the figure from <m>wax</m>, so that the mold is not ruined. Afterwards, then make make a small <m>lasagna of paste</m> as thick as you like, and, once you have greased the cavity of your first hollowed mold with <m>butter</m>, apply the <m>paste</m> to it [the cavity of the first concave mold] and then the upper part of the second mold. If you grease [the mold] with <m>oil</m>, it will be soaked up [into the mold] and it will not be as clean as <m>butter</m>.</ab>
</div>

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<page>131r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f267.item.r=</image>

<div>
<id>p131r_1</id>
<head><m>Herbs</m> difficult to burn in the mold</head>

<ab>Mold them in two or three castings; being annealed, your mold will open, and it will be easier to extract the <m>charcoal</m> from inside. </ab>

<ab>
<margin>left-top</margin>
Try [this]</ab>

</div>
<div>
<id>p131r_2</id>
<head>molded letter <m>paper</m></head>

<ab>Write with some <m>ink</m> well thickened with gum or any other <m>color dye</m> which has body, and which is not erased if dampened with brandy. Then put your paper on the sheet of clay, and dampen it with <m>brandy</m>. Cast both sides [of paper].</ab>

</div>
<div>
<id>p131r_3</id>
<head>Adorning beds, mirrors and similar things</head>

<ab>Model any drawing in half relief on a flat <m>slate</m> in order to set it on round or flat things. Cast with very fine <m>tin</m>, then you can gild it with <m>gold</m> leaf, and set it wherever you want. Fill the bottom of the relief with small <m>rubies</m>, <m>orpiment</m>, or any colored little <m>grains</m>.</ab>

</div>
<div>
<id>p131r_4</id>
<head>Training a dog </head>

<ab>You need to keep your dog attached; when it does what you command, to win its love, give it a piece of <m>cheese</m> which was held under the armpit. This <m>cheese</m> would be a bad drug if the master, or the apprentice, is red-haired. </ab>

</div>
<div>
<id>p131r_5</id>
<head>Molded <m>wax</m></head>

<ab>Wax representations made from <m>white lead</m> and <m>cerussite</m> are not made to be burned and melted in a mold where you want to cast <m>silver</m> and <m>gold</m>; this would make it brittle. Moreover if you use such a mold, the <m>wax</m> would be stripped thanks to the violence of the fire which makes it boil, and the mixture made of <m>cerussite</m> or any other metal would attack the mold. That's why if you want to sketch things</ab>

<ab>
<margin>left-bottom</margin>
<figure>+</figure>
<m>White wax</m> is more delicate than the other, and it doesn't leave any filth when you want to mold it as a core, or to mold hollow.</ab>

<cont/>
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<page>131v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f268.item.r=</image>

<div>
<cont/>
<id>p131r_5</id>

<ab>to be cast with <m>gold</m> and <m>silver</m>. Do not add to your wax all the drugs you are preparing, but only melted <m>sulphur</m>, as the candle melts, and also <m>candle smoke</m>. Melted <m>sulphur</m> falls down to the bottom of your mold, but lets the <m>wax</m> keeps some of its quality, melting it but leaving it a little firm. You can thus melt your <m>wax</m> representation on fire. If your wax is composed of <m>cerussite</m> or something similar, you need to put your mold into boiling water.</ab>

<ab>
<margin>left-top</margin>
You don't need all this to mold flat representations made of <m>wax</m>, because the <m>wax</m> can be removed completely from the mold, and won't remain inside it.</ab>

</div>
<div>
<id>p131v_1</id>
<head>When <m>lead</m> or <m>tin</m> casts fill with bubbles</head>

<ab>
If your casted work became porous, it can be repaired with very firm wax, and you can gild or silver it afterwards. If you cast fine, the alloy of one ⅓ of <m>lead</m> to one lb of tin is good. But if you cast with <m>lead</m><add>[ligue illegible]</add>, indeed you need one lb of <m>lead</m> to two ⅓ of tin, because lead is fat and sticky, goes over the alloy. I have cast a very small lizard that way. If you cast something thick, which keep the heat for a long time, it will make bubble if there is not enough <m>tin</m>. Your mold should not be cold, but cool enough that you can hold it without burning your hand, or that you can hold your finger into the cast without burning yourself. Your <m>tin</m> or <m>lead</m> must be like red. Cast your lead first, then cast tin. When you are going to cast, add a piece of <m>resin</m> to your mixture, then a little <m>looking-glass tin</m>. If you add too much <m>lead</m> to your <m>tin</m>, the mixture wouldn't be fluid enough. To know [the right proportion], If <m>tin</m> cries loudly, it means that there is not too much <m>lead</m>; if <m>tin</m> cries softly, it means that you added too much <m>lead</m>. Be wary</ab>

<ab>
<margin>left-middle</margin>
You can solder with the same matter, then rework it and [remove away all the unwanted thick parts] with a burin.</ab>

<ab>
<margin>left-middle</margin>

<m>Tin</m> is a metal that can[...], makes bubbles and burn, and more capricious to work with than <m>gold</m> and <m>silver</m>.</ab>

<ab>
<margin>left-bottom</margin>

<m>Lead</m> must not be only red, it must be quick, and as liquid as water, which is a sign of its perfect heat. Your mold must be so hot that you couldn't bear to put your finger into it. Do not open your mold until it has cooled down. Just as for <m>silver</m> and <m>gold</m>, it is a sign of good casting when <m>lead</m> comes out through the vent holes.</ab>

<cont/>
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<page>132r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f269.item.r=</image>

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<id>p131v_1</id>

<ab>not to heat your mold too hot because it makes it brittle, it makes bubbles and shrinks <m>tin</m>, and would also break the molded thing. Do not dip you hot mold into water because <m>tin</m> and <m>lead</m> would shrink.</ab>

</div>
<div>
<id>p132r_1</id>
<head>Mold made from two casts</head>

<ab>It opens once anealed. To clean it after, fix the clamps again, lute the mold, particularly the joints. Then let the earth slowly dry. When it doesn't smoke anymore, let the mold cool down until you can hold it without burning your skin. Then put your molds into the presses, or into a pot filled with sand, and cast with <m>tin</m>. If you cast <m>gold</m> or <m>silver</m> your mold needs to be cast twice, and cast when your mold becomes red-hot.</ab>

<ab>
<margin>left-top</margin>
If your heated mold [...], shrinks because of the bad quality of the <m>plaster</m>, open it after the first heating, tighten it, the lute it and clamp it again.</ab>

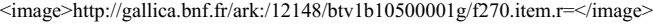
</div>
<div>
<id>p132r_2</id>
<head>How to anneal the molds</head>

<ab>Light your <m>charcoals</m> from the forge, then place them in a line in a corner of the forge according to the size of your mold, in order that it be surrounded with 3 or 4 fingers of <m>charcoals</m>, particularly at the bottom of the mole, where it is thicker. Then put your molds on these <m>charcoals</m> -- and not on a grate as some people do -- because they would get too strong a heat. Let your furnace heat up gradually for a while, always adding burning <m>charcoal</m>. As long as the mold is humid, your charcoal will be dampened at the bottom, but when it is dry, the charcoals start to blaze. When one side of the mold is dry, turn over; when both sides are dry and don't smoke anymore, increase the heat and cover your mold with burning coals. When the mold starts being red, do not blow on it with small bellows, because the mold would burst. Similarly, when your mold becomes red-hot, keep it well covered with burning charcoals and do not uncover it because it would crack. Prevent the draft

from a window from reaching it. Then let it cool down.

When you want to anneal your molds, luted or not, do mark the belly of your mold, in order to place that side at the bottom, against the burning charcoals, because if the intensity of the heat made the mold crack, it would be better that it do so on the bottom part instead of the top part.

If your mold is small, do not heat it as long, otherwise it would crack and be damaged.

132v


dry by itself. When your mold starts to redden, it will soon be become completely red-hot. But make sure that its bottom is red-hot too, without any black spot.

Common quarry sand

In a frame, If you cast it hot, sand will bubble; thus you have to just dry it.

I have tried to prevent molds blended with sand from cracking when annealed, and thus not get any with flaws that crack.

Crocum ferry

After I processed it with vinegar and reddened under heat, I had it ground very finely on the porphyry, until there could be no ruggedness felt under the nails. Then I soaked it in very good vinegar for 2 or 3 days, stirring the mixture several times a day. Then I boiled it and made it red-hot in a pot, and put the whole thing into an air furnace. I obtained a mass full of eyes, but that crumbled finely between the fingers. I mixed half as much as stone alum in the sand. I diluted the sand very thinly, and molded a very small lizard, which molded very cleanly and finely. Crocum ferry does not make the mold harder, but makes it firmer. When your mold is soft and fat under your nail, it means that the crocum is good, very fine and well prepared. You can add any quantity of crocum, your mold will not be damaged because it is a friend of gold. I think it would be the same with silver. And in order that molds made with it do not crack, make it with needle filings.

Sand is better made with distilled vinegar.

You can add this sand to the molds you want to use to cast silver, because it make molds firmer, and when you scratch it, you will find it a bit rougher than the other molds not made from this sand. You mold very clean with this sand.

You can use this one for all molds, because it prevents them from cracking and bursting when heated. This sand withstands several casts for molding flat medals. Sand from steel or needle filings is redder and better.

Gilding animals casted with silver

You can gild them with an amalgam, the lines won't be damaged if they are made from silver.

<div>
<id>p133r_1</id>

<head>Hard <m>wax</m> to imprint seals</head>

<ab>One makes these [seals] out of white <m>wax</m> which is harder than the other and one mixes in it some <m>cerussite</m> or <m>leaded white</m> very finely ground such that it is as hard as you like and mix in a drop of <m>turpentine</m> to bind it. Afterwards mix it in the color that you want. It is in <m>wax</m> that serves <pro>goldsmiths</pro> for modeling.</ab>

<div>
<id>p133r_2</id>

<head>Casting the feet of small <al>lizards</al> in <m>gold</m> and <m>silver</m></head>

<ab>Because the feet of these <al>[lizards]</al> that are very small are [also] very fine, once you have molded the top and you have uncover the vents and underneath the feet are covered lightly with <m>wax</m>, and then cast a second mold. And the <m>wax</m> that one removes, it leaves a little cast for these said feet, and again such that the underneath [part] comes out in one piece that you can repair well, And the scales on top of the toes come [out] well.</ab>

<ab>
<margin>left-top</margin>
Around the big <al>lizard</al> nails, put on each a small round of <m>wax</m> to make the cast like this.</ab>

<figure>
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<link>https://drive.google.com/open?id=0B9-oNrvWdIO5V2JZcWtNeEJmYWws</link>
<margin>left-middle</margin>
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<div>
<id>p133r_3</id>

<head>Marks [made by] the [pin] points of <m>iron</m> thread which one finds on the head of an animal</head>

<ab>With the points [or pins] with which you have dressed [your animal] all over, do same with the head, but [out of] the head which is pierced, a certain moisture seeps out around the wound and this little exudation prevents the sand from covering the point well and there always remain little pinpricks. But to avoid this, you can plant in the <m>plaster</m> of the said blunt pin point and around the upper part of the point, and over this point, there put a little hard <m>wax</m> or a little <m>mastic</m> or <m>cement</m>. And by the medium of the hot thread of <m>iron</m> there you can hold the throat of the animal which one removes [the <m>iron</m>] when you come to uncover the vents and the throat for doing the second cast.</ab>

</div>
<div>
<id>p133r_4</id>

<head>Clamps and the broken mold</head>

<ab>When you have done the first firing and taken off the clamps and you have opened your mold to clean it by taking off the bubbles with <m>quicksilver</m> which one cannot do with closed molds, reclose the mold and reattach the clamps but not in the same place as the first time. Reapply lute and dry it but if this is for casting <m>silver</m> or <m>gold</m> [use] lute with the rest of sand which has been used because this is the best netting. If your mold is broken in refiring you can augment it with clamps and lute.</ab>

<ab>
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<figure>X</figure>
</ab>
</div>

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<id>p133v_1</id>

<head> Porosity and little holes which are in the mold</head>

<ab>
It often happens when you cast two or three molds one after the other using the same dish, because the last casting comes from the bottom of the dish and is thicker, and then often the last cast which comes from the bottom and which is the thickest of the dish, may easily become porous. Cast thinner and neater, as long as you blow very strongly on the soaked sand while casting in order to clear the little air bubbles.</ab>

<ab>
<margin>left-top</margin>
When the sand is casted thick, it often becomes porous
</ab>
</div>

<div>
<id>p133v_2</id>
<head>Thing that cannot be released from the mold</head>

<ab>
You must cut it inside the mold in order to burn it. Cut it with scissors at once, that way <m>blood</m> doesn't flow on the mold, and doesn't make dirty. Because this dirt would be removed with difficulty when reheated. Then place your mold into a furnace, or something similar, in order to quickly dry the mold which must be removed before the forming of crusts or molds.@[a] </ab>

<ab>
<margin>left-middle</margin>
<figure>+</figure>
When you mold small <al>lizards</al>, and when you want to peel away the first molded part, start to peel away the heads, you will know its place thanks to the needle. Don't peel away the legs before peeling away the entire body, to avoid that the naked body, moving, doesn't remove the legs from their place, which would not be easy to put back into place. When you remove the small needles fixing the legs, block the legs or press the legs with something, that way while removing the needles the legs won't move; finally, you can fix the legs again with <m>wax</m>.</ab>
</div>

<div>
<id>p133v_3</id>

<head>Various animals entwined</head>

<ab>
You can entwine a <al>snake</al> with a <al>lizard</al>, each one biting the other, or a <al>snake</al> eating a <al>frog</al>, or a <al>wall lizard</al> or something similar. But since these interlacings cannot be stripped, cut what's possible, and allow the rest to burn. To make the mouth of the <al>snake</al>, which must stand high, hold a <al>wall lizard</al>, which is small and would need to be held in the air, because the <al>snake</al>'s head is higher, put under the <al>wall lizard</al> a mound of earth suitable to support it. And if you place the mold in a furnace, the animal drying quickly, it will quickly shrink, then it will burn more easily. These interlacings are also made to mask wounds or defects on the animal which happen when we catch them. Do not forget to join with <m>wax</m>, applied carefully with a hot <m>iron</m> wire, all the parts of the animal which are one on top of the other, or those which are not well pressed on the slab of clay, so that the soaked sand does not remove them. Do that in order to </ab>
<cont/>
</div>
[a]in this case, mold stands for "moisissure".

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<page>134r</page>
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<id>p133v_3</id>
<ab>
avoid putting pins that should not be put, only on the top of the head, the thickest parts of the body, and on the delicate parts like the middle of the <al>lizard</al>'s feet, the nails of which you will drive into the <m>clay</m>, thus they won't move. Once your first mold is casted, peel away the belly of the animal, but do not forget to put some tenuous little pieces of <m>wax</m> at the tip of the <al>lizard</al>'s feet. But if it is from a <al>wall lizard</al>, lightly cover all the sole of its feet with <m>wax</m>, which will be removed with the second cast. A cavity is left which will be fill these little fingers with metal. then rework it. Make sure that wax is fixed well on the parts of the animal which are one on top of the other, so that the soaked sand which comes in does not make them move. If that happened your jacket of <m>clay</m> would be spoiled and your mold wouldn't get the right thickness, would have holes, and would be spoiled. To obviate that you can make a thicker mold. If this occurs you could rework it in this fashion.</ab>
</div>
<div>
<id>p134r_1</id>

<head>To repair a pierced mold</head>

<ab>
If the molded animal pulls of in certain places, moves from the place you had fixed it, or has not completely died, or if you didn't fix it well with the <m>iron</m> pins or with the melted <m>wax</m>, or if the animal hasn't the right thickness on the outside at the relief, or if it is pierced, instead of removing the animal from the mold, peel away the weak parts, or the pierced part, and make holes all around these parts, then dip the back side of the mold, quickly cast onto this place with the same soaked sand which will go into the holes, the cast will adhere to the first cast. But you will have to lute this spot afterwards.</ab>
</div>
<div>
<id>p134r_2</id>

<head>Sand made of <m>crocum ferri</m></head>

<ab>
You can use the mixture made of <m>crocum ferri</m> to cast <m>silver</m> or <m>gold</m>, this mixture will make these casts firm, and prevent these from cracking and making burrs. And I believe that for flat things, it would withstand severa casts. Which however is not commonly done for <m>gold</m> and <m>silver</m>.</ab>

<ab>
<margin>left-bottom</margin>
Try <m>crocum ferri</m> for <m>lead</m> and <m>tin</m>.</ab>
</div>

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<page>134v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f274.image</image>

<div>
<id>p134v_1</id>

<head>
To make <m>gold</m> fluid </head>

<ab>
Put as much <m>verdigris</m> as <m>sal ammoniac</m>, as much <m>borax</m> as <m>salt peter</m>. But don't add a greater quantity of <m>borax</m> and <m>salt peter</m> than half the quantity of <m>verdegri</m> and <m>sal ammoniac</m>, in fact add as much <m>borax</m> and <m>salt petre</m> all together as <m>verdigris</m>. Grind into powder the ingredients on the <m>marble</m> slab, then grind again with very good <m>vinegar</m> until it is very fine, like a ground color, grind <m>verdigris</m> with <m>oil</m>, then leave to dry if you have time, if you haven't so much time dry it with the fire of the forge, make so that it is very dry.</ab>

<ab>
<margin>left-top</margin>
Grind</ab>

<ab>
<margin>left-top</margin>
One of the principal things for casting is to cast very hot, especially large metals</ab>
</div>
<div>
<id>p134v_2</id>

<head>
Casting big works with <m>gold</m></head>

<ab>
Not only should one clamp the mold very well, but also strengthen it with good iron wire links or adequate straps.</ab>

<ab>
<margin>left-middle</margin>
Before casting with gold, clean thoroughly your forge with <m>lead</m> and <m>tin</m>.</ab>
</div>
<div>
<id>p134v_3</id>

<head>Secret for soldering small works made of <m>gold</m> and <m>silver</m></head>

<ab>
If some little pieces of legs cannot be easily removed from <m>gold</m> or <m>silver</m> casting, it is because you didn't cast <m>wax</m> under these parts. Beat some soldering <m>gold</m> very finely, then cut it into very fine flakes, cut the quantity you need. Take some of this phlegm or white and thick <m>saliva</m> which is on the teeth, put some of this matter to the place you want to solder with the point of a burin, then place your flakes, then put to the opposite part a mixture made of a small quantity of soaked fat <m>earth</m> and <m>saliva</m>, thanks to the mixture the solder will better adhere. Then pulverize a bit of <m>borax</m> on it, and heat up <m>Quince pulp seed water</m> to a boil and others too.</ab>
</div>

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<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f275.item.r=</image>

<div>
<id>p135r_1</id>
<head>Colors for <m>gold</m>, or sauce</head>

<ab>
Take as much <m>verdigris</m> as <m>sal ammoniac</m>, & <m>salt petre</m> the size of a bean. Because if you put in too much, it would make the whole thing boil so much that it would all pour out. <m>Salt peter</m> is only put in to make it boil, so that you know when the material has boiled enough. This color put on sour melted <m>gold</m> makes it soft instantly.</ab>

<margin>left-top</margin>
<head>Softening <m>gold</m></head>
</div>

<div>

<id>p135r_2</id>
<head>Casting</head>

<ab>
It must be lengthened out with respect to the thing you have molded. And at a minimum for large molds it must be four or five fingers in length. You can make it large enough for half the mold & then divide it in three points of such a sort, and so the molded things will [each] be small. But your points must be small. And in the middle of the casting & of its points, make some holes to impede the fury of the <m>metal</m> & make it run more softly. It is also necessary to fix the bent parts of the animal with little notched lines, and similarly the end of the tail and other fine parts so that the <m>metal</m> goes throughout & flows from one part to the other.</ab>

<ab>
Do not forget to make a wax cast of the paws of [illeg] & things as subtle as this.</ab>

<ab>
<margin>left-top</margin>
When you have molded an animal in the first part of the mold, do not let this part dry out before you have removed the said animal, because when the mold dries out it will shrink and it will also make the animal shrink. Keep it therefore in a damp & humid place until you have done it.</ab>

<figure>
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<margin>left-middle</margin>
</figure>

<ab>
<margin>left-middle</margin>
Vine <m>leaf</m> is painted with a green made of <m>scudegrun</m> & <m>verd de terre</m>.</ab>

</div>
<div>
<id>p135r_3</id>
<head><m>Vine leaf</m> and small <m>frog</m></head>
<ab>

To mold, take <m>vine leaves</m> as they are blooming, just as with all <m>herbs</m> & <m>flowers</m>, or a young <m>vine</m> regrowing after it has been cut or when the leaves come off in autumn, because new <m>leaves</m> are knottier & have more evident veins. By contrast, old <m>leaves</m> are smooth all the way through & full of holes in many places. Place therefore your <m>leaf</m>, its back behind it against the <m>clay plaster</m>, and fasten it with little pins. Leave it nevertheless with its natural curling. And make a notch in the clay to hide the stem for the first cast. And when it's done and dry, take out the stem of the <m>leaf</m> and clean it well with your jacket and press it with a little scoop of clay. Afterwards, make many castings around the <m>leaf</m> with <m>wax</m>, as you know well, & make a slender casting & with many lines & knots.</ab>

<ab>
<margin>left-bottom</margin>

You can cast a spider or frog & whatever you like on the leaf.</ab>
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<page>135v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f276.item.r></image>

<div>
<id>p135v_a1</id>
<head>Casting <m>gold</m></head>

<ab>Take a crusol that can contain a certain amount, as you wish to place roughly crushed <m>borax</m> into it. Place [the <m>borax</m>] at the bottom and then the <m>gold</m> on top of it. Arrange the <m>coal</m> in the forge in such a way that the wind of the bellow will blow above the crusol and no on its side as it would cool it down and would not heat it enough. Also, make sure it is three good <ms><bp>fingers</bp></ms> away from the wall of the forge so that it is quite surrounded by <m>coal</m>. Allow the crusol to anneal until it is quite red. Then, have your boy strongly blow the bellow with long pushes, as in this manner it heats it better. Meanwhile, once the <m>gold</m> will be melted, blow a small bellow above it. Because [otherwise] the <m>gold</m> would turn sour and would acquire some kind of coat. As long as there will be some swirling smoke as you blow this way, it means that it is not soft enough for the cast. Therefore, uncover the melting pot and without displacing it and without stopping the bellow, tilt it towards you so you may well see what is inside. Throw a ferue of <m>borax</m> in it and blow wind towards it with the small bellow in a jerky fashion. Should there be <m>coal</m> inside, blow a bit stronger to make them come out. And if the <m>gold</m> still smokes and swirls, add <m>borax</m> again and blow above it. Beware that it is still sour and not soft enough, it will show a coat when you blow it as if it had cooled down. By contrast, if it does not show this sign, it is soft enough. Then, blow strongly so that it [the <m>gold</m>] is quite heated and once you think it is hot enough, throw colour on top of it in the crusol and the colour should be composed as following: <m>verdigris</m>, <m>saltpeter</m>, <m>sal ammoniac</m>, and a bit of <m>borax</m>. And continue blowing with the small bellows, and the <m>gold</m> will turn shiny like</ab>

<ab>
<margin>left-top</margin>
If you cast in <m>gold</m> some piece of important [dimensions], lute your mold with the same <m>sand</m> or put <m>crocum</m> on it.</ab>

<ab>
<margin>left-top</margin>

If you wish to cast some large work, or of important [dimensions], create at the bottom or at an extremity of the forge a vented furnace where you may hold your mold in the <m>sand</m> so red.</ab>

<ab>
<margin>left-middle</margin>
One may cast two or three pounds of <m>gold</m>.</ab>

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<page>136r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f277.item.r=</image>

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<id>p135v_1</id>

<ab>[FROM PREVIOUS PAGE: And continue blowing with the small bellows, and the <m>gold</m> will turn shiny like] a mirror. And once you will want to [start] casting, put a bit more colour in and let it rest a little, still blowing wind with both the small and the large bellows. Finally, place your mold quite red between the small molds [a] or in a crusol full of <m>sand</m> quite hot and [start] casting. And once it will have stuck, you may, if you wish, throw it into <m>water</m> as <m>gold</m> does not get damaged like <m>tin</m> which breaks.</ab>

<ab>To melt <m>gold</m> bullion, there is no need to blow above [the crusol] [b] with the small bellow.</ab>

<ab>For large works, the crusol made of <m>sand</m> has to be placed in a wind oven [c] so that it is all red at the end of the furnace.</ab>

<ab><m>Gold</m>, when exposed to wind or when forged becomes black. But a bit of <m>aqua forte</m> removes [the black] instantly.</ab>

<ab>If you have to forge a work made of <m>gold</m> on <m>brass</m> [d], as is done on small statues, put a <m>lead</m> sheet between the <m>gold</m> and the <m>brass</m>, and rather than anneal it and put it back in the fire, dip it in the <m>aqua forte</m> and it will be soft.</ab>

<ab>It is enough for the cast to be of the same thickness as the medal, that it from the middle of the cast, up to the medal. But if the medal is very thick, one should not thicken the cast as a result, because a thick cast never comes out well. It may be done large to embrace the medal as much as possible.</ab>

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<div>
<id>p136r_1</id>
<head>Enamelling thin works</head>

<ab><pro>Goldsmiths</pro> polish [e] the <m>gold</m> leaf using a burin and then they apply the <m>enamel</m> onto it.</ab>
</div>

[a] moulet?
[b] with reference to the process described in p135v
[c] The wind oven is part of the furnace, a specific hotter area within the furnace (see p135v)
[d] I think the process described is the gilding of brass objects to make them look like gold.
[e] a literal translation of the French text would be 'scrape' but I think polish works better to describe the process at work here.

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<id>p136v_1</id>

<head>
<m>latten</m> casting</head>

<ab>
Take <m>latten</m> from skillet pans, used to make baby's cereal, which is doulx and apt for the cast. Some people say that counter from <pl>Germany</pl> contain a lot of <m>calamine</m>. However, when they are tempered, <m>calamine</m> evaporates, <m>calamine</m> mixed with melted <m>latten</m> always evaporates, because <m>latten</m> becomes red again, thanks to the reiteration [a] of melting. However, fresh <m>calamine</m> and only fresh <m>calamine</m>, which is blended with remelted <m>latten</m>, makes <m>latten</m> run, and allow a clean cast. Because <m>calamine</m> which is added to remelted <m>latten</m> is half-damaged compared with its natural state. This half-damaged <m>calamine</m> make <m>latten</m> porous and agitated because it has a tendency to evaporate. Thus add only very fresh <m>calamine</m> to melted <m>latten</m>. Make sure to cast your mold very hot, you mold must become red-hot, like molds to cast <m>gold</m>, <m>silver</m>, <m>copper</m> and metal. If you pour fresh <m>calamine</m>, avoid its smoke which is pernicious. To cast a medal, I took thirty counters from <pl>Germany</pl> and xii <m>clous de rosette</m> made of doulx <m>latten</m>, like for claires [b] medals which are made of soft <m>latten</m>. This matter contains a lot of calamine like all kind of yellow <m>latten</m>, and make a lot of smoke which prevent <m>latten</m> from running, and make it porous, that is why it is necessary to make many vents and to cast very hot, <m>latten</m> must be as white as <m>water</m>, or as melted <m>silver</m> or as a polished mirror of <m>steel</m>. The second fusion will come out much better because evaporated <m>calamine</m> does not make as much smoke. If you use a molding frame that does not break and holds its own, the second [casting] will come more neatly out of the mold, because the frame is impregnated with the smoke of <m>calamine</m>. If you

add <m>sal ammoniac</m> to your <m>latten</m>, this <m>latten</m> will be clear and shiny, but will be even more shiny with <m>huile tingente[c]</m>. Do not use another sand than the above mentioned to mold a noyau, and the mold must become red-hot, like a mold to cast <m>gold</m>. Make a lot of vents. If you cast yellow latten with the prepared tutty you will not get any smoke.</ab>

<ab>
<margin>left-top</margin>
This metal is capricious to cast because of the smoke of the <m>calamine</m>. Do not leave it out of the fire, as some do with <m>silver</m>. Because when air or wind touch it, the mold gets cold at once. When you cast it, this metal always leaves burrs, like when casting <m>glass</m>. <pro>Foundry workers</pro> do not usually use this very yellow <m>latten</m> as much because of the <m>calamine</m> which is heated. They cast <m>red copper</m> and turn it yellow with fresh <m>calamine</m>, or with some prepared tutty. Before casting, clean it with <m>charcoal</m> with a quere[d] made of <m>copper</m> or <m>iron</m>. Then cover it with a cloth soaked in <m>lard</m> blended with <m>salt peter</m>, or <m>sal ammoniac</m>, in order to protect it from wind, which would cool it down.</ab>
</div>

[a]no translation found
[b]Claire is a refiner term, this is a mixture made of washed ashes and calcined bones to make small dishes
[c]Did not find translation. Here is a reference in French (18th century though). <https://goo.gl/yQn2NP>
[d]could not find a translation

.....
<page>137r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f279.image></image>

<div>
<id>p137r_1</id>
<head>Casting <m>red copper</m></head>

<ab>Pure cauldron <m>red copper</m>, or <m>copper</m> of any other small works, is good to cast. To make it runny, add some <m>sal ammoniac</m> to it, and when you are about to cast, add a little bit of <m>fine tin</m>. One must cast the <m>copper</m> very hot into the mold, so that it is burning and very red, like <m>gold</m>, <m>silver</m>, <m>latten</m>, and <m>metal</m>. You will know that it is hot enough when it is clear and shining like a newly polished <m>steel mirror</m>, or like <m>melted silver</m>. Protect it from wind so that it doesn't cool down too soon. Cover your cast with tow in order to prevent it from cooling down. <m>Red copper</m> is cleaner than <m>latten</m>, which smokes a lot, which prevents the metal from running. I have moulded it as a noyau very neatly, and as thin as a piece of <m>paper</m>. It must be very hot, white and shining like melted <m>silver</m>, or a mirror. I have moulded it as a noyau with the same <m>sand</m> mentioned above.</ab>

<ab>
<margin>left-top</margin>
It takes more time to melt <m>copper</m> and <m>latten</m> than any other metals, even <m>red copper</m>. It will run and will comes out well if you cast it very hot, and as thin as <m>water</m>.</ab>

<ab>
<margin>left-top</margin>
Remove your mold from the fire, and set it into embers, which should fill a pot or vessel.</ab>
</div>

<div>
<id>p137r_2</id>
<head><m>Huile tingente</m> to make metal runny</head>

<ab>Take some <m>mercury</m>, some real <m>sublimate of Venice</m>, and not <m>sublimate of arsenic</m>, <m>aes ustum</m> a poix, <m>sal ammoniac</m>, a poix. Ground everything separately, then blend them together in a <m>glass</m> bottle, and put it on warm ashes. You will see that everything melts like <m>wax</m>, and will have bright colors. Leave to set, add a small quantity of this mixture on every metal, it will run marvelously.</ab>

<ab>
<margin>left-middle</margin>
<pro>Foundry workers</pro> who make large casts to mold statues, clean the dirt from these molds with <m>calamine</m>, and a lot of <m>sal ammoniac</m> in order to make moulds clear and neat. When they want to cast they add a lot of <m>tin</m>. Dampness and cold could spoil their works, that is the reason why even a small source of <m>water</m> in the hole can damage the whole thing.</ab>
</div>

<div>
<id>p137r_3</id>
<head>Clamps</head>

<ab>To make clamps use flat tongs made from wire, which is reheated and bent, and hammered on anvil. Since those clamps are thinner, they burn more easily, being used in reheated molds. So use new clamps.</ab>

<figure>
<id>fig_p137r_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5aG5XMHdYTTYzbzA></link>

<div>
<figure>
<div>

<page>137v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f280.image</image>
<div>
<id>p137v_a1</id>
<head>Wetting <m>sand</m> to mold flat medals</head>

<ab>Flat medals, made of <m>wax</m> or <m>metal</m>, must be oiled very lightly and then touched with a brush of <m>spirits</m>. So that the already moistened mold takes well the oiled thing without rejecting it, one warms the water well in order to wet the <m>sand</m>, because with cold water it would be rejected. It is necessary to <m>oil</m> medals, because they do not expand[a] and in this way would break the mold. Animals, which expand, do not need oiling. The hot water much be of such a heat that you cannot keep your figure in it without it becoming too warm. Wet the thickest <m>sand</m> for flat and solid medals rather than [for] lizards and finer things. Once your sand has set, clean and scrape your mold on one side and the other, and on the bottom edge of the thrown side of the medal, make a notch so that it is easier to take out. Lower the mold,[b] scraping what surrounds the medal, so that on all sides you can take it out easily without it damaging the mold. Then make some gashes around the first mold used to throw so that the second time that you throw in it, it will be identical to the first one, and in the same way, make a notch at the bottom of the mold, where the held of the item cast lies. Also make some [gashes] in the sides of the cast. This done, wet the back of the first mold in <m>oiled water</m>, and after the second casting moisten it with warm <m>water</m> if you have molded your medal in a noyau mold.[c] Having done the first cast, and this one having set, cut around the edge of the medal with the point of a knife so that it comes out of the throw and is not buried in it. Then having made the second cast, and having opened your mold, take away the medal in one go, with two knife points, one on the side of the cast object and one on the side of the head, which is at the opposite end.</ab>

<ab>
<margin>left-middle</margin>
Take care not to oil your medal too much because if the seeps out, it prevents the <m>sand</m> from serving its purpose and coming together and renders it wavy and lumpy. You must only touch the middle of the medal with the tip of your oil brush and then smear the <m>oil</m> everywhere.</ab>

<ab>
<margin>left-bottom</margin>

Make especially sure that the throw is as wide as you can on the medal and holds it together well. The gate of your cast must always be big and always narrowing as you go towards the medal. Also do not forget to notch the gate of your cast.</ab>

<cont/>
</div>

[a]prester found in littré: V. n. En parlant des étoffes, du cuir, etc. s'étendre.
[b]This is not entirely clear. The "mold scraping" idea has been inserted above the line with no punctuation.
[c]There is an image of this on the page. It seems to be a certain kind/shape of mold?

<page>138r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f281.image</image>
<div>
<cont/>
<id>p137v_a1</id>

<ab>If you want to cast <m>gold</m>, <m>silver</m>, <m>copper</m> or <m>brass</m>, it is necessary that they be perfectly red, and fired on the inside when you cast, and perfectly reheated twice, if there is something inside to be burnt and cleaned.</ab>

<ab>The molds of animals that you burn must be reheated in such a manner that the animal burns up. But if it has big bones, it is frustrating to take out and <m>
<figure>✂</figure></m> often breaks delicate things because of its heaviness. One does not put <m><figure>✂</figure></m> in molds that can open. It is often molds for flowers that do not open because they are cast in one go.</ab>

<ab>When you want to reheat your molds, fix cramp-iron[a]s on the joints, so that when they reheat they will not bend, contract or break. This is done once the cast is made.</ab>

<ab>The scrapings of the mold can also be used instead of <m>bricks</m> after being reheated, and also the leftover bits of the mold that have already been used. You can also pat it over important things, like works in <m>gold</m> or <m>silver</m>. You can also reheat it and blend it with water diluted with <m>sal ammoniac</m>, like <m>spar</m> [b] from Germany, and is excellent <m>sand</m> for box molds for all metals.</ab>

<ab>The molds with things that need to be burnt out inside them, those ones will not open until the thing that is inside is burnt, like the molds for crayfish, crab, great-horned beetles. Do not open the molds in which you will burn something, not before the thing is burnt, it could be crayfishes, crabs, stag beetles, pictures, and pieces of wax and <m>black sulphur</m>, which do not come out of the mold [easily].</ab>

<ab>You must not mold on <m>brick</m> or <m>wood</m> because they absorb and attract water too early, and do not allow the sand to set. The best place for [the sand] is on a fresh layers of <m>clay</m>. And yet, I have experienced that <m>grey earth</m> dries out the mold too quickly. <m>Yellow [earth]</m> is better.</ab>

<ab>
<margin>left-middle</margin>
For medals, and flat things, the <m>sand</m> must be quite thick and moistened so that it sets quickly. And when the <m>sand</m> is thusly thick, you can shake and move the table where you mold lies, so that [the movement] makes the <m>sand</m> go everywhere. But when the <m>sand</m> is clear, as for flowers and herbs, you cannot shake it, nor can you do it when there is something attached to <m>wax</m> or other things which are prone to come off, such as the legs of crayfish or other similar things. And if by chance your <m>sand</m> is too thick, you must quickly add some water. And having put the <m>sand</m> in the water, examine it until it is thick on the bottom and clear on top. The clearest is thrown in immediately and then bubbles and the thickest is thrown in at the end so that it strengthens the mold.</ab>

</div>
[a]See Cotgrave: crampon
[b]In the Dictionnaire de l'académie: "Mot emprunté de l'Allemand, pour désigner une pierre feuilletée qui accompagne très-souvent les mines. Quelques-uns disent, Spar."

.....

<page>138v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f282.item.r=</image>

<div>
<id>p138v_1</id>
<head><m>Talcum</m> mixed with the molds</head>

<ab>I used the one from which I had extracted the <m>oil</m>. I crushed it very finely even more into a <m>steel</m> mortar with a pestle [which was] rough like a file. I crushed it very finely into a cottony, downy powder. I mixed it with the <m>sand</m> and soaked both matters together, and I casted it en noyau[a]. It molded very clean and there is no doubt that it will withstand fire.</ab>

</div>
<div>
<id>p138v_2</id>
<head><m>Gold</m> casted very thinly</head>

<ab>When goldsmiths have to solder something quickly but don't have enough time to forge it, they melt <m>gold</m> and then they pour it on a cloth or on anything else that withstands fire. Then they flatten it swiftly with a hammer or another similar tool, and it [the gold] is very tenuous and it even retains the impression of the cloth.</ab>

<ab>Unalloyed pure <m>gold</m> can be cast into a medal but not into herbs and lizards or other very delicate things if it is not alloyed.</ab>

<ab>A <m>pansy</m> which is <m>oiled</m> with <m>wheat oil</m> can be cast with alloyed <m>gold</m>.</ab>

<ab>And other fruit <m>leaves</m>[b], but with branch[es] that are not large and especially the leaves, once cast, can be soldered.</ab>

</div>
<div>
<id>p138v_3</id>
<head>Imitation <m>diamonds set into the work</m> [c]</head>

<ab>Spread a fine layer of imprinting black wax[d] on the inside of the frame [e]then coat it inside with <m>wheat oil</m>, then dust it with <m>lamp smoke</m> because this color should not be lustrous for fake stones. Having done this, set your stone with a piece of wax then, with a steel point, or a small finishing hammer, join the edge of the frame with the stone, so that light cannot get in but be careful not to hit the stone, which would break.</ab>
</div>
[a]No translation for "en noyau".
[b]"foeilles delices", not certain about this translation.
[c]See p_169r. "Mis en oeuvre" either means put into practice" or "put into the work". In this case it means "set into a frame", like setting a stone into a piece of jewelry
[d]suggestion: "black wax for modeling" (see notes Rowen/Landsman for translation 'esbaucher')
[e]"Chatton" Chaton, m. acut. Capsula. Est la petite chasse dans laquelle une pierre precieuse est enchassée, soit en anneau, ou autre joyau ou bague. Il semble estre sous-diminutif de chasse, Capsa, dont l'aisné diminutif est chasseton, Capsula

.....

<page>139r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f283.image</image>

<div>
<id>p139r_1</id>
<head>Casting with <m>lead</m> and <m>tin</m></head>

<ab>The way I have done it for lizards and serpents is with two ⅔ of <m>pure tin</m> for one lb. of <m>unrefined lead</m> that is impure[a]. The mold is made out of the aforementioned <m>sand</m>, common for all metals, when it is reheated, leave it to cool until you can put your finger in it without causing a breach in the gate of the cast. As for the <m>lead</m>, it is melted in a crucible with a bellows' fire until both the crucible and the <m>lead</m> are red. When it gets to that state, cleanse it anew with c<m>harcoal</m> or with a specially made scraper or with the puffs of the little bellows. This done, leave it to rest in it red state and you can even reheat it, then if you want, throw in a bit of <m>resin</m> to burn away the filth. However, some think it best not to put any in because it leaves filth. But when they are ready to cast, they cannot forget to also thrown in a bean-size quantity[b] of <m>bismuth</m> for each pound of <m>lead</m>. And it must be as red as the melted metal when it enters the mold. If the mold is large, it is much better to put in a press so that they join well and so that the <m>lead</m> does not come out of the mold. However, should this happen, and that for the first, second or third time your mold does not fill up, cast heartily because for as long as your metal is red it will mix and join with the other one and will come out very neatly like the main one. The same thing can be said for pure <m>tin</m> used for thin things. And that way for doing pure <m>tin</m> is one ⅓ of new <m>lead</m> for one lb. of the former.[c] Large molds must be put in a press held tightly between two layers of <m>copper</m>, and then bury the whole thing in <m>sand,</m> which is better than <m>ashes</m>, because being heavy, it will seal better. Otherwise large molds are prone to open because the metal is heavy. Some make square molds</ab>

<ab>
<margin>left-top</margin>
When the metal is thick, you do not need to cast as hot as when it is thin.</ab>

<ab>
<margin>top</margin>

When you melt this often, it becomes sour and fragile because it has to be cast very hot and it makes it half-calicanated. Use new stuff.</ab>

<margin>left-top</margin><figure>+</figure>

<ab>
<margin>left-top</margin>
Note that If you want to cast with <m>cuttlefish bone</m>, they cannot be too warm because that would burn the <m>cuttefish</m>. For this effect, try paper. It it becomes brown, it[d] is enough, and it is good for casting, but if it blackens the paper, then it is too hot.</ab>

<ab>
<margin>left-bottom</margin>
If you want to cast paper with writing on it, make a mixture with half <m>lead</m> and half <m>tin</m>. As soon as it is melted, throw it between two pieces of <m>cardboard</m> on a flat place that is level. And with a point or <m>gold</m> or <m>hard wood</m>, engrave on the left the writing that you want. And having poured the lead onto the <m>cardboard,</m> press onto it the other piece of <m>cardboard</m>.</ab>
<cont/>
</div>

[a]sophistiqué
[b]We couldn't find a definition for this word "ferve" but we think it may be some kind of quantity.
[c]I think he's referring to the tin.
[d]This means that it has become hot enough.

.....

<page>139v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f284.image</image>

<div>
<cont/>
<id>p139r_1</id>
<ab> are made of earth or <m>blades of copper</m>, or <m>iron</m>, or <m>wood</m> covered with <m>white iron</m>, in order to bury more easily these aforementioned molds between the thin sheets of <m>copper</m>, estrie[a] or one of <m>iron</m>.</ab>

</div>
<div>
<id>p139v_1</id>
<head>Casting <m>wax</m> to mold an animal that one has not got</head>

<ab>Take some <m>white wax</m> which is much more appropriate for this kind of work than anything else, because it is much firmer and does not leave as much filth, as much as you need to cast the animal that you propose, and no more. And [take] a half quantity of <m>ground coal</m> and neatly sieved through a cloth or coal sleeve, using it to give some color to your <m>wax</m>, that would otherwise be transparent and you will not be able to see your lines as clearly. Put your <m>wax</m> on the coal fire to melt. And when it is well-melted and well-liquified, take a full eared-porringer of melted wax, [and] as much <m>sulphur</m> as the amount of a large pulverized walnut. Melt all of over a slow fire and when it is melted, do not leave it on the fire because it will become too hot. But take it off and keep stirring it with a little stick and when it has finished bubbling and is as liquified as water, cast it into the <m>wax</m> that you will have previously removed from the fire. And mix and stir both of the them so that they join together. After stir in little by little while continuously mixing, the <m>charcoal</m> that has been repeatedly ground, and in this way it will be very well incorporated. This is how you will know that your <m>wax</m> has gone beyond its ideal heating point, it will release no more smoke, it will start to have lines appearing on the side and not in the middle, and those lines will be close to each other. If you cast too hot, you will not be able to separate your wax from your mold and it would stick to the cast. When it is at the right state, stir it with a little stick so that the pulverized <m>charcoal</m> is well mixed in and has not fallen to the bottom of the mixture. And in this way, throw it in your mold bit by bit and not in one go, because <m>wax</m> is not runny.</ab>

<ab>
<margi>left-top</margin>
This black sulphured <m>wax</m> is for fashioning[b] round figures that do not come out of the mold. And they need to be burned in the moule au noyau[c] rather than be opened like the ones that have something jutting out or an intertwining of legs and arms. And this <m>wax</m>, thanks to the <m>sulphur,</m> will melt with little heat and leave without leaving any filth. If by some misfortune the crushed <m>charcoal</m> remains in ashes, when you open the mold and blow on it, it will come clean.
</ab>

<ab>
<margin>left-bottom</margin>
To make wax serpents or other things to affix to candles, it is necessary to cast them with esbaucher[d] <m>wax</m> of all colors.</ab>
</div>
[a]We don't know what this is.
[b]suggestion: 'modeling' (see notes Rowen/Landsman on translation 'esbaucher')
[c]This is a certain type of mold that the author has drawn an image of on another page.
[d]This word refers to the kind of wax he just described in the text.

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<page>140r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f285.image</image>

<div>
<cont/>

<id>p139v_a1</id>
<ab>not like other things. And for this one, you can cast two or three times until your mold is full. Now, concerning this mold of pulverized <m>white plaster</m> & reheated in the manner of the <m>sand</m> from the preceding recipes, you should have made it long ago because it is used many times. But, before using it, soak it for a good hour in <m>cold water</m>, & at a minimum, at least as long in <m>hot water</m> that at first is so hot that you can't hold your finger in it. And not that it absorbs no more, but that it appears very wet overall without water seeping into it. In removing it closed from the <m>hot water</m>, cast your <m>wax</m> in such a state of heat as has been described. And neither the first nor the second casting will readily come out well, hardly, until the mold is soaked. Let it cool down before opening it so that the cast thing not break. You will know that the casting is good when the <m>wax</m> coming back out of the mold is thin and even. Remember to make several castings along the whole length of the mold so that in this way the <m>wax</m> runs better.</ab>

<ab>Make the first
<figure>
<id>fig_p140r_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5NFdsdU4tVFgxOTg</link>
</figure> casting twice as large as other molds. And if, in the first casting, your work fills with bubbles and in so doing does not come out neatly, it's all the same, because you have to face the fact that the three or four first do not readily come out well. Firstly, you will know whether there are a few barbs that keep it from stripping well. And you will remove them if, on their own, they do not remove themselves in the two or three first castings. And the more that you cast, the more you will do it neatly. And your mold will serve you more than one hundred times if it is well governed. But it is good to soak it one night or one day before casting so that it be well soaked. The same must be done for fruits made from sugar. This <m>wax</m> is very soft & friendly & pliant, like copper. And if it is hard [this is] because of <m>sulfur</m>, which makes it melt more easily than other [wax], so much that you can see evidence on a hot slate. And the <m>sulfur</m> that you put inside will be found the second time that you melt it, [as] cracks on the bottom. Having in this way passed through <m>wax</m>, it will not catch fire at all when put to a candle. And in this case, I believe that it will cast quite the medal [illegible]. One uses the same <m>wax</m> in place of <m>varnish</m> to [illegible].</ab>

<ab>
<margin>left-middle</margin>
When your animal is cast, cut away the broth & superfluous things with a hot pen knife. And if you want, plait and wrap it around some stick or candle, put it in some <m>hot water</m> to soften and hold it in turning it around.</ab>

<ab>
<margin>left-middle</margin>
Lower the protrusions of the castings so that they be even & that the <m>wax</m> has more silver so that it can run all in one go without turning through the windings of the snake.</ab>

<cont/>
</div>
.....

<page>140v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f286.item.r=</image>

<div>
<cont/>
<id>p139v_1</id>

<ab>
[f]a engrave on <m>silver</m> & <m>copper</m> with aqua fortis. With this [<m>wax</m>] too, one fills the cavity of a relief, & then casts in this cavity, with moistened <m>sand</m>, which immediately takes the relief very neatly. And then you can cast its cavity on it in <m>copper</m>, <m>gold</m>, and <m>silver</m>, and make really singular seals.</ab>

<ab>
<margin>left-top</margin>
Seals</ab>

</div>
<div>
<id>p140v_1</id>
<head>To cast in <m>sulfur</m></head>

<ab>
To cast neatly in <m>sulfur</m>, arrange the pith of <m>bread</m> under the brazier, as you know. Mold whatever you want into it & let it dry & you will have very neat work.</ab>

<ab>
<margin>left-top</margin>

Try <m>sulfur</m> passed through melted <m>wax</m>, because it will no longer ignite & and make eyelets.</ab>

</div>
<div>
<id>p140v_2</id>
<head>Molding and shrinking a large shape</head>

<ab>
Mold it with the pith of bread just out of the oven, or like that aforementioned, & in drying out it will shrink & consequently so will the medal that you will cast. By these means - lengthening out or enlarging the imprinted bread - you can vary the shape & from one face make several different ones. The <m>bread</m> straight from the oven is best. And the one which has been heated twice contracts more. You can cast <m>sulfur</m> without letting the imprint on the <m>bread</m> dry, if you want to cast it as large as it is. But, if you want to let it shrink, let it dry to a greater or lesser extent.</ab>

</div>
<div>
<id>p140v_3</id>
<head>Casting of <m>lead</m> and <m>tin</m> in <m>plaster</m></head>

<ab>
If you want to cast any flat medal or any animal that is not very heavy, and that must not be burnt inside the mold**[b]**, you can indeed cast in mixed <m>plaster</m> & <m>brique</m>, as above; and not in <m>plaster</m> alone, because it contracts too much feeling an ardent fire if it is not so mixed. But, with the <m>brick</m>, it holds well. However, take heed to dry out your mold at length & on a slow fire & with patience, because there is no need to reheat it. But when your work is of <m>flowers</m> or other things that want their mold to be reheated & set ablaze, mix the plaster with <m>stone alum</m> & even with <m>crocum</m>**[c]**. I have molded in very neat <m>plaster</m> & <m>brick</m> & it has sustained several castings.</ab>

<ab>
<margin>left-middle</margin>

I tried both <m>plaster</m> & <m>brick</m> alone and molded <fr>en noyau</fr> with like the others. My mold was very clean, having lightly <m>oiled</m> & rubbed my medal with <m>spirits</m>. I made my casting wide at the entrance, narrowing it as it goes along until it reaches the medal which is very thin. I notched the casting which embraced the medal well. I dried the mold out well on a slow fire & finally, I heated it well without turning it red. I let it cool in such a way that I could hold my finger to it without burning myself. I made a line of 4 $\frac{2}{3}$ of <m>tin</m>, & ix deniers of <m>lead</m>. I cast it red, and it was quite good and beautiful. Afterwards I put xii deniers of <m>lead</m> on top of 4 $\frac{2}{3}$ of <m>tin</m>. It is very good & beautiful.</ab>

<ab>
<margin>left-bottom</margin>

When there is nothing to burn in the mold, it is not necessary to reheat it for <m>lead</m> and <m>tin</m>. But for <m>flowers</m> and other things that must be burned, it is.</ab>
</div>

[a]This page has been checked, with the exception of this recipe stub, which continues from 139v-140r.
[b]Taking "molle" as a misspelling of "moule"
[c]translates "crocum," from "crocum ferri" = iron filings

.....

<page>141r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f287.image</image>

<div>
<id>p141r_1</id>
<head>Casting a <al>crayfish</al></head>

<ab>The<al>crayfish</al> is one of the most fantastical**[a]** to cast, but also provides an example for casting other difficult things. You can distinguish males and female by the eggs that the females carry, and by the four little white back legs that males have in their tails after behind main eight legs. It is good to dry them out a bit so that the bumps come out more hard and beautiful, because none of what is made of shell shrinks. It is true that should you dry the little inside legs for too long, they become diminished and have no body. If they are too dry, the flesh separates from the scales. Be sure then to maintain the mean.**[b]** They have certain hairs between the legs and at the end of the tail. And because all hair is difficult to cast, because it gets mixed up in the sand and does not release, you can burn it ^{by putting} a hot iron on the <al>crayfish</al> so that it ^{the hair} cannot be seen. As for animals that have hair or feathers, you must grease the hair with <m>wheat oil</m>, which is very dry and which will make the hair lay flat and straight. In this way, you will have form of your animal and you will have the hair. Should it ^{the hair} be wavy, it can be repaired. If your <al>crayfish</al> has eggs, which are delicate, and which will be difficult to take out of the mold without breaking, you would do better to make the first cast ^{with the crayfish} on its belly and eggs. In this way, you will open the mold with it ^{the crayfish} on its back, which will be hard and easy to manipulate. And the belly and the eggs will remain in the mold. And rather than opening your mold for the second time, you will have to burn it with them in it. This way, it will open easily, and what has been burned will come away cleanly. In this way, the second casting, which is done on the first one that has successfully taken hold, separates easily from the first, after having been refired. This is a singular secret for similar moldings.</ab>

<ab><margin>left-top</margin>The casting is made by the very thin tail.</ab>

<ab><margin>top</margin>If some little thing is missing, you can reattach it or else fill it with <m><pro>goldsmith</pro>'s cement**[c]**</m>.</ab>

<ab><margin>left-top</margin>Do not forget to rub with <m>spirits</m> before you mold.</ab>

<ab><margin>left-top</margin><al>Stag beetles</al>, <al>crayfish</al> and <al>crabs</al> are molded in the same way.</ab>

<ab><margin>left-middle</margin>If your <al>crayfish</al> has no eggs, mold the back on top and the belly on the bottom. You can give it some.</ab>

<ab><margin>left-middle</margin>It ^{the crayfish} can be molded hollow for the body, but not for the legs. And for a good result, ^{take note that} the <al>crayfish</al> is rather fantastical to mold hollow, this ought to be reserved for <al>turtles</al>.</ab>

<ab><margin>left-bottom</margin>When you open your mold, you will find the <al>crayfish</al> has white bones, but they will not be powder. And without opening it, the <m>mercury</m> will not have affected the process.</ab>

<ab><margin>bottom</margin>Apply very light oil colors.</ab>

<ab><margin>bottom</margin>To paint it, boil it with <m>wine</m> and a bit of <m>salt</m> so that they become very red and use this[d] as a model. Paint the back with <m>vermillion</m> mixed with <m>laquer</m>, and the sides and the underside of the belly and the legs with <m>vermillion</m> and <m>yellow ochre</m> and <m>white</m>.</ab>

<cont/>
</div>

[a]Fantasque could be translated also as new-fangled, but this carries important implications that we at this time are not comfortable making.
[b]We think this means the average time so that you are not drying them for too short or too long a time, but for an amount of time in between.
[c]At the end of this note there are several words hidden underneath the paper strip.
[d]Transcription and translation uncertain, but makes sense in the context.

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<page>141v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f288.image</image>

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<cont/>
<id>p141r_1</id>

<ab>Let your <al>crayfish</al> dry a little in the sun, by itself. If it has spawns, they will shrink while drying, and be all the more beautiful for it. Make your paste out of yellow <m><pro>potter</pro>'s earth</m>, just like for the other representations[a]. Lay your <al>crayfish</al> on it, the back side down, and the other parts which are more tricky to mold -- legs, belly, eggs - side up. Drive in the back side in the <m>clay</m> paste up to the legs, which is about the half part ^{of the crayfish's body}. Fix the body with a pointy <m>iron</m> thread in the middle and, if you feel you need it, you can also drive another one at the edge of the tail. And in order that the big legs be lower than the head, which is lower because it is linked from below, add a little <m>clay</m> ^{under the head}. Then add a bit of <m>clay</m> under the jacket in order to raise it. Hide the feelers under the <m>clay</m> and under the crusher claws, then arrange it as you like. Stretch out the walking legs to the joints, and for the first cast, bend the second part of the walking legs. Fix the joints of the legs with <m>wax</m> and a warm <m>wire</m> so they will not move. If you want to mould something fancier, fix the end of one walking legs on the body or on a crusher claw , using the same <m>wax</m>. If the female has spawns, bent half the tail over the spawns, and keep the shape by fixing the tail with a <m>wire</m>. Be sure that you will be able to clean this part. Finally rub your <al>crayfish</al> with <m>spirits</m>, and cast your sand. Once the sand is hardened, uncover the back side, the head, the eyes, and the small legs close to the crusher claws, and the walking legs all along. One must uncover these parts as much as possible. Then bend and cover the spawns again. The other animals</ab>

<ab><margin>left-top</margin>Fix what may raise ^{the two padded edges on each side of the mouth} with melted <m>wax</m> under them.</ab>

<figure>
<id>fig_p141v_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdIO5R0NIbHoyNTNQRIk</link>
<margin>left-top</margin>
</figure>

<ab><margin>left-middle</margin>Uncover as many parts as possible, but be sure the channel[b]s you make are well stripped.</ab>

<ab><margin>left-middle</margin>Inline your mould to the thickest side of the animal.</ab>

<ab><margin>left-bottom</margin>To paint the crayfish, one paints the middle of the back with <m>vermilion</m><m>,</m> mixed with a bit of <m>lacquer</m>. The sides, the belly, and below the legs with a mixture made of <m>vermilion</m>, <m>ceruse</m>, and a bit of yellow <m>ochre</m>. As with all things, always keep the real one in front of you in order to copy it ^{as realistically as possible}.</ab>

<ab><margin>left-bottom</margin>Lay the feelers on the crusher claws, or solder this part with a wire made of bleached <m>brass</m>.</ab>

<ab><margin>bottom</margin>When you arrange the legs, be sure that they do not get over the belly, and that they are well set against the belly otherwise [illegible]
</ab>

<cont/>
</div>
[a]figures
[b]encochement

.....

<page>142r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f289.image</image>

<div>
<cont/>
<id>p141r_1</id>

<ab>are not so uncovered. But since <al>crayfish</al> are troublesome to burn & clean, you have to uncover them as much as you can, even the little legs just until the tips, because they are troublesome & if, in order to not alter them, you have to take them out, you could break something of the mold. Having uncovered it with the point of a knife, like an engraving stick or another fitting thing, clean off the sand at the joints & elsewhere with your little brush really very scrupulously. And if something is loosed from the <al>crayfish</al>, attach it with <m>wax</m>, as is discussed. And also attach the horns this way. And also do[a] the casting the length of the extremes of the little legs with <m>wax</m>. And refill the holes that have been made with the needles of <m>iron</m> wire with it. And everything being quite clean & having uncovered more than half of it, <m>oil</m> your mold after having soaked the back side in <m>water</m>, & do not forget to <m>oil</m> all the fine parts that are

between the legs and the little beaks; then rub the `<al>crayfish</al>` with `<m>spirits</m>`. And do your second casting after having made the first mold even, so that the cramp-irons join better. Your mold must be larger on the side with the impression than on the back. Do not open your mold after the second casting, so that it not become `<m>clayed</m>` & reheated & that which is inside not be burned. Do not forget to mix `<m>crocum</m>` throughout your sand when you mold the `<al>crayfish</al>`, because you must really reheat it & the `<m>crocum</m>` withstands this marvelously. To mold a `<al>crab</al>` & to mold a `<al>crayfish</al>` are exactly the same. When you have uncovered your `<al>crayfish</al>` on one side do not delay at all before doing the second casting, because `<al>crayfish</al>` dry out. It is not enough to uncover the `<al>crayfish</al>` such that you see the whole half `<sup>the half you uncover</sup>`, but take heed that your mold, even when you are uncovering, be good and stripped, because if the `<al>crayfish</al>` gets well burnt, you will not be able to open your mold without breaking it.`</ab>`

`<ab><margin>top</margin>`Restore it with a pen knife, little chisels, or engraving sticks.`</ab>`

`<ab><margin>left-top</margin>`As you see here, but a `<m>wax</m>` casting all around & at the end of the tail, where you will do the principal casting, two or three more.`</ab>`

`<figure>`
`<id>fig_p142r_1</id>`
`<link>https://drive.google.com/open?id=0B9-oNrvWdlO5Qkh2MDhvUkFnUW8</link>`
`<margin>left-middle</margin>`
`</figure>`

`<ab><margin>left-middle</margin>`Make the casting with `<m>candle wax</m>` `<sup>shaped</sup>` in a thread like a large packthread, the whole length from the end of the claws to the extremity of the tail. If there is also some claw end or other part that is extended past the rank of the others or is raised up higher or turned up all alone, make `[b]` a `<m>wax</m>` casting for it from its end just barely not joined to the body or to one of the large claws or to some other place where lots of `<m>metal</m>` will be wedged in `[c]`.`</ab>`

`<ab><margin>left-bottom</margin>`Having uncovered it, attach & secure the two little bearded horns of the crayfish, and anything else that is not secured, with `<m>wax</m>`.`</ab>`
`</div>`

`[a]`"fais" but probably take as "attach" by analogy with previous sentence?
`[b]`Fr. "donne"
`[c]`"...endroit qui recoine force metal"

.....

`<page>142v</page>`
`<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f290.image</image>`

`<div>`
`<id>p142v_1</id>`

`<head>`Molding `<al>grasshoppers</al>` and things too thin`</head>`

`<ab>`If you have to mold a piece of written-on `<m>paper</m>` that is too thin, after you have done the first cast and it has set, give a little thickness to the reverse of your `<m>paper</m>` with melted `<m>butter</m>`, which is the most appropriate means there is, & for strengthening the wings of either a `<al>butterfly</al>` or a `<al>grasshopper</al>`, or any delicate part of an animal to which you need to give thickness. But take heed to apply this melted `<m>butter</m>` under the wing or in whichever place it will not be seen. For giving thickness to a pansy or other flowers, `<m>butter</m>` is not good, but rather `<m>wheat oil</m>`, which is soon dry & holds firm. `<m>Wax</m>` would not be appropriate here for it is too hot once melted, and it makes the thing to which it is applied contract. But `<m>butter</m>` is amiable and handleable.`</ab>`

`<ab><margin>left-top</margin>`If you write on `<m>paper</m>` or on common `<m>carton[a]</m>` and your letters are made with `<m>gum</m>`, the moisture of the `<m>clay</m>` plate (?) or the wet (?) sand for a noyau will moisten your letters `<sup>and</sup>` undo them. Write therefore with `<m>cinnabar</m>` mixed with `<m>oil</m>`, on oiled & stamped (?) `<m>paper[b]</m>`.`</ab>`

`<ab><margin>left-middle</margin>``<figure>+</figure>`Reheat your molds with `<m>charcoal</m>` firstly lit in the forge, so that the fire is not too hot and does not break the molds. And do not fire as strongly where the mold is thin as where it is thick.`</ab>`
`</div>`

`<div>`
`<id>p142v_2</id>`
`<head>`Molds`</head>`

`<ab>`Make some notches in them `<sup>the molds</sup>` that enter more into the inside of the mold than the outside because in this way, they have more strength. Take care to keep from reheating them suddenly in a burning hot fire, because this will make them break.`</ab>`

`<ab><margin>right-middle</margin>`Do not keep them `<sup>molds</sup>` in a humid place, nor an enclosed place if they are not well dry, because they will mold. The same with dried animals.`</ab>`

`<ab><margin>left-middle</margin>`It is not at all necessary to reheat the molds two times when the animal can pull away without burning, as do `<al>toads</al>`, which can very well be molded hollow like all thick animals. However, it is always good to redden the mold once.`</ab>`
`</div>`

`<div>`

<id>p142v_3</id>
<head><m>Lute</m> for <m>luting</m> your molds</head>

<ab>I have not found a ^{lute} that is made more quickly than this one, nor a better one. Take some of the lean <m>earth</m> that <pro>artillery founders</pro> and <pro>bell founders</pro> use to make their trusseaulx [c]and molds, which is lean and sandy. Soak it moderately like a very thick <m>mortar</m>. Mix into it about one half of <m>horse manure</m> and then beat it well. Afterwards mix in one third of <m>cloth waste</m> or <m>cloth shavings</m> and beat it again quite strongly. You can reheat your mold as soon as the <m>lute</m> is ready.</ab>
</div>

[a]in the sense of pasteboard thick paper?

[b]check annotation

[c]Another kind of mold. See "tasseau" in Encyclopédie, ou dictionnaire raisonné des sciences, des arts et ..., Volume 32, pps 725-726.

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<page>143r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f291.image</image>

<div>
<id>p143r_1</id>
<head>Moulding <al>turtles</al> and <al>tortoises[a]</al></head>

<ab>The <al>turtle</al> ^{the one that lives in water} is more beautiful to mold because it has prettier scales and straight legs. And the <al>tortoises</al> have crooked ones.[b] They eat a lot.[c] To get ready to kill them, open their mouth and pass a long penknife through all the intestines. And then make it swallow some <m>vinegar</m> mixed with <m>spirits</m> or <m>urine</m>, as one does for snakes. It is necessary to mould them in several pieces, and even the back sometimes because the sides of certain ones are narrower than others and in this instance do not release ^{from the cast}. To make them die, you must not boil them, because they come apart, even the shell, in the hot <m>water</m>. They ^{the shells} come out better ^{when} hollowed because their great thickness spoils the mold. <al>Turtles</al> are more lively. The <al>tortoises</al> are leaner, slower, and more sleepy. If you scratch <al>turtles</al> on their shells, no matter how sleepy they are, they move; <al>tortoises</al> do not.</ab>

<ab><margin>left-top</margin>They ^{<al>turtles/tortoises</al>} die rather quickly in <m>vinegar</m> mixed with a little <m>spirits</m> or <m>urine</m>, as with all other animals.</ab>

<ab><margin>left-top</margin>They ^{<al>turtles/tortoises</al>} keep their eyes closed in the winter, and appear as dead, having been numbed by the cold. They hide in the earth or under balls of grain [d]or warm manure, and survive only on the dampness of the <m>earth</m>.</ab>

<ab><margin>left-middle</margin>You must not mold them too soon after they die because they are still stiff. But on the day after, you will be able to manipulate them and bend their legs as you wish.</ab>
</div>

<div>
<id>p143r_2</id>
<head><pro>Moulders</pro> from <pl>Foix</pl></head>

<ab>Those who mould from across ^{the way} [e]make their sand from <m>crocum</m> ferri and <m>calcinated slate</m>, but <m>slate</m> always retains its bumpiness and swells because it is a fatty. In the noyau, it is not good; in sand, it can be put to use. They ^{the moulders from <pl>Foix</pl>} sieve their sand, grind it on <m>porphyry</m>, and soak it in <m>water</m>, and they keep the finest bits which are on the top, then they reheat it.</ab>
</div>

<div>
<id>p143r_3</id>
<head><al>Toad</al></head>

<ab>Having been molded, you can take it out it without burning it, because it releases ^{easily} from the mold. Choose the biggest and the lumpiest ^{<al>toads</al>}. Because they contain a lot of metal, it is best to mold them hollow, since they turn out better. You could make your cast in one great piece, but it would be troublesome to cut. It is best to do it in three or 4 parts, which will be[f][g] thicker close to the animal, and include the most amount of edge that is close to the cast/mold. Make as well some casting conduits all the way from the end ^{of the cast}.</ab>[h][i]

<ab><margin>left-bottom</margin>
<figure>
<id>fig_p143r_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5cnJFVXFuRURaY3c</link>
</figure>
Let the mold cool rather than opening it, even for thick things.</ab>

<cont/>
</div>

[a]I've put both here because in the paragraph, he distinguishes between tortoises (those that live on the garrigue) and turtles (those that live in water.)

[b]Meaning ... legs I think.

[c]This is a translation I found in an 1840 dictionary. I'm not sure if he means "They live a long time, or are lively or eat a lot?"

[d]I think these are like hay bales?

[e]Unsure as to whether I have the correct meaning here...
[f]I'm not sure this is the best translation here as I don't entirely understand the technique.
[g]Soersha agrees with this translation. June 2015.
[h]I'm not sure this is the best translation here as I don't entirely understand the technique.
[i]Soersha agrees with this translation. June 2015.

<page>143v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f292.image</image>

<div>
<cont/>
<id>p143r_3</id>

<ab>from one leg to another, so that the spread out limbs are in connection with one another due to the aforesaid casting conduits, which you will start in the waxless round parts that you have put at the edge of the legs to attract the metal. In this way, because you are casting in <m>wax</m>, there is no danger chipping any part of the molded animal. It's quicker to make the <m>wax</m> casting conduits thus, because in this way ^{you can} cast the second mold after ^{the first}, rather than wait to do them ^{the conduits} after the mold has been recooked, because then, if possible, you will need to have everything ready. You need to heat well, and even redden the molds where there are <m>wax</m> conduits so that they melt and leave nothing inside.</ab></div>

<div>
<id>p143v_1</id>
<head>Openwork carcanets</head>

<ab>You can mold them "en noyau" like <al>crayfish</al>, first with one side being made higher with <m>clay</m>, which is hollow, and then you open ^{it} from the other side and make the second casting. You can just as well cast ^{them} "en chassis", provided that they release well. If they do not, you can do a rough cast in <m>wax</m>, or you can fill up the cavities which do not release well with <m>wax</m> or <m>clay</m>.</ab>
</div>

<div>
<id>p143v_2</id>

<head><m>Iron</m> filings</head>

<ab>Because usually <m>filings</m> are mixed with impurities, it is good to heat them over fire to burn the filth and then wash them in clear <m>water</m>. In this way, the dirt will rise to the top of the <m>water</m>, which you will throw out, and the good filings will go at the bottom.</ab></div>

<div>
<id>p143v_3</id>
<head>Carnations</head>

<ab>Because those that you usually cast are generally quite big, they are heavy. And so for these, you make them with <m>silver</m> leaf or slivers.</ab></div>

<page>144r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f293.image</image>

<div>
<id>p144r_1</id>
<head><al>Turtles</al></head>

<ab>Turtles are much more beautiful and easier to mould than tortoises, being that they are flatter and have a longer head and tail, and straighter legs and release better than tortoises, who are scraggly[a], wrinkled, and deeply entrenched in their shells. They ^{tortoises} also have a rounder top shell which is however narrower on the sides than in the middle, which means that the top shell cannot be nicely molded in one go, like those ^{top shells} of turtles. Thus you will mould the top shell into two pieces, and the lower shell into two, three, or four pieces according to their appropriateness. If you want to release your turtle from the mold without burning it, you must mold it in several pieces. But if you want to burn what is inside, you can mold your top shell in one piece. So to get started, once you have completely killed it in vinegar and urine, as we know, leave it to fully dry out. And clean it well with a cloth. Then take your clay base place it on it, and scrape the base a bit so that the undershell fixes in. Having nevertheless previously arranged the head with a strong needle[b], as you do with other animals, holding it high by means of a bit of fresh clay that you will work with from beneath. Stretch the aforesaid head and legs with your little pincers. Once the head is in place, dextrously put a grain of yellow millet into each eye with the pincers, because as soon as they die, the eyes pop out and rot. You can do the same for any other animal with a large amaranth[c] seed, a small one, or a grain of rapeseed. This done, you will arrange the legs, securing them with little iron needles. And then plug with clay the holes all the empty spaces around the turtle, that is to say, what is in between the two shells, so that the sand does not hinder the releasing of it. And because the legs must be lower than the bottom shell, make a little dimple in your clay base to accommodate them. Your turtle</ab>

<ab><margin>left-top</margin>
When they are the well tender and mortifees & left aside, in two or three days you will be able to bend and manipulate them better.</ab>

<figure>
<id>fig_p144r_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5emg1MXZLREtSRk0</link>
<margin>left-middle</margin>
</figure>

<ab><margin>left-bottom</margin>
animal eyes of my own invention</ab>

<figure>
<id>fig_p144r_2</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5ZkN6Y3ZTM1pKRG8></link>
<margin>left-bottom</margin>
</figure>
<cont/>
</div>

[a]He uses *maigre* in French that carries a pejorative slant, perhaps best translated as scraggly, rather than lean or thin.
[b]This is a *pointine* in French, something like a piece of thin iron that can be used to support parts of the body.
[c]The reference can be found in *Classique Garnier, Dictionnaire La Curne*.

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<page>144v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f294.image></image>

<div>
<cont/>
<id>p144r_1</id>
<ab>having been thusly arranged and well cleaned with brushes, put the circle of <m>clay</m> around the base. Then, with a roll of <m>clay</m> placed on the middle of the back of the <al>turtle</al>, as you can see, and at the widest place, divide your mold. Next, <m>oil</m> with your <m>oiling</m> brush half the half of the shell you want to mold. If you sand is a bit thick, temper your sand with <m>warm water</m>, so that without burning yourself you can put your finger in it without burning it. And before casting it, rub the shell with <m>spirits</m>. And then cast your sand and let it take hold. And note that all things that have shells, or that are hard, or do not yield as <al>snakes</al> and <al>lizards</al> do, must be <m>oiled</m> to come out cleanly. The things that yield do not need this. <m>Warm water</m> means that the tempered sand sits better on the <m>oil</m>, which otherwise would not work. With the half <al>turtle</al> having been thusly molded, separate it with its mold from its <m>clay</m> base, which you will put aside to use when needed. Then clean your half-mold, join it and flatten it like for the others. But because there will be the roll of <m>clay</m>, the half mold will have more than just half the <al>turtle</al> in it, cut and crop cleanly the excess ^{clay} and clean everything well with the brushes. Then at the top of the edge of the mold on the side where it is cut, at the halfway mark, make two notches as you have done with other molds. And flip the <al>turtle</al> on its base, as it was, and secure the two legs that are not molded with <m>iron</m> pins and fill the empty areas around it, up to the edge of the top shell with some <m>clay</m>. Then place the circle around it and place a roll of <m>clay</m> on the first mold, a bit above the notches. And having <m>oiled</m> the first half-mold and its notches and the shell of the <al>turtle</al>, and having also sprinkled it with <m>spirits</m>, warm your <m>water</m>, temper your sand with a bit of <m>salt water</m>, <m>ammoniac</m>, and the aforesaid <m>hot water</m>, make your cast. Having taken hold, keep the sides of the two molds neatly together, and for each side secure the joints, which are not prone to do so naturally, with two clamps, so that when you uncover the bottom of the <al>turtle</al> the top joint does not open. Then open the side with the belly and the throat, in the same manner that you would open a <al>crayfish</al>, which is the hardest animal to mold of all animals that are molded in two parts.</ab>

<figure>
<id>fig_p144v_1</id>
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<margin>left-top</margin>
</figure>

<figure>
<id>fig_p144v_2</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5SIhGM2FYLUUteXc></link>
<margin>left-middle</margin>
</figure>

<ab><margin>left-bottom</margin>You can also mold your <al>turtle</al> in one piece for each side, but you will need to burn it. And then to make the hollow, the recooked mold cannot be well-<m>oiled</m> because it absorbs the <m>oil</m>.</ab>

<ab><margin>left-bottom</margin><figure>+</figure>
If there is <m>crocum</m> in your sand, the joints of the molds will not join well together. </ab>

<ab><margin>bottom</margin>
Turn to the second leaf.</ab>
</div>

.....
<page>145r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f295.image></image>

<div>
<id>p145r_1</id>
<head>Casting in three frames</head>

<ab>You can cast <m>bronze</m> mortars and similar things which release well in three frames. That is to say the body of the mortar in one, its noyau in the other, and the base of the mortar in the third one, because otherwise the cast which is there will not release well.</ab>

<ab><margin>top-left</margin>Inquire
<figure>
<id>fig_p145r_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5MXRaSldvRzY2UTg></link>

</figure>
A</ab>
</div>

<div>
<id>p145r_2</id>
<head><m>Cuttlefish bone</m></head>

<ab>You must not the the <m>tin</m> or <m>lead</m> too hotly, otherwise they will burn the <m>bone</m> and become lumpy. And to know when it is the right temperature, dip in it a little piece of twisted <m>paper</m>. If it turns black without catching fire, it is the right temperature. But if it burns & catches fire, it is too hot. You can cast <m>gold</m> & <m>silver</m>, but it never comes out neatly. To cast something delicate well, it is necessary that the <m>bone</m> not be extremely dry, because it is rough & does not release as well & crumbles & is brittle. In any case, before you cast, dry them ^{the <m>bones</m>} & especially for <m>gold</m>, which does not react well to humidity. You will know that they are dry enough when, after having exposed the insides and the imprint to the fire, they hiss & crackle when you bring them near your ear. Now join the <m>bone</m>s & daub the joints with a little <m>lute</m> and slowly dry it near the fire & cast & then shake the mold or scratch the rough crannies and let it cool down before opening it. Usually one cuts the <m>bone</m> in the middle, and the dull part that does not have half circles <figure/> is more delicate and smoother to mold with, and therefore one always uses it to do the main part of the casting, the other ^{part of the <m>cuttlefish bone</m>} is crumbly on the inside as on the outside. Therefore one does not use this to cast delicate things which need to be molded in two places. One evens out & flattens these two halves on some piece of wood, then one rubs some <m>charcoal</m> on top sot that it releases well. And to make sure the <m>charcoal</m> is evenly spread everywhere, tap the side of the hand which is holding the half <m>bone</m>. Both having been covered in <m>charcoal</m>, take the first <m>bone</m> that has been prepared & rounded on the edges, and having placed the medal above it, press the medal down strongly.</ab>

<cont/>
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<page>145v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f296.image</image>

<div>
<cont/>
<id>p145r_2</id>

<ab>But with fingers you cannot press evenly, because if you press on the edges, the middle will remain hollow. Start therefore with the middle, & then do the edges. But to do it better, put something flat & even on the medal, or some large square file, & press with that; because you will press evenly, sometimes with your knee, other times with your foot with your shoe taken off. & do it from above, so that you impress it without breaking it. If it is not well molded on the first go, do it again several times. Then restore and flatten your halves of molded bone from the sides. And to cut them well, always start from the most tractable part, coming out to the shell. And if your medal does not come out on its own, scratch the rough bone from behind and it will strip. When you want to cast, settle & affix the two bones with little pins of wood & in order to cast neatly, you have only to try it with sand.</ab>

</div>

<div>
<id>p145v_1</id>
<head>Moulding herbs and flowers</head>

<ab>You do not need thin sheets of <m>clay</m>, because the herbs and flowers will not lay on it, but as it were in the air, not touching any thing from any side, so you only have need of a circle & contour that is higher than for flat molds. And in this way, take heed to make it strong and thick, with the heft that you find suitable. Otherwise, it will sink in when the sand is inside. Fix & fortify it well with your foot & firmly attach all the joints. Then, have your flower well attached & assuredly fixed to the bottom of the casting with <m>wax</m> that is not at all rough, but smooth, so that it can strip off well. Then moisten your flower or herb in good <m>spirits</m> in a tall glass.</ab>

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<div>
<id>p145v_2</id>
<margin>left-top</margin>
<figure>+</figure>
<head>Flowers</head>

<ab>
When you mold them, if they are not strong enough to hold themselves upright, run a thread through the mold to keep them lest they do not hoist themselves up. & cast the sand little by little & continuously and blow on it strongly and constantly so that it covers everywhere. Otherwise, it will become lumpy. Since you won't put anything in thin sheets of <m>clay</m>, the mold sometimes sticks to the table on which you mold. To release it, strike a great blow on the side of the table with a hammer.</ab>

<figure>
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<link>https://drive.google.com/open?id=0B9-oNrvWdlO5b1FOQ1Z5b1J3TWc</link>
<margin>left-bottom</margin>
</figure>

<ab><margin>left-bottom</margin>

It's enough to reheat your clayed mold once for flowers & to heat it the second time if you cast in <m>tin</m> & <m>lead</m>, insomuch as[a] for <m>gold</m> & <m>silver</m>, you have to reheat it twice.</ab>

<ab><margin>bottom</margin>

When very red, cast your tin into the mold when it is so hot that you can[not?] hold your finger in the hole without [illeg].</ab>

<cont/>

</div>

[a]Taking "car" concessive, contrasting tin/lead with gold/silver and using "car" to explain why this specification is here necessary.

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<page>146r</page>

<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f297.image</image>

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<cont/>

<id>p145v_2</id>

<ab>so that it is wet everywhere. If not, wet it with a large brush dedicated to this. Then pass the end of an <m>iron</m> wire that holds the <m>wax</m> casting through the end of the mold that closes the circle and have the <m>iron</m> cleave the <m>clay</m> mold as you see; & position it on the bottom, attending that the herb or flower does not touch anything, and leave some thickness to your mold, because the moistened sand always elevates the herb or flower. Even so, if it isn't thick of its own accord, pass a slender thread through the flower with a needle to keep it elevated. Or, if you forget this, lower the flower with a little stick just until the cast sand starts to thicken. With your flower well-positioned, take a platter large enough to moisten all the sand that you will need to fill the mold. Put a little <m>sal ammoniac water</m> in it, and then some <m>spring water</m>, and when your platter is almost full, put a sprinkle of your sand on it, & mix it, & steep it until it becomes totally uniform, because if it were to clot it would consume the mold. For flowers it does not need to be very thick. And likewise, at the beginning, when you just barely pour it so it remains transparent on your flower & covers it sort of half way, blow strongly everywhere so that you eliminate the little bubbles, and afterwards finish filling it & blow, always slightly tilting your mold down towards its larger side. And if you find some remainder of thick sand that was not well soaked, set it rather close to the casting than otherwise. Then you can cast some thicker moistened sand to make it set sooner. I molded a marigold, with its leaves, in this manner. <m>Crocum ferri</m> is safer for flowers, and when there is some <m>crocum</m>, clay it with the same sand that has already been used and is the most excellent of all.</ab>

</div>

<ab><margin>left-top</margin>

The stem on which I cast a large branch of marigold with its flower, its buds and its leaves came out neatly, just like nature, was made of one lb. of fine <m>tin</m> mixed with two ounces of <m>lead</m>.</ab>

<ab><margin>left-top</margin>

If there are a few froths, fix it with a pen knife.</ab>

<ab><margin>left-top</margin>

Uncover the molded flower while gently breaking the mold with the point of a knife. And, even better, soak it well in <m>water</m>. Afterwards, try a pig brush[a][b] at the end of an <m>iron</m> wire, etc.</ab>

<ab><margin>left-middle</margin>

When the mold is reheated for the first time, leave it to cool down halfway. Then, run an <m>iron</m> wire gentle through the casting to make an opening for the burned ash inside. Afterwards, blow inside with a bellows and turn the mold over on the casting to make everything leave, & sometimes blow[c] and suck in with your mouth.</ab>

<ab><margin>left-bottom</margin>

Heed that you not attach the stem of the flower too firmly to the casting with <m>wax</m>, lest it consume something when you remove the <m>wax</m> from the casting. To remove it, you have to loosen it a bit all around & then draw it out by the <m>iron</m> wire with little pincers[d].</ab>

[a]Translates "sayet de porceau"; s.v. "saie," 2nd definition, at <http://www.cnrtl.fr/definition/saie>

[b]probably a brush made with pig bristles

[c]translates "suppe"; see "super," 1st (verbal) definition, at <http://www.cnrtl.fr/definition/super>

[d]s.v. "molet" at <http://www.cnrtl.fr/definition/molet>

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<page>146v</page>

<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f298.image</image>

<div>

<id>p146v_1</id>

<head>Molding<al>turtles</al></head>

<ab>This page is the continuation of the preceding second page. Since you have cleanly uncovered the part of the belly of <al>turtle</al>, its underparts, its throat, and its legs, and having cleaned it all well, do not forget to neatly uncover the cavity that these animals have in their shell between the neck and the shoulders. <al>Turtles</al> do not have as deep of a cavity as <al>tortoises</al>, which are more scraggly, and even amongst those, there are some that have a deeper cavity than others. Do not therefore forget to neatly uncover all of this in order to better represent its likeness. But if there is a cavity which spirals too much inside, and is not visible, nor releases well, apply some small lumps of <m>wax</m> and with heated needles of thick <m>iron</m> and work the aforesaid <m>wax</m>. Then plug the entrance to all of these cavities with <m>soft clay</m>. Put the circle of <m>clay</m> around the mold, which has already been made of two pieces and contains the back of the <al>turtle</al>. And to mold its belly, you must separate it in three as you can see. That is to say, you will put a roll of <m>clay</m> on top of the throat and another one on the edge of the shell of its tail, so that only the shell of the entire stomach will be accessible to be molded on its own. Having <m>oiled</m> the shell of the belly and having rubbed it with

<m>spirits</m>, thrown you slightly thick sand, which has been tempered with <m>hot water</m> as before. Having taken hold, remove the rolls of <m>clay</m>, then uncover the front and back legs, that is to say the bottom bit, and the throat and the tail, and remove the clay of all the cavities in one go if you can, because this will show you if the cavity will release well. And if the removed <m>clay</m> carries with it some of the <m>wax</m> you have applied, put it back into the places that did not release well, as has been said. And if some bit of the <al>turtle</al> has come away with the <m>clay</m>, or fallen off, attach and rejoin it to the mold with some <m>wax</m>, using a heated <m>iron</m> needle. Do not forget to put some melted <m>wax</m> at the edge of the <al>turtle</al>'s nails when you cast. Having molded and uncovered the shell of the belly, you will see the mold in this way.<figure>#</figure></ab>

<figure>
<id>fig_p146v_1</id>
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</figure>

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<margin>left-bottom</margin>
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<page>147r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f299.image</image>

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<cont>
<id>p146v_1</id>
<ab>Be warned to make an edge of <m>wax</m> melted with a warm <m>wire</m> all around the edge of the top shell which is close to the tail in order to make a better cast. And make it so that two small strings of the same <m>wax</m> goes from the end of the shell of the belly to the aforesaid <m>wax</m> edge, which is around the backbone of the shell, and the <al>turtle</al>'s tail should stay in between those two strings. This to make a better cast. This done, <m>oil</m> your mold and what you can see of the shell. Put the circle around this, and cast with <m>hot water</m> as before, and doing what you have done in the same manner to mold the third part which is the tail. And having taken hold, take away the <m>clay</m> contours, clean your mold, bring it together, and flatten it on all sides with a knife as you have done with other molds. Lightly dip it in <m>water</m>, then try to release it. Usually, one starts by the shell of the back, which is to say the top mold that is made of two halves. And this one is easier to release. The other follows directly after, but because it is next to the <al>turtle</al>'s shoulders which are deeply entrenched, it is sometimes troublesome to do this when lightly shaking and pulling the mold. Be advised to pull the easiest one first, because you will need to release them one after the other. The hardest one of all is the one molding the throat, the underside of the legs, and the cavities of the shoulders, which are so difficult to release, that if you do not plan accordingly by filling it with wax rather than molding what cannot be released, it will be difficult to take the pieces without breaking something. But if it happens to you again, there is a solution, provided that you keep the broken pieces, because you can reassemble them with stong iron needles, and fill whatever crack or flaw you have with some melted wax and hot iron as you have with others so that the noyau to mold hollow is easier to do. After</ab>

<ab><margin>bottom</margin>See the second page</ab>

<ab><margin>left-bottom</margin>To prevent breaking while releasing, be advised to uncover up to the point where it can still hold.</ab>

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<page>147v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f300.image</image>

<div>
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<id>p147v_1</id>
<figure>+</figure>
<head>Casting with <m>copper</m> alloyed with <m><figure>⧄</figure></m>, which is like very low <m>solder</m> and ard[...]es and <m>old K</m> and some Re out of xii[a]</head>

<ab>I have cast the cleanest I have ever seen, making very small figures as thin as paper. I have cast extremely hot in the very red mold, and have put into the melted mixture the two compositions which makes <m>silver</m> runny, and have done as if I wanted to throw with <m>pure silver</m>. This <m>league</m> [b] is white when it boils[c], as with every other alloy which contains even just a small bit of <m>silver</m>. To cast something very delicate, use this.</ab>

<div>
<id>p147v_2</id>
<head><m>Lute</m></head>

<ab>There is none better than the one you have used to mold, that is to say lumps from your molds. But you need to choose some of the better recooked ones.</ab></div>

<div>
<id>p147v_3</id>
<head>Crucible</head>

<ab>Beware that it should be three fingers away from the mouth of the bellows and that the aforesaid bellows is positioned underneath the bottom of the crucible, otherwise it will cool it down.</ab></div>

<div>
<id>p147v_4</id>
<head><m>Latten</m> casting</head>

<ab>I have taken the one you find in <m>latten</m> skillets, which have been beaten and forged quite thinly. Having well-heated it, I threw in two or three grains, like beads of <m>sal ammoniac</m>. This will clarify it like a mirror, having turned very white from all the heating, I threw in it some crushed raw and pure <m>calamine</m>[d]. The I casted it in its very red mold. It cast cleanly and thin like <m>paper</m>, and hollow on the back. Because it has become crusty I cooked it again, which means to say, I reddened it, and left it to cool, then put it in <m>bleach</m>, made partly with raw <m>tartar</m> and half of common <m>salt</m>. Having boiled it well, I brushed it in clear <m>water</m>, because the first time it was</ab>

<ab>
<margin>left-bottom</margin>Do not let it cool down when melting. It always makes into filaments, as you would find with melted <m>glass</m>, because of the <m>calamine</m>. Usually, the soft <m>letten</m> of skillets, becomes red from casting, even when it has been in the fire for a longtime, because the <m>calamine</m> evaporates. But the brittle <m>letten</m> of candleholders becomes yellow, as well as the[e] <m>filings</m> made of needles.</ab>

<cont/>
</div>

[a]These seem to be materials, but their exact nature is not clear. We are left somewhat baffled. See 86v for "old K".
[b]This can be found in Cotgrave: "solder, 8 parts whereof be fine gold, 3 of silver and 3 of copper."
[c]This could be an implement, like a pot in which one boils things: bullitoire
[d]See Cotgrave: "a certain yellow mineral substance, which fire consumes but melts not, that looks like gold; also the heavier soil of brass, or copper, which comes out of the sparkles and smoke that arise from the furnace and cleave to the roof and upper sides of the house, wherein it has melted."
[e]We think this must be the translation of "et" because otherwise the sentence does not make sense.

.....

<page>148r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f301.image></image>

<div>
<cont/>
<id>p147v_4</id>
<ab>not come out as clear as I wanted. I reheated it again, let it cool, put it into the <m>bleach</m> again, and brushed [it]. It came out very clear. In this way, you will do [the same] for <m>copper</m> and <m>silver</m>. And if you want to give...</ab>
</div>

<div>
<id>p148r_1</id>
<head>Beautiful color for <m>brass</m></head>

<ab>Having cleaned it well, as is stated, and brushed it well, boil it in water with <m>curcuma</m>[a] root</m> or <m>terra merita</m>, and it will become very beautiful.
</ab>
</div>

<div>
<id>p148r_2</id>
<head><m>Bleach</m></head>

<ab>If it is for <m>silver</m>, do not grind your <m>tartar</m>[b] with ^{something made of} <m>iron</m>, but with the handle of a hammer, because if you touch [your tartar with iron] and [end up] getting some <m>iron</m> in your <m>bleach</m>, the <m>silver</m> will come out red, like <m>copper</m>, and you will need to reheat it and put it back in the <m>bleach</m>. Mix in nothing with your <m>tartar</m> that should not enter the <m>bleach</m>, because this ruins it. Do not let it spill out when it first starts to boil because all its strength is in it [at this point]. <m>Bleach</m> is made with one part pulverized <m>tartar</m> and a half part of common <m>salt</m>.</ab>
</div>

<div>
<id>p148r_3</id>

<head>Moulding the foot of the bittern, or eagle, or other birds for the base of saltcellars or vases</head>

<ab>One commonly molds the foot and the legs up until the calf. And for this effect, one molds it in five pieces: the leg in two pieces, the top of the foot in two, and the bottom in one piece.</ab>

<ab><margin>left-bottom</margin>with[c] three pieces</ab>

</div>

<div>
<id>p148r_4</id>
<head><m>Crocum Ferri</m></head>

<ab>To make this quickly, redden the <m>filings</m> in an <m>iron</m> ladle[d], stirring [e]it often. Then sprinkle it with some good <m>vinegar</m> and let it evaporate. And without taking it away from above the fire, let it redden and take flame. Having been cooled, pulverise [it] on the <m>marble</m> and return it to the heat, sprinkle with <m>vinegar</m>, and then set it on fire. In this way it will soon be colored and made fine.</ab>

</div>

[a]And Indian root, curcurma longa, or tumeric.
[b]See Cotgrave.
[c]In the transcription I wasn't sure if this was "de" or "et." So either "with" or "and"....
[d]See Cotgrave: quaisse
[e]Or maybe shaking it if the caisse is really a ladle.

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<page>148v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f302.image</image>

<div>
<id>p148v_1</id>
<head>Molding <al>turtles</al></head>

<ab>Join the parts of your mold and fix very carefully clamps on all joints, which are on the top, the bottom and the sides of the mold; do not forget to notch the joints as you did with the other moulds. Once you have fixed all the clamps, remove the sides ones, not the others. Thus your mold, made of several pieces, will open as if it was only made of two parts. If you want to mould hollow, pierce a hole big enough to thread the end of your little finger in the middle of your mold's belly. Widen the outside of the hole, like clervois[a], in order to cast the core. You could avoid all these difficulties if you didn't mold hollow; you could then mold your <al>turtle</al> in two pieces, more quickly, and burn it inside of the mold like other animals. But a <al>turtle</al> is weighty, and would be heavy if not molded hollow. This is why it is considered better to mold it that way. It takes three days to make the proper ^{hollow} mould. To mold hollow and fantastic ^{forms}, you need very strong <m>plaster</m> which can withstand fire without bursting. But if you can't get this kind of <m>plaster</m>, add to it a little more <m>stone alum</m>, and also add <m>crocum</m>, which fortifies it ^{the <m>plaster</m>} and makes the flaws -- if there are any -- so fine that they can be easily removed. Do not forget to tighten your mould with a press in order to avoid flaws that happen when your mold is not tight enough, or when it bursts. To repair it, if the lines are not apparent enough retrace with a burin, then soften the lines with a ciselet[b]. You can remove the flaws with a chaple[c], a kind of burin. For the lumps and crumbly bits, they can be made with a gadget[d] or a small carving tool which isn't hammered, and hitting with a small file.</ab>

<ab><margin>left-top</margin>Make this hole before joining the mould.</ab></div>
[a]"claire voie", clerestory. Used in gothic architecture, but also refers to slated ventilation.
[b]finer burin.
[c]Thicker burin
[d]a "gad" in English is a type burin. Seems to be a similar root.

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<page>149r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f303.image</image>

<div>
<id>p149r_1</id>

<head>Softening <m>iron</m> and making it really soft</head>

<ab>Take some <m>sublimate</m>, some <m>quicklime</m>, <m>soap from Venice</m>, some <m>horse manure</m>, use less <m>sublimate</m> than the other ingredients. Moisten all of them together, forming a paste and apply this to the <m>iron</m>, and wrap the whole thing in a dirty rag. Then reheat it until it becomes very red. Or leave it overnight in a good fire until the morning. It will be very soft and you will be able to engrave whatever you wish on it.</ab></div>

<div>
<id>p149r_2</id>
<head>Gold of lesser quality from <pl>Germany</pl>, <m>Electre[a]</m></head>

<ab>One calls it <m>gold</m> from the <pl>Rhine</pl>, which you can find in the rivers, and it comes in gold flakes. <pl>German</pl> <pro>goldsmiths</pro> wanted to refine it, thinking that they could separate it from the whiteness, which they thought was <m>silver</m>. But it always stayed white, which proves that this whiteness is fixed and it's only fault lies in its color. This is <m>electre</m>, from which you can make cups, which will reveal poison.</ab></div>

<div>
<id>p149r_3</id>

<head>Various arts from <pl>Germany</pl></head>

<ab>They make great use of water mills, and most of these artisans working with <m>gold</m>, <m>silver</m>, and any other metal, bring their large works to the mills to be hammered. And to pull <m>iron thread</m>, they heat large amounts of <m>iron</m>, making it into a point, they hang it up still red, and very quickly pull it.</ab></div>

<div>
<id>p149r_4</id>
<head>Repairing medals</head>

<ab>If you want to mold any medal to use as a model, and at the same time repair it, cast it in a greater quantity of <m>tin</m>, and put in it a good amount of <m>tin alloy</m> to make it harder.</ab></div>

[a]See Cotgrave: "amber whereof beads, and bracelets are made, also, a mixture of gold, copper, and 1/5 part of silver."

.....
<page>149v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f304.image</image>

<div>
<id>p149v_1</id>
<head>Molding vases in several pieces</head>

<ab><pro>Goldsmiths</pro> who make large works and crockery use a turning lathe to get the model they wish in wood. Then with some <m>wax</m>, ^{<pro>the goldsmiths</pro>}model on top of this the masks, festoons, or anything that they wish. Then they mould the model in three, four, or several pieces.</ab></div>

<div>
<id>p149v_2</id>
<head><m>Gold</m> solder</head>

<ab>They take on a <m>gold</m> penny a small pinch of refined <m>aloc[a]</m>, as you would find in old sous.[b] </ab></div>

<div>
<id>p149v_3</id>
<head>Molding <al>flies</al></head>
<ab>Large <al>flies</al> can be molded & cast. But you must apply underneath their wings some <m>wheat oil</m>, which dries quickly and firms them up & gives them a little thickness. The same thing can be done with <al>butterflies</al>, <al>cicadas</al>, <al>grasshoppers</al> & similar things. But to cast them more easily, you must set them on some leaf or bouquet. The others should be placed by hand on a very slender <m>silver</m> blade. You cannot keep your insects when they are dead, since they will dry out and their legs will break.</ab>

<ab><margin>left-bottom</margin>
<m>Wheat oil</m> should be put underneath delicate things to make them more rigid & firm, & to make them hold their natural posture, since being weak & delicate, the tempered sand, adding weight to them, would change their form.</ab></div>

<div>
<id>p149v_4</id>
<head><al>Bats</al></head>

<ab>They have very thin wings and if they are large & dry and the wings are extended, you need a very big mold & it is not certain that the <m>metal</m> will run sufficiently; thus, when you want to mold some, choose the medium-sized ones & and arrange them so that they have their wings are half folded, since in this way they will come out better. But be sure to apply, as you known, some <m>wax</m> on the side of the wings, so that the cast flows well.</ab></div>

[a]We're not sure what this is.
[b]So the transcription says "vieux dozains" which we are taking to be old sous? Since a sol contained twelve pennies.

.....
<page>150r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f305.image</image>

<div>
<id>p150r_1</id>
<head>Very strong <m>wax</m></head>

<ab>Mix with this some <m>resin</m> and <m>bole</m>. Do not cast very hot.</ab></div>

<div>
<id>p150r_2</id>
<head>Diverse ligues [a]of <m>tin</m> and <m>lead</m></head>

<ab>For hollow things like <al>turtles</al>, ^{use} half refined <m>lead</m> and half <m>tin</m>.</ab>

<ab>For flowers, ^{use} almost all <m>lead</m>, that is to say, a lb of refined <m>tin</m>, and two <figure>3</figure> of refined <m>lead</m>.</ab>

<ab>For <al>lizards</al> that are as thick as one or two fingers, ^{use} almost all <m>lead</m>, that is to say, a lb of refined <m>lead</m> and 4 <figure>3</figure> of refined <m>tin</m>.</ab></div>

<div>
<id>p150r_3</id>
<head>Moulds</head>

<ab>They are less prone to cracking while reheating when they are very thick, rather than thin, provided that thickness is sufficient. Because when they are thin, they do not need to be fired for long, and prolonged exposure to heat corrupts them. The same thing will happen if you <m>plaster</m> is not strong.</ab>

<ab>It is better to reheat one mold on its own, rather than heating it with others, because should this be the case, you will need a stronger and hotter fire, than for just the one ^{mold}. And because <m>charcoal</m> will accumulate between multiple molds, the heat will transfer from mold to mold in such a way that you will often find that your molds have burst and crack on one side more than the other.</ab>

<ab>Therefore reheat your molds, not applying fire, except to cover it.</ab>

<ab>And because your <m>lead</m> and <m>tin</m> become brittle when melting the over high heat, in order to make them less brittle, heat them over low heat, and cast in stick form.</ab>

<ab>Brick cools the <m>metal</m>, and if this is not for binding, it is not necessary.</ab>

<ab><margin>left-middle</margin>The reheated moulds cannot withstand several casts, but the ones in which you only cast <m>tin</m> or <m>lead</m> and flat medals, these being dry, withstand a lot of casts.</ab>

<ab><margin>left-middle</margin><m>Iron filings</m> being finely ground, makes them ^{molds} very firm and strong.</ab></div>

<a>Alloy or solder, as per Pamela Smith

This may be a bit of a loose translation but it's the best we came up with.

<c>Perhaps ingots. Presumably to be used again.

.....

<page>150v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f306.image</image>

<div>
<id>p150v_1</id>
<head>Molding hollow</head>

<ab>This <m>iron</m> thread needle which you put thusly in the mold, is to better secure the noyau and keep it from shifting. All of this having been thusly arranged, lay your mold on the table, hole facing upwards, as it is explained at the side. And then fashion a circle made of soft <m>clay</m>, of about two fingers high around the hole, as you have done with other molds. And then temper your molding sand to an average thickness, and throw it in the hole until it ^{mold} is very very full, that is to say until you reach the <m>clay</m> circle. Be advised not to throw it ^{sand} in the middle of the hole, but from the sides, because the already made mold, will drink and suck the moisture. And the new sand that you throw on the sides of the mold, because drying quickly, will plug the mold and not ^{leave you enough time} to fill up. Whereas if you throw it ^{the sand} from the side, when you realize ^{it is drying too quickly} and plugs up, you can make a path for it. Having thrown it in, blow on the tempered sand, and insert the tip of your finger in the hole a few times, or some kind of pointy thing to keep it moist, in order to make sure that the hollow <m>wax</m> does not fill up. At the end throw in the thicker tempered sand to strengthen the mold, because when <m>water</m> rises up to the edge of the mold, it always makes it softer. Having taken hold, get rid of the <m>clay</m> circle and any raised parts that relate to the hole through which you threw the noyau, and you will not even realize the process took place. Once your mold has been thusly prepared, lute its outside with the same sand that you used to make the mold. Then reheat it on with a low heat to begin with, slant the mouth of the cast downwards so that the <m>wax</m> trickles out slowly, because if you reheat it over very high heat to make the <m>wax</m> come out, it will boil inside it ^{the mold}, and will leave blisters and lumpy bits on it ^{the cast}, as long as</ab>

<ab>
<margin>left-top</margin>
This <m>iron</m> thread needle is placed in the middle and the hole of the cast, when the mold is in many pieces, as is the one for <al>turtles</al>. And you do this so that the noyau does not shift. But when the mold is made out of only two pieces, it is not necessary to put needles in the middle.

<figure>
<id>fig_p150v_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5eVVJSjUwSFhpUzg</link>
</figure>

<figure>
<id>fig_p150v_2</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5eSUK5cUNtNnpqbXc</link>
</figure>
</ab>

<ab><margin>left-top</margin>Rather than cast the noyau, make some notches around the edge of the hole through which you would like to cast, so that your mold is even more secure.</ab>

<ab><margin>left-top</margin>Mixed <m>tallow</m> means that when you empty your <m>wax</m> when you are emptying the mold of <m>wax</m>, it does not clog.</ab>

<ab><margin>left-middle</margin>You will need half <m>tallow</m> and half <m>black wax</m>.</ab>

<ab><margin>left-middle</margin>Every thick piece comes out better when molded hollow, because a thick mass of mostly lead will stay warm for a long time and eat

away at the mold.</ab>

<ab><margin>left-bottom</margin>After the cast, you can mold the part of the shell of the belly where you made your hole on the real thing ^{animal} and fix it with <m>solder</m>.</ab>

<ab><margin>left-bottom</margin>

<figure>X</figure>Note that after your noyau has been cast, it is better to put your mold in hot <m>water</m> to open it, so that you can take away as much <m>wax</m> as you can, always softening it in the hot <m>water</m>. Because the less <m>wax</m> you have on it, the better it is, because it will reboil when you reheat your mold, and you will get blisters and lumps. And if you have little <m>wax</m> left ^{in the mold} you will not need to reheat it as much. When you open your mold the cast will break. But it can easily be repaired as long as you cast<figure>A^[a]</figure>

</ab>

<cont/>

</div>

[a]this note continues on the next folio (p151r)

.....

<page>151r</page>

<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f307.image</image>

<div>

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<id>p150v_1</id>

<ab>the <m>white wax</m> you have used to make <m>black sulfured wax^[a]</m> has so little substance, being dry and purified, that it leaves nothing with it ^{the mold mentioned in the first part of this recipe} and trickles very gently. As for the <m>coal</m> which is mixed in it, if there is some left, it burns and reduces to ash, and can be taken out afterwards by blowing into the cast. You can open your mold after the <m>wax</m> has been melted, if it does not contain a fantastical piece which could break, which are like pieces added on or similar. After having covered the clamps with <m>lute</m> from your sand, and having covered the entire mold, add to this a layer of common <m>luth</m>, and then sprinkle ground <m>brick</m> so that you can better manipulate it. Leave the <m>lute</m> to dry slowly, rather than putting it in the fire, which will melt the <m>wax</m>. The <m>wax</m>, having exited the cast, that which remains of it, mixed up with the <m>charcoal</m>, will burn. Having reheated it for the first time, <m>lute</m> it again, because should your <m>plaster</m> be bad, it will have eaten away at the mold and it will be found in the space between joints, which can result in large overflow. And for the second time, you must not reheat it if you want to cast only in <m>lead</m> or in <m>tin</m>, simply make sure that the <m>lute</m> is very dry and your mold attains an average heat. For those very large <al>turtles</al> that have enough thickness, it is better to cast using an alloy of half <m>tin</m> and half <m>lead</m>, as you do with hollow things. Because if you only need to cast very thin and fragile things like flowers, rely instead on using more <m>lead</m> than <m>tin</m>, which bubbles when cast a bit thickly and will easily some flaws to your cast. Mr. Alexander says he has never gone astray with using half <m>tin</m> and half <m>lead</m> for things that are heavy and as thick as the back of a knife. If there is <m>crocum</m>, you will not be able to clearly see the joints and as a consequence the metal overflow is very small and thin. You can repair this, and through the hole of the belly draw out the earth from the noyau, leaving it to soak for a long time.</ab>

<ab><margin>left-top</margin>When you have thrown your <m>black wax</m> in the mold and it has well-cooled, you will have to open half of your mold to make your cast. Should your <m>black wax</m> figure break, there is no danger, because it can always be reattached with a hot <m>iron</m> needle. Or if the pieces have not separated well from their mold, they will meet up and join as before, by joining and tightening the two halves of the mold. When your figure is big, you must interweave more <m>iron</m> threads inside to give support to the noyau.

<figure>

<id>fig_p151r_1</id>

<link>https://drive.google.com/open?id=0B9-oNrvWdlO5dEhadHEzVWlxcejg</link>

</figure>

</ab>

<ab><margin>left-bottom</margin>

<figure>A^[b]</figure>only in lead or in tin. You must however heat the mold as much as if you were about to burn an animal inside, so that the <m>wax</m> melts well and leaves nothing inside, and trickles out while holding the mold slanted towards the cast and at the same time reheating it. And when the mold has been reheated, leave it to cool slowly, then blow inside and suck out the wax ashes. Do this with a bellows.</ab>

<cont/>

</div>

[a]Although there is no sulphur mentioned in the rest of the recipe, the combining of sulphur and wax is very possible and appears in other recipes.

[b]This is the note that continues from the previous page.

.....

<page>151v</page>

<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f308.image</image>

<div>

<cont/>

<id>p150v_1</id>

<ab>However, do not remove it before you have prepared the whole <al>turtle</al>, because you will use this <m>earth</m> as <m>cement</m>, should you need embed something or make repairs with a chisel. If you have some metal overflow at the joints of your mold, you will remove it, either with a burin, called a chaple, or with a very sharp pen-knife, or with a small file followed by the chisel. Always keeping in front of you the real thing to do as best you can. You can curl on a file a chisel point, which is not at all moistened, to make something lumpy. For these <al>turtles</al> and any other hollow things which you can assume are quite big, create a <m>ligue</m> of half <m>lead</m> and half <m>refined tin</m>. There is more to do when molding a <al>turtle</al> than when molding a dozen flowers. If there is <m>crocum</m> in your sand, there will be no metal overflow and if you are unfortunate enough that there should be some on the sides, these will be as thin as <m>paper</m>, as long as your mold is very tightly pressed together. And to make the scales of <al>turtles</al> and other animals, you can make little chisel marks in the shape of round punch cutters, others in the shape of a gouge, or shaped like a <al>snake</al> or <al>lizard</al> scale, and other using a file used to make twisty bits and lumps. <al>Turtles</al> that do not need to be molded hollow are not as much work, because they are molded in two pieces and repair them with little chisels ^{that make} punch cutters, little couge,^[a] and serrated chisels.

<figure>
<id>fig_p151v_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdIO5bk1US2QxRTVwa2s</link>
<margin>left-middle</margin>
</figure>
</ab>
</div>

<div>
<id>p151v_1</id>
<head><m>Wheat oil</m></head>

<ab>You must make it between two empty <m>iron</m> blades, the bottom one in particular will be slanted downwards and on this one you will put an even and equal ^{layer} of <m>wheat</m>. Then you will place on it the other ^{blade}, which is very hot, and you will press it down until you see the black <m>oil</m> drip. Remove the top blade when you have enough. This <m>oil</m> dries very quickly. It gilds <m>silver</m> and varnished things, deepens the color of gilded things, ^{can be used} as a <m>varnish</m> on <m>iron</m> before you engrave it, and to varnish the tanned ^{<m>leather</m> parts} of sword hilts. And it could also be used by <pro>those who make gilded <m>leather</m></pro>.</ab>

<ab><margin>left-bottom</margin>You must use this, as has been done recently. And for things you need to mold, it must not be as thick as for dyeing.</ab>

<ab><margin>left-bottom</margin>You cannot use this to oil hairy animals, because it is too strong and rigid, but it is good to apply to the foot of a little animal, like a <al>fly</al> and other similar things. It is also excellent to dye white stones [illisible]</ab></div>

[a]This may be what the image is in the margin. This word is related to the English "gouge."

.....

<page>152r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f309.image</image>

<div>
<id>p152r_1</id>
<head>Molding a <al>rat</al></head>

<ab>The hairs of his whiskers are difficult to come out of the mold, you can therefore burn them away, as did the Tyrant of <pl>Syracuse[a]</pl>, which you can always replace with life-like <m>silver</m> ones. The tail comes out beautifully because it is made of little scale-like circles, almost like the ones of a <al>lizard</al>. And to make sure that the hair does not keep you from molding well, burn them away with a little candle. The rest of the body must be oiled with <m>wheat oil</m>, which will keep the hair rigid and flat, so that it will not be removed when it mixes with the sand, which will keep you cast from releasing cleanly. <m>Wheat oil</m>, being dry, will dry quickly, you will therefore be able to comb the hair with a very fine comb, so that the hair will be visible when cast. On a small <al>mouse</al> you can use <m>common oil</m> to flatten the hair. Rather than molding these animals immediately (except for <al>snakes</al> and <al>lizards</al>), keep them dead for a day a two so that they become easier to handle. Because they are very stiff after they die and cannot be shaped into a life-like looking position unless they are softened. However, if you mold it straight after it dies, and give it its shape before it becomes stiff, that will be good. And because the eyes die unless you mold them immediately, you can replace them with half a round <m>pea</m> taken out of its shell. Some cast their <al>rats</al> without any added preparations[b]; others oil them with <m>olive oil</m> which suits the purpose, because <m>wheat oil</m> is thick and has too much body. This is why we only use it ^{wheat oil} in light strokes on <m>fly</m>'s feet or beneath a wing, to keep them stiff. Others flatten the hair with an <m>egg yolk</m>, which dries quickly and does the job. Arrange it ^{the <al>rat</al>} on its <m>clay</m> base before oiling it. Raise its head with an <m>iron</m> needle, and also secure its feet with needles, and its tail. Rub it with <m>spirits</m> after the <m>oil</m> has dried. Then throw in your sand. Uncover it as you would other animals, then do your second casting. You must not open the mold until it is completely reheated and the <al>rat</al> has burned away.</ab>

<ab><margin>left-top</margin><figure>+</figure>If it is a big <al>rat</al>, it is better to cast it hollow, because it is thick. A <al>mouse</al> is easier to mold because it has shorter hair. It is impossible to mold an animal with big bones well, such as a bird or <al>rat</al>, without opening the mold to clean it well, because bones do not burn well.</ab>

<ab><margin>left-middle</margin>All kinds of feathers and hairs are difficult to mold realistically.</ab>

<ab><margin>left-middle</margin>It is better to use <m>butter</m> to grease hairy animals than <m>wheat oil</m>, because it is better suited to the purpose.</ab>

<ab><margin>left-bottom</margin>If the <al>rat</al> is big, it is better to remove its entrails, or having molded it, let it dry in an oven, because the large amount of moisture it contains would boil inside the mold, which would ruin it and make it lumpy.</ab>

<ab><margin>left-bottom</margin>Only put your mold in the oven after it has cooled, drying slowly, otherwise it will absorb the moisture and break.</ab></div>

[a]Denis de Syracuse, a Greek tyrant who burned off his beard rather than trust a barber to trim it. He was afraid that the barber would cut his throat.

[b]This reflects a bit of artistic license on our part but I think it makes the sentence clearer.

.....

<page>152v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f310.image</image>

<div>
<id>p152v_1</id>
<head>Repairing things that have been cast</head>

<ab>

It happens often that what you have cast has metal overflow, through a fault in the mold, the cast, or the `<m>metal</m>`. Faults in the mold arise from `<m>plaster</m>` which is not hard and strong and cannot withstand fire, which is corrected by mixing in `<m>crocum</m>`. If you use too much `<m>tin</m>` to cast things that are of an average thickness, it will not release well, because tin that is cast thickly, shrinks and bubbles. Similarly, the things you need to mold hollow can be fantastical to cast. If therefore you happen to have a flaw, have at your disposal a small `<m>leather</m>` pouch full of fine sand, and having placed it on the edge of a `<pro>goldsmith</pro>`'s bench, where one files, place upon it your work, and secure it there with a rope which passes underneath your foot. The sand in your pouch will respond ^{to} the pressure of your foot, and will immediately be thicker on one side and thinner on the other. Then immediately with a burin, outline and quickly make the subtle parts of your cast which did not come out, or which did not come out distinctly enough, or with a file or a chape, remove your `<m>metal</m>` overflow. If there is a flaw in your materials that has left a hole or something too hollow, scrape it with a burin and make notches around that hole, then make an imprint of this with `<m>wax</m>`. And place this imprint on a fine `<m>lead</m>` blade, and in this way mark the appropriate size of `<m>lead</m>`, or of any other alloy which resembles your casting material the most. Then place this piece of work on the notched part and attach it well with some with `<m>yellow latten</m>` or `<m>copper</m>` wire^[a] - and if you are dealing with `<m>tin</m>` or `<m>lead</m>` rub in some `<m>rosin</m>` around the edge - and apply all around it ^{the replacement piece} some little thin pieces of `<m>solder</m>` or `<m>tin</m>`, or any other thing. Then with a hot `<m>iron</m>`, or above the heat source of the forge, solder it together, and then repair all of this with the aforesaid tools and the appropriate chisels.</ab></sup></sup>

<ab><margin>left-top</margin>
For `<m>gold</m>` or a small work, you will need to attach them to a `<m>lead</m>` ball, which you will then place on the `<m>leather</m>` pouch and it will help secure the piece with the rope.</ab>

<figure>
<id>fig_p152v_1</id>
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<margin>left-middle</margin>
</figure>

<figure>
<id>fig_p152v_2</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5cWhfVmhvbWszZIE></link>
<margin>left-bottom</margin>
</figure>

</div>

[a]See the vocab for "archal."

.....

<page>153r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f311.image></image>

<div>
<id>p153r_1</id>
<head>Moulding the hollow part of a mold for seals or other things</head>

<ab>Firstly fashion your figure in `<m>wax</m>` on an even slab of `<m>glass</m>` or `<m>slate</m>`. After, mold it with our aforesaid sand, having beforehand lightly oiled it with `<m>oil</m>` as you know, and having rubbed it with `<m>spirits</m>`, while tempering your sand with `<m>hot water</m>`. Having made the hollow part, cast it in `<m>lead</m>`, using the same alloy as you do for other figures that come out very neatly. However, if it does not release cleanly, repair it. And then mold you `<m>lead</m>` relief with some `<m>wax</m>`. After, in your `<m>wax</m>` hollow part, throw in your tempered sand which will give you a relief, upon which you will cast your hollow part with `<m>silver</m>`, or any other metal used for seals. But note that you must not make your hollow part with melted `<m>wax</m>`, but only with `<m>wax</m>` warmed in hot `<m>water</m>`, or even better in hot `<m>urine</m>`. Or else, if you have a relief in `<m>gold</m>` or `<m>silver</m>`, or any other very neat and flawless metal, mold your `<m>wax</m>` hollow part directly upon it, and throw your sand in it ^{wax hollow}, which will form your relief, upon which you will cast your hollow part to make your seal or any other work. You could also cast a hollow part directly from the main part of your metal, even ^{should you want to cast} in `<m>lead</m>` or `<m>tin</m>`, which will not make the main part of your metal melt, if it has been lightly brushed with `<m>crushed chalk</m>`, or covered with `<m>candle soot</m>`, or with dried `<m>egg white</m>`.</ab></sup></sup>

<ab><margin>left-top</margin>When you have the `<m>wax</m>` imprint of your seal, mold in a noyau this piece of `<m>wax</m>`, which is hollow like your seal, and your noyau will make a mold similar to the imprint and the seal.</ab>

<ab><margin>left-middle</margin>Your `<m>wax</m>` must be mixed with a color that has little body, so that you may better see your imprint. The `<m>lamp black</m>` or `<m>soot black</m>` is good for this. `<m>White wax</m>` to do this is even better.</ab></div>

<div>
<id>p153r_2</id>
<head>Giving a medal the thickness that you wish</head>

<ab>Having cast your first mold, mix some `<m>crocum</m>` or well-ground `<m>iron scales</m>` to make it ^{your mould} firmer. And having taken hold, and made your hollow part very neat, take some paste made of^[a] fine</ab></sup>

<cont/>
</div>
[a]paste [made] of fine flour

.....

<page>153v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f312.image></image>

<div>
<cont/>
<id>p153r_1</id>
<ab><m>flour</m>. And to flatten it, take a small stick of <m>wood</m> that is round and of an even thickness, which has at both ends a small circle that sticks out from the rest, in order to give the necessary thickness to your <m>paste</m>. And then with this rolling pin, flatten it so that it becomes thin and tenuous. Then apply it on the hollow part of your mold and press it down with <m>cotton</m>, and your figure will impress itself on the paste, in relief on one side, and hollow on the other. This done, with a brush that is dipped in melted <m>butter</m>, grease the part of the <m>paste</m> that is hollow. Then trim off the extra <m>paste</m> which extends beyond the hollow of the mold. Also <m>oil</m> your mold with <m>olive oil</m>, as you have done with others. Apply your <m>clay</m> contour, and cast your second mold. And you will have a medal as thin and hollow on one side as you wish. You can have various rolling pins which some edges are more raised than others, in order to make various thicknesses, or you can use sheets <m>lead</m> and <m>copper</m> of various thicknesses, or <m>cardboard</m>, all cut with the rolling pin.[a]</ab>

<figure>
<id>fig_p153v_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5czR5TF9scDhsSG8></link>
<margin>left-top</margin>
</figure>

<ab><margin>left-top</margin>If you were to <m>oil</m> your <m>paste</m>, the <m>paste</m> would absorb the <m>oil</m> and it ^{the oil} would penetrate through the <m>paste</m>, right up to the edge of the mold and attack it. But <m>butter</m> remains on the surface of the <m>paste</m>, and does not penetrate it. For these back bits, you must not temper your sand with <m>hot water</m>, because it will melt the <m>butter</m>. <figure>

<id>fig_p153v_2</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5c3FRZnZSMnU1Njg></link>
</figure>
</ab></div>

<div>
<id>p153v_1</id>
<head>Testing the virtues and strength of sand that is to be reheated</head>

<ab>Once the sand has taken hold, you must find it soft and easy to cut, and not surly. It is better to test it in a large and fantastical mold, rather than a small one, because the big one will stay in the fire longer and the small one reheats very quickly.</ab></div>

<div>
<id>p153v_2</id>
<head>Softening <m>gold</m></head>

<ab>Sometimes, you can find <m>gold</m> that is so dry that neither <m>cement</m> nor <m>antimony</m> can soften it. Only <m>verdigris</m> can soften this <m>gold</m>.</ab>
</div>

[a]We think this means that you build towers made of sheets of lead, copper or cardboard at the edges of your paste and then roll the rolling pin across the top to get an even thickness, however thick you want it.

.....

<page>154r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f313.item.r></image>

<div>
<id>p154r_a1</id>
<head>Removing <m>gold</m>[a]</head>

<ab><m>Gold</m> as gilding goes away if the coin[b] is reheated and placed against a strong fire, even if there is <m>lead</m>, because <m>lead</m>, which contains much <m>mercury</m> will pierce a <m>gold</m> coin in a fire.</ab>

</div>
<div>
<id>p154r_a2</id>
<head>Cutting <m>lead</m></head>

<ab>Because it [<m>lead</m>] is fat, hard to catch[c] and corrosive[d] to the knife and the scissors, wet it up and you will [be able to] cut it like <m>glass</m>.</ab>

<ab>
<margin>left-top</margin>
Sometimes, <m>gold</m> and <m>silver</m>, exposed to a certain kind of smoke, take on a coloration during the casting. But these are neither flakes nor coat but a colour which will go away during the whitening[e].</ab>

</div>
<div>
<id>p154r_a3</id>
<head>Softening <m>silver</m></head>

<ab>When <pro>goldsmiths</pro> who work on a large scale[f] have forged their plates, to their loss [these plates] quite often break and crack because they have been sharpened [g]too much. In order to avoid this, when it is quite melted, throw in ground dry <m>mortar</m> made of <m>sand</m> and good <m>lime</m> which has been used before.</ab>

<ab>
<margin>left-middle</margin>

```
<pro>Goldsmiths</pro> do not work with <m>silver</m> from <m>real[h]</m> because it is rich in <m>lead[i]</m> and becomes sour when forged.</ab>

<ab>For small works and things that need to go only once to the fire, some weld with some old <m>sol</m>[j] or <m>carolus</m>[k] reheated and beaten. However if the
<m>sol</m> is not quite good [enough], the welding will not hold[i] as there is too much <m>copper</m> and one will have to weld twice. Others weld with [an alloy
made of] half <m>silver</m> and half fine <m>copper</m>.</ab>

</div>
<div>
<id>p154r_a4</id>
<head><m>Iron</m> flakes</head>

<ab>Once it is quite ground and refined on the <m>marble</m> slab and mixed with the <m>sand</m> from the core [mold] [m], dry it slowing without reheating it, and it
will allow for [n]several casting of <m>lead</m> and <m>tin</m>. <m>Copper</m> and <m>brass</m> come out fine of it. But if it is not ground as finely as
<m>crocum</m>, it sinks unless the <m>sand</m> is soaked and quite thick.</ab>

</div>
<div>
<id>p154r_a5</id>
<head><m>Metal file dust</m></head>

<ab>It does not melt by itself if it is not helped with some portion of similar <m>metal</m> melted to assemble it and bathe it[o], as it is more burnt than melted. <m>File
dust</m> from <m>tin</m> and <m>lead</m> are made with <m>tallow</m>, [<m>file dust</m>] from <m>gold</m> with <m>saltpeter</m>, [<m>file dust</m>]
from <m>silver</m> with <m>sandever[p][q][r].</m></ab>
</div>
```

[a]Very unsure about the meaning of this title in the ms.
[b]does "pièce" refer to a coin or simply a piece (as object) here?
[c]Gripper: Saisir, s'emparer (DMF)
Gripper: to seise, gripe, grasp, snatch (Cotgrave)
[d]I think that is what the author means by attaque
[e]I do not know how to translate bullitoire
[f]Grosserie: great worke (Cotgrave)
[g]Enaigri: sharpened, exasperated, angered (Cotgrave)
[h]coin
[i]is that the correct tranlation of plombens?
[j]coin
[k]coin
[l]I do not know what the author means exactly by 'mange' and I am deriving this proposed translation for the general meaning of the rest of the sentence.
[m]meaning unclear
[n]endure
[o]CHECK very unsure about this
[p]http://books.google.co.uk/books?id=k0IFNhxWj6gC&pg=PA139&lpg=PA139&dq=suin+de+verre&source=bl&ots=Zwaren5Lcl&sig=h_GoIiY2QxzDMfM6AvMJGpGIFLQ&hl=fr&sa=X&ei=FXa5U5HTI
[q]According to the above book about fake gem stones in the Middle Ages, 'suin de verre' or sandever in English is an obscure ingredient which seems to designate: les chlorures et les sulfates alcalins insolubles qui suragent sur le verre en fusion et qu'il est nécessaire d'écumer.
[r]Cotgrave: Sandever; the fattie substance floating on glasse when it is red-hot in the furnace, and which being cold is as hard as stone, yet brittle, and easily broken

```
<page>154v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f314.image</image>

<div>
<id>p154v_1</id>
<head><m>Latten</m></head>

<ab><m>Latten</m> is wonderful to cast. It is necessary to be cast very hot. And to master it better, throw in some <m>sal ammoniac</m> to clarify it. You can also add a
bit a <m>red copper</m>, and if you wish to, add in a bit of <m>tin</m> to make it runnier. It always leaves filaments like melted <m>glass</m>. It does not require as
long of a cast as others.</ab></div>

<div>
<id>p154v_2</id>

<head>Moulding</head>

<ab>It is a good thing that it be cooled on the outside, but nevertheless remain hot on the inside, because when it comes out of the fire, the air and the strong fumes from
the fire do not reheat the cast well.</ab></div>

<div>
<id>p154v_3</id>
<head>Casting <m>silver</m></head>
```

<ab><m>Silver</m> requires a long cast, whereas for <m>gold</m> it does not matter. <m>Silver</m>, when soldering, withstands the <m>water</m> you throw at it. Whereas <m>gold</m> sizzles, and because of this one throws its ?[a] on the <m>solder</m>.</ab>

<ab><margin>left-middle</margin>
Let the mold cool well after the casting, rather than opening it.</ab></div>

<div>
<id>p154v_4</id>
<head>Strengthening flowers and delicate things</head>

<ab>For flowers and small plants, do not use <m>wheat oil</m>. Instead, use some melted <m>butter</m> which you will apply lightly with a small brush on the back of the rose leaves, and pansies and any other flower that needs it. But one reinforces the little feet of <a>flies and other small animals with <m>wheat</m> oil to make them stronger and hold up.</ab>
</div>

<div>
<id>p154v_5</id>
<head>Strawberries </head>

<ab>The leaves are cast separately and then rejoined with solder. And because the strawberry fruit is cast solid, and because of this it is heavy, and the <m>tin</m> stem is slender, it will not be able to sustain ^{weight} without breaking quickly. One makes the stems with <m>latten</m> thread and then solder them back onto the fruit.</ab>
</div>

[a]We think that amongst all the crossings out that you find in the transcription, the author has forgotten to include a necessary word.

.....

<page>155r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f315.item.r=</image>

<div>
<id>p155r_a1</id>
<head>Moulding a rose</head>

<ab>Because of the little branches of the rose bush, which are around the flower, are sometimes very spread out, they would demand too big of a mold. We make and cast them separately, the rose and the rosebuds separately as well. And then one brings them together, soldering the little branches and leaves of the rose bush to the stem of the rose, on which you will have purposefully left bits of the small branches. Put your petal or rose as low as you can in your mould, because <m>sand</m> will always bring it up or raise it. You can also mould several petals together, arranged one on top of the other, separating them some thread. And for the look of the rose you can give a thin layer of <m>melted butter</m> on the back of the petals, but only on the outside petals, not the inside petals, to stiffen them and give them the strength to withstand, so that the wet sand does not stretch or spread them out more than necessary. You can also mold well the leaves of a rosebush, strawberry plant and similar things, that are flat and can be flattened without being spoiled. For two castings[a], to open your mould, when it has been reheated and then clean the ashes out, make some vents, and [you will be able to do] several casts. This is the easiest way and you can also do the other. And with little veins of <m>wax</m>[b] that has been adapted and joined from leaf to leaf[c], you can make casts. You can even make a little vein of <m>wax</m> from the back of the first petal, which will join up with the main cast. All of this will facilitate the casting process. The main thing is to let your reheated moulds cool down rather than cleaning them and blowing inside them to make the wax come out, because when the mold is hot, the ash almost attaches itself to it. But when it is cold it, it detaches[d] and leaves with air draft or when one draw in one's breath through the small opening.
</ab>

<figure>
<id>fig_p155r_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5b3IFZl8wbGducEk</link>
<margin>left-top</margin>
</figure>

<ab>
<margin>left-middle</margin>
You can also give a little thickness at the ends of the stems that are holding up the petals, by lightly oiling them underneath with <m>melted butter</m>, because the petals are big and weigh heavily, and the stem made of <m>lead</m> or <m>tin</m> will not have enough strength [to hold it].</ab>

<ab>
<margin>left-bottom</margin>
I would be of the opinion to mold the rose on its own with a bit of its stem close to its bud, and then to join the rose to a longer one [a stem] made of <m>glazed brass</m>, because the rose bloom is very big and heavy.</ab>

<ab>
<margin>bottom</margin>
Moisten your rose with <m>spirits</m> before placing it in the clay. Do not forget to oil the <m>wax</m> cast. And when you have thrown in your <m>wet sand</m>, blow heavily, until it begins to set. The rose came out well. But because the <m>sand</m> was mixed within the petals, soak your work in water for a long time so that when you shake it in the water, the earth comes off.</ab>

</div>

[a]This sentence is very awkward and we are still unsure of its meaning and proper translation.

[b]See annotation by Chiostrini and Palframan on Molding a Rose

[c]See Carlson Katz annotation on Roses.

[d]we could not find a proper translation for "s'esperd" but we feel this is the correct meaning

.....

<page>155v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f316.item.r=</image>

<div>
<id>p155v_1</id>
<head>Rose</head>

<ab>Because the rose bloom is rather wavy, and its petals are all mixed up and arranged in various ways, it will not be beautiful if it is not painted, and you must also consider that its weight cannot supported by the <m>tin</m> stem which is sour and fine. One moulds the flower of the rose in a separate mould, casting it thickly so that it comes out more easily. Then one cuts the cast at the edges of the stem of the bud, in which you graft and solder a stalk of brass wire to which you also solder the leaves. But because this <m>tin</m>, being so thin, is hard to solder, and may melt some of the leaves and also the cast flowers; [you should consider] that cast flowers, especially roses, are not beautiful without being painted, so one does not make the effort to solder them, but [instead] one grafts the pieces that you want to join together and glues them with <m>fish glue</m> that has been a little moistened and melted until thick. And so that it takes better, you heat the work in <m>tin</m> lightly and for a long time[a], because if it is cold, the <m>glue</m> will not take. Once your flower is thusly repaired, you follow the joints of the added parts with some <m>esbaucher wax</m>[b], which is a white <m>wax</m> mixed with much well-ground <m>ceruse</m> or even better, <m>white lead</m> melting it and placing it on your work with a small warm bit of <m>iron</m> needle. In the same way you can repair the little filaments that are in the middle of the rose, or the holes that may appear in some of the petals. Then paint your rose realistically. If you cast your rose in <m>gold</m> or <m>silver</m>, you can also rejoin [parts] and solder its. And in those materials, when you have join something very delicate together with the flower, such as a fly or other similar things, <m>fish glue</m> is excellent, and holds very well, fixing it with a few little needles that act as nails. The leaves and buds can be cast in two molds that can be opened once they have been reheated, but not before. Then these things join up [with the flower].</ab>

</div>
[a]We don't know if this French word is "long" or "loing."
[b]The definition of this follows in the text.

.....

<page>156r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f317.item.r=</image>

<div>
<id>p156r_1</id>
<head>Quickly molding and reducing a relief[a] to a hollow [mold]</head>

<ab>Make an impression in colored wax of the relief of your medal. And you will get a hollow mold, in which you can throw en noyau[b] a relief in sand. In this, you will cast your hollow in lead or tin. In this, you you will cast your wax relief. And then on this <m>wax</m>, you will make your hollow moule en noyau, in order to throw in it the relief in <m>gold</m> or <m>silver</m> or any other metal you would like. But to make this process go faster, if you are in a hurry, make the first impression and the first hollow out of the inside portion of the <m>bread</m> loaf[c], prepared as you know, and which will cast neatly. And inside this, throw in the melted <m>wax</m> which will give you a nice relief on which you can make your noyau.</ab>

</div>
<div>
<id>p156r_2</id>
<head>Work of <m>gold</m>, very light and hollow</head>

<ab>Cast a lizard or any medal you would like in <m>silver</m>, of low quality[d], then gild it lightly and once the first gilding is done and dried up in the fire, burnish it, brush it with a wirebrush. Apply another layer of light gilding, repeating the whole process. Do this three or four times or even more depending on how thick you want your <m>gold</m> to be. Then, in the most discrete place you can find, make a small opening, and submerge your entire work in some good <m>aqua fortis</m> which will penetrate into the <m>silver</m> through the opening and will corrode the <m>silver</m> without damaging the <m>gold</m>. In this way, you will have a lizard that is hollow until the tip of its nails or any type of work that is so light that when you blow into it, it moves. But be advised not to apply a thick layer of <m>amalgamated gold</m> in one go, or apply layers that are too thin each time, because this will block the finer details, but if you proceed as I have said, [i.e.] lightly and in many goes, and cleaning it well, then you will be fine.</ab>

<ab>
<margin>left-middle</margin>
If you want the aqua fortis to corrode well, you must cast in low quality silver.</ab>

</div>
<div>
<id>p156r_3</id>
<head>Chiseling</head>

<ab><m>Lead</m> is so fat and soft that one cannot hammer it harshly, one should have a light touch. Other big metals are easier to work with. With a graver[e] called an onglete, you make and retool the finest of details. With the ordinary graver, you make the larger ones, and with the chaple[f], you can remove the bits that have run, and with the [...] soften [...] some lines you flatten or raise the lines.</ab>
</div>

[a]suggestion: "Quickly molding and reducing a relief to a hollow [mold]" (see also field notes Landsman 12/12/14)
[b]We've consistently keep this in French as it refers to an image on another page and looks to be a special shape of a mold?
[c]mie de pain= the pith of the bread, according to Cotgrave, which makes sense here
[d]argent de billon it is a metal where the proportion of other metals is more important than silver
[e]See OED: Burin: a graver; the tool used by an engraver on copper; also attrib.
[f]A thicker graver/burin?

.....

<page>156v</page>

<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f318.item.r></image>

<div>
<id>p156v_1</id>
<head>Moulding a fly</head>

<ab>Take the fattest flies, that go to pantries, which are not hairy, if at all possible. If they are hairy, oil lightly their fur and their unmanageable hairs with olive oil to make them lie flat. Take them also and use them as quickly as you can after they have died, because if you leave them to dry out, their legs will break when you want to stretch them. You must also, to get a better cast, arrange them on some kind of leaf or other similar thing. This will help to cast their little legs, that are so fragile that unaided, they will not cast easily. They can be arranged on a sage leaf or something similar. They cast well in gold or silver but one usually the legs and wings separately and then join them [to the main body.] I have cast one on a bouquet of sage that had seven or eight leaves. And to arrange it, I affixed the legs on the largest leaf with a bit of melted wax and joined them with the pointed tip of a hot iron wire. And to make sure that the wings will [eventually] join more easily [to the body[a]], I applied on the underside some melted butter with a small brush. And with same pointed end of a warm iron wire I applied wheatgerm oil on the legs and feet. To tame and bed the downy hair that it has I also apply a bit of olive oil.</ab>

<ab>I do not let wheatgerm oil dry out a lot because it is very thick and has body, and the spirits cannot penetrate it.</ab>

<ab>For applications, tallow is too harsh, pork fat is too soft, and but is excellent because it is rather dry and curdles and instantly melts and is firmer. Be advised that the legs and feet are well set on the leaf, and not go beyond the leaf, because the whatever exceeds the leaf will not cast as well, even if cast in silver, because these parts are so fine that they will not cast. And truthfully, feet set on a leaf, cast well, but those set from leaf to leaf [b]which remain iffy did not come out well in tin. The rest of the fly was cast, and likewise the sage bouquet which cast very beautifully.</ab>

<ab>
<margin>left-top</margin>
You could also cast them well in gold or silver without the feet, and without attaching to a flower by casting it underneath the belly and then joining the feet with soldering. And if you need to, cast the wings in the same metal and it could be enameled if you fortify the wings and feet as previously said.</ab>

<ab>
<margin>left-top</margin>
If you sage leaf or your branch needs to be repaired, do as I have said underneath. And then paint in a realistic way.
<figure>
<id>fig_p156v_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5VjFPeWlJc05CbDQ></link>
</figure></ab>

<ab>
<margin>left-top</margin>
Having been cast in gold, one enamels the wings with window glass from Lorraine, which is and transparent.</ab>

<ab>
<margin>left-middle</margin>
It is necessary to cast the fly from the ass with some melted wax, fitted by an iron tip, as you know, to which the fly attaches itself, and then the wax, as it melts, serves as a cast for the fly.
<figure/>
<figure>+</figure>
</ab>

<ab>
<margin>left-bottom</margin>
If it happens that you have some defects with your fly's wings, hamer some very fine tin, or gold or silver, if you cast it, and shape with scissors the amount you need for your wings. And then apply it with tweezers and glue it with fish glue, applied like it is shown underneath. And before doing thing this heat your work lightly so that its coldness does not repel the glue, which will dry quickly, heating it from afar. Then cover lightly the joints of whatever you've attached to the cast with some esbaucher wax, which is a white wax that is mixed with a lot of ceruse or white lead, melting it with and warm iron tip. You will also cut little bits of harpsicord string and will glue them with the aforementioned glue when they are dry. That is the say, the feet, having been reworked thusly, you will make them bigger with this same melted wax so that they are equal in proportion [with the rest].</ab>
</div>

[a]This is the French verb "unir" and could mean that the wings stick together and don't fall apart as well. We're just not sure.
[b]Not quite sure what this means... feuille en feuille.....

.....
<page>157r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f319.item.r></image>
<div>
<id>p157r_1</id>
<head>The mode in which <pro>goldsmiths</pro> mold hollow molds</head>

<ab>
They mold their animal, or another solid work that they have already made or rough-hewn in <m>wax</m> and cast in <m>lead</m>, in two cuttlefish bones. When it is good and impressed in the two cuttlefish bones, they put a piece of <m>paper</m> between the two bones & join the bones and cast. The work comes in parts by halves. And then with a chisel[a] they dig into the two half castings of <m>gold</m> or <m>silver</m> as subtly as they like, and then they solder or braze them.</ab>

</div>
<div>
<id>p157r_2</id>
<head>Bright red</head>

<ab>
Some <pro>goldsmiths</pro> try it out on finely beaten gold that is well burnished with a chisel[b]. But it is better to try it out on a <m>gold</m> ingot, and to test it as well on a work or [illeg.] your materials fantastic, since if they look good & extraordinary on a big work, they will show excellently on a common work.</ab>

</div>
<div>
<id>p157r_3</id>
<head>Flies </head>

<ab>They mold better on a single leaf than on a bouquet or a flower or the branch of an herb, since the single leaf is molded in two halves that being reheated can be opened up and cleaned well, & the cast will be better. In picking up the fly, take heed that you do not break its legs or let it dry out too much, because then the legs will fall off. And if this happens to you, glue the same legs back on with <m>fish glue</m> and <m>wax</m>, or adapt the <m>hairs of a pig</m> or the <m>cords of a spinet</m>[c]. Fix the leaf on a flat piece of <m>clay</m> with two pins, and the fly with one pin in the middle of its body & the feet with wax.</ab>

</div>
<div>
<id>p157r_4</id>
<head>Cleaning and degreasing your cast <m>tin</m> works, and others</head>

<ab>Make some lye passed through scrupulously with good <m>ashes</m>, and it should be new & without any fat; boil your molded work in it. And the tallow and fat will leave.</ab>

</div>
<div>
<id>p157r_5</id>
<head>Arranging an herb or flower to cast</head>

<ab>
Always put the front side of the thing on the upper side because it will come out better and neater than the lower side. Make the casting large & prepare[d] the supraplus, which is the mass, after the <pro>founders</pro> of large castings[e], even for <m>silver</m>, which needs a very hot mold & is cast very hot. Very fine thing must be cast very hot.</ab>

<ab>
<margin>left-bottom</margin>

When you know that the stem of a flower burned in a mold that will not open at all is totally reduced to ashes, this is the sign that everything else is well-burned.</ab>
</div>
[a]or the head of a nail? translates "chaple"
[b]See note for "chisel"/Fr. "chaple" above
[c]despinette - does spinet refer to the instrument that is like a harpsichord?
[d]Very general translation of "fais"
[e]This translates the somewhat obscure French: "fais...que le supraplus (qui est la masse) aux fondeurs de grand gect..."

.....

<page>157v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f320.image</image>

<div>
<id>p157v_1</id>
<head>Ashes in the moulds</head>

<ab>You must let the mold cool down rather than blow into it to try to get the ashes out. Because when the mold is hot, the ashes stick to it, and when it cold, they separate from it more easily, and will exit by blowing with le vent a vapeur.[a]</ab></div>

<div>
<id>p157v_2</id>
<head>Moulding a <al>bird</al></head>

<ab>You must give it its pose. And to do so, place it on its side on a <m>clay</m> base and hide it in the aforesaid <m>clay</m> base and sink half the aforesaid <al>bird</al> in it, so that it is only showing one leg and foot, then you will apply <m>butter</m> to keep the feathers down and then apply some <m>spirits</m>. Then cast it. And having taken hold, clean and uncover what needs to be uncovered. Because the feet are made of bones which are hard to burn, and which will not be able to be cleaned neither with <m>quicksilver</m>, nor by any other means, if the mold does not open, uncover the bottom of the foot and the toe, or the nail of the claw, and mold

^{it with}a separate piece. Make a second cooking and open your mold and clean the bones. Feet generally remain ^{too} weak to support the a heavy <al>bird</al>. Because of this one adapts an <m>iron</m> thread in the foot mold which goes through the body before you cast. In this way they are reinforced. Some mold the wings in a frame. Some mold the head and the aforesaid wings separately en noyau, then reattach and repair them to start shaping ^{<al>the bird</al>}, which will firstly have a rigid shape and support the tempered sand.</ab>

<ab><margin>left-middle</margin>They are skinned and filled with <m>cotton cloth</m>. And for the best result, you must prepare it in the way that <pro>leather-workers</pro> do, so that they do not lose their comely feathers. The preparation is made of <m>alum</m> and <m>flour paste</m>.</ab></div>

<div>
<id>p157v_3</id>
<head>Po{ur} f{air}e h{...} vin f{...}rs [?]</head>

<ab>Prenez un chausson du pied droid <add>t</add> qui ait este porte et le faictes<lb>
trempier en <m>eau</m> et de l<m>eau</m> q{u}i en sortira la f{air}e boyre au mesme<lb>
ou autre et v{er}rez merveilles</ab></div>

<a>We think this means blowing very strongly but we don't want to assumer this.
See Godefroy: <http://micmap.org/dicfro/previous/dictionnaire-godefroy/110/1/adouber>.
<c>Marc is working on this.

.....

<page>158r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f321.item.zoom></image>

<div>
<id>p158r_a1</id>
<head><m>Mercury</m> in molds, for cleaning</head>

<ab>
Take heed that your mold be quite cold when you blow in order to get the <m>ashes</m> out. And afterwards put in <m>ŷ</m> in order to finish the cleaning. Because if the mold is hot, the <m>ŷ</m> will go inside & seek out the conduits in escaping, & will leave its own odor, which will aggravate the <m>tin</m> or other <m>metal</m> that you will cast, such that it will break in touching the mold at all. The heat of the mold also retains some little grains of <m>ŷ</m> that will make lumps & stick to the sides of the molded leaves and attach to them & make them frangible. The <m>ŷ</m> cleans ashes well. Likewise flat things where there are not delicate traces that it could break with its weight. And therefore, if you can cast neatly without putting it in, don't use it at all. But if you have need of it, mix your <m>crocum sand</m> so that it withstands fire well. And after having put in the <m>ŷ</m>, evacuate it, bouncing the end of the mold from below while moving it. Afterwards reheat your mold gently so that the <m>ŷ</m> is gone from everywhere. In this way I cast a branch of periwinkle leaves & flowers very neatly. Having put a branch of melted [illeg] in the flower, on the back of the flower [illeg].</ab>

<ab>
<margin>left-bottom</margin>

Reheat your mold until it is good and red before casting, so that the <m>ŷ</m> evaporates well.</ab></div>

.....

<page>158v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f322.image></image>

<div>
<id>p158v_1</id>
<head>Colors for green leafs</head>

<ab>One usually paints them with oil colors, because distemper colors do not stay on. For marigold flowers, lightly ground minium for some of them; for more yellowish ones, mix in a bit of massicot. For green, the vert-de-gris is dark<a> and too somber. If it is a yellowish-green, you can mix with the vert-de-gris a bit of yellow ochre and scudegrum. If the green is dark, mix in some coals made from peach pits, which makes a greenish-black, in the same way than the bone of an ox foot bone makes a bluish-black. And in such a manner, by judgement and discretion, put the color on the natural flower or leaf to see whether it is similar to the original. But paint it on very lightly so as not to cover the features of the work.</ab></div>

<div>
<id>p158v_2</id>
<head>Large molds</head>

<ab>Because of the heaviness of the metal, it is necessary to put them in a press so that they do not open.</ab></div>

<ab>It allows the piece to be taken out more easily, and even if it is slightly thick, it does not stick because it has no substance. I have perfumed my core molds this way for tin and lead, and I have casted in a cold mold, but drying it well beforehand, and the piece came out quite cleanly. It is true that lead mixed with half the quantity of tin, because the medal was very thick, was very hot.</ab>

</div>

<ab>If the medal is very thick, do not pour a cast that is as thick; the cast just needs to be half as thick as the medal. But if it is thin, make a cast that is of the same thickness. Do not forget to make vents in the cast or the whole matter will fill with bubbles.</ab>

[a]"a fonds" in the original
[b]I kept it as in the original: the translation in the vocabulary is a bit strange.

<div>
<id>p159r_a1</id>
<head>
Portraits in <m>Wax</m></head>

</div>
<div>
<id>p159r_a2</id>
<head>
Chasing tool</head>

```
</div>
<div>
<id>p159r_a3</id>
<head>
<m>Fish glue</m></head>
```

```
</div>
<div>
<id>p159r_a4</id>
<head>
<m>Tin</m> and <m>lead</m></head>
```

<ab>
They must be beaten out very finely and <m>tin</m> will not get brittle or break, <m>lead</m> is a bit fatter. But that is why it is necessary to reheat it under hot ashes, and to beat several leaves together.</ab>
</div>

[a]Cotgrave: "Ceruse, or, white lead, wherewith women paint; differs from Lithargie (called also, white lead) for this is made of the grossest lead, as it is in the mine; that, of lead refined, out of the mine."
[b]i.e. scratched the surface?

<page>159v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f324.item.r=</image>

<div>
<id>p159v_a1 [a]</id>
<head><m>
</m>Cleaning files</head>

<ab>Sometimes they [b] get fat from [c] the grease [d] of <m>lead</m>, which is naturally fat, or also filings corrode them. And you will be able to clean them with hot <m>charcoal</m> or a wire brush made from <m>brass.</m></ab>

</div>
<div>
<id>p159v_a2</id>
<head>Carnation</head>

<ab>Mold a carnation like a rose, marigold, and all other flowers as discussed, that is to say cast the flowers in one mold, and the leaves in two parts in order to solder them onto the flower, this is the better way. But you could mold the flower and the leaves together in one single closed mold. I tried one which came out well from the mold. But the <m>sand</m> must be very thin, and you must blow very strongly.</ab>

<figure>
<id>fig_p159v_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5LXNkdWdBLXZlRVk</link>
<margin>left-middle</margin>
</figure>

</div>
<div>
<id>p159v_a3</id>
<head>Molds</head>

<ab>Try to make one side as thick as the other, that way the two parts will be equally reheated. You can open this kind of mold in order to clean it, when you mold some animals, like crayfish, which burn. When your mold is reheated, do not wait too long before casting again, because it gets damp and loses its strength.</ab>

</div>
<div>
<id>p159v_a4</id>
<head>Unmixed <m>plaster</m></head>

<ab>If you reheat <m>plaster</m> as a stone it would crumble into <m>water</m>, but if you grind it before reheating it, it will not crumble into <m>water</m>, on the contrary it will harden in water, if it is good plaster like the one from Paris or from Spain which is as hard as a stone, and which is found through poor and dry earths, and which looks like white <m>salt</m>. German people use this <m>plaster</m> to make statues for their fountains, even if the statues are varnished they will not be damaged with <m>water</m>. On the contrary harden this plaster with <m>water</m>. When this <m>sand</m> is unmixed it sets much sooner. Oil very lightly your shapes of metal or of other materials otherwise you will not be able to strip <m>plaster</m> from the mold. And dip your mold into hot <m>water</m> in order to open it, if it does not open in hot <m>water</m>, it could open into boiling <m>water</m>.</ab>

</div>
<div>
<id>p159v_b4b</id>
<ab>It is sometimes even necessary to boil the mold in hot <m>water</m>, as you would with <m>wax</m> if you were afraid it would not strip well.

Cold water draws oil out to the surface of the plaster that has sucked it in and thus it strips.</ab>
</div>

[a]This translation should be checked by someone with better French skills than me.
[b]?corresponding to "elles"
[c]par le moyen de la = "by means of"
[d]graisse

<page>160r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f325.item.r=.zoom</image>

<div>
<id>p160r_1</id>
<head>Presses for large moulds</head>

<figure>
<id>fig_p160r_1</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5ZDNJT25PWjlUbGM</link>
</figure>

<figure>
<id>fig_p160r_2</id>
<link>https://drive.google.com/open?id=0B9-oNrvWdlO5MEUzMjhxUV11ZTA</link>
<margin>right-top</margin>
</figure>

<ab>It is made of two sheets of iron, fasten with four small iron pillars, in such a way that the upper sheet is able to move and run freely along the pillars, while the lower one is fixed. A St André cross, made of iron, stops at the end of the pillars; there is a screw in the middle of the cross, which squeezes the sheets against the molds placed between them.
For small molds, we make a frame; having put the mold between two metal sheets, we squeeze it in the frame with for small moulds, put the mould between two iron sheets, and squeeze it with wedges.
Big wooden presses, with a screw, apart from being heavy to move to the furnace for casting, make it difficult to see when they squeeze too much, and they often break the molds.</ab>

<ab>The screw is fixed to these nuts[a] which are nailed down the upper plate, this way it raises and presses the mould.</ab>

</div>
<div>
<id>p160r_2</id>
<head><m>Sand</m> to cast <m>flowers</m></head>

<ab>In casting with thick <m>sand</m>, <m>flowers</m> crumple; it squeezes them into a mass. For this reason, be sure to dilute your <m>sand</m> very thinly, and blow on it so the <m>flowers</m> are not entirely covered. When you dilute your <m>sand</m>, do not only stir it with the small shovel[b], but beat it as you would <m>egg white.</m></ab>
</div>

[a]escrou: Cotgrave says it can be the screw OR the whole in which the screw fits. In this context, "nut" makes more sense as the holes in which the screw moves.
[b]pallette. Cotgrave translates it as "a small shovel". Could also be a "palette".

.....

<page>160v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f326.item.r=</image>

<div>
<id>p160v_a1</id>
<head>Moulding a foot or a hand</head>

<ab>Mix <m>resin</m> to the <m>wax</m>. Once well blended, soften the mixture in hot <m>water</m> or <m>urine</m>. Then press [the foot or the hand] in the mixture in order to get an impress, then pour <m>plaster</m> the cast will which will be clean and come off easily.</ab>

</div>
<div>
<id>p160v_a2</id>
<head>Fine work made of <m>gold</m> or <m>silver</m></head>

<ab>You can mold flies by hand,without casting; the wings and the feet can be soldered, which you could not do with fine works of <m>lead</m> or <m>tin</m>, because they would melt. You can enamel the work made of <m>gold</m>.</ab>

</div>
<div>
<id>p160v_a3</id>
<head>Catching crayfishes</head>

<ab>On the end of a stick, attach pieces of liver or lung, from beef or mutton. Then hang a basket to the stick with a little string. Put the basket into water where crayfishes eat; they will soon aggregate on the flesh. Pull very gently your stick out of the water; the crayfishes which will try to return to the water will fall into the basket.</ab>

</div>
<div>
<id>p160v_a4</id>
<head>Catching passerines</head>

<ab>If they sleep in the farmyard, like they commonly do, close to the house, let the closest window open, and put a lighted candle close to the window. The candle itself must be hidden, only its glimmer must be seen through the room and through the window; all the passerines will then lock themselves into the room. But the night must be very dark, without any moonlight.</ab>

</div>
<div>
<id>p160v_a5</id>
<head>Catching birds</head>

<ab>During winter, when birds have molted, skin them and fill them with cloth and stuffing, or dry them well in an oven. Then arrange your birds on trees, have some [look

like they are] singing, [look like they have assembled] and are numerous.</ab>

<ab>the oven must not be too hot. This [method] is good in the summer, because flies would set into the eyes and make works before they [birds] could dry. You can work on the birds more easily in the winter, because they dry by themselves.</ab>
</div>

.....
<page>161r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f327.item.r=</image>

<div>
<id>p161r_1</id>
<head>Preparing <m>sand</m> for box molds</head>

<ab>After your <m>sand</m> [which is made] of <m>gip de lateribus</m>[a] and <m>alumen jameni</m>[b] is moulded as a core, gather the left-over pieces of your mould, and break them up on a long table with a big metal block until they are very well crushed. And for an even better result, sieve them neatly. Place them in the oven pots of <pro>tile-makers,</pro> in a bread oven, [firing them] several times, when it is well heated so that they become very red. When cooled down, rebreak them if you need to and sieve them. After, put them in a vaisse.[c] a semal[d][e] or a vessel that is appropriate and clean. And then wet it with clear <m>water.</m> And crush it and mix it with a big stick until it is very well-moistened and rinsed, and the dirty parts rise to the surface of the water. Keep stirring in this way the dulled <m>gip</m>, that one prepares to gilt burnished <m>gold</m> over five or six days, until it is in no way lumpy. And each time your stir it, leave it to rest and cull the clear <m>water</m> that will be on the top by tipping the pot, or with a porringer, or with a sponge. And put in some new clean <m>water</m> and in this way wash and stir until it is refined and purged of all filth. Then[f] leave to drain a little bit and to dry and make of it some balls, soaked in <m>sal ammoniac</m>. Having been thus dried, put them to be recooked in very high heat in a reverberatory furnace for a good amount of time, where they will become very red. Having been recooked in this way, crush them and try to mold something in a box mold, having moistened your crushed powder with <m>water of sal ammoniac</m>. And reheat and redden you box mold, and if your <m>sand</m> comes away, then you must crush it again and moisten it again with <m>water of sal ammoniac</m>. Remold it in two balls and recook it and redden it and continue this for as many times as necessary until it does not come away from the box mold when the box mold is reddened. The main thing is for it to be well recooked, because otherwise the <m>sal ammoniac</m> will not calcinate well and will make the <m>sand</m> coarse. And for this reason, you should not put in too much <m>sal ammoniac</m>.</ab>

<ab>
<margin>left-top</margin>
<figure>#</figure> Note that before drying it, you must first wet the whole thing and empty the liquid in another ordinary vessel, taking out the <m>litharge</m>[g][h], so that the <m>stones</m> and the <m>earth</m> remain at the bottom afterwards. You must leave it to rest and then empty the <m>water</m> by tipping it, and then leave it.</ab>

<ab>
<margin>left-bottom</margin>
nota</ab>
</div>

[a]Latin: brick mortar
[b]Latin: aka alumen album. "Alumen album fit iuxta mare in parte dextra seculi, et dicitur alumen iameni." See: Artis Chemicæ Principes, Avicenna Atque Geber
[c]We couldn't find this word but it must be some kind of vessel.
[d]This is a vessel: propre a porter la vendange et d'autres usages. See:http://books.google.com/books?id=gwdEAAAACAAJ&pg=PA535&lpg=PA535&dq=%22sema1%22+%22vaisseau%22&source=bl&ots=2oW7Nc821Y&sig=h6lou5b4XP6uMjMcR12vDiNCcsQ&hl=en&:YGIAw&ved=0CEMQ6AEwBQ#v=onepage&q=%22sema1%22%20%22vaisseau%22&f=false
[e]also found in Huguet : Benne pour porter la vendange.
[f]Need the superlinear ref. to the note encoded in the translation.
[g]Cotgrave says this is white lead.
[h]It probably is white lead, which is predominantly lead carbonate (2PbCO3-Pb(OH)2). Litharge also sometimes refers to PbO, i.e. lead(II) oxide.

.....
<page>161v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f328.item.r=</image>

<div>
<id>p161v_a1</id>
<head><m>Crocum ferri</m></head>

<ab>It can be made quickly if you put very filings that are very neat and have been washed in an <m>iron</m> drum that is only two fingers high, and cover it, and redden in a reverberation furnace, having washed it in good <m>vinegar</m> for a whole day. And it will be very well burnt and clean, mixed and crushed on a <m>marble</m> slab. Having been well crushed and when well mixed, the <m>vinegar</m> will corrode it easily and will redden it quickly and bind it. Wet it with the strongest one [a]you can find and burn it after. And after three or four wettings and burnings it will be ready. This one will be the color of columbine[b] and is considered firmer for casting than the one that is redder and more beautiful in color, done with rusty filings and has more color[c] in it. Some burn the filings many times on a shovel of <m>red iron</m>, watering each time with <m>vinegar</m>.</ab>
</div>

[a]implied: vinegar.
[b]Dictionnaire de l'Académie, 1798 : "Qui est d'une couleur mélangée entre le rouge et le violet, approchant du gris-de-lin. Taffetas colombin. Soie colombine. Couleur colombine. Ce mot est vieux: ont dit aujourd'hui Gorge de pigeon."
[c]Is this the meaning of taintures in this sense?

.....
<page>162r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f329.item.r=</image>

<div>
<id>p162r_a1</id>
<head>For La Boutique</head>

<ab>
According to Herodotus the Portuguese did not invent [deep sea] navigation as they put about. At the beginning of the first book entitled Clio, he says that the Phoenicians used to come navigating straight across [a] from the Red Sea to the Greek coast, mainly to Argos, where they would bring merchandise from Egypt and Assyria, which they usually sold out within 6 days[b].</ab>

<ab>
<margin>left-top</margin>

Navigation, commerce</ab>

<ab>
Herodotus, page 1 of Clio, says that the Greeks traveled in a long ship to the coast of Colchis and to the River Phasis, where they carried off Medea[c].</ab>

<ab>
<margin>left-top</margin>
Galleys.</ab>

<ab>
Herodotus, page 3, says that Gyges offered to the temple of Delphi "pateras[d] aureas sex pondo 30 talentorum".[e]</ab>

<ab>
<margin>left-bottom</margin>

Gold vases</ab>

<ab>
Halyattes, the father of Ctesias, "pateram ex ferro compactilem apud Delphos dedicavit spectatu dignam inter omnia quae sunt Delphis donaria, opus Glauci Chii qui solus omnium compactionem ferri excogitavit[f]." Compactile ferrum means made by assembling separate parts.</ab>

<ab>
<margin>left-bottom</margin>

Iron vase assembled and soldered</ab>

<ab><m>
</m>As small peddlers lay open small wares in order to buy more precious ones & to make more and more profits, I also, from a desire to learn, am laying open the little that is in my <fr>boutique</fr> in order to have through receive through a common[g] <fr>commerce des lettres</fr> much rarer secrets from my benevolent readers.</ab>

</div>
<div>
<id>p162r_a2</id>
<head>
Aes ustum </head>

<ab>
When it is finely ground [h]it moulds very cleanly. It has a certain fatness that makes it bind better than filings, I believe that it may not be ill suited for casting.</ab>

</div>
<div>
<id>p162r_a3</id>
<head>
Rotten wood</head>

<ab>
Wood that is white, being light as a sponge, if burnt in a close fire, may be used for casting lead in a box[i][j][k] and molds very cleanly, but these things [such material] do not withstand [open?] fire[l].</ab>

</div>
[a]"continuellement" here means non-stop navigation in deep sea waters, as opposed to the usual way of travelling along the coast
[b]Herodoti Halicarnassei Historiae lib. IX, & de vita Homeri libellus ; illi ex interpretatione Laur. Vallae adscripta, hic ex interpret. Conradi Heresbachii, vtraque ab Henr. Stephano recognita..., H. Estienne, 1566, p. 1
[c]Ibid.
[d](We could say "presented the temple with", but that would be awkward with the accusative)
[e]Ibid., p. 3.
[f]Ibid., p. 5. Maybe we should translate the Latin bits as footnotes? (here and elsewhere?)
[g]common, not in the modern sense of the word, but as in 'the common good'.
[h]is there a way of distinguishing pilé/pulvérisé? reduced to powder? Not to mention "subtilié" which is elsewhere.
[i]I believe he is making "sand" once more, this time from the ashes of this wood, is that possible? I don't understand how the material of the box might influence clean molding?
[j]The white wood ash "sand" will allow the lead object to come out of the mold cleanly.

[k]right, it's not about the box itself (what was I thinking?)
[l]if I understand this rightly, he means burn it in a close fire, not an open fire, which presumably would completely burn up this kind of wood leaving no ashes? Calling the wood "things" is strange though (I would have expected "things" to be casts).

.....

<page>162v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f330.item</image>

<div>
<id>p162v_1</id>
<head>Peach trees</head>
<ab>Because they cast [a]and blossom from the outset, and so are caught by the frost. We expose their roots during winter so that the cold slows them down, and they blossom later, which benefits their flowers.</ab>
</div>

<div>
<id>p162v_2</id>
<head>Olive trees</head>
<ab>In <pl>Spain</pl> they expose their roots during winter and in spring they earth they up again.</ab>
</div>

<div>
<id>p162v_3</id>
<head>Foot of <m>bittern</m></head>
<ab>To mold it it is better to extend its fingers and nails on something flat and let them dry this way for a few days, because in drying the scales will be rougher and the nerves and and tendons will be more apparent, and so the molded foot will be more artistic.</ab>
</div>

<div>
<id>p162v_4</id>
<head>Molds of things which have not been<lb>pelted[b]</head>
<ab>Animals who have not been pelted can only be molded in a double mold, which can only be opened after they have been refired and after the enclosed animal is completely burnt. Animals with bones or scales which are not reducible to powder thus remain in calcined pieces which will never removed from the cast, however much bellowing or whatever amount of quicksilver is put in, or by using a feather pipe while molding to make a gate. This is why you mold in two halves: so that once the mold is opened after having been refired, the thing you've burnt can be cleaned out. But I advise letting it cool so that in taking off the clamps (being hot) nothing cracks while it is still fragile. While it is hot the crust and cinders which it leaves and which stick to the mold can be more easily removed when it is cold. Also note that on the side of the lower mold when you reheat the animal, which in boiling leaves a crust, in this case, always mark on the mold where the back of the animal is so that in reheating it is to the top and above and will by this method stay cleaner.</ab>
</div>

[a]jectent
[b]en despouille

.....

<page>163r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f331.image</image>

<div>
<id>p163r_1</id>
<head><pro>Perfume-makers</pro></head>

<ab>They readily make their perfumes with half <m>amber</m> and half <m>musk</m>, and a little <m>civet</m>, because the scent of <m>amber</m> is stronger than the scent of <m>musk</m>.</ab>

<ab>To save using <m>amber[a]</m>, <pro>perfume-makers</pro> readily add add a small quantity of <m>musk</m> to the white layers,[b] in order to get a stronger scent. But to remove or hide the blackness of <m>musk</m>, they add a bit of <m>starch</m> from <pl>England</pl>, which is perfectly white.</ab>

<ab>To give off a scent of <m>white amber</m>, in the same way that the <pl>Portuguese</pl> do, take a chunk [c] of <m>amber</m> that you have broken up into pieces. Having put in a small <m>silver</m> porringer a spoon of <m>flower oil</m>, should you not have some, <m>linseed oil</m>.[d] That is to say, use the type of <m>silver</m> spoon one might use at the table. Pour into this your <m>ambergris</m>, then place it on a low fire, it will melt quickly if your <m>amber</m> is of good quality, and there will be no lumps. Having melted it all, put in a small bit of <m>civet</m>, and melt it all and blend it well together. Then take your gloves which you have cleaned and dried properly, and lightly dip the edge of your finger around the edge of the <m>oil</m>, with patience slowly spread the <m>oil</m> along the glove, and rub the glove between your hands and trace along the fingers and its stitches, one after the other. Leave it to dry. Then repeat the previous procedure until you have used up all of the <m>amber</m>.</ab>
</div>

<div>
<id>p163r_2</id>
<head><m>Spirits</m></head>

<ab><m>Spirits</m> are excellent for making molds clean, in such a way that where you do not apply some, common sand will attack the figure. But it needs to be passed three times. And because it does not take well on oiled things, I believe it may be best to put some into the <m>water</m> with which you have moistened your sand.</ab>
</div>

[a]See vocab for ambre blanc.
[b]We assume that here he is talking about building up a scent and the white layers are the first coats of white amber referred to above.

[c]Totally not sure about this translation. In French the word is "huchan" and we can only imagine that it means a chunk, since it is subsequently broken up into smaller pieces.
[d]The transcription seems to say "l'huile de bien" but we think that perhaps this should be "l'huile de lin"?

.....

<page>163v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f332.image></image>

<div>
<id>p163v_1</id>
<head><al>Crayfish</al></head>

<ab>The hairy little horns that it ^{the <al>crayfish</al>} has under its snout tend to come out lumpy and unclear because, after they are burnt, it is difficult to remove and clean away their ash, which, staying in the mold, prevents the <m>metal</m> from running well. And for this reason, one makes a small gate of <m>wax</m> thread at the tips of them, in order both to hold them up and to blow away the cumbersome ashes from inside.</ab>

<ab><margin>left-top</margin>Do not spare any pins, placing them not only in the middle of the body, but also on the big legs, and two or three on the tail, according to the positioning that you want to give it.</ab>
</div>

<div>
<id>p163v_2</id>
<head><m>Crocum</m></head>

<ab>That which has become rusty by itself, having been doused with <m>water</m>, <m>salt</m>, and either [a]<m>vinegar</m> or <m>urine</m>, and then well-dried and reddened by heating, is deep red when crushed on <m>porphyry</m> and is of the color of <m>bol de levant[b]</m>, almost like <m>minium</m>. But that which is doused with <m>urine</m> and dried acquires a deeper color and almost like crushed <m>aes ustum[c]</m> or like <m>vermillion</m>. But the first one, finely crushed, acquires a bright red color like <m>cinnabar</m> when placed in <m>spirits</m>. And the one and the other, when prepared as said above, finely ground and made very hot, produce a red smoke like an acid vapor[d] if, being thus hot and fine, one throws <m>vinegar</m>, <m>urine</m> or <m>spirits</m> on top. The <m>urine</m> makes a lot of color and the <m>spirits</m> as well.</ab>

<figure>
<id>fig_p163v_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5anJvd0FQQ3BkRFE></link>
</figure>
</div>

<div>
<id>p163v_3</id>
<head>Repairing</head>

<ab>It is much better to only lightly repair your work with a burin called an "onglet" [e]or the one called "chaple" or with a small file. Above all, avoid touching your work, but touch only the burr or lumpy parts that will have occurred during the casting. While reworking, moisten and rub your work with a small <m>bristle</m> brush.</ab>
</div>

[a]Seems to be the only possible distribution of ingredients. The original French "ou .. ou" is awkward and ambiguous,
[b]Bol seems to be a generic term for a type of clay earth with iron oxide. This type perhaps comes from the East? It's in a lot of 17th/18th-century works but I couldn't find the exact translation. For "bol" see: <http://www.cnrtl.fr/definition/bol>
[c]patina (often on bronze) predominantly composed of copper oxide.
[d]Un esprit volatil apparently can be made from a number of things, but I think it's basically a vapor or steam or distilled-like substance. For a definition of different types see http://books.google.fr/books?id=2NVTAAAAAYAAJ&pg=PA96&lpg=PA96&dq=%22esprit+volatil%22&source=bl&ots=Xne7LS2vbl&sig=bnNo5z6l_aOampDV81zDpe7tH6c&hl=fr&sa=X&ei=tXfLL
[e]For def. see: http://portail.atilf.fr/cgi-bin/getobject_?p.83:97./var/artfla/encyclopedie/textdata/image/

.....

<page>164r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f333.item></image>

<div>
<id>p164r_1</id>
<head>Casting</head>

<ab>
When you make a cast of delicate animals, like <al>snakes</al>, start by casting the belly side, or the under-tail, for this side has straighter lines which are easier to rework than the back side, which is more veined and is more marked.</ab>
</div>

<div>
<id>p164r_2</id>
<head>Repairing burrs</head>

<ab>
Remove burrs very carefully with the point of a chaple or a burin, then scrape the burrs with the side of a burin, or carefully with a small file, and rub with <m>willow coals</m> and small brushes. Continue, also, with the lines not interrupted by burrs.</ab>
</div>

<div>
<id>p164r_3</id>
<head><m>Leaded</m> <m>silver</m></head>

<ab>
A lot of ^{<m>silver</m>} is wasted because of the <m>lead</m> which is mixed with it, as occurs in cupellation, and makes it leap in small bits onto the edges of the crucible and onto the <m>charcoal</m> which covers it and is also gross. And by this means, it is good to melt <m>silver coins</m>, like reals and others, and make them into ingots prior to casting <al>lizards</al> or <al>animals</al>, for it comes out better. I cleanly molded a small, <m>silver</m> <al>viper</al>, like in the first ^{recipe above}. And at the end of the line of 4 <m>reals</m> (of 20 S of <pl>Spain</pl>), and one <m>coin</m> of 20 S from <pl>France</pl>.</ab>
</div>

<div>
<id>164r_4</id>
<head>Blowing</head>

<ab>
When you smelt <m>silver</m> and <m>gold</m> do not blow too hard or with force, because the <m>charcoal</m> will become spent and the crucible will sink and, potentially, spill over. But when your mold is ready, blow strongly in order to properly heat the <m>silver</m> or <m>gold</m>. When all are melted it is necessary to blow over the top with a small bellows, doing the same for <m>gold</m>, for it removes and reduces the smoke.</ab>
</div>

.....

<page>164v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f334.item</image>

<div>
<id>p164v_1</id>
<head>Molds</head>

<ab>
For molds wherein something must be burned (away), and do not open before being reheated: do not cast before refiring/reheating. For large molds that produce large and wide casts, it is necessary to score it with strong notches so that the <m>pitch</m> of the metal does not run as forcefully. These will not make many burrs if they are pressed. Prior to doing so, place between the mold and the press pieces of thick <m>felt for mittens</m>, which will fill the cavities and protect the mold from cracking. The clamps should be placed prior to heating or drying.</ab>

<ab>
<margin>top-left</margin>
When the ground from which you are making your earthen base to lay out your <m>animal</m> is too soft, the pins you place come out easily when you cast the sand and so the mold is spoilt and the design is defeated[a]. Therefore when you know that your slab is too tender and soft, spread some hot cinders over it and blow with the bellows.</ab>
</div>

<div>
<id>p164v_2</id>
<head>Sable</head>

<ab>
If you don't add enough <m>alum de plume</m> or <m>crocus</m> ^{your mold} will easily crack when heated. But if your ^{sand} is mixed as it ought to be, ^{the mold} will sustain itself. You'll know it's of good quality when, while still hot after casting, it immoderately soaks up the <m>water</m> when dipped and breaks easily, because the <m>alum</m> and the <m>crocus</m> render it spongy. To the contrary, bad sand, which is composed of <m>plaster</m> and <m>brick</m> and is not well mixed with <m>alum</m>, breaks easily in fire and hardens in <m>water</m>. If possible, cast all at once.</ab>
</div>

<div>
<id>p164v_3</id>
<head>Molds</head>

<ab>
When you cast <m>lead</m>, your molds must be very dry, even if the molds are large. The outside of the molds must be reddened, and it is necessary to keep them in continuous heat until they do not smoke anymore inside. Do not reheat your molds several times, but continue when you have begun and do not let them cool in the cold because they will crack. Let them cool near the fire. When your molds are still as warm as above, remark, you can put your finger into the casting gate without burning your skin. If your mold is large, it takes more time to cure and to dry, and you must also cast warmer.</ab>
</div>

[a]the original french is very confusing, and makes use of the word "seulement" in a way that was difficult to translate and makes little sense in the original context.

.....

<page>165r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b9059316c/f336.item</image>
<div>
<id>p165r_1</id>

<head>To repair</head>

<ab>If in your cast, there appear some little hole, fix it[a] with black or gray filling[b] <m>gum wax</m>.</ab>
</div>
<div>
<id>p165r_2</id>

<head><m>Lacquer</m></head>

<ab>If, having been ground, you let it[c] dry without keeping it in <m>water</m>, it will dry out and it will be quite difficult to crush it as before.</ab>
</div>
<div>
<id>p165r_3</id>

<head>Stretching a <m>canvas</m> portrait</head>

<ab>If it is crumpled or folded from when it was rolled up, moisten the back side with a wet sponge. Then stretch it very evenly without damaging it.</ab>
</div>
<div>
<id>p165r_4</id>

<head><m>Plaster</m> for molding</head>

<ab>When it is long to set and dry, it means that it is too fat and moist, which is what makes it shrink away from the fire and crack. [This is] where burrs come from. The one that comes from lean <m>earth</m> is better. You must not store your moulds made of fat <m>plaster</m> for a long time, because they resemble fat <m>earth</m>, which when drying, cracks. The best is to reheat them[d] soon after they have been made and cast. Moulds made of good <m>plaster</m> can be stored a long time.</ab>

<ab>
<margin>left-middle</margin>
It is necessary to <m>oil</m> it well, because it is more difficult to separate the two halves of these moulds than when it is mixed.</ab>
</div>
<div>
<id>p165r_5</id>

<head><m><al>Dragon</al>'s blood</m></head>

<ab>You can imitate <al>dragon</al>'s blood with <m>lacquer</m>, which surpasses it in beauty, if [when] diluted in <m>oil</m>, you use it to ice <m>gold</m> or <m>silver[e]</m>. Diluted with varnish, it fades.</ab>
</div>

<div>
<id>p165r_6</id>

<head>Repairing <al>snakes</al> and <al>lizards</al></head>

<ab>You could make some marks[f] according to the form of their scales, making some bigger ones and some smaller ones, according to the shape of the neck, the body and the tail. If there is something broken, you can attach[g] it with small <m>iron</m> or <m>steel</m> nails, and glue it with some <m>mouth glue</m>[h], then cover the whole thing with gray or black <m>wax</m>. But it is necessary to let the <m>glue</m> dry for one day without touching it.</ab>
</div>
[a]I think "with a knife" is implied here?
[b]esbaucher: cotgrave: trick. Meaning here to fill.
[c]...meaning the laquer...
[d]...meaning the molds I think...
[e]Not entirely sure this is the meaning here. Cotgrave seems to want to stick to the "cold" element of "glacer," not the meaning of icing a cake, but its seems that's what makes more sense here.
[f]Cotgrave says a poincon is a bodkin. He says it can also be a mark so I'm thinking here that these might be marks made with a bodkin? in order to replicate scales?
[g]hanter: recouvrir, [attacher]
[h]Colle de poisson: mouth glue, water glue (cotgrave)

.....

<page>165v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f336.image</image>

<div>
<id>p165v_a1</id>
<head>Positioning animals</head>

<ab>Not only with iron pins, but also with melted wax, are animals secured to the plate of earth. Wash them first in clear water, after they have died in the mixture of vinegar and urine. And with a sponge clean off their mouth, eyes and head of the froth coming out, because the sand would never set well there and would become lumpy.
</ab>

<ab>
<margin>left-top</margin>
If you want to keep their mouths open, insert a little bit of harpsichord string, which will not take up much space.</ab>

</div>
<div>
<id>p165v_a2</id>

<head>Keeping animals</head>

<ab>Having molded them once, if you want to keep them for molding again, put them back into the mixture of urine, vinegar and spirits where you killed the others, and they will keep for a long time without decaying. If you have a cut on your hand, take care that this mixture does not touch it.</ab>

</div>
<div>
<id>p165v_a3</id>
<head>Molds</head>

<ab>After molding them do not keep them for long, but use them as soon as you can, because when kept for a long time, they are prone to spoiling.</ab>

<ab>When your molds are reheated, the thing comes out better.</ab>

</div>
<div>
<id>p165v_a4</id>
<head>Lute</head>

<ab>When reheating, you might as well use common lute, But when you want to cast, strip it from the mold and, using a brush, quickly apply a layer of plaster lute that has been used for molds, because it will be dry as soon as it settles, and cast. This lute should be mainly placed on the joints.</ab>

</div>
<div>
<id>p165v_a5</id>
<head>Silver</head>

<ab>When it quivers after melting, that is a sign that it is very hot. And since it commonly contains much lead, it is good to throw in some lime mortar because that attracts and gathers in the lead.</ab>

</div>
<div>
<id>p165v_a6</id>
<head>Fly wings</head>

<ab>So that they come out best, place wax sprues from the gate up to the tips of the wings. Also make sure there is enough wax under the belly, because that is what makes the cast succeed.</ab>

<figure>
<id>fig_p165v_1</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdlO5UmhqY1RkWFRhYzg></link>
<margin>left-middle</margin>
</figure>

</div>
<div>
<id>p165v_a7</id>
<head>Attaching a fly wing or something similar</head>

<ab>They can be attached with fish glue mixed with spirits over slow heat, heating also the work. Then that is covered with wax "à ébaucher[a]", that should be grey.</ab>
</div>
[a]See p156v where the last note gives a description of it.

.....

<page>166r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f337.item.r></image>

<div>
<id>p166r_a1</id>
<head>For The Workshop</head>

<ab>
If anything has been omitted: the harvester is not blamed for leaving some corn ears.
If some craftsmen complain that their craft secrets are being discovered, as the goddesses [a]of Eleusis complained to N... the Pythagorean that their mysteries were disclosed, excuse yourself in the words of the ancient Hebrews quoted by Josephus[b].</ab>

<ab>
As a hen will carefully search the rubbish thrown out of a house for a crumb or a seed to distribute to her chicks, so will the orphans receive whatever has been discovered in those arts that are considered vile and despicable.</ab>

<ab>
Tablet of Cebes: idle. But The Workshop [c]represents all things active. </ab>

<ab>

If the jealous will disapprove picking up these things and taking them from others, reply that since "Nullum est jam dictum quod non dictum aut factum sit prius[d]" and it is commonly believed that all arts were invented within a thousand years, do not imagine you could invent anything new but rather rediscover the long-buried, unknown and forgotten books of Numa in order to reveal them to those who are ignorant of them; and that, just as the day before teaches the day after, you also needed to learn from those before you in order to teach those after you. The Latins learnt from the Greeks, as Cicero did from Plato and Virgil from Homer (Aulus Gellius[e]). Valerius Maximus only reawakened various authors who had written previously, and Livy's 23rd book [revived] the 2nd or 3rd book of Polybius. Among the Egyptians, Serapis was deemed the inventor of agriculture, but among the Sicilians, it was Ceres. Pythagoras and others from Tyana learnt during their travels those disciplines which, back in their countries, they claimed to have invented. Does one not call a weaver the maker of his precious cloth or fabric, even though he did not himself dye and twine [the yarn] nor wind and prepare[f] the reels and balls of thread? Is a mason not the builder of the house, for picking up ready-made stones? "Apollo a Chirone medicari didicit & tamen deus medicinae habitus[g]." "Homerus Orphei poema imitatus est, nam cum Orpheus sic prius exorsus esset, 'Iram cane dea Cereris frugiferentis', ita reddit Homerus, 'Iram cane dea Pelidae Achillis'". In Justino martire, Circa princip[h].

[a]Here the author seems to have mistaken "deae" for a nom. pl. instead of a gen. sing. (see the Latin quote on page 1r). Possible evidence of his scarce command of Latin. See also 162r: his quotes from Herodotus are all from pages 1 to 5. At least his Latin spellings are always correct.
[b]Quote needs identification in Flavius J.
[c]I prefer caps here, since in this context it is clearly a title (and we can't use italics because they are reserved for markup).
[d]"There is not one word that was never said or done before". Terence, Eunuchus, Prologue. The author adds "aut factum", since his is a book of facts, not words.
[e]Identify quote in A.G.
[f]for a more technical meaning?
[g]Source? "Apollo learnt medication from Chiron, yet he is considered the god of medicine."
[h](Find reference.) Homer imitated the poem of Orpheus, since Orpheus had begun, 'Iram cane dea Cereris frugiferentis', and Homer rendered this as, 'Iram cane dea Pelidae Achillis' (Justinus martyr, Circa Princ.).

<page>166v</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f338.image</image>

<div>
<id>p166v_1</id>
<ab>In addition to Greek examples, there is a psalm of David mentioned by Saint Athanasius in the Synopsis. It begins "Parvus eram" and in the second verse the royal prophet says, "Manus meae fecerunt organum et digiti mei aptarunt psalterium[a]".</ab>
</div>

<div>
<id>p166v_2</id>
<head>Keeping <m>chestnuts</m></head>
<ab>They must be left in their burs until you want to eat them. And in their burs, store them in a humid place, or in the drain[b] of some roof.</ab>
</div>

<div>
<id>p166v_3</id>
<head>Scented <m>candle</m> from <pl>Le Mans</pl></head>
<ab>They melt in with <m>wax</m> and <m>candles</m> <m>rosin</m> that is called in <pl>France</pl> [rosin] from <pl>Burgundy</pl>, which is white and not as rough or dry as the other <m>rosin</m>. They also perfume the wick and the end of the wick[c].</ab>
</div>

<div>
<id>p166v_4</id>
<head>Softening <m>gold</m></head>
<ab>Dissolve some <m>caput mortuum[d]</m> in distilled <m>vinegar</m>. Filter and congeal in salt and cement the <m><figure>☉</figure></m> with this, and it will soften wonderfully.</ab>
</div>

<div>
<id>p166v_5</id>
<head>Improve the color of <m>gold</m></head>
<ab>Put some <m>aqua fortis made with vitriol[e]</m> on hot embers. Cover the bottle, that is, the vent, with a tile. Then uncover it and a thick smoke will come out in which you put the <m>gold</m> and it will take on an excellent color.</ab>
</div>

[a]Ps. [151] (= Add. Ps. 1), 1-2: "I was a child"; "My hands have made the organ and my fingers have prepared (tuned?) the psalterion".
[b]"agout" - Installation permettant de recueillir les eaux pluviales et les eaux usées, canalisation, égout (DMF)
In Cotgrave, there is "agoutter" (to drop, as rain from a house eave) and "agouttis" (the eavings of a house).

Unless it is constantly raining, the drains would presumably not be wet all the time. Perhaps the idea of a "cistern" is better here, or some other receptacle for catching rain water?
[c]I can't find a good word in English for "lumignon" (Bout de mèche allumée d'une bougie ou d'une lampe à huile). The "snuff" of the wick is the burnt part on the end, but it doesn't seem logical that a candlemaker would burn a candle before selling it or add scent to the burnt part.
[d]It is unclear what the caput mortuum is referring to.
[e]"eau fort de vitriol" - aqua fortis seems to have been made by mixing either sand, alum, or vitriol with alabaster and then distilling it. I am assuming he is saying that aqua fortis made with vitriol is the kind to be used here.

<page>167r</page>
<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f339.image</image>

<div>
<id>p167r_1</id>
<head>Mortars</head>
<figure>
<id>fig_p167r_1</id>

<link><https://drive.google.com/open?id=0B9-oNrvWdlO5QTU2dDJtWHU5Nmc></link>
</figure>

<ab>Mortars are made of the best alloy of <m>"métal"</m> and the finest possible <m>rosette[a]</m> so that they do not burst, that is to say one part <m>"métal"</m> and two [parts] fine <m>rosette</m> or old caudron, which is even better. Some of them are of fifteen or seventeen lb. and are loaded with two and a half lb. of <m>grain powder</m>, and they are put on common windows and doors. Others are of a weight of 25 to 27 lb. and are loaded with eight lb. of powder. And in such a way, they are made according to the stress they have to resist. On the outside, they are all the same shape, but in the inside, they are made like a crucible, </figure>

<id>fig_p167r_2</id>

<link><https://drive.google.com/open?id=0B9-oNrvWdlO5OUZCVWpfdmlzbWc></link>

</figure> narrower on the inside of the bottom and getting wider towards the mouth. And this is to reinforce the bottom, given the quantity of powder they hold, and so that they do not burst. Those of 4 s. lb. are eight lines[b] thick at the bottom, and get thinner in the inside towards the opening, which is of 4 lines. They have xiii "poulsses" long and seven in diameter at the mouth and opening.</ab>

<cont/>

</div>

[a]Cotgrave: a kind of fine copper that comes from Hungary.

[b]"lignes" in the original

.....

<page>167v</page>

<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f340.image></image>

<div>

<cont/>

<id>p167r_1</id>

<ab>

The ones weighing about 27 lb. are ix "pousses[a]" long and the mouth is 6 [pousses] in diameter. At their bottom, they are 6 "lignes[b]" thick, and 3 for the mouth. The ones weighing 17 lb are 8 "poulsses" long and the mouth is five [poulsses]. They are all covered together, all the forks, rods, and tools which one uses, with a big canvas or big cloth so that they do not make any noise[c]. It is good that they sit loaded for a while. After the <m>powder</m> has been put in, it is grinded very hard and then tightly pressed <m>paper</m> is put in, then a cake of well pressed <m>wax</m> and, on the <m>wax</m>, another slab[d] of <m>cork</m> that fits in very snugly and that you have to force in. In this way, the <m>powder</m> remains well packed and gives much greater force, and if you wish, you can put on the <m>cork</m> slab one of wood pierced in the middle, if the mortar has a pierced bottom, which is believed to be the easiest and which makes a greater hole[e] because it cannot recoil. And for these that have a pierced bottom, one must have a good gimlet to first make the hole so that the rod, which is made by the point of the gimlet[f], has before and without noise attached the mortar[g]. And for these, one must cover the button, which is quite sharply filed, with <m>waxed canvas</m>, or add <m>wax</m> to it so that it completely plugs the hole in the bottom. But because the doors are sometimes iron and the iron rods cannot pierce it, an iron fork made like pincers and another iron stick made with three claws, like you see painted, are use. And in this way, the mortar stays in place well and is immediately steadied. The iron rods must come out of the mortar from every point made by the gimlet, namely by three or 4 finger lengths, which all enter in the door. When the mortars are placed, the touch-hole[h] is filled with good <m>powder</m> and inserted in it is a <m>feather shaft</m> filled with tightly pressed <m>powder</m> and moistened with <m>vinegar</m>, or if needed, the powder is firmly pressed with the palms of the hands and, having wet it with some <m>saliva</m>, you shape it like a cocoon or</ab>

<cont/>

</div>

[a]translation?

[b]translation?

[c]"affin qu'il ne fassent point de bruit" - literal translation, but unsure if it conveys the actual meaning.

[d]verify translation - "platine"

Littre: Proprement, pièce plate, dans divers instruments ou ustensiles.

Plaque de fer attachée à une porte au devant de la serrure, et percée pour le passage de la clef.

Verrou à platine, verrou monté sur une plaque de fer.

On dit aussi loquet à platine.

[e]verify translation - "faulcée" (fossé?)

[f]par la pointe EN guimbelet??

[g]Awkward - "aye plustost & sans bruit attaché le mortier"

[h]"lumière" - This is obviously a sort of chamber that holds powder, but I am unsure of the technical name for it

.....

<page>168r</page>

<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f341.image></image>

<div>

<cont/>

<id>p167r_1</id>

<figure>

<id>fig_p168r_1</id>

<link><https://drive.google.com/open?id=0B9-oNrvWdlO5RUdzVWVNM3l6QVU></link>

<margin>left-top</margin>

</figure>

<ab>rod that we put into the touch-hole[a], and then, with a rope wrapped around the end of a baston, we ignite it and promptly retreat to take cover. The small mortars of 16 or 17 lb. are attached[b] promptly with a short, hooked rod made like a wimble[c], and if it is of a good alloy, we can ignite it by hand. It is done in this way for mortars of 27 lb., but then we only put three lb. of powder. The door really needs to be strong so as not to be blown off for seventeenth lb. mortars. Some people use little bells worn by oxen or mules as an door for the mortars[d]. Others load two canons of guns and, with a gimlet and a string, tie them with the opening against the door. If the mortars are good and do not explode, they only recoil and cannot harm if one is next to them.</ab>

<ab>A is a small rod, hooked like the hinge of a door and made as a biron[e] on one end in order to quickly attach a small mortar with handles [f]of fifteen or xvii lb. for a common door or window.</ab>

<ab>B is a small mortar with handles.</ab>

<ab>C is a common mortar of 27 or 30 lb. pierced at the bottom, where a large <m>iron</m> rod passes, button-like on one end made like a biron on the tip so as to quickly hang the mortar against a door that will not be covered with <m>iron</m> plates.</ab>

<ab>D These are slabs of <m>wax</m>, <m>cork</m> and <m>wood</m> for loading the mortar and well packing the powder.</ab>

<ab>E <m>iron</m> fork made in the form of pincers as tall as a man, used to quickly set the mortar without a rod. It must be of soft <m>iron</m> so that the sides fold easily, if needed, and to accommodate the height or width of the door.</ab>

<ab>F is another <m>iron</m> fork of the same height that supports the fork made as pincers and also sustains the mortar, and with its low claw prevents the pincers from recoiling.</ab>

<cont/>

</div>

[a]"lumi re" in the original

[b]"s'attachent" in the original.

[c]"like a" or "with a"?

[d]"pour un porte de paysant" in the original. I don't know how to understand the word "paysant."

[e]in the original

[f]"mortier   oreilles" in the original

.....

<page>168v</page>

<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f342.image</image>

<div>

<cont/>

<id>p167r_1</id>

<ab>G mortar with handles placed with its rod.</ab>

<ab>H knife-like saws to cut, if needed, a portcullis.</ab>

<ab>J the big <m>iron</m> rods that cross and tie the mortar.</ab>

<ab>K are like <m>iron</m> rods, all round and covered with cloths, like all the rest, so as to make the big <m>iron</m> rods, pierced close to the button, turn.</ab>

<ab>L axes that one must always carry to chop and to break, in the fortuitous case that the mortar has left anything entirely.</ab>

<ab>M a large <m>wood</m> mallet for knocking down what was begun and weakened by the axes and "birons[a]".</ab>

<ab>N are big wheelwright augers and birons for easily cutting a door or window by making large holes close to one another.</ab>

<ab>O crutches as tall as a man that must be carried to put under a portcullis immediately after the canon has been shot and to prevent the portcullis from falling.</ab>

<ab>P are small <m>iron</m> pincers for putting any low mortar against the bolt of a door.</ab>

</div>

[a]?

.....

<page>169r</page>

<image>http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f343.item</image>

<underline>

<ab><underline>Molding a tortoise</underline></ab>

<ab><underline>Furry animals</underline></ab>

<ab><underline>Birds</underline></ab>

<ab><underline>Leaf of vine - put</underline></ab><pl>

</pl><ab>Of the strawberry</ab>

<ab><underline>Piece molding</underline></ab><pl>

</pl><ab>Molding a vase in a chassis</ab>

<ab>Molding <la>capilli veneris</la></ab>

<ab>Molds that need to be <underline>supported by nets</underline></ab><pl>

</pl><ab>Molding a bouquet</ab>

<ab><underline>Rose carnations</underline></ab><pl>

</pl><ab><underline>Hollow molding</underline></ab>

<ab>Hollow gilded <m>silver</m></ab>

<ab><underline>Molding wort plant put</underline></ab><pl>

</pl><ab><figure>+</figure> <underline>Crayfish put and seen</underline></ab><pl>

</pl><ab>Grasshoppers</ab>

<ab>Barbots</ab>

<ab>Butterflies</ab>

<ab><underline>Flies</underline></ab><pl>

</pl><ab>Bats</ab>

<ab>Enameling thin <m>gold</m></ab>

<ab>Molding in <m>sugar</m></ab>

<ab>Molding fish</ab>

<ab><underline>Molding in <m>sulfur</m> put</underline></ab>

<ab><m>Cardboard</m></ab>

<ab><underline><figure>+</figure> <m>Wax</m> put</underline></ab>

<ab>Roughing out</ab>

<ab>Stamped medals full of <m>lead</m></ab>

<ab>Molding large carved <m>wooden</m> pieces</ab>

<ab><underline>Molding and making casts of <m>metal</m> animals</underline></ab>

<ab><figure>+</figure> <underline><m>Wheat oil</m> put and seen</underline></ab><pl>
</pl><ab><underline>Stag beetle</underline></ab>
<ab>Foot of a bittern molded</ab>
<ab><underline>Thickness of a medal</underline></ab>
<ab><underline>Very hard <m>wax</m></underline></ab>
<ab><underline>Molding the engraving on a ring</underline></ab><pl>
</pl><ab>Molding pieces of carcanets</ab>
<ab>Things that do not release from the mold ...[a][b] </ab>
<ab>Fountains</ab>
<ab>Grottoes</ab>
<ab>Sundry mosses applied[c]</ab>
<ab><underline>Molding in three chassis</underline></ab>
<ab>Tempering <m>iron</m> in order to engrave it</ab>
<ab>Etching[d] on <m>silver</m> and <m>iron</m></ab>
<ab>Stamping</ab>
<ab>White <m>enamel</m>, softening it with black calcinated <m>river pebbles</m></ab>
</underline>

[a]Text not fully legible

[b]"Choses qui ne sont point de despoui[lle]" [lle] is the no-longer illegible part. He is discussing how to deal with things that are "undercut" in the mold and thus would not release well from the mold.

[c]as in, "applied to a grotto wall"

[d]The French of which "etching" is a translation is "Graver deau forte" and therefore implies the material: aqua fortis. Should we change the translation to reflect this?

.....

<page>169v</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f344.image></image>

<div>
<id>p169v_1</id>
<head>Molding very thin like<pro>bimbalotiers[a]</pro> do</head>

<ab>They mould in <m>paper</m> or <m>cardboard</m>, because neither sandy <m>earth</m> nor <m>metal</m> would mold so neatly and would not receive. Or they engrave with <m>stone</m> from <pl>lstre</pl>, for the main side, and the reverse is in <m>paper</m> or <m>cardboard</m>.</ab>
</div>

<div>
<id>p169v_2</id>
<head>Reducing a round form into a hollow one</head>

<ab>You must firstly mold it with <m>plaster</m> as has been said before, and in the <m>plaster</m> throw in some <m>wax</m> from which will come a relief that you will repair very neatly, and will shape at your pleasure. Then you will mold this <m>wax</m> relief once more in <m>plaster</m>, or even better with your tempered sand, <figure>#</figure>, as if for a noyau. And in the latter throw in some <m>lead</m> and from this you will have a very neat relief, and if it is not, repair it on <m>cement</m> or a similar material, and beware that it should always release well. And this one will always serve you as a model for making as many hollow molds as you would like. In this way, place half of your <m>lead</m> figure which you have <m>oiled</m>, and moisten it with <m>spirits</m> as we have seen above, on your <m>clay</m> base, then mold this half with a <m>white plaster</m>. Having taken hold, and having smoothed and flattened the <m>plaster</m>, take your <m>black wax</m> mixed with <m>sulphur</m> as we have seen before, because it leaves nothing in the mold and melts quicker. Dip it in hot <m>urine</m>, and once it is softened enough, make an imprint with it on your half figure in <m>lead</m>. And you will have a very neat hollow mold. Then adapt your <m>wax</m> hollow mold into a square shape, as it pleases you, as long as you have sufficient thickness, and make sure it is well even on all sides, and then mold this hollow <m>wax</m>; and having it imprinted on one side, and having it flat and even on the other, in the same way that you would mold a flat medal in <m>wax</m> or <m>metal</m> in a noyau. And the noyau mold will give you, whether in <m>silver</m> or any other metal you wish, a figure similar to the <m>wax</m> one, which is to say, flat and even on one side, and hollow on the other. And having cleaned this hollow, and having put it in the <m>bleach</m>, if it is made of <m>silver</m>, you can put into it some <m>sugar</m> work</ab>

<ab><margin>left-top</margin>Having made a hole in your <m>clay</m> base and having placed there your half figure in <m>wax</m>, <m>oil</m> it a bit so that it comes out more easily.</ab>

<ab><margin>left-middle</margin>Transparent <m>wax</m> is not good as opposed to one that has body, for making good imprints.</ab>

<figure>
<id>fig_p169v_3</id>
<link><https://drive.google.com/open?id=0B9-oNrvWdIO5MGtBY2MyYTNKTUE></link>
<margin>left-bottom</margin>
</figure>

<ab><margin>left-bottom</margin>Because you have molded one of your halves with <m>plaster</m> and <m>wax</m>, you can do the same thing to the other half.
</ab>
<cont/>
</div>

[a]Cotgrave says this is a "paltry peddler". So is it the peddler who is paltry or a peddler who sells paltry things? We think the latter.

.....

<page>170r</page>
<image><http://gallica.bnf.fr/ark:/12148/btv1b10500001g/f345.image></image>
<div>
<cont/>
<id>p169_2</id>

<ab><m>perfume</m> and similar things. But be sure that the <m>black</m> <m>wax</m> with which you cast is hard, so it will come off rapidly when dry so you can

<ab>They have also received, as part of the revenue of the year fifteen hundred seventy nine, three hundred forty three and one third ecus.</ab>

<ab>And for the first year, they remain [to be paid] eighty livres and fifteen sous, | and for the second settlement, they have not yet received anything.</ab>

</div>
<div>
<id>p170v_a2</id>
<head>Against plague</head>

<ab>Against pestilential fever or poison or plague, take an ecu's weight of your <m>opiate</m> soaked in <m>scabious[d]</m> or <m>bugloss extract,</m> or heavily diluted <m>wine</m>, if you are already ill. After taking it, you must make yourself sweat.</ab>

</div>
<div>
<id>p170v_a3</id>
<head>To preserve oneself</head>

<ab><m>Acetum</m> paratum ex <m>ruta</m> <m>baccis juniperi</m> simul tuis Eo<lb><m>aceto</m> extinguantur <m>lateres</m> igniti. Et vapor excipiat ore [e]<lb><m>naribus. <m>Rue</m> <m>vinegar</m> together with crushed <m>juniper berries</m>. Pour the <m>vinegar</m> over red hot <m>bricks</m> and inhale the vapor through the mouth and nostrils. This is to preserve oneself when going into noxious air: a garment can be perfumed with this vapor in order to remove infection from a room, house, etc. And if you find yourself in a place where you do not have this preparation, carry <m>rue</m> and <m>berries</m>[e] crushed together, then, if need be, boil them in <m>vinegar</m> and use as described.</ab>

<ab>Ottonis episcopi | Frisigensis | Ab orbe condito[f]</ab>

<ab>Abbatis | Urspergensis | Chronicon[g]</ab>

<ab>Hieronymus | Mercurialis, | Variarum[h]</ab>

<ab>Rerum Scoticarum Historia, Georgio Bucanano autore[i]</ab>

<ab>Rembertus Dodonæus, Mechliniensis medicus, | De stirpium historia[j]</ab>

<ab>De L'Orme, De l'invention de bien bastir[k], and other works</ab>

<ab>Telesius, De coloribus, Vascosan[l]</ab>

<ab>Marbodius, De lapillis prætiosis[m]</ab>
</div>

[a]This whole page is bound in upside down. The first text block is struck through with diagonal lines (it is in a different hand and seems to be part of a legal document). The author thus seems to have re-used this sheet.

[b]Ouvrier or d'Ouvrier, a prominent family of Toulouse (or two?) from the late 14th to the 18th c. Several members served as capitouls (on the city council), others (same family?) as magistrates in the Parlement of Toulouse. I couldn't find any who had died in 1578 but In 1572, Benoit Ouvrier, a merchant and burgher of Toulouse, made his wil: <https://archive.org/stream/histoiresdesinsti04dume#page/488/mode/2up> (p. 488)

Dictionnaire universel de la noblesse de France. Vol.4 (capitouls): <http://gallica.bnf.fr/ark:/12148/bpt6k368576/f51.image.r=d%27OUVRIER.hl>

Nobiliaire Toulousain, v. 2, p. 220-222: <https://archive.org/stream/nobiliairetoulou00brgoog#page/n732/mode/2up>

[c]Haute-Garonne, arrondissement de Saint-Gaudens, canton d'Isle-en-Dodon (approx. 60 km SW of Toulouse)

[d]"scabieuse" seems to refer to several kinds of plants, not all of which belong to the genus Scabiosa in modern botany.

[e]We've assumed that "grains" translates "baccae juniperi" from the Latin above.

[f]A 12th-c. chronicle: Otto episcopus Frisingensis, Rerum ab origine mundi ad ipsius usque tempora gestarum, libri octo, Strasburg, Matthias Schürer, for Leonhard Alantsee & Lukas Alantsee, 1515... ; Chronicon, sive Rerum ab orbe condito ad sua usque tempora gestarum, libri octo.... Basel, Petrus Perna, 1569. The reference here points to the more recent edition.

[g]Chronicon Abbatis Urspergensis a Nino ... usque ad Fridericum II, Augsburg, Johannes Miller, 1515

(<http://bildsuche.digitale-sammlungen.de/index.html?c=viewer&lv=1&bandnummer=bsb00001918&pimage=00001918&suchbegriff=&l=fr>); Strasburg 1537, 1538, 1540 (Worldcat)

Earlier edn: Burchardus Urspergensis, Historia Friderici Imperatoris, sive Chronicon, Augsburg, Monastery of SS. Ulrich & Afra, 1472.(USTC)

Later: Conradi a Lichtenaw abbas Urspergensis Chronicon absolutissimum a Nino Assyriorum Rege usque ad tempora Friderici II. Imp., Basel, P. Perna, 1569. (Worldcat & Google Books)

Early 13-c. text by Burchard of Biberach, provost of Ursberg (Bavaria), rather than the abbot, Conrad of Lichtenau. (<http://www.mirabileweb.it/author/-author/20163>)

[h]Girolamo Mercuriale, Variarum lectionum libri quatuor..., Venice, Grazioso Percacino for Paolo & Antonio Meietti, 1570, etc. (USTC)

[i]George Buchanan,

Rerum Scoticarum historia, auctore Georgio Buchanano Scoto, Edinburgh, Alexander Arbuthnet, 1582. (USTC)

[j]Rembert Dodoens, Trium priorum de stirpium historia commentariorum imagines ad vivum... Antwerp, 1553 (USTC)

[k]Philibert De L’Orme, Les Nouvelles inventions pour bien bastir et a petits frais, Paris : F.Morel, 1561

[l]This seems to mix up two books (seen bound together?): Antonio Telesio, Libellus de coloribus..., Venice, Bernardino Vitali, 1528, etc.; Aristotelis seu Theophrasti De coloribus libellus, a Simone Portio Neapolitano Latinitate donatus..., Paris, Michel de Vascosan, 1549.

Telesius is often associated (bound, printed) with works of Lazare de Baïf. See: <http://crm.revues.org/12315>

[m]Marbodius or Marbodius. 11th/12th-c. Liber de lapidibus, printed Vienna 1511, etc., often reprinted throughout the 16th c. (USTC)

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